

4. (c) Kinetic energy $k = \frac{1}{2}mv^2 \Rightarrow k \propto v^2$

It means the graph between the speed and kinetic energy will be parabola

5. (c) Friction is a non-conservative external force to the system, it decreases momentum and kinetic energy both.

6. (a) Initial K.E. of block when bullet strikes to it

$$= \frac{1}{2}(m+M)V^2$$

Due to this K.E. block will rise to a height h .

Its potential energy = $(m+M)gh$.

By the law of conservation of energy

$$\frac{1}{2}(m+M)V^2 = (m+M)gh \quad \therefore V = \sqrt{2gh}$$

7. (c)

8. (c) $E = \frac{1}{2}mv^2$. Differentiating w.r.t. x , we get

$$\frac{dE}{dx} = \frac{1}{2}m \times 2v \frac{dv}{dx} = mv \times \frac{dv}{dx} \times \frac{dt}{dx} = mv \times \frac{a}{v} = ma$$

9. (d) Kinetic energy for first condition

$$= \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}m(20^2 - 10^2) = 150 \text{ mJ}$$

$$\text{K.E. for second condition} = \frac{1}{2}m(10^2 - 0^2) = 50 \text{ mJ}$$

$$\therefore \frac{(K.E.)I}{(K.E.)II} = \frac{150m}{50m} = 3$$

10. (a) By conservation of energy, $mgh = \frac{1}{2}mv^2$

$$\Rightarrow v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 1} = \sqrt{19.6} = 4.43 \text{ m/s}$$

11. (b) $P = \sqrt{2mE} \therefore P \propto \sqrt{E}$

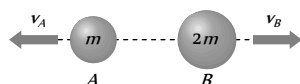
In given problem K.E. becomes 64% of the original value.

$$\frac{P_2}{P_1} = \sqrt{\frac{E_2}{E_1}} = \sqrt{\frac{64E}{100E}} = 0.8 \Rightarrow P_2 = 0.8 P$$

$\therefore P_2 = 80\%$ of the original value.

i.e. decrease in momentum is 20%.

12. (a)



By the conservation of momentum, $m_A v_A = m_B v_B$

$$\Rightarrow m \times 16 = 2m \times v_B \Rightarrow v_B = 8 \text{ m/s}$$

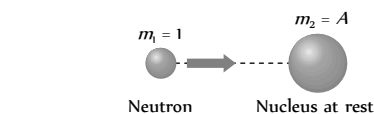
$$\text{Kinetic energy of system} = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$$

$$= \frac{1}{2} \times m \times (16)^2 + \frac{1}{2} \times (2m) \times 8^2 = 192 \text{ mJ}$$

13. (b,d) When $m_1 > m_2$ and m_2 at rest, after collision the ball of mass m_2 moves with double the velocity of u_1 . So option (b) is incorrect.

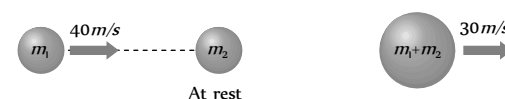
When collision is oblique and m_2 at rest with $m_1 = m_2$, after collision the ball moves in perpendicular direction. So option (d) is also incorrect.

14. (a)



$$\left(\frac{\Delta k}{k}\right)_{\text{retained}} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 = \left(\frac{1 - A}{1 + A}\right)^2$$

15. (c)



Initial momentum of the system = $m_1 \times 40 + m_2 \times 0$

Final momentum of the system = $(m_1 + m_2) \times 30$

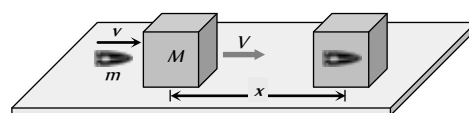
By the law of conservation of momentum

$$m_1 \times 40 + m_2 \times 0 = (m_1 + m_2) \times 30$$

$$\Rightarrow 40m_1 = 30m_1 + 30m_2 \Rightarrow 10m_1 = 30m_2 = \frac{m_1}{m_2} = 3$$

16. (b) Momentum and kinetic energy is conserved only in this case.

17. (c)



Let speed of the bullet = v

Speed of the system after the collision = V

By conservation of momentum $mv = (m+M)V$

$$\Rightarrow V = \frac{mv}{M+m}$$

So the initial K.E. acquired by the system

$$= \frac{1}{2}(M+m)V^2 = \frac{1}{2}(m+M)\left(\frac{mv}{M+m}\right)^2 = \frac{1}{2}\frac{m^2v^2}{(m+M)}$$

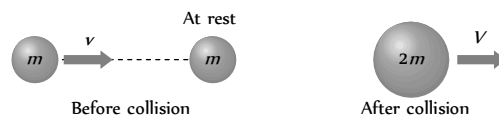
This kinetic energy goes against friction work done by friction = $\mu R \times x = \mu(m+M)g \times x$

By the law of conservation of energy

$$\frac{1}{2}\frac{m^2v^2}{(m+M)} = \mu(m+M)g \times x \Rightarrow v^2 = 2\mu g x \left(\frac{m+M}{m}\right)^2$$

$$\therefore v = \sqrt{2\mu g x \left(\frac{m+M}{m}\right)^2}$$

18. (a)



Initial momentum = mv

Final momentum = $2mV$

By the conservation of momentum, $mv = 2mV$

$$\Rightarrow V = \frac{v}{2}$$

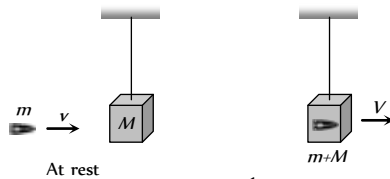
$$\text{K.E. of the system after the collision} = \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2$$

$$\therefore \text{loss in K.E.} = \frac{1}{2}mv^2 - \frac{1}{4}mv^2 = \frac{1}{4}mv^2$$

This loss in K.E. will increase the temperature

$$\therefore 2m \times s \times \Delta t = \frac{1}{4}mv^2 \Rightarrow \Delta t = \frac{v^2}{8s}$$

19. (d)



$$\text{Initial kinetic energy of bullet} = \frac{1}{2}mv^2$$

After inelastic collision system moves with velocity V

By the conservation of momentum

$$mv + 0 = (m + M)V \Rightarrow V = \frac{mv}{m + M}$$

$$\text{Kinetic energy of system} = \frac{1}{2}(m + M)V^2$$

$$= \frac{1}{2}(m + M)\left(\frac{mv}{m + M}\right)^2$$

$$\text{Loss of kinetic energy} = \frac{1}{2}mv^2 - \frac{1}{2}(m + M)\left(\frac{mv}{m + M}\right)^2$$

$$= \frac{1}{2}mv^2\left(\frac{M}{m + M}\right)$$



Chapter 8 Gravitation

Introduction

Newton at the age of twenty-three is said to have seen an apple falling down from tree in his orchid. This was the year 1665. He started thinking about the role of earth's attraction in the motion of moon and other heavenly bodies.



Fig. 8.1

By comparing the acceleration due to gravity due to earth with the acceleration required to keep the moon in its orbit around the earth, he was able to arrive the Basic Law of Gravitation.

Newton's law of Gravitation

Newton's law of gravitation states that every body in this universe attracts every other body with a force, which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres. The direction of the force is along the line joining the particles.

Thus the magnitude of the gravitational force F that two particles of masses m_1 and m_2 are separated by a distance r exert on each other is

$$\text{given by } F \propto \frac{m_1 m_2}{r^2}$$

$$\text{or } F = G \frac{m_1 m_2}{r^2}$$

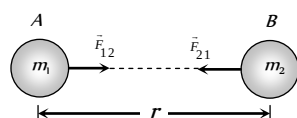


Fig. 8.2

Vector form : According to Newton's law of gravitation

$$\vec{F}_{12} = \frac{-Gm_1 m_2}{r^2} \hat{r}_{21} = \frac{-Gm_1 m_2}{r^3} \vec{r}_{21} = \frac{-Gm_1 m_2}{|\vec{r}_{21}|^3} \vec{r}_{21}$$

Here negative sign indicates that the direction of $\vec{F}_{12}$ is opposite to that of $\hat{r}_{21}$.

$$\text{Similarly } \vec{F}_{21} = \frac{-Gm_1 m_2}{r^2} \hat{r}_{12} = \frac{-Gm_1 m_2}{r^3} \vec{r}_{12} = \frac{-Gm_1 m_2}{|\vec{r}_{12}|^3} \vec{r}_{12}$$

$$= \frac{Gm_1 m_2}{r^2} \hat{r}_{21} \quad [\because \hat{r}_{12} = -\hat{r}_{21}]$$

$\therefore$ It is clear that $\vec{F}_{12} = -\vec{F}_{21}$. Which is Newton's third law of motion.

Here G is constant of proportionality which is called 'Universal gravitational constant'.

If $m_1 = m_2$ and $r = 1$ then $G = F$

i.e. universal gravitational constant is equal to the force of attraction between two bodies each of unit mass whose centres are placed unit distance apart.

(i) The value of G in the laboratory was first determined by Cavendish using the torsional balance.

(ii) The value of G is $6.67 \times 10^{-11} \text{ N-m kg}^{-2}$ in S.I. and $6.67 \times 10^{-8} \text{ dyne-cm-g}^{-2}$ in C.G.S. system.

(iii) Dimensional formula $[M^{-1}L^3T^{-2}]$.

(iv) The value of G does not depend upon the nature and size of the bodies.

(v) It also does not depend upon the nature of the medium between the two bodies.

(vi) As G is very small, hence gravitational forces are very small, unless one (or both) of the mass is huge.

Properties of Gravitational Force

(1) It is always attractive in nature while electric and magnetic force can be attractive or repulsive.

(2) It is independent of the medium between the particles while electric and magnetic force depend on the nature of the medium between the particles.

(3) It holds good over a wide range of distances. It is found true for interplanetary to inter atomic distances.

(4) It is a central force *i.e.* acts along the line joining the centres of two interacting bodies.

(5) It is a two-body interaction *i.e.* gravitational force between two particles is independent of the presence or absence of other particles; so the principle of superposition is valid *i.e.* force on a particle due to number of particles is the resultant of forces due to individual particles *i.e.*

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

While nuclear force is many body interaction

(6) It is the weakest force in nature : As $F_{\text{grav}} > F_{\text{elect}} > F_{\text{nuc}}$.

(7) The ratio of gravitational force to electrostatic force between two electrons is of the order of 10^{-43} .

(8) It is a conservative force *i.e.* work done by it is path independent or work done in moving a particle round a closed path under the action of gravitational force is zero.

(9) It is an action reaction pair *i.e.* the force with which one body (say earth) attracts the second body (say moon) is equal to the force with which moon attracts the earth. This is in accordance with Newton's third law of motion.

Note : The law of gravitation is stated for two point masses, therefore for any two arbitrary finite size bodies, as shown in the figure, It can not be applied as there is not unique value for the separation.

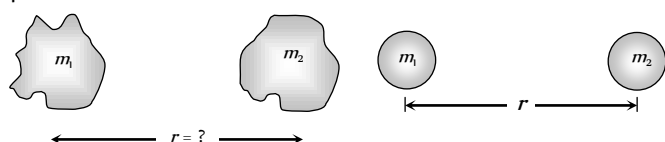


Fig. 8.3

But if the two bodies are uniform spheres then the separation r may be taken as the distance between their centres because a sphere of uniform mass behave as a point mass for any point lying outside it.

Acceleration Due to Gravity

The force of attraction exerted by the earth on a body is called gravitational pull or gravity.

We know that when force acts on a body, it produces acceleration. Therefore, a body under the effect of gravitational pull must accelerate.

The acceleration produced in the motion of a body under the effect of gravity is called acceleration due to gravity, it is denoted by g .

Consider a body of mass m is lying on the surface of earth then gravitational force on the body is given by

$$F = \frac{GMm}{R^2} \quad \dots(i)$$

Where M = mass of the earth and R = radius of the earth.

If g is the acceleration due to gravity, then the force on the body due to earth is given by

$$\text{Force} = \text{mass} \times \text{acceleration}$$

$$\text{or } F = mg \quad \dots(ii)$$

$$\text{From (i) and (ii) we have } mg = \frac{GMm}{R^2}$$

$$\therefore g = \frac{GM}{R^2} \quad \dots(iii)$$

$$\Rightarrow g = \frac{G}{R^2} \left(\frac{4}{3} \pi R^3 \rho \right)$$

$$[\text{As mass } (M) = \text{volume} \left(\frac{4}{3} \pi R^3 \right) \times \text{density } (\rho)]$$

$$\therefore g = \frac{4}{3} \pi \rho G R \quad \dots(iv)$$

$$(i) \text{ From the expression } g = \frac{GM}{R^2} = \frac{4}{3} \pi \rho G R \text{ it is clear that its}$$

value depends upon the mass radius and density of planet and it is independent of mass, shape and density of the body placed on the surface of the planet. *i.e.* a given planet (reference body) produces same acceleration in a light as well as heavy body.

(ii) The greater the value of (M/R^2) or ρR , greater will be value of g for that planet.

(iii) Acceleration due to gravity is a vector quantity and its direction is always towards the centre of the planet.

(iv) Dimension $[g] = [LT^{-2}]$

(v) its average value is taken to be 9.8 m/s or 981 cm/sec or 32 feet/sec , on the surface of the earth at mean sea level.

(vi) The value of acceleration due to gravity vary due to the following factors : (a) Shape of the earth, (b) Height above the earth surface, (c) Depth below the earth surface and (d) Axial rotation of the earth.

Variation in g Due to Shape of Earth

Earth is elliptical in shape. It is flattened at the poles and bulged out at the equator. The equatorial radius is about 21 km longer than polar radius,

$$\text{from } g = \frac{GM}{R^2}$$

$$\text{At equator } g_e = \frac{GM}{R_e^2}$$

...(i)

$$\text{At poles } g_p = \frac{GM}{R_p^2}$$

...(ii)

$$\text{From (i) and (ii) } \frac{g_e}{g_p} = \frac{R_p^2}{R_e^2}$$

$$\text{Since } R_{\text{equator}} > R_{\text{pole}}$$

$$\therefore g_{\text{pole}} > g_{\text{equator}} \text{ and } g_p = g_e + 0.018 \text{ ms}^{-2}$$

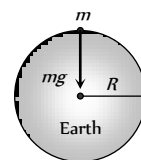


Fig. 8.4

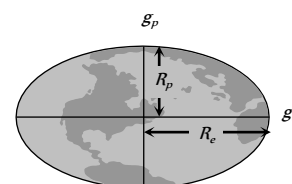


Fig. 8.5

Therefore the weight of body increases as it is taken from equator to the pole.

Variation in g With Height

Acceleration due to gravity at the surface of the earth

$$g = \frac{GM}{R^2} \quad \dots(i)$$

Acceleration due to gravity at height h from the surface of the earth

$$g' = \frac{GM}{(R+h)^2} \quad \dots(ii)$$

$$\text{From (i) and (ii) } g' = g \left(\frac{R}{R+h} \right)^2 \quad \dots(iii)$$

$$= g \frac{R^2}{r^2} \quad \dots(iv)$$

[As $r = R + h$]

(i) As we go above the surface of the earth, the value of g decreases because $g' \propto \frac{1}{r^2}$.

(ii) If $r = \infty$ then $g' = 0$, i.e., at infinite distance from the earth, the value of g becomes zero.

(iii) If $h \ll R$ i.e., height is negligible in comparison to the radius then from equation (iii) we get

$$g' = g \left(\frac{R}{R+h} \right)^2 = g \left(1 + \frac{h}{R} \right)^{-2} = g \left[1 - \frac{2h}{R} \right]$$

[As $h \ll R$]

(iv) If $h \ll R$ then decrease in the value of g with height :

$$\text{Absolute decrease } \Delta g = g - g' = \frac{2hg}{R}$$

$$\text{Fractional decrease } \frac{\Delta g}{g} = \frac{g - g'}{g} = \frac{2h}{R}$$

$$\text{Percentage decrease } \frac{\Delta g}{g} \times 100\% = \frac{2h}{R} \times 100\%$$

Variation in g With Depth

Acceleration due to gravity at the surface of the earth

$$g = \frac{GM}{R^2} = \frac{4}{3} \pi \rho G R \quad \dots(i)$$

Acceleration due to gravity at depth d from the surface of the earth

$$g' = \frac{4}{3} \pi \rho G (R-d) \quad \dots(ii)$$

$$\text{From (i) and (ii) } g' = g \left[1 - \frac{d}{R} \right]$$

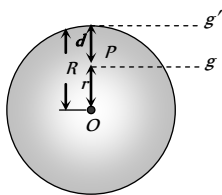


Fig. 8.7

(i) The value of g decreases on going below the surface of the earth. From equation (ii) we get $g' \propto (R-d)$.

So it is clear that if d increase, the value of g decreases.

(ii) At the centre of earth $d = R \therefore g' = 0$, i.e., the acceleration due to gravity at the centre of earth becomes zero.

(iii) Decrease in the value of g with depth

$$\text{Absolute decrease } \Delta g = g - g' = \frac{dg}{R}$$

$$\text{Fractional decrease } \frac{\Delta g}{g} = \frac{g - g'}{g} = \frac{d}{R}$$

$$\text{Percentage decrease } \frac{\Delta g}{g} \times 100\% = \frac{d}{R} \times 100\%$$

(iv) The rate of decrease of gravity outside the earth (if $h \ll R$) is double to that of inside the earth.

Variation in g Due to Rotation of Earth

As the earth rotates, a body placed on its surface moves along the circular path and hence experiences centrifugal force, due to it, the apparent weight of the body decreases.

Since the magnitude of centrifugal force varies with the latitude of the place, therefore the apparent weight of the body varies with latitude due to variation in the magnitude of centrifugal force on the body.

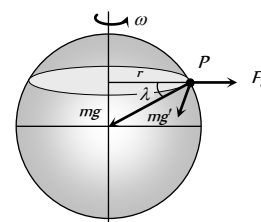


Fig. 8.8

If the body of mass m lying at point P , whose latitude is λ , then due to rotation of earth its apparent weight can be given by $\vec{mg}' = \vec{mg} + \vec{F}_c$

$$\text{or } mg' = \sqrt{(mg)^2 + (F_c)^2 + 2mg F_c \cos(180^\circ - \lambda)}$$

$$\Rightarrow mg' = \sqrt{(mg)^2 + (m\omega^2 R \cos \lambda)^2 + 2mg m\omega^2 R \cos \lambda (-\cos \lambda)}$$

$$[\text{As } F_c = m\omega^2 r = m\omega^2 R \cos \lambda]$$

$$\text{By solving we get } g' = g - \omega^2 R \cos^2 \lambda$$

Note : The latitude at a point on the surface of the earth is defined as the angle, which the line joining that point to the centre of earth makes with equatorial plane. It is denoted by λ .

For the poles $\lambda = 90^\circ$ and for equator $\lambda = 0^\circ$

(i) Substituting $\lambda = 90^\circ$ in the above expression we get $g_{\text{pole}} = g - \omega^2 R \cos^2 90^\circ$

$$\therefore g_{\text{pole}} = g \quad \dots(i)$$

i.e., there is no effect of rotational motion of the earth on the value of g at the poles.

(ii) Substituting $\lambda = 0^\circ$ in the above expression we get $g_{\text{equator}} = g - \omega^2 R \cos^2 0^\circ$

$$\therefore g_{\text{equator}} = g - \omega^2 R \quad \dots(ii)$$

i.e., the effect of rotation of earth on the value of g at the equator is maximum.

From equation (i) and (ii)

$$g_{\text{pole}} - g_{\text{equator}} = R\omega^2 = 0.034m/s^2$$

(iii) When a body of mass m is moved from the equator to the poles, its weight increases by an amount

$$m(g_p - g_e) = m\omega^2 R$$

(iv) Weightlessness due to rotation of earth : As we know that apparent weight of the body decreases due to rotation of earth. If ω is the angular velocity of rotation of earth for which a body at the equator will become weightless

$$g' = g - \omega^2 R \cos^2 \lambda$$

$$\Rightarrow 0 = g - \omega^2 R \cos^2 0^\circ \quad [\text{As } \lambda = 0^\circ \text{ for equator}]$$

$$\Rightarrow g - \omega^2 R = 0$$

$$\therefore \omega = \sqrt{\frac{g}{R}}$$

$$\text{or time period of rotation of earth } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R}{g}}$$

Substituting the value of $R = 6400 \times 10^3 \text{ m}$ and $g = 10 \text{ m/s}^2$ we get

$$\omega = \frac{1}{800} = 1.25 \times 10^{-3} \frac{\text{rad}}{\text{sec}} \text{ and } T = 5026.5 \text{ sec} = 1.40 \text{ hr.}$$

Note : $\square$ This time is about $\frac{1}{17}$ times the present time

period of earth. Therefore if earth starts rotating 17 times faster then all objects on equator will become weightless.

$\square$ If earth stops rotation about its own axis then at the equator the value of g increases by $\omega^2 R$ and consequently the weight of body lying there increases by $m\omega^2 R$.

$\square$ After considering the effect of rotation and elliptical shape of the earth, acceleration due to gravity at the poles and equator are related as

$$g_p = g_e + 0.034 + 0.018 \text{ m/s}^2 \quad \therefore g_p = g_e + 0.052 \text{ m/s}^2$$

Mass and Density of Earth

Newton's law of gravitation can be used to estimate the mass and density of the earth.

$$\text{As we know } g = \frac{GM}{R^2}, \text{ so we have } M = \frac{gR^2}{G}$$

$$\therefore M = \frac{9.8 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.98 \times 10^{24} \text{ kg} \approx 10^{25} \text{ kg}$$

$$\text{and as we know } g = \frac{4}{3} \pi \rho GR, \text{ so we have } \rho = \frac{3g}{4\pi GR}$$

$$\therefore \rho = \frac{3 \times 9.8}{4 \times 3.14 \times 6.67 \times 10^{-11} \times 6.4 \times 10^6} = 5478.4 \text{ kg/m}^3$$

Inertial and Gravitational Masses

(i) **Inertial mass** : It is the mass of the material of the body, which measures its inertia.

If an external force F acts on a body of mass m , then according to Newton's second law of motion

$$F = m_i a \text{ or } m_i = \frac{F}{a}$$

Hence inertial mass of a body may be measured as the ratio of the magnitude of the external force applied on it to the magnitude of acceleration produced in its motion.

(i) It is the measure of ability of the body to oppose the production of acceleration in its motion by an external force.

(ii) Gravity has no effect on inertial mass of the body.

(iii) It is proportional to the quantity of matter contained in the body.

(iv) It is independent of size, shape and state of body.

(v) It does not depend on the temperature of body.

(vi) It is conserved when two bodies combine physically or chemically.

(vii) When a body moves with velocity v , its inertial mass is given by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}, \text{ where } m_0 = \text{rest mass of body, } c = \text{velocity of light}$$

in vacuum,

(2) **Gravitational Mass** : It is the mass of the material of body, which determines the gravitational pull acting upon it.

If M is the mass of the earth and R is the radius, then gravitational pull on a body of mass m_g is given by

$$F = \frac{GMm_g}{R^2} \text{ or } m_g = \frac{F}{GM/R^2} = \frac{F}{I}$$

Here m_g is the gravitational mass of the body, if $I = 1$ then $m_g = F$

Thus the gravitational mass of a body is defined as the gravitational pull experienced by the body in a gravitational field of unit intensity,

(3) **Comparison between inertial and gravitational mass**

(i) Both are measured in the same units.

(ii) Both are scalar.

(iii) Both do not depend on the shape and state of the body

(iv) Inertial mass is measured by applying Newton's second law of motion where as gravitational mass is measured by applying Newton's law of gravitation.

(v) Spring balance measure gravitational mass and inertial balance measure inertial mass.

(4) **Comparison between mass and weight of the body**

Mass (m)	Weight (W)
It is a quantity of matter contained in a body.	It is the attractive force exerted by earth on any body.
Its value does not change with g	Its value changes with g .
Its value can never be zero for any material particle.	At infinity and at the centre of earth its value is zero.
Its unit is kilogram and its	Its unit is Newton or $kg\text{-wt}$ and

dimension is $[M]$.	dimension are $[MLT^{-2}]$
It is determined by a physical balance.	It is determined by a spring balance.
It is a scalar quantity.	It is a vector quantity.

Gravitational Field

The space surrounding a material body in which gravitational force of attraction can be experienced is called its gravitational field.

Gravitational field intensity : The intensity of the gravitational field of a material body at any point in its field is defined as the force experienced by a unit mass (test mass) placed at that point, provided the unit mass (test mass) itself does not produce any change in the field of the body.

So if a test mass m at a point in a gravitational field experiences a force $\vec{F}$ then

$$\vec{I} = \frac{\vec{F}}{m}$$

(i) It is a vector quantity and is always directed towards the centre of gravity of body whose gravitational field is considered.

(ii) Units : *Newton/kg* or *m/s*

(iii) Dimension : $[MLT]$

(iv) If the field is produced by a point mass M and the test mass m is at a distance r from it then by Newton's law of gravitation $F = \frac{GMm}{r^2}$, then intensity of gravitational field

$$I = \frac{F}{m} = \frac{GMm/r^2}{m}$$

$$\therefore I = \frac{GM}{r^2}$$

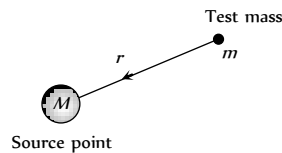


Fig. 8.9

(v) As the distance (r) of test mass from the point mass (M), increases, intensity of gravitational field decreases

$$I = \frac{GM}{r^2};$$

$$\therefore I \propto \frac{1}{r^2}$$

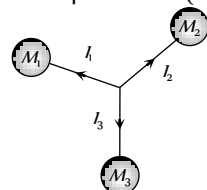


Fig. 8.10

(vi) Intensity of gravitational field $I = 0$, when $r = \infty$.

(vii) Intensity at a given point (P) due to the combined effect of different point masses can be calculated by vector sum of different intensities

$$\vec{I}_{net} = \vec{I}_1 + \vec{I}_2 + \vec{I}_3 + \dots$$

(viii) Point of zero intensity : If two bodies A and B of different masses m_1 and m_2 are d distance apart.

Let P be the point of zero intensity *i.e.*, the intensity at this point is equal and opposite due to two bodies A and B and if any test mass placed at this point it will not experience any force.

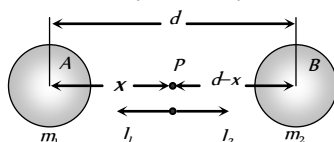
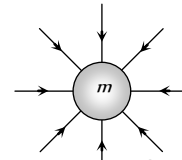


Fig. 8.11

$$\text{For point } P, \vec{I}_1 + \vec{I}_2 = 0 \Rightarrow \frac{-Gm_1}{x^2} + \frac{Gm_2}{(d-x)^2} = 0$$

$$\text{By solving } x = \frac{\sqrt{m_1} d}{\sqrt{m_1} + \sqrt{m_2}} \text{ and } (d-x) = \frac{\sqrt{m_2} d}{\sqrt{m_1} + \sqrt{m_2}}$$

(ix) Gravitational field line is a line, straight or curved such that a unit mass placed in the field of another mass would always move along this line. Field lines for an isolated mass m are radially inwards.



$$(x) \text{ As } I = \frac{GM}{r^2} \text{ and also } \frac{GM}{R^2} \therefore I = g$$

Thus the intensity of gravitational field at a point in the field is equal to acceleration of test mass placed at that point.

Gravitational Field Intensity for Different Bodies

(1) Intensity due to uniform solid sphere

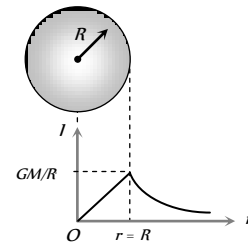


Fig. 8.13

Outside the surface $r > R$	On the surface $r = R$	Inside the surface $r < R$
$I = \frac{GM}{r^2}$	$I = \frac{GM}{R^2}$	$I = \frac{GM r}{R^3}$

(2) Intensity due to spherical shell

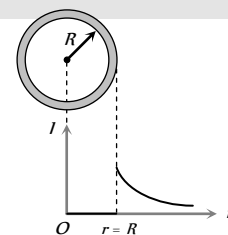


Fig. 8.14

Outside the surface $r > R$	On the surface $r = R$	Inside the surface $r < R$
$I = \frac{GM}{r^2}$	$I = \frac{GM}{R^2}$	$I = 0$

(3) Intensity due to uniform circular ring

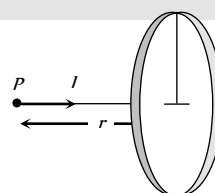
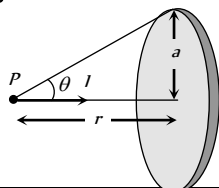


Fig. 8.15

At a point on its axis	At the centre of the ring
$I = \frac{GMr}{(a^2 + r^2)^{3/2}}$	$I = 0$

(4) Intensity due to uniform disc



At a point on its axis	Fig. 8.16	At the centre of the disc
$I = \frac{2GM}{a^2} \left[\frac{1}{r} - \frac{1}{\sqrt{r^2 + a^2}} \right]$		$I = 0$
or $I = \frac{2GM}{a^2} (1 - \cos \theta)$		

Gravitational Potential

At a point in a gravitational field potential V is defined as negative of work done per unit mass in shifting a test mass from some reference point (usually at infinity) to the given point i.e.,

$$V = -\frac{W}{m} = -\int \frac{\vec{F} \cdot d\vec{r}}{m} = -\int \vec{I} \cdot d\vec{r} \quad [\text{As } \frac{F}{m} = I]$$

$$\therefore I = -\frac{dV}{dr}$$

i.e., negative gradient of potential gives intensity of field or potential is a scalar function of position whose space derivative gives intensity. Negative sign indicates that the direction of intensity is in the direction where the potential decreases.

(i) It is a scalar quantity because it is defined as work done per unit mass.

(ii) Unit : *Joule/kg* or *m/sec*

(iii) Dimension : $[MLT^{-2}]$

(iv) If the field is produced by a point mass then

$$V = -\int I dr = -\int \left(-\frac{GM}{r^2} \right) dr \quad [\text{As } I = -\frac{GM}{r^2}]$$

$$\therefore V = -\frac{GM}{r} + c \quad [\text{Here } c = \text{constant of integration}]$$

Assuming reference point at ∞ and potential to be zero there we get

$$0 = -\frac{GM}{\infty} + c \Rightarrow c = 0$$

$$\therefore \text{Gravitational potential } V = -\frac{GM}{r}$$

(v) Gravitational potential difference : It is defined as the work done to move a unit mass from one point to the other in the gravitational field. The gravitational potential difference in bringing unit test mass m from point A to point B under the gravitational influence of source mass M is

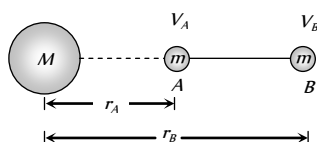


Fig. 8.17

$$\Delta V = V_B - V_A = \frac{W_{A \rightarrow B}}{m} = -GM \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$$

(vi) Potential due to large numbers of particle is given by scalar addition of all the potentials.

$$V = V_1 + V_2 + V_3 + \dots$$

$$= -\frac{GM}{r_1} - \frac{GM}{r_2} - \frac{GM}{r_3} - \dots$$

$$= -G \sum_{i=1}^{i=n} \frac{M_i}{r_i}$$

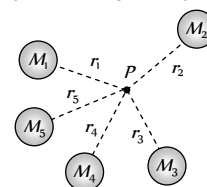


Fig. 8.18

Gravitational Potential for Different Bodies

(1) Potential due to uniform ring

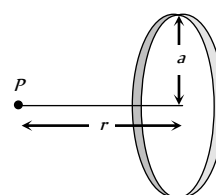


Fig. 8.19

At a point on its axis	At the centre
$V = -\frac{GM}{\sqrt{a^2 + r^2}}$	$V = -\frac{GM}{a}$

(2) Potential due to spherical shell

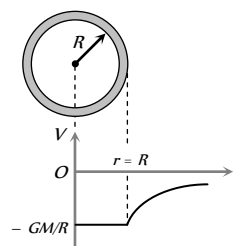


Fig. 8.20

Outside the surface $r > R$	On the surface $r = R$	Inside the surface $r < R$
$V = -\frac{GM}{r}$	$V = -\frac{GM}{R}$	$V = -\frac{GM}{R}$

(3) Potential due to uniform solid sphere

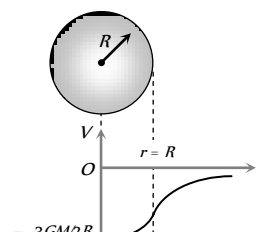
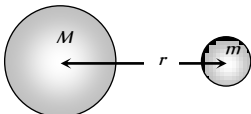


Fig. 8.21

Outside the surface $r > R$	On the surface $r = R$	Inside the surface $r < R$
$V = \frac{-GM}{r}$	$V_{\text{surface}} = \frac{-GM}{R}$	$V = \frac{-GM}{2R} \left[3 - \left(\frac{r}{R} \right)^2 \right]$ at the centre ($r = 0$) $V_{\text{centre}} = \frac{-3}{2} \frac{GM}{R}$ (max.) $V_{-} = \frac{3}{2} V_{\text{surface}}$

Gravitational Potential Energy

The gravitational potential energy of a body at a point is defined as the amount of work done in bringing the body from infinity to that point against the gravitational force.

$$W = \int_{\infty}^r \frac{GMm}{x^2} dx = -GMm \left[\frac{1}{x} \right]_{\infty}^r$$


$$W = -\frac{GMm}{r}$$

Fig. 8.22

This work done is stored inside the body as its gravitational potential energy

$$\therefore U = -\frac{GMm}{r}$$

(i) Potential energy is a scalar quantity.

(ii) Unit : Joule

(iii) Dimension : $[MLT^2]$

(iv) Gravitational potential energy is always negative in the gravitational field because the force is always attractive in nature.

(v) As the distance r increases, the gravitational potential energy becomes less negative i.e., it increases.

(vi) If $r = \infty$ then it becomes zero (maximum)

(vii) In case of discrete distribution of masses

Gravitational potential energy

$$U = \sum u_i = - \left[\frac{Gm_1m_2}{r_{12}} + \frac{Gm_2m_3}{r_{23}} + \dots \right]$$

(viii) If the body of mass m is moved from a point at a distance r_1 to a point at distance r_2 ($r_1 > r_2$) then change in potential energy

$$\Delta U = \int_{r_1}^{r_2} \frac{GMm}{x^2} dx = -GMm \left[\frac{1}{x} \right]_{r_1}^{r_2}$$

$$\text{or } \Delta U = GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

As r_1 is greater than r_2 , the change in potential energy of the body will be negative. It means that if a body is brought closer to earth its potential energy decreases.

(ix) Relation between gravitational potential energy and potential

$$U = -\frac{GMm}{r} = m \left[\frac{-GM}{r} \right]$$

$$\therefore U = mV$$

(x) Gravitational potential energy at the centre of earth relative to infinity.

$$U_{\text{centre}} = m V_{\text{centre}} = m \left(-\frac{3}{2} \frac{GM}{R} \right) = -\frac{3}{2} \frac{GMm}{R}$$

(xi) Gravitational potential energy of a body at height h from the earth surface is given by

$$U_h = -\frac{GMm}{R+h} = -\frac{gR^2m}{R+h} = -\frac{mgR}{1+\frac{h}{R}}$$

Work Done Against Gravity

If the body of mass m is moved from the surface of earth to a point at distance h above the surface of earth, then change in potential energy or work done against gravity will be

$$W = \Delta U = GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$\Rightarrow W = GMm \left[\frac{1}{R} - \frac{1}{R+h} \right] \quad [\text{As } r_1 = R \text{ and } r_2 = R+h]$$

$$\Rightarrow W = \frac{GMmh}{R^2 \left(1 + \frac{h}{R} \right)} = \frac{mgh}{1 + \frac{h}{R}} \quad [\text{As } \frac{GM}{R^2} = g]$$

(i) When the distance h is not negligible and is comparable to radius of the earth, then we will use above formula.

$$(ii) \text{ If } h = nR \text{ then } W = mgR \left(\frac{n}{n+1} \right)$$

$$(iii) \text{ If } h = R \text{ then } W = \frac{1}{2} mgR$$

(iv) If h is very small as compared to radius of the earth then term h/R can be neglected

$$\text{From } W = \frac{mgh}{1+h/R} = mgh \quad \left[\text{As } \frac{h}{R} \rightarrow 0 \right]$$

Escape Velocity

The minimum velocity with which a body must be projected up so as to enable it to just overcome the gravitational pull, is known as escape velocity.

The work done to displace a body from the surface of earth ($r = R$) to infinity ($r = \infty$) is

$$W = \int_R^{\infty} \frac{GMm}{x^2} dx = -GMm \left[\frac{1}{x} \right]_R^{\infty}$$

$$\Rightarrow W = \frac{GMm}{R}$$

This work required to project the body so as to escape the gravitational pull is performed on the body by providing an equal amount of kinetic energy to it at the surface of the earth.

If v_e is the required escape velocity, then kinetic energy which should be given to the body is $\frac{1}{2}mv_e^2$

$$\therefore \frac{1}{2}mv_e^2 = \frac{GMm}{R} \Rightarrow v_e = \sqrt{\frac{2GM}{R}}$$

$$\Rightarrow v_e = \sqrt{2gR} \quad [\text{As } GM = gR^2]$$

$$\text{or } v_e = \sqrt{2 \times \frac{4}{3}\pi\rho GR \times R} \Rightarrow v_e = R\sqrt{\frac{8}{3}\pi G\rho}$$

$$[\text{As } g = \frac{4}{3}\pi\rho GR]$$

(i) Escape velocity is independent of the mass and direction of projection of the body.

(ii) Escape velocity depends on the reference body. Greater the value of (M/R) or (gR) for a planet, greater will be escape velocity.

(iii) For the earth as $g = 9.8 \text{ m/s}^2$ and $R = 6400 \text{ km}$

$$\therefore v_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.2 \text{ km/sec}$$

(iv) A planet will have atmosphere if the velocity of molecule in its atmosphere $\left[v_{rms} = \sqrt{\frac{3RT}{M}} \right]$ is lesser than escape velocity. This is why earth has atmosphere (as at earth $v_{rms} < v_e$) while moon has no atmosphere (as at moon $v_{rms} > v_e$)

(v) If a body projected with velocity lesser than escape velocity ($v < v_e$), it will reach a certain maximum height and then may either move in an orbit around the planet or may fall down back to the planet.

(vi) Maximum height attained by body : Let a projection velocity of body (mass m) is v , so that it attains a maximum height h . At maximum height, the velocity of particle is zero, so kinetic energy is zero.

By the law of conservation of energy

Total energy at surface = Total energy at height h .

$$\Rightarrow -\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h}$$

$$\Rightarrow \frac{v^2}{2} = GM \left[\frac{1}{R} - \frac{1}{R+h} \right] = \frac{GMh}{R(R+h)}$$

$$\Rightarrow \frac{2GM}{v^2 R} = \frac{R+h}{h} = 1 + \frac{R}{h}$$

$$\Rightarrow h = \frac{R}{\left(\frac{2GM}{v^2 R} - 1 \right)} = \frac{R}{\frac{v_e^2}{v^2} - 1} = R \left[\frac{v^2}{v_e^2 - v^2} \right]$$

$$[\text{As } v_e = \sqrt{\frac{2GM}{R}} \therefore \frac{2GM}{R} = v_e^2]$$

(vii) If a body is projected with velocity greater than escape velocity ($v > v_e$) then by conservation of energy.

Total energy at surface = Total energy at infinite

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}m(v')^2 + 0$$

$$\text{i.e., } (v')^2 = v^2 - \frac{2GM}{R} \Rightarrow v'^2 = v^2 - v_e^2 \quad [\text{As } \frac{2GM}{R} = v_e^2]$$

$$\therefore v' = \sqrt{v^2 - v_e^2}$$

i.e., the body will move in interplanetary or inter stellar space with velocity $\sqrt{v^2 - v_e^2}$.

(viii) Energy to be given to a stationary object on the surface of earth so that its total energy becomes zero, is called escape energy.

$$\text{Total energy at the surface of the earth} = KE + PE = 0 - \frac{GMm}{R}$$

$$\therefore \text{Escape energy} = \frac{GMm}{R}$$

(ix) If the escape velocity of a body is equal to the velocity of light then from such bodies nothing can escape, not even light. Such bodies are called black holes.

The radius of a black hole is given as

$$R = \frac{2GM}{C^2}$$

$$[\text{As } C = \sqrt{\frac{2GM}{R}}, \text{ where } C \text{ is the velocity of light}]$$

Kepler's Laws of Planetary Motion

Planets are large natural bodies rotating around a star in definite orbits. The planetary system of the star sun called solar system consists of nine planets, viz., Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune and Pluto. Out of these planets Mercury is the smallest and closest to the sun and so hottest. Jupiter is largest and has maximum moons (12). Venus is closest to Earth and brightest. Kepler after a life time study, work out three empirical laws which govern the motion of these planets and are known as *Kepler's laws of planetary motion*. These are,

(1) **The law of Orbits** : Every planet moves around the sun in an elliptical orbit with sun at one of the foci.

(2) **The law of Area** : The line joining the sun to the planet sweeps out equal areas in equal interval of time. i.e. areal velocity is constant. According to this law planet will move slowly when it is farthest from sun and more rapidly when it is nearest to sun. It is similar to law of conservation of angular momentum.

$$\text{Areal velocity} = \frac{dA}{dt} = \frac{1}{2} \frac{r(vdt)}{dt} = \frac{1}{2}rv$$

$$\therefore \frac{dA}{dt} = \frac{L}{2m}$$

$$[\text{As } L = mvr ; rv = \frac{L}{m}]$$

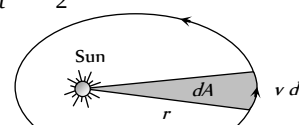


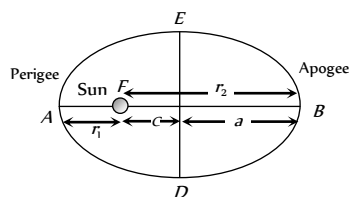
Fig. 8.23

(3) **The law of periods** : The square of period of revolution (T) of any planet around sun is directly proportional to the cube of the semi-major axis of the orbit.

$$T^2 \propto a^3 \text{ or } T^2 \propto \left(\frac{r_1 + r_2}{2} \right)^3$$

Proof : From the figure $AB = AF + FB$

$$2a = r_1 + r_2 \quad \therefore a = \frac{r_1 + r_2}{2}$$



where a = semi-major axis **Fig. 8.24**

r_1 = Shortest distance of planet from sun (perigee).

r_2 = Largest distance of planet from sun (apogee).

Important data

Planet	Semi-major axis a (10 ⁶ meter)	Period T (year)	T/a (10 ⁶ year/meter)
Mercury	5.79	0.241	2.99
Venus	10.8	0.615	3.00
Earth	15.0	1.00	2.96
Mars	22.8	1.88	2.98
Jupiter	77.8	11.9	3.01
Saturn	143	29.5	2.98
Uranus	287	84.0	2.98
Neptune	450	165	2.99
Pluto	590	248	2.99

Note: □

Kepler's laws are valid for satellites also.

Solar System



Velocity of a Planet in Terms of Eccentricity

Applying the law of conservation of angular momentum at perigee and apogee

$$mv_p r_p = mv_a r_a$$

$$\Rightarrow \frac{v_p}{v_a} = \frac{r_a}{r_p} = \frac{a+c}{a-c} = \frac{1+e}{1-e}$$

$$[\text{As } r_p = a - c, \quad r_a = a + c \text{ and eccentricity } e = \frac{c}{a}]$$

Applying the conservation of mechanical energy at perigee and apogee

$$\frac{1}{2}mv_p^2 - \frac{GMm}{r_p} = \frac{1}{2}mv_a^2 - \frac{GMm}{r_a}$$

$$\Rightarrow v_p^2 - v_a^2 = 2GM \left[\frac{1}{r_p} - \frac{1}{r_a} \right]$$

$$\Rightarrow v_a^2 \left[\frac{r_a^2 - r_p^2}{r_p^2} \right] = 2GM \left[\frac{r_a - r_p}{r_a r_p} \right] \quad [\text{As } v_p = \frac{v_a r_a}{r_p}]$$

$$\Rightarrow v_a^2 = \frac{2GM}{r_a + r_p} \left[\frac{r_p}{r_a} \right] \Rightarrow v_a^2 = \frac{2GM}{a} \left(\frac{a-c}{a+c} \right) = \frac{GM}{a} \left(\frac{1-e}{1+e} \right)$$

Thus the speeds of planet at apogee and perigee are

$$v_a = \sqrt{\frac{GM}{a} \left(\frac{1-e}{1+e} \right)},$$

$$v_p = \sqrt{\frac{GM}{a} \left(\frac{1+e}{1-e} \right)}$$

Note: □ The gravitational force is a central force so torque on planet relative to sun is always zero, hence angular momentum of a planet or satellite is always constant irrespective of shape of orbit.

Some

	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	Pluto
Mean distance from sun, km	149600	228	778	1430	2870	4500	5900
Period of revolution, year	1.00	1.88	11.9	29.5	84.0	165	248
Orbital speed, km/s	29.8	24.1	13.1	9.64	6.81	5.43	4.74
Equatorial diameter, km	12800	6790	143000	120000	51800	49500	2300
Mass (Earth =1)	1.000	0.107	318	95.1	14.5	17.2	0.002
Density (Water =1)	5.52	3.95	1.31	0.704	1.21	1.67	2.03
Surface value of g , m/s ²	9.78	3.72	22.9	9.05	7.77	11.0	0.5
Escape velocity, km/s	11.2	5.0	59.5	35.6	21.2	23.6	1.1
Known satellites	1	2	16+ring	18+rings	17+rings	8+rings	1

Fig. 8.25

Orbital Velocity of Satellite

Satellites are natural or artificial bodies describing orbit around a planet under its gravitational attraction. Moon is a natural satellite while INSAT-IB is an artificial satellite of earth. Condition for establishment of artificial satellite is that the centre of orbit of satellite must coincide with centre of earth or satellite must move around great circle of earth.

Orbital velocity of a satellite is the velocity required to put the satellite into its orbit around the earth.

For revolution of satellite around the earth, the gravitational pull provides the required centripetal force.

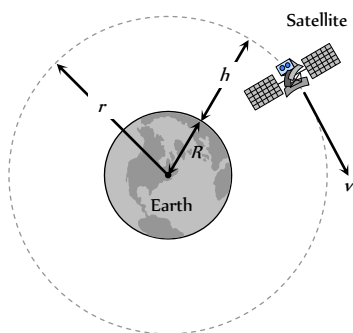


Fig. 8.26

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$\Rightarrow v = \sqrt{\frac{GM}{r}}$$

$$v = \sqrt{\frac{gR^2}{R+h}} = R\sqrt{\frac{g}{R+h}}$$

$$[\text{As } GM = gR^2 \text{ and } r = R + h]$$

(i) Orbital velocity is independent of the mass of the orbiting body and is always along the tangent of the orbit *i.e.*, satellites of different masses have same orbital velocity, if they are in the same orbit.

(ii) Orbital velocity depends on the mass of central body and radius of orbit.

(iii) For a given planet, greater the radius of orbit, lesser will be the orbital velocity of the satellite ($v \propto 1/\sqrt{r}$).

(iv) Orbital velocity of the satellite when it revolves very close to the surface of the planet

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{R+h}} \quad \therefore v = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

$$[\text{As } h = 0 \text{ and } GM = gR^2]$$

$$\text{For the earth } v = \sqrt{9.8 \times 6.4 \times 10^6} = 7.9 \text{ km/s} \approx 8 \text{ km/sec}$$

$$(v) \text{ Close to the surface of planet } v = \sqrt{\frac{GM}{R}}$$

$$[\text{As } v_e = \sqrt{\frac{2GM}{R}}]$$

$$\therefore v = \frac{v_e}{\sqrt{2}} \text{ i.e., } v_{\text{escape}} = \sqrt{2} v_{\text{orbital}}$$

It means that if the speed of a satellite orbiting close to the earth is made $\sqrt{2}$ times (or increased by 41%) then it will escape from the gravitational field.

(vi) If the gravitational force of attraction of the sun on the planet varies as $F \propto \frac{1}{r^n}$ then the orbital velocity varies as $v \propto \frac{1}{\sqrt{r^{n-1}}}$.

Time Period of Satellite

It is the time taken by satellite to go once around the earth.

$$\therefore T = \frac{\text{Circumference of the orbit}}{\text{orbital velocity}}$$

$$\Rightarrow T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r}{GM}} \quad [\text{As } v = \sqrt{\frac{GM}{r}}]$$

$$\Rightarrow T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{r^3}{gR^2}} \quad [\text{As } GM = gR^2]$$

$$\Rightarrow T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} = 2\pi \sqrt{\frac{R}{g}} \left(1 + \frac{h}{R}\right)^{3/2} \quad [\text{As } r = R + h]$$

(i) From $T = 2\pi \sqrt{\frac{r^3}{GM}}$, it is clear that time period is independent

of the mass of orbiting body and depends on the mass of central body and radius of the orbit

$$(ii) T = 2\pi \sqrt{\frac{r^3}{GM}}$$

$$\Rightarrow T^2 = \frac{4\pi^2}{GM} r^3 \text{ i.e., } T^2 \propto r^3$$

This is in accordance with Kepler's third law of planetary motion r becomes a (semi major axis) if the orbit is elliptic.

(iii) Time period of nearby satellite,

$$\text{From } T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{R^3}{gR^2}} = 2\pi \sqrt{\frac{R}{g}}$$

$$[\text{As } h = 0 \text{ and } GM = gR^2]$$

$$\text{For earth } R = 6400 \text{ km and } g = 9.8 \text{ m/s}^2$$

$$T = 84.6 \text{ minute} \approx 1.4 \text{ hr}$$

(iv) Time period of nearby satellite in terms of density of planet can be given as

$$T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{R^3}{GM}} = \frac{2\pi(R^3)^{1/2}}{\left[G \cdot \frac{4}{3}\pi R^3 \rho\right]^{1/2}} = \sqrt{\frac{3\pi}{G\rho}}$$

(v) If the gravitational force of attraction of the sun on the planet varies as $F \propto \frac{1}{r^n}$ then the time period varies as $T \propto r^{\frac{n+1}{2}}$

(vi) If there is a satellite in the equatorial plane rotating in the direction of earth's rotation from west to east, then for an observer, on the earth, angular velocity of satellite will be $(\omega_s - \omega_e)$. The time interval between the two consecutive appearances overhead will be

$$T = \frac{2\pi}{\omega_s - \omega_e} = \frac{T_s T_e}{T_e - T_s} \quad \left[\text{As } T = \frac{2\pi}{\omega}\right]$$

If $\omega_s = \omega_e$, $T = \infty$ *i.e.* satellite will appear stationary relative to earth. Such satellites are called geostationary satellites.

Height of Satellite

As we know, time period of satellite

$$T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}}$$

By squaring and rearranging both sides

$$\frac{gR^2T^2}{4\pi^2} = (R+h)^3$$

$$\Rightarrow h = \left(\frac{T^2 g R^2}{4\pi^2} \right)^{1/3} - R$$

By knowing the value of time period we can calculate the height of satellite from the surface of the earth.

Geostationary Satellite

The satellite which appears stationary relative to earth is called geostationary or geosynchronous satellite, communication satellite.

A geostationary satellite always stays over the same place above the earth such a satellite is never at rest. Such a satellite appears stationary due to its zero relative velocity *w.r.t.* that place on earth.

The orbit of a geostationary satellite is known as the parking orbit.

(i) It should revolve in an orbit concentric and coplanar with the equatorial plane.

(ii) Its sense of rotation should be same as that of earth about its own axis *i.e.*, in anti-clockwise direction (from west to east).

(iii) Its period of revolution around the earth should be same as that of earth about its own axis.

$$\therefore T = 24 \text{ hr} = 86400 \text{ sec}$$

(iv) Height of geostationary satellite

$$\text{As } T = 2\pi \sqrt{\frac{r^3}{GM}} \Rightarrow 2\pi \sqrt{\frac{(R+h)^3}{GM}} = 24 \text{ hr}$$

Substituting the value of G and M we get
 $R+h=r=42000 \text{ km} = 7R$

$\therefore$ height of geostationary satellite from the surface of earth
 $h=6R=36000 \text{ km}$

(v) Orbital velocity of geo stationary satellite can be calculated by

$$v = \sqrt{\frac{GM}{r}}$$

Substituting the value of G and M we get
 $v = 3.08 \text{ km/sec}$

Angular Momentum of Satellite

Angular momentum of satellite $L = mvr$

$$\Rightarrow L = m \sqrt{\frac{GM}{r}} r \quad [\text{As } v = \sqrt{\frac{GM}{r}}]$$

$$\therefore L = \sqrt{m^2 GM r}$$

i.e., Angular momentum of satellite depends on both the mass of orbiting and central body as well as the radius of orbit.

(i) In case of satellite motion, force is central so torque = 0 and hence angular momentum of satellite is conserved *i.e.*, $L = \text{constant}$

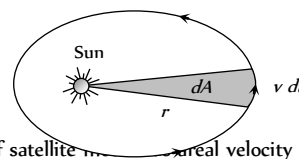


Fig. 8.27

(ii) In case of satellite motion, areal velocity

$$\frac{dA}{dt} = \frac{1}{2} \frac{(r)(v dt)}{dt} = \frac{1}{2} rv$$

$$\Rightarrow \frac{dA}{dt} = \frac{L}{2m} \quad [\text{As } L = mvr]$$

But as $L = \text{constant}$, $\therefore$ areal velocity $(dA/dt) = \text{constant}$ which is Kepler's II law

i.e., Kepler's II law or constancy of areal velocity is a consequence of conservation of angular momentum.

Energy of Satellite

When a satellite revolves around a planet in its orbit, it possesses both potential energy (due to its position against gravitational pull of earth) and kinetic energy (due to orbital motion).

(1) Potential energy : $U = mV = \frac{-GMm}{r} = \frac{-L^2}{mr^2}$

$$\left[\text{As } V = \frac{-GM}{r}, L^2 = m^2 GMr \right]$$

(2) Kinetic energy : $K = \frac{1}{2} mv^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2}$

$$\left[\text{As } v = \sqrt{\frac{GM}{r}} \right]$$

(3) Total energy :

$$E = U + K = \frac{-GMm}{r} + \frac{GMm}{2r} = \frac{-GMm}{2r} = \frac{-L^2}{2mr^2}$$

(i) Kinetic energy, potential energy or total energy of a satellite depends on the mass of the satellite and the central body and also on the radius of the orbit.

(ii) From the above expressions we can say that

Kinetic energy (K) = - (Total energy)

Potential energy (U) = 2 (Total energy)

Potential energy (K) = - 2 (Kinetic energy)

(iii) Energy graph for a satellite

(iv) Energy distribution in elliptical orbit

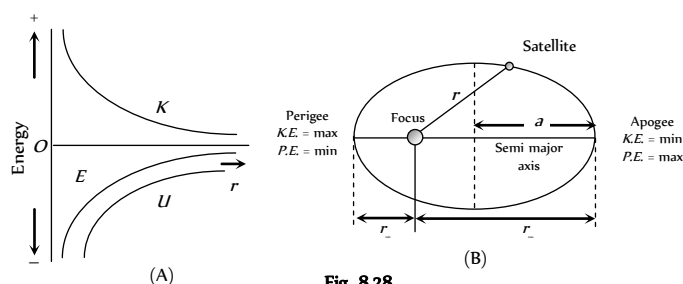


Fig. 8.28

(v) If the orbit of a satellite is elliptical then

(a) Total energy (E) = $\frac{-GMm}{2a}$ = constant ; where a is semi-major

axis .

(b) Kinetic energy (K) will be maximum when the satellite is closest to the central body (at perigee) and minimum when it is farthest from the central body (at apogee)

(c) Potential energy (U) will be minimum when kinetic energy = maximum i.e., the satellite is closest to the central body (at perigee) and maximum when kinetic energy = minimum i.e., the satellite is farthest from the central body (at apogee).

(vi) Binding Energy : Total energy of a satellite in its orbit is negative. Negative energy means that the satellite is bound to the central body by an attractive force and energy must be supplied to remove it from the orbit to infinity. The energy required to remove the satellite from its orbit to infinity is called Binding Energy of the system, i.e.,

$$\text{Binding Energy (B.E.)} = -E = \frac{GMm}{2r}$$

Change in the Orbit of Satellite

When the satellite is transferred to a higher orbit ($r_2 > r_1$) then variation in different quantities can be shown by the following table

Quantities	Variation	Relation with r
Orbital velocity	Decreases	$v \propto \frac{1}{\sqrt{r}}$
Time period	Increases	$T \propto r^{3/2}$
Linear momentum	Decreases	$P \propto \frac{1}{\sqrt{r}}$
Angular momentum	Increases	$L \propto \sqrt{r}$
Kinetic energy	Decreases	$K \propto \frac{1}{r}$
Potential energy	Increases	$U \propto -\frac{1}{r}$
Total energy	Increases	$E \propto -\frac{1}{r}$
Binding energy	Decreases	$BE \propto \frac{1}{r}$

Note: □

Work done in changing the orbit

$$W = E_2 - E_1$$

$$W = \left(-\frac{GMm}{2r_2} \right) - \left(-\frac{GMm}{2r_1} \right)$$

$$W = \frac{GMm}{2} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

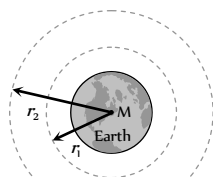


Fig. 8.29

Weightlessness

The weight of a body is the force with which it is attracted towards the centre of earth. When a body is stationary with respect to the earth, its weight equals the gravity. This weight of the body is known as its static or true weight.

We become conscious of our weight, only when our weight (which is gravity) is opposed by some other object. Actually, the secret of measuring the weight of a body with a weighing machine lies in the fact that as we place the body on the machine, the weighing machine opposes the weight of the body. The reaction of the weighing machine to the body gives the measure of the weight of the body.

The state of weightlessness can be observed in the following situations.

(1) **When objects fall freely under gravity** : For example, a lift falling freely, or an airship showing a feat in which it falls freely for a few seconds during its flight, are in state of weightlessness.

(2) **When a satellite revolves in its orbit around the earth** : Weightlessness poses many serious problems to the astronauts. It becomes quite difficult for them to control their movements. Everything in the satellite has to be kept tied down. Creation of artificial gravity is the answer to this problem.

(3) **When bodies are at null points in outer space** : On a body projected up, the pull of the earth goes on decreasing, but at the same time the gravitational pull of the moon on the body goes on increasing. At one particular position, the two gravitational pulls may be equal and opposite and the net pull on the body becomes zero. This is zero gravity region or the null point and the body in question is said to appear weightless.

Weightlessness in a Satellite

A satellite, which does not produce its own gravity moves around the earth in a circular orbit under the action of gravity. The acceleration of satellite is $\frac{GM}{r^2}$ towards the centre of earth.

If a body of mass m placed on a surface inside a satellite moving around the earth. Then force on the body are

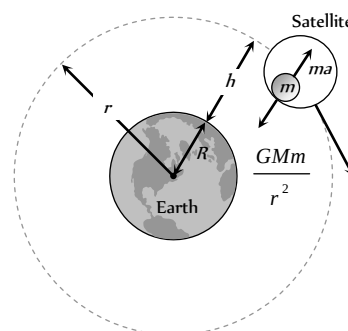


Fig. 8.30

(i) The gravitational pull of earth = $\frac{GMm}{r^2}$

(ii) The reaction by the surface = R

By Newton's law $\frac{GmM}{r^2} - R = m a$

$$\frac{GmM}{r^2} - R = m \left(\frac{GM}{r^2} \right)$$

$$\therefore R = 0$$

Thus the surface does not exert any force on the body and hence its apparent weight is zero.

A body needs no support to stay at rest in the satellite and hence all position are equally comfortable. Such a state is called weightlessness.

(i) One will find it difficult to control his movement, without weight he will tend to float freely. To get from one spot to the other he will have to push himself away from the walls or some other fixed objects.

(ii) As everything is in free fall, so objects are at rest relative to each other, *i.e.*, if a table is withdrawn from below an object, the object will remain where it was without any support.

(iii) If a glass of water is tilted and glass is pulled out, the liquid in the shape of container will float and will not flow because of surface tension.

(iv) If one tries to strike a match, the head will light but the stick will not burn. This is because in this situation convection currents will not be set up which supply oxygen for combustion

(v) If one tries to perform simple pendulum experiment, the pendulum will not oscillate. It is because there will not be any restoring torque and so $T = 2\pi\sqrt{L/g'} = \infty$. [As $g' = 0$]

(vi) Condition of weightlessness can be experienced only when the mass of satellite is negligible so that it does not produce its own gravity.

e.g. Moon is a satellite of earth but due to its own weight it applies gravitational force of attraction on the body placed on its surface and hence weight of the body will not be equal to zero at the surface of the moon.

Tips & Tricks

✍ The reference frame attached to the earth is non-inertial, because the earth revolves about its own axis as well as about the sun.

✍ Gravity holds the atmosphere around to the earth.

✍ If the earth were at one fourth the present distance from the sun, the duration of the year will be one eighth of the present year.

✍ If a packet is just released from an artificial satellite, it does not fall to the earth. On the other hand it will continue orbiting along with the satellite.

✍ Astronauts orbiting around the earth cannot use a pendulum clock, however, they can use spring clock

✍ To the astronauts in space, the sky appears black due to the absence of atmosphere above them.

✍ The gravitational force is much smaller than the electrical force because the value of G is very very small.

✍ The dimensional formula of gravitational field is same as that of acceleration due to gravity.

✍ A body in gravitational field has maximum binding energy when it is at rest.

✍ The moon is the natural satellite of the earth, but a man does not feel weightlessness on the surface of the moon. This is because, the mass of the moon is very large and it exerts a gravitational force on the man. On the other hand, the mass of the artificial satellite is very small and it exerts negligible or no gravitational force on the astronaut, so astronaut

feels weightlessness in the artificial satellite but not on the moon.

✍ The planets are heavenly bodies revolving around the sun. The sun and the nine planets, revolving around it, constitute the solar system.

✍ All other planets except mercury and pluto revolve around the sun in almost circular orbits.

✍ If the radius of planet decreases by $x\%$ keeping the mass constant. The acceleration due to gravity on its surface increases by $2x\%$.

✍ If the mass of a planet increases by $x\%$ keeping radius constant, the acceleration due to gravity on its surface increases by $x\%$.

✍ If the density of the planet decreases by $x\%$, keeping the radius constant, the acceleration due to gravity decreases by $x\%$.

✍ If the radius of the planet decreases by $x\%$, keeping the density constant, the acceleration due to gravity decreases by $x\%$.

✍ For the planets orbiting around the sun, angular speed, linear speed, kinetic energy etc. change with time but angular momentum remains constant.

✍ The ratio of inertial mass to gravitational mass is 1.

✍ Inertial mass m becomes infinite if the body moves with velocity of light.

✍ Intensity of gravitational field inside a shell is zero.

✍ If two spheres of same material, mass and radius are put in contact, the gravitational attraction between them is directly proportional to the fourth power of their radius.

✍ (a) There is no atmosphere on the moon because escape velocity on the moon is less than the *rms* velocity of the gas molecules.

(b) Two satellites are orbiting in circular orbits of radii r_1 and r_2 . Their orbital speeds are in the ratio : $v_1 / v_2 = (r_2 / r_1)^{1/2}$. It is independent to their masses

✍ Planets describe equal area around the sun in equal intervals of time.

✍ If the gravitational attraction of the sun on the planets varies as n th power of distance (of the planet from the sun), then year of the planet will be proportional to R^{-n} .

✍ An object will experience weightlessness at equator, if the angular speed of the earth about its axis becomes more than $(1/800)$ rad s .

✍ Orbital velocity very near the surface of the earth is about 7.92 *kms*.

✍ Greater the height of the satellite, smaller is the orbital velocity.

✍ Orbital velocity independent of the mass of the satellite.

✍ Orbital velocity is depends on the mass of the planet as well as radius of the orbit.

✍ If the altitude of the satellite is n times the radius of the earth, then the orbital velocity will be $(1/\sqrt{1+n})$ times the orbital velocity near the surface of the earth.

✍ If the radius of the orbit of a satellite is n times the radius of the earth, then its orbital velocity will be $(1/\sqrt{n})$ the orbital velocity near the surface of the earth.

✍ The centripetal acceleration of the satellite is equal to the

acceleration due to gravity.

✍ The gravitational potential energy of a satellite of mass m is

$$U_p = -GMm/r, \text{ where } r \text{ is the radius of the orbit of satellite.}$$

✍ Kinetic energy of the satellite

$$= \frac{GMm}{2r}$$

✍ Total energy of the satellite

$$E = U + K = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}$$

✍ When velocity of the satellite increases, its kinetic energy increases and hence total energy becomes less negative. That is the satellite begins to revolve in orbit of greater radius.

✍ If the total energy of the satellite becomes +ve, the satellite escapes from the gravitational pull of the earth.

✍ When the satellite is taken to greater height the potential energy increases (becomes less negative) and kinetic energy decreases.

✍ For the orbiting satellite, the kinetic energy is less than potential energy. When $KE = PE$, the satellite escapes away from the gravitation pull of the earth.

✍ No energy is dissipated in keeping the satellite in orbit around a planet

✍ Time period of the satellite very near the surface of the earth is about 84.6 minutes or 1.4 hr.

✍ Geo-stationary satellite is a satellite which appear stationary to the observers on the earth. It is also called geosynchronous satellite.

✍ The orbit of a geostationary satellite is known as the parking orbit.

✍ To throw an ant or an elephant out of the gravitational field, the required velocity of projection is same !

✍ Escape velocity depends on the mass and size of the planet. That is why escape velocity on the planet Jupiter is more than on the earth and escape velocity on the Moon is less than that on the earth.

✍ If a body is orbiting around the earth, then it will escape away, when its velocity is increased by 41.8%.

✍ If the radius of earth is doubled keeping the density unchanged the escape velocity will be doubled.

✍ Escape velocity $= \sqrt{2} \times$ orbital velocity.

✍ If the body is at a height h above the surface of the earth, then escape velocity is given by

$$v_{es} = \sqrt{2g(R+h)}$$

✍ It is the least velocity required by a body to escape away from the gravitational pull of the earth.

✍ Escape velocity from the surface of the earth $= \sqrt{2gR} = 11.20 \text{ kms}^{-1}$

✍ Body does not return to the earth when fired with escape velocity, irrespective of the angle of projection

✍ The escape velocity from the moon is 2.4 kms.

✍ When a projectile is fired with velocity less than the escape velocity, the sum of its gravitational potential and kinetic energy is negative.

✍ If ratio of the radii of two planets is r and the ratio of the acceleration due to gravity on their surface is a , then ratio of escape velocities is $\sqrt{ar}$.

✍ If the radius of the earth is doubled keeping the mass unchanged, the escape velocity will become $(1/\sqrt{2})$ times the present value

✍ If a body falls freely from infinite height, then it reaches the surface of the earth with velocity 11.2 km/s

✍ When a body falls from a height h to the surface of the earth, its velocity on reaching the surface of the earth is given by

$$= \left[2gR \left(\frac{h}{R+h} \right) \right]^{1/2}$$

When $h \ll R$, we find : $v = \sqrt{2gh}$

✍ The tail of the comets points away from the sun due to the radiation pressure the sun.

Ordinary Thinking

Objective Questions

Newton's Law of Gravitation

- The tidal waves in the sea are primarily due to
 - The gravitational effect of the moon on the earth
 - The gravitational effect of the sun on the earth
 - The gravitational effect of venus on the earth
 - The atmospheric effect of the earth itself
- If there were a smaller gravitational effect, which of the following forces do you think would alter in some respect [NCERT 1978]
 - Viscous forces
 - Archimedes uplift
 - Electrostatic force
 - None of the above
- A satellite of the earth is revolving in a circular orbit with a uniform speed v . If the gravitational force suddenly disappears, the satellite will [AIIMS 1982; AIEEE 2002]
 - Continue to move with velocity v along the original orbit
 - Move with a velocity v , tangentially to the original orbit
 - Fall down with increasing velocity
 - Ultimately come to rest somewhere on the original orbit
- The atmosphere is held to the earth by [IIT 1986]
 - Winds
 - Gravity
 - Clouds
 - None of the above
- The weight of a body at the centre of the earth is [AFMC 1988]
 - Zero
 - Infinite
 - Same as on the surface of earth
 - None of the above
- If the distance between two masses is doubled, the gravitational attraction between them [CPMT 1973; AMU (Med.) 2000]
 - Is doubled
 - Becomes four times
 - Is reduced to half
 - Is reduced to a quarter
- Which of the following is the evidence to show that there must be a force acting on earth and directed towards the sun [AIIMS 1980]
 - Deviation of the falling bodies towards east
 - Revolution of the earth round the sun
 - Phenomenon of day and night
 - Apparent motion of sun round the earth
- The gravitational force between two stones of mass 1 kg each separated by a distance of 1 metre in vacuum is [DPMT 1984]
 - Zero
 - $6.675 \times 10^{-5}\text{ newton}$
 - $6.675 \times 10^{-11}\text{ newton}$
 - $6.675 \times 10^{-8}\text{ newton}$
- Two particles of equal mass go round a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is [CBSE PMT 1995; RPMT 2003]
 - $v = \frac{1}{2R} \sqrt{\frac{1}{Gm}}$
 - $v = \sqrt{\frac{Gm}{2R}}$
 - $v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$
 - $v = \sqrt{\frac{4Gm}{R}}$
- The earth (mass $= 6 \times 10^{24}\text{ kg}$) revolves round the sun with angular velocity $2 \times 10^{-7}\text{ rad/s}$ in a circular orbit of radius $1.5 \times 10^8\text{ km}$. The force exerted by the sun on the earth in newtons, is [CBSE PMT 1995; AFMC 1999; Pb. PMT 2003]
 - 18×10^{25}
 - Zero
 - 27×10^{39}
 - 36×10^{21}
- Gravitational mass is proportional to gravitational [AIIMS 1998]
 - Field
 - Force
 - Intensity
 - All of these
- The gravitational force between two point masses m_1 and m_2 at separation r is given by $F = k \frac{m_1 m_2}{r^2}$

The constant k [CPMT 1993]

 - Depends on system of units only
 - Depends on medium between masses only
 - Depends on both (a) and (b)
 - Is independent of both (a) and (b)
- The distance of the centres of moon and earth is D . The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force will be zero
 - $\frac{D}{2}$
 - $\frac{2D}{3}$
 - $\frac{4D}{3}$
 - $\frac{9D}{10}$
- Who among the following gave first the experimental value of G
 - Cavendish
 - Copernicus
 - Brook Teylor
 - None of these
- The mass of the moon is $7.34 \times 10^{22}\text{ kg}$ and the radius is $1.74 \times 10^6\text{ m}$. The value of gravitation force will be [AMU 1999]
 - 1.45 N/kg
 - 1.55 N/kg
 - 1.75 N/kg
 - 1.62 N/kg
- The centripetal force acting on a satellite orbiting round the earth and the gravitational force of earth acting on the satellite both equal F . The net force on the satellite is [AMU 1999]
 - Zero
 - F
 - $F\sqrt{2}$
 - $2F$
- Reason of weightlessness in a satellite is [RPMT 2000]
 - Zero gravity
 - Centre of mass
 - Zero reaction force by satellite surface



- (d) None
18. Mass M is divided into two parts xM and $(1-x)M$. For a given separation, the value of x for which the gravitational attraction between the two pieces becomes maximum is [EAMCET 2001]
- (a) $\frac{1}{2}$ (b) $\frac{3}{5}$
(c) 1 (d) 2
19. The force of gravitation is [AIIMS 2002]
- (a) Repulsive (b) Electrostatic
(c) Conservative (d) Non-conservative
20. The gravitational force F_g between two objects does not depend on [RPET 2003]
- (a) Sum of the masses
(b) Product of the masses
(c) Gravitational constant
(d) Distance between the masses
21. Two sphere of mass m and M are situated in air and the gravitational force between them is F . The space around the masses is now filled with a liquid of specific gravity 3. The gravitational force will now be [CBSE PMT 2003]
- (a) F (b) $\frac{F}{3}$
(c) $\frac{F}{9}$ (d) $3F$
22. Earth binds the atmosphere because of [J&K CET 2005]
- (a) Gravity
(b) Oxygen between earth and atmosphere
(c) Both (a) and (b)
(d) None of these
23. Which of the following statements about the gravitational constant is true [Kerala PET 2005]
- (a) It is a force
(b) It has no unit
(c) It has same value in all systems of units
(d) It depends on the value of the masses
(e) It does not depend on the nature of the medium in which the bodies are kept.
24. Two identical solid copper spheres of radius R placed in contact with each other. The gravitational attraction between them is proportional to [Kerala PET 2005]
- (a) R (b) R^{-}
(c) R (d) R^+
- (c) $x = \frac{h}{2}$ (d) $x = h^2$
3. The time period of a simple pendulum on a freely moving artificial satellite is [CPMT 1984; AFMC 2002]
- (a) Zero (b) 2 sec
(c) 3 sec (d) Infinite
4. Two planets have the same average density but their radii are R_1 and R_2 . If acceleration due to gravity on these planets be g_1 and g_2 respectively, then [AIIMS 1985]
- (a) $\frac{g_1}{g_2} = \frac{R_1}{R_2}$ (b) $\frac{g_1}{g_2} = \frac{R_2}{R_1}$
(c) $\frac{g_1}{g_2} = \frac{R_1^2}{R_2^2}$ (d) $\frac{g_1}{g_2} = \frac{R_1^3}{R_2^3}$
5. An iron ball and a wooden ball of the same radius are released from a height ' H ' in vacuum. The time taken by both of them to reach the ground is [NCERT 1975; AFMC 1998]
- (a) Unequal (b) Exactly equal
(c) Roughly equal (d) Zero
6. The correct answer to above question is based on [NCERT 1975]
- (a) Acceleration due to gravity in vacuum is same irrespective of size and mass of the body
(b) Acceleration due to gravity in vacuum depends on the mass of the body
(c) There is no acceleration due to gravity in vacuum
(d) In vacuum there is resistance offered to the motion of the body and this resistance depends on the mass of the body
7. When a body is taken from the equator to the poles, its weight
- (a) Remains constant
(b) Increases
(c) Decreases
(d) Increases at N -pole and decreases at S -pole
8. A body of mass m is taken to the bottom of a deep mine. Then
- (a) Its mass increases (b) Its mass decreases
(c) Its weight increases (d) Its weight decreases
9. A body weighs 700 gm wt on the surface of the earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius is half that of the earth [CMC Vellore 1984; AFMC 2000]
- (a) 200 gm wt (b) 400 gm wt
(c) 50 gm wt (d) 300 gm wt
10. In order to find time, the astronaut orbiting in an earth satellite should use [DPMT 1982]
- (a) A pendulum clock
(b) A watch having main spring to keep it going
(c) Either a pendulum clock or a watch
(d) Neither a pendulum clock nor a watch
11. A spherical planet far out in space has a mass M_0 and diameter D_0 . A particle of mass m falling freely near the surface of this planet will experience an acceleration due to gravity which is equal to [NCERT 1983; BHU 2002]
- (a) GM_0 / D_0^2 (b) $4mGM_0 / D_0^2$

Acceleration Due to Gravity

1. Weightlessness experienced while orbiting the earth in space-ship, is the result of [NCERT 1978; DPMT 1982]
- (a) Inertia (b) Acceleration
(c) Zero gravity (d) Free fall towards earth
2. If the change in the value of ' g ' at a height h above the surface of the earth is the same as at a depth x below it, then (both x and h being much smaller than the radius of the earth)
- (a) $x = h$ (b) $x = 2h$

- (c) $4GM_0/D_0^2$ (d) GmM_0/D_0^2
12. If the earth stops rotating, the value of ' g ' at the equator will [CPMT 1986]
(a) Increase (b) Remain same
(c) Decrease (d) None of the above
13. The mass and diameter of a planet have twice the value of the corresponding parameters of earth. Acceleration due to gravity on the surface of the planet is [NCERT 1971; Pb. PMT 2000]
(a) 9.8 m/sec^2 (b) 4.9 m/sec^2
(c) 980 m/sec^2 (d) 19.6 m/sec^2
14. As we go from the equator to the poles, the value of g [CPMT 1975; AFMC 1995; AFMC 2004]
(a) Remains the same
(b) Decreases
(c) Increases
(d) Decreases upto a latitude of 45°
15. Force of gravity is least at [CPMT 1992]
(a) The equator
(b) The poles
(c) A point in between equator and any pole
(d) None of these
16. The radius of the earth is 6400 km and $g = 10 \text{ m/sec}^2$. In order that a body of 5 kg weighs zero at the equator, the angular speed of the earth is [MP PMT 1985]
(a) $1/80 \text{ radian/sec}$ (b) $1/400 \text{ radian/sec}$
(c) $1/800 \text{ radian/sec}$ (d) $1/1600 \text{ radian/sec}$
17. The value of ' g ' at a particular point is 9.8 m/s^2 . Suppose the earth suddenly shrinks uniformly to half its present size without losing any mass. The value of ' g ' at the same point (assuming that the distance of the point from the centre of earth does not shrink) will now be [NCERT 1984; DPMT 1999]
(a) 4.9 m/sec^2 (b) 3.1 m/sec^2
(c) 9.8 m/sec^2 (d) 19.6 m/sec^2
18. If R is the radius of the earth and g the acceleration due to gravity on the earth's surface, the mean density of the earth is MH CET (Med.) 1999; CBSE PMT 1995
(a) $4\pi G/3gR$ (b) $3\pi R/4gG$
(c) $3g/4\pi RG$ (d) $\pi RG/12G$
19. The weight of an object in the coal mine, sea level, at the top of the mountain are W_1 , W_2 and W_3 respectively, then [EAMCET 1990]
(a) $W_1 < W_2 > W_3$ (b) $W_1 = W_2 = W_3$
(c) $W_1 < W_2 < W_3$ (d) $W_1 > W_2 > W_3$
20. The radii of two planets are respectively R_1 and R_2 and their densities are respectively ρ_1 and ρ_2 . The ratio of the accelerations due to gravity at their surfaces is [MP PET 1994]
(a) $g_1 : g_2 = \frac{\rho_1}{R_1^2} : \frac{\rho_2}{R_2^2}$ (b) $g_1 : g_2 = R_1 R_2 : \rho_1 \rho_2$
(c) $g_1 : g_2 = R_1 \rho_2 : R_2 \rho_1$ (d) $g_1 : g_2 = R_1 \rho_1 : R_2 \rho_2$
21. The mass of the earth is 81 times that of the moon and the radius of the earth is 3.5 times that of the moon. The ratio of the acceleration due to gravity at the surface of the moon to that at the surface of the earth is [MP PMT 1994]
(a) 0.15 (b) 0.04
(c) 1 (d) 6
22. Spot the *wrong* statement :
The acceleration due to gravity ' g ' decreases if [MP PMT 1994]
(a) We go down from the surface of the earth towards its centre
(b) We go up from the surface of the earth
(c) We go from the equator towards the poles on the surface of the earth
(d) The rotational velocity of the earth is increased
23. Which of the following statements is true [Manipal MEE 1995]
(a) g is less at the earth's surface than at a height above it or a depth below it
(b) g is same at all places on the surface of the earth
(c) g has its maximum value at the equator
(d) g is greater at the poles than at the equator
24. A spring balance is graduated on sea level. If a body is weighed with this balance at consecutively increasing heights from earth's surface, the weight indicated by the balance
(a) Will go on increasing continuously
(b) Will go on decreasing continuously
(c) Will remain same
(d) Will first increase and then decrease
25. The value of g on the earth's surface is 980 cm/sec^2 . Its value at a height of 64 km from the earth's surface is [MP PMT 1995]
(a) 960.40 cm/sec^2 (b) 984.90 cm/sec^2
(c) 982.45 cm/sec^2 (d) 977.55 cm/sec^2
(Radius of the earth $R = 6400 \text{ kilometers}$)
26. Choose the correct statement from the following :
[CPMT 1990; BHU 1998; Kerala PMT 2002]
Weightlessness of an astronaut moving in a satellite is a situation of
(a) Zero g (b) No gravity
(c) Zero mass (d) Free fall
27. If the earth rotates faster than its present speed, the weight of an object will [Haryana CEE 1996]
(a) Increase at the equator but remain unchanged at the poles
(b) Decrease at the equator but remain unchanged at the poles
(c) Remain unchanged at the equator but decrease at the poles
(d) Remain unchanged at the equator but increase at the poles
28. If the earth suddenly shrinks (without changing mass) to half of its present radius, the acceleration due to gravity will be [MNR 1998]
(a) $g/2$ (b) $4g$
(c) $g/4$ (d) $2g$
29. The moon's radius is $1/4$ that of the earth and its mass is $1/80$ times that of the earth. If g represents the acceleration due to gravity on the surface of the earth, that on the surface of the moon is



MP PET 2000, 01; RPET 2000; Pb. PET 2001]

- (a) $g/4$ (b) $g/5$
(c) $g/6$ (d) $g/8$
30. R is the radius of the earth and ω is its angular velocity and g_p is the value of g at the poles. The effective value of g at the latitude $\lambda = 60^\circ$ will be equal to [MP PMT 1999]
- (a) $g_p - \frac{1}{4} R\omega^2$ (b) $g_p - \frac{3}{4} R\omega^2$
(c) $g_p - R\omega^2$ (d) $g_p + \frac{1}{4} R\omega^2$
31. The depth d at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value at the surface, is [R = radius of the earth] [MP PMT 1999; Kerala PMT 2005]
- (a) $\frac{R}{n}$ (b) $R\left(\frac{n-1}{n}\right)$
(c) $\frac{R}{n^2}$ (d) $R\left(\frac{n}{n+1}\right)$
32. At what height over the earth's pole, the free fall acceleration decreases by one percent (assume the radius of earth to be 6400 km) [KCET 1994]
- (a) 32 km (b) 80 km
(c) 1.253 km (d) 64 km
33. The diameters of two planets are in the ratio 4 : 1 and their mean densities in the ratio 1 : 2. The acceleration due to gravity on the planets will be in ratio [ISM Dhanbad 1994]
- (a) 1 : 2 (b) 2 : 3
(c) 2 : 1 (d) 4 : 1
34. At what altitude in metre will the acceleration due to gravity be 25% of that at the earth's surface (Radius of earth = R metre)
- (a) $\frac{1}{4} R$ (b) R
(c) $\frac{3}{8} R$ (d) $\frac{R}{2}$
35. If the angular speed of the earth is doubled, the value of acceleration due to gravity (g) at the north pole [EAMCET (Med.) 1995]
- (a) Doubles (b) Becomes half
(c) Remains same (d) Becomes zero
36. At the surface of a certain planet, acceleration due to gravity is one-quarter of that on earth. If a brass ball is transported to this planet, then which one of the following statements is not correct
- (a) The mass of the brass ball on this planet is a quarter of its mass as measured on earth
(b) The weight of the brass ball on this planet is a quarter of the weight as measured on earth
(c) The brass ball has the same mass on the other planet as on earth
(d) The brass ball has the same volume on the other planet as on earth
37. Weight of 1 kg becomes $1/6$ on moon. If radius of moon is 1.768×10^6 m, then the mass of moon will be [RPET 1997]
- (a) 1.99×10^{30} kg (b) 7.56×10^{22} kg
(c) 5.98×10^{24} kg (d) 7.65×10^{22} kg
38. Radius of earth is around 6000 km. The weight of body at height of 6000 km from earth surface becomes [RPMT 1997]
- (a) Half (b) One-fourth
(c) One third (d) No change
39. Let g be the acceleration due to gravity at earth's surface and K be the rotational kinetic energy of the earth. Suppose the earth's radius decreases by 2% keeping all other quantities same, then
- (a) g decreases by 2% and K decreases by 4%
(b) g decreases by 4% and K increases by 2%
(c) g increases by 4% and K increases by 4%
(d) g decreases by 4% and K increases by 4%
40. Where will it be profitable to purchase 1 kilogram sugar [RPET 1996]
- (a) At poles (b) At equator
(c) At 45° latitude (d) At 40° latitude
41. If the radius of the earth shrinks by 1.5% (mass remaining same), then the value of acceleration due to gravity changes by
- (a) 1% (b) 2%
(c) 3% (d) 4%
42. If radius of the earth contracts 2% and its mass remains the same, then weight of the body at the earth surface [CPMT 1997; KCET (Engg./Med.) 2001]
- (a) Will decrease (b) Will increase
(c) Will remain the same (d) None of these
43. If mass of a body is M on the earth surface, then the mass of the same body on the moon surface is [AIIMS 1997; RPMT 1997; JIPMER 2000]
- (a) $M/6$ (b) Zero
(c) M (d) None of these
44. Mass of moon is 7.34×10^{22} kg. If the acceleration due to gravity on the moon is 1.47 m/s^2 , the radius of the moon is [ISM Dhanbad 1994]
- ($G = 6.667 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$) [AFMC 1998]
- (a) 0.56×10^4 m (b) 1.87×10^6 m
(c) 1.92×10^6 m (d) 1.01×10^8 m
45. What should be the velocity of earth due to rotation about its own axis so that the weight at equator become $3/5$ of initial value. Radius of earth on equator is 6400 km [AMU 1999]
- (a) $7.4 \times 10^{-4} \text{ rad/sec}$ (b) $6.7 \times 10^{-4} \text{ rad/sec}$
(c) $7.8 \times 10^{-4} \text{ rad/sec}$ (d) $8.7 \times 10^{-4} \text{ rad/sec}$ [SCRA 1994]
46. Acceleration due to gravity is ' g ' on the surface of the earth. The value of acceleration due to gravity at a height of 32 km above earth's surface is (Radius of the earth = 6400 km) [KCET (Engg./Med.) 1999]
- (a) $0.9 g$ (b) $0.99 g$
(c) $0.8 g$ (d) $1.01 g$
47. At what height from the ground will the value of ' g ' be the same as that in 10 km deep mine below the surface of earth [RPET 1999]
- (a) 20 km (b) 10 km
(c) 15 km (d) 5 km
48. If the Earth losses its gravity, then for a body [BHU 1999; MHCET 2003]

- (a) Weight becomes zero, but not the mass
(b) Mass becomes zero, but not the weight
(c) Both mass and weight become zero
(d) Neither mass nor weight become zero
49. The height of the point vertically above the earth's surface, at which acceleration due to gravity becomes 1% of its value at the surface is (Radius of the earth = R)
[EAMCET (Engg.) 2000]
(a) $8R$ (b) $9R$
(c) $10R$ (d) $20R$
50. An object weights 72 N on earth. Its weight at a height of $R/2$ from earth is [AIIMS 2000]
(a) 32 N (b) 56 N
(c) 72 N (d) Zero
51. The angular velocity of the earth with which it has to rotate so that acceleration due to gravity on 60° latitude becomes zero is (Radius of earth = 6400 km . At the poles $g = 10\text{ ms}^{-2}$)
(a) $2.5 \times 10^{-3}\text{ rad/s}$ (b) $5.0 \times 10^{-1}\text{ rad/s}$
(c) $10 \times 10^1\text{ rad/s}$ (d) $7.8 \times 10^{-2}\text{ rad/s}$
52. Assuming earth to be a sphere of a uniform density, what is the value of gravitational acceleration in a mine 100 km below the earth's surface (Given $R = 6400\text{ km}$)
[AFMC 2000; Pb. PMT 2000]
(a) 9.66 m/s^2 (b) 7.64 m/s^2
(c) 5.06 m/s (d) 3.10 m/s^2
53. If radius of earth is R then the height ' h ' at which value of ' g ' becomes one-fourth is [BHU 2000]
(a) $\frac{R}{4}$ (b) $\frac{3R}{4}$
(c) R (d) $\frac{R}{8}$
54. R and r are the radii of the earth and moon respectively. ρ_e and ρ_m are the densities of earth and moon respectively. The ratio of the accelerations due to gravity on the surfaces of earth and moon is
(a) $\frac{R}{r} \frac{\rho_e}{\rho_m}$ (b) $\frac{r}{R} \frac{\rho_e}{\rho_m}$
(c) $\frac{r}{R} \frac{\rho_m}{\rho_e}$ (d) $\frac{R}{r} \frac{\rho_e}{\rho_m}$
55. If the mass of earth is 80 times of that of a planet and diameter is double that of planet and ' g ' on earth is 9.8 m/s^2 , then the value of ' g ' on that planet is
[Pb. PMT 1999; CPMT 2000]
(a) 4.9 m/s^2 (b) 0.98 m/s^2
(c) 0.49 m/s^2 (d) 49 m/s^2
56. Assume that the acceleration due to gravity on the surface of the moon is 0.2 times the acceleration due to gravity on the surface of the earth. If R_e is the maximum range of a projectile on the earth's surface, what is the maximum range on the surface of the moon for the same velocity of projection [Kerala (Engg.) 2001]
(a) $0.2 R_e$ (b) $2 R_e$
(c) $0.5 R_e$ (d) $5 R_e$
57. The angular speed of earth, so that the object on equator may appear weightless, is ($g = 10\text{ m/s}^2$, radius of earth 6400 km)
(a) $1.25 \times 10^{-3}\text{ rad/sec}$ (b) $1.56 \times 10^{-3}\text{ rad/sec}$
(c) $1.25 \times 10^{-1}\text{ rad/sec}$ (d) 1.56 rad/sec
58. At what distance from the centre of the earth, the value of acceleration due to gravity g will be half that on the surface ($R =$ radius of earth) [MP PMT 2001]
(a) $2R$ (b) R
(c) $1.414R$ (d) $0.414R$
59. If density of earth increased 4 times and its radius become half of what it is, our weight will [AMU (Engg.) 2001]
(a) Be four times its present value
(b) Be doubled
(c) Remain same
(d) Be halved [EAMCET 2000]
60. A man can jump to a height of 1.5 m on a planet A . What is the height he may be able to jump on another planet whose density and radius are, respectively, one-quarter and one-third that of planet A
(a) 1.5 m (b) 15 m
(c) 18 m (d) 28 m
61. Weight of a body is maximum at [AFMC 2001]
(a) Moon (b) Poles of earth
(c) Equator of earth (d) Centre of earth
62. What will be the acceleration due to gravity at height h if $h \gg R$. Where R is radius of earth and g is acceleration due to gravity on the surface of earth [RPET 2001]
(a) $\frac{g}{\left(1 + \frac{h}{R}\right)^2}$ (b) $g\left(1 - \frac{2h}{R}\right)$
(c) $\frac{g}{\left(1 - \frac{h}{R}\right)^2}$ (d) $g\left(1 - \frac{h}{R}\right)$
63. The acceleration due to gravity near the surface of a planet of radius R and density ρ is proportional to [EAMCET 2000]
[MP PET 2002; AIEEE 2004]
(a) $\frac{d}{R^2}$ (b) dR^2
(c) dR (d) $\frac{d}{R}$
64. The acceleration due to gravity is g at a point distant r from the centre of earth of radius R . If $r < R$, then [CPMT 2002]
(a) $g \propto r$ (b) $g \propto r^2$
(c) $g \propto r^{-1}$ (d) $g \propto r^{-2}$
65. A body weight W newton at the surface of the earth. Its weight at a height equal to half the radius of the earth will be [UPSEAT 2002]
(a) $\frac{W}{2}$ (b) $\frac{2W}{3}$
(c) $\frac{4W}{9}$ (d) $\frac{8W}{27}$

66. If the density of the earth is doubled keeping its radius constant then acceleration due to gravity will be ($g = 9.8 \text{ m/s}^2$) [Pb. PMT 2002; Orissa 2002] (a) 2% decrease (b) 0.5% decrease (c) 1% increase (d) 0.5% increase
67. The acceleration due to gravity at pole and equator can be related as (a) $g_p < g_e$ (b) $g_p = g_e = g$ (c) $g_p = g_e < g$ (d) $g_p > g_e$
68. If the value of 'g' acceleration due to gravity, at earth surface is 10 m/s^2 , its value in m/s^2 at the centre of the earth, which is assumed to be a sphere of radius 'R' metre and uniform mass density is [AIIMS 2002] (a) 5 (b) $10/R$ (c) $10/2R$ (d) Zero
69. A research satellite of mass 200 kg circles the earth in an orbit of average radius $3R/2$ where R is the radius of the earth. Assuming the gravitational pull on a mass of 1 kg on the earth's surface to be 10 N , the pull on the satellite will be [Kerala (Engg.) 2002] (a) 880 N (b) 889 N (c) 890 N (d) 892 N
70. Acceleration due to gravity on moon is $1/6$ of the acceleration due to gravity on earth. If the ratio of densities of earth (ρ_e) and moon (ρ_m) is $\left(\frac{\rho_e}{\rho_m}\right) = \frac{5}{3}$ then radius of moon R_m in terms of R_e will be (a) $\frac{5}{18} R_e$ (b) $\frac{1}{6} R_e$ (c) $\frac{3}{18} R_e$ (d) $\frac{1}{2\sqrt{3}} R_e$
71. The acceleration of a body due to the attraction of the earth (radius R) at a distance $2R$ from the surface of the earth is (g = acceleration due to gravity at the surface of the earth) [MP PET 2003] (a) $\frac{g}{9}$ (b) $\frac{g}{3}$ (c) $\frac{g}{4}$ (d) g
72. The depth at which the effective value of acceleration due to gravity is $\frac{g}{4}$ is [MP PET 2003] (a) R (b) $\frac{3R}{4}$ (c) $\frac{R}{2}$ (d) $\frac{R}{4}$
73. Weight of a body of mass m decreases by 1% when it is raised to height h above the earth's surface. If the body is taken to a depth h in a mine, change in its weight is [KCET 2003; MP PMT 2003] (a) 2% decrease (b) 0.5% decrease (c) 1% increase (d) 0.5% increase
74. If both the mass and the radius of the earth decrease by 1%, the value of the acceleration due to gravity will [MP PET 2004] (a) Decrease by 1% (b) Increase by 1% (c) Increase by 2% (d) Remain unchanged
75. The density of a newly discovered planet is twice that of earth. The acceleration due to gravity at the surface of the planet is equal to that at the surface of the earth. If the radius of the earth is R , the radius of the planet would be [CBSE PMT 2004] (a) $2R$ (b) $4R$ (c) $\frac{1}{4} R$ (d) $\frac{1}{2} R$
76. Two planets of radii in the ratio $2 : 3$ are made from the material of density in the ratio $3 : 2$. Then the ratio of acceleration due to gravity g_1 / g_2 at the surface of the two planets will be (a) 1 (b) 2.25 (c) $4/9$ (d) 0.12
77. A person will get more quantity of matter in kg-wt. at [J & K CET 2004] (a) Poles (b) At latitude of 60° (c) Equator (d) Satellite
78. At what depth below the surface of the earth, acceleration due to gravity g will be half its value 1600 km above the surface of the earth [Pb. PMT 2004] (a) $4.2 \times 10^6 \text{ m}$ (b) $3.19 \times 10^6 \text{ m}$ (c) $1.59 \times 10^6 \text{ m}$ (d) None of these
79. What should be the angular speed of earth, so that body lying on equator may appear weightlessness ($g = 10 \text{ m/s}^2$, $R = 6400 \text{ km}$) [Pb. PET 2000] (a) $\frac{1}{800} \text{ rad/s}$ (b) $\frac{1}{400} \text{ rad/s}$ (c) $\frac{1}{600} \text{ rad/s}$ (d) $\frac{1}{100} \text{ rad/s}$
80. A body weight 500 N on the surface of the earth. How much would it weigh half way below the surface of the earth [Pb. PET 2001; BHU 2004] (a) 125 N (b) 250 N (c) 500 N (d) 1000 N
81. If the density of a small planet is the same as that of earth, while the radius of the planet is 0.2 times that of the earth, the gravitational acceleration on the surface of that planet is [UPSEAT 2004; CBSE PMT 2005] (a) $0.2 g$ (b) $0.4 g$ (c) $2 g$ (d) $4 g$
82. Acceleration due to gravity 'g' for a body of mass 'm' on earth's surface is proportional to (Radius of earth= R , mass of earth= M) (a) GM/R^2 (b) m^0 (c) mM (d) $1/R^{3/2}$
83. A body has a weight 90 kg on the earth's surface, the mass of the moon is $1/9$ that of the earth's mass and its radius is $1/2$ that of the earth's radius. On the moon the weight of the body is

- (a) 45 kg (b) 202.5 kg
(c) 90 kg (d) 40 kg
84. If it is assumed that the spinning motion of earth increases, then the weight of a body on equator [RPMT 2003]
(a) Decreases (b) Remains constant
(c) Increases (d) Becomes more at poles
85. The masses of two planets are in the ratio 1 : 2. Their radii are in the ratio 1 : 2. The acceleration due to gravity on the planets are in the ratio [MH CET 2004]
(a) 1 : 2 (b) 2 : 1
(c) 3 : 5 (d) 5 : 3
86. If earth is supposed to be a sphere of radius R , if g_p is value of acceleration due to gravity at latitude of 30° and g at the equator, the value of $g - g_{30^\circ}$ is [DCE 2005]
(a) $\frac{1}{4}\omega^2 R$ (b) $\frac{3}{4}\omega^2 R$
(c) $\omega^2 R$ (d) $\frac{1}{2}\omega^2 R$
87. If M the mass of the earth and R its radius, the ratio of the gravitational acceleration and the gravitational constant is [J&K CET 2005]
(a) $\frac{R^2}{M}$ (b) $\frac{M}{R^2}$
(c) MR^2 (d) $\frac{M}{R}$
- (c) 7.60×10^4 metres (d) 1.90×10^2 metres
5. The change in potential energy, when a body of mass m is raised to a height nR from the earth's surface is (R = Radius of earth)
(a) $mgR \frac{n}{n-1}$ (b) $nmgR$
(c) $mgR \frac{n^2}{n^2+1}$ (d) $mgR \frac{n}{n+1}$
6. The masses and radii of the earth and moon are M_1, R_1 and M_2, R_2 respectively. Their centres are distance d apart. The minimum velocity with which a particle of mass m should be projected from a point midway between their centres so that it escapes to infinity is [MP PET 1997]
(a) $2\sqrt{\frac{G}{d}(M_1 + M_2)}$ (b) $2\sqrt{\frac{2G}{d}(M_1 + M_2)}$
(c) $2\sqrt{\frac{Gm}{d}(M_1 + M_2)}$ (d) $2\sqrt{\frac{Gm(M_1 + M_2)}{d(R_1 + R_2)}}$
7. If mass of earth is M , radius is R and gravitational constant is G , then work done to take 1 kg mass from earth surface to infinity will be [RPET 1997]
(a) $\sqrt{\frac{GM}{2R}}$ (b) $\frac{GM}{R}$
(c) $\sqrt{\frac{2GM}{R}}$ (d) $\frac{GM}{2R}$

Gravitation Potential, Energy and Escape Velocity

1. A body of mass m rises to height $h = R/5$ from the earth's surface, where R is earth's radius. If g is acceleration due to gravity at earth's surface, the increase in potential energy is [CPMT 1989; SCRA 1996; DPMT 2001]
(a) mgh (b) $\frac{4}{5}mgh$
(c) $\frac{5}{6}mgh$ (d) $\frac{6}{7}mgh$
2. In a gravitational field, at a point where the gravitational potential is zero [CPMT 1990]
(a) The gravitational field is necessarily zero
(b) The gravitational field is not necessarily zero
(c) Nothing can be said definitely about the gravitational field
(d) None of these
3. The gravitational field due to a mass distribution is $E = K/x^3$ in the x -direction. (K is a constant). Taking the gravitational potential to be zero at infinity, its value at a distance x is
(a) K/x (b) $K/2x$
(c) K/x^2 (d) $K/2x^2$
4. The mass of the earth is 6.00×10^{24} kg and that of the moon is 7.40×10^{22} kg. The constant of gravitation $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$. The potential energy of the system is -7.79×10^{28} joules. The mean distance between the earth and moon is [MP PMT 1995]
(a) 3.80×10^8 metres (b) 3.37×10^6 metres
8. A rocket is launched with velocity 10 km/s. If radius of earth is R , then maximum height attained by it will be [RPET 1997]
(a) $2R$ (b) $3R$
(c) $4R$ (d) $5R$
9. There are two bodies of masses 100 kg and 10000 kg separated by a distance 1 m. At what distance from the smaller body, the intensity of gravitational field will be zero [BHU 1997]
(a) $\frac{1}{9}m$ (b) $\frac{1}{10}m$
(c) $\frac{1}{11}m$ (d) $\frac{10}{11}m$
10. What is the intensity of gravitational field of the centre of a spherical shell [MP PET 1994]
(a) Gm/r^2 (b) g
(c) Zero (d) None of these
11. The gravitational potential energy of a body of mass ' m ' at the earth's surface $-mgR_e$. Its gravitational potential energy at a height R_e from the earth's surface will be (Here R_e is the radius of the earth) [AIIMS 2000; MP PET 2000; Pb. PMT 2004]
(a) $-2mgR_e$ (b) $2mgR_e$



- (c) $\frac{1}{2}mgR_e$ (d) $-\frac{1}{2}mgR_e$
12. Escape velocity of a body of 1 kg mass on a planet is 100 m/sec. Gravitational Potential energy of the body at the Planet is
(a) $-5000 J$ (b) $-1000 J$
(c) $-2400 J$ (d) $5000 J$
13. A body of mass m is placed on the earth's surface. It is taken from the earth's surface to a height $h = 3R$. The change in gravitational potential energy of the body is
[CBSE PMT 2002]
(a) $\frac{2}{3}mgR$ (b) $\frac{3}{4}mgR$
(c) $\frac{mgR}{2}$ (d) $\frac{mgR}{4}$
14. A body of mass m kg. starts falling from a point $2R$ above the Earth's surface. Its kinetic energy when it has fallen to a point ' R ' above the Earth's surface [R -Radius of Earth, M -Mass of Earth, G -Gravitational Constant]
[MP PMT 2002]
(a) $\frac{1}{2} \frac{GMm}{R}$ (b) $\frac{1}{6} \frac{GMm}{R}$
(c) $\frac{2}{3} \frac{GMm}{R}$ (d) $\frac{1}{3} \frac{GMm}{R}$
15. A body is projected vertically upwards from the surface of a planet of radius R with a velocity equal to half the escape velocity for that planet. The maximum height attained by the body is [KCET (Engg./Med.) 2002]
(a) $R/3$ (b) $R/2$
(c) $R/4$ (d) $R/5$
16. Energy required to move a body of mass m from an orbit of radius $2R$ to $3R$ is
[AIEEE 2002]
(a) $GMm/12R^2$ (b) $GMm/3R^2$
(c) $GMm/8R$ (d) $GMm/6R$
17. The kinetic energy needed to project a body of mass m from the earth surface (radius R) to infinity is
[AIEEE 2002]
(a) $mgR/2$ (b) $2 mgR$
(c) mgR (d) $mgR/4$
18. Radius of orbit of satellite of earth is R . Its kinetic energy is proportional to
[BHU 2003; CPMT 2004]
(a) $\frac{1}{R}$ (b) $\frac{1}{\sqrt{R}}$
(c) R (d) $\frac{1}{R^{3/2}}$
19. In some region, the gravitational field is zero. The gravitational potential in this region
[BVP 2003]
(a) Must be variable (b) Must be constant
(c) Cannot be zero (d) Must be zero
20. A particle falls towards earth from infinity. Its velocity on reaching the earth would be
[Orissa JEE 2003]
(a) Infinity (b) $\sqrt{2gR}$
(c) $2\sqrt{gR}$ (d) Zero
21. Gas escapes from the surface of a planet because it acquires an escape velocity. The escape velocity will depend on which of the following factors :
I. Mass of the planet
II. Mass of the particle escaping
III. Temperature of the planet
IV. Radius of the planet
Select the correct answer from the codes given below :
[SCRA 1994]
(a) I and II (b) II and IV
(c) I and IV (d) I, III and IV
22. v_e and v_p denotes the escape velocity from the earth and another planet having twice the radius and the same mean density as the earth. Then
[NCERT 1974; MP PMT 1994]
(a) $v_e = v_p$ (b) $v_e = v_p / 2$
(c) $v_e = 2v_p$ (d) $v_e = v_p / 4$
23. The escape velocity of a sphere of mass m from earth having mass M and radius R is given by
[NCERT 1981, 84; CBSE PMT 1999]
(a) $\sqrt{\frac{2GM}{R}}$ (b) $2\sqrt{\frac{GM}{R}}$
(c) $\sqrt{\frac{2GMm}{R}}$ (d) $\sqrt{\frac{GM}{R}}$
24. The escape velocity for a rocket from earth is 11.2 km/sec. Its value on a planet where acceleration due to gravity is double that on the earth and diameter of the planet is twice that of earth will be in km/sec
[NCERT 1983; CPMT 1990; MP PMT 2000; UPSEAT 1999]
(a) 11.2 (b) 5.6
(c) 22.4 (d) 53.6
25. The escape velocity from the earth is about 11 km/second. The escape velocity from a planet having twice the radius and the same mean density as the earth, is
[NCERT 1980; MP PMT 1987; MP PET 2001, 2003; AIIMS 2001; UPSEAT 1999]
(a) 22 km/sec (b) 11 km/sec
(c) 5.5 km/sec (d) 15.5 km/sec
26. A missile is launched with a velocity less than the escape velocity. The sum of its kinetic and potential energy is
[MNR 1986; MP PET 1995]
(a) Positive
(b) Negative
(c) Zero
(d) May be positive or negative depending upon its initial velocity
27. If g is the acceleration due to gravity at the earth's surface and r is the radius of the earth, the escape velocity for the body to escape out of earth's gravitational field is
[NCERT 1975; RPET 2003]
(a) gr (b) $\sqrt{2gr}$
(c) g/r (d) r/g
28. The escape velocity of a projectile from the earth is approximately [DPMT 1982, 84]
(a) 11.2 m/sec (b) 112 km/sec
(c) 11.2 km/sec (d) 11200 km/sec

29. The escape velocity of a particle of mass m varies as
[CPMT 1978; RPMT 1999; AIEEE 2002]
- (a) m^2 (b) m
(c) m^0 (d) m^{-1}
30. For the moon to cease to remain the earth's satellite, its orbital velocity has to increase by a factor of [MP PET 1994]
- (a) 2 (b) $\sqrt{2}$
(c) $1/\sqrt{2}$ (d) $\sqrt{3}$
31. The escape velocity of an object from the earth depends upon the mass of the earth (M), its mean density (ρ), its radius (R) and the gravitational constant (G). Thus the formula for escape velocity is
- (a) $v = R\sqrt{\frac{8\pi}{3}G\rho}$ (b) $v = M\sqrt{\frac{8\pi}{3}GR}$
(c) $v = \sqrt{2GMR}$ (d) $v = \sqrt{\frac{2GM}{R^2}}$
32. Escape velocity on a planet is v_e . If radius of the planet remains same and mass becomes 4 times, the escape velocity becomes [MP PMT 1996; DPMT 1999]
- (a) $4v_e$ (b) $2v_e$
(c) v_e (d) $\frac{1}{2}v_e$
33. The mass of the earth is 81 times that of the moon and the radius of the earth is 3.5 times that of the moon. The ratio of the escape velocity on the surface of earth to that on the surface of moon will be [MP PMT/PET 1998; JIPMER 2000]
- (a) 0.2 (b) 2.57
(c) 4.81 (d) 0.39
34. The escape velocity from the surface of earth is V_e . The escape velocity from the surface of a planet whose mass and radius are 3 times those of the earth will be
[MP PMT/PET 1998; JIPMER 2001, 02; Pb. PMT 2004]
- (a) V_e (b) $3V_e$
(c) $9V_e$ (d) $27V_e$
35. How much energy will be necessary for making a body of 500 kg escape from the earth
[$g = 9.8 \text{ m/s}^2$, radius of earth = $6.4 \times 10^6 \text{ m}$]
[MP PET 1999]
- (a) About $9.8 \times 10^6 \text{ J}$ (b) About $6.4 \times 10^8 \text{ J}$
(c) About $3.1 \times 10^{10} \text{ J}$ (d) About $27.4 \times 10^{12} \text{ J}$
36. The escape velocity for the earth is 11.2 km/sec . The mass of another planet is 100 times that of the earth and its radius is 4 times that of the earth. The escape velocity for this planet will be [MP PMT 1999; Pb. PMT 2002]
- (a) 112.0 km/s (b) 5.6 km/s
(c) 280.0 km/s (d) 56.0 km/s
37. The escape velocity of a planet having mass 6 times and radius 2 times as that of earth is
[CPMT 1999; MP PET 2003; Pb. PET 2002]
- (a) $\sqrt{3} V_e$ (b) $3 V_e$
(c) $\sqrt{2} V_e$ (d) $2 V_e$
38. The escape velocity of an object on a planet whose g value is 9 times on earth and whose radius is 4 times that of earth in km/s is
- (a) 67.2 (b) 33.6
(c) 16.8 (d) 25.2
39. The escape velocity on earth is 11.2 km/s . On another planet having twice radius and 8 times mass of the earth, the escape velocity will be [Bihar CMEET 1995]
- (a) 3.7 km/s (b) 11.2 km/s
(c) 22.4 km/s (d) 43.2 km/s
40. The escape velocity of a body on the surface of the earth is 11.2 km/s . If the earth's mass increases to twice its present value and the radius of the earth becomes half, the escape velocity would become [MP PMT 1998]
- (a) 5.6 km/s
(b) 11.2 km/s (remain unchanged)
(c) 22.4 km/s
(d) 44.8 km/s
41. Given mass of the moon is $1/81$ of the mass of the earth and corresponding radius is $1/4$ of the earth. If escape velocity on the earth surface is 11.2 km/s , the value of same on the surface of the moon is
[CPMT 1997; AIIMS 2000; Pb. PMT 2001]
- (a) 0.14 km/s (b) 0.5 km/s
(c) 2.5 km/s (d) 5 km/s
42. The angular velocity of rotation of star (of mass M and radius R) at which the matter start to escape from its equator will be
- (a) $\sqrt{\frac{2GM^2}{R}}$ (b) $\sqrt{\frac{2GM}{g}}$
(c) $\sqrt{\frac{2GM}{R^3}}$ (d) $\sqrt{\frac{2GR}{M}}$
43. The least velocity required to throw a body away from the surface of a planet so that it may not return is (radius of the planet is $6.4 \times 10^6 \text{ m}$, $g = 9.8 \text{ m/sec}^2$) [AMU (Engg.) 1999]
- (a) $9.8 \times 10^{-3} \text{ m/sec}$ (b) $12.8 \times 10^3 \text{ m/sec}$
(c) $9.8 \times 10^3 \text{ m/sec}$ (d) $11.2 \times 10^3 \text{ m/sec}$
44. How many times is escape velocity (V_e), of orbital velocity (V_0) for a satellite revolving near earth [RPMT 2000]
- (a) $\sqrt{2}$ times (b) 2 times
(c) 3 times (d) 4 times
45. Escape velocity on earth is 11.2 km/s . What would be the escape velocity on a planet whose mass is 1000 times and radius is 10 times that of earth [DCE 2001; DPMT 2004]
- (a) 112 km/s (b) 11.2 km/s
(c) 1.12 km/s (d) 3.7 km/s
46. If the radius of a planet is R and its density is ρ , the escape velocity from its surface will be [MP PMT 2001]
- (a) $v_e \propto \rho R$ (b) $v_e \propto \sqrt{\rho R}$
(c) $v_e \propto \frac{\sqrt{\rho}}{R}$ (d) $v_e \propto \frac{1}{\sqrt{\rho R}}$
47. Escape velocity on the earth [BHU 2001]
- (a) Is less than that on the moon
(b) Depends upon the mass of the body [EAMCET 1994]



- (c) Depends upon the direction of projection
(d) Depends upon the height from which it is projected
48. If acceleration due to gravity on the surface of a planet is two times that on surface of earth and its radius is double that of earth. Then escape velocity from the surface of that planet in comparison to earth will be [RPET 2001]
(a) $2 v_e$ (b) $3 v_e$
(c) $4 v_e$ (d) None of these
49. The escape velocity of a rocket launched from the surface of the earth [UPSEAT 2001]
(a) Does not depend on the mass of the rocket
(b) Does not depend on the mass of the earth
(c) Depends on the mass of the planet towards which it is moving
(d) Depends on the mass of the rocket
50. The ratio of the radii of planets A and B is k_1 and ratio of acceleration due to gravity on them is k_2 . The ratio of escape velocities from them will be [BHU 2002]
(a) $k_1 k_2$ (b) $\sqrt{k_1 k_2}$
(c) $\sqrt{\frac{k_1}{k_2}}$ (d) $\sqrt{\frac{k_2}{k_1}}$
51. A mass of $6 \times 10^{24} \text{ kg}$ is to be compressed in a sphere in such a way that the escape velocity from the sphere is $3 \times 10^8 \text{ m/s}$. Radius of the sphere should be ($G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$)
(a) 9 km (b) 9 m
(c) 9 cm (d) 9 mm
52. The escape velocity of a body on an imaginary planet which is thrice the radius of the earth and double the mass of the earth is (v_e is the escape velocity of earth) [Kerala (Med.) 2002]
(a) $\sqrt{2/3} v_e$ (b) $\sqrt{3/2} v_e$
(c) $\sqrt{2/3} v_e$ (d) $2/\sqrt{3} v_e$
53. Escape velocity on the surface of earth is 11.2 km/s . Escape velocity from a planet whose mass is the same as that of earth and radius $1/4$ that of earth is [CBSE PMT 2000; JIPMER 2002; BHU 2004]
(a) 2.8 km/s (b) 15.6 km/s
(c) 22.4 km/s (d) 44.8 km/s
54. The velocity with which a projectile must be fired so that it escapes earth's gravitation does not depend on [AIIMS 2003]
(a) Mass of the earth
(b) Mass of the projectile
(c) Radius of the projectile's orbit
(d) Gravitational constant
55. The radius of a planet is $\frac{1}{4}$ of earth's radius and its acceleration due to gravity is double that of earth's acceleration due to gravity. How many times will the escape velocity at the planet's surface be as compared to its value on earth's surface [BCECE 2003; MH CET 2000]
(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$
(c) $2\sqrt{2}$ (d) 2
56. The escape velocity for the earth is v_e . The escape velocity for a planet whose radius is four times and density is nine times that of the earth, is [MP PET 2003]
(a) $36 v_e$ (b) $12 v_e$
(c) $6 v_e$ (d) $20 v_e$
57. The escape velocity for a body projected vertically upwards from the surface of earth is 11 km/s . If the body is projected at an angle of 45° with the vertical, the escape velocity will be
(a) $\frac{11}{\sqrt{2}} \text{ km/s}$ (b) $11\sqrt{2} \text{ km/s}$
(c) 22 km/s (d) 11 km/s
58. If V , R and g denote respectively the escape velocity from the surface of the earth, radius of the earth, and acceleration due to gravity, then the correct equation is [MP PMT 2004]
(a) $V = \sqrt{gR}$ (b) $V = \sqrt{\frac{4}{3} gR^3}$
(c) $V = R\sqrt{g}$ (d) $V = \sqrt{2gR}$
59. The escape velocity for a body of mass 1 kg from the earth surface is 11.2 km/s . The escape velocity for a body of mass 100 kg [UPSEAT 2002] would be [DCE 2003]
(a) $11.2 \times 10^2 \text{ km/s}$ (b) 11.2 km/s
(c) $11.2 \times 10^{-2} \text{ km/s}$ (d) None of these
60. The acceleration due to gravity on a planet is same as that on earth and its radius is four times that of earth. What will be the value of escape velocity on that planet if it is v_e on earth
(a) v_e (b) $2v_e$
(c) $4v_e$ (d) $\frac{v_e}{2}$
61. If the radius of a planet is four times that of earth and the value of g is same for both, the escape velocity on the planet will be
(a) 11.2 km/s (b) 5.6 km/s
(c) 22.4 km/s (d) None
62. If the radius and acceleration due to gravity both are doubled, escape velocity of earth will become [RPMT 2002]
(a) 11.2 km/s (b) 22.4 km/s
(c) 5.6 km/s (d) 44.8 km/s
63. A planet has twice the radius but the mean density is $\frac{1}{4}$ th as compared to earth. What is the ratio of escape velocity from earth to that from the planet [MH CET 2004]
(a) 3 : 1 (b) 1 : 2
(c) 1 : 1 (d) 2 : 1
64. The escape velocity from earth is v_{es} . A body is projected with velocity $2v_{es}$ with what constant velocity will it move in the inter planetary space [DCE 2002]

- (a) v_{es} (b) $3v_{es}$
(c) $\sqrt{3}v_{es}$ (d) $\sqrt{5}v_{es}$
65. A particle of mass 10 g is kept on the surface of a uniform sphere of mass 100 kg and radius 10 cm . Find the work to be done against the gravitational force between them to take the particle far away from the sphere (you may take $G = 6.67 \times 10^{-11}\text{ Nm}^2/\text{kg}^2$)
(a) $6.67 \times 10^{-7}\text{ J}$ (b) $6.67 \times 10^{-8}\text{ J}$
(c) $13.34 \times 10^{-7}\text{ J}$ (d) $3.33 \times 10^{-7}\text{ J}$
66. For a satellite moving in an orbit around the earth, the ratio of kinetic energy to potential energy is [CBSE PMT 2005]
(a) 2 (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$
67. 3 particles each of mass m are kept at vertices of an equilateral triangle of side L . The gravitational field at centre due to these particles is [DCE 2005]
(a) Zero (b) $\frac{3GM}{L^2}$
(c) $\frac{9GM}{L^2}$ (d) $\frac{12}{\sqrt{3}} \frac{GM}{L^2}$
68. The value of escape velocity on a certain planet is 2 km/s . Then the value of orbital speed for a satellite orbiting close to its surface is
(a) 12 km/s (b) 1 km/s
(c) $\sqrt{2}\text{ km/s}$ (d) $2\sqrt{2}\text{ km/s}$
69. Four particles each of mass M , are located at the vertices of a square with side L . The gravitational potential due to this at the centre of the square is [Kerala PET 2005]
(a) $-\sqrt{32} \frac{GM}{L}$ (b) $-\sqrt{64} \frac{GM}{L^2}$
(c) Zero (d) $\sqrt{32} \frac{GM}{L}$
70. There are two planets. The ratio of radius of the two planets is K but ratio of acceleration due to gravity of both planets is g . What will be the ratio of their escape velocity [BHU 2005]
(a) $(Kg)^{1/2}$ (b) $(Kg)^{-1/2}$
(c) $(Kg)^2$ (d) $(Kg)^{-2}$
2. If r represents the radius of the orbit of a satellite of mass m moving around a planet of mass M , the velocity of the satellite is given by [CPMT 1974; MP PMT 1987; RPMT 1999]
(a) $v^2 = g \frac{M}{r}$ (b) $v^2 = \frac{GMm}{r}$
(c) $v = \frac{\sqrt{GM}}{r}$ [AIEEE 2005] (d) $v^2 = \frac{GM}{r}$
3. Select the correct statement from the following [MP PMT 1993]
(a) The orbital velocity of a satellite increases with the radius of the orbit
(b) Escape velocity of a particle from the surface of the earth depends on the speed with which it is fired
(c) The time period of a satellite does not depend on the radius of the orbit
(d) The orbital velocity is inversely proportional to the square root of the radius of the orbit
4. An earth satellite of mass m revolves in a circular orbit at a height h from the surface of the earth. R is the radius of the earth and g is acceleration due to gravity at the surface of the earth. The velocity of the satellite in the orbit is given by [NCERT 1983; AIEEE 2004]
(a) $\frac{gR^2}{R+h}$ (b) gR
(c) $\frac{gR}{R+h}$ [DCE 2005] (d) $\sqrt{\frac{gR^2}{R+h}}$
5. Consider a satellite going round the earth in an orbit. Which of the following statements is wrong [NCERT 1966]
(a) It is a freely falling body
(b) It suffers no acceleration
(c) It is moving with a constant speed
(d) Its angular momentum remains constant
6. Two satellites of masses m_1 and m_2 ($m_1 > m_2$) are revolving round the earth in circular orbits of radius r_1 and r_2 ($r_1 > r_2$) respectively. Which of the following statements is true regarding their speeds v_1 and v_2 ? [NCERT 1984; MNR 1995; BHU 1998]
(a) $v_1 = v_2$ (b) $v_1 < v_2$
(c) $v_1 > v_2$ (d) $\frac{v_1}{r_1} = \frac{v_2}{r_2}$
7. A satellite which is geostationary in a particular orbit is taken to another orbit. Its distance from the centre of earth in new orbit is 2 times that of the earlier orbit. The time period in the second orbit is [NCERT 1997]
(a) 4.8 hours (b) $48\sqrt{2}$ hours
(c) 24 hours (d) $24\sqrt{2}$ hours
8. The ratio of the K.E. required to be given to the satellite to escape earth's gravitational field to the K.E. required to be given so that the satellite moves in a circular orbit just above earth atmosphere is
(a) One (b) Two
(c) Half (d) Infinity

Motion of Satellite

1. If v_e and v_o represent the escape velocity and orbital velocity of a satellite corresponding to a circular orbit of radius R , then [CPMT 1982; MP PMT 1997; KCET (Engg./Med.) 1999; AIIMS 2002]
(a) $v_e = v_o$
(b) $\sqrt{2}v_o = v_e$
(c) $v_e = v_o/\sqrt{2}$
(d) v_e and v_o are not related



9. An astronaut orbiting the earth in a circular orbit 120 km above the surface of earth, gently drops a spoon out of space-ship. The spoon will [NCERT 1971]
- Fall vertically down to the earth
 - Move towards the moon
 - Will move along with space-ship
 - Will move in an irregular way then fall down to earth
10. The period of a satellite in a circular orbit around a planet is independent of [NCERT 1974; AIEEE 2004]
- The mass of the planet
 - The radius of the planet
 - The mass of the satellite
 - All the three parameters (a), (b) and (c)
11. If a satellite is orbiting the earth very close to its surface, then the orbital velocity mainly depends on [NCERT 1982]
- The mass of the satellite only
 - The radius of the earth only
 - The orbital radius only
 - The mass of the earth only
12. The relay satellite transmits the T.V. programme continuously from one part of the world to another because its
- Period is greater than the period of rotation of the earth
 - Period is less than the period of rotation of the earth about its axis
 - Period has no relation with the period of the earth about its axis
 - Period is equal to the period of rotation of the earth about its axis
 - Mass is less than the mass of the earth
13. Two satellites *A* and *B* go round a planet *P* in circular orbits having radii $4R$ and R respectively. If the speed of the satellite *A* is $3V$, the speed of the satellite *B* will be. [MNR 1991; AIIMS 1995; UPSEAT 2000]
- $12V$
 - $6V$
 - $\frac{4}{3}V$
 - $\frac{3}{2}V$
14. A geostationary satellite [CPMT 1990]
- Revolves about the polar axis
 - Has a time period less than that of the near earth satellite
 - Moves faster than a near earth satellite
 - Is stationary in the space
15. A small satellite is revolving near earth's surface. Its orbital velocity will be nearly [CPMT 1987; Orissa JEE 2002; JIPMER 2001, 02]
- 8 km/sec
 - 11.2 km/sec
 - 4 km/sec
 - 6 km/sec
16. A satellite revolves around the earth in an elliptical orbit. Its speed [NCERT 1981; MP PET 2001]
- Is the same at all points in the orbit
 - Is greatest when it is closest to the earth
 - Is greatest when it is farthest from the earth
 - Goes on increasing or decreasing continuously depending upon the mass of the satellite
17. The orbital velocity of an artificial satellite in a circular orbit just above the earth's surface is v . For a satellite orbiting at an altitude of half of the earth's radius, the orbital velocity is [MNR 1994]
- $\frac{3}{2}v$
 - $\sqrt{\frac{3}{2}}v$
 - $\sqrt{\frac{2}{3}}v$
 - $\frac{2}{3}v$
18. In a satellite if the time of revolution is T , then K.E. is proportional to [BHU 1995]
- $\frac{1}{T}$
 - $\frac{1}{T^2}$
 - $\frac{1}{T^3}$
 - $T^{-2/3}$
19. If the height of a satellite from the earth is negligible in comparison to the radius of the earth R , the orbital velocity of the satellite is [MP PET 1995;]
- gR
 - $gR/2$
 - $\sqrt{g/R}$
 - $\sqrt{gR}$
20. Choose the correct statement from the following : The radius of the orbit of a geostationary satellite depends upon [MNR 1984, 93]
- Mass of the satellite, its time period and the gravitational constant
 - Mass of the satellite, mass of the earth and the gravitational constant
 - Mass of the earth, mass of the satellite, time period of the satellite and the gravitational constant
 - Mass of the earth, time period of the satellite and the gravitational constant
21. Out of the following, the only incorrect statement about satellites is
- A satellite cannot move in a stable orbit in a plane passing through the earth's centre
 - Geostationary satellites are launched in the equatorial plane
 - We can use just one geostationary satellite for global communication around the globe
 - The speed of a satellite increases with an increase in the radius of its orbit
22. A satellite is moving around the earth with speed v in a circular orbit of radius r . If the orbit radius is decreased by 1%, its speed will [MP PET 1995]
- Increase by 1%
 - Increase by 0.5%
 - Decrease by 1%
 - Decrease by 0.5%
23. Orbital velocity of an artificial satellite does not depend upon
- Mass of the earth
 - Mass of the satellite
 - Radius of the earth
 - Acceleration due to gravity
24. The time period of a geostationary satellite is [EAMCET 1994; MP PMT 1999]
- 24 hours
 - 12 hours
 - 365 days
 - One month

25. Orbital velocity of earth's satellite near the surface is 7 km/s . When the radius of the orbit is 4 times than that of earth's radius, then orbital velocity in that orbit is
[EAMCET (Engg.) 1995]
- (a) 3.5 km/s (b) 7 km/s
(c) 72 km/s (d) 14 km/s
26. Two identical satellites are at R and $7R$ away from earth surface, the wrong statement is (R = Radius of earth)
[RPM T 1997]
- (a) Ratio of total energy will be 4
(b) Ratio of kinetic energies will be 4
(c) Ratio of potential energies will be 4
(d) Ratio of total energy will be 4 but ratio of potential and kinetic energies will be 2
27. For a satellite escape velocity is 11 km/s . If the satellite is launched at an angle of 60° with the vertical, then escape velocity will be [CBSE PM T 1993; RPM T 1997]
- (a) 11 km/s (b) $11\sqrt{3} \text{ km/s}$
(c) $\frac{11}{\sqrt{3}} \text{ km/s}$ (d) 33 km/s
28. The mean radius of the earth is R , its angular speed on its own axis is ω and the acceleration due to gravity at earth's surface is g . The cube of the radius of the orbit of a geostationary satellite will be
- (a) $R^2 g / \omega$ (b) $R^2 \omega^2 / g$
(c) Rg / ω^2 (d) $R^2 g / \omega^2$
29. Which one of the following statements regarding artificial satellite of the earth is incorrect [NDA 1995; MP PM T 2000]
- (a) The orbital velocity depends on the mass of the satellite
(b) A minimum velocity of 8 km/sec is required by a satellite to orbit quite close to the earth
(c) The period of revolution is large if the radius of its orbit is large
(d) The height of a geostationary satellite is about 36000 km from earth
30. A ball is dropped from a spacecraft revolving around the earth at a height of 120 km . What will happen to the ball
[CBSE PM T 1996; CPMT 2001; BHU 1999]
- (a) It will continue to move with velocity v along the original orbit of spacecraft
(b) It will move with the same speed tangentially to the spacecraft
(c) It will fall down to the earth gradually
(d) It will go very far in the space
31. A satellite whose mass is M , is revolving in circular orbit of radius r around the earth. Time of revolution of satellite is
[AMU 1999]
- (a) $T \propto \frac{r^5}{GM}$ (b) $T \propto \sqrt{\frac{r^3}{GM}}$
(c) $T \propto \sqrt{\frac{r}{GM^2/3}}$ (d) $T \propto \sqrt{\frac{r^3}{GM^1/4}}$
32. An artificial satellite is placed into a circular orbit around earth at such a height that it always remains above a definite place on the surface of earth. Its height from the surface of earth is
- (a) 6400 km (b) 4800 km
(c) 32000 km (d) 36000 km
33. The weight of an astronaut, in an artificial satellite revolving around the earth, is [BHU 1999]
- (a) Zero
(b) Equal to that on the earth
(c) More than that on the earth
(d) Less than that on the earth
34. In the following four periods [AMU 2000]
- (i) Time of revolution of a satellite just above the earth's surface (T_{st})
(ii) Period of oscillation of mass inside the tunnel bored along the diameter of the earth (T_{ma})
(iii) Period of simple pendulum having a length equal to the earth's radius in a uniform field of 9.8 N/kg (T_{sp})
(iv) Period of an infinite length simple pendulum in the earth's real gravitational field (T_{is})
- (a) $T_{st} > T_{ma}$ (b) $T_{ma} > T_{st}$
(c) $T_{sp} > T_{is}$ [CBSE PM T 1992] (d) $T_{st} = T_{ma} = T_{sp} = T_{is}$
35. The periodic time of a communication satellite is [MP PM T 2000]
- (a) 6 hours (b) 12 hours
(c) 18 hours (d) 24 hours
36. The orbital speed of an artificial satellite very close to the surface of the earth is V_o . Then the orbital speed of another artificial satellite at a height equal to three times the radius of the earth is
- (a) $4 V_o$ (b) $2 V_o$
(c) $0.5 V_o$ (d) $4 V_o$
37. Which of the following statements is correct in respect of a geostationary satellite [MP PET 2001]
- (a) It moves in a plane containing the Greenwich meridian
(b) It moves in a plane perpendicular to the celestial equatorial plane
(c) Its height above the earth's surface is about the same as the radius of the earth
(d) Its height above the earth's surface is about six times the radius of the earth
38. The distance of a geo-stationary satellite from the centre of the earth (Radius $R = 6400 \text{ km}$) is nearest to [AFMC 2001]
- (a) $5 R$ (b) $7 R$
(c) $10 R$ (d) $18 R$
39. If Gravitational constant is decreasing in time, what will remain unchanged in case of a satellite orbiting around earth
- (a) Time period (b) Orbiting radius
(c) Tangential velocity (d) Angular velocity
40. Periodic time of a satellite revolving above Earth's surface at a height equal to R , radius of Earth, is

[g is acceleration due to gravity at Earth's surface]

[MP PMT 2002]

- (a) $2\pi\sqrt{\frac{2R}{g}}$ (b) $4\sqrt{2}\pi\sqrt{\frac{R}{g}}$
(c) $2\pi\sqrt{\frac{R}{g}}$ (d) $8\pi\sqrt{\frac{R}{g}}$

41. Given radius of Earth ' R ' and length of a day ' T ' the height of a geostationary satellite is [G—Gravitational Constant, M—Mass of Earth] [MP PMT 2002]

- (a) $\left(\frac{4\pi^2 GM}{T^2}\right)^{1/3}$ (b) $\left(\frac{4\pi GM}{R^2}\right)^{1/3} - R$
(c) $\left(\frac{GMT^2}{4\pi^2}\right)^{1/3} - R$ (d) $\left(\frac{GMT^2}{4\pi^2}\right)^{1/3} + R$

42. A geo-stationary satellite is orbiting the earth at a height of $6R$ above the surface of earth, R being the radius of earth. The time period of another satellite at a height of $2.5R$ from the surface of earth is

[UPSEAT 2002; AMU (Med.) 2002; Pb. PET 2003]

- (a) 10 hr (b) $(6/\sqrt{2})\text{ hr}$
(c) 6 hr (d) $6\sqrt{2}\text{ hr}$

43. The distance between centre of the earth and moon is 384000 km . If the mass of the earth is $6 \times 10^{24}\text{ kg}$ and $G = 6.66 \times 10^{-11}\text{ Nm}^2/\text{kg}^2$. The speed of the moon is nearly

[MH CET 2002]

- (a) 1 km/sec (b) 4 km/sec
(c) 8 km/sec (d) 11.2 km/sec

44. A satellite is launched into a circular orbit of radius ' R ' around earth while a second satellite is launched into an orbit of radius $1.02R$. The percentage difference in the time periods of the two satellites is

- (a) 0.7 (b) 1.0
(c) 1.5 (d) 3

45. Where can a geostationary satellite be installed

[MP PMT 2004]

- (a) Over any city on the equator
(b) Over the north or south pole
(c) At height R above earth
(d) At the surface of earth

46. Distance of geostationary satellite from the surface of earth $\text{radius}(R_e = 6400\text{ km})$ in terms of R_e is [Pb. PET 2000]

- (a) $13.76 R_e$ (b) $10.76 R_e$
(c) $6.56 R_e$ (d) $2.56 R_e$

47. A satellite is to revolve round the earth in a circle of radius 8000 km . The speed at which this satellite be projected into an orbit, will be [Pb. PET 2002]

- (a) 3 km/s (b) 16 km/s
(c) 7.15 km/s (d) 8 km/s

48. Two satellite A and B , ratio of masses $3:1$ are in circular orbits of radii r and $4r$. Then ratio of total mechanical energy of A to B is

- (a) $1:3$ (b) $3:1$
(c) $3:4$ (d) $12:1$

49. The orbital velocity of a planet revolving close to earth's surface is

- (a) $\sqrt{2gR}$ (b) $\sqrt{gR}$
(c) $\sqrt{\frac{2g}{R}}$ (d) $\sqrt{\frac{g}{R}}$

50. If the gravitational force between two objects were proportional to $1/R$ (and not as $1/R^2$) where R is separation between them, then a particle in circular orbit under such a force would have its orbital speed v proportional to [CBSE PMT 1994; JIPMER 2001, 02]

- (a) $1/R^2$ (b) R^0
(c) R^1 (d) $1/R$

51. A satellite moves around the earth in a circular orbit of radius r with speed v . If the mass of the satellite is M , its total energy is

- (a) $-\frac{1}{2}Mv^2$ (b) $\frac{1}{2}Mv^2$
(c) $\frac{3}{2}Mv^2$ (d) Mv^2

52. A satellite with kinetic energy E_k is revolving round the earth in a circular orbit. How much more kinetic energy should be given to it so that it may just escape into outer space

- (a) E_k (b) $2E_k$
(c) $\frac{1}{2}E_k$ (d) $3E_k$

53. Potential energy of a satellite having mass ' m ' and rotating at a height of $6.4 \times 10^6\text{ m}$ from the earth surface is

[AIIMS 2000; CBSE PMT 2001; BHU 2001]

- (a) $-0.5mgR_e$ (b) $-mgR_e$
(c) $-2mgR_e$ (d) $4mgR_e$

54. When a satellite going round the earth in a circular orbit of radius r and speed v loses some of its energy, then r and v change as [JIPMER 2002; EAMCET 2003]

- (a) r and v both with increase
(b) r and v both will decrease
(c) r will decrease and v will increase
(d) r will decrease and v will decrease

55. An earth satellite S has an orbit radius which is 4 times that of a communication satellite C . The period of revolution of S is

- (a) 4 days (b) 8 days
(c) 16 days (d) 32 days

56. Which is constant for a satellite in orbit

[Bihar CMEET 1995]

- (a) Velocity (b) Angular momentum
(c) Potential energy (d) Acceleration
(e) Kinetic energy

57. If satellite is shifted towards the earth. Then time period of satellite will be [RPMT 2000]

- (a) Increase (b) Decrease
(c) Unchanged (d) Nothing can be said

58. Which of the following quantities does not depend upon the orbital radius of the satellite [DCE 2000,03]

(a) $\frac{T}{R}$ (b) $\frac{T^2}{R}$
(c) $\frac{T^2}{R^2}$ (d) $\frac{T^2}{R^3}$

59. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to four times the previous value, the new time period will become

(a) 20 hours (b) 10 hours
(c) 80 hours (d) 40 hours

60. A satellite moves round the earth in a circular orbit of radius R making one revolution per day. A second satellite moving in a circular orbit, moves round the earth once in 8 days. The radius of the orbit of the second satellite is

(a) $8R$ (b) $4R$
(c) $2R$ (d) R

61. A person sitting in a chair in a satellite feels weightless because

- (a) The earth does not attract the objects in a satellite
(b) The normal force by the chair on the person balances the earth's attraction
(c) The normal force is zero
(d) The person in satellite is not accelerated

62. Two satellites A and B go round a planet in circular orbits having radii $4R$ and R , respectively. If the speed of satellite A is $3v$, then speed of satellite B is [Pb. PET 2004]

(a) $\frac{3v}{2}$ (b) $\frac{4v}{2}$
(c) $6v$ (d) $12v$

63. If $g \propto \frac{1}{R^3}$ (instead of $\frac{1}{R^2}$), then the relation between time period of a satellite near earth's surface and radius R will be

(a) $T^2 \propto R^3$ (b) $T \propto R^2$
(c) $T^2 \propto R$ (d) $T \propto R$

64. To an astronaut in a spaceship, the sky appears

(a) Black (b) White
(c) Green (d) Blue

65. A geostationary satellite is revolving around the earth. To make it escape from gravitational field of earth, its velocity must be increased

(a) 100% (b) 41.4%
(c) 50% (d) 59.6%

66. A satellite moves in a circle around the earth. The radius of this circle is equal to one half of the radius of the moon's orbit. The satellite completes one revolution in

(a) $\frac{1}{2}$ lunar month (b) $\frac{2}{3}$ lunar month
(c) $2^{-3/2}$ lunar month (d) $2^{3/2}$ lunar month

67. A satellite of mass m is placed at a distance r from the centre of earth (mass M). The mechanical energy of the satellite is

(a) $-\frac{GMm}{r}$ (b) $\frac{GMm}{r}$

(c) $\frac{GMm}{2r}$ (d) $-\frac{GMm}{2r}$

Kepler's Laws of Planetary Motion

1. The distance of neptune and saturn from sun are nearly 10^{13} and 10^{12} meters respectively. Assuming that they move in circular orbits, their periodic times will be in the ratio [NCERT 1975; CBSE PMT 1994; MP PMT 1995]

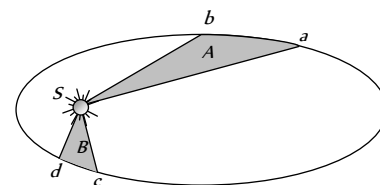
(a) $\sqrt{10}$ (b) 100
(c) $10\sqrt{10}$ (d) $1/\sqrt{10}$

2. The figure shows the motion of a planet around the sun in an elliptical orbit with sun at the focus. The shaded areas A and B are also shown in the figure which can be assumed to be equal. If t_1 and t_2 represent the time for the planet to move from a to b and d to c respectively, then

[CPMT 1986, 88]

[UPSEAT 2004]

- (a) $t_1 < t_2$
(b) $t_1 > t_2$
(c) $t_1 = t_2$
(d) $t_1 \leq t_2$



3. The period of a satellite in a circular orbit of radius R is T , the period of another satellite in a circular orbit of radius $4R$ is

[AIIMS 2000; CBSE PMT 2002]

(a) $4T$ (b) $7/4$
(c) $8T$ (d) $7/8$

4. Orbit of a planet around a star is [CPMT 1982]

- (a) A circle (b) An ellipse
(c) A parabola (d) A straight line

5. If a body describes a circular motion under inverse square field, the time taken to complete one revolution T is related to the radius of the circular orbit as

[NCERT 1975; RPMT 2000]

(a) $T \propto r$ (b) $T \propto r^2$
(c) $T^2 \propto r^3$ (d) $T \propto r^4$

6. If the earth is at one-fourth of its present distance from the sun, the duration of the year will be [EAMCET 1987]

- (a) Half the present year
(b) One-eighth the present year
(c) One-fourth the present year
(d) One-sixth the present year

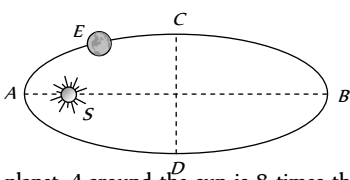
7. The earth revolves about the sun in an elliptical orbit with mean radius 9.3×10^7 m in a period of 1 year. Assuming that there are no outside influences

- (a) The earth's kinetic energy remains constant
(b) The earth's angular momentum remains constant
(c) The earth's potential energy remains constant
(d) All are correct

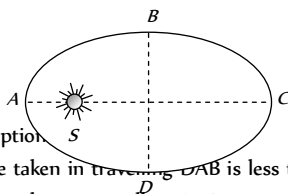
8. Venus looks brighter than other planets because

[MNR 1985]

- (a) It is heavier than other planets

- (b) It has higher density than other planets
(c) It is closer to the earth than other planets
(d) It has no atmosphere
9. A planet moves around the sun. At a given point P , it is closest from the sun at a distance d_1 and has a speed v_1 . At another point Q , when it is farthest from the sun at a distance d_2 , its speed will be [MP PMT 1987; DCE 2002]
- (a) $\frac{d_1^2 v_1}{d_2^2}$ (b) $\frac{d_2 v_1}{d_1}$
(c) $\frac{d_1 v_1}{d_2}$ (d) $\frac{d_2^2 v_1}{d_1^2}$
10. The orbital speed of Jupiter is [MNR 1986; UPSEAT 2000]
(a) Greater than the orbital speed of earth
(b) Less than the orbital speed of earth
(c) Equal to the orbital speed of earth
(d) Zero
11. Two planets move around the sun. The periodic times and the mean radii of the orbits are T_1, T_2 and r_1, r_2 respectively. The ratio T_1 / T_2 is equal to [CPMT 1978]
(a) $(r_1 / r_2)^{1/2}$ (b) r_1 / r_2
(c) $(r_1 / r_2)^2$ (d) $(r_1 / r_2)^{3/2}$
12. Kepler's second law regarding constancy of aerial velocity of a planet is a consequence of the law of conservation of [CPMT 1990; AIIMS 2002]
(a) Energy (b) Angular momentum
(c) Linear momentum (d) None of these
13. The largest and the shortest distance of the earth from the sun are r_1 and r_2 , its distance from the sun when it is at the perpendicular to the major axis of the orbit drawn from the sun
(a) $\frac{r_1 + r_2}{4}$ (b) $\frac{r_1 r_2}{r_1 + r_2}$
(c) $\frac{2r_1 r_2}{r_1 + r_2}$ (d) $\frac{r_1 + r_2}{3}$
14. The rotation period of an earth satellite close to the surface of the earth is 83 minutes. The time period of another earth satellite in an orbit at a distance of three earth radii from its surface will be
(a) 83 minutes (b) $83 \times \sqrt{8}$ minutes
(c) 664 minutes (d) 249 minutes
15. A satellite of mass m is circulating around the earth with constant angular velocity. If radius of the orbit is R_0 and mass of the earth M , the angular momentum about the centre of the earth is [MP PMT 1996; RPMT 2000]
(a) $m\sqrt{GMR_0}$ (b) $M\sqrt{GmR_0}$
(c) $m\sqrt{\frac{GM}{R_0}}$ (d) $M\sqrt{\frac{GM}{R_0}}$
16. According to Kepler, the period of revolution of a planet (T) and its mean distance from the sun (r) are related by the equation [EAMCET (Med.) 1995; MH CET 2000; Pb. PET 2001]
(a) $T^3 r^3 = \text{constant}$ (b) $T^2 r^{-3} = \text{constant}$
(c) $Tr^3 = \text{constant}$ (d) $T^2 r = \text{constant}$
17. A planet revolves around sun whose mean distance is 1.588 times the mean distance between earth and sun. The revolution time of planet will be [RPET 1997]
(a) 1.25 years (b) 1.59 years
(c) 0.89 years (d) 2 years
18. A satellite A of mass m is at a distance of r from the centre of the earth. Another satellite B of mass $2m$ is at a distance of $2r$ from the earth's centre. Their time periods are in the ratio of
(a) 1 : 2 (b) 1 : 16
(c) 1 : 32 (d) 1 : $2\sqrt{2}$
19. The earth E moves in an elliptical orbit with the sun S at one of the foci as shown in figure. Its speed of motion will be maximum at the point [BHU 1994; CPMT 1997]
- 
- (a) C (b) A
(c) B (d) D
20. The period of revolution of planet A around the sun is 8 times that of B . The distance of A from the sun is how many times greater than that of B from the sun [CBSE PMT 1997; BHU 2001]
(a) 2 (b) 3
(c) 4 (d) 5
21. If the radius of earth's orbit is made $1/4$, the duration of an year will become [BHU 1998; JIPMER 2001, 2002]
(a) 8 times (b) 4 times
(c) $1/8$ times (d) $1/4$ times
22. Planetary system in the solar system describes [DCE 1999]
(a) Conservation of energy
(b) Conservation of linear momentum
(c) Conservation of angular momentum [CBSE PMT 1991]
(d) None of these
23. If mass of a satellite is doubled and time period remain constant the ratio of orbit in the two cases will be [RPET 2000]
(a) 1 : 2 (b) 1 : 1
(c) 1 : 3 (d) None of these
24. The earth revolves round the sun in one year. If the distance between them becomes double, the new period of revolution will be [MP PMT 1994]
(a) $1/2$ year (b) $2\sqrt{2}$ years
(c) 4 years (d) 8 years
25. Kepler discovered [DCE 2000]
(a) Laws of motion
(b) Laws of rotational motion
(c) Laws of planetary motion
(d) Laws of curvilinear motion
26. In the solar system, which is conserved [DCE 2001]
(a) Total Energy (b) K.E.
(c) Angular Velocity (d) Linear Momentum
27. The maximum and minimum distances of a comet from the sun are $8 \times 10^{12} m$ and $1.6 \times 10^{12} m$. If its velocity when nearest to the sun is $60 m/s$, what will be its velocity in m/s when it is farthest
(a) 12 (b) 60
(c) 112 (d) 6
28. A body revolved around the sun 27 times faster than the earth what is the ratio of their radii [DPMT 2002]

- (a) $1/3$ (b) $1/9$
(c) $1/27$ (d) $1/4$
29. The period of moon's rotation around the earth is nearly 29 days. If moon's mass were 2 fold, its present value and all other things remained unchanged, the period of moon's rotation would be nearly [Kerala (Engg.) 2002]
- (a) $29\sqrt{2}$ days (b) $29/\sqrt{2}$ days
(c) 29×2 days (d) 29 days
30. Two planets at mean distance d_1 and d_2 from the sun and their frequencies are n_1 and n_2 respectively then [Kerala (Med.) 2002]
- (a) $n_1^2 d_1^2 = n_2^2 d_2^2$ (b) $n_2^2 d_2^3 = n_1^2 d_1^3$
(c) $n_1 d_1^2 = n_2 d_2^2$ (d) $n_1^2 d_1 = n_2^2 d_2$
31. Which of the following astronomer first proposed that sun is static and earth rounds sun [AFMC 2002]
- (a) Copernicus (b) Kepler
(c) Galileo (d) None
32. The distance of a planet from the sun is 5 times the distance between the earth and the sun. The time period of the planet is
- (a) $5^{3/2}$ years (b) $5^{2/3}$ years
(c) $5^{1/3}$ years (d) $5^{1/2}$ years
33. A planet is revolving around the sun as shown in elliptical path



The correct option is

- (a) The time taken in travelling DAB is less than that for BCD
(b) The time taken in travelling DAB is greater than that for BCD
(c) The time taken in travelling CDA is less than that for ABC
(d) The time taken in travelling CDA is greater than that for ABC
34. In the previous question the orbital velocity of the planet will be minimum at [UPSEAT 2003; RPET 2002]
- (a) A (b) B
(c) C (d) D
35. The radius of orbit of a planet is two times that of the earth. The time period of planet is [BHU 2003; CPMT 2004]
- (a) 4.2 years (b) 2.8 years
(c) 5.6 years (d) 8.4 years
36. The orbital angular momentum of a satellite revolving at a distance r from the centre is L . If the distance is increased to $16r$, then the new angular momentum will be [MP PET 2003]
- (a) $16 L$ (b) $64 L$
(c) $\frac{L}{4}$ (d) $4 L$
37. According to Kepler's law the time period of a satellite varies with its radius as [Orissa JEE 2003]
- (a) $T^2 \propto R^3$ (b) $T^3 \propto R^2$
(c) $T^2 \propto (1/R^3)$ (d) $T^3 \propto (1/R^2)$
38. In planetary motion the areal velocity of position vector of a planet depends on angular velocity (ω) and the distance of the planet from sun (r). If so the correct relation for areal velocity is

- (a) $\frac{dA}{dt} \propto \omega r$ (b) $\frac{dA}{dt} \propto \omega^2 r$
(c) $\frac{dA}{dt} \propto \omega r^2$ (d) $\frac{dA}{dt} \propto \sqrt{\omega r}$

39. The ratio of the distances of two planets from the sun is 1.38. The ratio of their period of revolution around the sun is
- (a) 1.38 (b) $1.38^{3/2}$
(c) $1.38^{1/2}$ (d) 1.38^3
(e) 1.38^2
40. Kepler's second law (law of areas) is nothing but a statement of
- (a) Work energy theorem
(b) Conservation of linear momentum
(c) Conservation of angular momentum
(d) Conservation of energy
41. In an elliptical orbit under gravitational force, in general [UPSEAT 2003] [UPSEAT 2004]
- (a) Tangential velocity is constant
(b) Angular velocity is constant
(c) Radial velocity is constant
(d) Areal velocity is constant
42. If a new planet is discovered rotating around Sun with the orbital radius double that of earth, then what will be its time period (in earth's days) [DCE 2004]
- (a) 1032 (b) 1023
(c) 1024 (d) 1043
43. Suppose the law of gravitational attraction suddenly changes and becomes an inverse cube law i.e. $F \propto 1/r^3$, but still remaining a central force. Then [UPSEAT 2002]
- (a) Keplers law of areas still holds
(b) Keplers law of period still holds
(c) Keplers law of areas and period still hold
(d) Neither the law of areas, nor the law of period still holds
44. What does not change in the field of central force [MP PMT 2004]
- (a) Potential energy (b) Kinetic energy
(c) Linear momentum (d) Angular momentum
45. The eccentricity of earth's orbit is 0.0167. The ratio of its maximum speed in its orbit to its minimum speed is [NCERT 1973]
- (a) 2.507 (b) 1.033
(c) 8.324 (d) 1.000
46. The mass of a planet that has a moon whose time period and orbital radius are T and R respectively can be written as [AMU 1995]
- (a) $4\pi^2 R^3 G^{-1} T^{-2}$ (b) $8\pi^2 R^3 G^{-1} T^{-2}$
(c) $12\pi^2 R^3 G^{-1} T^{-2}$ (d) $16\pi^2 R^3 G^{-1} T^{-2}$
47. If orbital velocity of planet is given by $v = G^a M^b R^c$, then [EAMCET 2003] [EAMCET 1994]

- (a) $a = 1/3, b = 1/3, c = -1/3$
 (b) $a = 1/2, b = 1/2, c = -1/2$
 (c) $a = 1/2, b = -1/2, c = 1/2$
 (d) $a = 1/2, b = -1/2, c = -1/2$
48. Hubble's law states that the velocity with which milky way is moving away from the earth is proportional to

[Kerala PMT 2004]

- (a) Square of the distance of the milky way from the earth
 (b) Distance of milky way from the earth
 (c) Mass of the milky way
 (d) Product of the mass of the milky way and its distance from the earth
 (e) Mass of the earth
49. Two satellite are revolving around the earth with velocities v_1 and v_2 and in radii r_1 and r_2 ($r_1 > r_2$) respectively. Then

[BHU 2005]

- (a) $v_1 = v_2$ (b) $v_1 > v_2$
 (c) $v_1 < v_2$ (d) $\frac{v_1}{r_1} = \frac{v_2}{r_2}$
50. The condition for a uniform spherical mass m of radius r to be a black hole is [G = gravitational constant and g = acceleration due to gravity] [AIIMS 2005]
- (a) $(2Gm/r)^{1/2} \leq c$ (b) $(2Gm/r)^{1/2} = c$
 (c) $(2Gm/r)^{1/2} \geq c$ (d) $(gm/r)^{1/2} \geq c$
51. Earth is revolving around the sun if the distance of the Earth from the Sun is reduced to $1/4$ of the present distance then the present day length reduced by [BHU 2005]
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{8}$ (d) $\frac{1}{6}$

Critical Thinking

Objective Questions

1. Imagine a light planet revolving around a very massive star in a circular orbit of radius R with a period of revolution T . If the gravitational force of attraction between planet and star is proportional to $R^{-5/2}$, then T^2 is proportional to [IIT 1989; RPMT 1997]
- (a) R^3 (b) $R^{7/2}$
 (c) $R^{5/2}$ (d) $R^{3/2}$
2. The magnitudes of the gravitational force at distances r_1 and r_2 from the centre of a uniform sphere of radius R and mass M are F_1 and F_2 respectively. Then [IIT 1994]

- (a) $\frac{F_1}{F_2} = \frac{r_1}{r_2}$ if $r_1 < R$ and $r_2 < R$
 (b) $\frac{F_1}{F_2} = \frac{r_1^2}{r_2^2}$ if $r_1 > R$ and $r_2 > R$
 (c) $\frac{F_1}{F_2} = \frac{r_1}{r_2}$ if $r_1 > R$ and $r_2 > R$
 (d) $\frac{F_1}{F_2} = \frac{r_2^2}{r_1^2}$ if $r_1 < R$ and $r_2 < R$

3. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of earth
- (a) The acceleration of S is always directed towards the centre of the earth
 (b) The angular momentum of S about the centre of the earth changes in direction but its magnitude remains constant
 (c) The total mechanical energy of S varies periodically with time
 (d) The linear momentum of S remains constant in magnitude
4. A mass M is split into two parts, m and $(M-m)$, which are then separated by a certain distance. What ratio of m/M maximizes the gravitational force between the two parts

[AMU 2000]

- (a) $1/3$ (b) $1/2$
 (c) $1/4$ (d) $1/5$
5. Suppose the gravitational force varies inversely as the n^{th} power of distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to
- (a) $R^{\left(\frac{n+1}{2}\right)}$ (b) $R^{\left(\frac{n-1}{2}\right)}$
 (c) R^n (d) $R^{\left(\frac{n-2}{2}\right)}$
6. If the radius of the earth were to shrink by 1% its mass remaining the same, the acceleration due to gravity on the earth's surface would

[IIT 1981; CPMT 1981; MP PMT 1996, 97; Roorkee 1992; MP PET 1999; Kerala PMT 2004]

- (a) Decrease by 2% (b) Remain unchanged
 (c) Increase by 2% (d) Increase by 1%
7. The radius and mass of earth are increased by 0.5%. Which of the following statements are true at the surface of the earth
- (a) g will increase
 (b) g will decrease
 (c) Escape velocity will remain unchanged
 (d) Potential energy will remain unchanged
8. In order to make the effective acceleration due to gravity equal to zero at the equator, the angular velocity of rotation of the earth about its axis should be ($g = 10 \text{ ms}^{-2}$ and radius of earth is 6400 kms) [Roorkee 2000]
- (a) 0 radsec^{-1} (b) $\frac{1}{800} \text{ radsec}^{-1}$
 (c) $\frac{1}{80} \text{ radsec}^{-1}$ (d) $\frac{1}{8} \text{ radsec}^{-1}$
9. A simple pendulum has a time period T_1 when on the earth's surface and T_2 when taken to a height R above the earth's surface, where R is the radius of the earth. The value of T_2 / T_1 is
- (a) 1 (b) $\sqrt{2}$

- (c) 4 (d) 2
10. A body of mass m is taken from earth surface to the height h equal to radius of earth, the increase in potential energy will be

CBSE PMT 1991; Kurukshetra CEE 1996;
CMEET Bihar 1995; MNR 1998; AIEEE 2004]

- (a) mgR (b) $\frac{1}{2}mgR$
- (c) $2mgR$ (d) $\frac{1}{4}mgR$

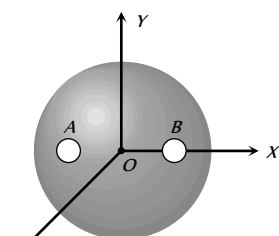
11. An artificial satellite moving in a circular orbit around the earth has a total (kinetic + potential) energy E_0 . Its potential energy is [IIT 1997 Canceled; MH CET 2002; MP PMT 2000]

- (a) $-E_0$ (b) $1.5 E_0$
- (c) $2 E_0$ (d) E_0

12. A rocket of mass M is launched vertically from the surface of the earth with an initial speed V . Assuming the radius of the earth to be R and negligible air resistance, the maximum height attained by the rocket above the surface of the earth is

- (a) $R \left(\frac{gR}{2V^2} - 1 \right)$ (b) $R \left(\frac{gR}{2V^2} - 1 \right)$
- (c) $R \left(\frac{2gR}{V^2} - 1 \right)$ (d) $R \left(\frac{2gR}{V^2} - 1 \right)$

13. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit with their centres at $A(-2, 0, 0)$ and $B(2, 0, 0)$ respectively are taken out of the solid leaving behind spherical cavities as shown in figure



[IIT 1993]

- (a) The gravitational force due to this object at the origin is zero
- (b) The gravitational force at the point $B(2, 0, 0)$ is zero
- (c) The gravitational potential is the same at all points of the circle $y^2 + z^2 = 36$
- (d) The gravitational potential is the same at all points on the circle $y^2 + z^2 = 4$

14. Two bodies of masses m_1 and m_2 are initially at rest at infinite distance apart. They are then allowed to move towards each other under mutual gravitational attraction. Their relative velocity of approach at a separation distance r between them is

- (a) $\left[2G \frac{(m_1 - m_2)}{r} \right]^{1/2}$ (b) $\left[\frac{2G}{r} (m_1 + m_2) \right]^{1/2}$
- (c) $\left[\frac{r}{2G(m_1 m_2)} \right]^{1/2}$ (d) $\left[\frac{2G}{r} m_1 m_2 \right]^{1/2}$

15. A projectile is projected with velocity kv_e in vertically upward direction from the ground into the space. (v_e is escape velocity and $k < 1$). If air resistance is considered to be negligible then the maximum height from the centre of earth to which it can go, will be : (R = radius of earth)

[Roorkee 1999; RPET 1999]

- (a) $\frac{R}{k^2 + 1}$ (b) $\frac{R}{k^2 - 1}$

- (c) $\frac{R}{1 - k^2}$ (d) $\frac{R}{k + 1}$

16. [NCERT 1971; CPMT 1971, 97; IIT 1983; IIT 1995] A satellite is launched into a circular orbit of radius R around the earth. A second satellite is launched into an orbit of radius $(1.01)R$. The period of the second satellite is larger than that of the first one by approximately

- (a) 0.5% (b) 1.0%
- (c) 1.5% (d) 3.0%

17. If the distance between the earth and the sun becomes half its present value, the number of days in a year would have been

- (a) 64.5 (b) 129
- (c) 182.5 (d) 730

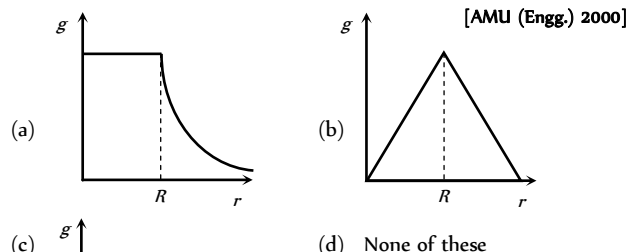
18. A geostationary satellite orbits around the earth in a circular orbit of radius 36000 km. Then, the time period of a satellite orbiting a few hundred kilometres above the earth's surface ($R_{\text{Earth}} = 6400 \text{ km}$) will approximately be

[IIT-JEE (Screening) 2002]

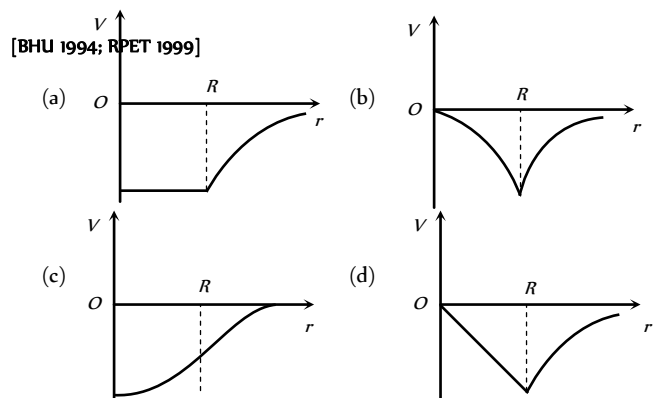
- (a) $\frac{1}{2} h$ (b) $1 h$
- (c) $2 h$ (d) $4 h$

Graphical Questions

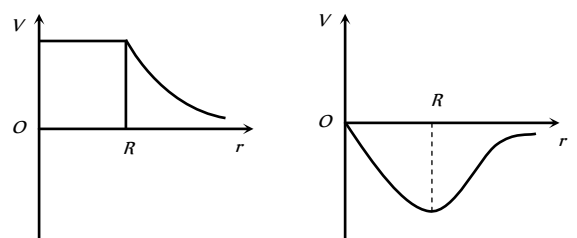
1. Assuming the earth to have a constant density, point out which of the following curves show the variation of acceleration due to gravity from the centre of earth to the points far away from the surface of earth



2. The diagram showing the variation of gravitational potential of earth with distance from the centre of earth is

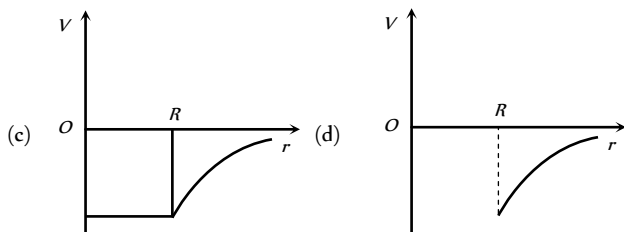


3. By which curve will the variation of gravitational potential of a hollow sphere of radius R with distance be depicted

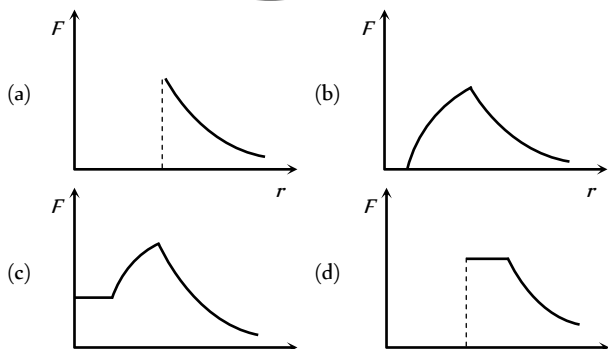
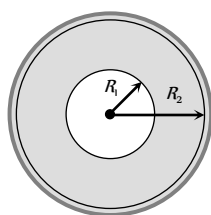


(a)

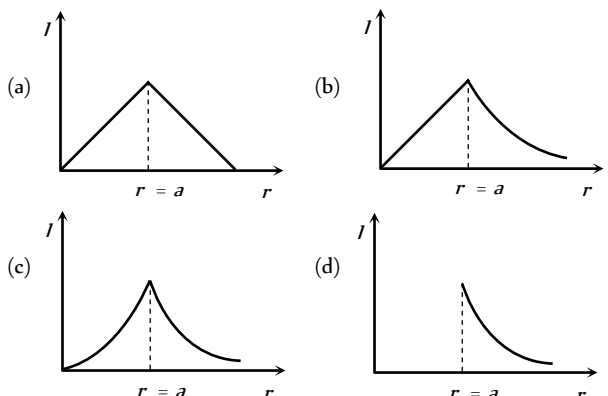
(b)



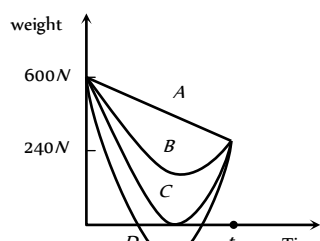
4. A sphere of mass M and radius R_1 has a concentric cavity of radius R_2 as shown in figure. The force F exerted by the sphere on a particle of mass m located at a distance r from the centre of sphere varies as ($0 \leq r \leq \infty$)



5. Which one of the following graphs represents correctly the variation of the gravitational field (F) with the distance (r) from the centre of a spherical shell of mass M and radius a

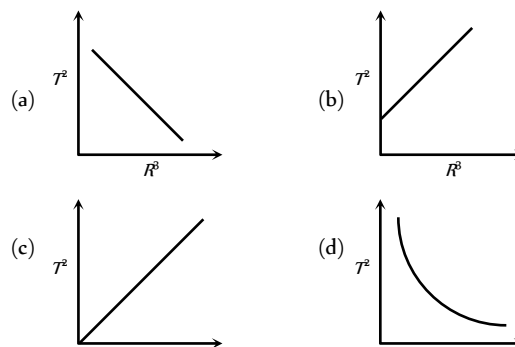


6. Suppose, the acceleration due to gravity at the earth's surface is 10 m/s^2 and at the surface of Mars it is 4.0 m/s^2 . A 60 kg passenger goes from the earth to the Mars in a spaceship moving with a constant velocity. Neglect all other objects in the sky. Which part of figure best represents the weight (net gravitational force) of the passenger as a function of time.



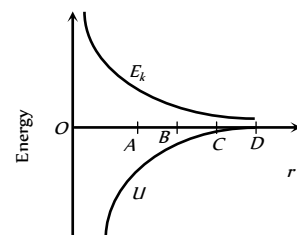
- (a) A
(b) B
(c) C
(d) D

7. Which of the following graphs represents the motion of a planet moving about the sun [NCERT 1983]

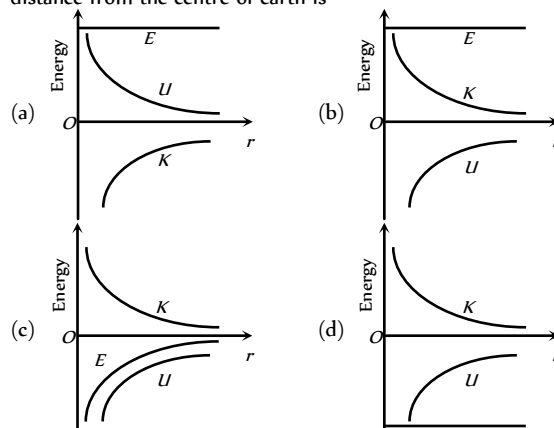


8. The curves for potential energy (U) and kinetic energy (E_k) of a two particle system are shown in figure. At what points the system will be bound?

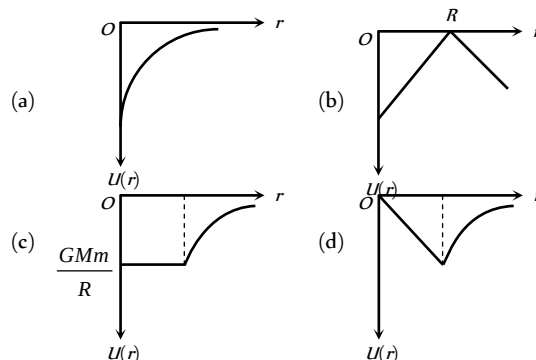
- (a) Only at point D
(b) Only at point A
(c) At point D and A
(d) At points A, B and C



9. The correct graph representing the variation of total energy (E_t) kinetic energy (E_k) and potential energy (U) of a satellite with its distance from the centre of earth is



10. A shell of mass M and radius R has a point mass m placed at a distance r from its centre. The gravitational potential energy $U(r)$ vs r will be



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : Smaller the orbit of the planet around the sun, shorter is the time it takes to complete one revolution.
Reason : According to Kepler's third law of planetary motion, square of time period is proportional to cube of mean distance from sun.
2. Assertion : Gravitational force between two particles is negligibly small compared to the electrical force.
Reason : The electrical force is experienced by charged particles only.
3. Assertion : The universal gravitational constant is same as acceleration due to gravity.
Reason : Gravitational constant and acceleration due to gravity have same dimensional formula.
4. Assertion : The value of acceleration due to gravity does not depend upon mass of the body on which force is applied.
Reason : Acceleration due to gravity is a constant quantity.
5. Assertion : If a pendulum is suspended in a lift and lift is falling freely, then its time period becomes infinite.
Reason : Free falling body has acceleration equal to acceleration due to gravity.
6. Assertion : If earth suddenly stops rotating about its axis, then the value of acceleration due to gravity will become same at all the places.
Reason : The value of acceleration due to gravity is independent of rotation of earth.
7. Assertion : The difference in the value of acceleration due to gravity at pole and equator is proportional to square of angular velocity of earth.
Reason : The value of acceleration due to gravity is minimum at the equator and maximum at the pole.
8. Assertion : There is no effect of rotation of earth on acceleration due to gravity at poles.
Reason : Rotation of earth is about polar axis.
9. Assertion : A force act upon the earth revolving in a circular orbit about the sun. Hence work should be done on the earth.
Reason : The necessary centripetal force for circular motion of earth comes from the gravitational force between earth and sun.
10. Assertion : The ratio of inertial mass to gravitational mass is equal to one.

- Reason : The inertial mass and gravitational mass of a body are equivalent.
11. Assertion : Gravitational potential of earth at every place on it is negative.
Reason : Every body on earth is bound by the attraction of earth.
12. Assertion : Even when orbit of a satellite is elliptical, its plane of rotation passes through the centre of earth.
Reason : According to law of conservation of angular momentum plane of rotation of satellite always remain same.
13. Assertion : A planet moves faster, when it is closer to the sun in its orbit and vice versa.
Reason : Orbital velocity in orbital of planet is constant.
14. Assertion : Orbital velocity of a satellite is greater than its escape velocity.
Reason : Orbit of a satellite is within the gravitational field of earth whereas escaping is beyond the gravitational field of earth.
15. Assertion : If an earth satellite moves to a lower orbit, there is some dissipation of energy but the satellite speed increases.
Reason : The speed of satellite is a constant quantity.
16. Assertion : Earth has an atmosphere but the moon does not.
Reason : Moon is very small in comparison to earth.
17. Assertion : The time period of geostationary satellite is 24 hours.
Reason : Geostationary satellite must have the same time period as the time taken by the earth to complete one revolution about its axis.
18. Assertion : The principle of superposition is not valid for gravitational force.
Reason : Gravitational force is a conservative force.
19. Assertion : Two different planets have same escape velocity.
Reason : Value of escape velocity is a universal constant.
20. Assertion : The time period of revolution of a satellite close to surface of earth is smaller than that revolving away from surface of earth.
Reason : The square of time period of revolution of a satellite is directly proportional to cube of its orbital radius.
21. Assertion : When distance between two bodies is doubled and also mass of each body is also doubled, gravitational force between them remains the same.
Reason : According to Newton's law of gravitation, force is directly proportional to mass of bodies and inversely proportional to square of distance between them.
22. Assertion : Generally the path of a projectile from the earth is parabolic but it is elliptical for projectiles going to a very large height.
Reason : The path of a projectile is independent of the gravitational force of earth.
23. Assertion : A body becomes weightless at the centre of earth.
Reason : As the distance from centre of earth decreases, acceleration due to gravity increases.
24. Assertion : Space rockets are usually launched in the equatorial line from west to east.

- Reason : The acceleration due to gravity is minimum at the equator.
25. Assertion : The binding energy of a satellite does not depend upon the mass of the satellite.
- Reason : Binding energy is the negative value of total energy of satellite.
26. Assertion : We can not move even a finger without disturbing all the stars.
- Reason : Every body in this universe attracts every other body with a force which is inversely proportional to the square of distance between them.
27. Assertion : If earth were a hollow sphere, gravitational field intensity at any point inside the earth would be zero.
- Reason : Net force on a body inside the sphere is zero.
28. Assertion : For a satellite revolving very near to earth's surface the time period of revolution is given by 1 hour 24 minutes.
- Reason : The period of revolution of a satellite depends only upon its height above the earth's surface.
29. Assertion : A person sitting in an artificial satellite revolving around the earth feels weightless.
- Reason : There is no gravitational force on the satellite.
30. Assertion : The speed of satellite always remains constant in an orbit.
- Reason : The speed of a satellite depends on its path.
31. Assertion : The speed of revolution of an artificial satellite revolving very near the earth is 8 kms^{-1} .
- Reason : Orbital velocity of a satellite, become independent of height of near satellite.
32. Assertion : Gravitational field is zero both at centre and infinity.
- Reason : The dimensions of gravitational field is $[LT^{-2}]$.
33. Assertion : For the planets orbiting around the sun, angular speed, linear speed, KE changes with time, but angular momentum remains constant.
- Reason : No torque is acting on the rotating planet. So its angular momentum is constant.

Answers

Newton's Law of Gravitation

1	a	2	b	3	b	4	b	5	a
6	d	7	b	8	c	9	c	10	d
11	d	12	a	13	d	14	a	15	d
16	b	17	c	18	a	19	c	20	a
21	a	22	a	23	e	24	c		

Acceleration Due to Gravity

1	d	2	b	3	d	4	a	5	b
6	a	7	b	8	d	9	b	10	b
11	c	12	a	13	b	14	c	15	a

16	c	17	c	18	c	19	a	20	d
21	a	22	c	23	d	24	b	25	a
26	d	27	b	28	b	29	b	30	a
31	b	32	a	33	c	34	b	35	c
36	a	37	d	38	b	39	c	40	b
41	c	42	b	43	c	44	b	45	c
46	b	47	a	48	a	49	b	50	a
51	a	52	a	53	c	54	a	55	c
56	d	57	a	58	d	59	b	60	c
61	b	62	a	63	c	64	a	65	c
66	a	67	d	68	d	69	a	70	a
71	a	72	b	73	b	74	b	75	d
76	a	77	d	78	a	79	a	80	b
81	a	82	a	83	d	84	a	85	b
86	b	87	b						

Gravitation Potential, Energy and Escape Velocity

1	c	2	a	3	d	4	a	5	d
6	a	7	b	8	c	9	c	10	c
11	d	12	a	13	b	14	b	15	a
16	d	17	c	18	a	19	b	20	b
21	c	22	b	23	a	24	c	25	a
26	b	27	b	28	c	29	c	30	b
31	a	32	b	33	c	34	a	35	c
36	d	37	a	38	a	39	c	40	c
41	c	42	c	43	d	44	a	45	a
46	b	47	d	48	a	49	a	50	b
51	d	52	a	53	c	54	b	55	a
56	b	57	d	58	d	59	b	60	b
61	c	62	b	63	c	64	c	65	b
66	b	67	a	68	c	69	a	70	a

Motion of Satellite

1	b	2	d	3	d	4	d	5	b
6	b	7	b	8	b	9	c	10	c
11	b	12	d	13	b	14	a	15	a
16	b	17	c	18	d	19	d	20	d
21	d	22	b	23	b	24	a	25	a
26	d	27	a	28	d	29	a	30	a
31	b	32	d	33	a	34	c	35	d
36	c	37	d	38	b	39	c	40	b
41	c	42	d	43	a	44	d	45	a
46	c	47	c	48	d	49	b	50	b
51	a	52	a	53	a	54	c	55	b



426 Gravitation

56	b	57	b	58	d	59	d	60	b
61	c	62	c	63	b	64	a	65	b
66	c	67	d						

Kepler's laws of Planetary Motion

1	c	2	c	3	c	4	b	5	c
6	b	7	b	8	c	9	c	10	b
11	d	12	b	13	c	14	c	15	a
16	b	17	d	18	d	19	b	20	c
21	c	22	c	23	b	24	b	25	c
26	a	27	a	28	b	29	d	30	b
31	a	32	a	33	a	34	c	35	b
36	d	37	a	38	c	39	b	40	c
41	d	42	a	43	d	44	d	45	b
46	a	47	b	48	b	49	c	50	c
51	c								

Critical Thinking Questions

1	b	2	ab	3	a	4	b	5	a
6	c	7	bcd	8	b	9	d	10	b
11	c	12	c	13	acd	14	b	15	c
16	c	17	b	18	c				

Graphical Questions

1	c	2	c	3	c	4	b	5	d
6	c	7	c	8	d	9	c	10	c

Assertion and Reason

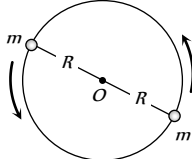
1	a	2	b	3	d	4	c	5	a
6	c	7	b	8	a	9	e	10	a
11	a	12	a	13	c	14	e	15	c
16	b	17	b	18	e	19	d	20	a
21	a	22	c	23	c	24	b	25	e
26	a	27	a	28	a	29	c	30	e
31	a	32	b	33	a				

AS Answers and Solutions

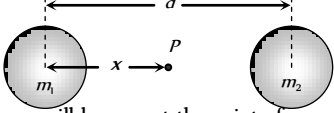
Newton's Law of Gravitation

- (a)
- (b) As it depends on the weight of the body.
- (b) Due to inertia of direction.
- (b)
- (a)
- (d) $F \propto \frac{1}{r^2}$. If r becomes double then F reduces to $\frac{F}{4}$
- (b)
- (c) $F = G \frac{m_1 m_2}{r^2} = 6.675 \times \frac{1 \times 1}{1^2} \times 10^{-11} = 6.675 \times 10^{-11} \text{ N}$
- (c) Centripetal force provided by the gravitational force of attraction between two particles

$$i.e. \frac{mv^2}{R} = \frac{Gm \times m}{(2R)^2}$$

$$\Rightarrow v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$$

- (d) $m = 6 \times 10^{24} \text{ kg}$, $\omega = 2 \times 10^{-7} \text{ rad/s}$, $R = 1.5 \times 10^{11} \text{ m}$
The force exerted by the sun on the earth $F = m\omega^2 R$
By substituting the value we can get, $F = 36 \times 10^{21} \text{ N}$
- (d)

- (a) k represents gravitational constant which depends only on the system of units.

- (d) 
Force will be zero at the point of zero intensity

$$x = \frac{\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} d = \frac{\sqrt{81M}}{\sqrt{81M} + \sqrt{M}} D = \frac{9}{10} D.$$

- (a)
- (d) $g = \frac{GM}{R^2} = \frac{6.67 \times 10^{-11} \times 7.34 \times 10^{22}}{(1.74 \times 10^6)^2} = 1.62 \text{ N/kg}$
- (b) Actually gravitational force provides the centripetal force.
- (c)
- (a) $F \propto xm \times (1-x)m = xm^2(1-x)$

For maximum force $\frac{dF}{dx} = 0$

$$\Rightarrow \frac{dF}{dx} = m^2 - 2xm^2 = 0 \Rightarrow x = 1/2$$

- (c)
- (a)
- (a) Gravitational force does not depend on the medium.
- (a)
- (e)

$$24. (c) F = \frac{G \times m \times m}{(2R)^2} = \frac{G \times \left(\frac{4}{3} \pi R^3 \rho\right)^2}{4R^2} = \frac{4}{9} \pi^2 \rho^2 R^4$$

$$\therefore F \propto R^4$$

Acceleration Due to Gravity

- (d)
- (b) The value of g at the height h from the surface of earth

$$g' = g \left(1 - \frac{2h}{R}\right)$$
The value of g at depth x below the surface of earth

$$g' = g \left(1 - \frac{x}{R}\right)$$
These two are given equal, hence $\left(1 - \frac{2h}{R}\right) = \left(1 - \frac{x}{R}\right)$
On solving, we get $x = 2h$
- (d) Time period of simple pendulum $T = 2\pi \sqrt{\frac{l}{g'}}$
In artificial satellite $g' = 0 \therefore T = \text{infinite}$.
- (a) $g = \frac{4}{3} \pi \rho G R$. If $\rho = \text{constant}$ then $\frac{g_1}{g_2} = \frac{R_1}{R_2}$
- (b) Time of decent $t = \sqrt{\frac{2h}{g}}$. In vacuum no other force works except gravity so time period will be exactly equal.
- (a)

7. (b) Because acceleration due to gravity increases
8. (d) Because acceleration due to gravity decreases
9. (b) We know that $g = \frac{GM}{R^2}$
On the planet $g_p = \frac{GM/7}{R^2/4} = \frac{4g}{7} = \frac{4}{7}g$
Hence weight on the planet $= 700 \times \frac{4}{7} = 400 \text{ gm wt}$
10. (b) In pendulum clock the time period depends on the value of g , while in spring watch, the time period is independent of the value of g .
11. (c) $g = \frac{GM}{R^2} = \frac{GM_0}{(D_0/2)^2} = \frac{4GM_0}{D_0^2}$
12. (a)
13. (b) $\frac{g'}{g} = \frac{M'}{M} \left(\frac{R}{R'} \right)^2 = \left(\frac{2M}{M} \right) \left(\frac{R}{2R} \right)^2 = \frac{1}{2}$
 $\Rightarrow g' = \frac{g}{2} = \frac{9.8}{2} = 4.9 \text{ m/s}^2$
14. (c)
15. (a)
16. (c) For the condition of weightlessness at equator
 $\omega = \sqrt{\frac{g}{R}} \therefore \omega = \sqrt{\frac{1}{640 \times 10^3}} = \frac{1}{800} \text{ rad/s}$
17. (c) $g = \frac{GM}{r^2}$. Since M and r are constant, so $g = 9.8 \text{ m/s}^2$
18. (c) $g = \frac{GM}{R^2}$ and $M = \frac{4}{3}\pi R^3 \times \rho$
 $\therefore g = \frac{4}{3} \frac{\pi R^3 \times G \rho}{R^2} \Rightarrow \rho = \frac{3g}{4\pi R G}$
19. (a) Because value of g decreases when we move either in coal mine or at the top of mountain.
20. (d) $g = \frac{4}{3}\pi \rho G R \therefore \frac{g_1}{g_2} = \frac{R_1 \rho_1}{R_2 \rho_2}$
21. (a) $g = \frac{GM}{R^2}$ (Given $M_e = 81M_m$, $R_e = 3.5R_m$)
Substituting the above values, $\frac{g_m}{g_e} = 0.15$
22. (c) Value of g decreases when we go from poles to equator.
23. (d)
24. (b) Because value of g decreases with increasing height.
25. (a) $\frac{g'}{g} = \left(\frac{R}{R+h} \right)^2 = \left(\frac{6400}{6400+64} \right)^2 \Rightarrow g' = 960.40 \text{ cm/s}^2$
26. (d)
27. (b) $g' = g - \omega^2 R \cos^2 \lambda$
Rotation of the earth results in the decreased weight apparently. This decrease in weight is not felt at the poles as the angle of latitude is 90° .
28. (b) $g = \frac{GM}{R^2}$. If radius shrinks to half of its present value then g will become four times.
29. (b) Using $g = \frac{GM}{R^2}$ we get $g_m = g/5$
30. (a) $g = g_p - R\omega^2 \cos^2 \lambda = g_p - \omega^2 R \cos^2 60^\circ = g_p - \frac{1}{4}R\omega^2$
31. (b) $g' = g \left(1 - \frac{d}{R} \right) \Rightarrow \frac{g}{n} = g \left(1 - \frac{d}{R} \right) \Rightarrow d = \left(\frac{n-1}{n} \right) R$
32. (a) $g \propto \frac{GM}{r^2} \therefore g \propto \frac{1}{r^2}$ or $r \propto \frac{1}{\sqrt{g}}$
If g decrease by one percent then r should be increase by $\frac{1}{2}\%$ i.e. $R = \frac{1}{2 \times 100} \times 6400 = 32 \text{ km}$
33. (c) $g = \frac{4}{3}G\pi R\rho \Rightarrow \frac{g_1}{g_2} = \frac{\rho_1 R_1}{\rho_2 R_2} = \frac{1}{2} \times \frac{4}{1} = \frac{2}{1}$
34. (b) $g' = g \left(\frac{R}{R+h} \right)^2 \Rightarrow \frac{g}{4} = g \left(\frac{R}{R+h} \right)^2 \Rightarrow \frac{1}{2} = \frac{R}{R+h}$
 $\Rightarrow R+h = 2R \therefore h = R$
35. (c) Acceleration due to gravity at poles is independent of the angular speed.
36. (a) Mass of the ball always remain constant. It does not depend upon the acceleration due to gravity
37. (d) $g_m = \frac{GM_m}{R_m^2}$ and $g_m = \frac{g_e}{6} = \frac{9.8}{6} \text{ m/s}^2 = 1.63 \text{ m/s}^2$
Substituting $R_m = 1.768 \times 10^6 \text{ m}$, $g_m = 1.63 \text{ m/s}^2$
and $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$ We get
 $M_m = 7.65 \times 10^{22} \text{ kg}$
38. (b) $g' = g \left(\frac{R}{R+h} \right)^2 \Rightarrow$ when $h = R$ then $g' = \frac{g}{4}$
So the weight of the body at this height will become one-fourth.
39. (c) $g = \frac{GM}{R^2}$ and $K = \frac{L^2}{2I}$
If mass of the earth and its angular momentum remains constant then $g \propto \frac{1}{R^2}$ and $K \propto \frac{1}{R^2}$
i.e. if radius of earth decreases by 2% then g and K both increases by 4%.
40. (b) Weight is least at the equator.
41. (c) $g \propto \frac{1}{R^2}$
Percentage change in $g = 2(\text{percentage change in } R)$
 $= 2 \times 1.5 = -3\%$
42. (b) $g \propto \frac{1}{R^2}$. If radius of earth decreases by 2% then g will increase by 4% i.e. weight of the body at earth surface will increase by 4%
43. (c) Mass does not vary from place to place.
44. (b) $g = \frac{GM}{R^2} \Rightarrow R = \sqrt{\frac{GM}{g}}$
Substituting the above values we get $R = 1.87 \times 10^6 \text{ m}$.

45. (c) Weight of the body at equator = $\frac{3}{5}$ of initial weight

$$\therefore g' = \frac{3}{5}g \text{ (because mass remains constant)}$$

$$g' = g - \omega^2 R \cos^2 \lambda \Rightarrow \frac{3}{5}g = g - \omega^2 R \cos^2(0^\circ)$$

$$\Rightarrow \omega^2 = \frac{2g}{5R} \Rightarrow \omega = \sqrt{\frac{2g}{5R}} = \sqrt{\frac{2 \times 10}{5 \times 6400 \times 10^3}}$$

$$= 7.8 \times 10^{-4} \frac{\text{rad}}{\text{sec}}$$

46. (b) $h = 32 \text{ km}$, $R = 6400 \text{ km}$, so $h \ll R$

$$g' = g \left(1 - \frac{2h}{R}\right) = g \left(1 - \frac{2 \times 32}{6400}\right) \Rightarrow g' = \frac{99}{100}g = 0.99g$$

47. (a) Same change in the value of g can be observed at a depth x and height $2x$

$$\text{given } d = x = 10 \text{ km} \therefore h = 2x = 20 \text{ km}$$

48. (a)

49. (b) $\frac{g'}{g} = \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{100} = \left(\frac{R}{R+h}\right)^2 \Rightarrow h = 9R$

50. (a) $g' = g \left(\frac{R}{R+h}\right)^2 = g \left(\frac{R}{R+\frac{R}{2}}\right)^2 = \frac{4}{9}g$

$$\therefore W' = \frac{4}{9} \times W = \frac{4}{9} \times 72 = 32 \text{ N}$$

51. (a) $g' = g - \omega^2 R \cos^2 \lambda \Rightarrow 0 = g - \omega^2 R \cos^2 60^\circ$

$$0 = g - \frac{\omega^2 R}{4} \Rightarrow \omega = 2\sqrt{\frac{g}{R}} = \frac{1}{400} \frac{\text{rad}}{\text{sec}} = 2.5 \times 10^{-3} \frac{\text{rad}}{\text{sec}}$$

52. (a) $g' = g \left(1 - \frac{d}{R}\right) = 9.8 \left(1 - \frac{100}{6400}\right) = 9.66 \text{ m/s}^2$

53. (c) $g' = g \left(\frac{R}{R+h}\right)^2 = \frac{g}{4}$. By solving $h = R$

54. (a) $g = \frac{4}{3}\pi\rho GR \therefore g \propto r\rho \therefore \frac{g_e}{g_m} = \frac{R}{r} \times \frac{\rho_e}{\rho_m}$

55. (c) $g_p = g_e \left(\frac{M_p}{M_e}\right) \left(\frac{R_e}{R_p}\right)^2 = 9.8 \left(\frac{1}{80}\right)(2)^2$
 $= 9.8/20 = 0.49 \text{ m/s}^2$

56. (d) Range of projectile $R = \frac{u^2 \sin 2\theta}{g}$

$$\text{if } u \text{ and } \theta \text{ are constant then } R \propto \frac{1}{g}$$

$$\frac{R_m}{R_e} = \frac{g_e}{g_m} \Rightarrow \frac{R_m}{R_e} = \frac{1}{0.2} \Rightarrow R_m = \frac{R_e}{0.2} \Rightarrow R_m = 5R_e$$

57. (a) For condition of weightlessness of equator

$$\omega = \sqrt{\frac{g}{R}} = \frac{1}{800} = 1.25 \times 10^{-3} \frac{\text{rad}}{\text{s}}$$

58. (d) $g' = g \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{\sqrt{2}} = \frac{R}{R+h}$

$$\Rightarrow R+h = \sqrt{2}R \Rightarrow h = (\sqrt{2}-1)R = 0.414R$$

59. (b) $g \propto \rho R$

60. (c) $H = \frac{u^2}{2g} \Rightarrow H \propto \frac{1}{g} \Rightarrow \frac{H_B}{H_A} = \frac{g_A}{g_B}$

$$\text{Now } g_B = \frac{g_A}{12} \text{ as } g \propto \rho R$$

$$\therefore \frac{H_B}{H_A} = \frac{g_A}{g_B} = 12 \Rightarrow H_B = 12 \times H_A = 12 \times 1.5 = 18 \text{ m}$$

61. (b)

62. (a) $g' = g \left(\frac{R}{R+h}\right)^2 = \frac{g}{\left(1+\frac{h}{R}\right)^2}$

63. (c) $g = \frac{4}{3}\pi\rho GR \Rightarrow g \propto dR$ ($\rho = d$ given in the problem)

64. (a) Inside the earth $g' = \frac{4}{3}\pi\rho Gr \therefore g' \propto r$

65. (c) $g' = g \left(\frac{R}{R+h}\right)^2 = \frac{4}{9}g \therefore W' = \frac{4}{9}W$

66. (a) $g \propto \rho$

67. (d)

68. (d)

69. (a) $g' = g \left(\frac{R}{R+h}\right)^2 = g \left(\frac{R}{3R/2}\right)^2 = \frac{4}{9}g$

$$\therefore W' = \frac{4}{9} \times mg = \frac{4 \times 200 \times 9.8}{9} = 880 \text{ N}$$

70. (a) $g = \frac{4}{3}\pi G \rho R \Rightarrow g \propto \rho R \Rightarrow \frac{g_e}{g_m} = \frac{\rho_e}{\rho_m} \times \frac{R_e}{R_m}$

$$\Rightarrow \frac{6}{1} = \frac{5}{3} \times \frac{R_e}{R_m} \Rightarrow R_m = \frac{5}{18}R_e$$

71. (a) $g' = g \left(\frac{R}{R+h}\right)^2 = g \left(\frac{R}{R+2R}\right)^2 = \frac{g}{9}$

72. (b) $g' = g \left(1 - \frac{d}{R}\right) \Rightarrow \frac{g}{4} = g \left(1 - \frac{d}{R}\right) \Rightarrow d = \frac{3}{4}R$

73. (b) For height $\frac{\Delta g}{g} \times 100\% = \frac{2h}{R} = 1\%$

$$\text{For depth } \frac{\Delta g}{g} \times 100\% = \frac{d}{R} = \frac{h}{R} = \frac{1}{2}\% = 0.5\%$$

74. (b) As $g = \frac{GM}{R^2}$ therefore 1% decrease in mass will decrease the value of g by 1%.

But 1% decrease in radius will increase the value of g by 2%.

As a whole value of g increase by 1%.

$$75. \quad (d) \quad g = \frac{4}{3} \pi \rho G R \Rightarrow \frac{R_p}{R_e} = \left(\frac{g_p}{g_e} \right) \left(\frac{\rho_e}{\rho_p} \right) = (1) \times \left(\frac{1}{2} \right)$$

$$\Rightarrow R_p = \frac{R_e}{2} = \frac{R}{2}$$

$$76. \quad (a) \quad \frac{g_1}{g_2} = \frac{\rho_1}{\rho_2} \times \frac{R_1}{R_2} = \frac{3}{2} \times \frac{2}{3} = 1$$

77. (d) Because the body weighs zero in satellite

$$78. \quad (a) \quad \text{Radius of earth } R = 6400 \text{ km} \therefore h = \frac{R}{4}$$

Acceleration due to gravity at a height h

$$g_h = g \left(\frac{R}{R+h} \right)^2 = g \left(\frac{R}{R+\frac{R}{4}} \right)^2 = \frac{16}{25} g$$

At depth ' d ' value of acceleration due to gravity

$$g_d = \frac{1}{2} g_h \quad (\text{According to problem})$$

$$\Rightarrow g_d = \frac{1}{2} \left(\frac{16}{25} g \right) \Rightarrow g \left(1 - \frac{d}{R} \right) = \frac{1}{2} \left(\frac{16}{25} g \right)$$

By solving we get $d = 4.3 \times 10^6 \text{ m}$

$$79. \quad (a) \quad g' = g - \omega^2 R \cos^2 \lambda$$

For weightlessness at equator $\lambda = 0^\circ$ and $g' = 0$

$$\therefore 0 = g - \omega^2 R \Rightarrow \omega = \sqrt{\frac{g}{R}} = \frac{1}{800} \frac{\text{rad}}{\text{sec}}$$

$$80. \quad (b) \quad \text{Weight on surface of earth, } mg = 500 \text{ N}$$

and weight below the surface of earth at $d = \frac{R}{2}$

$$mg' = mg \left(1 - \frac{d}{R} \right) = mg \left(1 - \frac{1}{2} \right) = \frac{mg}{2} = 250 \text{ N}$$

$$81. \quad (a) \quad g = \frac{4}{3} \pi G R \rho \text{ and } g' = \frac{4}{3} \pi G R' \rho$$

$$\therefore \frac{g'}{g} = \frac{R'}{R} = 0.2 \Rightarrow g' = 0.2 g$$

$$82. \quad (a)$$

$$83. \quad (d) \quad \frac{g_m}{g_e} = \frac{M_m}{M_e} \times \left(\frac{R_e}{R_m} \right)^2 = \left(\frac{1}{9} \right) \left(\frac{2}{1} \right)^2 = \frac{4}{9} \Rightarrow g_m = \frac{4}{9} g_e$$

$$\therefore W_m = \frac{4}{9} \times W_e = \frac{4}{9} \times 90 = 40 \text{ kg}$$

$$84. \quad (a) \quad g' = g - \omega^2 R, \text{ when } \omega \text{ increases } g' \text{ decreases.}$$

$$85. \quad (b) \quad \frac{g'}{g} = \frac{M'}{M} \times \frac{R^2}{R'^2} = \frac{1}{2} \times \frac{4}{1} = \frac{2}{1}$$

$$86. \quad (b) \quad \text{Acceleration due to gravity at latitude } \lambda \text{ is given by}$$

$$g' = g - R \omega^2 \cos^2 \lambda$$

$$\text{At } 30^\circ, g_{30^\circ} = g - R \omega^2 \cos^2 30^\circ = g - \frac{3}{4} R \omega^2$$

$$\therefore g - g_{30} = \frac{3}{4} \omega^2 R.$$

$$87. \quad (b) \quad \text{Acceleration due to gravity } g = \frac{GM}{R^2} \therefore \frac{g}{G} = \frac{M}{R^2}$$

Gravitational Potential, Energy and Escape Velocity

$$1. \quad (c) \quad \Delta U = \frac{mgh}{1 + h/R}$$

$$\text{Substituting } R = 5h \text{ we get } \Delta U = \frac{mgh}{1 + 1/5} = \frac{5}{6} mgh$$

$$2. \quad (a) \quad I = \frac{-dV}{dx}$$

If $V = 0$ then gravitational field is necessarily zero.

$$3. \quad (d) \quad \text{Gravitational potential} = \int I dx = \int_x^\infty \frac{K}{x^3} dx$$

$$= K \left(\frac{x^{-3+1}}{-3+1} \right)_x^\infty = \left(\frac{-K}{2x^2} \right)_x^\infty = \frac{K}{2x^2}$$

$$4. \quad (a) \quad U = -\frac{GMm}{r}$$

$$\Rightarrow 7.79 \times 10^{28} = \frac{6.67 \times 10^{-11} \times 7.4 \times 10^{22} \times 6 \times 10^{24}}{r}$$

$$\Rightarrow r = 3.8 \times 10^8 \text{ m}$$

$$5. \quad (d) \quad \Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{mg nR}{1 + \frac{nR}{R}} = \frac{nm gR}{n+1}$$

$$6. \quad (a) \quad \text{Gravitational potential at mid point}$$

$$V = \frac{-GM_1}{d/2} + \frac{-GM_2}{d/2}$$

$$\text{Now, } PE = m \times V = \frac{-2Gm}{d} (M_1 + M_2)$$

[m = mass of particle]

So, for projecting particle from mid point to infinity

$$KE = PE$$

$$\Rightarrow \frac{1}{2} mv^2 = \frac{2Gm}{d} (M_1 + M_2) \Rightarrow v = 2 \sqrt{\frac{G(M_1 + M_2)}{d}}$$

$$7. \quad (b) \quad \text{Potential energy of the } 1 \text{ kg mass which is placed at the earth surface} = -\frac{GM}{R}$$

its potential energy at infinite = 0

$$\therefore \text{Work done} = \text{change in potential energy} = \frac{GM}{R}$$

$$8. \quad (c)$$

$$9. \quad (c) \quad \frac{G \times 100}{x^2} = \frac{G \times 10000}{(1-x)^2} \Rightarrow \frac{10}{x} = \frac{100}{1-x} \Rightarrow x = \frac{1}{11} \text{ m}$$

10. (c)
11. (d) $\Delta U = U_2 - U_1 = \frac{mgh}{1 + \frac{h}{R_e}} = \frac{mgR_e}{1 + \frac{R_e}{R_e}} = \frac{mgR_e}{2}$
 $\Rightarrow U_2 - (-mgR_e) = \frac{mgR_e}{2} \Rightarrow U_2 = -\frac{1}{2}mgR_e$
12. (a) $v_e = \sqrt{\frac{2GM}{R}} = 100 \Rightarrow \frac{GM}{R} = 5000$
 Potential energy $U = -\frac{GMm}{R} = -5000 J$
13. (b) $\Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{mg \times 3R}{1 + \frac{3R}{R}} = \frac{3}{4}mgR$
14. (b) Potential energy $U = \frac{-GMm}{r} = -\frac{GMm}{R+h}$
 $U_{initial} = -\frac{GMm}{3R}$ and $U_{final} = -\frac{GMm}{2R}$
 Loss in $PE =$ gain in $KE = \frac{GMm}{2R} - \frac{GMm}{3R} = \frac{GMm}{6R}$
15. (a) If body is projected with velocity v ($v < v_e$) then
 height up to which it will rise, $h = \frac{R}{\frac{v_e^2}{v^2} - 1}$
 $v = \frac{v_e}{2}$ (given) $\therefore h = \frac{R}{\left(\frac{v_e}{v_e/2}\right)^2 - 1} = \frac{R}{4-1} = \frac{R}{3}$
16. (d) Change in potential energy in displacing a body from r_1 to r_2 is given by
 $\Delta U = GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = GMm \left(\frac{1}{2R} - \frac{1}{3R} \right) = \frac{GMm}{6R}$
17. (c) $\frac{1}{2}mv_e^2 = \frac{1}{2}m \cdot 2gR = mgR$
18. (a) $K.E. = \frac{GMm}{2R}$
19. (b) $I = \frac{-dV}{dr}$. If $I = 0$ then $V = \text{constant}$
20. (b) This should be equal to escape velocity *i.e.* $\sqrt{2gR}$
21. (c) $v_e = \sqrt{\frac{2GM}{R}}$ *i.e.* escape velocity depends upon the mass and radius of the planet.
22. (b) $v_e = \sqrt{\frac{2GM}{R}} = R\sqrt{\frac{8}{3}\pi G\rho}$
 If mean density is constant then $v_e \propto R$
 $\frac{v_e}{v_p} = \frac{R_e}{R_p} = \frac{1}{2} \Rightarrow v_e = \frac{v_p}{2}$
23. (a) Escape velocity does not depend on the mass of the projectile
24. (c) $\frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_p}{R_e}} = \sqrt{2 \times 2} = 2$
 $\Rightarrow v_p = 2 \times v_e = 2 \times 11.2 = 22.4 \text{ km/s}$
25. (a) $v_e = \sqrt{\frac{2GM}{R}} = R\sqrt{\frac{8}{3}\pi G\rho} \therefore v_e \propto R$ if $\rho = \text{constant}$
 Since the planet having double radius in comparison to earth therefore the escape velocity becomes twice *i.e.* 22 km/s .
26. (b) If missile launched with escape velocity then it will escape from the gravitational field and at infinity its total energy becomes zero.
 But if the velocity of projection is less than escape velocity then sum of energies will be negative. This shows that attractive force is working on the satellite.
27. (b)
28. (c)
29. (c) Because it does not depend on the mass of projectile
30. (b) $v_e = \sqrt{2}v_0$, *i.e.* if the orbital velocity of moon is increased by factor of $\sqrt{2}$ then it will escape out from the gravitational field of earth.
31. (a)
32. (b) $v_e = \sqrt{\frac{2GM}{R}} \Rightarrow v_e \propto \sqrt{M}$ if $R = \text{constant}$
 If the mass of the planet becomes four times then escape velocity will become 2 times.
33. (c) Escape velocity $v_e = \sqrt{\frac{2GM}{R}}$
 $\therefore \frac{v_e}{v_m} = \sqrt{\frac{M_e R_m}{M_m R_e}} = \sqrt{\frac{81}{3.5}} = 4.81$
34. (a) $v_e = \sqrt{\frac{2GM}{R}} \therefore v_e \propto \sqrt{\frac{M}{R}}$
 If mass and radius of the planet are three times than that of earth then escape velocity will be same.
35. (c) Potential energy of a body at the surface of earth
 $PE = -\frac{GMm}{R} = -\frac{gR^2m}{R} = -mgR$
 $= -500 \times 9.8 \times 6.4 \times 10^6 = -3.1 \times 10^{10} J$
 So if we give this amount of energy in the form of kinetic energy then body escape from the earth.
36. (d) Escape velocity $v = \sqrt{\frac{2GM}{R}} \Rightarrow \frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}}$
 $\Rightarrow v_p = 5v_e = 5 \times 11.2 = 56 \text{ km/s}$
37. (a) $\frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}} = \sqrt{6 \times \frac{1}{2}} = \sqrt{3} \therefore v_p = \sqrt{3}v_e$

38. (a) $\frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_p}{R_e}} = \sqrt{9 \times 4} = 6 \therefore v_p = 6 \times v_e = 67.2 \text{ km/s}$

39. (c) $\frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_p}{R_e}} = \sqrt{8 \times \frac{1}{2}} = 2 \therefore v_p = 2 \times v_e = 22.4 \text{ km/s}$

40. (c) $v_e = \sqrt{\frac{2GM}{R}} \therefore v_e \propto \sqrt{\frac{M}{R}}$

If M becomes double and R becomes half then escape velocity becomes two times.

41. (c) On earth $v_e = \sqrt{\frac{2GM}{R}} = 11.2 \text{ km/s}$

On moon $v_m = \sqrt{\frac{2GM \times 4}{81 \times R}} = \frac{2}{9} \sqrt{\frac{2GM}{R}}$
 $= \frac{2}{9} \times 11.2 = 2.5 \text{ km/s}$

42. (c) Escape velocity $v = \sqrt{\frac{2GM}{R}}$

If star rotates with angular velocity ω

then $\omega = \frac{v}{R} = \frac{1}{R} \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2GM}{R^3}}$

43. (d) Escape velocity from surface of earth $v_e = \sqrt{2gR}$
 $= \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.2 \times 10^3 \text{ m/s}$

44. (a)

45. (a) $\frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}} = \sqrt{(1000) \times \left(\frac{1}{10}\right)} = 10$
 $v_p = 10 \times 11.2 = 112 \text{ km/s}$

46. (b) $v_e = R \sqrt{\frac{8}{3} G \pi \rho} \therefore v_e \propto R \sqrt{\rho}$

47. (d) $v_e = \sqrt{\frac{2GM}{(R+h)}}$

48. (a) $v = \sqrt{2gR}$. If acceleration due to gravity and radius of the planet, both are double that of earth then escape velocity will be two times. i.e. $v_p = 2v_e$

49. (a)

50. (b) $v = \sqrt{2gR} \Rightarrow \frac{v_A}{v_B} = \sqrt{\frac{g_A}{g_B} \times \frac{R_A}{R_B}} = \sqrt{k_1 \times k_2} = \sqrt{k_1 k_2}$

51. (d) $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6 \times 10^{24}}{R}} = 3 \times 10^8$

By solving $R = 9 \text{ mm}$

52. (a) $\frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}} = \sqrt{2 \times \frac{1}{3}} = \sqrt{\frac{2}{3}} \therefore v_p = \sqrt{\frac{2}{3}} v_e$

53. (c) $v_e \propto \frac{1}{\sqrt{R}}$. If R becomes $\frac{1}{4}$ then v_e will be 2 times.

54. (b)

55. (a) $v = \sqrt{2gR} \Rightarrow \frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_e}{R_p}} = \sqrt{2 \times \frac{1}{4}} = \frac{1}{\sqrt{2}}$
 $\therefore v_p = \frac{v_e}{\sqrt{2}}$

56. (b) $v \propto R \sqrt{\rho} \therefore \frac{v_p}{v_e} = \frac{R_p}{R_e} \times \sqrt{\frac{\rho_p}{\rho_e}} = 4 \times \sqrt{9} = 12$
 $\Rightarrow v_p = 12v_e$

57. (d) Escape velocity does not depend upon the angle of projection.

58. (d)

59. (b) Escape velocity is independent of mass of object.

60. (b) $v = \sqrt{2gR} \Rightarrow \frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_p}{R_e}} = \sqrt{1 \times 4} = 2$
 $\therefore v_p = 2v_e$

61. (c) $v = \sqrt{2gR} \Rightarrow \frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_p}{R_e}} = \sqrt{1 \times 4} = 2$
 $\Rightarrow v_p = 2 \times v_e = 2 \times 11.2 = 22.4 \text{ km/s}$

62. (b) $v = \sqrt{2gR}$. If g and R both are doubled then v will become two times i.e. $11.2 \times 2 = 22.4 \text{ km/s}$

63. (c) $v = R \sqrt{\frac{8}{3} \pi \rho G} \Rightarrow \frac{v_p}{v_e} = \frac{R_p}{R_e} \sqrt{\frac{\rho_p}{\rho_e}} = 2 \sqrt{\frac{1}{4}} = 1$

64. (c) Velocity of body in inter planetary space $v' = \sqrt{v^2 - v_{es}^2}$
 where v_{es} = escape velocity and
 v = velocity of projection
 $\therefore v' = \sqrt{(2v_{es})^2 - v_{es}^2} = \sqrt{3v_{es}^2} \Rightarrow v' = \sqrt{3} v_{es}$

65. (b) Potential energy of system of two mass

$$U = \frac{-GMm}{R} = \frac{-6.67 \times 10^{-11} \times 100 \times 10 \times 10^{-3}}{10 \times 10^{-2}}$$

$$U = -6.67 \times 10^{-10} \text{ J}$$

So, the amount of work done to take the particle up to infinite will be $6.67 \times 10^{-10} \text{ J}$

66. (b) For a moving satellite kinetic energy = $\frac{GMm}{2r}$

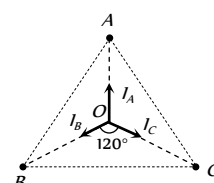
$$\text{Potential energy} = \frac{-GMm}{r}$$

$$\therefore \frac{\text{Kinetic energy}}{\text{Potential energy}} = \frac{1}{2}$$

67. (a) Due to three particles net intensity at the centre

$$I = \vec{I}_A + \vec{I}_B + \vec{I}_C = 0$$

because out of these three intensities one equal in magnitude and the angle between each other is 120° .



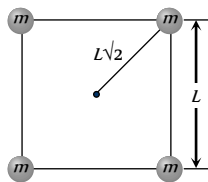
68. (c) $v_0 = \frac{v_e}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ km/s}$

69. (a) Potential at the centre due to single mass = $\frac{-GM}{L/\sqrt{2}}$

Potential at the centre due to all four masses

$$= -4 \frac{GM}{L/\sqrt{2}} = -4\sqrt{2} \frac{GM}{L}$$

$$= -\sqrt{32} \times \frac{GM}{L}$$



70. (a) $v = \sqrt{2gR} \therefore \frac{v_1}{v_2} = \sqrt{\frac{g_1}{g_2} \times \frac{R_1}{R_2}} = \sqrt{g \times K} = (Kg)^{1/2}$

Motion of Satellite

1. (b) $v_e = \sqrt{2gR}$ and $v_0 = \sqrt{gR} \therefore \sqrt{2} v_0 = v_e$

2. (d)

3. (d) $v_0 = \sqrt{\frac{GM}{r}}$

4. (d) $v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R+h}}$

5. (b) Centripetal acceleration works on it.

6. (b) $v = \sqrt{\frac{GM}{r}}$ if $r_1 > r_2$ then $v_1 < v_2$

Orbital speed of satellite does not depend upon the mass of the satellite

7. (b) $T \propto r^{3/2}$. If r becomes double then time period will become $(2)^{3/2}$ times.

So new time period will be $24 \times 2\sqrt{2} \text{ hr}$ i.e. $T = 48\sqrt{2}$

8. (b) $K.E.$ required for satellite to escape from earth's gravitational

$$\text{field } \frac{1}{2} m v_e^2 = \frac{1}{2} m \left(\sqrt{\frac{2GM}{R}} \right)^2 = \frac{GMm}{R}$$

$K.E.$ required for satellite to move in circular orbit

$$\frac{1}{2} m v_0^2 = \frac{1}{2} m \left(\sqrt{\frac{GM}{R}} \right)^2 = \frac{GMm}{2R}$$

The ratio between these two energies = 2

9. (c) The velocity of the spoon will be equal to the orbital velocity when dropped out of the space-ship.

10. (c)

11. (b) $v_0 = \sqrt{gR}$

12. (d) Telecommunication satellites are geostationary satellite

13. (b) $v = \sqrt{\frac{GM}{R}} \Rightarrow \frac{v_A}{v_B} = \sqrt{\frac{R_B}{R_A}} = \sqrt{\frac{R}{4R}} = \frac{1}{2}$

$$\therefore \frac{v_A}{v_B} = \frac{3V}{v_B} = \frac{1}{2} \therefore v_B = 6V$$

14. (a)

15. (a)

16. (b)

17. (c) $v = \sqrt{\frac{GM}{R+h}}$

For first satellite $h=0$, $v_1 = \sqrt{\frac{GM}{R}}$

For second satellite $h = \frac{R}{2}$, $v_2 = \sqrt{\frac{2GM}{3R}}$

$$v_2 = \sqrt{\frac{2}{3}} v_1 = \sqrt{\frac{2}{3}} v$$

18. (d) $v = \sqrt{\frac{GM}{r}} \therefore K.E. \propto v^2 \propto \frac{1}{r}$ and $T^2 \propto r^3$

$$\therefore K.E. \propto T^{-2/3}$$

19. (d)

20. (d) $T = 2\pi \sqrt{\frac{r^3}{GM}} \Rightarrow r^3 = \frac{GMT^2}{4\pi^2} \Rightarrow r = \left[\frac{GMT^2}{4\pi^2} \right]^{1/3}$

21. (d) $v \propto \frac{1}{\sqrt{r}}$. The speed of satellite decreases with an increase in the radius of its orbit.

22. (b) $v \propto \frac{1}{\sqrt{r}}$

$$\% \text{ increase in speed} = \frac{1}{2} (\% \text{ decrease in radius})$$

$$= \frac{1}{2} (1\%) = 0.5\%$$

i.e. speed will increase by 0.5%

23. (b) $v = \sqrt{\frac{GM}{r}}$

24. (a)

25. (a) $v \propto \frac{1}{\sqrt{r}}$. If orbital radius becomes 4 times then orbital

$$\text{velocity will become half, i.e. } \frac{7}{2} = 3.5 \text{ km/s}$$

26. (d) Orbital radius of satellites $r_1 = R + R = 2R$

$$r_2 = R + 7R = 8R$$

$$U_1 = \frac{-GMm}{r_1} \text{ and } U_2 = \frac{-GMm}{r_2}$$

$$K_1 = \frac{GMm}{2r_1} \text{ and } K_2 = \frac{GMm}{2r_2}$$

$$E_1 = \frac{GMm}{2r_1} \text{ and } E_2 = \frac{GMm}{2r_2}$$

$$\therefore \frac{U_1}{U_2} = \frac{K_1}{K_2} = \frac{E_1}{E_2} = 4$$

27. (a) Escape velocity is same for all angles of projection.

28. (d) Orbital velocity $v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{r}}$ and $v_0 = r\omega$

This gives $r^3 = \frac{R^2 g}{\omega^2}$

29. (a)
30. (a) Due to inertia it will continue to move along the original path of the space craft.

31. (b)
32. (d)
33. (a)

34. (c) (i) $T_{st} = 2\pi \sqrt{\frac{(R+h)^3}{GM}} = 2\pi \sqrt{\frac{R}{g}}$

[As $h \ll R$ and $GM = gR^2$]

(ii) $T_{ma} = 2\pi \sqrt{\frac{R}{g}}$

(iii) $T_{sp} = 2\pi \sqrt{\frac{1}{g\left(\frac{1}{l} + \frac{1}{R}\right)}} = 2\pi \sqrt{\frac{R}{2g}}$

[As $l = R$]

(iv) $T_{is} = 2\pi \sqrt{\frac{R}{g}}$ [As $l = \infty$]

35. (d)

36. (c) $v \propto \frac{1}{\sqrt{r}}$, If $r = R$ then $v = V_0$

If $r = R + h = R + 3R = 4R$ then $v = \frac{V_0}{2} = 0.5V_0$

37. (d)

38. (b) $6R$ from the surface of earth and $7R$ from the centre.

39. (c) $T^2 = \frac{4\pi^2}{GM} r^3$. If G is variable then time period, angular velocity and orbital radius also changes accordingly.

40. (b) $T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} = 2\pi \sqrt{\frac{(2R)^3}{gR^2}} = 4\sqrt{2}\pi \sqrt{\frac{R}{g}}$

41. (c) $T = 2\pi \sqrt{\frac{r^3}{GM}} \Rightarrow T^2 = \frac{4\pi^2}{GM} (R+h)^3$

$\Rightarrow R+h = \left[\frac{GMT^2}{4\pi^2} \right]^{1/3} \Rightarrow h = \left[\frac{GMT^2}{4\pi^2} \right]^{1/3} - R$

42. (d) Distances of the satellite from the centre are $7R$ and $3.5R$ respectively.

$\frac{T_2}{T_1} = \left(\frac{R_2}{R_1} \right)^{3/2} \Rightarrow T_2 = 24 \left(\frac{3.5R}{7R} \right)^{3/2} = 6\sqrt{2} \text{ hr}$

43. (a) $v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{384000 \times 10^3}} = 1 \text{ km/s}$

44. (d) % change in $T = \frac{3}{2}$ (% change in R) = $\frac{3}{2} \times (2)\% = 3\%$

45. (a)
46. (c)

47. (c) $v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{r}} = \sqrt{\frac{10 \times (64 \times 10^5)^2}{8000 \times 10^3}}$

$= 71.5 \times 10^2 \text{ m/s} = 7.15 \text{ km/s}$

48. (d) Total mechanical energy of satellite

$E = \frac{-GMm}{2r} \Rightarrow \frac{E_A}{E_B} = \frac{m_A}{m_B} \times \frac{r_B}{r_A} \Rightarrow \frac{3}{1} \times \frac{4r}{r} = \frac{12}{1}$

49. (b)

50. (b) Gravitational force provides the required centripetal force for orbiting the satellite

$\frac{mv^2}{R} = \frac{K}{R}$ because $\left(F \propto \frac{1}{R} \right)$

$\therefore v \propto R^0$

51. (a) Total energy = - (kinetic energy) = $-\frac{1}{2}Mv^2$

52. (a) Binding energy = - kinetic energy
And if this amount of energy (E_k) given to satellite then it will escape into outer space

53. (a) Potential energy = $\frac{-GMm}{r} = \frac{GMm}{R_e + h} = \frac{-GMm}{2R_e}$

$= -\frac{gR_e^2 m}{2R_e} = -\frac{1}{2}mgR_e = -0.5mgR_e$

54. (c) $B.E. = -\frac{GMm}{r}$. If $B.E.$ decreases then r also decreases and v increases as $v \propto \frac{1}{\sqrt{r}}$

55. (b) Time period of communication satellite $T_c = 1$ day
Time period of another satellite = T_s

$\frac{T_s}{T_c} = \left(\frac{r_s}{r_c} \right)^{3/2} = (4)^{3/2} \Rightarrow T_s = T_c \times (4)^{3/2} = 8 \text{ days.}$

56. (b) Angular momentum is conserved in central field.

57. (b) $T^2 \propto r^3$

58. (d) $T^2 \propto R^3 \therefore \frac{T^2}{R^3} = \text{constant}$

59. (d) $T_2 = T_1 \left(\frac{R_2}{R_1} \right)^{3/2} = T_1 (4)^{3/2} = 8T_1 = 40 \text{ hr}$

60. (b) Given that, $T_1 = 1$ day and $T_2 = 8$ days

$\therefore \frac{T_2}{T_1} = \left(\frac{r_2}{r_1} \right)^{3/2} \Rightarrow \frac{r_2}{r_1} = \left(\frac{T_2}{T_1} \right)^{2/3} = \left(\frac{8}{1} \right)^{2/3} = 4$

$\Rightarrow r_2 = 4r_1 = 4R$

61. (c)

62. (c) $\frac{v_B}{v_A} = \sqrt{\frac{r_A}{r_B}} = \sqrt{\frac{4R}{R}} = 2$

$\Rightarrow v_B = 2 \times v_A = 2 \times 3v = 6v$

63. (b) Gravitational force provides the required centripetal force

$m\omega^2 R = \frac{GMm}{R^3} \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{R^4} \Rightarrow T \propto R^2$

64. (a)

65. (b) $v_e = \sqrt{2}v_0 = 1.414 v_0$

Fractional increase in orbital velocity $\left(\frac{\Delta v}{v}\right)$

$$= \frac{v_e - v_0}{v_0} = 0.414$$

$\therefore$ Percentage increase = 41.4%

66. (c) Time period of revolution of moon around the earth
 = 1 lunar month.

$$\frac{T_s}{T_m} = \left(\frac{r_s}{r_m}\right)^{3/2} = \left(\frac{1}{2}\right)^{3/2} \Rightarrow T_s = 2^{-3/2} \text{ lunar month.}$$

67. (d)

Kepler's Laws of Planetary Motion

1. (c) $\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{3/2} = \left(\frac{10^{13}}{10^{12}}\right)^{3/2} = (1000)^{1/2} = 10\sqrt{10}$

2. (c) Areal velocity of the planet remains constant. If the areas A and B are equal then $t_1 = t_2$.

3. (c) $\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{3/2} = \left(\frac{R}{4R}\right)^{3/2} \Rightarrow T_2 = 8T_1$

4. (b)

5. (c)

6. (b) Since $T^2 \propto r^3 \therefore \left(\frac{T'}{T}\right)^2 = \left(\frac{1}{4}\right)^3 \Rightarrow T' = \frac{1}{8}T$

7. (b) Kinetic and potential energies varies with position of earth w.r.t. sun. Angular momentum remains constant every where.

8. (c)

9. (c) Angular momentum remains constant

$$mv_1d_1 = mv_2d_2 \Rightarrow v_2 = \frac{v_1d_1}{d_2}$$

10. (b) Orbital radius of Jupiter > Orbital radius of Earth

$$\frac{v_J}{v_e} = \frac{r_e}{r_J}. \text{ As } r_J > r_e \text{ therefore } v_J < v_e$$

11. (d) $T^2 \propto r^3 \Rightarrow \frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2}$

12. (b) $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$

13. (c) The earth moves around the sun in elliptical path. so by using the properties of ellipse

$$r_1 = (1+e)a \text{ and } r_2 = (1-e)a$$

$$\Rightarrow a = \frac{r_1 + r_2}{2} \text{ and } r_1 r_2 = (1-e^2)a^2$$

where a = semi major axis

b = semi minor axis

e = eccentricity

Now required distance = semi latusrectum $= \frac{b^2}{a}$

$$= \frac{a^2(1-e^2)}{a} = \frac{(r_1 r_2)}{(r_1 + r_2)/2} = \frac{2r_1 r_2}{r_1 + r_2}$$

14. (c) For first satellite $r_1 = R$ and $T_1 = 83 \text{ minute}$

For second satellite $r_2 = 4R$

$$T_2 = T_1 \left(\frac{r_2}{r_1}\right)^{3/2} = T_1(4)^{3/2} = 8T_1 = 8 \times 83 = 664 \text{ minutes}$$

15. (a) Angular momentum = Mass $\times$ Orbital velocity $\times$ Radius

$$= m \times \left(\sqrt{\frac{GM}{R_0}}\right) \times R_0 = m\sqrt{GMR_0}$$

16. (b) $\frac{T^2}{r^3} = \text{constant} \Rightarrow T^2 r^{-3} = \text{constant}$

17. (d) $\frac{T_{\text{planet}}}{T_{\text{earth}}} = \left(\frac{r_{\text{planet}}}{r_{\text{earth}}}\right)^{3/2} = (1.588)^{3/2} = 2 \therefore T_{\text{planet}} = 2 \text{ year}$

18. (d) Mass of the satellite does not effects on time period

$$\frac{T_A}{T_B} = \left(\frac{r_1}{r_2}\right)^{3/2} = \left(\frac{r}{2r}\right)^{3/2} = \left(\frac{1}{8}\right)^{1/2} = \frac{1}{2\sqrt{2}}$$

19. (b) Speed of the earth will be maximum when its distance from the sun is minimum because $mvr = \text{constant}$

20. (c) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{3/2} \Rightarrow 8 = \left(\frac{r_A}{r_B}\right)^{3/2} \Rightarrow r_A = (8)^{2/3} r_B = 4r_B$

21. (c) $T^2 \propto r^3$. If r made half then T will become $\frac{T}{8}$.

22. (c)

23. (b) Mass of satellite does not affects on orbital radius.

24. (b) $\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = (2)^{3/2} = 2\sqrt{2} \Rightarrow T_2 = 2\sqrt{2} \text{ years.}$

25. (c)

26. (a)

27. (a) By conservation of angular momentum $mvr = \text{constant}$

$$v_{\min} \times r_{\max} = v_{\max} \times r_{\min}$$

$$\therefore v_{\min} = \frac{60 \times 1.6 \times 10^{12}}{8 \times 10^{12}} = \frac{60}{5} = 12 \text{ m/s}$$

28. (b) $\omega_{\text{body}} = 27\omega_{\text{earth}}$

$$T^2 \propto r^3 \Rightarrow \omega^2 \propto \frac{1}{r^3} \Rightarrow \omega \propto \frac{1}{r^{3/2}} \therefore r \propto \frac{1}{\omega^{2/3}}$$

$$\Rightarrow \frac{r_{\text{body}}}{r_{\text{earth}}} = \left(\frac{\omega_{\text{earth}}}{\omega_{\text{body}}}\right)^{2/3} = \left(\frac{1}{27}\right)^{2/3} = \frac{1}{9}$$

29. (d) Time period does not depends upon the mass of satellite.

30. (b) $\frac{T^2}{R^3} = \frac{T^2}{d^3} = \frac{1}{n^2 d^3} = \text{constant}$

$$\therefore n_1^2 d_1^3 = n_2^2 d_2^3 \quad [\text{where } n = \text{frequency}]$$

31. (a)

32. (a)

33. (a) During path DAB planet is nearer to sun as comparison with path BCD . So time taken in travelling DAB is less than that for BCD because velocity of planet will be more in region DAB .
34. (c) Because distance of point C is maximum from the sun.
35. (b) $T_2 = T_1 \left(\frac{R_2}{R_1} \right)^{3/2} = 1 \times (2)^{3/2} = 2.8 \text{ year}$
36. (d) $L = mvr = m \sqrt{\frac{GM}{r}} r = m \sqrt{GM} r \therefore L \propto \sqrt{r}$
37. (a)
38. (c) $\frac{dA}{dt} = \frac{L}{2m} = \frac{dA}{dt} \propto vr \propto \omega r^2$
39. (b)
40. (c)
41. (d)
42. (a) $T^2 \propto R^3 \Rightarrow \left(\frac{T_P}{T_E} \right)^2 = \left(\frac{R_P}{R_E} \right)^3 = \left(\frac{2R_E}{R_E} \right)^3$
 $\Rightarrow \frac{T_P}{T_E} = (2)^{3/2} = 2\sqrt{2}$
 $\Rightarrow T_P = 2\sqrt{2} \times 365 = 1032.37 \approx 1032 \text{ days}$
43. (d)
44. (d) For central force, torque is zero.
 $\therefore \tau = \frac{dL}{dt} = 0 \Rightarrow L = \text{constant}$
i.e. Angular momentum is constant.
45. (b) $\frac{v_{\max}}{v_{\min}} = \frac{1+e}{1-e} = \frac{1+0.0167}{1-0.0167} = 1.033$
46. (a) $m\omega^2 R = \frac{GMm}{R^2} \Rightarrow \left(\frac{2\pi}{T} \right)^2 R = \frac{GM}{R^2} \Rightarrow M = \frac{4\pi^2 R^3}{GT^2}$
47. (b) $v = \sqrt{\frac{GM}{R}} = G^{1/2} M^{1/2} R^{-1/2}$
48. (b)
49. (c) $v = \sqrt{\frac{GM}{R}}$ if $r_1 > r_2$ then $v_1 < v_2$
50. (c) Escape velocity for that body $v_e = \sqrt{\frac{2Gm}{r}}$
 v_e should be more than or equal to speed of light
i.e. $\sqrt{\frac{2Gm}{r}} \geq c$
51. (c) $\frac{T_2}{T_1} = \left(\frac{r_2}{r_1} \right)^{3/2} = \left(\frac{1}{4} \right)^{3/2} = \frac{1}{8}$
2. (a, b) $g = \frac{4}{3} \pi \rho Gr \therefore g \propto r$ if $r < R$
 $g = \frac{GM}{r^2} \therefore g \propto \frac{1}{r^2}$ if $r > R$
 If $r_1 < R$ and $r_2 < R$ then $\frac{F_1}{F_2} = \frac{g_1}{g_2} = \frac{r_1}{r_2}$
 If $r_1 > R$ and $r_2 > R$ then $\frac{F_1}{F_2} = \frac{g_1}{g_2} = \left(\frac{r_2}{r_1} \right)^2$
3. (a)
4. (b) $F = \frac{Gm(M-m)}{r^2}$
 For maximum force $\frac{dF}{dm} = 0$
 $\Rightarrow \frac{d}{dm} \left(\frac{GmM}{r^2} - \frac{Gm^2}{r^2} \right) = 0$
 $\Rightarrow M - 2m = 0 \Rightarrow \frac{m}{M} = \frac{1}{2}$
5. (a) $m\omega^2 R \propto \frac{1}{R^n} \Rightarrow m \left(\frac{4\pi^2}{T^2} \right) R \propto \frac{1}{R^n} \Rightarrow T^2 \propto R^{n+1}$
 $\therefore T \propto R^{\left(\frac{n+1}{2} \right)}$
6. (c) $g = \frac{GM}{R^2}$. If mass remains constant then $g \propto \frac{1}{R^2}$
 % increase in $g = 2$ (% decrease in R) $= 2 \times 1\% = 2\%$.
7. (b, c, d) $g = \frac{GM}{R^2}, v_e = \sqrt{\frac{2GM}{R}}$ and $U = \frac{-GMm}{R}$
 $\therefore g \propto \frac{M}{R^2}, v_e \propto \sqrt{\frac{M}{R}}$ and $U \propto \frac{M}{R}$
 If both mass and radius are increased by 0.5% then v_e and U remains unchanged where as g decrease by 0.5%.
8. (b) $g' = g - \omega^2 R \cos^2 \lambda$
 For weightlessness at equator $\lambda = 0$ and $g' = 0$
 $\therefore 0 = g - \omega^2 R \Rightarrow \omega = \sqrt{\frac{g}{R}} = \frac{1}{800} \frac{\text{rad}}{\text{s}}$
9. (d) If acceleration due to gravity is g at the surface of earth then at height R it value becomes $g' = g \left(\frac{R}{R+h} \right)^2 = \frac{g}{4}$
 $T_1 = 2\pi \sqrt{\frac{l}{g}}$ and $T_2 = 2\pi \sqrt{\frac{l}{g/4}} \therefore \frac{T_2}{T_1} = 2$
10. (b) $\Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{1}{2} mgR$ ($\because h = R$)
11. (c) Potential energy $= 2 \times (\text{Total energy}) = 2E_0$
 Because we know $U = \frac{-GMm}{r}$ and $E_0 = \frac{-GMm}{2r}$
12. (c) $\Delta K.E. = \Delta U$

Critical Thinking Questions

1. (b) For revolution of planet centripetal force is provided by gravitational force of attraction
 $m\omega^2 R \propto R^{-5/2} \Rightarrow \frac{1}{T^2} \propto R^{-7/2} \Rightarrow T^2 \propto R^{7/2}$
11. (c) Potential energy $= 2 \times (\text{Total energy}) = 2E_0$
 Because we know $U = \frac{-GMm}{r}$ and $E_0 = \frac{-GMm}{2r}$
12. (c) $\Delta K.E. = \Delta U$

$$\Rightarrow \frac{1}{2} MV^2 = GM_e M \left(\frac{1}{R} - \frac{1}{R+h} \right) \quad \dots(i)$$

$$\text{Also } g = \frac{GM_e}{R^2} \quad \dots(ii)$$

$$\text{On solving (i) and (ii) } h = \frac{R}{\left(\frac{2gR}{V^2} - 1 \right)}$$

13. (a, c, d) Since cavities are symmetrical w.r.t. O . So the gravitational force at the centre is zero.

The radius of the circle $z^2 + y^2 = 36$ is 6.

For all points for $r \geq 6$, the body behaves as if whole of the mass is concentrated at the centre. So the gravitational potential is same.

Above is true for $z^2 + y^2 = 4$ as well.

14. (b) Let velocities of these masses at r distance from each other be v_1 and v_2 respectively.

By conservation of momentum

$$m_1 v_1 - m_2 v_2 = 0$$

$$\Rightarrow m_1 v_1 = m_2 v_2 \quad \dots(i)$$

By conservation of energy

change in P.E. = change in K.E.

$$\frac{Gm_1 m_2}{r} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\Rightarrow \frac{m_1^2 v_1^2}{m_1} + \frac{m_2^2 v_2^2}{m_2} = \frac{2Gm_1 m_2}{r} \quad \dots(ii)$$

On solving equation (i) and (ii)

$$v_1 = \sqrt{\frac{2Gm_2^2}{r(m_1 + m_2)}} \quad \text{and} \quad v_2 = \sqrt{\frac{2Gm_1^2}{r(m_1 + m_2)}}$$

$$\therefore v_{\text{app}} = |v_1| + |v_2| = \sqrt{\frac{2G}{r}} (m_1 + m_2)$$

15. (c) Kinetic energy = Potential energy

$$\frac{1}{2} m (kv_e)^2 = \frac{mgh}{1 + \frac{h}{R}} \Rightarrow \frac{1}{2} mk^2 2gR = \frac{mgh}{1 + \frac{h}{R}}$$

$$\Rightarrow h = \frac{Rk^2}{1 - k^2}$$

Height of Projectile from the earth's surface = h

$$\text{Height from the centre } r = R + h = R + \frac{Rk^2}{1 - k^2}$$

$$\text{By solving } r = \frac{R}{1 - k^2}$$

16. (c) In the problem orbital radius is increased by 1%.

Time period of satellite $T \propto r^{3/2}$

Percentage change in time period

$$= \frac{3}{2} (\% \text{ change in orbital radius}) = \frac{3}{2} (1\%) = 1.5\%$$

17. (b) According to Kepler's third law, the ratio of the squares of the periods of any two planets revolving about the sun is equal to the ratio of the cubes of their average distances from the sun i.e.

$$\left(\frac{T_1}{T_2} \right)^2 = \left(\frac{r_1}{r_2} \right)^3 = \left[\frac{r_1}{\frac{1}{2} r_1} \right]^3 = 8 \Rightarrow \frac{T_1}{T_2} = 2\sqrt{2}$$

$$\therefore T_2 = \frac{T_1}{2\sqrt{2}} = \frac{365 \text{ days}}{2\sqrt{2}} = 129 \text{ days}$$

$$18. (c) \frac{T_2}{T_1} = \left(\frac{r_2}{r_1} \right)^{3/2} \Rightarrow T_2 = 24 \left(\frac{6400}{36000} \right)^{3/2} \approx 2 \text{ hour}$$

Graphical Questions

- (c) $g \propto r$ (if $r < R$) and $g \propto \frac{1}{r^2}$ (if $r > R$)
- (c) $V_{\text{in}} = \frac{-Gm}{2R} \left[3 - \left(\frac{r}{R} \right)^2 \right]$, $V_{\text{surface}} = \frac{-GM}{R}$, $V_{\text{out}} = \frac{-GM}{r}$
- (c) For hollow sphere
 $V_{\text{in}} = \frac{-GM}{R}$, $V_{\text{surface}} = \frac{-GM}{R}$, $V_{\text{out}} = \frac{-GM}{r}$
 i.e. potential remain constant inside the sphere and it is equal to potential at the surface and increase when the point moves away from the surface of sphere.
- (b) $F = 0$ when $0 \leq r \leq R_1$
 because intensity is zero inside the cavity.
 F increase when $R_1 \leq r \leq R_2$
 $F \propto \frac{1}{r^2}$ when $r > R_2$
- (d) Intensity will be zero inside the spherical shell.
 $I = 0$ upto $r = a$ and $I \propto \frac{1}{r^2}$ when $r > a$
- (c) Initially the weight of the passenger = $60 \times 10 = 600 \text{ N}$
 Finally the weight of the passenger = $60 \times 4 = 240 \text{ N}$
 and during the flight in between some where its weight will be zero because at that point gravitational pull of earth and mars will be equal.
- (c) Kepler's law $T^2 \propto R^3$
- (d) The system will be bound at points where total energy is negative. In the given curve at point A, B and C the P.E. is more than K.E.
- (c) $U = \frac{-GMm}{r}$, $K = \frac{GMm}{2r}$ and $E = \frac{-GMm}{2r}$
 For a satellite U, K and E varies with r and also U and E remains negative whereas K remain always positive.
- (c) Gravitational P.E. = $m \times$ gravitational potential
 $U = mV$ So the graph of U will be same as that of V for a spherical shell.

Assertion and Reason

1. (a) According to Kepler's third law $T^2 \propto r^3$
If r is small then T will also be small.
2. (b) For two electron $\frac{F_g}{F_e} = 10^{-43}$ i.e. gravitational force is negligible in comparison to electrostatic force of attraction.
3. (d) The universal gravitational constant G is totally different from g .
$$G = \frac{FR^2}{Mm}$$

The constant G is scalar and posses the dimensions $[M^{-1}L^3T^{-2}]$.
$$g = \frac{GM}{R^2}$$

 g is a vector and has got the dimensions $[M^0LT^{-2}]$.
It is not a universal constant.
4. (c) Acceleration due to gravity is given by $g = \frac{GM}{R^2}$. Thus it does not depend on mass of body on which it is acting. Also it is not a constant quantity it changes with change in value of both M and R (distance between two bodies).
5. (a) If a pendulum is suspended in a lift and lift is moving downward with some acceleration a , then time period of pendulum is given by, $T = 2\pi\sqrt{\frac{l}{g-a}}$.
In the case of free fall, $a = g$ then $T = \infty$
i.e., the time period of pendulum becomes infinite.
6. (c) The value of g at any place is given by the relation,
 $g' = g - \omega^2 R_e \cos^2 \lambda$
When λ is angle of latitude and ω is the angular velocity of earth. If earth suddenly stops rotating, then $\omega = 0$
 $\therefore g' = g$ i.e., the value of g will be same at all places.
7. (b) Acceleration due to gravity,
 $g' = g - R\omega^2 \cos^2 \lambda$
At equator, $\lambda = 0^\circ$ i.e. $\cos 0^\circ = 1 \therefore g_e = g = R\omega^2$
At poles, $\lambda = 90^\circ$ i.e. $\cos 90^\circ = 0 \therefore g_p = g$
Thus, $g_p - g_e = g - g + R\omega^2 = R\omega^2$
Also, the value of g is maximum at poles and minimum at equators.
8. (a) As the rotation of earth takes place about polar axis therefore body placed at poles will not feel any centrifugal force and its weight or acceleration due to gravity remains unaffected.
9. (e) Earth revolves around the sun in circular path and required centripetal force is provided by gravitational force between earth and sun but the work done by this centripetal force is zero.
10. (a) Inertial mass and gravitational mass are equivalent. Both are scalar quantities and measured in the same unit. They are quite different in the method of their measurement. Also gravitational mass of a body is affected by the presence of other bodies near it where as internal mass remain unaffected.
11. (a) Because gravitational force is always attractive in nature and every body is bound by this gravitational force of attraction of earth.
12. (a) As no torque is acting on the planet, its angular momentum must remain constant in magnitude as well as direction. Therefore, plane of rotation must pass through the centre of earth.
13. (c) According to Kepler's law of planetary motion, a planet revolves around the sun in such a way that its areal velocity is constant. i.e., it move faster, when it is closer the sun and vice-versa.
14. (e) Escape velocity = $\sqrt{2} \times$ orbital velocity.
15. (c) Due to resistance force of atmosphere, the satellite revolving around the earth losses kinetic energy. Therefore in a particular orbit the gravitational attraction of earth on satellite becomes greater than that required for circular orbit there. Therefore satellite moves down to a lower orbit. In the lower orbit as the potential energy ($U = -GMm/r$) becomes more negative, Hence kinetic energy ($E_K = GMm/2r$) increases, and hence speed of satellite increases.
16. (b) If root mean square velocity of the gas molecules is less than escape velocity from that planet (or satellite) then atmosphere will remain attached with that planet and if $v_{rms} > v_{escape}$ then there will be no atmosphere on the planet. This is the reason for no atmosphere at moon.
17. (b) As the geostationary satellite is established in an orbit in the plane of the equator at a particular place, so it move in the same sense as the earth and hence its time period of revolution is equal to 24 hours, which is equal to time period of revolution of earth about its axis.
18. (e) The total gravitational force on one particle due to number of particles is the resultant force of attraction (or gravitational force) exerted on the given particle due to individual particles.
i.e., $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$ It means the principle of superposition is valid.
19. (d) As, escape velocity = $\sqrt{\frac{2GM}{R}}$, so its value depends on mass of planet and radius of the planet. The two different planets have same escape velocity, when these quantities (mass and radius) are equal.
20. (a) According to kepler's law $T^2 \propto r^3 \propto (R+h)^3$
i.e. if distance of satellite is more then its time period will be more.
21. (a) According to Newton's law of gravitation,
 $F = \frac{Gm_1m_2}{r^2}$. When m_1, m_2 and r_2 all are doubled,
 $F = \frac{G(2m_1)(2m_2)}{(2r)^2} = \frac{Gm_1m_2}{r^2}$, i.e. F remains the same.
22. (c) Upto ordinary heights, the change in the distance of a projectile from the centre of earth is negligible compared to the radius of earth. Hence the projectile moves under a nearly uniform gravitational force and the path is parabolic. But for the projectiles moving to a large height, the gravitational force decreases quite decreasing variable force, the path of the projectile becomes elliptical.

23. (c) As the distance from centre of earth decreases, acceleration due to gravity decreases and at the centre of earth it becomes zero.

$$g' = g \left(1 - \frac{d}{R} \right). \text{ If } d = R \text{ then } g' = 0$$

24. (b) We know that earth revolves from west to east about its polar axis. Therefore, all the particles on the earth have velocity from west to east.

This velocity is maximum in the equatorial line, as $v = R\omega$, where R is the radius of earth and ω is the angular velocity of revolution of earth about its polar axis.

When a rocket is launched from west to east in equatorial plane, the maximum linear velocity is added to the launching velocity of the rocket, due to which launching becomes easier.

25. (e) Binding energy $= \frac{GMm}{R} = - [\text{Total energy of satellite}]$

and it is clear that it depends upon the mass of the satellite.

26. (a) According to Newton's law of gravitation, every body in this universe attracts every other body with a force which is inversely proportional to the square of the distance between them. When we move our finger, the distance of the objects with respect to finger changes, hence the force of attraction changes, disturbing the entire universe, including stars.

27. (a) Intensity inside a hollow sphere is zero, so force is also equal to zero. $\vec{F} = m\vec{E}$

28. (a) The time period of satellite which is very near to earth is given by

$$T = 2\pi \sqrt{\frac{R}{g}} \cong 84 \text{ min} = 1 \text{ hr. } 24 \text{ min}$$

29. (c) A person feels his weight only when the surface on which he is standing exerts a reactionary force on him. Because the acceleration of the person and that of the satellite revolving round the earth are equal ($= g$), hence acceleration of the person with respect to the satellite is zero.

Therefore person feels weightless on satellite, although the gravitational force is acting on a satellite.

30. (e) If the orbital path of a satellite is circular, then its speed is constant and if the orbital path of a satellite is elliptical, then its speed in its orbit is not constant. In that case its areal velocity is constant.

31. (a) $v_0 = R_e \sqrt{\frac{g}{R_e + h}}$

For satellite revolving very near to earth $R_e + h = R_e$

As ($h \ll R$)

$$v_o = \sqrt{R_e g} \simeq \sqrt{64 \times 10^5 \times 10} = 8 \times 10^3 \text{ m/s} = 8 \text{ kms}^{-1}$$

Which is independent of height of a satellite.

32. (b)

33. (a) The torque on a body is given by $\vec{\tau} = \frac{d\vec{L}}{dt}$

In case of planet orbiting around sun no torque is acting on it.

$$\frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L} = \text{constant.}$$

Gravitation

Self Evaluation Test - 8

- Two identical spheres are placed in contact with each other. The force of gravitation between the spheres will be proportional to (R = radius of each sphere)
 - R
 - R^2
 - R^4
 - None of these
- Suppose that the force of earth's gravity suddenly disappears, choose the correct answer out of the following statements
 - The weight of the body will become zero but mass remains the same
 - The mass of the body will become zero but the weight remains the same
 - Both the mass and weight will be the same
 - Mass and weight will remain the same
- An earth satellite is moved from one stable circular orbit to a further stable circular orbit, which one of the following quantities increase
 - Gravitational force
 - Gravitational P.E.
 - Linear orbital speed
 - Centripetal acceleration
- Two planets revolve round the sun with frequencies N_1 and N_2 revolutions per year. If their average orbital radii be R_1 and R_2 respectively, then R_1 / R_2 is equal to
 - $(N_1 / N_2)^{3/2}$
 - $(N_2 / N_1)^{3/2}$
 - $(N_1 / N_2)^{2/3}$
 - $(N_2 / N_1)^{2/3}$
- There is no atmosphere on the moon because
 - It is closer to the earth
 - It revolves round the earth
 - It gets light from the sun
 - The escape velocity of gas molecules is lesser than their root mean square velocity here
- Two heavenly bodies S_1 and S_2 , not far off from each other are seen to revolve in orbits
 - Around their common centre of mass
 - Which are arbitrary
 - With S_1 fixed and S_2 moving round S_1
 - With S_2 fixed and S_1 moving round S_2
- The mass of the moon is about 1.2% of the mass of the earth. Compared to the gravitational force the earth exerts on the moon, the gravitational force the moon exerts on earth
 - Is the same
 - Is smaller
 - Is greater
 - Varies with its phase
- A clock S is based on oscillation of a spring and a clock P is based on pendulum motion. Both clocks run at the same rate on earth. On a planet having the same density as earth but twice the radius
 - S will run faster than P
 - P will run faster than S
 - They will both run at the same rate as on the earth
 - None of these
- Consider earth to be a homogeneous sphere. Scientist A goes deep down in a mine and scientist B goes high up in a balloon. The value of g measured by
 - A goes on decreasing and that by B goes on increasing
 - B goes on decreasing and that by A goes on increasing
 - Each decreases at the same rate
 - Each decreases at different rates
- The mass of the moon is $\frac{1}{81}$ of the earth but the gravitational pull is $\frac{1}{6}$ of the earth. It is due to the fact that
 - The radius of the moon is $\frac{81}{6}$ of the earth
 - The radius of the earth is $\frac{9}{\sqrt{6}}$ of the moon
 - Moon is the satellite of the earth
 - None of the above
- A weight is suspended from the ceiling of a lift by a spring balance. When the lift is stationary the spring balance reads W . If the lift suddenly falls freely under gravity, the reading on the spring balance will be
 - W
 - $2W$
 - $W/2$
 - 0
- If a planet consists of a satellite whose mass and radius were both half that of the earth, the acceleration due to gravity at its surface would be (g on earth = 9.8 m/sec^2)
 - 4.9 m/sec^2
 - 8.9 m/sec^2
 - 19.6 m/sec^2
 - 29.4 m/sec^2
- At a given place where acceleration due to gravity is ' g ' m/sec^2 , a sphere of lead of density ' d ' kg/m^3 is gently released in a column of liquid of density ' ρ ' kg/m^3 . If $d > \rho$, the sphere will
 - Fall vertically with an acceleration ' g ' m/sec^2
 - Fall vertically with no acceleration
 - Fall vertically with an acceleration $g \left(\frac{d-\rho}{d} \right)$
 - Fall vertically with an acceleration $g \left(\frac{\rho}{d} \right)$
- g_e and g_p denote the acceleration due to gravity on the surface of the earth and another planet whose mass and radius are twice as that of earth. Then
 - $g_p = g_e$
 - $g_p = g_e / 2$
 - $g_p = 2g_e$
 - $g_p = g_e / 4$
- If the value of g at the surface of the earth is 9.8 m/sec^2 , then the value of g at a place 480 km above the surface of the earth will be (Radius of the earth is 6400 km)
 - 9.8 m/sec^2
 - 9.8 m/sec^2
 - 9.8 m/sec^2
 - 9.8 m/sec^2



- (a) 8.4 m/sec^2 (b) 9.8 m/sec^2
 (c) 7.2 m/sec^2 (d) 4.2 m/sec^2
16. The acceleration due to gravity about the earth's surface would be half of its value on the surface of the earth at an altitude of ($R = 4000 \text{ mile}$)
 (a) 1200 mile (b) 2000 mile
 (c) 1600 mile (d) 4000 mile
17. A pendulum clock is set to give correct time at the sea level. This clock is moved to hill station at an altitude of 2500 m above the sea level. In order to keep correct time of the hill station, the length of the pendulum
 (a) Has to be reduced
 (b) Has to be increased
 (c) Needs no adjustment
 (d) Needs no adjustment but its mass has to be increased
18. At some point the gravitational potential and also the gravitational field due to earth is zero. The point is
 (a) On earth's surface
 (b) Below earth's surface
 (c) At a height R_e from earth's surface ($R_e =$ radius of the earth)
 (d) At infinity
19. A body falls freely under gravity. Its speed is v when it has lost an amount U of the gravitational energy. Then its mass is
 (a) $\frac{Ug}{v^2}$ (b) $\frac{U^2}{g}$
 (c) $\frac{2U}{v^2}$ (d) $2Ugv^2$
20. The ratio of the radius of the earth to that of the moon is 10. The ratio of acceleration due to gravity on the earth and on the moon is 6. The ratio of the escape velocity from the earth's surface to that from the moon is
 (a) 10 (b) 6
 (c) Nearly 8 (d) 1.66
21. Escape velocity from the moon surface is less than that on the earth surface, because
 (a) Moon has no atmosphere while the earth has
 (b) Radius of moon is less than that of the earth
 (c) Moon is nearer to the sun
 (d) Moon is attracted by other planets
22. The ratio of the radius of a planet 'A' to that of planet 'B' is ' r '. The ratio of acceleration due to gravity on the planets is ' x '. The ratio of the escape velocities from the two planets is
 (a) xr (b) $\sqrt{\frac{r}{x}}$
 (c) $\sqrt{rx}$ (d) $\sqrt{\frac{x}{r}}$
23. Time period of revolution of a nearest satellite around a planet of radius R is T . Period of revolution around another planet, whose radius is $3R$ but having same density is
 (a) T (b) $3T$
 (c) $9T$ (d) $3\sqrt{3}T$
24. The maximum possible velocity of a satellite orbiting round the earth in a stable orbit is
 (a) $\sqrt{2R_e g}$ (b) $\sqrt{R_e g}$
 (c) $\sqrt{\frac{R_e g}{2}}$ (d) Infinite
25. A man inside an artificial satellite feels weightlessness because the force of attraction due to earth is
 (a) Zero at that place
 (b) Is balanced by the force of attraction due to moon
 (c) Equal to the centripetal force
 (d) Non-effective due to particular design of the satellite
26. Two identical satellites A and B are circulating round the earth at the height of R and $2R$ respectively, (where R is radius of the earth). The ratio of kinetic energy of A to that of B is
 (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
 (c) 2 (d) $\frac{3}{2}$
27. The mean radius of the earth's orbit round the sun is $1.5 \times 10^{11} \text{ m}$. The mean radius of the orbit of mercury round the sun is $6 \times 10^{10} \text{ m}$. The mercury will rotate around the sun in
 (a) A year (b) Nearly 4 years
 (c) Nearly $\frac{1}{4}$ year (d) 2.5 years

AS Answers and Solutions

(SET - 8)

$$1. \quad (c) \quad F = G \frac{M \times M}{R^2} = \frac{G \times \left(\frac{4}{3} \pi R^3 \rho \right)^2}{(2R)^2} \Rightarrow F \propto \frac{R^6}{R^2} \propto R^4$$

2. (a) In the absence of gravity weight of the bodies will become zero but mass will not change.

3. (b) $U = \frac{-GMm}{r}$. If r increases then U also increases.

4. (d) According to Kepler's law $T^2 \propto R^3$

If N is the frequencies then $N^2 \propto (R)^{-3}$

$$\text{or } \frac{N_2}{N_1} = \left(\frac{R_2}{R_1}\right)^{-3/2} \Rightarrow \frac{R_1}{R_2} = \left(\frac{N_2}{N_1}\right)^{2/3}$$

5. (d)

6. (a)

7. (a) Force between earth and moon $F = \frac{Gm_m m_e}{r^2}$

This amount of force, both earth and moon will exert on each other i.e. they exert same force on each other.

8. (b) $g = \frac{4}{3}\pi\rho GR$. If density is same then $g \propto R$

According to problem $R_p = 2R_e \therefore g_p = 2g_e$

For clock P (based on pendulum motion) $T = 2\pi\sqrt{\frac{l}{g}}$

Time period decreases on planet so it will run faster because $g_p > g_e$

For clock S (based on oscillation of spring) $T = 2\pi\sqrt{\frac{m}{k}}$

So it does not change.

9. (d) For scientist A which goes down in a mine

$$g' = g\left(1 - \frac{d}{R}\right)$$

For scientist B , which goes up in a air $g' = g\left(1 - \frac{2h}{R}\right)$

So it is clear that value of g measured by each will decrease at different rates.

10. (b) Gravitational pull depends upon the acceleration due to gravity on that planet.

$$M_m = \frac{1}{81}M_e, g_m = \frac{1}{6}g_e$$

$$g = \frac{GM}{R^2} \Rightarrow \frac{R_e}{R_m} = \left(\frac{M_e}{M_m} \times \frac{g_m}{g_e}\right)^{1/2} = \left(81 \times \frac{1}{6}\right)^{1/2}$$

$$\therefore R_e = \frac{9}{\sqrt{6}}R_m$$

11. (d) Reading of spring balance $R = m(g - a)$

If the lift falls freely then $a = g \therefore R = 0$

12. (c) $g = \frac{GM}{R^2} \therefore g \propto \frac{M}{R^2}$

According to problem $M_p = \frac{M_e}{2}$ and $R_p = \frac{R_e}{2}$

$$\therefore \frac{g_p}{g_e} = \left(\frac{M_p}{M_e}\right)\left(\frac{R_e}{R_p}\right)^2 = \left(\frac{1}{2}\right) \times (2)^2 = 2$$

$$\Rightarrow g_p = 2g_e = 2 \times 9.8 = 19.6 \text{ m/s}^2$$

13. (c) Apparent weight = actual weight - upthrust force

$$Vdg' = Vdg - V\rho g$$

$$\Rightarrow g' = \left(\frac{d - \rho}{d}\right)g$$

14. (b) $g \propto \frac{M}{R^2}$. If mass and radius of the planet are twice then g_p

will be half that of g_e i.e. $g_p = \frac{g_e}{2}$

15. (a) The value of g on the surface of the earth $g \propto \frac{1}{R^2}$

At height h from the surface of the earth $g' \propto \frac{1}{(R+h)^2}$

$$\therefore g' = g \frac{R^2}{(R+h)^2} = \frac{9.8 \times (6400)^2}{(6400 + 480)^2} = 8.4 \text{ m/s}^2$$

16. (c) $\frac{g'}{g} = \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{2} = \left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{2} = \left(\frac{4000}{4000+h}\right)^2$

By solving we get $h = 1656.85 \text{ mile} \approx 1600 \text{ mile}$

17. (a) $T = 2\pi\sqrt{\frac{l}{g}}$. At the hill g will decrease so to keep the time period same the length of pendulum has to be reduced.

18. (d) $V = \frac{-GM}{r}$ and $I = \frac{GM}{r^2}$

$V = 0$ and $I = 0$ at $r = \infty$

19. (c) $U = \text{Loss in gravitational energy} = \text{gain in K.E.}$

$$\text{So, } U = \frac{1}{2}mv^2 \Rightarrow m = \frac{2U}{v^2}$$

20. (c) $\frac{v_e}{v_m} = \sqrt{\frac{g_e}{g_m} \frac{R_e}{R_m}} = \sqrt{6 \times 10} = \sqrt{60} \approx 8 \text{ (nearly)}$

21. (b)

22. (c) $v_e = \sqrt{2gR} \Rightarrow \frac{v_A}{v_B} = \sqrt{\frac{g_A}{g_B} \times \frac{R_A}{R_B}} = \sqrt{x \times r} \therefore \frac{v_A}{v_B} = \sqrt{rx}$

23. (a) Time period of satellite which is very near to planet

$$T = 2\pi\sqrt{\frac{R^3}{GM}} = 2\pi\sqrt{\frac{R^3}{G\frac{4}{3}\pi R^3\rho}} \therefore T \propto \sqrt{\frac{1}{\rho}}$$

i.e. time period of nearest satellite does not depend upon the radius of planet, it only depends upon the density of the planet.
In the problem, density is same so time period will be same.

24. (b) Otherwise centrifugal force exceeds the force of attraction or we can say that gravitational force won't be able to keep the satellite in circular motion.

25. (c)

26. (d) $\frac{K_A}{K_B} = \frac{r_B}{r_A} = \left(\frac{R+h_B}{R+h_A}\right) = \left(\frac{R+2R}{R+R}\right) = \frac{3}{2}$

27. (c) $\frac{T_{\text{mercury}}}{T_{\text{earth}}} = \left(\frac{r_{\text{mercury}}}{r_{\text{earth}}}\right)^{3/2} = \left(\frac{6 \times 10^{10}}{1.5 \times 10^{11}}\right)^{3/2} = \frac{1}{4}$
(approx.)

$$\therefore T_{\text{mercury}} = \frac{1}{4} \text{ year}$$



Chapter 9 Elasticity

Interatomic Forces

The forces between the atoms due to electrostatic interaction between the charges of the atoms are called interatomic forces. These forces are electrical in nature and these are active if the distance between the two atoms is of the order of atomic size *i.e.* 10^{-10} metre.

(1) Every atom is electrically neutral, the number of electrons (negative charge) orbiting around the nucleus is equal to the number of protons (positive charge) in the nucleus. So if two atoms are placed at a very large distance from each other then there will be a very small (negligible) interatomic force working between them.

(2) When two atoms are brought closer to each other to a distance of the order of 10^{-10} m, the distances between their positive nuclei and negative electron clouds get disturbed, and due to this, attractive interatomic force is produced between two atoms.

(3) This attractive force increases continuously with decrease in r and becomes maximum for one value of r called critical distance, represented by x (as shown in the figure). Beyond this the attractive force starts decreasing rapidly with further decrease in the value of r .

(4) When the distance between the two atoms becomes r_0 , the interatomic force will be zero. This distance r_0 is called normal or equilibrium distance.

($r_0 = 0.74 \text{ \AA}$ for hydrogen).

(5) When the distance between the two atoms further decreased, the interatomic force becomes repulsive in nature and increases very rapidly with decrease in distance between two atoms.

(6) The potential energy U is related with the interatomic force F by the following relation.

$$F = -\frac{dU}{dr}$$

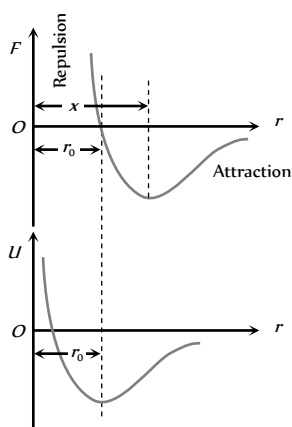


Fig. 9.1

(i) When two atoms are at very large distance, the potential energy is negative and becomes more negative as r is decreased.

(ii) When the distance between the two atoms becomes r_0 , the potential energy of the system of two atoms becomes minimum (*i.e.* attains maximum negative value). As the state of minimum potential energy is the state of equilibrium, hence the two atoms at separation r_0 will be in a state of equilibrium.

$$(U_0 = -7.2 \times 10^{-19} \text{ Joule for hydrogen}).$$

(iii) When the distance between the two atoms is further decreased (*i.e.* $r < r_0$) the negative value of potential energy of the system starts decreasing. It becomes zero and then attains positive value with further decrease in r (as shown in the figure).

Intermolecular Forces

The forces between the molecules due to electrostatic interaction between the charges of the molecules are called intermolecular forces. These forces are also called Vander Waal forces and are quite weak as compared to inter-atomic forces. These forces are also electrical in nature and these are active if the separation between two molecules is of the order of molecular size *i.e.* $\approx 10^{-9}$ m.

(1) It is found that the force of attraction between molecules varies inversely as seventh power of the distance between them *i.e.*

$$F_{\text{att}} \propto \frac{1}{r^7} \quad \text{or} \quad F_{\text{rep}} = \frac{-a}{r^7}$$

The negative sign indicates that the force is attractive in nature.

(2) When the distance between molecules becomes less than r_0 , the forces becomes repulsive in nature and is found to vary inversely as ninth power of the distance between them *i.e.*

$$F_{\text{rep}} \propto \frac{1}{r^9} \quad \text{or} \quad F_{\text{rep}} = \frac{b}{r^9}$$

Therefore force between two molecules is given by

$$F = F_{\text{att}} + F_{\text{rep}} = \frac{-a}{r^7} + \frac{b}{r^9}$$

The value of constants a and b depend upon the structure and nature of molecules.



(3) Intermolecular forces between two molecules has the same general nature as shown in the figure for interatomic forces.

(4) Potential Energy : Potential energy can be approximately expressed by the formula $U = \frac{A}{r^n} - \frac{B}{r^m}$

where the term $\frac{A}{r^n}$ represents repulsive contribution and term $\frac{B}{r^m}$ represents the attractive contribution. Constants A , B and numbers m and n are different for different molecules.

For majority of solids $n = 12$ and $m = 6$.

So potential energy can be expressed as $U = \frac{A}{r^{12}} - \frac{B}{r^6}$

Comparison Between Interatomic and Intermolecular Forces

(i) Similarities

- (i) Both the forces are electrical in origin.
- (ii) Both the forces are active over short distances.
- (iii) General shape of force-distance graph is similar for both the forces.

(iv) Both the forces are attractive up to certain distance between atoms/molecules and become repulsive when the distance between them become less than that value.

(2) Dissimilarities

(i) Interatomic force depends upon the distance between the two atoms, whereas the intermolecular force depends upon the distance between the two molecules as well as their relative orientation.

(ii) Interatomic forces are about 50 to 100 times stronger than intermolecular forces.

(iii) The value of r for two atoms is smaller than the corresponding value for the molecules. Therefore one molecule is not restricted to attract only one molecule, but can attract many molecule. It is not so in case of atoms, since the atoms of one molecule cannot bind the atoms of other molecules.

States of Matter

The three states of matter differ from each other due to the following two factors.

(1) The different magnitudes of the interatomic and intermolecular forces.

(2) The extent of random thermal motion of atoms and molecules of a substance (which depends upon temperature).

Comparison Chart of Solid, Liquid and Gaseous States

Property	Solid	Liquid	Gas
Shape	Definite	Not definite	Not definite
Volume	Definite	Definite	Not definite
Density	Maximum	Less than solids but more than gases.	Minimum
Compressibility	Incompressible	Less than gases but more than solids.	Compressible
Crystallinity	Crystalline	Non-crystalline	
Interatomic or intermolecular distance	Constant	Not constant	Not constant
Relation between kinetic energy K and potential energy (U)	$K < U$	$K > U$	$K \gg U$
Intermolecular force	Strongest	Less than solids but more than gases.	Weakest
Freedom of motion	Molecules vibrate about their mean position but cannot move freely.	Molecules have limited free motion.	Molecules are free to move.
Effect of temperature	Matter remains in solid form below a certain temperature.	Liquids are found at temperatures more than that of solid.	These are found at temperatures greater than that of solids and liquids.

Note : □ The fourth state of matter in which the medium is in the form of positive and negative ions, is known as plasma. Plasma occurs in the atmosphere of stars (including the sun) and in discharge tubes.

Types of Solids

A solid is that state of matter in which its constituent atoms or molecules are held strongly at the position of minimum potential energy

and it has a definite shape and volume. The solids can be classified into two categories, crystalline and glassy or amorphous solids.

Comparison chart of Crystalline and Amorphous Solids

Crystalline solids	Amorphous or glassy solids
The constituent atoms, ions or molecules are arranged in a regular repeated three dimensional pattern, within the	The constituent atoms, ions or molecules are not arranged in a regular repeated three dimensional pattern, within the

solid.	solid.
Definite external geometric shape.	No regularity in external shape.
All the bonds in ions, or atoms or molecules are equally strong.	All the bonds are not equally strong.
They are anisotropic.	They are isotropic.
They have sharp melting point.	They don't have sharp melting point.
They have a long-range order of atoms or ions or molecules in them.	They don't have a long-range order.
They are considered true and stable solids.	They are not regarded as true and stable solids.

Elastic Property of Matter

(1) **Elasticity** : The property of matter by virtue of which a body tends to regain its original shape and size after the removal of deforming force is called elasticity.

(2) **Plasticity** : The property of matter by virtue of which it does not regain its original shape and size after the removal of deforming force is called plasticity.

(3) **Perfectly elastic body** : If on the removal of deforming forces the body regain its original configuration completely it is said to be perfectly elastic.

A quartz fibre and phosphor bronze (an alloy of copper containing 4% to 10% tin, 0.05% to 1% phosphorus) is the nearest approach to the perfectly elastic body.

(4) **Perfectly plastic body** : If the body does not have any tendency to recover its original configuration, on the removal of deforming force, it is said to be perfectly plastic.

Paraffin wax, wet clay are the nearest approach to the perfectly plastic body.

Practically there is no material which is either perfectly elastic or perfectly plastic and the behaviour of actual bodies lies between the two extremes.

(5) **Reason of elasticity** : In a solids, atoms and molecules are arranged in such a way that each molecule is acted upon by the forces due to neighbouring molecules. These forces are known as intermolecular forces.

For simplicity, the two molecules in their equilibrium positions (at inter-molecular distance $r = r_0$) are shown by connecting them with a spring.

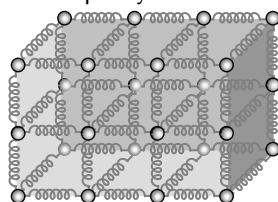


Fig. 9.2

In fact, the spring connecting the two molecules represents the inter-molecular force between them. On applying the deforming forces, the molecules either come closer or go far apart from each other and restoring forces are developed. When the deforming force is removed, these restoring forces bring the molecules of the solid to their respective equilibrium position ($r = r_0$) and hence the body regains its original form.

(6) **Elastic limit** : Elastic bodies show their property of elasticity upto a certain value of deforming force. If we go on increasing the deforming force then a stage is reached when on removing the force, the body will not return to its original state. The maximum deforming force upto which a body retains its property of elasticity is called elastic limit of the material of body.

Elastic limit is the property of a body whereas elasticity is the property of material of the body.

(7) **Elastic fatigue** : The temporary loss of elastic properties because of the action of repeated alternating deforming force is called elastic fatigue.

Due to elastic fatigue :

(i) Bridges are declared unsafe after a long time of their use.

(ii) Spring balances show wrong readings after they have been used for a long time.

(iii) We are able to break the wire by repeated bending.

(8) **Elastic after effect** : The time delay in which the substance regains its original condition after the removal of deforming force is called elastic after effect. It is the time for which restoring forces are present after the removal of the deforming force, it is negligible for perfectly elastic substance, like quartz, phosphor bronze and large for glass fibre.

Stress

When a force is applied on a body, there will be relative displacement of the particles and due to property of elasticity, an internal restoring force is developed which tends to restore the body to its original state.

The internal restoring force acting per unit area of cross section of the deformed body is called stress.

At equilibrium, restoring force is equal in magnitude to external force, stress can therefore also be defined as external force per unit area on a body that tends to cause it to deform.

If external force F is applied on the area A of a body then,

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} = \frac{F}{A}$$

$$\text{Unit : } N/m^2 \text{ (S.I.) , } dyne/cm^2 \text{ (C.G.S.)}$$

$$\text{Dimension : } [ML^{-1}T^{-2}]$$

Stress developed in a body depends upon how the external forces are applied over it.

On this basis there are two types of stresses : Normal and Shear or tangential stress

(1) **Normal stress** : Here the force is applied normal to the surface.

It is again of two types : Longitudinal and Bulk or volume stress

(i) **Longitudinal stress**

(a) It occurs only in solids and comes in to picture when one of the three dimensions *viz.* length, breadth, height is much greater than other two.

(b) Deforming force is applied parallel to the length and causes increase in length.

(c) Area taken for calculation of stress is the area of cross section.

(d) Longitudinal stress produced due to increase in length of a body under a deforming force is called tensile stress.

(e) Longitudinal stress produced due to decrease in length of a body under a deforming force is called compressive stress.

(ii) **Bulk or Volume stress**

(a) It occurs in solids, liquids or gases.

(b) In case of fluids only bulk stress can be found.

(c) It produces change in volume and density, shape remaining same.

(d) Deforming force is applied normal to surface at all points.

(e) Area for calculation of stress is the complete surface area perpendicular to the applied forces.

(f) It is equal to change in pressure because change in pressure is responsible for change in volume.

(2) **Shear or tangential stress** : It comes into picture when successive layers of solid move on each other *i.e.* when there is a relative displacement between various layers of solid.

(i) Here deforming force is applied tangential to one of the faces.

(ii) Area for calculation is the area of the face on which force is applied.

(iii) It produces change in shape, volume remaining the same.

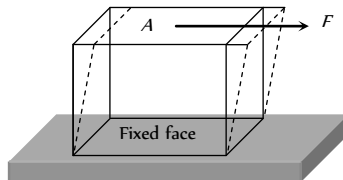


Fig. 9.3

Difference between Pressure and Stress

Pressure	Stress
Pressure is always normal to the area.	Stress can be normal or tangential.
Always compressive in nature.	May be compressive or tensile in nature.

Strain

The ratio of change in configuration to the original configuration is called strain.

Being the ratio of two like quantities, it has no dimensions and units.

Strain are of three types :

(1) **Linear strain** : If the deforming force produces a change in length alone, the strain produced in the body is called linear strain or tensile strain.

$$\text{Linear strain} = \frac{\text{Change in length}(\Delta l)}{\text{Original length}(l)}$$

Linear strain in the direction of deforming force is called longitudinal strain and in a direction perpendicular to force is called lateral strain.

(2) **Volumetric strain** : If the deforming force produces a change in volume alone the strain produced in the body is called volumetric strain.

$$\text{Volumetric strain} = \frac{\text{Change in volume}(\Delta V)}{\text{Original volume}(V)}$$

(3) **Shearing strain** : If the deforming force

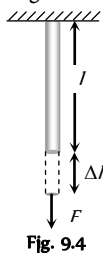


Fig. 9.4

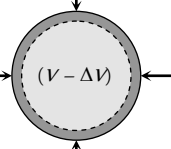


Fig. 9.5

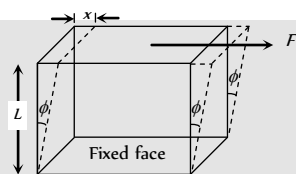


Fig. 9.6

produces a change in the shape of the body without changing its volume, strain produced is called shearing strain.

It is defined as angle in radians through which a plane perpendicular to the fixed surface of the cubical body gets turned under the effect of tangential force.

$$\phi = \frac{x}{L}$$

Note : □ When a beam is bent both compression strain as well as an extension strain is produced.

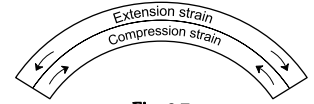
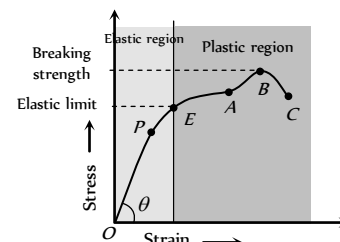


Fig. 9.7

Stress-strain Curve

If by gradually increasing the load on a vertically suspended metal wire, a graph is plotted between stress (or load) and longitudinal strain (or elongation) we get the curve as shown in figure. From this curve it is clear that :



(1) When the strain is small ($< 2\%$) (*i.e.*, in region OP) stress is proportional to strain. This is the region where the so called Hooke's law is obeyed. The point P is called limit of proportionality and slope of line OP gives the Young's modulus Y of the material of the wire. If θ is the angle of OP from strain axis then $Y = \tan \theta$.

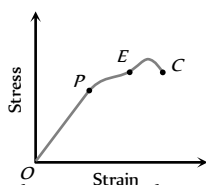
(2) If the strain is increased a little bit, *i.e.*, in the region PE , the stress is not proportional to strain. However, the wire still regains its original length after the removal of stretching force. This behaviour is shown up to point E known as elastic limit or yield-point. The region OPE represents the elastic behaviour of the material of wire.

(3) If the wire is stretched beyond the elastic limit E , *i.e.*, between EA , the strain increases much more rapidly and if the stretching force is removed the wire does not come back to its natural length. Some permanent increase in length takes place.

(4) If the stress is increased further, by a very small increase in it a very large increase in strain is produced (region AB) and after reaching point B , the strain increases even if the wire is unloaded and ruptures at C . In the region BC the wire literally flows. The maximum stress corresponding to B after which the wire begins to flow and breaks is called breaking or ultimate tensile strength. The region $EABC$ represents the plastic behaviour of the material of wire.

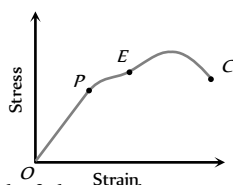
(5) Stress-strain curve for different materials are as follows :

Brittle material	Ductile material	Elastomers
------------------	------------------	------------



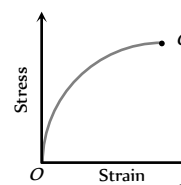
The plastic region between E and C is small for brittle material and it will break soon after the elastic limit is crossed.

Example : Glass, cast iron.



The material of the wire have a good plastic range and such materials can be easily changed into different shapes and can be drawn into thin wires

Example. Mild steel



Stress-strain curve is not a straight line within the elastic limit for elastomers and strain produced is much larger than the stress applied. Such materials have no plastic range and the breaking point lies very close to elastic limit. Example rubber

Hooke's law and Modulus of Elasticity

According to this law, within the elastic limit, stress is proportional to the strain.

$$\text{i.e. stress} \propto \text{strain or } \frac{\text{stress}}{\text{strain}} = \text{constant} = E$$

The constant E is called modulus of elasticity.

(1) It's value depends upon the nature of material of the body and the manner in which the body is deformed.

(2) It's value depends upon the temperature of the body.

(3) It's value is independent of the dimensions (length, volume etc.) of the body.

There are three moduli of elasticity namely Young's modulus (Y), Bulk modulus (K) and modulus of rigidity (η) corresponding to three types of the strain.

Young's Modulus (Y)

It is defined as the ratio of normal stress to longitudinal strain within limit of proportionality.

$$Y = \frac{\text{Normal stress}}{\text{longitudinal strain}} = \frac{F/A}{l/L} = \frac{FL}{Al}$$

If force is applied on a wire of radius r by hanging a weight of mass M , then

$$Y = \frac{MgL}{\pi r^2 l}$$

(i) If the length of a wire is doubled,

$$\text{Then longitudinal strain} = \frac{\text{change in length}(l)}{\text{initial length}(L)}$$

$$= \frac{\text{final length} - \text{initial length}}{\text{Initial length}} = \frac{2L - L}{L} = 1$$

$$\therefore \text{Young's modulus} = \frac{\text{stress}}{\text{strain}} \Rightarrow Y = \text{stress}$$

[As strain = 1]

So young's modulus is numerically equal to the stress which will double the length of a wire.

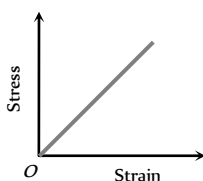


Fig. 9.9

$$(ii) \text{ Increment in the length of wire } l = \frac{FL}{\pi r^2 Y}$$

$$\left[\text{As } Y = \frac{FL}{Al} \right]$$

So if same stretching force is applied to different wires of same material, $l \propto \frac{L}{r^2}$ [As F and Y are constant]

i.e., greater the ratio $\frac{L}{r^2}$, greater will be the elongation in the wire.

(iii) Elongation in a wire by its own weight : The weight of the wire Mg act at the centre of gravity of the wire so that length of wire which is stretched will be $L/2$.

$$\therefore \text{Elongation } l = \frac{FL}{AY} = \frac{Mg(L/2)}{AY} = \frac{MgL}{2AY} = \frac{L^2 dg}{2Y}$$

[As mass (M) = volume (AL) $\times$ density (d)]

(iv) Thermal stress : If a rod is fixed between two rigid supports, due to change in temperature its length will change and so it will exert a normal stress (compressive if temperature increases and tensile if temperature decreases) on the supports. This stress is called thermal stress.

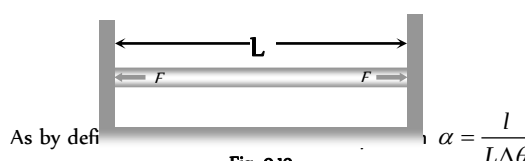


Fig. 9.10

$$\text{As by def } \alpha = \frac{l}{L\Delta\theta}$$

$$\Rightarrow \text{thermal strain } \frac{l}{L} = \alpha\Delta\theta$$

So thermal stress = $Y\alpha\Delta\theta$ [As Y = stress/strain]

And tensile or compressive force produced in the body = $Y\alpha\Delta\theta$

Note : In case of volume expansion Thermal stress = $K\gamma\Delta\theta$

Where K = Bulk modulus, γ = coefficient of cubical expansion

(v) **Force between the two rods** : Two rods of different metals, having the same area of cross section A , are placed end to end between two massive walls as shown in figure. The first rod has a length L_1 , coefficient of linear expansion α_1 and young's modulus Y_1 . The corresponding quantities for second rod are L_2 , α_2 and Y_2 . If the temperature of both the rods is now raised by T degrees.

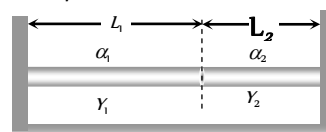


Fig. 9.11

Increase in length of the composite rod (due to heating) will be equal to

$$l_1 + l_2 = [L_1\alpha_1 + L_2\alpha_2]T \quad [\text{As } l = L \alpha \Delta\theta]$$

and due to compressive force F from the walls due to elasticity, decrease in length of the composite rod will be equal to

$$\left[\frac{L_1}{Y_1} + \frac{L_2}{Y_2} \right] \frac{F}{A} \quad \left[\text{As } l = \frac{FL}{AY} \right]$$

as the length of the composite rod remains unchanged the increase in length due to heating must be equal to decrease in length due to

$$\text{compression i.e. } \frac{F}{A} \left[\frac{L_1}{Y_1} + \frac{L_2}{Y_2} \right] = [L_1\alpha_1 + L_2\alpha_2]T$$

$$\text{or } F = \frac{A[L_1\alpha_1 + L_2\alpha_2]T}{\left[\frac{L_1}{Y_1} + \frac{L_2}{Y_2} \right]}$$

(vi) **Force constant of wire** : Force required to produce unit elongation in a wire is called force constant of material of wire. It is denoted by k .

$$\therefore k = \frac{F}{l} \quad \dots(i)$$

but from the definition of young's modulus

$$\frac{F}{l} = \frac{YA}{L} \quad \dots(ii)$$

$$\text{from (i) and (ii) } k = \frac{YA}{L}$$

It is clear that the value of force constant depends upon the dimension (length and area of cross section) and material of a substance.

(vii) **Actual length of the wire** : If the actual length of the wire is L , then under the tension T_1 its length becomes L_1 and under the tension T_2 its length becomes L_2 .

$$L_1 = L + l_1 \Rightarrow L_1 = L + \frac{T_1}{k} \quad \dots(i)$$

$$\text{and } L_2 = L + l_2 \Rightarrow L_2 = L + \frac{T_2}{k} \quad \dots(ii)$$

$$\text{From (i) and (ii) we get } L = \frac{L_1 T_2 - L_2 T_1}{T_2 - T_1}$$

Work Done in Stretching a Wire

In stretching a wire work is done against internal restoring forces. This work is stored in the wire as elastic potential energy or strain energy.

If a force F acts along the length L of the wire of cross-section A and stretches it by x then

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{x/L} = \frac{FL}{Ax} \Rightarrow F = \frac{YA}{L} x$$

So the work done for an additional small increase dx in length,

$$dW = Fdx = \frac{YA}{L} x \cdot dx$$

Hence the total work done in increasing the length by l ,

$$W = \int_0^l dW = \int_0^l Fdx = \int_0^l \frac{YA}{L} x \cdot dx = \frac{1}{2} \frac{YA}{L} l^2$$

This work done is stored in the wire.

$$\therefore \text{Energy stored in wire } U = \frac{1}{2} \frac{YAl^2}{L} = \frac{1}{2} Fl \quad \left[\text{As } F = \frac{YAl}{L} \right]$$

Dividing both sides by volume of the wire we get energy stored in unit volume of wire.

$$U_v = \frac{1}{2} \times \frac{F}{A} \times \frac{l}{L} = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{1}{2} \times Y \times (\text{strain})^2$$

$$= \frac{1}{2Y} (\text{stress})^2 \quad [\text{As } AL = \text{volume of wire}]$$

Total energy stored in wire (U)	Energy stored in per unit volume of wire (U_v)
$\frac{1}{2} Fl$	$\frac{1}{2} \frac{Fl}{\text{volume}}$
$\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$	$\frac{1}{2} \times \text{stress} \times \text{strain}$
$\frac{1}{2} \times Y \times (\text{strain})^2 \times \text{volume}$	$\frac{1}{2} \times Y \times (\text{strain})^2$
$\frac{1}{2Y} \times (\text{stress})^2 \times \text{volume}$	$\frac{1}{2Y} \times (\text{stress})^2$

Note : $\square$ If the force on the wire is increased from F_1 to F_2 and the elongation in wire is l then energy stored in the wire

$$U = \frac{1}{2} \frac{(F_1 + F_2)}{2} l$$

$$\square \text{ Thermal energy density} = \text{Thermal energy per unit volume} = \frac{1}{2} \times$$

Thermal stress $\times$ strain

$$= \frac{1}{2} \frac{F}{A} \frac{l}{L} = \frac{1}{2} (Y\alpha\Delta\theta)(\alpha\Delta\theta) = \frac{1}{2} Y\alpha^2 (\Delta\theta)^2$$

Breaking of Wire

When the wire is loaded beyond the elastic limit, then strain increases much more rapidly. The maximum stress corresponding to B (see stress-strain curve) after which the wire begin to flow and breaks, is called breaking stress or tensile strength and the force by application of which the wire breaks is called the breaking force.

(i) Breaking force depends upon the area of cross-section of the wire i.e., Breaking force $\propto A$

$$\therefore \text{Breaking force} = P \times A$$

Here P is a constant of proportionality and known as breaking stress.

(ii) Breaking stress is a constant for a given material and it does not depend upon the dimension (length or thickness) of wire.

(iii) If a wire of length L is cut into two or more parts, then again it's each part can hold the same weight. Since breaking force is independent of the length of wire.

(iv) If a wire can bear maximum force F , then wire of same material but double thickness can bear maximum force $4F$

(v) The working stress is always kept lower than that of a breaking stress.

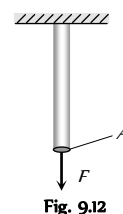


Fig. 9.12

So that safety factor = $\frac{\text{breaking stress}}{\text{working stress}}$, may have large value.

(vi) Breaking of wire under its own weight.

Breaking force = Breaking stress $\times$ Area of cross section

Weight of wire = $Mg = ALdg = PA$ [P = Breaking stress]

[As mass = volume $\times$ density = ALd]

$$\Rightarrow Ldg = P \therefore L = \frac{P}{dg}$$

This is the length of wire if it breaks by its own weight.

Bulk Modulus

When a solid or fluid (liquid or gas) is subjected to a uniform pressure all over the surface, such that the shape remains the same, then there is a change in volume.

Then the ratio of normal stress to the volumetric strain within the elastic limits is called as Bulk modulus. This is denoted by K .

$$K = \frac{\text{Normal stress}}{\text{volumetric strain}}$$

$$K = \frac{F/A}{-\Delta V/V} = \frac{-pV}{\Delta V}$$

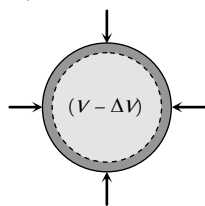


Fig. 9.13

where p = increase in pressure; V = original volume; ΔV = change in volume

The negative sign shows that with increase in pressure p , the volume decreases by ΔV i.e. if p is positive, ΔV is negative. The reciprocal of bulk modulus is called compressibility.

$$C = \text{compressibility} = \frac{1}{K} = \frac{\Delta V}{pV}$$

S.I. unit of compressibility is Nm^{-1} and C.G.S. unit is $\text{dyne}^{-1} \text{cm}$.

Gases have two bulk moduli, namely isothermal elasticity E_θ and adiabatic elasticity E_ϕ .

(1) Isothermal elasticity (E_θ) : Elasticity possess by a gas in isothermal condition is defined as isothermal elasticity.

For isothermal process, $PV = \text{constant}$ (Boyle's law)

Differentiating both sides

$$P dV + V dP = 0 \Rightarrow P dV = -V dP$$

$$P = \frac{dP}{(-dV/V)} = \frac{\text{stress}}{\text{strain}} = E_\theta \therefore E_\theta = P$$

i.e., Isothermal elasticity is equal to pressure.

(2) Adiabatic elasticity (E_ϕ) : Elasticity possess by a gas in adiabatic condition is defined as adiabatic elasticity.

For adiabatic process, $PV^\gamma = \text{constant}$ (Poisson's law)

Differentiating both sides,

$$P \gamma V^{\gamma-1} dV + V^\gamma dP = 0 \Rightarrow \gamma P dV + V dP = 0$$

$$\gamma P = \frac{dP}{\left(-\frac{dV}{V}\right)} = \frac{\text{stress}}{\text{strain}} = E_\phi$$

$$\therefore E_\phi = \gamma P$$

i.e., adiabatic elasticity is equal to γ times pressure.

$$\left[\text{Where } \gamma = \frac{C_p}{C_v} \right]$$

Note : $\square$ Ratio of adiabatic to isothermal elasticity

$$\frac{E_\phi}{E_\theta} = \frac{\gamma P}{P} = \gamma > 1 \therefore E_\phi > E_\theta$$

i.e., adiabatic elasticity is always more than isothermal elasticity.

Density of Compressed Liquid

If a liquid of density ρ , volume V and bulk modulus K is compressed, then its density increases.

$$\text{As density } \rho = \frac{m}{V} \text{ so } \frac{\Delta \rho}{\rho} = \frac{-\Delta V}{V} \quad \dots(i)$$

But by definition of bulk modulus

$$K = \frac{-V \Delta P}{\Delta V} \Rightarrow -\frac{\Delta V}{V} = \frac{\Delta P}{K} \quad \dots(ii)$$

$$\text{From (i) and (ii) } \frac{\Delta \rho}{\rho} = \frac{\rho' - \rho}{\rho} = \frac{\Delta P}{K} \quad [\text{As } \Delta \rho = \rho' - \rho]$$

$$\text{or } \rho' = \rho \left[1 + \frac{\Delta P}{K} \right] = \rho [1 + C \Delta P] \quad \left[\text{As } \frac{1}{K} = C \right]$$

Fractional Change in the Radius of Sphere

A solid sphere of radius R made of a material of bulk modulus K is surrounded by a liquid in a cylindrical container.

A massless piston of area A floats on the surface of the liquid.

$$\text{Volume of the spherical body } V = \frac{4}{3} \pi R^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta R}{R}$$

$$\therefore \frac{\Delta R}{R} = \frac{1}{3} \frac{\Delta V}{V} \quad \dots(i)$$

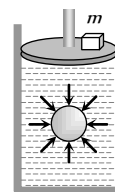


Fig. 9.14

$$\text{Bulk modulus } K = -V \frac{\Delta P}{\Delta V}$$

$$\therefore \left| \frac{\Delta V}{V} \right| = \frac{\Delta P}{K} = \frac{mg}{AK} \quad \dots(ii)$$

$$\left[\text{As } \Delta P = \frac{mg}{A} \right]$$

Substituting the value of $\frac{\Delta V}{V}$ from equation (ii) in equation (i) we

$$\text{get } \frac{\Delta R}{R} = \frac{1}{3} \frac{mg}{AK}$$

Modulus of Rigidity

Within limits of proportionality, the ratio of tangential stress to the shearing strain is called modulus of rigidity of the material of the body and

is denoted by η , i.e. $\eta = \frac{\text{Shearing stress}}{\text{Shearing strain}}$

In this case the shape of a body changes but its volume remains unchanged.

Consider a cube of material fixed at its lower face and acted upon by a tangential force F at its upper surface having area A . The shearing stress, then, will be

$$\text{Shearing stress} = \frac{F}{A}$$

This shearing force causes the consecutive horizontal layers of the cube to be slightly displaced or sheared relative to one another, each line such as PQ or RS in the cube is rotated through an angle ϕ by this shear. The shearing strain is defined as the angle ϕ in radians through which a line normal to a fixed surface has turned. For small values of angle,

$$\text{Shearing strain} = \phi = \frac{QQ'}{PQ} = \frac{x}{L}$$

$$\text{So } \eta = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F/A}{\phi} = \frac{F}{A\phi}$$

Only solids can exhibit a shearing as these have definite shape.

Poisson's Ratio

When a long bar is stretched by a force along its length then its length increases and the radius decreases as shown in the figure.

Lateral strain : The ratio of change in radius or diameter to the original radius or diameter is called lateral strain.

Longitudinal strain : The ratio of change in length to the original length is called longitudinal strain.

The ratio of lateral strain to longitudinal strain is called Poisson's ratio (σ).

$$\text{i.e. } \sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$\sigma = \frac{-dr/r}{dL/L}$$

Negative sign indicates that the radius of the bar decreases when it is stretched.

Poisson's ratio is a dimensionless and a unitless quantity.

Relation Between Volumetric Strain, Lateral Strain and Poisson's Ratio

If a long bar have a length L and radius r then volume $V = \pi r^2 L$

Differentiating both the sides $dV = \pi r^2 dL + \pi 2rLdr$

Dividing both the sides by volume of bar

$$\frac{dV}{V} = \frac{\pi r^2 dL}{\pi r^2 L} + \frac{\pi 2rLdr}{\pi r^2 L} = \frac{dL}{L} + 2 \frac{dr}{r}$$

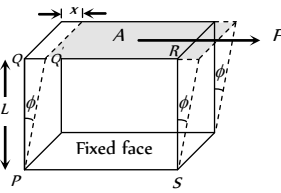


Fig. 9.15

$$\Rightarrow \text{Volumetric strain} = \text{longitudinal strain} + 2(\text{lateral strain})$$

$$\Rightarrow \frac{dV}{V} = \frac{dL}{L} + 2\sigma \frac{dL}{L} = (1 + 2\sigma) \frac{dL}{L}$$

$$\left[\text{As } \sigma = \frac{dr/r}{dL/L} \Rightarrow \frac{dr}{r} = \sigma \frac{dL}{L} \right]$$

$$\text{or } \sigma = \frac{1}{2} \left[\frac{dV}{AdL} - 1 \right]$$

[where A = cross-section of bar]

$$(i) \text{ If a material having } \sigma = -0.5 \text{ then } \frac{dV}{V} = [1 + 2\sigma] \frac{dL}{L} = 0$$

$\therefore$ Volume = constant or $K = \infty$ i.e. the material is incompressible.

(ii) If a material having $\sigma = 0$, then lateral strain is zero i.e. when a substance is stretched its length increases without any decrease in diameter e.g. cork. In this case change in volume is maximum.

(iii) Theoretical value of Poisson's ratio $-1 < \sigma < 0.5$.

(iv) Practical value of Poisson's ratio $0 < \sigma < 0.5$

Relation between Y , k , η and σ

Moduli of elasticity are three, viz. Y , K and η while elastic constants are four, viz. Y , K , η and σ . Poisson's ratio σ is not modulus of elasticity as it is the ratio of two strains and not of stress to strain. Elastic constants are found to depend on each other through the relations :

$$Y = 3K(1 - 2\sigma) \quad \dots(i)$$

$$Y = 2\eta(1 + \sigma) \quad \dots(ii)$$

Eliminating σ or Y between these, we get

$$Y = \frac{9K\eta}{3K + \eta} \quad \dots(iii)$$

$$\sigma = \frac{3K - 2\eta}{6K + 2\eta} \quad \dots(iv)$$

Torsion of Cylinder

If the upper end of a cylinder is clamped and a torque is applied at the lower end the cylinder gets twisted by angle θ . Simultaneously shearing strain ϕ is produced in the cylinder.

(i) The angle of twist θ is directly proportional to the distance from the fixed end of the cylinder.

At fixed end $\theta = 0^\circ$ and at free end $\theta =$ maximum.

(ii) The value of angle of shear ϕ is directly proportional to the radius of the cylindrical shell.

At the axis of cylinder $\phi = 0$ and at the outermost shell $\phi =$ maximum.

(iii) Relation between angle of twist (θ) and angle of shear (ϕ)

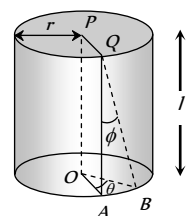


Fig. 9.17

$$AB = r\theta = \phi l \quad \therefore \phi = \frac{r\theta}{l}$$

(iv) Twisting couple per unit twist or torsional rigidity or torque required to produce unit twist.

$$C = \frac{\pi \eta r^4}{2l} \quad \therefore C \propto r^4 \propto A^2$$

(v) Work done in twisting the cylinder through an angle θ is

$$W = \frac{1}{2} C \theta^2 = \frac{\pi \eta r^4 \theta^2}{4l}$$

Interatomic Force Constant

Behaviour of solids with respect to external forces is such that if their atoms are connected to springs. When an external force is applied on a solid, this distance between its atoms changes and interatomic force works to restore the original dimension.

The ratio of interatomic force to that of change in interatomic distance is defined as the interatomic force constant. $K = \frac{F}{\Delta r}$

It is also given by $K = Y \times r_0$ [Where Y = Young's modulus, r_0 = Normal distance between the atoms of wire]

Unit of interatomic force constant is N/m and Dimension MT^{-2}

Note : □ The number of atoms having interatomic distance r_i in length l of a wire, $N = l/r_i$.

The number of atoms in area A of wire having interatomic separation r_i is $N = A/r_0^2$.

Elastic Hysteresis

When a deforming force is applied on a body then the strain does not change simultaneously with stress rather it lags behind the stress. The lagging of strain behind the stress is defined as elastic hysteresis. This is the reason why the values of strain for same stress are different while increasing the load and while decreasing the load.

Hysteresis loop : The area of the stress-strain curve is called the hysteresis loop and it is numerically equal to the work done in loading the material and then unloading it.

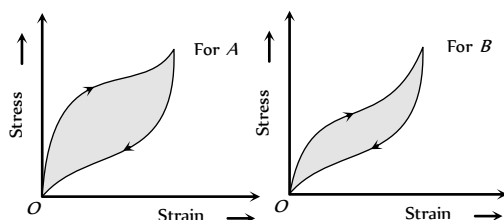


Fig. 9.18 Young's modulus (Y) $10^9 N/m$

If we have two tyres of rubber having different hysteresis loop then rubber B should be used for making the car tyres. It is because of the reason that area under the curve *i.e.* work done in case of rubber B is lesser and hence the car tyre will not get excessively heated and rubber A should be used to absorb vibration of the machinery because of the large area of the curve, a large amount of vibrational energy can be dissipated.

Factors Affecting Elasticity

(1) Hammering and rolling : Crystal grains break up into smaller units by hammering and rolling. This results in increase in the elasticity of material.

(2) Annealing : The metals are annealed by heating and then cooling them slowly. Annealing results in decrease in the elasticity of material.

(3) Temperature : Intermolecular forces decreases with rise in temperature. Hence the elasticity decreases with rise in temperature but the elasticity of invar steel (alloy) does not change with change of temperature.

(4) Impurities : Due to impurities in a material, elasticity can increase or decrease. The type of effect depends upon the nature of impurities present in the material.

Important Facts About Elasticity

(1) The body which requires greater deforming force to produce a certain change in dimension is more elastic.

Example : Ivory and steel balls are more elastic than rubber.

(2) When equal deforming force is applied on different bodies then the body which shows less deformation is more elastic.

Example : (i) For same load, more elongation is produced in rubber wire than in steel wire hence steel is more elastic than rubber.

(ii) Water is more elastic than air as volume change in water is less for same applied pressure.

(iii) Four identical balls of different materials are dropped from the same height then after collision, balls rises upto different heights.

The order of their height can be given by $h_{\text{steel}} > h_{\text{ivory}} > h_{\text{rubber}} > h_{\text{air}}$ because $Y_{\text{steel}} > Y_{\text{ivory}} > Y_{\text{rubber}} > Y_{\text{air}}$.

(3) The value of moduli of elasticity is independent of the magnitude of the stress and strain. It depends only on the nature of material of the body.

(4) For a given material there can be different moduli of elasticity depending on the type of stress applied and resulting strain.

Name of substance	Fig. 9.18 Young's modulus (Y) $10^9 N/m$	Bulk modulus (K) $10^9 N/m$	Modulus of rigidity (η) $10^9 N/m$
Aluminium	6.9	7.0	2.6
Brass	9.0	6.7	3.4
Copper	11.0	13.0	4.5
Iron	19.0	14.0	4.6
Steel	20.0	16.0	8.4
Tungsten	36.0	20.0	15.0
Diamond	83.0	55.0	34.0

Water	–	0.22	–
Glycerin	–	0.45	–
Air	–	1.01	–

(5) The moduli of elasticity has same dimensional formula and units as that of stress since strain is dimensionless. $\therefore$ Dimensional formula is $[ML^{-1}T^{-2}]$ while units *dynes/cm* or *Newton/m*.

(6) Greater the value of moduli of elasticity more elastic is the material. But as $Y \propto (1/l)$, $K \propto (1/\Delta V)$ and $\eta \propto (1/\phi)$ for a constant stress, so smaller change in shape or size for a given stress corresponds to greater elasticity.

(7) The moduli of elasticity Y and η exist only for solids as liquids and gases cannot be deformed along one dimension only and also cannot sustain shear strain. However K exist for all states of matter *viz.* solid, liquid or gas.

(8) Gases being most compressible are least elastic while solids are most *i.e.* the bulk modulus of gas is very low while that for liquids and solids is very high. $K_{\text{sol}} > K_{\text{liq}} > K_{\text{gas}}$

(9) For a rigid body l , ΔV or $\phi = 0$ so Y , K or η will be ∞ , *i.e.* elasticity of a rigid body is infinite.

Diamond and carborundum are nearest approach to rigid bodies.

(10) In a suspension bridge there is a stretch in the ropes by the load of the bridge. Due to which length of rope changes. Hence Young's modulus of elasticity is involved.

(11) In an automobile tyre as the air is compressed, volume of the air in tyre changes, hence the bulk modulus of elasticity is involved.

(12) In transmitting power, an automobile shaft is sheared as it rotates, so shearing strain is set up, hence modulus of rigidity is involved.

(13) The shape of rubber heels changes under stress, so modulus of rigidity is involved.

Practical Applications of Elasticity

(i) The metallic parts of machinery are never subjected to a stress beyond elastic limit, otherwise they will get permanently deformed.

(ii) The thickness of the metallic rope used in the crane in order to lift a given load is decided from the knowledge of elastic limit of the material of the rope and the factor of safety.

(iii) The bridges are declared unsafe after long use because during its long use, a bridge under goes quick alternating strains continuously. It results in the loss of elastic strength.

(iv) Maximum height of a mountain on earth can be estimated from the elastic behaviour of earth.

At the base of the mountain, the pressure is given by $P = h\rho g$ and it must be less than elastic limit (K) of earth's supporting material.

$$K > P > h\rho g \quad \therefore h < \frac{K}{\rho g} \quad \text{or} \quad h_{\text{max}} = \frac{K}{\rho g}$$

(v) In designing a beam for its use to support a load (in construction of roofs and bridges), it is advantageous to increase its depth rather than the breadth of the beam because the depression in rectangular beam.

$$\delta = \frac{Wl^3}{4Ybd^3}$$

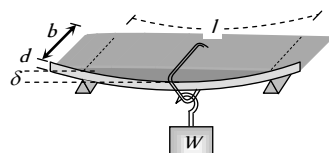


Fig. 9.19

To minimize the depression in the beam, it is designed as *I*-shaped girder.

(vi) For a beam with circular cross-section depression is given by
$$\delta = \frac{WL^3}{12\pi r^4 Y}$$

(vii) A hollow shaft is stronger than a solid shaft made of same mass, length and material.

Torque required to produce a unit twist in a solid shaft

$$\tau_{\text{solid}} = \frac{\pi \eta r^4}{2l} \quad \dots(i)$$

and torque required to produce a unit twist in a hollow shaft

$$\tau_{\text{hollow}} = \frac{\pi \eta (r_2^4 - r_1^4)}{2l} \quad \dots(ii)$$

From (i) and (ii),

$$\frac{\tau_{\text{hollow}}}{\tau_{\text{solid}}} = \frac{r_2^4 - r_1^4}{r^4} = \frac{(r_2^2 + r_1^2)(r_2^2 - r_1^2)}{r^4} \quad \dots(iii)$$

Since two shafts are made from equal volume $\therefore \pi r^2 l = \pi (r_2^2 - r_1^2) l \Rightarrow r^2 = r_2^2 - r_1^2$

Substituting this value in equation (iii) we get,

$$\frac{\tau_{\text{hollow}}}{\tau_{\text{solid}}} = \frac{r_2^2 + r_1^2}{r^2} > 1 \quad \therefore \tau_{\text{hollow}} > \tau_{\text{solid}}$$

i.e., the torque required to twist a hollow shaft is greater than the torque necessary to twist a solid shaft of the same mass, length and material through the same angle. Hence, a hollow shaft is stronger than a solid shaft.

Tips & Tricks

✍ Metals are polycrystalline materials.

✍ Metals are elastic for small strains and for large strains, metals become plastic.

✍ The substances having large molecular structure (formed by the union of two to several thousand simple molecules) are called polymers.

✍ Rubber is a polymer.

✍ Rubber is elastic for very large strains.

✍ It stretches easily at first but then becomes stiffer.

✍ Young's modulus is defined only for the solids.

✍ Bulk modulus was first defined by Maxwell.

✍ Bulk modulus is defined for all types of materials, solids, liquids and gases.

- ✍ Reciprocal of bulk modulus is called compressibility.
 - ✍ Hooke's law is obeyed only for small values of strain.
 - ✍ Higher value of the elasticity (modulus) means greater force is required for producing a given change.
 - ✍ The material which break as soon as the stress goes beyond the elastic limit are called brittle.
 - ✍ The material which do not break well beyond the elastic limit are called ductile.
 - ✍ The deformation beyond elastic limit is called plasticity.
 - ✍ Rubber sustains elasticity even when stretched several times its length.
- However it is not ductile. It breaks down as soon as the elastic limit is crossed.
- ✍ Within elastic limit, the force constant for a spring is given by $K = \frac{YA}{L}$
 - ✍ Elastic after effect is a temporary absence of the elastic properties.
 - ✍ Quartz is the best available example of perfectly elastic materials.
 - ✍ Isothermal elasticity = pressure (P)
 - ✍ Adiabatic elasticity = Ratio of specific heats $\times$ pressure $= \gamma P$
 - ✍ Elasticity is meaningless for the rigid bodies. It is the property of the non rigid bodies.
 - ✍ Diamond and carborundum are the nearest approach to the rigid body.
 - ✍ Elastic fatigue occurs, when a metal is subjected to repeated loading and unloading.
 - ✍ Theoretical value of Poisson's ratio lies between -1 and $+1/2$ but practical value lies between zero and $+1/2$.
 - ✍ Negative value of Poisson's ratio means that if length increases then radius decreases.
 - ✍ Stress and pressure have the same units and dimensions, but the pressure is always normal to the surface but the stress may be parallel or perpendicular to the surface.
 - ✍ Normal stress is also called tensile stress when the length of the body tends to increase.
 - ✍ Normal stress is also called compressive stress when length of the body tends to decrease.
 - ✍ Tangential stress is also called shearing stress.
 - ✍ When the deforming force is inclined to the surface, both the tangential as well as normal stress are produced.
 - ✍ When a body is sheared, two mutually perpendicular strains are produced. They are called longitudinal strain and compressional strain. Both are equal in magnitude.
 - ✍ When a beam is bent, both extensional as well as compressional strain is produced.
 - ✍ The energy stored by an elastic material is the area under the force-extension graph. The area under the stress-strain graph gives the

energy stored per unit volume.

- ✍ Thermal stress in a rod $= Y\alpha \Delta\theta$. It is independent of the area of cross section or length of the wire.
- ✍ Breaking stress for a wire of unit cross-section is called tensile strength.
- ✍ Breaking stress does not depend on the length or area of cross section of the wire. However it depends on the material of the wire.
- ✍ Breaking force depends on the area of cross section. Breaking stress of a wire is called tensile strength.
- ✍ If we double the radius of rope its breaking force becomes four times. But the breaking stress remains unchanged.
- ✍ If a beam of rectangular cross-section is loaded its depression at the beam is inversely proportional to the cube of thickness.
- ✍ If a beam of circular cross-section is loaded, its depression is inversely proportional to the fourth power of radius. i.e. $\delta \propto \frac{1}{r^4}$

Ordinary Thinking

Objective Questions

Young's Modulus and Breaking Stress

1. The length of an iron wire is L and area of cross-section is A . The increase in length is l on applying the force F on its two ends. Which of the statement is correct [NCERT 1976]
 - (a) Increase in length is inversely proportional to its length L
 - (b) Increase in length is proportional to area of cross-section A
 - (c) Increase in length is inversely proportional to A
 - (d) Increase in length is proportional to Young's modulus
2. The increase in length is l of a wire of length L by the longitudinal stress. Then the stress is proportional to [MP PET 1986]
 - (a) L/l
 - (b) l/L
 - (c) $l \times L$
 - (d) $l^2 \times L$
3. The dimensions of four wires of the same material are given below. In which wire the increase in length will be maximum when the same tension is applied [IIT 1981; NCERT 1976; MP PET/PMT 1998; CPMT 1983, 90; MP PMT 1992, 94, 97; MP PET 1989, 90, 99]
 - (a) Length 100 cm, Diameter 1 mm
 - (b) Length 200 cm, Diameter 2 mm
 - (c) Length 300 cm, Diameter 3 mm
 - (d) Length 50 cm, Diameter 0.5 mm

4. The ratio of the lengths of two wires A and B of same material is 1 : 2 and the ratio of their diameter is 2 : 1. They are stretched by the same force, then the ratio of increase in length will be [MP PMT 1986; MP PET/PMT 1988]
- (a) 2 : 1 (b) 1 : 4
(c) 1 : 8 (d) 8 : 1
5. The Young's modulus of a wire of length L and radius r is YN/m . If the length and radius are reduced to $L/2$ and $r/2$, then its Young's modulus will be [MP PMT 1985; MP PET 1997; KCET 1999]
- (a) $Y/2$ (b) Y
(c) $2Y$ (d) $4Y$
6. A beam of metal supported at the two ends is loaded at the centre. The depression at the centre is proportional to [CPMT 1983, 84]
- (a) Y^2 (b) Y
(c) $1/Y$ (d) $1/Y^2$
7. When a certain weight is suspended from a long uniform wire, its length increases by one cm. If the same weight is suspended from another wire of the same material and length but having a diameter half of the first one then the increase in length will be
- (a) 0.5 cm (b) 2 cm
(c) 4 cm (d) 8 cm
8. Hook's law defines [MP PMT/PET 1988]
- (a) Stress (b) Strain
(c) Modulus of elasticity (d) Elastic limit
9. A wire is loaded by 6 kg at its one end, the increase in length is 12 mm. If the radius of the wire is doubled and all other magnitudes are unchanged, then increase in length will be [MP PMT 1987; AI SSCE 1982]
- (a) 6 mm (b) 3 mm
(c) 24 mm (d) 48 mm
10. The area of cross-section of a wire of length 1.1 metre is 1 mm². It is loaded with 1 kg. If Young's modulus of copper is $1.1 \times 10^{11} \text{ N/m}^2$, then the increase in length will be (If $g = 10 \text{ m/s}^2$) [MP PET 1989]
- (a) 0.01 mm (b) 0.075 mm
(c) 0.1 mm (d) 0.15 mm
11. On increasing the length by 0.5 mm in a steel wire of length 2 m and area of cross-section 2 mm^2 , the force required is [Y for steel = $2.2 \times 10^{11} \text{ N/m}^2$] [MP PET/PMT 1988]
- (a) $1.1 \times 10^5 \text{ N}$ (b) $1.1 \times 10^4 \text{ N}$
(c) $1.1 \times 10^3 \text{ N}$ (d) $1.1 \times 10^2 \text{ N}$
12. If Young's modulus of iron is $2 \times 10^{11} \text{ N/m}^2$ and the interatomic spacing between two molecules is $3 \times 10^{-10} \text{ metre}$, the interatomic force constant is [JIPMER 1978]
- (a) 60 N/m (b) 120 N/m
(c) 30 N/m (d) 180 N/m
13. In CGS system, the Young's modulus of a steel wire is 2×10^{12} . To double the length of a wire of unit cross-section area, the force required is [MP PMT 1989]
- (a) $4 \times 10^6 \text{ dynes}$ (b) $2 \times 10^{12} \text{ dynes}$
(c) $2 \times 10^{12} \text{ newtons}$ (d) $2 \times 10^8 \text{ dynes}$
14. The material which practically does not show elastic after effect is [JIPMER 1997]
- (a) Copper (b) Rubber
(c) Steel (d) Quartz
15. If the temperature increases, the modulus of elasticity
- (a) Decreases (b) Increases
(c) Remains constant (d) Becomes zero
16. A force F is needed to break a copper wire having radius R . The force needed to break a copper wire of radius $2R$ will be [MP PET 1990]
- (a) $F/2$ (b) $2F$
(c) $4F$ (d) $F/4$
17. The relationship between Young's modulus Y , Bulk modulus K and modulus of rigidity η is [MP PET 1991; MP PMT 1997]
- (a) $Y = \frac{9\eta K}{3K + 2Y}$ (b) $\frac{9YK}{Y + 3K}$
(c) $Y = \frac{9\eta K}{3 + K}$ (d) $Y = \frac{3\eta K}{9\eta + K}$
18. The diameter of a brass rod is 4 mm and Young's modulus of brass is $9 \times 10^{10} \text{ N/m}^2$. The force required to stretch by 0.1% of its length is [MP PET 1991; BVP 2003]
- (a) $360 \pi \text{ N}$ (b) 36 N
(c) $144 \pi \times 10^3 \text{ N}$ (d) $36 \pi \times 10^5 \text{ N}$
19. If x longitudinal strain is produced in a wire of Young's modulus y , then energy stored in the material of the wire per unit volume is [MP PMT 1987, 89, 92; CPMT 1997; Pb. PMT 1999; KCET 2000; AIIMS 2001]
- (a) yx^2 (b) $2yx^2$
(c) $\frac{1}{2}y^2x$ (d) $\frac{1}{2}yx^2$
20. In a wire of length L , the increase in its length is l . If the length is reduced to half, the increase in its length will be
- (a) l (b) $2l$
(c) $\frac{l}{2}$ (d) None of the above
21. The Young's modulus of a rubber string 8 cm long and density 1.5 kg/m^3 is $5 \times 10^8 \text{ N/m}^2$, is suspended on the ceiling in a room. The increase in length due to its own weight will be
- (a) $9.6 \times 10^{-5} \text{ m}$ (b) $9.6 \times 10^{-11} \text{ m}$
(c) $9.6 \times 10^{-3} \text{ m}$ (d) 9.6 m
22. A and B are two wires. The radius of A is twice that of B . They are stretched by the same load. Then the stress on B is [MP PMT 1993]
- (a) Equal to that on A (b) Four times that on A
(c) Two times that on A (d) Half that on A

23. If the length of a wire is reduced to half, then it can hold the load
 (a) Half (b) Same
 (c) Double (d) One fourth
24. To double the length of a iron wire having 0.5 cm^2 area of cross-section, the required force will be ($Y = 10^{12} \text{ dyne / cm}^2$)
 (a) $1.0 \times 10^{-7} \text{ N}$ (b) $1.0 \times 10^7 \text{ N}$
 (c) $0.5 \times 10^{-7} \text{ N}$ (d) $0.5 \times 10^{12} \text{ dyne}$
25. The spring balance does not read properly after its long use, because
 (a) The elasticity of spring increases
 (b) The elasticity decreases
 (c) Its plastic power decreases
 (d) Its plastic power increases
26. Two wires of equal lengths are made of the same material. Wire A has a diameter that is twice as that of wire B. If identical weights are suspended from the ends of these wires, the increase in length is
 [EAMCET 1983; MP PMT 1990; MP PET 1995]
 (a) Four times for wire A as for wire B
 (b) Twice for wire A as for wire B
 (c) Half for wire A as for wire B
 (d) One-fourth for wire A as for wire B
27. Why the spring is made up of steel in comparison of copper
 (a) Copper is more costly than steel
 (b) Copper is more elastic than steel
 (c) Steel is more elastic than copper
 (d) None of the above
28. Steel and copper wires of same length are stretched by the same weight one after the other. Young's modulus of steel and copper are $2 \times 10^{11} \text{ N / m}^2$ and $1.2 \times 10^{11} \text{ N / m}^2$. The ratio of increase in length
 [MP PET 1984]
 (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
 (c) $\frac{5}{4}$ (d) $\frac{5}{2}$
29. An area of cross-section of rubber string is 2 cm^2 . Its length is doubled when stretched with a linear force of $2 \times 10^5 \text{ dynes}$. The Young's modulus of the rubber in dyne / cm^2 will be
 (a) 4×10^5 (b) 1×10^5
 (c) 2×10^5 (d) 1×10^4
30. Increase in length of a wire is 1 mm when suspended by a weight. If the same weight is suspended on a wire of double its length and double its radius, the increase in length will be
 [CPMT 1976]
 (a) 2 mm (b) 0.5 mm
 (c) 4 mm (d) 0.25 mm
31. The temperature of a wire of length 1 metre and area of cross-section 1 cm^2 is increased from 0°C to 100°C . If the rod is not allowed to increase in length, the force required will be ($\alpha = 10^{-5} / ^\circ \text{C}$ and $Y = 10^{11} \text{ N / m}^2$)
 [NCERT 1976; CPMT 1982, 91]
 (a) 10^3 N (b) 10^4 N
 (c) 10^5 N (d) 10^9 N
32. A rod of length l and area of cross-section A is heated from 0°C to 100°C . The rod is so placed that it is not allowed to increase in length, then the force developed is proportional to
 (a) l (b) l^{-1}
 (c) A [MP PMT 1987] (d) A^{-1}
33. An aluminum rod (Young's modulus $= 7 \times 10^9 \text{ N / m}^2$) has a breaking strain of 0.2% . The minimum cross-sectional area of the rod in order to support a load of 10^4 Newton's is
 [MP PMT 1991]
 (a) $1 \times 10^{-2} \text{ m}^2$ (b) $1.4 \times 10^{-3} \text{ m}^2$
 (c) $3.5 \times 10^{-3} \text{ m}^2$ (d) $7.1 \times 10^{-4} \text{ m}^2$
34. Two wires of copper having the length in the ratio $4 : 1$ and their radii ratio as $1 : 4$ are stretched by the same force. The ratio of longitudinal strain in the two will be
 (a) $1 : 16$ (b) $16 : 1$
 (c) $1 : 64$ (d) $64 : 1$
35. A weight of 200 kg is suspended by vertical wire of length 600.5 cm . The area of cross-section of wire is 1 mm^2 . When the load is removed, the wire contracts by 0.5 cm . The Young's modulus of the material of wire will be
 (a) $2.35 \times 10^{12} \text{ N / m}^2$ (b) $1.35 \times 10^{10} \text{ N / m}^2$
 (c) $13.5 \times 10^{11} \text{ N / m}^2$ (d) $23.5 \times 10^9 \text{ N / m}^2$
36. If a load of 9 kg is suspended on a wire, the increase in length is 4.5 mm . The force constant of the wire is
 (a) $0.49 \times 10^4 \text{ N / m}$ (b) $1.96 \times 10^4 \text{ N / m}$
 (c) $4.9 \times 10^4 \text{ N / m}$ (d) $0.196 \times 10^4 \text{ N / m}$
37. The ratio of diameters of two wires of same material is $n : 1$. The length of wires are 4 m each. On applying the same load, the increase in length of thin wire will be
 (a) n^2 times (b) n times
 (c) $2n$ times (d) None of the above
38. Longitudinal stress of 1 kg / mm^2 is applied on a wire. The percentage increase in length is ($Y = 10^{11} \text{ N / m}^2$)
 [MP PET 1985]
 (a) 0.002 (b) 0.001
 (c) 0.003 (d) 0.01
39. A steel wire is stretched with a definite load. If the Young's modulus of the wire is Y . For decreasing the value of Y
 (a) Radius is to be decreased
 (b) Radius is to be increased
 (c) Length is to be increased
 (d) None of the above
40. The interatomic distance for a metal is $3 \times 10^{-10} \text{ m}$. If the interatomic force constant is $3.6 \times 10^{-9} \text{ N / \AA}$, then the Young's modulus in N / m^2 will be
 (a) 1.2×10^{11} (b) 4.2×10^{11}
 (c) 10.8×10^{-19} (d) 2.4×10^{10}

41. Two identical wires of rubber and iron are stretched by the same weight, then the number of atoms in the iron wire will be
(a) Equal to that of rubber
(b) Less than that of the rubber
(c) More than that of the rubber
(d) None of the above
42. The force constant of a wire does not depend on
(a) Nature of the material (b) Radius of the wire
(c) Length of the wire (d) None of the above
43. The elasticity of *invar*
(a) Increases with temperature rise
(b) Decreases with temperature rise
(c) Does not depend on temperature
(d) None of the above
44. After effects of elasticity are maximum for
(a) Glass (b) Quartz
(c) Rubber (d) Metal
45. In suspended type moving coil galvanometer, quartz suspension is used because
(a) It is good conductor of electricity
(b) Elastic after effects are negligible
(c) Young's modulus is greater
(d) There is no elastic limit
46. A force of 200 N is applied at one end of a wire of length 2 m and having area of cross-section 10^{-2} cm^2 . The other end of the wire is rigidly fixed. If coefficient of linear expansion of the wire $\alpha = 8 \times 10^{-6} / ^\circ\text{C}$ and Young's modulus $Y = 2.2 \times 10^{11} \text{ N/m}^2$ and its temperature is increased by 5°C , then the increase in the tension of the wire will be
(a) 4.2 N (b) 4.4 N
(c) 2.4 N (d) 8.8 N
47. When compared with solids and liquids, the gases have
(a) Minimum volume elasticity
(b) Maximum volume elasticity
(c) Maximum Young's modulus
(d) Maximum modulus of rigidity
48. The length of a wire is 1.0 m and the area of cross-section is $1.0 \times 10^{-2} \text{ cm}^2$. If the work done for increase in length by 0.2 cm is 0.4 joule, then Young's modulus of the material of the wire is
(a) $2.0 \times 10^{10} \text{ N/m}^2$ (b) $4 \times 10^{10} \text{ N/m}^2$
(c) $2.0 \times 10^{11} \text{ N/m}^2$ (d) $2 \times 10^{10} \text{ N/m}^2$
49. The quality of the material which opposes the change in shape, volume or length is called
(a) Intermolecular repulsion
(b) Intermolecular behaviour
(c) Viscosity
(d) Elasticity
50. For silver, Young's modulus is $7.25 \times 10^{10} \text{ N/m}^2$ and Bulk modulus is $11 \times 10^{10} \text{ N/m}^2$. Its Poisson's ratio will be
(a) -1 (b) 0.5
(c) 0.39 (d) 0.25
51. The longitudinal strain is only possible in
(a) Gases (b) Fluids
(c) Solids (d) Liquids
52. If the density of the material increases, the value of Young's modulus
(a) Increases
(b) Decreases
(c) First increases then decreases
(d) First decreases then increases
53. Young's modulus of rubber is 10^4 N/m^2 and area of cross-section is 2 cm^2 . If force of 2×10^5 dynes is applied along its length, then its initial length l becomes
(a) $3l$ (b) $4l$
(c) $2l$ (d) None of the above
54. The elastic limit for a gas
(a) Exists
(b) Exists only at absolute zero
(c) Exists for a perfect gas
(d) Does not exist
55. If Young's modulus for a material is zero, then the state of material should be
(a) Solid (b) Solid but powder
(c) Gas (d) None of the above
56. Liquids have no Poisson's ratio, because
(a) It has no definite shape
(b) It has greater volume
(c) It has lesser density than solid
(d) None of the above
57. A wire of length L and radius r is rigidly fixed at one end. On stretching the other end of the wire with a force F , the increase in its length is l . If another wire of same material but of length $2L$ and radius $2r$ is stretched with a force of $2F$, the increase in its length will be
[NCERT 1980; AIIMS 1980; MP PET 1989, 92; MP PET/PMT 1988; MP PMT 1996, 2002; UPSEAT 2002]
(a) l (b) $2l$
(c) $\frac{l}{2}$ (d) $\frac{l}{4}$
58. In steel, the Young's modulus and the strain at the breaking point are $2 \times 10^{11} \text{ Nm}^{-2}$ and 0.15 respectively. The stress at the breaking point for steel is therefore
[MP PET 1990; MP PMT 1992; DPMT 2001]
(a) $1.33 \times 10^{11} \text{ Nm}^{-2}$ (b) $1.33 \times 10^{12} \text{ Nm}^{-2}$
(c) $7.5 \times 10^{-13} \text{ Nm}^{-2}$ (d) $3 \times 10^{10} \text{ Nm}^{-2}$
59. Which of the following statements is correct [MP PET 1992]
(a) Hooke's law is applicable only within elastic limit

- (b) The adiabatic and isothermal elastic constants of a gas are equal
(c) Young's modulus is dimensionless
(d) Stress multiplied by strain is equal to the stored energy
60. The force required to stretch a steel wire of 1 cm^2 cross-section to 1.1 times its length would be ($Y = 2 \times 10^{11}\text{ Nm}^{-2}$)
[MP PET 1992]
- (a) $2 \times 10^6\text{ N}$ (b) $2 \times 10^3\text{ N}$
(c) $2 \times 10^{-6}\text{ N}$ (d) $2 \times 10^{-7}\text{ N}$
61. Which one of the following substances possesses the highest elasticity
[MP PMT 1992; RPMT 1999; RPET 2000; MH CET (Med.) 2001]
- (a) Rubber (b) Glass
(c) Steel (d) Copper
62. Which one of the following quantities does not have the unit of force per unit area
[MP PMT 1992]
- (a) Stress
(b) Strain
(c) Young's modulus of elasticity
(d) Pressure
63. A copper wire and a steel wire of the same diameter and length are connected end to end and a force is applied, which stretches their combined length by 1 cm. The two wires will have
- (a) Different stresses and strains
(b) The same stress and strain
(c) The same strain but different stresses
(d) The same stress but different strains
64. A steel ring of radius r and cross-section area ' A ' is fitted on to a wooden disc of radius R ($R > r$). If Young's modulus be E , then the force with which the steel ring is expanded is
[EAMCET 1986]
- (a) $AE \frac{R}{r}$ (b) $AE \left(\frac{R-r}{r} \right)$
(c) $\frac{E}{A} \left(\frac{R-r}{A} \right)$ (d) $\frac{Er}{AR}$
65. A wire extends by 1 mm when a force is applied. Double the force is applied to another wire of same material and length but half the radius of cross-section. The elongation of the wire in mm will be
- (a) 8 (b) 4
(c) 2 (d) 1
66. Two wires of the same material have lengths in the ratio 1 : 2 and their radii are in the ratio $1 : \sqrt{2}$. If they are stretched by applying equal forces, the increase in their lengths will be in the ratio
- (a) $2 : \sqrt{2}$ (b) $\sqrt{2} : 2$
(c) 1 : 1 (d) 1 : 2
67. When a weight of 10 kg is suspended from a copper wire of length 3 metres and diameter 0.4 mm, its length increases by 2.4 cm. If the diameter of the wire is doubled, then the extension in its length will be
[MP PMT 1994]
- (a) 9.6 cm (b) 4.8 cm (c) 1.2 cm (d) 0.6 cm
68. A force of 10^3 newton stretches the length of a hanging wire by 1 millimetre. The force required to stretch a wire of same material and length but having four times the diameter by 1 millimetre is
- (a) $4 \times 10^3\text{ N}$ (b) $16 \times 10^3\text{ N}$
(c) $\frac{1}{4} \times 10^3\text{ N}$ (d) $\frac{1}{16} \times 10^3\text{ N}$
69. Two wires 'A' and 'B' of the same material have radii in the ratio 2 : 1 and lengths in the ratio 4 : 1. The ratio of the normal forces required to produce the same change in the lengths of these two wires is
[Haryana CEE 1996]
- (a) 1 : 1 (b) 2 : 1
(c) 1 : 4 (d) 1 : 2
70. Density of rubber is d . A thick rubber cord of length L and cross-section area A undergoes elongation under its own weight on suspending it. This elongation is proportional to
- (a) dL (b) Ad/L
(c) Ad/L^2 (d) dL^2
71. The ratio of two specific heats of gas C_p/C_v for argon is 1.6 and for hydrogen is 1.4. Adiabatic elasticity of argon at pressure P is E . Adiabatic elasticity of hydrogen will also be equal to E at the pressure
- (a) P [MP PMT 1992] (b) $\frac{8}{7}P$
(c) $\frac{7}{8}P$ (d) $1.4P$
72. The relation between γ, η and K for a elastic material is
- (a) $\frac{1}{\eta} = \frac{1}{3\gamma} + \frac{1}{9K}$ (b) $\frac{1}{K} = \frac{1}{3\gamma} + \frac{1}{9\eta}$
(c) $\frac{1}{\gamma} = \frac{1}{3K} + \frac{1}{9\eta}$ (d) $\frac{1}{\gamma} = \frac{1}{3\eta} + \frac{1}{9K}$
73. A fixed volume of iron is drawn into a wire of length L . The extension x produced in this wire by a constant force F is proportional to
[MP PMT 1999]
- (a) $\frac{1}{L^2}$ (b) $\frac{1}{L}$
(c) L^2 (d) L
74. A wire of cross-sectional area 3 mm^2 is first stretched between two fixed points at a temperature of 20°C . Determine the tension when the temperature falls to 10°C . Coefficient of linear expansion $\alpha = 10^{-5}^\circ\text{C}^{-1}$ and $Y = 2 \times 10^{11}\text{ N/m}^2$
[EAMCET 1986]
- (a) 20 N (b) 30 N
(c) 60 N [MP PET 1994] (d) 120 N
75. To keep constant time, watches are fitted with balance wheel made of
[EAMCET 1994]
- (a) Invar (b) Stainless steel
(c) Tungsten (d) Platinum
76. A wire is stretched by 0.01 m by a certain force F . Another wire of same material whose diameter and length are double to the original wire is stretched by the same force. Then its elongation will be [EAMCET (Engg.

- (a) 0.005 m (b) 0.01 m [AFMC 1999]
(c) 0.02 m (d) 0.002 m
77. The possible value of Poisson's ratio is [EAMCET (Med.) 1995]
(a) 1 (b) 0.9
(c) 0.8 (d) 0.4
78. The coefficient of linear expansion of brass and steel are α_1 and α_2 . If we take a brass rod of length l_1 and steel rod of length l_2 at 0°C , their difference in length ($l_2 - l_1$) will remain the same at a temperature if [EAMCET (Med.) 1995]
(a) $\alpha_1 l_2 = \alpha_2 l_1$ (b) $\alpha_1 l_2^2 = \alpha_2 l_1^2$
(c) $\alpha_1^2 l_1 = \alpha_2^2 l_2$ (d) $\alpha_1 l_1 = \alpha_2 l_2$
79. A rod is fixed between two points at 20°C . The coefficient of linear expansion of material of rod is $1.1 \times 10^{-5} / ^\circ\text{C}$ and Young's modulus is $1.2 \times 10^{11} \text{ N/m}$. Find the stress developed in the rod if temperature of rod becomes 10°C [RPET 1997]
(a) $1.32 \times 10^7 \text{ N/m}^2$ (b) $1.10 \times 10^{15} \text{ N/m}^2$
(c) $1.32 \times 10^8 \text{ N/m}^2$ (d) $1.10 \times 10^6 \text{ N/m}^2$
80. The extension of a wire by the application of load is 3 mm . The extension in a wire of the same material and length but half the radius by the same load is [CMEET Bihar 1995]
(a) 12 mm (b) 0.75 mm
(c) 15 mm (d) 6 mm
81. A rubber pipe of density $1.5 \times 10^3 \text{ N/m}^2$ and Young's modulus $5 \times 10^6 \text{ N/m}^2$ is suspended from the roof. The length of the pipe is 8 m . What will be the change in length due to its own weight [RPET 1996]
(a) 9.6 m (b) $9.6 \times 10^3 \text{ m}$
(c) $19.2 \times 10^{-2} \text{ m}$ (d) $9.6 \times 10^{-2} \text{ m}$
82. In which case there is maximum extension in the wire, if same force is applied on each wire [AFMC 1997]
(a) $L = 500 \text{ cm}$, $d = 0.05 \text{ mm}$
(b) $L = 200 \text{ cm}$, $d = 0.02 \text{ mm}$
(c) $L = 300 \text{ cm}$, $d = 0.03 \text{ mm}$
(d) $L = 400 \text{ cm}$, $d = 0.01 \text{ mm}$
83. If a spring is extended to length l , then according to Hook's law
(a) $F = kl$ (b) $F = \frac{k}{l}$
(c) $F = k^2 l$ (d) $F = \frac{k^2}{l}$
84. Which of the following affects the elasticity of a substance [AIIMS 1999]
(a) Hammering and annealing
(b) Change in temperature
(c) Impurity in substance
(d) All of these
85. An iron rod of length 2 m and cross section area of 50 mm^2 , stretched by 0.5 mm , when a mass of 250 kg is hung from its lower end. Young's modulus of the iron rod is
(a) $19.6 \times 10^{10} \text{ N/m}^2$ (b) $19.6 \times 10^{15} \text{ N/m}^2$
(c) $19.6 \times 10^{18} \text{ N/m}^2$ (d) $19.6 \times 10^{20} \text{ N/m}^2$
86. In solids, inter-atomic forces are [DCE 1999]
(a) Totally repulsive
(b) Totally attractive
(c) Combination of (a) and (b)
(d) None of these
87. A force F is applied on the wire of radius r and length L and change in the length of wire is l . If the same force F is applied on the wire of the same material and radius $2r$ and length $2L$, Then the change in length of the other wire is [RPMT 1999]
(a) l (b) $2l$
(c) $l/2$ (d) $4l$
88. The modulus of elasticity is dimensionally equivalent to [MH CET (Med.) 1999]
(a) Surface tension (b) Stress
(c) Strain (d) None of these
89. Under elastic limit the stress is [MH CET 1999; KCET 1999]
(a) Inversely, proportional to strain
(b) Directly proportional to strain
(c) Square root of strain
(d) Independent of strain
90. A steel wire of 1 m long and 1 mm^2 cross section area is hang from rigid end. When weight of 1 kg is hung from it then change in length will be (given $Y = 2 \times 10^{11} \text{ N/m}^2$) [RPMT 2000]
(a) 0.5 mm (b) 0.25 mm
(c) 0.05 mm (d) 5 mm
91. A load W produces an extension of 1 mm in a thread of radius r . Now if the load is made $4W$ and radius is made $2r$ all other things remaining same, the extension will become [RPET 2000]
(a) 4 mm [CPMT 1997] (b) 16 mm
(c) 1 mm (d) 0.25 mm
92. The units of Young 's modulus of elasticity are [CPMT 2000; KCET 2000]
(a) Nm^{-1} (b) N-m
(c) Nm^{-2} (d) N-m^2
93. Two similar wires under the same load yield elongation of 0.1 mm and 0.05 mm respectively. If the area of cross- section of the first wire is 4 mm^2 , then the area of cross section of the second wire is [CPMT 2000]
(a) 6 mm^2 (b) 8 mm^2
(c) 10 mm^2 (d) 12 mm^2



94. A 5 m long aluminium wire ($Y = 7 \times 10^{10} \text{ N/m}^2$) of diameter 3 mm supports a 40 kg mass. In order to have the same elongation in a copper wire ($Y = 12 \times 10^{10} \text{ N/m}^2$) of the same length under the same weight, the diameter should now be, in mm.
- (a) 1.75 (b) 1.5
(c) 2.5 (d) 5.0
95. How much force is required to produce an increase of 0.2% in the length of a brass wire of diameter 0.6 mm
- [MP PMT 2000]
- (Young's modulus for brass = $0.9 \times 10^{11} \text{ N/m}^2$)
- (a) Nearly 17 N (b) Nearly 34 N
(c) Nearly 51 N (d) Nearly 68 N
96. On applying a stress of $20 \times 10^8 \text{ N/m}^2$ the length of a perfectly elastic wire is doubled. Its Young's modulus will be
- [MP PET 2000]
- (a) $40 \times 10^8 \text{ N/m}^2$ (b) $20 \times 10^8 \text{ N/m}^2$
(c) $10 \times 10^8 \text{ N/m}^2$ (d) $5 \times 10^8 \text{ N/m}^2$
97. When a uniform wire of radius r is stretched by a 2 kg weight, the increase in its length is 2.00 mm. If the radius of the wire is $r/2$ and other conditions remain the same, the increase in its length is [EAMCET (Engg.) 2000]
- (a) 2.00 mm (b) 4.00 mm
(c) 6.00 mm (d) 8.00 mm
98. The length of an elastic string is a metre when the longitudinal tension is 4 N and b metre when the longitudinal tension is 5 N. The length of the string in metre when the longitudinal tension is 9 N is [EAMCET 2001]
- (a) $a - b$ (b) $5b - 4a$
(c) $2b - \frac{1}{4}a$ (d) $4a - 3b$
99. Stress to strain ratio is equivalent to [RPET 2001]
- (a) Modulus of elasticity
(b) Poisson's Ratio
(c) Rayhold number
(d) Fund number
100. Which is correct relation [RPET 2001]
- (a) $Y < \sigma$ (b) $Y > \sigma$
(c) $Y = \sigma$ (d) $\sigma = +1$
101. If the interatomic spacing in a steel wire is $3.0 \text{ \AA}$ and $Y_{\text{steel}} = 20 \times 10^{10} \text{ N/m}^2$ then force constant is
- [RPET 2001]
- (a) $6 \times 10^{-2} \text{ N/\AA}$ (b) $6 \times 10^{-9} \text{ N/\AA}$
(c) $4 \times 10^{-5} \text{ N/\AA}$ (d) $6 \times 10^{-5} \text{ N/\AA}$
102. A copper wire of length 4.0 m and area of cross-section 1.2 cm^2 is stretched with a force of $4.8 \times 10^3 \text{ N}$. If Young's modulus for copper is $1.2 \times 10^{11} \text{ N/m}^2$, the increase in the length of the wire will be
- [MP PET 2001]
- (a) 1.33 mm (b) 1.33 cm
(c) 2.66 mm (d) 2.66 cm
103. A metal bar of length L and area of cross-section A is clamped between two rigid supports. For the material of the rod, its Young's modulus is Y and coefficient of linear expansion is α . If the temperature of the rod is increased by $\Delta t^\circ \text{C}$, the force exerted by the rod on the supports is
- [MP PMT 2001]
- (a) $Y A L \Delta t$ (b) $Y A \alpha \Delta t$
(c) $\frac{Y L \alpha \Delta t}{A}$ (d) $Y \alpha A L \Delta t$
104. According to Hook's law of elasticity, if stress is increased, the ratio of stress to strain [KCET 2000 AIIMS 2001]
- (a) Increases (b) Decreases
(c) Becomes zero (d) Remains constant
105. A pan with set of weights is attached with a light spring. When disturbed, the mass-spring system oscillates with a time period of 0.6 s. When some additional weights are added then time period is 0.7 s. The extension caused by the additional weights is approximately given by
- [UPSEAT 2002]
- (a) 1.38 cm (b) 3.5 cm
(c) 1.75 cm (d) 2.45 cm
106. A uniform plank of Young's modulus Y is moved over a smooth horizontal surface by a constant horizontal force F . The area of cross section of the plank is A . The compressive strain on the plank in the direction of the force is
- [Kerala PET 2002]
- (a) F / AY (b) $2F / AY$
(c) $\frac{1}{2}(F / AY)$ (d) $3F / AY$
107. The mean distance between the atoms of iron is $3 \times 10^{-10} \text{ m}$ and interatomic force constant for iron is 7 N/m . The Young's modulus of elasticity for iron is [JIPMER 2002]
- (a) $2.33 \times 10^5 \text{ N/m}^2$ (b) $23.3 \times 10^{10} \text{ N/m}^2$
(c) $233 \times 10^{10} \text{ N/m}^2$ (d) $2.33 \times 10^{10} \text{ N/m}^2$
108. Two wires A and B are of same materials. Their lengths are in the ratio 1 : 2 and diameters are in the ratio 2 : 1 when stretched by force F_A and F_B respectively they get equal increase in their lengths. Then the ratio F_A / F_B should be
- [Orissa JEE 2002]
- (a) 1 : 2 (b) 1 : 1
(c) 2 : 1 (d) 8 : 1
109. The breaking stress of a wire depends upon [AIIMS 2002]
- (a) Length of the wire
(b) Radius of the wire
(c) Material of the wire
(d) Shape of the cross section

110. The area of cross section of a steel wire ($Y = 2.0 \times 10^{11} \text{ N/m}^2$) is 0.1 cm^2 . The force required to double its length will be

(a) $2 \times 10^{12} \text{ N}$ (b) $2 \times 10^{11} \text{ N}$
(c) $2 \times 10^{10} \text{ N}$ (d) $2 \times 10^6 \text{ N}$

111. A rubber cord catapult has cross-sectional area 25 mm^2 and initial length of rubber cord is 10 cm . It is stretched to 5 cm , and then released to project a missile of mass 5 gm . Taking $Y_{\text{rubber}} = 5 \times 10^8 \text{ N/m}^2$ velocity of projected missile is

[CPMT 2002]

(a) 20 ms^{-1} (b) 100 ms^{-1}
(c) 250 ms^{-1} (d) 200 ms^{-1}

112. According to Hook's law force is proportional to

[RPET 2003]

(a) $\frac{1}{x}$ (b) $\frac{1}{x^2}$
(c) x (d) x^2

113. In the Young's experiment, If length of wire and radius both are doubled then the value of Y will become

[RPET 2003]

(a) 2 times (b) 4 times
(c) Remains same (d) Half

114. Minimum and maximum values of Poisson's ratio for a metal lies between

(a) $-\infty$ to $+\infty$ (b) 0 to 1
(c) $-\infty$ to 1 (d) 0 to 0.5

115. A wire of diameter 1 mm breaks under a tension of 1000 N . Another wire, of same material as that of the first one, but of diameter 2 mm breaks under a tension of

[Orissa JEE 2003]

(a) 500 N (b) 1000 N
(c) 10000 N (d) 4000 N

116. Young's modulus of perfectly rigid body material is

[KCET 2003]

(a) Zero (b) Infinity
(c) $1 \times 10^{10} \text{ N/m}^2$ (d) $10 \times 10^{10} \text{ N/m}^2$

117. A wire of length 2 m is made from 10 cm^3 of copper. A force F is applied so that its length increases by 2 mm . Another wire of length 8 m is made from the same volume of copper. If the force F is applied to it, its length will increase by

(a) 0.8 cm (b) 1.6 cm
(c) 2.4 cm (d) 3.2 cm

118. A wire of cross section 4 mm is stretched by 0.1 mm by a certain weight. How far (length) will be wire of same material and length but of area 8 mm stretch under the action of same force

(a) 0.05 mm (b) 0.10 mm
(c) 0.15 mm (d) 0.20 mm
(e) 0.25 mm

119. A substance breaks down by a stress of 10^7 N/m^2 . If the density of the material of the wire is $3 \times 10^4 \text{ kg/m}^3$, then the length of the wire of the

substance which will break under its own weight when suspended vertically, is

[DPMT 2004]

(a) 66.6 m (b) 60.0 m
(c) 33.3 m (d) 30.0 m

120. A rubber cord 10 m long is suspended vertically. How much does it stretch under its own weight (Density of rubber is 1500 kg/m^3 , $Y = 5 \times 10^8 \text{ N/m}^2$, $g = 10 \text{ m/s}^2$)

[Pb. PET 2001]

(a) $15 \times 10^{-3} \text{ m}$ (b) $7.5 \times 10^{-3} \text{ m}$
(c) $12 \times 10^{-3} \text{ m}$ (d) $25 \times 10^{-3} \text{ m}$

121. The value of Poisson's ratio lies between

[AIIMS 1985; MP PET 1986; DPMT 2002]

(a) -1 to $\frac{1}{2}$ (b) $-\frac{3}{4}$ to $-\frac{1}{2}$
(c) $-\frac{1}{2}$ to 1 (d) 1 to 2

122. The Poisson's ratio cannot have the value

[EAMCET 1989]

(a) 0.7 (b) 0.2
(c) 0.1 (d) 0.5

123. There is no change in the volume of a wire due to change in its length on stretching. The Poisson's ratio of the material of the wire is

[MH CET 2004]

(a) $+0.50$ (b) -0.50
(c) 0.25 (d) -0.25

124. A material has Poisson's ratio 0.50. If a uniform rod of it suffers a longitudinal strain of 2×10^{-3} , then the percentage change in volume is

[EAMCET 1987]

(a) 0.6 (b) 0.4
(c) 0.2 (d) Zero

125. Four identical rods are stretched by same force. Maximum extension is produced in

(a) $L = 10 \text{ cm}, D = 1 \text{ mm}$ (b) $L = 100 \text{ cm}, D = 2 \text{ mm}$
(c) $L = 200 \text{ cm}, D = 3 \text{ mm}$ (d) $L = 300 \text{ cm}, D = 4 \text{ mm}$

Bulk Modulus

1. The isothermal elasticity of a gas is equal to

[CPMT 1981; MP PMT 2004]

(a) Density (b) Volume
(c) Pressure (d) Specific heat

2. The adiabatic elasticity of a gas is equal to

[CPMT 1982]

(a) $\gamma \times \text{density}$ (b) $\gamma \times \text{volume}$
(c) $\gamma \times \text{pressure}$ (d) $\gamma \times \text{specific heat}$

3. The specific heat at constant pressure and at constant volume for an ideal gas are C_p and C_v and its adiabatic and isothermal elasticities are E_ϕ and E_θ respectively. The ratio of E_ϕ to E_θ is

(a) C_v / C_p (b) C_p / C_v
(c) $C_p C_v$ (d) $1 / C_p C_v$

4. The only elastic modulus that applies to fluids is

[BCECE 2003]

(a) Young's modulus (b) Shear modulus

KCET 1999; Pb. PMT 2003]

- (c) Modulus of rigidity (d) Bulk modulus
5. The ratio of the adiabatic to isothermal elasticities of a triatomic gas is [MP PET 1991]
 (a) $3/4$ (b) $4/3$
 (c) 1 (d) $5/3$
6. If the volume of the given mass of a gas is increased four times, the temperature is raised from 27°C to 127°C . The elasticity will become
 (a) 4 times (b) $1/4$ times
 (c) 3 times (d) $1/3$ times
7. The compressibility of water is 4×10^{-5} per unit atmospheric pressure. The decrease in volume of 100 cubic centimeter of water under a pressure of 100 atmosphere will be
 (a) 0.4 cc (b) 4×10^{-5} cc
 (c) 0.025 cc (d) 0.004 cc
8. If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is $1 \times 10^3 \text{ kg/m}^3$ and $g = 10 \text{ m/s}^2$, then the volume elasticity in N/m^2 will be
 (a) 10^8 (b) 2×10^8
 (c) 10^9 (d) 2×10^9
9. The compressibility of a material is
 (a) Product of volume and its pressure
 (b) The change in pressure per unit change in volume strain
 (c) The fractional change in volume per unit change in pressure
 (d) None of the above
10. When a pressure of 100 atmosphere is applied on a spherical ball, then its volume reduces to 0.01%. The bulk modulus of the material of the rubber in dyne/cm^2 is [MP PET 1985; DPMT 2002]
 (a) 10×10^{12} (b) 100×10^{12}
 (c) 1×10^{12} (d) 20×10^{12}
11. In the three states of matter, the elastic coefficient can be
 (a) Young's modulus
 (b) Coefficient of volume elasticity
 (c) Modulus of rigidity
 (d) Poisson's ratio
12. Bulk modulus was first defined by [CPMT 1987]
 (a) Young (b) Bulk
 (c) Maxwell (d) None of the above
13. A uniform cube is subjected to volume compression. If each side is decreased by 1%, then bulk strain is [EAMCET (Engg.) 1995; DPMT 2000]
 (a) 0.01 (b) 0.06
 (c) 0.02 (d) 0.03
14. A ball falling in a lake of depth 200 m shows 0.1% decrease in its volume at the bottom. What is the bulk modulus of the material of the ball [AFMC 1997]
 (a) $19.6 \times 10^8 \text{ N/m}^2$ (b) $19.6 \times 10^{-10} \text{ N/m}^2$
 (c) $19.6 \times 10^{10} \text{ N/m}^2$ (d) $19.6 \times 10^{-8} \text{ N/m}^2$
15. The isothermal bulk modulus of a gas at atmospheric pressure is
 (a) 1 mm of Hg (b) 13.6 mm of Hg
 (c) $1.013 \times 10^5 \text{ N/m}^2$ (d) $2.026 \times 10^5 \text{ N/m}^2$
16. Coefficient of isothermal elasticity E_θ and coefficient of adiabatic elasticity E_ϕ are related by ($\gamma = C_p / C_v$) [MP PET 2000]
 (a) $E_\theta = \gamma E_\phi$ (b) $E_\phi = \gamma E_\theta$
 (c) $E_\theta = \gamma / E_\phi$ (d) $E_\theta = \gamma^2 E_\phi$
17. The bulk modulus of an ideal gas at constant temperature [MP PMT 1990] [MP PMT 2004]
 (a) Is equal to its volume V (b) Is equal to $p/2$
 (c) Is equal to its pressure p (d) Can not be determined
18. The Bulk modulus for an incompressible liquid is [BHU 2004]
 (a) Zero [MP PMT 1991] (b) Unity
 (c) Infinity (d) Between 0 to 1
19. The pressure applied from all directions on a cube is P . How much its temperature should be raised to maintain the original volume? The volume elasticity of the cube is β and the coefficient of volume expansion is α
 (a) $\frac{P}{\alpha\beta}$ (b) $\frac{P\alpha}{\beta}$
 (c) $\frac{P\beta}{\alpha}$ (d) $\frac{\alpha\beta}{P}$
20. The pressure of a medium is changed from $1.01 \times 10^5 \text{ Pa}$ to $1.165 \times 10^5 \text{ Pa}$ and change in volume is 10% keeping temperature constant. The Bulk modulus of the medium is
 (a) $204.8 \times 10^5 \text{ Pa}$ (b) $102.4 \times 10^5 \text{ Pa}$
 (c) $51.2 \times 10^5 \text{ Pa}$ (d) $1.55 \times 10^5 \text{ Pa}$
21. For a constant hydraulic stress on an object, the fractional change in the object's volume $\left(\frac{\Delta V}{V}\right)$ and its bulk modulus (B) are related as
 (a) $\frac{\Delta V}{V} \propto B$ (b) $\frac{\Delta V}{V} \propto \frac{1}{B}$
 (c) $\frac{\Delta V}{V} \propto B^2$ (d) $\frac{\Delta V}{V} \propto B^{-2}$

Rigidity Modulus

1. Modulus of rigidity of diamond is
 (a) Too less
 (b) Greater than all matters
 (c) Less than all matters
 (d) Zero
2. The ratio of lengths of two rods A and B of same material is 1 : 2 and the ratio of their radii is 2 : 1, then the ratio of modulus of rigidity of A and B will be
 (a) 4 : 1 [AIIMS 2000] (b) 16 : 1

- (c) 8 : 1 (d) 1 : 1
3. Which statement is true for a metal [DPMT 2001]
(a) $Y < \eta$ (b) $Y = \eta$
(c) $Y > \eta$ (d) $Y < 1/\eta$
4. Which of the following relations is true [CPMT 1984]
(a) $3Y = K(1 - \sigma)$ (b) $K = \frac{9\eta Y}{Y + \eta}$
(c) $\sigma = (6K + \eta)Y$ (d) $\sigma = \frac{0.5Y - \eta}{\eta}$
5. Two wires A and B of same length and of the same material have the respective radii r_1 and r_2 . Their one end is fixed with a rigid support, and at the other end equal twisting couple is applied. Then the ratio of the angle of twist at the end of A and the angle of twist at the end of B will be [AIIMS 1980]
(a) $\frac{r_1^2}{r_2^2}$ (b) $\frac{r_2^2}{r_1^2}$
(c) $\frac{r_2^4}{r_1^4}$ (d) $\frac{r_1^4}{r_2^4}$
6. When a spiral spring is stretched by suspending a load on it, the strain produced is called
(a) Shearing (b) Longitudinal
(c) Volume (d) Transverse
7. The Young's modulus of the material of a wire is $6 \times 10^{12} \text{ N/m}^2$ and there is no transverse strain in it, then its modulus of rigidity will be
(a) $3 \times 10^{12} \text{ N/m}^2$ (b) $2 \times 10^{12} \text{ N/m}^2$
(c) 10^{12} N/m^2 (d) None of the above
8. If the Young's modulus of the material is 3 times its modulus of rigidity, then its volume elasticity will be
(a) Zero (b) Infinity
(c) $2 \times 10^{10} \text{ N/m}^2$ (d) $3 \times 10^{10} \text{ N/m}^2$
9. Modulus of rigidity of a liquid [RPET 2000]
(a) Non zero constant
(b) Infinite
(c) Zero
(d) Can not be predicted
10. For a given material, the Young's modulus is 2.4 times that of rigidity modulus. Its Poisson's ratio is [EAMCET 1990; RPET 2001]
(a) 2.4 (b) 1.2
(c) 0.4 (d) 0.2
11. A cube of aluminium of sides 0.1 m is subjected to a shearing force of 100 N. The top face of the cube is displaced through 0.02 cm with respect to the bottom face. The shearing strain would be
(a) 0.02 (b) 0.1
(c) 0.005 (d) 0.002
12. The reason for the change in shape of a regular body is [EAMCET 1980]
(a) Volume stress (b) Shearing strain
(c) Longitudinal strain (d) Metallic strain
13. The lower surface of a cube is fixed. On its upper surface, force is applied at an angle of 30° from its surface. The change will be of the type
(a) Shape (b) Size
(c) None (d) Shape and size
14. The upper end of a wire of radius 4 mm and length 100 cm is clamped and its other end is twisted through an angle of 30° . Then angle of shear is [NCERT 1990; MP PMT 1996]
(a) 12° (b) 0.12°
(c) 1.2° (d) 0.012°
15. Mark the wrong statement [MP PMT 2003]
(a) Sliding of molecular layer is much easier than compression or expansion
(b) Reciprocal of bulk modulus of elasticity is called compressibility
(c) It is difficult to twist a long rod as compared to small rod
(d) Hollow shaft is much stronger than a solid rod of same length and same mass
16. A 2 m long rod of radius 1 cm which is fixed from one end is given a twist of 0.8 radians. The shear strain developed will be
(a) 0.002 (b) 0.004
(c) 0.008 (d) 0.016
17. A rod of length l and radius r is joined to a rod of length $l/2$ and radius $r/2$ of same material. The free end of small rod is fixed to a rigid base and the free end of larger rod is given a twist of θ° , the twist angle at the joint will be [RPET 1997]
(a) $\theta/4$ (b) $\theta/2$
(c) $5\theta/6$ (d) $8\theta/9$
18. Shearing stress causes change in [RPET 2002; BCECE 2001, 04]
(a) Length (b) Breadth
(c) Shape (d) Volume

Work Done in Stretching a Wire

1. If the potential energy of a spring is V on stretching it by 2 cm, then its potential energy when it is stretched by 10 cm will be
(a) $V/25$ (b) $5V$
(c) $V/5$ (d) $25V$
2. The work done in stretching an elastic wire per unit volume is or strain energy in a stretched string is [NCERT 1981; EAMCET (Med.) 1995; MNR 1981; MP PET 1984; RPMT 1999; DCE 2003]
(a) Stress $\times$ Strain (b) $\frac{1}{2} \times$ Stress $\times$ Strain
(c) $2 \times$ strain $\times$ stress (d) Stress/Strain
3. Calculate the work done, if a wire is loaded by ' Mg ' weight and the increase in length is ' l '



[CPMT 1999; DCE 1999, 2001; Pb. PET 2000, 01]

- (a) Mgl (b) Zero
(c) $Mgl/2$ (d) $2Mgl$
4. Two wires of same diameter of the same material having the length l and $2l$. If the force F is applied on each, the ratio of the work done in the two wires will be [MP PET 1989]
(a) 1 : 2 (b) 1 : 4
(c) 2 : 1 (d) 1 : 1
5. A 5 metre long wire is fixed to the ceiling. A weight of 10 kg is hung at the lower end and is 1 metre above the floor. The wire was elongated by 1 mm. The energy stored in the wire due to stretching is [MP PET 1989]
(a) Zero (b) 0.05 joule
(c) 100 joule (d) 500 joule
6. If the force constant of a wire is K , the work done in increasing the length of the wire by l is [MP PMT 1989]
(a) $Kl/2$ (b) Kl
(c) $Kl^2/2$ (d) Kl^2
7. If the tension on a wire is removed at once, then
(a) It will break
(b) Its temperature will reduce
(c) There will be no change in its temperature
(d) Its temperature increases
8. When strain is produced in a body within elastic limit, its internal energy
(a) Remains constant (b) Decreases
(c) Increases (d) None of the above
9. When shearing force is applied on a body, then the elastic potential energy is stored in it. On removing the force, this energy
(a) Converts into kinetic energy
(b) Converts into heat energy
(c) Remains as potential energy
(d) None of the above
10. A brass rod of cross-sectional area 1 cm^2 and length 0.2 m is compressed lengthwise by a weight of 5 kg. If Young's modulus of elasticity of brass is $1 \times 10^{11}\text{ N/m}^2$ and $g = 10\text{ m/sec}^2$, then increase in the energy of the rod will be
(a) 10^{-5} J (b) $2.5 \times 10^{-5}\text{ J}$
(c) $5 \times 10^{-5}\text{ J}$ (d) $2.5 \times 10^{-4}\text{ J}$
11. If one end of a wire is fixed with a rigid support and the other end is stretched by a force of 10 N, then the increase in length is 0.5 mm. The ratio of the energy of the wire and the work done in displacing it through 1.5 mm by the weight is
(a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) 1
12. A wire is suspended by one end. At the other end a weight equivalent to 20 N force is applied. If the increase in length is 1.0 mm, the increase in energy of the wire will be
(a) 0.01 J (b) 0.02 J
(c) 0.04 J (d) 1.00 J
13. In the above question, the ratio of the increase in energy of the wire to the decrease in gravitational potential energy when load moves downwards by 1 mm, will be
(a) 1 (b) $\frac{1}{4}$
(c) $\frac{1}{3}$ (d) $\frac{1}{2}$
14. The Young's modulus of a wire is Y . If the energy per unit volume is E , then the strain will be
(a) $\sqrt{\frac{2E}{Y}}$ (b) $\sqrt{2EY}$
(c) EY (d) $\frac{E}{Y}$
15. The ratio of Young's modulus of the material of two wires is 2 : 3. If the same stress is applied on both, then the ratio of elastic energy per unit volume will be
(a) 3 : 2 (b) 2 : 3
(c) 3 : 4 (d) 4 : 3
16. The length of a rod is 20 cm and area of cross-section 2 cm^2 . The Young's modulus of the material of wire is $1.4 \times 10^{11}\text{ N/m}^2$. If the rod is compressed by 5 kg-wt along its length, then increase in the energy of the rod in joules will be
(a) 8.57×10^{-6} (b) 22.5×10^{-4}
(c) 9.8×10^{-5} (d) 45.0×10^{-5}
17. If a spring extends by x on loading, then the energy stored by the spring is (if T is tension in the spring and k is spring constant)
(a) $\frac{T^2}{2x}$ (b) $\frac{T^2}{2k}$
(c) $\frac{2x}{T^2}$ (d) $\frac{2T^2}{k}$
18. On stretching a wire, the elastic energy stored per unit volume is
(a) $Fl/2AL$ (b) $FA/2L$
(c) $FL/2A$ (d) $FL/2$
19. When a force is applied on a wire of uniform cross-sectional area $3 \times 10^{-6}\text{ m}^2$ and length 4 m, the increase in length is 1 mm. Energy stored in it will be ($Y = 2 \times 10^{11}\text{ N/m}^2$) [MP PMT 1991]
(a) 6250 J (b) 0.177 J
(c) 0.075 J (d) 0.150 J
20. K is the force constant of a spring. The work done in increasing its extension from l_1 to l_2 will be [MP PET 1995; MP PMT 1996]
(a) $K(l_2 - l_1)$ (b) $\frac{K}{2}(l_2 + l_1)$
(c) $K(l_2^2 - l_1^2)$ (d) $\frac{K}{2}(l_2^2 - l_1^2)$

21. When a 4 kg mass is hung vertically on a light spring that obeys Hooke's law, the spring stretches by 2 cms . The work required to be done by an external agent in stretching this spring by 5 cms will be ($g = 9.8 \text{ metres / sec}^2$)

[MP PMT 1995]

- (a) 4.900 joule (b) 2.450 joule
(c) 0.495 joule (d) 0.245 joule

22. A wire of length L and cross-sectional area A is made of a material of Young's modulus Y . It is stretched by an amount x . The work done is

[MP PET 1996; BVP 2003; UPSEAT 2001]

- (a) $\frac{YxA}{2L}$ (b) $\frac{Yx^2 A}{L}$
(c) $\frac{Yx^2 A}{2L}$ (d) $\frac{2Yx^2 A}{L}$

23. The elastic energy stored in a wire of Young's modulus Y is

[MP PMT 1999]

- (a) $Y \times \frac{\text{Strain}^2}{\text{Volume}}$
(b) $\text{Stress} \times \text{Strain} \times \text{Volume}$
(c) $\frac{\text{Stress}^2 \times \text{Volume}}{2Y}$
(d) $\frac{1}{2} Y \times \text{Stress} \times \text{Strain} \times \text{Volume}$

24. A wire of length 50 cm and cross sectional area of 1 sq. mm is extended by 1 mm . The required work will be ($Y = 2 \times 10^{10} \text{ Nm}^{-2}$)

[RPET 1999]

- (a) $6 \times 10^{-2} \text{ J}$ (b) $4 \times 10^{-2} \text{ J}$
(c) $2 \times 10^{-2} \text{ J}$ (d) $1 \times 10^{-2} \text{ J}$

25. The work per unit volume to stretch the length by 1% of a wire with cross sectional area of 1 mm^2 will be. [$Y = 9 \times 10^{11} \text{ N / m}^2$]

- (a) $9 \times 10^{11} \text{ J}$ (b) $4.5 \times 10^7 \text{ J}$
(c) $9 \times 10^7 \text{ J}$ (d) $4.5 \times 10^{11} \text{ J}$

26. When load of 5 kg is hung on a wire then extension of 3 m takes place, then work done will be [RPMT 2000]

- (a) 75 joule (b) 60 joule
(c) 50 joule (d) 100 joule

27. A stretched rubber has

[AIIMS 2000]

- (a) Increased kinetic energy
(b) Increased potential energy
(c) Decreased kinetic energy
(d) Decreased potential energy

28. Which of the following is true for elastic potential energy density

- (a) $\text{Energy density} = \frac{1}{2} \times \text{strain} \times \text{stress}$
(b) $\text{Energy density} = (\text{strain})^2 \times \text{volume}$
(c) $\text{Energy density} = (\text{strain}) \times \text{volume}$
(d) $\text{Energy density} = (\text{stress}) \times \text{volume}$

29. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm . Then the elastic energy stored in the wire is

- (a) 0.1 J (b) 0.2 J
(c) 10 J (d) 20

30. Wires A and B are made from the same material. A has twice the diameter and three times the length of B . If the elastic limits are not reached, when each is stretched by the same tension, the ratio of energy stored in A to that in B is

[Kerala PMT 2004]

- (a) $2 : 3$ (b) $3 : 4$
(c) $3 : 2$ (d) $6 : 1$

Critical Thinking

Objective Questions

1. An Indian rubber cord L metre long and area of cross-section $A \text{ metre}^2$ is suspended vertically. Density of rubber is $D \text{ kg / metre}^3$ and Young's modulus of rubber is $E \text{ newton / metre}^2$. If the wire extends by l metre under its own weight, then extension l is

- (a) $L^2 Dg / E$ (b) $L^2 Dg / 2E$
(c) $L^2 Dg / 4E$ (d) L

2. To break a wire, a force of 10^6 N / m^2 is required. If the density of the material is $3 \times 10^3 \text{ kg / m}^3$, then the length of the wire which will break by its own weight will be

[Roorkee 1979; DPMT 2004]

- (a) 34 m (b) 30 m
(c) 300 m (d) 3 m

3. Two rods of different materials having coefficients of linear expansion α_1, α_2 and Young's moduli Y_1 and Y_2 respectively are fixed between two rigid massive walls. The rods are heated such that they undergo the same increase in temperature. There is no bending of rods. If $\alpha_1 : \alpha_2 = 2 : 3$, the thermal stresses developed in the two rods are equally provided $Y_1 : Y_2$ is equal to

- (a) $2 : 3$ (b) $1 : 1$
(c) $3 : 2$ (d) $4 : 9$

4. The extension in a string obeying Hooke's law is x . The speed of sound in the stretched string is v . If the extension in the string is increased to $1.5x$, the speed of sound will be

[IIT 1996]

- (a) $1.22 v$ (b) $0.61 v$
(c) $1.50 v$ (d) $0.75 v$

5. One end of a uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height $3L/4$ from its lower end is

- (a) $\frac{W_1}{S}$ (b) $\frac{W_1 + (W/4)}{S}$
(c) $\frac{W_1 + (3W/4)}{S}$ (d) $\frac{W_1 + W}{S}$

6. There are two wires of same material and same length while the diameter of second wire is 2 times the diameter of first wire, then ratio of extension produced in the wires by applying same load will be

[DCE 2000; Roorkee 2000; DCE 2003]

- (a) 1 : 1 (b) 2 : 1
(c) 1 : 2 (d) 4 : 1
7. A particle of mass m is under the influence of a force F which varies with the displacement x according to the relation $F = -kx + F_0$ in which k and F_0 are constants. The particle when disturbed will oscillate

[UPSEAT 2001]

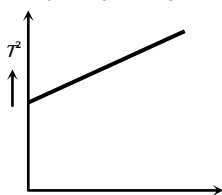
- (a) about $x = 0$, with $\omega \neq \sqrt{k/m}$
(b) about $x = 0$, with $\omega = \sqrt{k/m}$
(c) about $x = F_0/k$ with $\omega = \sqrt{k/m}$
(d) about $x = F_0/k$ with $\omega \neq \sqrt{k/m}$
8. An elastic material of Young's modulus Y is subjected to a stress S . The elastic energy stored per unit volume of the material is

MP PMT 1990, 96; IIT 1992; AIIMS 1997]

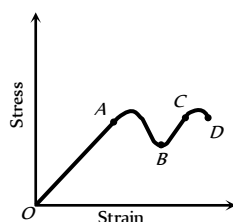
- (a) $\frac{2Y}{S^2}$ (b) $\frac{S^2}{2Y}$
(c) $\frac{S}{2Y}$ (d) $\frac{S^2}{Y}$

Graphical Questions

1. The graph shown was obtained from experimental measurements of the period of oscillations T for different masses M placed in the scale pan on the lower end of the spring balance. The most likely reason for the line not passing through the origin is that the



- (a) Spring did not obey Hooke's Law
(b) Amplitude of the oscillations was too large
(c) Clock used needed regulating
(d) Mass of the pan was neglected
2. A graph is shown between stress and strain for a metal. The part in which Hooke's law holds good is



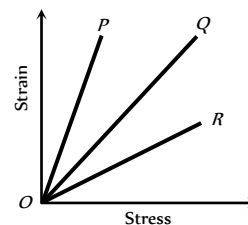
- (a) OA
(b) AB
(c) BC
(d) CD
3. In the above graph, point B indicates
- (a) Breaking point (b) Limiting point

- (c) Yield point (d) None of the above

4. In the above graph, point D indicates

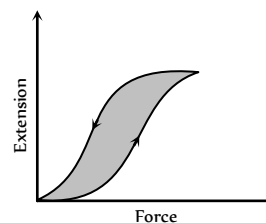
- (a) Limiting point (b) Yield point
(c) Breaking point (d) None of the above

5. The strain-stress curves of three wires of different materials are shown in the figure. P , Q and R are the elastic limits of the wires. The figure shows that



- (a) Elasticity of wire P is maximum
(b) Elasticity of wire Q is maximum
(c) Tensile strength of R is maximum
(d) None of the above is true

6. The diagram shows a force-extension graph for a rubber band. Consider the following statements

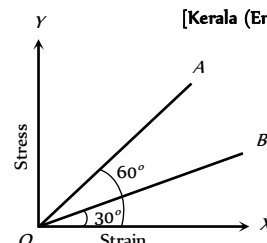


- I. It will be easier to compress this rubber than expand it
II. Rubber does not return to its original length after it is stretched
III. The rubber band will get heated if it is stretched and released

Which of these can be deduced from the graph

- (a) III only (b) II and III
(c) I and III (d) I only

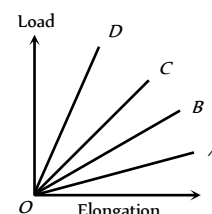
7. The stress versus strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then



[Kerala (Engg.) 2001]

- (a) $Y_B = 2Y_A$
(b) $Y_A = Y_B$
(c) $Y_B = 3Y_A$
(d) $Y_A = 3Y_B$

8. The load versus elongation graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line



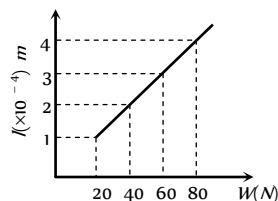
- (a) OD
(b) OC

[KCET 2001]

- (c) OB
(d) OA

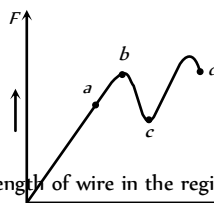
9. The adjacent graph shows the extension (Δl) of a wire of length 1 m suspended from the top of a roof at one end with a load W connected to the other end. If the cross sectional area of the wire is 10^{-6} m^2 , calculate the young's modulus of the material of the wire [IIT-JEE (Screening) 2003]

- (a) $2 \times 10^{11}\text{ N/m}^2$
(b) $2 \times 10^{-11}\text{ N/m}^2$
(c) $3 \times 10^{-12}\text{ N/m}^2$
(d) $2 \times 10^{-13}\text{ N/m}^2$



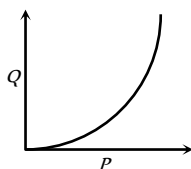
10. The graph is drawn between the applied force F and the strain (x) for a thin uniform wire. The wire behaves as a liquid in the part [CPMT 1988]

- (a) ab
(b) bc
(c) cd
(d) oa

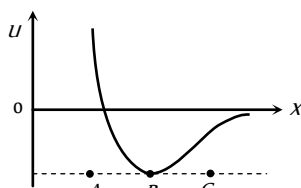


11. The graph shows the behaviour of a length of wire in the region for which the substance obeys Hook's law. P and Q represent [AMU 2001]

- (a) $P = \text{applied force, } Q = \text{extension}$
(b) $P = \text{extension, } Q = \text{applied force}$
(c) $P = \text{extension, } Q = \text{stored elastic energy}$
(d) $P = \text{stored elastic energy, } Q = \text{extension}$



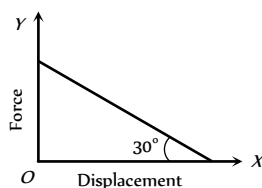
12. The potential energy U between two molecules as a function of the distance X between them has been shown in the figure. The two molecules are [CPMT 1986, 88, 91]



- (a) Attracted when x lies between A and B and are repelled when x lies between B and C
(b) Attracted when x lies between B and C and are repelled when x lies between A and B
(c) Attracted when they reach B
(d) Repelled when they reach B

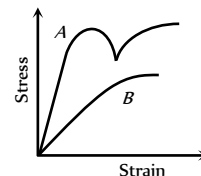
13. The value of force constant between the applied elastic force F and displacement will be

- (a) $\sqrt{3}$
(b) $\frac{1}{\sqrt{3}}$
(c) $\frac{1}{2}$
(d) $\frac{\sqrt{3}}{2}$



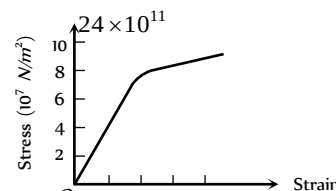
14. The diagram shows stress v/s strain curve for the materials A and B . From the curves we infer that [AIIMS 1987]

- (a) A is brittle but B is ductile
(b) A is ductile and B is brittle
(c) Both A and B are ductile
(d) Both A and B are brittle



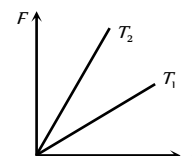
15. Which one of the following is the Young's modulus (in N/m) for the wire having the stress-strain curve shown in the figure

- (a)
(b) 8.0×10^{11}
(c) 10×10^{11}
(d) 2.0×10^{11}



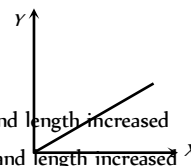
- The diagram shows the change x in the length of a thin uniform wire caused by the application of stress F at two different temperatures T_1 and T_2 . The variations shown suggest that

- (a) $T_1 > T_2$
(b) $T_1 < T_2$
(c) $T_1 = T_2$
(d) None of these



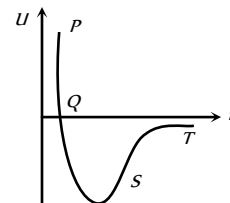
17. A student plots a graph from his reading on the determination of Young's modulus of a metal wire but forgets to label. The quantities on X and Y axes may be respectively.

- (a) Weight hung and length increased
(b) Stress applied and length increased
(c) Stress applied and strain developed
(d) Length increased and weight hung



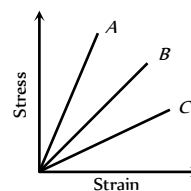
18. The points of maximum and minimum attraction in the curve between potential energy (U) and distance (r) of a diatomic molecules are respectively

- (a) S and R
(b) T and S
(c) R and S
(d) S and T



19. The stress-strain curves for brass, steel and rubber are shown in the figure. The lines A , B and C are for

- (a) Rubber, brass and steel respectively
(b) Brass, steel and rubber respectively
(c) Steel, brass and rubber respectively
(d) Steel, rubber and brass respectively



A R Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : The stretching of a coil is determined by its shear modulus.
 Reason : Shear modulus change only shape of a body keeping its dimensions unchanged.
2. Assertion : Spring balances show correct readings even after they had been used for a long time interval.
 Reason : On using for long time, spring balances losses its elastic strength.
3. Assertion : Steel is more elastic than rubber.
 Reason : Under given deforming force, steel is deformed less than rubber.
4. Assertion : Glassy solids have sharp melting point.
 Reason : The bonds between the atoms of glassy solids get broken at the same temperature.
5. Assertion : A hollow shaft is found to be stronger than a solid shaft made of same material.
 Reason : The torque required to produce a given twist in hollow cylinder is greater than that required to twist a solid cylinder of same size and material.
6. Assertion : Bulk modulus of elasticity (K) represents incompressibility of the material.
 Reason : Bulk modulus of elasticity is proportional to change in pressure.
7. Assertion : Strain is a unitless quantity.
 Reason : Strain is equivalent to force.
8. Assertion : The bridges declared unsafe after a long use.
 Reason : Elastic strength of bridges losses with time.
9. Assertion : Two identical solid balls, one of ivory and the other of wet-clay are dropped from the same height on the floor. Both the balls will rise to same height after bouncing.
 Reason : Ivory and wet-clay have same elasticity.
10. Assertion : Young's modulus for a perfectly plastic body is zero.
 Reason : For a perfectly plastic body, restoring force is zero.
11. Assertion : Identical springs of steel and copper are equally stretched. More work will be done on the steel spring.
 Reason : Steel is more elastic than copper.
12. Assertion : Stress is the internal force per unit area of a body.
 Reason : Rubber is less elastic than steel.

Answers

Young's Modulus and Breaking Stress

1	c	2	b	3	d	4	c	5	b
6	c	7	c	8	c	9	b	10	c
11	d	12	a	13	b	14	d	15	a
16	c	17	a	18	a	19	d	20	c
21	b	22	b	23	b	24	d	25	b
26	d	27	c	28	b	29	b	30	b
31	b	32	c	33	d	34	b	35	a
36	b	37	a	38	b	39	d	40	a
41	c	42	d	43	c	44	a	45	b
46	d	47	a	48	c	49	d	50	c
51	c	52	a	53	c	54	a	55	b
56	a	57	a	58	d	59	a	60	a
61	c	62	b	63	d	64	b	65	a
66	c	67	d	68	b	69	a	70	d
71	b	72	d	73	c	74	c	75	a
76	a	77	d	78	d	79	a	80	a
81	d	82	d	83	a	84	d	85	a
86	c	87	c	88	b	89	b	90	c
91	c	92	c	93	b	94	c	95	c
96	b	97	d	98	b	99	a	100	b
101	b	102	a	103	b	104	d	105	b
106	a	107	d	108	d	109	c	110	d
111	c	112	c	113	c	114	d	115	d
116	b	117	d	118	a	119	c	120	a
121	a	122	a	123	b	124	b	125	b

Bulk Modulus

1	c	2	c	3	b	4	d	5	b
6	d	7	a	8	d	9	c	10	c
11	b	12	c	13	d	14	a	15	c
16	b	17	c	18	c	19	a	20	d
21	b								

Rigidity Modulus

1	b	2	d	3	c	4	d	5	c
6	a	7	a	8	b	9	c	10	d
11	d	12	b	13	d	14	b	15	c
16	b	17	d	18	c				

Work Done in Stretching a Wire

1	d	2	b	3	c	4	a	5	b
6	c	7	d	8	c	9	b	10	b
11	c	12	a	13	d	14	a	15	a
16	a	17	b	18	a	19	c	20	d
21	b	22	c	23	c	24	c	25	b
26	a	27	b	28	a	29	a	30	b

Critical Thinking Questions

1	b	2	a	3	c	4	a	5	c
6	d	7	c	8	b				

Graphical Questions

1	d	2	a	3	c	4	c	5	d
6	a	7	d	8	a	9	a	10	b
11	c	12	b	13	b	14	b	15	d
16	a	17	c	18	d	19	c		

Assertion and Reason

1	a	2	e	3	a	4	d	5	a
6	a	7	c	8	a	9	d	10	a
11	a	12	b						

AS Answers and Solutions

Young's Modulus and Breaking Stress

1. (c) $l = \frac{FL}{YA} \Rightarrow l \propto \frac{1}{A}$

2. (b) $\text{Stress} \propto \text{Strain} \Rightarrow \text{Stress} \propto \frac{l}{L}$

3. (d) $Y = \frac{F}{A} \frac{L}{\Delta L} \Rightarrow l \propto \frac{L}{A} \propto \frac{L}{\pi d^2}$

$\therefore l \propto \frac{L}{d^2}$ [As F and Y are constant]

The ratio of $\frac{L}{d^2}$ is maximum for case (d)

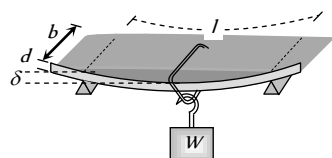
4. (c) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{L}{d^2} \Rightarrow \frac{l_1}{l_2} = \frac{L_1}{L_2} \times \left(\frac{d_2}{d_1}\right)^2 = \frac{1}{2} \times \left(\frac{1}{2}\right)^2 = \frac{1}{8}$

5. (b) Young's modulus of wire does not vary with dimension of wire. It is the property of given material.

6. (c) Depression in beam

$$\delta = \frac{WL^3}{4Ybd^3}$$

$$\therefore \delta \propto \frac{1}{Y}$$



7. (c) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{1}{r^2}$ (F, L and Y are constant)

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2}\right)^2 = (2)^2 = 4 \Rightarrow l_2 = 4l_1 = 4\text{ cm}$$

8. (c)

9. (b) $l \propto \frac{1}{r^2}$. If radius of the wire is doubled then increment in

length will become $\frac{1}{4}$ times i.e. $\frac{12}{4} = 3\text{ mm}$

10. (c) $l = \frac{mgL}{AY} = \frac{1 \times 10 \times 1.1}{1.1 \times 10^{11} \times 10^{-6}} \text{ m} = 0.1 \text{ mm}$

11. (d) $F = \frac{YAL}{L} = \frac{2.2 \times 10^{11} \times 2 \times 10^{-6} \times 5 \times 10^{-4}}{2} = 1.1 \times 10^2 \text{ N}$

12. (a) Interatomic force constant $K = Y \times r_0$
 $= 2 \times 10^{11} \times 3 \times 10^{-10} = 60 \text{ N/m}$

13. (b) To double the length of wire, Stress = Young's modulus

$$\therefore \frac{F}{A} = 2 \times 10^{12} \frac{\text{dyne}}{\text{cm}^2}.$$

If $A = 1$ then $F = 2 \times 10^{12} \text{ dyne}$

14. (d)

15. (a) Because due to increase in temperature intermolecular forces decreases.

16. (c) Breaking Force $\propto$ Area of cross section of wire (πr)
 If radius of wire is double then breaking force will become four times.

17. (a) $Y = 3K(1 - 2\sigma)$ and $Y = 2\eta(1 + \sigma)$

$$\text{Eliminating } \sigma \text{ we get } Y = \frac{9\eta K}{\eta + 3K}$$

18. (a) $F = \frac{YAL}{L} = \frac{9 \times 10^{10} \times \pi \times 4 \times 10^{-6} \times 0.1}{100} = 360 \pi \text{ N}$

19. (d) Energy stored per unit volume $= \frac{1}{2} \times \text{Stress} \times \text{Strain}$

$$= \frac{1}{2} \times \text{Young's modulus} \times (\text{Strain})^2 = \frac{1}{2} \times Y \times x^2$$

20. (c) $l \propto L$ i.e. if length is reduced to half then increase in length will be $\frac{l}{2}$.

21. (b) $l = \frac{L^2 dg}{2Y} = \frac{(8 \times 10^{-2})^2 \times 1.5 \times 9.8}{2 \times 5 \times 10^8} = 9.6 \times 10^{-11} \text{ m}$

22. (b) Stress $= \frac{\text{force}}{\text{Area}} \therefore \text{Stress} \propto \frac{1}{\pi^2}$

$$\frac{S_B}{S_A} = \left(\frac{r_A}{r_B}\right)^2 = (2)^2 \Rightarrow S_B = 4S_A$$

23. (b) Breaking force $\propto$ Area of cross section of wire

i.e. load hold by the wire does not depend upon the length of the wire.

24. (d) If length of wire doubled then strain = 1

$$Y = \text{stress} \Rightarrow F = Y \times A = 10^{12} \times 0.5 = 0.5 \times 10^{12} \text{ dyne}$$

25. (b) Due to elastic fatigue its elastic property decreases.

26. (d) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{1}{r^2}$ (F, L and Y are same)

$$\frac{l_A}{l_B} = \left(\frac{r_B}{r_A}\right)^2 = \left(\frac{r_B}{2r_B}\right)^2 = \frac{1}{4} \Rightarrow l_A = 4l_B \text{ or } l_B = \frac{l_A}{4}$$

27. (c)

28. (b) $l = \frac{FL}{AY} \Rightarrow \frac{l_s}{l_{cu}} = \frac{Y_{cu}}{Y_s}$ (F, L and Y are constant)

$$\therefore \frac{l_s}{l_{cu}} = \frac{1.2 \times 10^{11}}{2 \times 10^{11}} = \frac{3}{5}$$

29. (b) If length of the wire is doubled then strain = 1

$$\therefore Y = \text{Stress} = \frac{\text{Force}}{\text{Area}} = \frac{2 \times 10^5}{2} = 10^5 \frac{\text{dyne}}{\text{cm}^2}$$

30. (b) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{L}{r^2}$ (F and Y are same)

$$\therefore \frac{l_2}{l_1} = \frac{L_2}{L_1} \left(\frac{r_1}{r_2}\right)^2 = 2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{2} \Rightarrow l_2 = \frac{l_1}{2} = \frac{l}{2} = 0.5\text{ mm}.$$

31. (b) $F = \text{force developed} = YA \alpha(\Delta\theta)$
 $= 10^{11} \times 10^{-4} \times 10^{-5} \times 100 = 10^4 \text{ N}$

32. (c) $F = YA \alpha \Delta\theta \therefore F \propto A$

33. (d) $Y = \frac{F/A}{\text{strain}} \Rightarrow A = \frac{F}{Y \times \text{strain}} = \frac{10^4}{7 \times 10^9 \times 0.002}$
 $= \frac{1}{14} \times 10^{-2} = 7.1 \times 10^{-4} \text{ m}^2$

34. (b) strain $\propto$ stress $\propto \frac{F}{A}$

$$\text{Ratio of strain} = \frac{A_2}{A_1} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{4}{1}\right)^2 = \frac{16}{1}$$

35. (a) $F = 2000 \text{ N}, L = 6 \text{ m}, l = 0.5 \text{ cm}, A = 10^{-6} \text{ m}^2$

$$Y = \frac{FL}{Al} = \frac{2000 \times 6}{10^{-6} \times 0.5 \times 10^{-2}} = 2.35 \times 10^{12} \text{ N/m}^2$$

36. (b) $F = Kx \Rightarrow K = \frac{F}{x} = \frac{9 \times 9.8}{4.5 \times 10^{-3}} = 1.96 \times 10^4 \text{ N/m}$

37. (a) $l \propto \frac{FL}{r^2 Y} \Rightarrow l \propto \frac{1}{r^2}$ (F, L and Y are constant)

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2}\right)^2 = (n)^2 \Rightarrow l_2 = n^2 l_1$$

38. (b) Longitudinal strain $\frac{l}{L} = \frac{\text{stress}}{Y} = \frac{10^6}{10^{11}} = 10^{-5}$

$$\text{Percentage increase in length} = 10^{-5} \times 100 = 0.001\%$$

39. (d) It is the specific property of a particular metal at a given temperature which can be changed only by temperature variations.

40. (a) $Y = \frac{3.6 \times 10^{-9} \text{ N}/\text{\AA}}{3 \times 10^{-10} \text{ m}} = 1.2 \times 10^{11} \text{ N/m}^2$

41. (c)

42. (d) $K = \frac{YA}{L} = \frac{Y \times \pi r^2}{L} \Rightarrow K \propto \frac{Yr^2}{L}$

i.e. force constant of a wire depends on young's modulus (nature of the material), radius of the wire and length of the wire.

43. (c)

44. (a)

45. (b)

46. (d) Increase in tension of wire = $YA \alpha \Delta \theta$

$$= 8 \times 10^{-6} \times 2.2 \times 10^{11} \times 10^{-2} \times 10^{-4} \times 5 = 8.8 \text{ N}$$

47. (a) A small change in pressure produces a large change in volume.

48. (c) $W = \frac{1}{2} \frac{YAL^2}{L} \Rightarrow 0.4 = \frac{1}{2} \times \frac{Y \times 1^{-6} \times (0.2 \times 10^{-2})^2}{1}$

$$\therefore Y = 2 \times 10^{11} \text{ N/m}^2$$

49. (d)

50. (c) $Y = 3K(1 - 2\sigma)$

$$\sigma = \frac{3K - Y}{6K} = \frac{3 \times 11 \times 10^{10} - 7.25 \times 10^{10}}{6 \times 11 \times 10^{10}} \Rightarrow \sigma = 0.39$$

51. (c)

52. (a) If density of the material increases then more force (stress) is required for same deformation i.e. the value of young's modulus increases.

53. (c) $Y = 10^4 \text{ N/m}^2, A = 2 \times 10^{-4} \text{ m}^2, F = 2 \times 10^5 \text{ dyne} = 2 \text{ N}$

$$l = \frac{FL}{AY} = \frac{2 \times L}{2 \times 10^{-4} \times 10^4} = L$$

$$\therefore \text{Final length} = \text{initial length} + \text{increment} = 2L$$

54. (a)

55. (b) Y is defined for solid only and for powders, $Y = 0$

56. (a)

57. (a) $l = \frac{FL}{AY} = \frac{FL}{\pi^2 Y} \therefore l \propto \frac{FL}{r^2} \quad (Y = \text{constant})$

$$\therefore \frac{l_2}{l_1} = \frac{F_2}{F_1} \times \frac{L_2}{L_1} \left(\frac{r_1}{r_2} \right)^2 = 2 \times 2 \times \left(\frac{1}{2} \right)^2 = 1$$

$$\therefore l_2 = l_1 \text{ i.e. increment in its length will be } l$$

58. (d) Breaking stress = strain $\times$ Young's modulus

$$= 0.15 \times 2 \times 10^{11} = 3 \times 10^{10} \text{ Nm}^{-2}$$

59. (a) In accordance with Hooke's law.

60. (a) $F = A \times Y \times \text{strain} = 1 \times 10^{-4} \times 2 \times 10^{11} \times 0.1 = 2 \times 10^6 \text{ N}$

61. (c)

62. (b) Because strain is a dimensionless and unitless quantity.

63. (d) $\text{Stress} = \frac{\text{Force}}{\text{area}}$

In the present case, force applied and area of cross-section of wires are same, therefore stress has to be the same.

$$\text{Strain} = \frac{\text{Stress}}{Y}$$

Since the Young's modulus of steel wire is greater than the copper wire, therefore, strain in case of steel wire is less than that in case of copper wire.

64. (b) Initial length (circumference) of the ring = $2\pi r$

Final length (circumference) of the ring = $2\pi R$

Change in length = $2\pi R - 2\pi r$.

$$\text{strain} = \frac{\text{change in length}}{\text{original length}} = \frac{2\pi(R-r)}{2\pi r} = \frac{R-r}{r}$$

$$\text{Now Young's modulus } E = \frac{F/A}{l/L} = \frac{F/A}{(R-r)r}$$

$$\therefore F = AE \left(\frac{R-r}{r} \right)$$

65. (a) $l = \frac{FL}{\pi^2 r} \Rightarrow l \propto \frac{F}{r^2} \quad (Y \text{ and } L \text{ are constant})$

$$\frac{l_2}{l_1} = \frac{F_2}{F_1} \times \left(\frac{r_1}{r_2} \right)^2 = 2 \times (2)^2 = 8 \therefore l_2 = 8l_1 = 8 \times 1 = 8 \text{ mm}$$

66. (c) $l = \frac{FL}{\pi^2 Y} \Rightarrow l \propto \frac{L}{r^2} \quad (F \text{ and } Y \text{ are constant})$

$$\frac{l_1}{l_2} = \frac{L_1}{L_2} \left(\frac{r_2}{r_1} \right)^2 = \frac{1}{2} (\sqrt{2})^2 \therefore \frac{l_1}{l_2} = 1:1$$

67. (d) $l \propto \frac{1}{r^2} \quad (F, L \text{ and } Y \text{ are constant})$

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2} \right)^2 = \left(\frac{1}{2} \right)^2 \Rightarrow l_2 = \frac{l_1}{4} = \frac{2.4}{4} \Rightarrow l_2 = 0.6 \text{ cm}$$

68. (b) $F = Y \times A \times \frac{l}{L} \Rightarrow F \propto r^2 \quad (Y, l \text{ and } L \text{ are constant})$

If diameter is made four times then force required will be 16 times. i.e. $16 \times 10^5 \text{ N}$

69. (a) $F = Y \times A \times \frac{l}{L} \Rightarrow F \propto \frac{r^2}{L} \quad (Y \text{ and } l \text{ are constant})$

$$\therefore \frac{F_1}{F_2} = \left(\frac{r_1}{r_2} \right)^2 \left(\frac{L_2}{L_1} \right) = \left(\frac{2}{1} \right)^2 \left(\frac{1}{4} \right) = 1 \Rightarrow \frac{F_1}{F_2} = 1:1$$

70. (d) Increment in length $l = \frac{L^2 dg}{2Y} \therefore l \propto L^2 d$

71. (b) Adiabatic elasticity $E = \gamma P$

For argon $E_{Ar} = 1.6 P$ (i)

For hydrogen $E_{H_2} = 1.4 P'$ (ii)

As elasticity of hydrogen and argon are equal

$$\therefore 1.6P = 1.4P' \Rightarrow P' = \frac{8}{7} P$$

72. (d)

73. (c) $l = \frac{FL}{AY} = \frac{FL^2}{(AL)Y} = \frac{FL^2}{VY}$

If volume is fixed then $l \propto L^2$

74. (c) $F = YA \alpha \Delta t = 2 \times 10^{11} \times 3 \times 10^{-6} \times 10^{-5} \times (20-10) = 60 \text{ N}$

75. (a) Because dimension of invar does not varies with temperature.

76. (a) $l = \frac{FL}{\pi^2 Y} \therefore l \propto \frac{L}{r^2} \quad (Y \text{ and } F \text{ are constant})$

$$\frac{l_2}{l_1} = \frac{L_2}{L_1} \times \left(\frac{r_1}{r_2}\right)^2 = (2) \times \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

$$\Rightarrow l_2 = \frac{l_1}{2} = \frac{0.01m}{2} = 0.005m$$

77. (d) Poisson's ratio varies between -1 and 0.5

78. (d) $L_2 = l_2(1 + \alpha_2 \Delta\theta)$ and $L_1 = l_1(1 + \alpha_1 \Delta\theta)$

$$\Rightarrow (L_2 - L_1) = (l_2 - l_1) + \Delta\theta(l_2 \alpha_2 - l_1 \alpha_1)$$

$$\text{Now } (L_2 - L_1) = (l_2 - l_1) \text{ so, } l_2 \alpha_2 - l_1 \alpha_1 = 0$$

79. (a) Thermal stress = $Y \alpha \Delta\theta$

$$= 1.2 \times 10^{11} \times 1.1 \times 10^{-5} \times (20 - 10) = 1.32 \times 10^7 \text{ N/m}^2$$

80. (a) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{1}{r^2}$ (FL and Y are constant)

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2}\right)^2 = (2)^2 \Rightarrow l_2 = 4l_1 = 4 \times 3 = 12mm$$

81. (d) $l = \frac{L^2 dg}{2Y} = \frac{(8)^2 \times 1.5 \times 10^3 \times 10}{2 \times 5 \times 10^6} = 9.6 \times 10^{-2} m$

82. (d) $l \propto \frac{L}{r^2}$ (Y and F are constant)

Maximum extension takes place in that wire for which the ratio of $\frac{L}{r^2}$ will be maximum.

83. (a)

84. (d)

85. (a) $Y = \frac{MgL}{Al} = \frac{250 \times 9.8 \times 2}{50 \times 10^{-6} \times 0.5 \times 10^{-3}} = 19.6 \times 10^{10} \text{ N/m}^2$

86. (c)

87. (c) $l = \frac{FL}{AY} \Rightarrow l \propto \frac{L}{r^2}$ (F and Y are constant)

$$\frac{l_2}{l_1} = \frac{L_2}{L_1} \times \left(\frac{r_1}{r_2}\right)^2 = 2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{2} \therefore l_2 = \frac{l_1}{2}$$

i.e. the change in the length of other wire is $\frac{l}{2}$

88. (b)

89. (b)

90. (c) $l = \frac{MgL}{YA} = \frac{1 \times 10 \times 1}{2 \times 10^{11} \times 10^{-6}} = 0.05 \text{ mm}$

91. (c) $l = \frac{FL}{AY} \therefore l \propto \frac{F}{r^2}$

$$\frac{l_1}{l_2} = \frac{F_2}{F_1} \left(\frac{r_1}{r_2}\right)^2 = (4) \times \left(\frac{1}{2}\right)^2 = 1 \therefore l_2 = l_1 = 1mm$$

92. (c)

93. (b) $l = \frac{FL}{AY} \therefore l \propto \frac{1}{A}$ (FL and Y are constant)

$$\frac{A_2}{A_1} = \frac{l_1}{l_2} \Rightarrow A_2 = A_1 \left(\frac{0.1}{0.05}\right) = 2A_1 = 2 \times 4 = 8mm^2$$

94. (c) $l = \frac{FL}{\pi r^2 Y} \Rightarrow r^2 \propto \frac{1}{Y}$ (FL and l are constant)

$$\frac{r_2}{r_1} = \left(\frac{Y_1}{Y_2}\right)^{1/2} = \left(\frac{7 \times 10^{10}}{12 \times 10^{10}}\right)^{1/2}$$

$$\Rightarrow r_2 = 1.5 \times \left(\frac{7}{12}\right)^{1/2} = 1.145 \text{ mm} \therefore \text{dia} = 2.29 \text{ mm}$$

95. (c) $F = \frac{YAl}{L} = 0.9 \times 10^{11} \times \pi \times (0.3 \times 10^{-3})^2 \times \frac{0.2}{100} = 51 \text{ N}$

96. (b) Young's modulus = $\frac{\text{stress}}{\text{strain}}$

As the length of wire get doubled therefore strain = 1

$$\therefore Y = \text{strain} = 20 \times 10^9 \text{ N/m}$$

97. (d) $l = \frac{FL}{\pi r^2 Y} \therefore l \propto \frac{1}{r^2}$ (FL and Y are constant)

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2}\right)^2 = (2)^2 \Rightarrow l_2 = 4l_1 = 4 \times 2 = 8 \text{ mm.}$$

98. (b) Let L is the original length of the wire and K is force constant of wire.

Final length = initial length + elongation

$$L' = L + \frac{F}{K}$$

$$\text{For first condition } a = L + \frac{4}{K} \quad \dots(i)$$

$$\text{For second condition } b = L + \frac{5}{K} \quad \dots(ii)$$

By solving (i) and (ii) equation we get

$$L = 5a - 4b \text{ and } K = \frac{1}{b - a}$$

Now when the longitudinal tension is $9N$, length of the string =

$$L + \frac{9}{K} = 5a - 4b + 9(b - a) = 5b - 4a.$$

99. (a)

100. (b)

101. (b) $K = Yr_0 = 20 \times 10^{10} \times 3 \times 10^{-10} = 60 \text{ N/m}$
 $= 6 \times 10^{-9} \text{ N/\AA}$

102. (a) $l = \frac{FL}{AY} = \frac{4.8 \times 10^3 \times 4}{1.2 \times 10^{-4} \times 1.2 \times 10^{11}} = 1.33 \text{ mm}$

103. (b)

104. (d) $Y = \frac{\text{Stress}}{\text{Strain}} = \text{Constant}$

It depends only on nature of material.

105. (b) $2\pi\sqrt{\frac{m}{k}} = 0.6 \quad \dots(i) \text{ and } 2\pi\sqrt{\frac{m+m'}{k}} = 0.7 \quad \dots(ii)$

$$\text{Dividing (ii) by (i) we get } \left(\frac{7}{6}\right)^2 = \frac{m+m'}{m} = \frac{49}{36}$$

$$\frac{m+m'}{m} - 1 = \frac{49}{36} - 1 \Rightarrow \frac{m'}{m} = \frac{13}{36} \Rightarrow m' = \frac{13m}{36}$$

$$\text{Also } \frac{k}{m} = \frac{4\pi^2}{(0.6)^2}$$

$$\text{Desired extension} = \frac{m'g}{k} = \frac{13}{36} \times \frac{mg}{k}$$

$$= \frac{13}{36} \times 10 \times \frac{0.36}{4\pi^2} \approx 3.5 \text{ cm}$$

106. (a) $Y = \frac{F/A}{\text{Strain}} \Rightarrow \text{strain} = \frac{F}{AY}$

107. (d) $Y = \frac{k}{r_0} = \frac{7}{3 \times 10^{-10}} = 2.33 \times 10^{10} \text{ N/m}^2$

108. (d) $F = Y \times A \times \frac{l}{L} \Rightarrow F \propto \frac{r^2}{L}$ (Y and l are constant)

$$\frac{F_A}{F_B} = \left(\frac{r_A}{r_B}\right)^2 \times \left(\frac{L_B}{L_A}\right) = \left(\frac{2}{1}\right)^2 \times \left(\frac{2}{1}\right) = \frac{8}{1}$$

109. (c)

110. (d) When the length of wire is doubled then $l = L$ and strain = 1

$$\therefore Y = \text{strain} = \frac{F}{A}$$

$$\therefore \text{Force} = Y \times A = 2 \times 10^{11} \times 0.1 \times 10^{-4} = 2 \times 10^6 \text{ N}$$

111. (c) Potential energy stored in the rubber cord catapult will be converted into kinetic energy of mass.

$$\frac{1}{2}mv^2 = \frac{1}{2} \frac{YAL^2}{L} \Rightarrow v = \sqrt{\frac{YAL^2}{mL}}$$

$$= \sqrt{\frac{5 \times 10^8 \times 25 \times 10^{-6} \times (5 \times 10^{-2})^2}{5 \times 10^{-3} \times 10 \times 10^{-2}}} = 250 \text{ m/s}$$

112. (c)

113. (c)

114. (d)

115. (d) Breaking force $\propto r$

If diameter becomes double then breaking force will become four times i.e. $1000 \times 4 = 4000 \text{ N}$

116. (b)

117. (d) $l = \frac{FL}{AY} = \frac{FL^2}{(AL)Y} = \frac{FL^2}{VY}$

$$\therefore l \propto L^2 \text{ If volume of the wire remains constant}$$

$$\frac{l_2}{l_1} = \left(\frac{L_2}{L_1}\right)^2 = \left(\frac{8}{2}\right)^2 = 16$$

$$\therefore l_2 = 16 \times l_1 = 16 \times 2 = 32 \text{ mm} = 3.2 \text{ cm}$$

118. (a) $l = \frac{FL}{AY} \therefore l \propto \frac{1}{A}$ (FL and Y are constant)

$$\frac{l_2}{l_1} = \frac{A_1}{A_2} = \frac{4}{8} = \frac{1}{2} \Rightarrow l_2 = \frac{l_1}{2} = \frac{0.1}{2} = 0.05 \text{ mm}$$

119. (c) $L = \frac{p}{dg} = \frac{10^6}{3 \times 10^3 \times 10} = \frac{100}{3} = 33.3 \text{ m}$

120. (a) $l = \frac{L^2 dg}{2Y} = \frac{(10)^2 \times 1500 \times 10}{2 \times 5 \times 10^8} = 15 \times 10^{-4} \text{ m}$

121. (a) $Y = 3K(1 - 2\sigma)$, $Y = 2\eta(1 + \sigma)$

For $Y = 0$, we get $1 - 2\sigma = 0$, also $1 + \sigma = 0$

$$\Rightarrow \sigma \text{ lies between } \frac{1}{2} \text{ and } -1.$$

122. (a) Value of Poisson's ratio lie in range of -1 to $\frac{1}{2}$

123. (b) We know that $\frac{dV}{V} = (1 + 2\sigma) \frac{dL}{L}$

If $\sigma = -\frac{1}{2}$ then $\frac{dV}{V} = 0$

i.e. there is no change in volume.

124. (b) $\frac{dV}{V} = (1 + 2\sigma) \frac{dL}{L}$

$$\frac{dV}{V} = 2 \times 2 \times 10^{-3} = 4 \times 10^{-3} \left[\because \sigma = 0.5 = \frac{1}{2} \right]$$

$$\therefore \text{Percentage change in volume} = 4 \times 10^{-1} = 0.4\%$$

125. (b) $l = \frac{FL}{\pi^2 Y} \therefore l \propto \frac{L}{r^2}$

Ratio of $\frac{L}{r^2}$ is maximum for wire in option (b).

Bulk Modulus

1. (c) Isothermal elasticity $K_i = P$

2. (c) Adiabatic elasticity $K_a = \gamma P$

3. (b) Ratio of adiabatic and isothermal elasticities

$$\frac{E\phi}{E\theta} = \frac{\gamma P}{P} = \gamma = \frac{C_p}{C_v}$$

4. (d)

5. (b) For triatomic gas $\gamma = \frac{4}{3}$

6. (d) From the ideal gas equation $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\frac{E_2}{E_1} = \frac{P_2}{P_1} = \frac{V_1}{V_2} \times \frac{T_2}{T_1} = \left(\frac{1}{4}\right) \times \left(\frac{400}{300}\right) = \frac{1}{3} \Rightarrow E_2 = \frac{E_1}{3}$$

i.e. elasticity will become $\frac{1}{3}$ times.

7. (a) $C = \frac{1}{K} = \frac{\Delta V/V}{\Delta P} \Rightarrow \Delta V = C \times \Delta P \times V$

$$= 4 \times 10^{-5} \times 100 \times 100 = 0.4 \text{ cc}$$

8. (d) $K = \frac{\Delta P}{\Delta V/V} = \frac{h\rho g}{\Delta V/V} = \frac{200 \times 10^3 \times 10}{0.1/100} = 2 \times 10^9$

9. (c) $\frac{1}{K} = \text{compressibility} = \left(\frac{-\Delta V/V}{\Delta P}\right)$

10. (c) $K = \frac{100}{0.01/100} = 10^6 \text{ atm} = 10^{11} \text{ N/m}^2 = 10^{12} \text{ dyne/cm}^2$

11. (b)

12. (c)

13. (d) If side of the cube is L then $V = L^3 \Rightarrow \frac{dV}{V} = 3 \frac{dL}{L}$

$$\therefore \% \text{ change in volume} = 3 \times (\% \text{ change in length})$$

$$= 3 \times 1\% = 3\% \therefore \text{Bulk strain } \frac{\Delta V}{V} = 0.03$$

14. (a) $B = \frac{\Delta p}{\Delta V/V} = \frac{h\rho g}{0.1/100} = \frac{200 \times 10^3 \times 9.8}{1/1000}$

$$= 19.6 \times 10^8 \text{ N/m}^2$$

15. (c) Isothermal elasticity $K_i = P = 1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$

16. (b)

17. (c) Isothermal bulk modulus = Pressure of gas

18. (c)

19. (a) If coefficient of volume expansion is α and rise in temperature is $\Delta\theta$ then $\Delta V = V\alpha\Delta\theta \Rightarrow \frac{\Delta V}{V} = \alpha\Delta\theta$

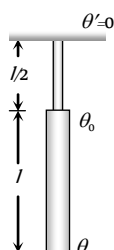
$$\text{Volume elasticity } \beta = \frac{P}{\Delta V/V} = \frac{P}{\alpha\Delta\theta} \Rightarrow \Delta\theta = \frac{P}{\alpha\beta}$$

20. (d) $K = \frac{\Delta p}{\Delta V/V} = \frac{(1.165 - 1.01) \times 10^5}{10/100} = \frac{0.155 \times 10^5}{1/10}$
 $= 1.55 \times 10^5 \text{ pa}$

21. (b) $B = \frac{\Delta p}{\Delta V/V} \Rightarrow \frac{1}{B} \propto \frac{\Delta V}{V} \quad [\Delta p = \text{constant}]$

Rigidity Modulus

1. (b)
 2. (d) Modulus of rigidity is the property of material.
 3. (c) $Y = 2\eta(1 + \sigma)$
 4. (d) $Y = 2\eta(1 + \sigma) \Rightarrow \sigma = \frac{0.5Y - \eta}{\eta}$
 5. (c) Twisting couple $C = \frac{\pi\eta r^4 \theta}{2l}$
 If material and length of the wires A and B are equal and equal twisting couple are applied then
 $\theta \propto \frac{1}{r^4} \therefore \frac{\theta_1}{\theta_2} = \left(\frac{r_2}{r_1}\right)^4$
 6. (a) A small part of the spring bear tangential stress, causing straining strain.
 7. (a) $Y = 2\eta(1 + \sigma)$
 For no transverse strain ($\sigma = 0$)
 $Y = 2\eta \Rightarrow \eta = \frac{Y}{2} = 3 \times 10^{12} \text{ N/m}^2$
 8. (b) $Y = 2\eta(1 + \sigma) \Rightarrow 3\eta = 2\eta(1 + \sigma) \Rightarrow \sigma = \frac{3}{2} - 1 = \frac{1}{2}$
 Now substituting the value of σ in the following expression.
 $Y = 3K(1 - 2\sigma) \Rightarrow K = \frac{Y}{3(1 - 2\sigma)} = \infty$
 9. (c)
 10. (d) $Y = 2\eta(1 + \sigma)$
 $2.4\eta = 2\eta(1 + \sigma) \Rightarrow 1.2 = 1 + \sigma \Rightarrow \sigma = 0.2$
 11. (d) Shearing strain $\phi = \frac{x}{L} = \frac{0.02 \text{ cm}}{10 \text{ cm}} \therefore \phi = 0.002$
 12. (b)
 13. (d) There will be both shear stress and normal stress.
 14. (b) Angle of shear $\phi = \frac{r\theta}{L} = \frac{4 \times 10^{-1}}{100} \times 30^\circ = 0.12^\circ$
 15. (c) For twisting, Angle of shear $\phi \propto \frac{1}{L}$
 i.e. if L is more then ϕ will be small.
 16. (b) $r\theta = L\phi \Rightarrow 10^{-2} \times 0.8 = 2 \times \phi \Rightarrow \phi = 0.004$
 17. (d) $\tau = C.\theta = \frac{\pi\eta r^4 \theta}{2L} = \text{Constant}$



$$\Rightarrow \frac{\pi\eta r^4(\theta - \theta_0)}{2l} = \frac{\pi\eta(r/2)^4(\theta_0 - \theta')}{2(l/2)}$$

$$\Rightarrow \frac{(\theta - \theta_0)}{2} = \frac{\theta_0}{16} \Rightarrow \theta_0 = \frac{8}{9}\theta$$

18. (c)

Work Done in Stretching a Wire

1. (d) $U = \frac{1}{2} \left(\frac{YA}{L} \right) l^2 \therefore U \propto l^2$
 $\frac{U_2}{U_1} = \left(\frac{l_2}{l_1} \right)^2 = \left(\frac{10}{2} \right)^2 = 25 \Rightarrow U_2 = 25U_1$
 i.e. potential energy of the spring will be 25 J
 2. (b)
 3. (c) Work done $= \frac{1}{2} Fl = \frac{Mgl}{2}$
 4. (a) $W = \frac{1}{2} Fl \therefore W \propto l$ (F is constant)
 $\therefore \frac{W_1}{W_2} = \frac{l_1}{l_2} = \frac{l}{2l} = \frac{1}{2}$
 5. (b) $W = \frac{1}{2} \times F \times l = \frac{1}{2} mgl$
 $= \frac{1}{2} \times 10 \times 10 \times 1 \times 10^{-1} = 0.05 \text{ J}$
 6. (c) $K = \frac{F}{l}$ and $W = \frac{1}{2} Fl = \frac{1}{2} Kl \times l = \frac{1}{2} Kl^2$
 7. (d) Due to tension, intermolecular distance between atoms is increased and therefore potential energy of the wire is increased and with the removal of force interatomic distance is reduced and so is the potential energy. This change in potential energy appears as heat in the wire and thereby increases the temperature.
 8. (c) Due to increase in intermolecular distance.
 9. (b)
 10. (b) $U = \frac{1}{2} \times \frac{(\text{stress})^2}{Y} \times \text{volume} = \frac{1}{2} \times \frac{F^2 \times A \times L}{A^2 \times Y}$
 $= \frac{1}{2} \times \frac{F^2 L}{AY} = \frac{1}{2} \times \frac{(50)^2 \times 0.2}{1 \times 10^{-4} \times 1 \times 10^{11}} = 2.5 \times 10^{-5} \text{ J}$
 11. (c) Work done in stretching a wire
 $W = \frac{1}{2} Fl = \frac{1}{2} \times 10 \times 0.5 \times 10^{-3} = 2.5 \times 10^{-3} \text{ J}$
 Work done to displace it through 1.5 mm
 $W = F \times l = 5 \times 10^{-3} \text{ J}$
 The ratio of above two work = 1 : 2
 12. (a) Increase in energy $= \frac{1}{2} \times 20 \times 1 \times 10^{-3} = 0.01 \text{ J}$
 13. (d) Ratio of work done $= \frac{1/2 Fl}{Fl} = \frac{1}{2}$
 14. (a) Energy per unit volume $= \frac{1}{2} \times Y \times (\text{strain})^2$
 $\therefore \text{strain} = \sqrt{\frac{2E}{Y}}$
 15. (a) Energy per unit volume $= \frac{(\text{stress})^2}{2Y}$

$$\frac{E_1}{E_2} = \frac{Y_2}{Y_1} \text{ (Stress is constant)} \therefore \frac{E_1}{E_2} = \frac{3}{2}$$

$$\begin{aligned} 16. \quad (a) \quad \text{Energy} &= \frac{1}{2} Fl = \frac{1}{2} \times F \times \left(\frac{FL}{AY} \right) = \frac{1}{2} \times \frac{F^2 L}{AY} \\ &= \frac{1}{2} \times \frac{(50)^2 \times 20 \times 10^{-2}}{2 \times 10^{-4} \times 1.4 \times 10^{11}} = 8.57 \times 10^{-6} \text{ J} \end{aligned}$$

$$17. \quad (b) \quad U = \frac{F^2}{2K} = \frac{T^2}{2K}$$

$$18. \quad (a) \quad \text{Energy stored per unit volume} = \frac{1}{2} \left(\frac{F}{A} \right) \left(\frac{l}{L} \right) = \frac{Fl}{2AL}$$

$$19. \quad (c) \quad U = \frac{1}{2} \times \frac{YAl^2}{L} = \frac{1}{2} \times \frac{2 \times 10^{11} \times 3 \times 10^{-6} \times (1 \times 10^{-3})^2}{4} = 0.075 \text{ J}$$

$$20. \quad (d) \quad \text{At extension } l_1, \text{ the stored energy} = \frac{1}{2} Kl_1^2$$

$$\text{At extension } l_2, \text{ the stored energy} = \frac{1}{2} Kl_2^2$$

Work done in increasing its extension from l_1 to l_2

$$= \frac{1}{2} K(l_2^2 - l_1^2)$$

$$21. \quad (b) \quad K = \frac{F}{x} = \frac{40}{2 \times 10^{-2}} = 0.2 \text{ N/m}$$

$$\text{Work done} = \frac{1}{2} Kx^2 = \frac{1}{2} \times (0.2) \times (0.05)^2 = 2.5 \text{ J}$$

$$22. \quad (c)$$

$$23. \quad (c)$$

$$24. \quad (c) \quad W = \frac{YAl^2}{2L} = \frac{2 \times 10^{10} \times 10^{-6} \times (10^{-3})^2}{2 \times 50 \times 10^{-2}} = 2 \times 10^{-2} \text{ J}$$

$$25. \quad (b) \quad U = \frac{1}{2} \times Y \times (\text{Strain})^2 = \frac{1}{2} \times 9 \times 10^{11} \times \left(\frac{1}{100} \right)^2 = 4.5 \times 10^7 \text{ J}$$

$$26. \quad (a) \quad W = \frac{1}{2} Fl = \frac{1}{2} \times Mg \times l = \frac{1}{2} \times 5 \times 10 \times 3 = 75 \text{ J}$$

$$27. \quad (b)$$

$$28. \quad (a)$$

$$29. \quad (a) \quad U = \frac{1}{2} \times F \times l = \frac{1}{2} \times 200 \times 10^{-3} = 0.1 \text{ J}$$

$$30. \quad (b) \quad U = \frac{1}{2} Fl = \frac{F^2 L}{2AY}. \quad U \propto \frac{L}{r^2} \quad (F \text{ and } Y \text{ are constant})$$

$$\therefore \frac{U_A}{U_B} = \left(\frac{L_A}{L_B} \right) \times \left(\frac{r_A}{r_B} \right)^2 = (3) \times \left(\frac{1}{2} \right)^2 = \frac{3}{4}$$

Critical Thinking Questions

$$1. \quad (b)$$

$$2. \quad (a) \quad L = \frac{P}{dg} = \frac{6}{3 \times 10^3 \times 10} = \frac{100}{3} = 34 \text{ m}$$

$$3. \quad (c) \quad \text{Thermal stress} = Y \alpha \Delta \theta.$$

If thermal stress and rise in temperature are equal then

$$Y \propto \frac{1}{\alpha} \Rightarrow \frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$$

$$4. \quad (a) \quad \text{Speed of sound in a stretched string } v = \sqrt{\frac{T}{\mu}} \quad \dots(i)$$

Where T is the tension in the string and μ is mass per unit length.

According to Hooke's law, $F \propto x \therefore T \propto x \quad \dots(ii)$

From (i) and (ii) $v \propto \sqrt{x} \therefore v' = \sqrt{1.5} v = 1.22 v$

$$5. \quad (c) \quad \text{Total force at height } 3L/4 \text{ from its lower end} \\ = \text{Weight suspended} + \text{Weight of } 3/4 \text{ of the chain} \\ = W_1 + (3W/4)$$

$$\text{Hence stress} = \frac{W_1 + (3W/4)}{S}$$

$$6. \quad (d) \quad l = \frac{FL}{AY} \therefore l \propto \frac{1}{r^2} \quad (F, L \text{ and } Y \text{ are constant})$$

$$\frac{l_1}{l_2} = \left(\frac{r_2}{r_1} \right)^2 = (2)^2 = 4$$

$$7. \quad (c) \quad \text{Restoring force is zero at mean position}$$

$$F = -Kx + F_0 \Rightarrow 0 = -Kx + F_0 \Rightarrow x = \frac{F_0}{K}$$

i.e. the particle will oscillate about $x = \frac{F_0}{K}$

$$8. \quad (b)$$

Graphical Questions

$$1. \quad (d) \quad T = 2\pi \sqrt{\frac{M}{K}} \Rightarrow T^2 \propto M$$

If we draw a graph between T^2 and M then it will be straight line.

and for $M = 0$, $T = 0$

i.e. the graph should pass through the origin.

but from the it is not reflected it means the mass of pan was neglected.

$$2. \quad (a) \quad \text{In the region } OA, \text{ stress} \propto \text{strain } i.e. \text{ Hooke's law hold good.}$$

$$3. \quad (c)$$

$$4. \quad (c)$$

$$5. \quad (d) \quad \text{As stress is shown on } x\text{-axis and strain on } y\text{-axis}$$

$$\text{So we can say that } Y = \cot \theta = \frac{1}{\tan \theta} = \frac{1}{\text{slope}}$$

So elasticity of wire P is minimum and of wire R is maximum

$$6. \quad (a) \quad \text{Area of hysteresis loop gives the energy loss in the process of stretching and unstretching of rubber band and this loss will appear in the form of heating.}$$

$$7. \quad (d) \quad \frac{Y_A}{Y_B} = \frac{\tan \theta_A}{\tan \theta_B} = \frac{\tan 60}{\tan 30} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3 \Rightarrow Y_A = 3Y_B$$

$$8. \quad (a) \quad l = \frac{FL}{AY} \therefore l \propto \frac{1}{r^2} \quad (Y, L \text{ and } F \text{ are constant})$$

i.e. for the same load, thickest wire will show minimum elongation. So graph D represent the thickest wire.

$$9. \quad (a) \quad \text{From the graph } l = 10^{-4} \text{ m}, F = 20 \text{ N}$$

$$A = 10^{-6} \text{ m}^2, L = 1 \text{ m}$$

$$\therefore Y = \frac{FL}{Al} = \frac{20 \times 1}{10^{-6} \times 10^{-4}} = 20 \times 10^{10} = 2 \times 10^{11} \text{ N/m}^2$$

10. (b) At point b, yielding of material starts.
11. (c) Graph between applied force and extension will be straight line because in elastic range,
Applied force $\propto$ extension
but the graph between extension and stored elastic energy will be parabolic in nature
As $U = 1/2 kx^2$ or $U \propto x^2$.
12. (b) $F = -\left(\frac{dU}{dx}\right)$.
In the region BC slope of the graph is positive
 $\therefore F =$ negative i.e. force is attractive in nature
In the region AB slope of the graph is negative
 $\therefore F =$ positive i.e. force is repulsive in nature
13. (b) Force constant, $K = \tan 30^\circ = 1/\sqrt{3}$
14. (b) In ductile materials, yield point exist while in Brittle material, failure would occur without yielding.
15. (d) Young's modulus is defined only in elastic region and
$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{8 \times 10^7}{4 \times 10^{-4}} = 2 \times 10^{11} \text{ N/m}^2$$
16. (a) Elasticity of wire decreases at high temperature i.e. at higher temperature slope of graph will be less.
So we can say that $T_1 > T_2$
17. (c)
18. (d) Attraction will be minimum when the distance between the molecule is maximum.
Attraction will be maximum at that point where the positive slope is maximum because $F = -\frac{dU}{dx}$
19. (c) $Y = \tan \theta$. According to figure $\theta_A > \theta_B > \theta_C$
i.e. $\tan \theta_A > \tan \theta_B > \tan \theta_C$
or $Y_A > Y_B > Y_C$
 $\therefore A, B,$ and C graph are for steel, brass and rubber respectively.

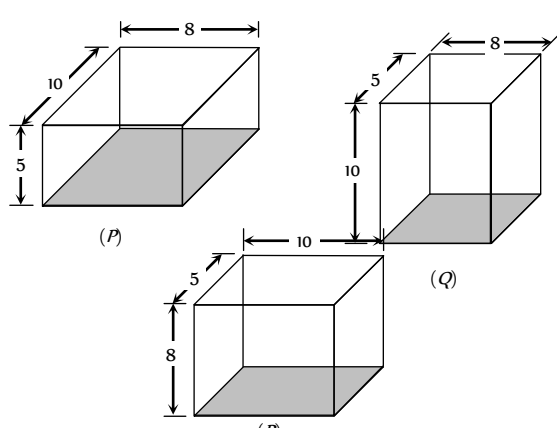
5. (a)
6. (a) Bulk modulus of elasticity measures how good the body is to regain its original volume on being compressed. Therefore, it represents incompressibility of the material. $K = \frac{-P\Delta V}{\Delta V}$ where P is increase in pressure, ΔV is change in volume.
7. (c) Strain is the ratio of change in dimensions of the body to the original dimensions. Because this is a ratio, therefore it is dimensionless quantity.
8. (a) A bridge during its use undergoes alternating strains for a large number of times each day, depending upon the movement of vehicles on it when a bridge is used for long time, it losses its elastic strength. Due to which the amount of strain in the bridge for a given stress will become large and ultimately, the bridge may collapse. This may not happen, if the bridges are declared unsafe after long use.
9. (d) Ivory is more elastic than wet-clay. Hence the ball of ivory will rise to a greater height. In fact the ball of wet-clay will not rise at all, it will be somewhat flattened permanently.
10. (a) Young's modulus of a material, $Y = \frac{\text{Stress}}{\text{Strain}}$
Here, stress = $\frac{\text{Restoring force}}{\text{Area}}$.
As restoring force is zero $\therefore Y = 0$.
11. (a) Work done = $\frac{1}{2} \times \text{Stress} \times \text{Strain} = \frac{1}{2} \times Y \times (\text{Strain})^2$.
Since, elasticity of steel is more than copper, hence more work has to be done in order to stretch the steel.
12. (b) Stress is defined as internal force (restoring force) per unit area of a body. Also, rubber is less elastic than steel, because restoring force is less for rubber than steel.

Assertion and Reason

1. (a) Because, the stretching of coil simply changes its shape without any change in the length of the wire used in coil. Due to which shear modulus of elasticity is involved.
2. (e) When a spring balance has been used for a long time, the spring in the balance fatigued and there is loss of strength of the spring. In such a case, the extension in the spring is more for a given load and hence the balance gives wrong readings.
3. (a) Elasticity is a measure of tendency of the body to regain its original configuration. As steel is deformed less than rubber therefore steel is more elastic than rubber.
4. (d) In a glassy solid (i.e., amorphous solid) the various bonds between the atoms or ions or molecules of a solid are not equally strong. Different bonds are broken at different temperatures. Hence there is no sharp melting point for a glassy solid.

Elasticity

SET Self Evaluation Test -9

- Two wires A and B of same length, same area of cross-section having the same Young's modulus are heated to the same range of temperature. If the coefficient of linear expansion of A is $3/2$ times of that of wire B . The ratio of the forces produced in two wires will be
 (a) $2/3$ (b) $9/4$
 (c) $4/9$ (d) $3/2$
- A wire of area of cross-section 10^{-6} m^2 is increased in length by 0.1% . The tension produced is 1000 N . The Young's modulus of wire is
 (a) 10^{12} N/m^2 (b) 10^{11} N/m^2
 (c) 10^{10} N/m^2 (d) 10^9 N/m^2
- To break a wire of one meter length, minimum 40 kg wt. is required. Then the wire of the same material of double radius and 6 m length will require breaking weight
 (a) 80 kg-wt (b) 240 kg-wt
 (c) 200 kg-wt (d) 160 kg-wt
- The breaking stress of a wire of length L and radius r is 5 kg-wt/m^2 . The wire of length $2l$ and radius $2r$ of the same material will have breaking stress in kg-wt/m^2
 (a) 5 (b) 10
 (c) 20 (d) 80
- The increase in length on stretching a wire is 0.05% . If its Poisson's ratio is 0.4 , then its diameter
 (a) Reduce by 0.02% (b) Reduce by 0.1%
 (c) Increase by 0.02% (d) Decrease by 0.4%
- If Poisson's ratio σ is $-\frac{1}{2}$ for a material, then the material is
 (a) Uncompressible (b) Elastic fatigue
 (c) Compressible (d) None of the above
- If the breaking force for a given wire is F , then the breaking force of two wires of same magnitude will be
 (a) F (b) $4F$
 (c) $8F$ (d) $2F$
- If the thickness of the wire is doubled, then the breaking force in the above question will be
 (a) $6F$ (b) $4F$
 (c) $8F$ (d) F
- On all the six surfaces of a unit cube, equal tensile force of F is applied. The increase in length of each side will be (Y = Young's modulus, σ = Poisson's ratio)
 (a) $\frac{F}{Y(1-\sigma)}$ (b) $\frac{F}{Y(1+\sigma)}$
 (c) $\frac{F(1-2\sigma)}{Y}$ (d) $\frac{F}{Y(1+2\sigma)}$
- The mass and length of a wire are M and L respectively. The density of the material of the wire is d . On applying the force F on the wire, the increase in length is l , then the Young's modulus of the material of the wire will be
 (a) $\frac{Fdl}{Ml}$ (b) $\frac{FL}{Mdl}$
 (c) $\frac{FMl}{dl}$ (d) $\frac{FdL^2}{Ml}$
- Two exactly similar wires of steel and copper are stretched by equal forces. If the difference in their elongations is 0.5 cm , the elongation (l) of each wire is
 $Y_s(\text{steel}) = 2.0 \times 10^{11} \text{ N/m}^2$
 $Y_c(\text{copper}) = 1.2 \times 10^{11} \text{ N/m}^2$
 (a) $l_s = 0.75 \text{ cm}, l_c = 1.25 \text{ cm}$
 (b) $l_s = 1.25 \text{ cm}, l_c = 0.75 \text{ cm}$
 (c) $l_s = 0.25 \text{ cm}, l_c = 0.75 \text{ cm}$
 (d) $l_s = 0.75 \text{ cm}, l_c = 0.25 \text{ cm}$
- If the compressibility of water is σ per unit atmospheric pressure, then the decrease in volume V due to P atmospheric pressure will be
 (a) $\sigma P / V$ (b) σPV
 (c) σ / PV (d) $\sigma V / P$
- A rectangular block of size $10 \text{ cm} \times 8 \text{ cm} \times 5 \text{ cm}$ is kept in three different positions P , Q and R in turn as shown in the figure. In each case, the shaded area is rigidly fixed and a definite force F is applied tangentially to the opposite face to deform the block. The displacement of the upper face will be

 (a) Same in all the three cases
 (b) Maximum in P position
 (c) Maximum in Q position
 (d) Maximum in R position

1. (d) $F = YA \alpha \Delta \theta$

If Y , A and $\Delta \theta$ are constant then $\frac{F_A}{F_B} = \frac{\alpha_A}{\alpha_B} = \frac{3}{2}$

2. (a) $Y = \frac{FL}{Al} = \frac{1000 \times 100}{10^{-6} \times 0.1} = 10^{12} \text{ N/m}^2$

3. (d) Breaking force = Breaking stress $\times$ Area of cross section of wire

$\therefore$ Breaking force $\propto r$ (Breaking distance is constant)

If radius becomes doubled then breaking force will become 4 times i.e. $40 \times 4 = 160 \text{ kg wt}$

4. (a) Breaking stress depends on the material of wire.

5. (a) Poisson's ratio = $\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$\therefore$ Lateral strain = $0.4 \times \frac{0.05}{100}$

So reduced by 0.02%.

6. (a) $\frac{dV}{V} = (1 + 2\sigma) \frac{dL}{dL}$

if $\sigma = -\frac{1}{2}$ then $\frac{dV}{V} = 0$ i.e. $K = \infty$

7. (d) Breaking force $\propto$ Area of cross section

If area is double then breaking force will become two times.

8. (b) Breaking force $\propto \pi r^2$

If thickness (radius) of wire is doubled then breaking force will become four times.

10. (d) $Y = \frac{F}{A} \frac{L}{l} = \frac{FdL^2}{Ml}$

As $M = \text{volume} \times \text{density} = A \times L \times d \therefore A = \frac{M}{Ld}$

11. (a) $l \propto \frac{1}{Y} \Rightarrow \frac{Y_s}{Y_c} = \frac{l_c}{l_s} \Rightarrow \frac{l_c}{l_s} = \frac{2 \times 10^{11}}{1.2 \times 10^{11}} = \frac{5}{3} \dots(i)$

Also $l_c - l_s = 0.5 \dots(ii)$

On solving (i) and (ii) $l_c = 1.25 \text{ cm}$ and $l_s = 0.75 \text{ cm}$.

12. (b) Compressibility = $\frac{\Delta V/V}{P} \Rightarrow \sigma = \frac{\Delta V}{PV} \Rightarrow \Delta V = \sigma PV$

13. (d) $\eta = \frac{F/A}{x/L} \Rightarrow x = \frac{L}{\eta} \times \frac{F}{A}$

If η and F are constant then $x \propto \frac{L}{A}$

For maximum displacement area at which force applied should be minimum and vertical side should be maximum, this is given in the R position of rectangular block.

9. (c) Tensile strain on each face = $\frac{F}{Y}$

Lateral strain due to the other two forces acting on

perpendicular faces = $\frac{-2\sigma F}{Y}$

Total increase in length = $(1 - 2\sigma) \frac{F}{Y}$



Chapter 10 Surface Tension

Intermolecular Force

The force of attraction or repulsion acting between the molecules are known as intermolecular force. The nature of intermolecular force is electromagnetic.

The intermolecular forces of attraction may be classified into two types.

Cohesive force	Adhesive force
The force of attraction between molecules of same substance is called the force of cohesion. This force is lesser in liquids and least in gases.	The force of attraction between the molecules of the different substances is called the force of adhesion.
Ex. (i) Two drops of a liquid coalesce into one when brought in mutual contact. (ii) It is difficult to separate two sticky plates of glass welded with water. (iii) It is difficult to break a drop of mercury into small droplets because of large cohesive force between the mercury molecules.	Ex. (i) Adhesive force enables us to write on the blackboard with a chalk. (ii) A piece of paper sticks to another due to large force of adhesion between the paper and gum molecules. (iii) Water wets the glass surface due to force of adhesion.

Note : ☐ Cohesive or adhesive forces are inversely proportional to the eighth power of distance between the molecules.

Surface Tension

The property of a liquid due to which its free surface tries to have minimum surface area and behaves as if it were under tension somewhat like a stretched elastic membrane is called surface tension. A small liquid drop has spherical shape, as due to surface tension the liquid surface tries to have minimum surface area and for a given volume, the sphere has minimum surface area.

Surface tension of a liquid is measured by the force acting per unit length on either side of an imaginary line drawn on the free surface of liquid, the direction of this force being perpendicular to the line and tangential to the free surface of liquid. So if F is the force acting on one side of imaginary line of length L , then $T = (F/L)$

(1) It depends only on the nature of liquid and is independent of the area of surface or length of line considered.

(2) It is a scalar as it has a unique direction which is not to be specified.

(3) Dimension : $[MT^{-1}]$. (Similar to force constant)

(4) Units : N/m (S.I.) and $Dyne/cm$ [C.G.S.]

(5) It is a molecular phenomenon and its root cause is the electromagnetic forces.

Force Due to Surface Tension

If a body of weight W is placed on the liquid surface, whose surface tension is T . If F is the minimum force required to pull it away from the water then value of F for different bodies can be calculated by the following table.

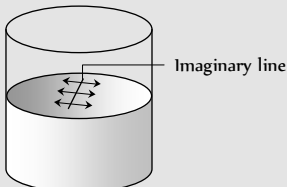
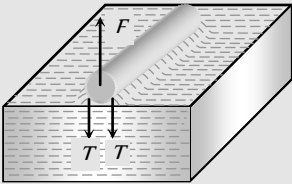
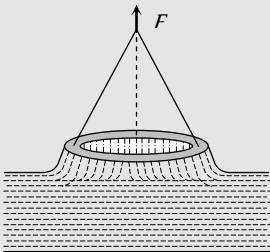
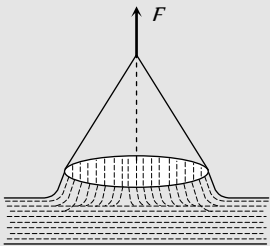
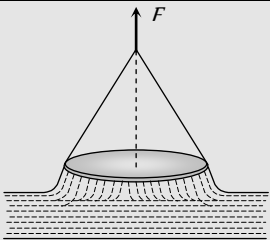
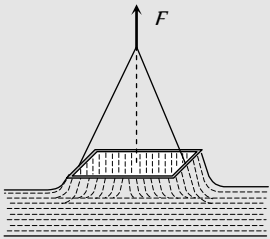
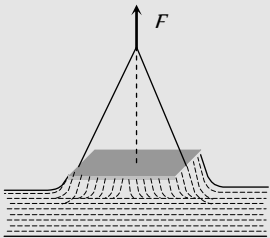
Body	Figure	Force
		

Fig. 10.1

Needle (Length = l)		$F = 2lT + W$
Hollow disc (Inner radius = r_1 Outer radius = r_2)		$F = 2\pi(r_1 + r_2)T + W$
Thin ring (Radius = r)		$F = 2\pi(r_1 + r_2)T + W$ $F = 4\pi rT + W$
Circular plate or disc (Radius = r)		$F = 2\pi rT + W$
Square frame (Side = l)		$F = 8lT + W$
Square plate		$F = 4lT + W$

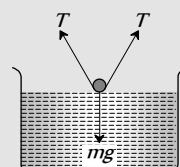
Examples of Surface Tension

(1) When mercury is split on a clean glass plate, it forms globules. Tiny globules are spherical on the account of surface tension because force of gravity is negligible. The bigger globules get flattened from the middle but have round shape near the edges.



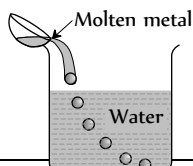
(2) When a greased iron needle is placed gently on the surface of water at rest, so that it does not prick the water surface, the needle floats on the surface of

water despite it being heavier because the weight of needle is balanced by the vertical components of the forces of surface tension. If the water

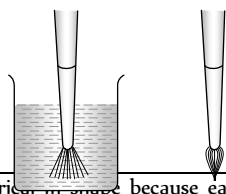


surface is pricked by one end of the needle, the needle sinks down.

(3) When a molten metal is poured into water from a suitable height, the falling stream of metal breaks up and the detached portion of the liquid in small quantity acquire the spherical shape.



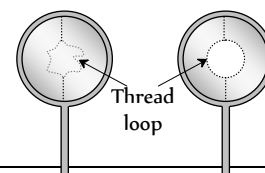
(5) Hair of shaving brush/painting brush when dipped in water spread out, but as soon as it is taken out, its hair stick together.



(7) Rain drops are spherical in shape because each drop tends to acquire minimum surface area due to surface tension, and for a given volume, the surface area of sphere is minimum.

(4) Take a frame of wire and dip it in soap solution and take it out, a soap film will be formed in the frame. Place a loop of wet thread gently on the film. It will remain in the form, we place it on the film according to

figure. Now, piercing the film with a pin at any point inside the loop, It immediately takes the circular form as shown in figure.



(6) If a small irregular piece of camphor is floated on the surface of pure water, it does not remain steady but dances about on the surface. This is because, irregular shaped camphor dissolves unequally and decreases the surface tension of the water locally. The unbalanced forces make it to move haphazardly in different directions.

(8) Oil drop spreads on cold water. Whereas it may remain as a drop on hot water. This is due to the fact that the surface tension of oil is less than that of cold water and is more than that of hot water.

Factors Affecting Surface Tension

(1) **Temperature** : The surface tension of liquid decreases with rise of temperature. The surface tension of liquid is zero at its boiling point and it vanishes at critical temperature. At critical temperature, intermolecular forces for liquid and gases becomes equal and liquid can expand without any restriction. For small temperature differences, the variation in surface tension with temperature is linear and is given by the relation

$$T_t = T_0(1 - \alpha t)$$

where T_t , T_0 are the surface tensions at $t^\circ\text{C}$ and 0°C respectively and α is the temperature coefficient of surface tension.

Examples : (i) Hot soup tastes better than the cold soup.

(ii) Machinery parts get jammed in winter.

(2) **Impurities** : The presence of impurities either on the liquid surface or dissolved in it, considerably affect the surface tension, depending upon the degree of contamination. A highly soluble substance like sodium chloride when dissolved in water, increases the surface tension of water. But the sparingly soluble substances like phenol when dissolved in water, decreases the surface tension of water.

Applications of Surface Tension

(1) The oil and grease spots on clothes cannot be removed by pure water. On the other hand, when detergents (like soap) are added in water, the surface tension of water decreases. As a result of this, wetting power of soap solution increases. Also the force of adhesion between soap solution and oil or grease on the clothes increases. Thus, oil, grease and dirt particles get mixed with soap solution easily. Hence clothes are washed easily.

(2) The antiseptics have very low value of surface tension. The low value of surface tension prevents the formation of drops that may otherwise block the entrance to skin or a wound. Due to low surface tension, the antiseptics spreads properly over wound.

(3) Surface tension of all lubricating oils and paints is kept low so that they spread over a large area.

(4) Oil spreads over the surface of water because the surface tension of oil is less than the surface tension of cold water.

(5) A rough sea can be calmed by pouring oil on its surface.

(6) In soldering, addition of 'flux' reduces the surface tension of molten tin, hence, it spreads.

Molecular Theory of Surface Tension

The maximum distance upto which the force of attraction between two molecules is appreciable is called molecular range ($\approx 10^{-9}\text{m}$). A sphere with a molecule as centre and radius equal to molecular range is called the sphere of influence. The liquid enclosed between free surface (PQ) of the liquid and an imaginary plane (RS) at a distance r (equal to molecular range) from the free surface of the liquid form a liquid film.

To understand the concept of tension acting on the free surface of a liquid, let us consider four liquid molecules like A, B, C and D. Their sphere of influence are shown in the figure.

(1) Molecule A is well within the liquid, so it is attracted equally in all directions. Hence the net force on this molecule is zero and it moves freely inside the liquid.

(2) Molecule B is little below the free surface of the liquid and it is also attracted equally in all directions. Hence the resultant force acts on it is also zero.

(3) Molecule C is just below the upper surface of the liquid film and the part of its sphere of influence is outside the free liquid surface. So the number of molecules in the upper half (attracting the molecules upward) is less than the number of molecule in the lower half (attracting the molecule downward). Thus the molecule C experiences a net downward force.

(4) Molecule D is just on the free surface of the liquid. The upper half of the sphere of influence has no liquid molecule. Hence the molecule D experiences a maximum downward force.

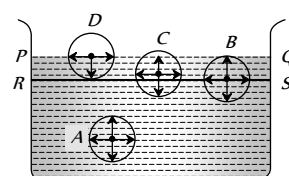


Fig. 10.2

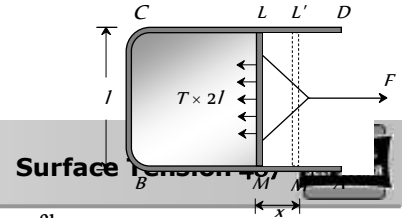


Fig. 10.3

Thus all molecules lying on surface film experiences a net downward force. Therefore, free surface of the liquid behaves like a stretched membrane.

Surface Energy

The molecules on the liquid surface experience net downward force. So to bring a molecule from the interior of the liquid to the free surface, some work is required to be done against the intermolecular force of attraction, which will be stored as potential energy of the molecule on the surface. The potential energy of surface molecules per unit area of the surface is called surface energy.

Unit : *Joule/m* (S.I.) *erg/cm* (C.G.S.)

Dimension : $[MT^{-2}]$

If a rectangular wire frame $ABCD$, equipped with a sliding wire LM dipped in soap solution, a film is formed over the frame. Due to the surface tension, the film will have a tendency to shrink and thereby, the sliding wire LM will be pulled in inward direction. However, the sliding wire can be held in this position under a force F , which is equal and opposite to the force acting on the sliding wire LM all along its length due to surface tension in the soap film.

If T is the force due to surface tension per unit length, then $F = T \times 2l$

Here l is length of the sliding wire LM . The length of the sliding wire has been taken as $2l$ for the reason that the film has got two free surfaces.

Suppose that the sliding wire LM is moved through a small distance x , so as to take the position $L'M'$. In this process, area of the film increases by $2l \times x$ (on the two sides) and to do so, the work done is given by

$$W = F \times x = (T \times 2l) \times x = T \times (2lx) = T \times \Delta A$$

$$\therefore W = T \times \Delta A \quad [\Delta A = \text{Total increase in area of the film}]$$

If temperature of the film remains constant in this process, this work done is stored in the film as its surface energy.

$$\text{From the above expression } T = \frac{W}{\Delta A} \text{ or } T = \frac{W}{\Delta A} \quad [\text{If } \Delta A = 1]$$

i.e. surface tension may be defined as the amount of work done in increasing the area of the liquid surface by unity against the force of surface tension at constant temperature.

Work Done in Blowing a Liquid Drop or Soap Bubble

(1) If the initial radius of liquid drop is r_1 and final radius of liquid drop is r_2 then

$$W = T \times \text{Increment in surface area}$$

$$W = T \times 4\pi[r_2^2 - r_1^2] \quad [\text{drop has only one free surface}]$$

(2) In case of soap bubble

$$W = T \times 8\pi[r_2^2 - r_1^2] \quad [\text{Bubble has two free surfaces}]$$

Splitting of Bigger Drop

When a drop of radius R splits into n smaller drops, (each of radius r) then surface area of liquid increases. Hence the work is to be done against surface tension.

Since the volume of liquid remains constant therefore

$$\frac{4}{3}\pi R^3 = n \frac{4}{3}\pi r^3 \quad \therefore R^3 = nr^3$$

$$\text{Work done} = T \times \Delta A = T \times [\text{Total final surface area of } n \text{ drops} - \text{surface area of big drop}] = T[n4\pi r^2 - 4\pi R^2]$$

Various formulae of work done				
$4\pi T[nr^2 - R^2]$	$4\pi R^2 T[n^{1/3} - 1]$	$4\pi T r^2 n^{2/3}[n^{1/3} - 1]$	$4\pi T R^3 \left[\frac{1}{r} - \frac{1}{R} \right]$	

If the work is not done by an external source then internal energy of liquid decreases, subsequently temperature decreases. This is the reason why spraying causes cooling.

By conservation of energy, Loss in thermal energy = work done against surface tension

$$JQ = W$$

$$\Rightarrow JmS\Delta\theta = 4\pi TR^3 \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\Rightarrow J \frac{4}{3}\pi R^3 d S \Delta\theta = 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$[\text{As } m = V \times d = \frac{4}{3}\pi R^3 \times d]$$

$$\therefore \text{Decrease in temperature } \Delta\theta = \frac{3T}{JSd} \left[\frac{1}{r} - \frac{1}{R} \right]$$

where J = mechanical equivalent of heat, S = specific heat of liquid, d = density of liquid.

Formation of Bigger Drop

If n small drops of radius r coalesce to form a big drop of radius R then surface area of the liquid decreases.

Amount of surface energy released = Initial surface energy – final surface energy

$$E = n4\pi r^2 T - 4\pi R^2 T$$

Various formulae of released energy				
$4\pi T[nr^2 - R^2]$	$4\pi R^2 T(n^{1/3} - 1)$	$4\pi T r^2 n^{2/3}(n^{1/3} - 1)$	$4\pi T R^3 \left[\frac{1}{r} - \frac{1}{R} \right]$	

(i) If this released energy is absorbed by a big drop, its temperature increases and rise in temperature can be given by

$$\Delta\theta = \frac{3T}{Js d} \left[\frac{1}{r} - \frac{1}{R} \right]$$

(ii) If this released energy is converted into kinetic energy of a big drop without dissipation then by the law of conservation of energy.

$$\frac{1}{2}mv^2 = 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{4}{3} \pi R^3 d \right] v^2 = 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right]$$

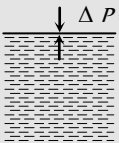
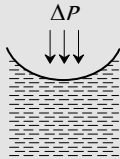
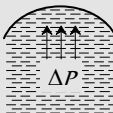
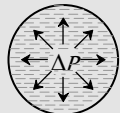
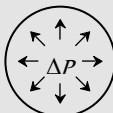
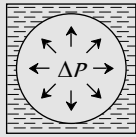
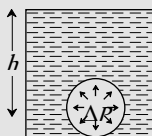
$$\Rightarrow v^2 = \frac{6T}{d} \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\therefore v = \sqrt{\frac{6T}{d} \left(\frac{1}{r} - \frac{1}{R} \right)}$$

Excess Pressure

Due to the property of surface tension a drop or bubble tends to contract and so compresses the matter enclosed. This in turn increases the internal pressure which prevents further contraction and equilibrium is achieved. So in equilibrium the pressure inside a bubble or drop is greater than outside and the difference of pressure between two sides of the liquid surface is called excess pressure. In case of a drop, excess pressure is provided by hydrostatic pressure of the liquid within the drop while in case of bubble the gauge pressure of the gas confined in the bubble provides it.

Excess pressure in different cases is given in the following table :

Plane surface	Concave surface
 $\Delta P = 0$	 $\Delta P = \frac{2T}{R}$
Convex surface	Drop
 $\Delta P = \frac{2T}{R}$	 $\Delta P = \frac{2T}{R}$
Bubble in air	Bubble in liquid
 $\Delta P = \frac{4T}{R}$	 $\Delta P = \frac{2T}{R}$
Bubble at depth h below the free surface of liquid of density d	Cylindrical liquid surface
 $\Delta P = \frac{2T}{R} + h d g$	 $\Delta P = \frac{T}{R}$
Liquid surface of unequal radii	Liquid film of unequal radii
 $\Delta P = T \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$	 $\Delta P = 2T \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$

Note : □ Excess pressure is inversely proportional to the radius of bubble (or drop), i.e., pressure inside a smaller bubble (or drop) is higher than inside a larger bubble (or drop). That is why when two bubbles of different sizes are put in communication with each other, the air will rush from smaller to larger bubble, so that the smaller will shrink while the larger will expand till the smaller bubble reduces to droplet.

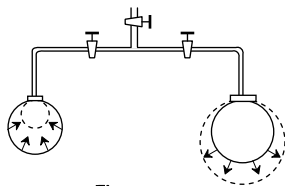


Fig. 10.4

Shape of Liquid Meniscus

We know that a liquid assumes the shape of the vessel in which it is contained i.e. it can not oppose permanently any force that tries to change its shape. As the effect of force is zero in a direction perpendicular to it, the free surface of liquid at rest adjusts itself at right angles to the resultant force.

When a capillary tube is dipped in a liquid, the liquid surface becomes curved near the point of contact. This curved surface is due to the resultant of two forces i.e. the force of cohesion and the force of adhesion. The curved surface of the liquid is called meniscus of the liquid.

If liquid molecule A is in contact with solid (i.e. wall of capillary tube) then forces acting on molecule A are

(i) Force of adhesion F_a (acts outwards at right angle to the wall of the tube).

(ii) Force of cohesion F_c (acts at an angle 45° to the vertical).

Resultant force F_r depends upon the value of F_a and F_c .

If resultant force F_r make an angle α with F_a .

$$\text{Then } \tan \alpha = \frac{F_c \sin 135^\circ}{F_a + F_c \cos 135^\circ} = \frac{F_c}{\sqrt{2} F_a - F_c}$$

By knowing the direction of resultant force we can find out the shape of meniscus because the free surface of the liquid adjust itself at right angle to this resultant force.

If $F_c = \sqrt{2} F_a$ $\tan \alpha = \infty \quad \therefore \alpha = 90^\circ$ i.e. the resultant force acts vertically downwards. Hence the liquid meniscus must be horizontal.	$F_c < \sqrt{2} F_a$ $\tan \alpha = \text{positive} \quad \therefore \alpha$ is acute angle i.e. the resultant force directed outside the liquid. Hence the liquid meniscus must be concave upward.	$F_c > \sqrt{2} F_a$ $\tan \alpha = \text{negative} \quad \therefore \alpha$ is obtuse angle i.e. the resultant force directed inside the liquid. Hence the liquid meniscus must be convex upward.
Example: Pure water in silver coated capillary tube.	Example: Water in glass capillary tube.	Example: Mercury in glass capillary tube.

Angle of Contact

Angle of contact between a liquid and a solid is defined as the angle enclosed between the tangents to the liquid surface and the solid surface inside the liquid, both the tangents being drawn at the point of contact of the liquid with the solid.

$\theta < 90^\circ$ $F_a > \frac{F_c}{\sqrt{2}}$ concave meniscus. Liquid wets the solid surface	
$\theta = 90^\circ$ $F_a = \frac{F_c}{\sqrt{2}}$ plane meniscus. Liquid does not wet the solid surface.	
$\theta > 90^\circ$ $F_a < \frac{F_c}{\sqrt{2}}$	

convex meniscus.

Liquid does not wet the solid surface.

(i) Its value lies between 0° and 180°

$\theta = 0^\circ$ for pure water and glass, $\theta = 8^\circ$ for tap water and glass, $\theta = 90^\circ$ for water and silver

$\theta = 138^\circ$ for mercury and glass, $\theta = 160^\circ$ for water and chromium

(ii) It is particular for a given pair of liquid and solid. Thus the angle of contact changes with the pair of solid and liquid.

(iii) It does not depend upon the inclination of the solid in the liquid.

(iv) On increasing the temperature, angle of contact decreases.

(v) Soluble impurities increase the angle of contact.

(vi) Partially soluble impurities decrease the angle of contact.

Capillarity

If a tube of very narrow bore (called capillary) is dipped in a liquid, it is found that the liquid in the capillary either ascends or descends relative to the surrounding liquid. This phenomenon is called capillarity.

The root cause of capillarity is the difference in pressures on two sides of (concave and convex) curved surface of liquid.

Examples of capillarity :

- (i) Ink rises in the fine pores of blotting paper leaving the paper dry.
- (ii) A towel soaks water.
- (iii) Oil rises in the long narrow spaces between the threads of a wick.
- (iv) Wood swells in rainy season due to rise of moisture from air in the pores.
- (v) Ploughing of fields is essential for preserving moisture in the soil.
- (vi) Sand is drier soil than clay. This is because holes between the sand particles are not so fine as compared to that of clay, to draw up water by capillary action.

Ascent Formula

When one end of capillary tube of radius r is immersed into a liquid of density d which wets the sides of the capillary tube (water and capillary tube of glass), the shape of the liquid meniscus in the tube becomes concave upwards.

R = radius of curvature of liquid meniscus.

T = surface tension of liquid

P = atmospheric pressure

Pressure at point $A = P$, Pressure at point $B = P - \frac{2T}{R}$

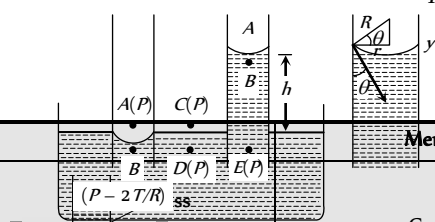


Fig. 10.5

Meniscus	Angle of contact	Level
Concave	$\theta < 90^\circ$	Rises
Plane	$\theta = 90^\circ$	No rise, no fall
Convex	$\theta > 90^\circ$	Fall

- (iii) For a given liquid and solid at a given place

$$h \propto \frac{1}{r} \quad [\text{As } T, \theta, d \text{ and } g \text{ are constant}]$$

i.e. lesser the radius of capillary greater will be the rise and vice-versa. This is called **Jurin's law**.

- (iv) If the weight of the liquid contained in the meniscus is taken into consideration then more accurate ascent formula is given by

$$h = \frac{2T \cos \theta}{rdg} - \frac{r}{3}$$

- (v) In case of capillary of insufficient length i.e. $L < h$, the liquid will neither overflow from the upper end like a fountain nor will it tickle along the vertical sides of the tube. The liquid after reaching the upper end will increase the radius of its meniscus without changing nature such that :

$$hr = Lr' \quad \therefore L < h \therefore r' > r$$

Pressure at points C and D just above and below the plane surface of liquid in the vessel is also P (atmospheric pressure). The points B and D are in the same horizontal plane in the liquid but the pressure at these points is different.

In order to maintain the equilibrium the liquid level rises in the capillary tube upto height h .

Pressure due to liquid column = pressure difference due to surface tension

$$\Rightarrow h d g = \frac{2T}{R}$$

$$\therefore h = \frac{2T}{R d g} = \frac{2T \cos \theta}{r d g} \quad \left[\text{As } R = \frac{r}{\cos \theta} \right]$$

- (i) The capillary rise depends on the nature of liquid and solid both i.e. on T, d, θ and R .

- (ii) Capillary action for various liquid-solid pair.

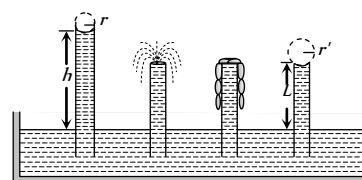


Fig. 10.6

- (vi) If a capillary tube is dipped into a liquid and tilted at an angle α from vertical, then the vertical height of liquid column remains same whereas the length of liquid column (l) in the capillary tube increases.

$$h = l \cos \alpha \quad \text{or} \quad l = \frac{h}{\cos \alpha}$$

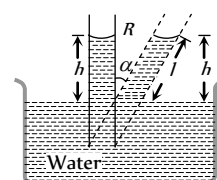


Fig. 10.7

(vii) It is important to note that in equilibrium, the height h is independent of the shape of capillary if the radius of meniscus remains the same. That is why the vertical height h of a liquid column in capillaries of different shapes and sizes will be same if the radius of meniscus remains the same.

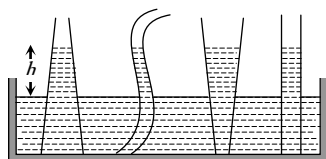
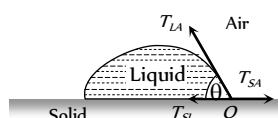


Fig. 10.8

Shape of Drops

Whether the liquid will be in equilibrium in the form of a drop or it will spread out; depends on the relative strength of the force due to surface tension at the three interfaces.



Special Cases

Fig. 10.9

$T_{SL} > T_{SA} \cos \theta$ is positive i.e. $0^\circ < \theta < 90^\circ$.

This condition is fulfilled when the molecules of liquid are strongly attracted to that of solid.

Example : (i) Water on glass.

(ii) Kerosene oil on any surface.

$T_{SL} < T_{SA} \cos \theta$ is negative i.e. $90^\circ < \theta < 180^\circ$.

This condition is fulfilled when the molecules of the liquid are strongly attracted to themselves and weakly w.r.t. that of solid.

Example : (i) Mercury on glass surface.

(ii) Water on lotus leaf (or a waxy or oily surface)

$(T_{SL} + T_{SA} \cos \theta) > T_{LA}$

In this condition, the molecule of liquid will not be in equilibrium and experience a net force at the interface. As a result, the liquid spreads.

Example : (i) Water on a clean glass plate.

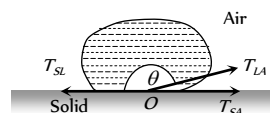


Fig. 10.10

$$T_{SL} + T_{SA} \cos \theta = T_{LA} \text{ or } \cos \theta = \frac{T_{SA} - T_{SL}}{T_{LA}}$$

Useful Facts and Formulae

(1) **Formation of double bubble** : If r_1 and r_2 are the radii of smaller and larger bubble and P_0 is the atmospheric pressure, then the pressure inside them will be $P_1 = P_0 + \frac{4T}{r_1}$ and $P_2 = P_0 + \frac{4T}{r_2}$.

Now as $r_1 < r_2 \therefore P_1 > P_2$

So for interface

$$\Delta P = P_1 - P_2 = 4T \left[\frac{1}{r_1} - \frac{1}{r_2} \right] \quad \dots(i)$$

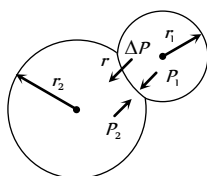


Fig. 10.11

As excess pressure acts from concave to convex side, the interface will be concave towards the smaller bubble and convex towards larger bubble and if r is the radius of interface.

$$\Delta P = \frac{4T}{r} \quad \dots(ii)$$

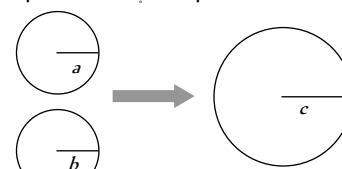
$$\text{From (i) and (ii)} \quad \frac{1}{r} = \frac{1}{r_1} - \frac{1}{r_2}$$

$$\therefore \text{Radius of the interface } r = \frac{r_1 r_2}{r_2 - r_1}$$

(2) **Formation of a single bubble**

(i) Under isothermal condition two soap bubble of radii ' a ' and ' b ' coalesce to form a single bubble of radius ' c '.

If the external pressure is P_0 then pressure inside bubbles



$$P_a = \left(P_0 + \frac{4T}{a} \right), \quad P_b = \left(P_0 + \frac{4T}{b} \right) \text{ and } P_c = \left(P_0 + \frac{4T}{c} \right)$$

and volume of the bubbles

$$V_a = \frac{4}{3} \pi a^3, \quad V_b = \frac{4}{3} \pi b^3, \quad V_c = \frac{4}{3} \pi c^3$$

Now as mass is conserved $\mu_a + \mu_b = \mu_c$

$$\Rightarrow \frac{P_a V_a}{RT_a} + \frac{P_b V_b}{RT_b} = \frac{P_c V_c}{RT_c}$$

$$\left[\text{As } PV = \mu RT, \text{ i.e., } \mu = \frac{PV}{RT} \right]$$

$$\Rightarrow P_a V_a + P_b V_b = P_c V_c \quad \dots(i)$$

[As temperature is constant, i.e., $T_a = T_b = T_c$]

Substituting the value of pressure and volume

$$\Rightarrow \left[P_0 + \frac{4T}{a} \right] \left[\frac{4}{3} \pi a^3 \right] + \left[P_0 + \frac{4T}{b} \right] \left[\frac{4}{3} \pi b^3 \right]$$

$$= \left[P_0 + \frac{4T}{c} \right] \left[\frac{4}{3} \pi c^3 \right]$$

$$\Rightarrow 4T(a^2 + b^2 - c^2) = P_0(c^3 - a^3 - b^3)$$

$$\therefore \text{Surface tension of the liquid } T = \frac{P_0(c^3 - a^3 - b^3)}{4(a^2 + b^2 - c^2)}$$

(ii) If two bubble coalesce in vacuum then by substituting $P_0 = 0$ in the above expression we get

$$a^2 + b^2 - c^2 = 0 \quad \therefore c^2 = a^2 + b^2$$

Radius of new bubble $= c = \sqrt{a^2 + b^2}$ or can be expressed as

$$r = \sqrt{r_1^2 + r_2^2}.$$

(3) The difference of levels of liquid column in two limbs of U-tube of unequal radii r_1 and r_2 is

$$h = h_1 - h_2 = \frac{2T \cos \theta}{dg} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

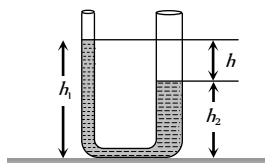


Fig. 10.13

(4) A large force (F) is required to draw apart two glass plate normally enclosing a thin water film because the thin water film formed between the two glass plates will have concave surface all around. Since on the concave side of a liquid surface, pressure is more, work will have to be done in drawing the plates apart.

$$F = \frac{2AT}{t} \text{ where } T = \text{surface tension of water film, } t = \text{thickness of}$$

film, A = area of film.

(5) When a soap bubble is charged, then its size increases due to outward force on the bubble.

(6) The materials, which when coated on a surface and water does not enter through that surface are known as water proofing agents. For example wax etc. Water proofing agent increases the angle of contact.

(7) Values of surface tension of some liquids.

Glycerine	0.063
Carbon tetrachloride	0.027
Ethyl alcohol	0.022

Tips & Tricks

- ✍ Surface tension does not depend on the area of the surface.
- ✍ When there is no external force, the shape of a liquid drop is determined by the surface tension of the liquid.
- ✍ Soap helps in better cleaning of clothes because it reduces the surface tension of the liquid.
- ✍ If a beaker is filled with liquid of density ρ upto a height h , then the mean pressure on the walls of the beaker is $h\rho g/2$.
- ✍ The pressure on the concave side of a curved surface is always greater than that on its convex side.
- ✍ Molecular forces do not obey the inverse square law of distance.
- ✍ The molecular forces are of electrical origin.
- ✍ Work done in forming a soap bubble of radius R is $8\pi R^2 T$, where T = surface tension.
- ✍ Energy is always required to split a drop of liquid into a number of small drops. It is because, the surface area of the small drops formed is greater than the surface area of the original single drop.
- ✍ Work done in breaking a drop of radius R into n drops of equal size $= 4\pi R^2 T (n^{1/3} - 1)$.
- Same amount of energy is liberated in combining n drops into a single drop.
- ✍ When the liquid drops merge into each other to form a larger drop, energy is released.
- ✍ Surface tension of molten cadmium increases with the increases in temperature.
- ✍ Detergents decrease both the angle of contact as well as surface tension.
- ✍ Angle of contact is independent of the angle of inclination of the walls.
- ✍ The materials used for water proofing increases the angle of contact as well as surface tension.
- ✍ A liquid does not wet the containing vessel if its angle of contact is obtuse.
- ✍ In case of liquids which do not wet the walls of the containing vessel, the force of adhesion is less than $1/\sqrt{2}$ times the force of cohesion.
- ✍ The liquid rises in a capillary tube, when the angle of contact is acute.
- ✍ The height of the liquid column in a capillary tube on the moon is six times that on the earth.
- ✍ Angle of contact between a liquid and a solid surface. Increases



with increase in temperature of the liquid and decreases on adding impurity to the liquid.

✍ For a liquid – solid interface, if the angle of contact is acute, then

- (i) The liquid will wet the solid.
- (ii) The liquid will rise in the capillary tube made of such a solid and
- (iii) Meniscus of the liquid will be concave.

✍ In case the angle of contact is obtuse, then

- (i) The liquid will not wet the solid.
- (ii) The liquid will get depressed in the tube and
- (iii) Meniscus of the liquid will be convex.

✍ When the capillary tube is of insufficient length, the liquid will not overflow. It rises upto the top end of the tube and then adjusts the radius of curvature of its meniscus.

Ordinary Thinking

Objective Questions

Surface Tension

- The value of surface tension of a liquid at critical temperature is
 - Zero
 - Infinite
 - Between 0 and ∞
 - Can not be determined
- The spherical shape of rain-drop is due to

[CPMT 1976, 90; NCERT 1982; AIIMS 1998; MH CET 2000; DCE 1999; AFMC 1999; CPMT 2001; AFMC 2001]

 - Density of the liquid
 - Surface tension
 - Atmospheric pressure
 - Gravity
- Surface tension is due to
 - Frictional forces between molecules
 - Cohesive forces between molecules
 - Adhesive forces between molecules
 - Gravitational forces
- When there is no external force, the shape of a liquid drop is determined by

[CPMT 1988, 86; DPMT 1982]

 - Surface tension of the liquid
 - Density of liquid
 - Viscosity of liquid
 - Temperature of air only
- Soap helps in cleaning clothes, because

[DPMT 1983, 2001]

 - Chemicals of soap change
 - It increases the surface tension of the solution
 - It absorbs the dirt
 - It lowers the surface tension of the solution
- A pin or a needle floats on the surface of water, the reason for this is

[MP PET/PMT 1988; CPMT 1975]

 - Surface tension
 - Less weight
 - Upthrust of liquid
 - None of the above
- Coatings used on raincoat are waterproof because
 - Water is absorbed by the coating
 - Cohesive force becomes greater
 - Water is not scattered away by the coating
 - Angle of contact decreases
- If temperature increases, the surface tension of a liquid

[MP PMT 1994; EAMCET (Engg.) 1995; RPET 2003]

 - Increases
 - Decreases
 - Remains the same
 - Increases then decreases
- A drop of oil is placed on the surface of water. Which of the following statement is correct

[NCERT 1976; DPMT 1982]

 - It will remain on it as a sphere
 - It will spread as a thin layer
 - It will be partly as spherical droplets and partly as thin film
 - It will float as a distorted drop on the water surface
- The temperature at which the surface tension of water is zero
 - 0°C
 - 277 K
 - 370°C
 - Slightly less than 647 K
- A small air bubble is at the inner surface of the bottom of a beaker filled with cold water. Now water of the beaker is heated. The size of bubble increases. The reason for this may be
 - Increase in the saturated vapour pressure of water
 - Root mean square velocity of air molecules inside the bubble increases
 - Decrease in surface tension of water
 - All of the above
- The spiders and insects move and run about on the surface of water without sinking because
 - Elastic membrane is formed on water due to property of surface tension [AIIMS 1980]
 - Spiders and insects are lighter
 - Spiders and insects swim on water
 - Spider and insects experience upthrust
- Small droplets of a liquid are usually more spherical in shape than larger drops of the same liquid because

[EAMCET 1988]

 - Force of surface tension is equal and opposite to the force of gravity
 - Force of surface tension predominates the force of gravity
 - Force of gravity predominates the force of surface tension
 - Force of gravity and force of surface tension act in the same direction and are equal
- Hairs of shaving brush cling together when it is removed from water due to
 - Force of attraction between hair
 - Surface tension
 - Viscosity of water
 - Characteristic property of hairs
- A square frame of side L is dipped in a liquid. On taking out, a membrane is formed. If the surface tension of the liquid is T , the force acting on the frame will be

[MP PMT 1990; DPMT 2004]

 - $2\ TL$
 - $4\ TL$
 - $8\ TL$
 - $10\ TL$
- Water does not wet an oily glass because
 - Cohesive force of oil > adhesive force between oil and glass
 - Cohesive force of oil > cohesive force of water
 - Oil repels water
 - Cohesive force for water > adhesive force between water and oil molecules
- A water drop takes the shape of a sphere in a oil while the oil drop spreads in water, because
 - C.F. for water > A.F. for water and oil
 - C.F. for oil > A.F. for water and oil
 - C.F. for oil < A.F. for water and oil
 - None of the above

(A.F. = adhesive force C.F. = cohesive force)
- Which of the fact is not due to surface tension
 - Dancing of a camphor piece over the surface of water
 - Small mercury drop itself becomes spherical
 - A liquid surface comes at rest after stirring
 - Mercury does not wet the glass vessel
- In the glass capillary tube, the shape of the surface of the liquid depends upon

[MP PMT 1989]

 - Only on the cohesive force of liquid molecules

- (b) Only on the adhesive force between the molecules of glass and liquid
(c) Only on relative cohesive and adhesive force between the atoms
(d) Neither on cohesive nor on adhesive force
- 20.** Force necessary to pull a circular plate of 5 cm radius from water surface for which surface tension is 75 dynes/cm, is [MP PMT 1991]
(a) 30 dyne (b) 60 dynes
(c) 750 dynes (d) 750 π dynes
- 21.** The property of surface tension is obtained in
(a) Solids, liquids and gases (b) Liquids
(c) Gases (d) Matter
- 22.** The surface tension of a liquid [MNR 1990]
(a) Increases with area
(b) Decreases with area
(c) Increase with temperature
(d) Decrease with temperature
- 23.** If two glass plates are quite nearer to each other in water, then there will be force of
(a) Attraction (b) Repulsion
(c) Attraction or repulsion (d) None of the above
- 24.** On mixing the salt in water, the surface tension of water will
(a) Increase (b) Decrease
(c) Remain unchanged (d) None of the above
- 25.** The maximum force, in addition to the weight required to pull a wire of 5.0 cm long from the surface of water at temperature 20°C, is 728 dynes. The surface tension of water is
(a) 7.28 N/cm (b) 7.28 dyne/cm
(c) 72.8 dyne/cm (d) 7.28 $\times 10$ dyne/cm
- 26.** Consider a liquid contained in a vessel. The liquid solid adhesive force is very weak as compared to the cohesive force in the liquid. The shape of the liquid surface near the solid shall be
(a) Horizontal (b) Almost vertical
(c) Concave (d) Convex
- 27.** At which of the following temperatures, the value of surface tension of water is minimum [MP PMT/PET 1998]
(a) 4°C (b) 25°C
(c) 50°C (d) 75°C
- 28.** If a glass rod is dipped in mercury and withdrawn out, the mercury does not wet the rod because [MP PET 1995]
(a) Angle of contact is acute
(b) Cohesion force is more
(c) Adhesion force is more
(d) Density of mercury is more
- 29.** Mercury does not wet glass, wood or iron because [MP PET 1997]
(a) Cohesive force is less than adhesive force
(b) Cohesive force is greater than adhesive force
(c) Angle of contact is less than 90°
(d) Cohesive force is equal to adhesive force
- 30.** Surface tension of a liquid is found to be influenced by [ISM Dhanbad 1994]
(a) It increases with the increase of temperature
(b) Nature of the liquid in contact
(c) Presence of soap that increases it
(d) Its variation with the concentration of the liquid
- 31.** When a drop of water is dropped on oil surface, then [RPMT 1997]
(a) It will mix up with oil
(b) It spreads in the form of a film
(c) It will deform
(d) It remains spherical
- 32.** Two pieces of glass plate one upon the other with a little water in between them cannot be separated easily because of
(a) Inertia (b) Pressure
(c) Surface tension (d) Viscosity
- 33.** Small liquid drops assume spherical shape because [JIPMER 1997]
(a) Atmospheric pressure exerts a force on a liquid drop
(b) Volume of a spherical drop is minimum
(c) Gravitational force acts upon the drop
(d) Liquid tends to have the minimum surface area due to surface tension
- 34.** A thin metal disc of radius r floats on water surface and bends the surface downwards along the perimeter making an angle θ with vertical edge of the disc. If the disc displaces a weight of water W and surface tension of water is T , then the weight of metal disc is
(a) $2\pi rT + W$ (b) $2\pi rT \cos \theta - W$
(c) $2\pi rT \cos \theta + W$ (d) $W - 2\pi rT \cos \theta$
- 35.** A 10 cm long wire is placed horizontally on the surface of water and is gently pulled up with a force of 2×10^{-2} N to keep the wire in equilibrium. The surface tension, in N/m, of water is
(a) 0.1 (b) 0.2
(c) 0.001 (d) 0.002
- 36.** It is easy to wash clothes in hot water because its [RPMT 2000]
(a) Surface tension is more
(b) Surface tension is less
(c) Consumes less soap
(d) None of these
- 37.** Due to [MNR 1994] property of water, tiny particles of camphor dance on the surface of water [RPMT 1999]
(a) Viscosity (b) Surface tension
(c) Weight (d) Floating force
- 38.** The force required to separate two glass plates of area 10^{-2} m^2 with a film of water 0.05 mm thick between them, is (Surface tension of water is $70 \times 10^{-3} \text{ N/m}$) [KCET 2000; Pb. PET 2001; RPET 2002]
(a) 28 N (b) 14 N
(c) 50 N (d) 38 N
- 39.** Oil spreads over the surface of water whereas water does not spread over the surface of the oil, due to [MH CET 2001]
(a) Surface tension of water is very high
(b) Surface tension of water is very low
(c) Viscosity of oil is high
(d) Viscosity of water is high
- 40.** Cohesive force is experienced between [MH CET 2001]
(a) Magnetic substances
(b) Molecules of different substances
(c) Molecules of same substances
(d) None of these



41. The property utilized in the manufacture of lead shots is [AIIMS 2002]

(a) Specific weight of liquid lead
(b) Specific gravity of liquid lead
(c) Compressibility of liquid lead
(d) Surface tension of liquid lead

42. The dimensions of surface tension are [MH CET 2002]

(a) $[MLT^{-1}]$ (b) $[ML^2T^{-2}]$
(c) $[ML^0T^{-2}]$ (d) $[ML^{-1}T^{-2}]$

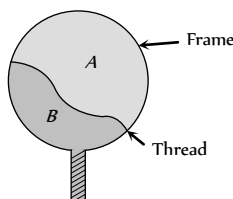
43. A wooden stick 2 m long is floating on the surface of water. The surface tension of water 0.07 N/m. By putting soap solution on one side of the sticks the surface tension is reduced to 0.06 N/m. The net force on the stick will be

[Pb. PMT 2002]

(a) 0.07 N (b) 0.06 N
(c) 0.01 N (d) 0.02 N

44. A thread is tied slightly loose to a wire frame as in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A punctured with a pin, the thread.

[KCET 2004]



- (a) Becomes concave toward A
(b) Becomes convex towards A
(c) Remains in the initial position
(d) Either (a) or (b) depending on the size of A w.r.t. B
45. The force required to take away a flat circular plate of radius 2 cm from the surface of water, will be (the surface tension of water is 70 dyne/cm) [Pb. PET 2001]
- (a) 280π dyne (b) 250π dyne
(c) 140π dyne (d) 210π dyne
46. Surface tension may be defined as [CPMT 1990]
- (a) The work done per unit area in increasing the surface area of a liquid under isothermal condition
(b) The work done per unit area in increasing the surface area of a liquid under adiabatic condition
(c) The work done per unit area in increasing the surface area of a liquid under both isothermal and adiabatic conditions
(d) Free surface energy per unit volume

Surface Energy

1. Energy needed in breaking a drop of radius R into n drops of radii r is given by [CPMT 1982, 97]
- (a) $4\pi T(nr^2 - R^2)$ (b) $\frac{4}{3}\pi(r^3n - R^2)$
(c) $4\pi T(R^2 - nr^2)$ (d) $4\pi T(nr^2 + R^2)$
2. The potential energy of a molecule on the surface of liquid compared to one inside the liquid is [MP PMT 1993]

(a) Zero (b) Smaller
(c) The same (d) Greater

3. Two droplets merge with each other and forms a large droplet. In this process

[CBSE PMT 1993; RPMT 1997, 2000; CPMT 2001; BHU 2001; AFMC 2002]

(a) Energy is liberated
(b) Energy is absorbed
(c) Neither liberated nor absorbed
(d) Some mass is converted into energy

4. A drop of liquid of diameter 2.8 mm breaks up into 125 identical drops. The change in energy is nearly (S.T. of liquid = 75 dynes/cm)

(a) Zero (b) 19 erg
(c) 46 erg (d) 74 erg

5. Radius of a soap bubble is r , surface tension of soap solution is T . Then without increasing the temperature, how much energy will be needed to double its radius

[CPMT 1991; Pb. PMT 2000; RPET 2001]

(a) $4\pi r^2 T$ (b) $2\pi r^2 T$
(c) $12\pi r^2 T$ (d) $24\pi r^2 T$

6. Work done in splitting a drop of water of 1 mm radius into 10^6 droplets is (Surface tension of water = $72 \times 10^{-3} \text{ J/m}^2$)

[MP PET/PMT 1988; CPMT 1989; RPET 2001]

(a) $9.58 \times 10^{-5} \text{ J}$ (b) $8.95 \times 10^{-5} \text{ J}$
(c) $5.89 \times 10^{-5} \text{ J}$ (d) $5.98 \times 10^{-6} \text{ J}$

7. A spherical liquid drop of radius R is divided into eight equal droplets. If surface tension is T , then the work done in this process will be [CPMT 1990]

(a) $2\pi R^2 T$ (b) $3\pi R^2 T$
(c) $4\pi R^2 T$ (d) $2\pi R T^2$

8. The amount of work done in blowing a soap bubble such that its diameter increases from d to D is (T = surface tension of the solution) [MP PMT 1996]

(a) $4\pi(D^2 - d^2)T$ (b) $8\pi(D^2 - d^2)T$
(c) $\pi(D^2 - d^2)T$ (d) $2\pi(D^2 - d^2)T$

9. If T is the surface tension of soap solution, the amount of work done in blowing a soap bubble from a diameter D to $2D$ is

(a) $2\pi D^2 T$ (b) $4\pi D^2 T$
(c) $6\pi D^2 T$ (d) $8\pi D^2 T$

10. The radius of a soap bubble is increased from $\frac{1}{\sqrt{\pi}} \text{ cm}$ to $\frac{2}{\sqrt{\pi}} \text{ cm}$.

If the surface tension of water is 30 dynes per cm, then the work done will be [MP PMT 1986]

(a) 180 ergs (b) 360 ergs
(c) 720 ergs (d) 960 ergs

11. The surface tension of a liquid is 5 N/m. If a thin film of the area 0.02 m² is formed on a loop, then its surface energy will be

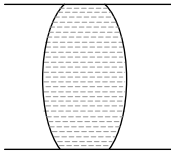
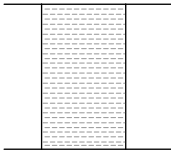
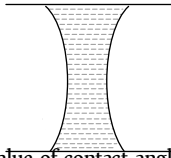
- (a) $5 \times 10^2 J$ (b) $2.5 \times 10^{-2} J$
- (c) $2 \times 10^{-1} J$ (d) $5 \times 10^{-1} J$
12. If work W is done in blowing a bubble of radius R from a soap solution, then the work done in blowing a bubble of radius $2R$ from the same solution is [MP PET 1990]
- (a) $W/2$ (b) $2W$
- (c) $4W$ (d) $2\frac{1}{3}W$
13. A spherical drop of oil of radius 1 cm is broken into 1000 droplets of equal radii. If the surface tension of oil is 50 dynes/cm , the work done is [MP PET 1990]
- (a) $18\pi\text{ ergs}$ (b) $180\pi\text{ ergs}$
- (c) $1800\pi\text{ ergs}$ (d) $8000\pi\text{ ergs}$
14. The work done in blowing a soap bubble of radius r of the solution of surface tension T will be [DPMT 1999; MP PMT 2003]
- (a) $8\pi^2 T$ (b) $2\pi^2 T$
- (c) $4\pi^2 T$ (d) $\frac{4}{3}\pi^2 T$
15. If two identical mercury drops are combined to form a single drop, then its temperature will [RPET 2000]
- (a) Decrease (b) Increase
- (c) Remains the same (d) None of the above
16. If the surface tension of a liquid is T , the gain in surface energy for an increase in liquid surface by A is [MP PET 1991; RPMT 2002]
- (a) AT^{-1} (b) AT
- (c) $A^2 T$ (d) $A^2 T^2$
17. The surface tension of a soap solution is $2 \times 10^{-2} N/m$. To blow a bubble of radius 1 cm , the work done is [MP PMT 1989]
- (a) $4\pi \times 10^{-6} J$ (b) $8\pi \times 10^{-6} J$
- (c) $12\pi \times 10^{-6} J$ (d) $16\pi \times 10^{-6} J$
18. A mercury drop of 1 cm radius is broken into 10^6 small drops. The energy used will be (surface tension of mercury is $35 \times 10^{-3} N/cm$) [Roorkee 1984]
- (a) $4.4 \times 10^{-3} J$ (b) $2.2 \times 10^{-4} J$
- (c) $8.8 \times 10^{-4} J$ (d) $10^4 J$
19. The surface tension of a liquid at its boiling point [MP PMT 1980]
- (a) Becomes zero
- (b) Becomes infinity
- (c) is equal to the value at room temperature
- (d) is half to the value at the room temperature
20. Surface tension of a soap solution is $1.9 \times 10^{-2} N/m$. Work done in blowing a bubble of 2.0 cm diameter will be [MP PMT 1991]
- (a) $7.6 \times 10^{-6} \pi\text{ joule}$ (b) $15.2 \times 10^{-6} \pi\text{ joule}$
- (c) $1.9 \times 10^{-6} \pi\text{ joule}$ (d) $1 \times 10^{-4} \pi\text{ joule}$
21. The surface tension of liquid is 0.5 N/m . If a film is held on a ring of area 0.02 m , its surface energy is [CPMT 1977]
- (a) $5 \times 10^{-4}\text{ joule}$ (b) $2.0 \times 10^{-4}\text{ joule}$
- (c) $4 \times 10^{-4}\text{ joule}$ (d) $0.8 \times 10^{-4}\text{ joule}$
22. What is ratio of surface energy of 1 small drop and 1 large drop, if 1000 small drops combined to form 1 large drop [CPMT 1990]
- (a) $100 : 1$ (b) $1000 : 1$
- (c) $10 : 1$ (d) $1 : 100$
23. The amount of work done in forming a soap film of size $10\text{ cm} \times 10\text{ cm}$ is (Surface tension $T = 3 \times 10^{-2} N/m$) [MP PET 1994; MP PET 2000]
- (a) $6 \times 10^{-4} J$ (b) $3 \times 10^{-4} J$
- (c) $6 \times 10^{-3} J$ (d) $3 \times 10^{-4} J$
24. The work done in blowing a soap bubble of 10 cm radius is (Surface tension of the soap solution is $\frac{3}{100} N/m$) [MP PMT 1995; MH CET 2002]
- (a) $75.36 \times 10^{-4}\text{ joule}$ (b) $37.68 \times 10^{-4}\text{ joule}$
- (c) $150.72 \times 10^{-4}\text{ joule}$ (d) 75.36 joule
25. A liquid drop of diameter D breaks upto into 27 small drops of equal size. If the surface tension of the liquid is σ , then change in surface energy is [DCE 2005]
- (a) $\pi D^2 \sigma$ (b) $2\pi D^2 \sigma$
- (c) $3\pi D^2 \sigma$ (d) $4\pi D^2 \sigma$
26. One thousand small water drops of equal radii combine to form a big drop. The ratio of final surface energy to the total initial surface energy is [MP PET 1997; KCET 1999]
- (a) $1000 : 1$ (b) $1 : 1000$
- (c) $10 : 1$ (d) $1 : 10$
27. The work done in increasing the size of a soap film from $10\text{ cm} \times 6\text{ cm}$ to $10\text{ cm} \times 11\text{ cm}$ is $3 \times 10^{-4}\text{ joule}$. The surface tension of the film is [MP PET 1999; JIPMER 2001, 02; MP PMT 2000; AIIMS 2000]
- (a) $1.5 \times 10^{-2} N/m$ (b) $3.0 \times 10^{-2} N/m$
- (c) $6.0 \times 10^{-2} N/m$ (d) $11.0 \times 10^{-2} N/m$
28. If σ be the surface tension, the work done in breaking a big drop of radius R in n drops of equal radius is [Bihar CEET 1995]
- (a) $Rn^{2/3}\sigma$ (b) $(n^{2/3} - 1)R\sigma$
- (c) $(n^{1/3} - 1)R\sigma$ (d) $4\pi R^2(n^{1/3} - 1)\sigma$
- (e) $\frac{1}{n^{1/3} - 1} \sigma R$
29. A big drop of radius R is formed by 1000 small droplets of water, then the radius of small drop is [AFMC 1998; Pb. PMT 2000]
- (a) $R/2$ (b) $R/5$



- (c) $R/6$ (d) $R/10$
30. When 10^6 small drops coalesce to make a new larger drop then the drop [RPM T 1999]
 (a) Density increases
 (b) Density decreases
 (c) Temperature increases
 (d) Temperature decreases
31. Which of the following statements are true in case when two water drops coalesce and make a bigger drop [Roorkee 1999]
 (a) Energy is released
 (b) Energy is absorbed
 (c) The surface area of the bigger drop is greater than the sum of the surface areas of both the drops
 (d) The surface area of the bigger drop is smaller than the sum of the surface areas of both the drops
32. 8000 identical water drops are combined to form a big drop. Then the ratio of the final surface energy to the initial surface energy of all the drops together is [EAMCET (Engg.) 2000]
 (a) 1 : 10 (b) 1 : 15
 (c) 1 : 20 (d) 1 : 25
33. The surface energy of liquid film on a ring of area 0.15 m^2 is (Surface tension of liquid = 5 Nm^{-1}) [EAMCET (Engg.) 2000]
 (a) 0.75 J (b) 1.5 J
 (c) 2.25 J (d) 3.0 J
34. 8 mercury drops coalesce to form one mercury drop, the energy changes by a factor of [DCE 2000]
 (a) 1 (b) 2
 (c) 4 (d) 6
35. If work done in increasing the size of a soap film from $10 \text{ cm} \times 6 \text{ cm}$ to $10 \text{ cm} \times 11 \text{ cm}$ is $2 \times 10^{-4} \text{ J}$, then the surface tension is [AIIMS 2000]
 (a) $2 \times 10^{-2} \text{ Nm}^{-1}$ (b) $2 \times 10^{-4} \text{ Nm}^{-1}$
 (c) $2 \times 10^{-6} \text{ Nm}^{-1}$ (d) $2 \times 10^{-8} \text{ Nm}^{-1}$
36. A mercury drop of radius 1 cm is sprayed into 10^6 drops of equal size. The energy expended in joules is (surface tension of Mercury is $460 \times 10^{-3} \text{ N/m}$) [EAMCET 2001]
 (a) 0.057 (b) 5.7
 (c) 5.7×10^{-4} (d) 5.7×10^{-6}
37. When two small bubbles join to form a bigger one, energy is [BHU 2001]
 (a) Released (b) Absorbed
 (c) Both (a) and (b) (d) None of these
38. A film of water is formed between two straight parallel wires of length 10 cm each separated by 0.5 cm . If their separation is increased by 1 mm while still maintaining their parallelism, how much work will have to be done (Surface tension of water = $7.2 \times 10^{-2} \text{ N/m}$)
- (a) $7.22 \times 10^{-6} \text{ Joule}$ (b) $1.44 \times 10^{-5} \text{ Joule}$
 (c) $2.88 \times 10^{-5} \text{ Joule}$ (d) $5.76 \times 10^{-5} \text{ Joule}$
39. A drop of mercury of radius 2 mm is split into 8 identical droplets. Find the increase in surface energy. (Surface tension of mercury is 0.465 J/m^2) [UPSEAT 2002]
 (a) $23.4 \mu\text{J}$ (b) $18.5 \mu\text{J}$
 (c) $26.8 \mu\text{J}$ (d) $16.8 \mu\text{J}$
40. Two small drops of mercury, each of radius R , coalesce to form a single large drop. The ratio of the total surface energies before and after the change is [AIIMS 2003; DCE 2003]
 (a) $1 : 2^{1/3}$ (b) $2^{1/3} : 1$
 (c) $2 : 1$ (d) $1 : 2$
41. Radius of a soap bubble is increased from R to $2R$ work done in this process in terms of surface tension is [BHU 2003, RPET 2001; CPMT 2004]
 (a) $24\pi R^2 S$ (b) $48\pi R^2 S$
 (c) $12\pi R^2 S$ (d) $36\pi R^2 S$
42. The work done in blowing a soap bubble of radius 0.2 m is (the surface tension of soap solution being 0.06 N/m) [Pb. PET 2002]
 (a) $192\pi \times 10^{-4} \text{ J}$ (b) $280\pi \times 10^{-4} \text{ J}$
 (c) $200\pi \times 10^{-3} \text{ J}$ (d) None of these
43. A liquid film is formed in a loop of area 0.05 m . Increase in its potential energy will be ($T = 0.2 \text{ N/m}$) [RPM T 2002]
 (a) $5 \times 10^{-2} \text{ J}$ (b) $2 \times 10^{-2} \text{ J}$
 (c) $3 \times 10^{-2} \text{ J}$ (d) None of these
44. In order to float a ring of area 0.04 m in a liquid of surface tension 75 N/m , the required surface energy will be [RPM T 2003]
 (a) 3 J (b) 6.5 J
 (c) 1.5 J (d) 4 J
45. If two soap bubbles of equal radii r coalesce then the radius of curvature of interface between two bubbles will be [J&K CET 2005]
 (a) r (b) 0
 (c) Infinity (d) $1/2r$

Angle of Contact

1. A liquid does not wet the sides of a solid, if the angle of contact is [MP PAT 1990; AFMC 1988; MNR 1998; RPM T 1999, 2003; Pb. PMT 2002 KCET 2005]
 (a) Zero (b) Obtuse (More than 90°)
 (c) Acute (Less than 90°) (d) 90°
2. The meniscus of mercury in the capillary tube is [MP PET/PMT 1988]
 (a) Convex (b) Concave
 (c) Plane (d) Uncertain
3. When the radius of curvature is increased the angle of contact of a liquid [MP PET 2001]

- (a) Increases
(b) Decreases
(c) Remains the same
(d) First increases and then decreases
4. The angle of contact between glass and mercury is [MP PMT 1987]
(a) 0° (b) 30°
(c) 90° (d) 135°
5. A mercury drop does not spread on a glass plate because the angle of contact between glass and mercury is [MP PMT 1984]
(a) Acute (b) Obtuse
(c) Zero (d) 90°
6. A liquid is coming out from a vertical tube. The relation between the weight of the drop W , surface tension of the liquid T and radius of the tube r is given by, if the angle of contact is zero
(a) $W = \pi^2 T$ (b) $W = 2\pi r T$
(c) $W = 2r^2 \pi T$ (d) $W = \frac{3}{4} \pi^3 T$
7. The parts of motor cars are polished by chromium because the angle of contact between water and chromium is
(a) 0° (b) 90°
(c) Less than 90° (d) Greater than 90°
8. A glass plate is partly dipped vertically in the mercury and the angle of contact is measured. If the plate is inclined, then the angle of contact will
(a) Increase (b) Remain unchanged
(c) Increase or decrease (d) Decrease
9. The liquid meniscus in capillary tube will be convex, if the angle of contact is [EAMCET (Med.) 1995; KCET 2001; Pb. PET 2000]
(a) Greater than 90° (b) Less than 90°
(c) Equal to 90° (d) Equal to 0°
10. If a water drop is kept between two glass plates, then its shape is
(a)  (b) 
(c)  (d) None of these
11. The value of contact angle for kerosene with solid surface. [RPMT 2000]
(a) 0° (b) 90°
(c) 45° (d) 33°
12. Nature of meniscus for liquid of 0° angle of contact [RPET 2001]
(a) Plane (b) Parabolic
(c) Semi-spherical (d) Cylindrical
13. A liquid wets a solid completely. The meniscus of the liquid in a sufficiently long tube is [Kerala (Engg.) 2002]

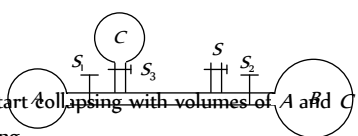
- (a) Flat (b) Concave
(c) Convex (d) Cylindrical

14. What is the shape when a non-wetting liquid is placed in a capillary tube [AFMC 2004]
(a) Concave upward (b) Convex upward
(c) Concave downward (d) Convex downward
15. For which of the two pairs, the angle of contact is same [J & K CET 2004]
(a) Water and glass; glass and mercury
(b) Pure water and glass; glass and alcohol
(c) Silver and water; mercury and glass
(d) Silver and chromium; water and chromium
16. If the surface of a liquid is plane, then the angle of contact of the liquid with the walls of container is [MH CET 2004]
(a) Acute angle (b) Obtuse angle
(c) 90° (d) 0°

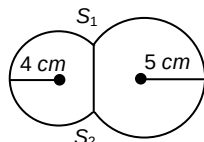
Pressure Difference

1. A soap bubble assumes a spherical surface. Which of the following statement is wrong [NCERT 1976]
(a) The soap film consists of two surface layers of molecules back to back
(b) The bubble encloses air inside it
(c) The pressure of air inside the bubble is less than the atmospheric pressure; that is why the atmospheric pressure has compressed it equally from all sides to give it a spherical shape
(d) Because of the elastic property of the film, it will tend to shrink to as small a surface area as possible for the volume it has enclosed
2. If two soap bubbles of different radii are in communication with each other [NCERT 1980; MP PMT/PET 1988; AIEEE 2004]
(a) Air flows from larger bubble into the smaller one
(b) The size of the bubbles remains the same
(c) Air flows from the smaller bubble into the large one and the larger bubble grows at the expense of the smaller one [CPMT 1997]
(d) The air flows from the larger
3. The surface tension of soap solution is $25 \times 10^{-3} \text{ Nm}^{-1}$. The excess pressure inside a soap bubble of diameter 1 cm is
(a) 10 Pa (b) 20 Pa
(c) 5 Pa (d) None of the above
4. When two soap bubbles of radius r_1 and r_2 ($r_2 > r_1$) coalesce, the radius of curvature of common surface is [MP PMT 1996]
(a) $r_2 - r_1$ (b) $\frac{r_2 - r_1}{r_1 r_2}$
(c) $\frac{r_1 r_2}{r_2 - r_1}$ (d) $r_2 + r_1$
5. The excess pressure due to surface tension in a spherical liquid drop of radius r is directly proportional to [MP PMT 1987; KCET 2000]
(a) r (b) r^2



- (c) r^{-1} (d) r^{-2}
6. A long cylindrical glass vessel has a small hole of radius ' r ' at its bottom. The depth to which the vessel can be lowered vertically in the deep water bath (surface tension T) without any water entering inside is [MP PMT 1990]
- (a) $4\pi prg$ (b) $3\pi prg$
(c) $2\pi prg$ (d) T/prg
7. If the surface tension of a soap solution is 0.03 MKS units, then the excess of pressure inside a soap bubble of diameter 6 mm over the atmospheric pressure will be
- (a) Less than 40 N/m (b) Greater than 40 N/m
(c) Less than 20 N/m (d) Greater than 20 N/m
8. The excess of pressure inside a soap bubble than that of the outer pressure is [MP PMT 1989; BHU 1995; MH CET 2002; RPET 2003; AMU (Engg.) 2000]
- (a) $\frac{2T}{r}$ (b) $\frac{4T}{r}$
(c) $\frac{T}{2r}$ (d) $\frac{T}{r}$
9. The pressure of air in a soap bubble of 0.7 cm diameter is 8 mm of water above the pressure outside. The surface tension of the soap solution is [MP PET 1991; MP PMT 1997]
- (a) 100 dyne/cm (b) 68.66 dyne/cm
(c) 137 dyne/cm (d) 150 dyne/cm
10. Pressure inside two soap bubbles are 1.01 and 1.02 atmospheres. Ratio between their volumes is [MP PMT 1991]
- (a) 102 : 101 (b) (102)³ : (101)³
(c) 8 : 1 (d) 2 : 1
11. A capillary tube of radius r is dipped in a liquid of density ρ and surface tension S . If the angle of contact is θ , the pressure difference between the two surfaces in the beaker and the capillary
- (a) $\frac{S}{r} \cos \theta$ (b) $\frac{2S}{r} \cos \theta$
(c) $\frac{S}{r \cos \theta}$ (d) $\frac{2S}{r \cos \theta}$
12. The radii of two soap bubbles are r_1 and r_2 . In isothermal conditions, two meet together in vacuum. Then the radius of the resultant bubble is given by [MP PMT 2001; RPET 1999; EAMCET 2003]
- (a) $R = (r_1 + r_2)/2$ (b) $R = r_1(r_1 r_2 + r_2)$
(c) $R^2 = r_1^2 + r_2^2$ (d) $R = r_1 + r_2$
13. The adjoining diagram shows three soap bubbles A , B and C prepared by blowing the capillary tube fitted with stop cocks, S_1 , S_2 and S_3 . With stop cock S_3 closed and stop cocks S_1 , S_2 and S_3 opened
- 
- (a) B will start collapsing with volumes of A and C increasing
(b) C will start collapsing with volumes of A and B increasing
- (c) C and A both will start collapsing with the volume of B increasing
(d) Volumes of A , B and C will become equal at equilibrium
14. When a large bubble rises from the bottom of a lake to the surface, its radius doubles. If atmospheric pressure is equal to that of column of water height H , then the depth of lake is [AIIMS 1995; AFMC 1997]
- (a) H (b) $2H$
(c) $7H$ (d) $8H$
15. A soap bubble in vacuum has a radius of 3 cm and another soap bubble in vacuum has a radius of 4 cm. If the two bubbles coalesce under isothermal condition, then the radius of the new bubble is [MP PMT/PT 1997]
- (a) 2.3 cm (b) 4.5 cm
(c) 5 cm (d) 7 cm
16. The volume of an air bubble becomes three times as it rises from the bottom of a lake to its surface. Assuming atmospheric pressure to be 75 cm of Hg and the density of water to be 1/10 of the density of mercury, the depth of the lake is
- (a) 5 m (b) 10 m
(c) 15 m (d) 20 m
17. Excess pressure of one soap bubble is four times more than the other. Then the ratio of volume of first bubble to another one is [CPMT 1997; MP PMT 1997]
- (a) 1 : 64 (b) 1 : 4
(c) 64 : 1 (d) 1 : 2
18. There are two liquid drops of different radii. The excess pressure inside over the outside is [JIPMER 1999]
- (a) More in the big drop
(b) More in the small drop
(c) Equal in both drops
(d) There is no excess pressure inside the drops
19. If pressure at half the depth of a lake is equal to 2/3 pressure at the bottom of the lake then what is the depth of the lake [RPET 2000]
- (a) 10 m (b) 20 m
(c) 60 m (d) 30 m
20. If the radius of a soap bubble is four times that of another, then the ratio of their pressures will be [AIIMS 2000]
- (a) 1 : 4 (b) 4 : 1
(c) 16 : 1 (d) 1 : 16
21. A spherical drop of water has radius 1 mm. If surface tension of water is 70×10^{-3} N/m difference of pressures between inside and out side of the spherical drop is [CPMT 2000; AIIMS 2000]
- (a) 35 N/m² (b) 70 N/m²
(c) 140 N/m² (d) Zero
22. The pressure at the bottom of a tank containing a liquid does not depend on [CPMT 1988; Kerala (Engg.) 2001]
- (a) Acceleration due to gravity
(b) Height of the liquid column
(c) Area of the bottom surface
(d) Nature of the liquid
23. In capillary pressure below the curved surface of water will be

- (a) Equal to atmospheric
(b) Equal to upper side pressure
(c) More than upper side pressure
(d) Lesser than upper side pressure
24. Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be [MP PMT 2002]



- (a) 4 cm
(b) 20 cm
(c) 5 cm
(d) 4.5 cm
25. The pressure inside a small air bubble of radius 0.1 mm situated just below the surface of water will be equal to [Take surface tension of water $70 \times 10^{-3} \text{ Nm}^{-1}$ and atmospheric pressure = $1.013 \times 10^5 \text{ Nm}^{-2}$] [AMU (Med.) 2002]
- (a) $2.054 \times 10^3 \text{ Pa}$ (b) $1.027 \times 10^3 \text{ Pa}$
(c) $1.027 \times 10^5 \text{ Pa}$ (d) $2.054 \times 10^5 \text{ Pa}$
26. Two bubbles A and B ($A > B$) are joined through a narrow tube. Then [UPSEAT 2001; Kerala (Med.) 2002]
- (a) The size of A will increase
(b) The size of B will increase
(c) The size of B will increase until the pressure equals
(d) None of these
27. Two soap bubbles have different radii but their surface tension is the same. Mark the correct statement [MP PMT 2004]
- (a) Internal pressure of the smaller bubble is higher than the internal pressure of the larger bubble
(b) Pressure of the larger bubble is higher than the smaller bubble
(c) Both bubbles have the same internal pressure
(d) None of the above
28. If the excess pressure inside a soap bubble is balanced by oil column of height 2 mm, then the surface tension of soap solution will be ($r = 1 \text{ cm}$ and density $d = 0.8 \text{ gm/cc}$) [J & K CET 2004]
- (a) 3.9 N/m (b) $3.9 \times 10^{-4} \text{ N/m}$
(c) $3.9 \times 10^{-4} \text{ dyne/m}$ (d) 3.9 dyne/m
29. In Jager's method, at the time of bursting of the bubble [RPET 2002]
- (a) The internal pressure of the bubble is always greater than external pressure
(b) The internal pressure of the bubble is always equal to external pressure
(c) The internal pressure of the bubble is always less than external pressure
(d) The internal pressure of the bubble is always slightly greater than external pressure

30. The excess pressure in a soap bubble is thrice that in other one. Then the ratio of their volume is [RPMT 2003; CPMT 2001]

- (a) 1 : 3 (b) 1 : 9
(c) 27 : 1 (d) 1 : 27

Capillarity

1. When two capillary tubes of different diameters are dipped vertically, the rise of the liquid is [NCERT 1978]
- (a) Same in both the tubes
(b) More in the tube of larger diameter
(c) Less in the tube of smaller diameter
(d) More in the tube of smaller diameter
2. Due to capillary action, a liquid will rise in a tube, if the angle of contact is [DPMT 1984; AFMC 1988; BHU 2001]
- (a) Acute (b) Obtuse
(c) 90° (d) Zero
3. In the state of weightlessness, a capillary tube is dipped in water, then water
- (a) Will not rise at all
(b) Will rise to same height as at atmospheric pressure
(c) Will rise to less height than at atmospheric pressure
(d) Will rise up to the upper end of the capillary tube of any length
4. Two parallel glass plates are dipped partly in the liquid of density ' d ' keeping them vertical. If the distance between the plates is ' x ', surface tension for liquids is T and angle of contact is θ , then rise of liquid between the plates due to capillary will be
- (a) $\frac{T \cos \theta}{xd}$ (b) $\frac{2T \cos \theta}{xdg}$
(c) $\frac{2T}{xdg \cos \theta}$ (d) $\frac{T \cos \theta}{xdg}$
5. Water rises in a capillary tube to a certain height such that the upward force due to surface tension is balanced by $75 \times 10^{-4} \text{ N}$ force due to the weight of the liquid. If the surface tension of water is $6 \times 10^{-2} \text{ Nm}^{-1}$, the inner circumference of the capillary must be [CPMT 1995]
- (a) $1.25 \times 10^{-2} \text{ m}$ (b) $0.50 \times 10^{-2} \text{ m}$
(c) $6.5 \times 10^{-2} \text{ m}$ (d) $12.5 \times 10^{-2} \text{ m}$
6. It is not possible to write directly on blotting paper or newspaper with ink pen
- (a) Because of viscosity (b) Because of inertia
(c) Because of friction (d) Because of capillarity
7. Two capillary tubes P and Q are dipped in water. The height of water level in capillary P is 2/3 to the height in Q capillary. The ratio of their diameters is [MP PMT 1985]
- (a) 2 : 3 (b) 3 : 2
(c) 3 : 4 (d) 4 : 3



8. Two capillaries made of same material but of different radii are dipped in a liquid. The rise of liquid in one capillary is 2.2 cm and that in the other is 6.6 cm . The ratio of their radii is
(a) $9 : 1$ (b) $1 : 9$
(c) $3 : 1$ (d) $1 : 3$
9. Two capillaries made of the same material with radii $r_1 = 1\text{ mm}$ and $r_2 = 2\text{ mm}$. The rise of the liquid in one capillary ($r_1 = 1\text{ mm}$) is 30 cm , then the rise in the other will be
(a) 7.5 cm (b) 60 cm
(c) 15 cm (d) 120 cm
10. When a capillary is dipped in water, water rises to a height h . If the length of the capillary is made less than h , then
(a) The water will come out
(b) The water will not come out
(c) The water will not rise
(d) The water will rise but less than height of capillary
11. Water rises upto 10 cm height in a long capillary tube. If this tube is immersed in water so that the height above the water surface is only 8 cm , then [MP PMT 1991]
(a) Water flows out continuously from the upper end
(b) Water rises upto upper end and forms a spherical surface
(c) Water only rises upto 6 cm height
(d) Water does not rise at all
12. A vessel, whose bottom has round holes with diameter of 0.1 mm , is filled with water. The maximum height to which the water can be filled without leakage is
(S.T. of water $= 75\text{ dyne/cm}$, $g = 1000\text{ cm/s}^2$) [CPMT 1989; J&K CET 2004]
(a) 100 cm (b) 75 cm
(c) 50 cm (d) 30 cm
13. Water rises in a capillary tube when its one end is dipped vertically in it, is 3 cm . If the surface tension of water is $75 \times 10^{-3}\text{ N/m}$, then the diameter of capillary will be [MP PET 1989]
(a) 0.1 mm (b) 0.5 mm
(c) 1.0 mm (d) 2.0 mm
14. A capillary tube made of glass is dipped into mercury. Then [MP PET 1996]
(a) Mercury rises in the capillary tube
(b) Mercury rises and flows out of the capillary tube
(c) Mercury descends in the capillary tube
(d) Mercury neither rises nor descends in the capillary tube
15. By inserting a capillary tube upto a depth l in water, the water rises to a height h . If the lower end of the capillary is closed inside water and the capillary is taken out and closed end opened, to what height the water will remain in the tube [RPET 1996; DPMT 2000]
(a) Zero (b) $l + h$
(c) $2h$ (d) h
16. If the diameter of a capillary tube is doubled, then the height of the liquid that will rise is [MP PET 1990]
(a) Twice (b) Half
(c) Same as earlier (d) None of these
17. If the surface tension of water is 0.06 N/m , then the capillary rise in a tube of diameter 1 mm is ($\theta = 0^\circ$) [AFMC 1998]
(a) 1.22 cm (b) 2.44 cm
(c) 3.12 cm [MP PET 1991] (d) 3.86 cm
18. Two capillary tubes of radii 0.2 cm and 0.4 cm are dipped in the same liquid. The ratio of heights through which liquid will rise in the tubes is [MNR 1998]
(a) $1 : 2$ (b) $2 : 1$
(c) $1 : 4$ (d) $4 : 1$
19. A capillary tube when immersed vertically in liquid records a rise of 3 cm . If the tube is immersed in the liquid at an angle of 60° with the vertical. The length of the liquid column along the tube is
(a) 9 cm (b) 6 cm
(c) 3 cm (d) 2 cm
20. The action of a nib split at the top is explained by [JIPMER 1999]
(a) Gravity flow (b) Diffusion of fluid
(c) Capillary action (d) Osmosis of liquid
21. The correct relation is [RPMT 2002]
(a) $r = \frac{2T \cos \theta}{hdg}$ (b) $r = \frac{hdg}{2T \cos \theta}$
(c) $r = \frac{2T dgh}{\cos \theta}$ (d) $r = \frac{T \cos \theta}{2hdg}$
22. Water rises upto a height h in a capillary on the surface of earth in stationary condition. Value of h increases if this tube is taken
(a) On sun
(b) On poles
(c) In a lift going upward with acceleration
(d) In a lift going downward with acceleration
23. During capillary rise of a liquid in a capillary tube, the surface of contact that remains constant is of [Pb. PMT 2000]
(a) Glass and liquid (b) Air and glass
(c) Air and liquid (d) All of these
24. A shell having a hole of radius r is dipped in water. It holds the water upto a depth of h then the value of r is [RPMT 2000]
(a) $r = \frac{2T}{hdg}$ (b) $r = \frac{T}{hdg}$
(c) $r = \frac{Tg}{hd}$ (d) None of these
25. In a capillary tube, water rises by 1.2 mm . The height of water that will rise in another capillary tube having half the radius of the first, is [CPMT 2001; Pb. PET 2002]
(a) 1.2 mm (b) 2.4 mm
(c) 0.6 mm (d) 0.4 mm
26. If capillary experiment is performed in vacuum then for a liquid there [RPET 2001]

- (a) It will rise (b) Will remain same
(c) It will fall (d) Rise to the top
27. If liquid level falls in a capillary then radius of capillary will [RPET 2001]
(a) Increase (b) Decrease
(c) Unchanged (d) None of these
28. Water rises to a height h in a capillary at the surface of earth. On the surface of the moon the height of water column in the same capillary will be [MP PMT 2001]
(a) $6h$ (b) $\frac{1}{6}h$
(c) h (d) Zero
29. Two capillary tubes of same diameter are put vertically one each in two liquids whose relative densities are 0.8 and 0.6 and surface tensions are 60 and 50 dyne/cm respectively Ratio of heights of liquids in the two tubes $\frac{h_1}{h_2}$ is [MP PMT 2002]
(a) $\frac{10}{9}$ (b) $\frac{3}{10}$
(c) $\frac{10}{3}$ (d) $\frac{9}{10}$
30. Water rises in a vertical capillary tube upto a height of 2.0 cm . If the tube is inclined at an angle of 60° with the vertical, then upto what length the water will rise in the tube [UPSEAT 2002]
(a) 2.0 cm (b) 4.0 cm
(c) $\frac{4}{\sqrt{3}}\text{cm}$ (d) $2\sqrt{2}\text{cm}$
31. The surface tension for pure water in a capillary tube experiment is [MH CET 2002]
(a) $\frac{\rho g}{2hr}$ (b) $\frac{2}{hr\rho g}$
(c) $\frac{r\rho g}{2h}$ (d) $\frac{hr\rho g}{2}$
32. In a capillary tube experiment, a vertical 30 cm long capillary tube is dipped in water. The water rises up to a height of 10 cm due to capillary action. If this experiment is conducted in a freely falling elevator, the length of the water column becomes [Orissa JEE 2003; AIEEE 2005]
(a) 10 cm (b) 20 cm
(c) 30 cm (d) Zero
33. Radius of a capillary is $2 \times 10^{-3}\text{m}$. A liquid of weight $6.28 \times 10^{-4}\text{N}$ may remain in the capillary then the surface tension of liquid will be [RPET 2003]
(a) $5 \times 10^{-3}\text{N/m}$ (b) $5 \times 10^{-2}\text{N/m}$
(c) 5 N/m (d) 50 N/m
34. Two long capillary tubes A and B of radius $R_1 > R_2$ dipped in same liquid. Then [Orissa PMT 2004]
(a) Water rise is more in A than B
(b) Water rises more in B than A
(c) Same water rise in both
(d) All of these according to the density of water
35. If water rises in a capillary tube upto 3 cm . What is the diameter of capillary tube (Surface tension of water = $7.2 \times 10^{-2}\text{N/m}$)
(a) $9.6 \times 10^{-2}\text{m}$ (b) $9.6 \times 10^{-3}\text{m}$
(c) $9.6 \times 10^{-4}\text{m}$ (d) $9.6 \times 10^{-5}\text{m}$
36. When a capillary is dipped in water, water rises 0.015 m in it. If the surface tension of water is $75 \times 10^{-2}\text{N/m}$, the radius of capillary is
(a) 0.1 mm (b) 0.5 mm
(c) 1 mm (d) 2 mm
37. In a capillary tube, water rises to 3 mm . The height of water that will rise in another capillary tube having one-third radius of the first is [BHU 2004]
(a) 1 mm (b) 3 mm
(c) 6 mm (d) 9 mm
38. Kerosene oil rises up the wick in a lantern [NCERT 1980; MNR 1985]
(a) Due to surface tension of the oil
(b) The wick attracts the kerosene oil
(c) Of the diffusion of the oil through the wick
(d) None of the above
39. Water rises against gravity in a capillary tube when its one end is dipped into water because
(a) Pressure below the meniscus is less than atmospheric pressure
(b) Pressure below the meniscus is more than atmospheric pressure
(c) Capillary attracts water
(d) Of viscosity
40. A capillary tube of radius R is immersed in water and water rises in it to a height H . Mass of water in the capillary tube is M . If the radius of the tube is doubled, mass of water that will rise in the capillary tube will now be [RPMT 1997; RPET 1999; CPMT 2002]
(a) M (b) $2M$
(c) $M/2$ (d) $4M$
41. Water rises up to a height h in a capillary tube of certain diameter. This capillary tube is replaced by a similar tube of half the diameter. Now the water will rise to the height of [Kerala PMT 2005]
(a) $4h$ (b) $3h$
(c) $2h$ (d) h

Critical Thinking

Objective Questions

1. There is a horizontal film of soap solution. On it a thread is placed in the form of a loop. The film is pierced inside the loop and the thread becomes a circular loop of radius R . If the surface tension of the loop be T , then what will be the tension in the thread
(a) $\pi R^2 / T$ (b) $\pi R^2 T$
(c) $2\pi RT$ (d) $2RT$
2. A large number of water drops each of radius r combine to have a drop of radius R . If the surface tension is T and the mechanical equivalent of heat is J , then the rise in temperature will be [MP PET 1994; DPMT 2002]
(a) $\frac{2T}{rJ}$ (b) $\frac{3T}{RJ}$
(c) $\frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$ (d) $\frac{2T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$

3. An air bubble in a water tank rises from the bottom to the top. Which of the following statements are true

[Roorkee 2000]

- (a) Bubble rises upwards because pressure at the bottom is less than that at the top.
 (b) Bubble rises upwards because pressure at the bottom is greater than that at the top.
 (c) As the bubble rises, its size increases
 (d) As the bubble rises, its size decreases

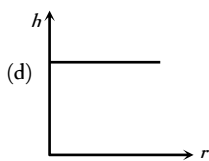
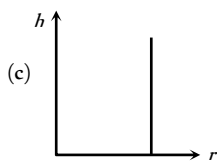
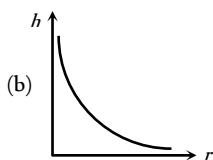
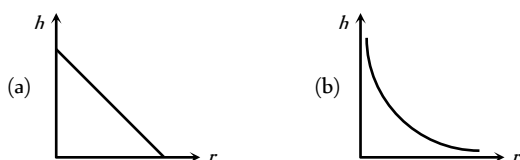
4. In a surface tension experiment with a capillary tube water rises upto 0.1 m . If the same experiment is repeated on an artificial satellite, which is revolving around the earth, water will rise in the capillary tube upto a height of

[Roorkee 1992]

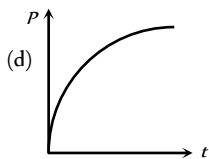
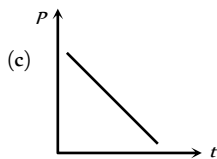
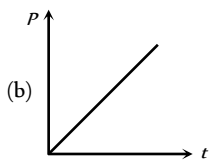
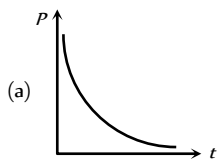
- (a) 0.1 m
 (b) 0.2 m
 (c) 0.98 m
 (d) Full length of the capillary tube

Graphical Questions

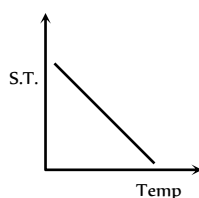
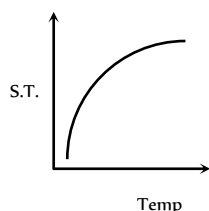
1. The correct curve between the height or depression h of liquid in a capillary tube and its radius is



2. A soap bubble is blown with the help of a mechanical pump at the mouth of a tube. The pump produces a certain increase per minute in the volume of the bubble, irrespective of its internal pressure. The graph between the pressure inside the soap bubble and time t will be-

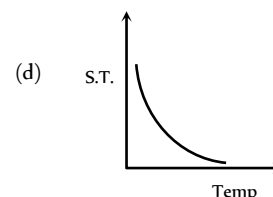
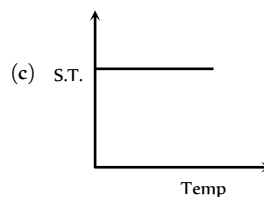


3. Which graph represents the variation of surface tension with temperature over small temperature ranges for water



(a)

(b)



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

- Assertion : It is easier to spray water in which some soap is dissolved.
 Reason : Soap is easier to spread.
- Assertion : It is better to wash the clothes in cold soap solution.
 Reason : The surface tension of cold solution is more than the surface tension of hot solution.
- Assertion : When height of a tube is less than liquid rise in the capillary tube, the liquid does not overflow.
 Reason : Product of radius of meniscus and height of liquid in capillary tube always remains constant.
- Assertion : A needle placed carefully on the surface of water may float, whereas a ball of the same material will always sink.
 Reason : The buoyancy of an object depends both on the material and shape of the object.
- Assertion : A large force is required to draw apart normally two glass plates enclosing a thin water film.
 Reason : Water works as glue and sticks two glass plates.
- Assertion : The impurities always decrease the surface tension of a liquid.
 Reason : The change in surface tension of the liquid depends upon the degree of contamination of the impurity.
- Assertion : The angle of contact of a liquid decrease with increase in temperature.



504 Surface Tension

- Reason : With increase in temperature, the surface tension of liquid increase.
8. Assertion : The concept of surface tension is held only for liquids.
- Reason : Surface tension does not hold for gases.
9. Assertion : At critical temperature, surface tension of a liquid becomes zero.
- Reason : At this temperature, intermolecular forces for liquids and gases become equal. Liquid can expand without any restriction.
10. Assertion : A large soap bubble expands while a small bubble shrinks, when they are connected to each other by a capillary tube.
- Reason : The excess pressure inside bubble (or drop) is inversely proportional to the radius.
11. Assertion : Tiny drops of liquid resist deforming forces better than bigger drops.
- Reason : Excess pressure inside a drop is directly proportional to surface tension.
12. Assertion : The water rises higher in a capillary tube of small diameter than in the capillary tube of large diameter.
- Reason : Height through which liquid rises in a capillary tube is inversely proportional to the diameter of the capillary tube.
13. Assertion : Hot soup tastes better than the cold soup.
- Reason : Hot soup has high surface tension and it does not spread properly on our tongue.
14. Assertion : The shape of a liquid drop is spherical.
- Reason : The pressure inside the drop is greater than that of outside.

Answers

Surface Tension

1	a	2	b	3	b	4	a	5	d
6	a	7	b	8	b	9	b	10	cd
11	d	12	a	13	b	14	b	15	c
16	d	17	a	18	c	19	c	20	d
21	b	22	d	23	a	24	a	25	c
26	d	27	d	28	b	29	b	30	d
31	d	32	c	33	d	34	c	35	a
36	b	37	b	38	a	39	a	40	c
41	d	42	c	43	d	44	a	45	a
46	a								

Surface Energy

1	a	2	d	3	a	4	d	5	d
6	b	7	c	8	d	9	c	10	c
11	c	12	c	13	c	14	a	15	b
16	b	17	d	18	a	19	a	20	b
21	b	22	d	23	a	24	a	25	b
26	d	27	b	28	d	29	d	30	c
31	ad	32	c	33	b	34	c	35	a
36	a	37	a	38	b	39	a	40	b
41	a	42	a	43	b	44	a	45	c

Angle of Contact

1	b	2	a	3	b	4	d	5	b
6	b	7	d	8	b	9	a	10	c
11	a	12	c	13	b	14	b	15	b
16	d								

Pressure Difference

1	c	2	c	3	b	4	c	5	c
6	c	7	b	8	b	9	b	10	c
11	b	12	c	13	c	14	c	15	c
16	c	17	a	18	b	19	b	20	a
21	c	22	c	23	d	24	b	25	c
26	a	27	a	28	b	29	a	30	d

Capillarity

1	d	2	a	3	d	4	b	5	d
6	d	7	b	8	c	9	c	10	b
11	b	12	d	13	c	14	c	15	d
16	b	17	b	18	b	19	b	20	c
21	a	22	d	23	c	24	a	25	b
26	a	27	a	28	a	29	d	30	b
31	d	32	c	33	b	34	a	35	a
36	c	37	d	38	a	39	a	40	b
41	c								

Critical Thinking Questions

1	d	2	c	3	bc	4	d		
---	---	---	---	---	----	---	---	--	--



Graphical Questions

1	b	2	a	3	b				
---	---	---	---	---	---	--	--	--	--

Assertion and Reason

1	c	2	e	3	a	4	c	5	c
6	e	7	c	8	b	9	a	10	a
11	b	12	a	13	c	14	b		

AS

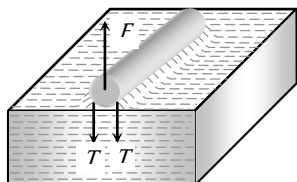
Answers and Solutions

Surface Tension

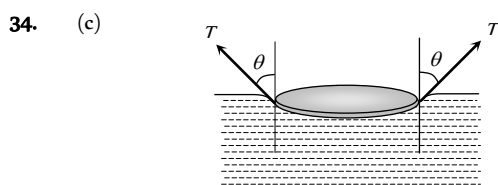
1. (a)
2. (b)
3. (b)
4. (a)
5. (d) Soap helps to lower the surface tension of solution, thus soap get stick to the dust particles and grease and these are removed by action of water.
6. (a)
7. (b)
8. (b)
9. (b)
10. (c,d) At critical temperature ($T_c = 370^\circ C = 643 K$), the surface tension of water is zero.
11. (d)
12. (a) Weight of spiders or insects can be balanced by vertical component of force due to surface tension.
13. (b)
14. (b)
15. (c) Force on each side $= 2TL$ (due to two surfaces)
 $\therefore$ Force on the frame $= 4(2TL) = 8TL$
16. (d)
17. (a)
18. (c) This happens due to viscosity.
19. (c)
20. (d) The total length of the circular plate on which the force will act $= 2\pi R$
 Force to pull $= 2\pi RT = 2 \times \pi \times 5 \times 75 = 750\pi$ dynes

21. (b)
22. (d) $T = T_0(1 - \alpha t)$
23. (a) Due to force of attraction it is not easier to separate the two glass plates.
24. (a) Soluble impurities increases the surface tension.

25. (c) $T = \frac{F}{2l} = \frac{728}{2 \times 5}$
 $\therefore T = 72.8 \text{ dyne/cm}$



26. (d) Cohesive force > Adhesive force, so shape of liquid surface near the solid would be convex.
 For example mercury surface in glass capillary is convex.
27. (d) Surface tension decreases with increase in temperature.
28. (b)
29. (b)
30. (d)
31. (d) Because surface tension of water > surface tension of oil
32. (c) Surface tension pulls the plates towards each other.
33. (d) Sphere has the minimum surface area for the given volume of the liquid.



Weight of metal disc = total upward force

= upthrust force + force due to surface tension

= weight of displaced water + $T \cos \theta (2\pi r)$

$$= W + 2\pi r T \cos \theta$$

35. (a) $T = \frac{F}{2l} = \frac{2 \times 10^{-2}}{2 \times 10 \times 10^{-2}} = 0.1 \text{ N/m}$

36. (b) Surface tension of water decrease with rise in temperature.

37. (b)

38. (a) Force required to separate the plates

$$F = \frac{2TA}{t} = \frac{2 \times 70 \times 10^{-3} \times 10^{-2}}{0.05 \times 10^{-3}} = 28 \text{ N}$$

39. (a)

40. (c) The cohesive force is the force of attraction between the molecules of same substance.

41. (d)

42. (c) $T = \frac{F}{l} = \frac{[MLT^{-2}]}{[L]} = [ML^0T^{-2}]$

43. (d) Net force on stick = $F_1 - F_2 = (T_1 - T_2)l$
 $= (0.07 - 0.06)l = 0.01 \times 2 = 0.02 \text{ N}$

44. (a) Because film tries to cover minimum surface area.

45. (a) Force required, $F = 2\pi r T = 2\pi \times 2 \times 70 = 280\pi \text{ Dyne}$

46. (a)

Surface Energy

1. (a) Energy needed = Increment in surface energy
 $= (\text{surface energy of } n \text{ small drops}) - (\text{surface energy of one big drop})$

$$= n4\pi r^2 T - 4\pi R^2 T = 4\pi T(nr^2 - R^2)$$

2. (d)

3. (a) When two droplets merge with each other, their surface energy decreases.

$$W = T(\Delta A) = (\text{negative}) \text{ i.e. energy is released.}$$

4. (d) $E = 4\pi R^2 T(n^{1/3} - 1)$

$$= 4 \times 3.14 \times (1.4 \times 10^{-1})^2 \times 75(125^{1/3} - 1) = 74 \text{ erg}$$

5. (d) $W = 8\pi T(R_2^2 - R_1^2) = 8\pi T[(2r)^2 - (r)^2] = 24\pi r^2 T$

6. (b) Work done in splitting a water drop of radius R into n drops of equal size = $4\pi R^2 T(n^{1/3} - 1)$

$$= 4\pi \times (10^{-3})^2 \times 72 \times 10^{-3} \times (10^{6/3} - 1)$$

$$= 4\pi \times 10^{-6} \times 72 \times 10^{-3} \times 99 = 8.95 \times 10^{-5} \text{ J}$$

7. (c) $W = 4\pi R^2 T(r^{1/3} - 1) = 4\pi R^2 T(8^{1/3} - 1) = 4\pi R^2 T$

8. (d) $W = T \times 8\pi(r_2^2 - r_1^2) = T \times 8\pi\left(\frac{D^2}{4} - \frac{d^2}{4}\right)$

$$= 2\pi(D^2 - d^2)T$$

9. (c) Work done to increase the diameter of bubble from d to D

$$W = 2\pi(D^2 - d^2)T = 2\pi[(2D)^2 - (D)^2]T = 6\pi D^2 T$$

10. (c) $W = 8\pi T(r_2^2 - r_1^2) = 8\pi T\left[\left(\frac{2}{\sqrt{\pi}}\right)^2 - \left(\frac{1}{\sqrt{\pi}}\right)^2\right]$

$$\therefore W = 8 \times \pi \times 30 \times \frac{3}{\pi} = 720 \text{ erg}$$

11. (c) $W = T \times \Delta A = 5 \times 2 \times (0.02)$ (Film has two free surfaces)

$$= 2 \times 10^{-1} \text{ J}$$

12. (c) $W = 8\pi R^2 T \therefore W \propto R^2$ (T is constant)

If radius becomes double then work done will become four times.

13. (c) $W = 4\pi R^2 T(n^{1/3} - 1) = 4\pi \times 1 \times 50(10^{3/3} - 1)$

$$= 1800\pi \text{ erg}$$

14. (a)

15. (b) Surface energy of combined drop will be lowered, so excess surface energy will raise the temperature of the drop.
16. (b) Surface energy = surface tension $\times$ increment in area
 $= T \times A$
17. (d) $W = 8\pi R^2 T = 8 \times \pi \times (10^{-2})^2 \times 2 \times 10^{-2} = 16\pi \times 10^{-6} J$
18. (a) $E = 4\pi R^2 T(n^{1/3} - 1)$
 $= 4 \times 3.14 \times 10^{-4} \times 35 \times 10^{-1} (10^{6/3} - 1) = 4.4 \times 10^{-3} J$
19. (a)
20. (b) $W = 8\pi R^2 T = 8\pi \times (1 \times 10^{-2})^2 \times 1.9 \times 10^{-2} = 15.2 \times 10^{-6} \pi J$
21. (b) Surface energy = $T \times \Delta A = 0.5 \times 2 \times (0.02) = 2 \times 10^{-2} J$
22. (d) Volume of liquid remain same i.e. volume of 1000 small drops will be equal to volume of one big drop
 $n \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow 1000r^3 = R^3 \Rightarrow R = 10r \therefore \frac{r}{R} = \frac{1}{10}$
 $\frac{\text{surface energy of one small drop}}{\text{surface energy of one big drop}} = \frac{4\pi r^2 T}{4\pi R^2 T} = \frac{1}{100}$
23. (a) $E = T \times \Delta A = 3 \times 10^{-2} \times 2(100 \times 10^{-4}) = 6 \times 10^{-4} J$
24. (a) $W = 8\pi R^2 T = 8 \times 3.14 \times (10 \times 10^{-2})^2 \times \frac{3}{100}$
 $= 7.536 \times 10^{-3} J$
25. (b) Work done = $4\pi R^2 T(n^{1/3} - 1) = 4\pi \left(\frac{D}{2}\right)^2 \sigma(n^{1/3} - 1)$
 $= \pi D^2 \sigma(27^{1/3} - 1) = 2\pi D^2 \sigma$
26. (d) As volume remain constant therefore $R = n^{1/3} r$
 $\frac{\text{surface energy of one big drop}}{\text{surface energy of } n \text{ drop}} = \frac{4\pi R^2 T}{n \times 4\pi r^2 T}$
 $\frac{R^2}{nr^2} = \frac{n^{2/3} r^2}{nr^2} = \frac{1}{n^{1/3}} = \frac{1}{(1000)^{1/3}} = \frac{1}{10}$
27. (b) $W = T \times \Delta A \therefore T = \frac{W}{\Delta A}$
 $T = \frac{3 \times 10^{-4}}{2 \times (110 - 60) \times 10^{-4}} \text{ (Soap film has two surfaces)}$
 $= 3 \times 10^{-2} N/m$
28. (d)
29. (d) $\frac{4}{3} \pi R^3 = 1000 \times \frac{4}{3} \pi r^3 \text{ (As volume remains constant)}$
 $R^3 = 1000r^3 \Rightarrow R = 10r \Rightarrow r = \frac{R}{10}$
30. (c) Because energy is liberated
31. (a,d)
32. (c) As volume remains constant $R^3 = 8000r^3 \therefore R = 20r$
 $\frac{\text{Surface energy of one big drop}}{\text{Surface energy of 8000 small drop}} = \frac{4\pi R^2 T}{8000 \times 4\pi r^2 T}$
 $= \frac{R^2}{8000r^2} = \frac{(20r)^2}{8000r^2} = \frac{1}{20}$
33. (b) Surface energy = $T \times A = 5 \times 2 \times (0.15) = 1.5 J$
34. (c) As volume remains constant therefore $R = n^{1/3} r$
 $\frac{\text{Energy of big drop}}{\text{Energy of small drop}} = \frac{4\pi R^2 T}{4\pi r^2 T} = \frac{R^2}{r^2} = (8)^{2/3} = 4$
35. (a) $T = \frac{W}{\Delta A} = \frac{2 \times 10^{-4}}{2 \times (50 \times 10^{-4})} = 2 \times 10^{-2} N/m$
36. (a) $W = T \Delta A = 4\pi R^2 T(n^{1/3} - 1)$
 $= 4 \times 3.14 \times (10^{-2})^2 \times 460 \times 10^{-3} \times [(10^6)^{1/3} - 1] = 0.057$
37. (a)
38. (b) Increment in area of soap film = $A_2 - A_1$
 $= 2 \times [(10 \times 0.6) - (10 \times 0.5)] \times 10^{-4} = 2 \times 10^{-4} m^2$
 Work done = $T \times \Delta A$
 $= 7.2 \times 10^{-2} \times 2 \times 10^{-4} = 1.44 \times 10^{-5} J$
39. (a) Increase in surface energy or work done in splitting a big drop
 $= 4\pi R^2 T(n^{1/3} - 1)$
 $\Rightarrow W = 4\pi \times (2 \times 10^{-3})^2 \times 0.465(8^{1/3} - 1) = 23.4 \mu J$
40. (b) The ratio of the total surface energies before and after the change = $n^{1/3} : 1 = 2^{1/3} : 1$
41. (a) $W = 8\pi S(R_2^2 - R_1^2) = 8\pi S[(2R)^2 - R^2] = 24\pi R^2 S$
42. (a) $W = 8\pi r^2 \times T = 8\pi \times (0.2)^2 \times 0.06 = 192\pi \times 10^{-4} J$
43. (b) Increment in Potential energy = $T \times \Delta A$
 $= 0.02 \times 2 \times 0.05 = 2 \times 10^{-2} J$
44. (a) $E = T \times \Delta A = 75 \times 0.04 = 3 J$
45. (c) $r = \frac{r_1 r_2}{r_2 - r_1} = \infty$ since $r_1 = r_2$

Angle of Contact

1. (b)
2. (a)
3. (b) Cohesive force decreases so angle of contact decreases.
4. (d)



5. (b)
6. (b)
7. (d)
8. (b)
9. (a)
10. (c) Angle of contact is acute.
11. (a)
12. (c)
13. (b)
14. (b) Since for such liquid (Non-wetting) angle of contact is obtuse.
15. (b) Both liquids water and alcohol have same nature (i.e. wet the solid). Hence angle of contact for both is acute.
16. (d) Tangent drawn at point of contact makes 0° with wall of container.

Pressure Difference

1. (c)
2. (c) Since $\Delta P \propto \frac{1}{R}$
3. (b) Excess pressure $\Delta P = \frac{4T}{r}$

$$= \frac{4 \times 2 \times 25 \times 10^{-3}}{1 \times 10^{-2}} = 20 \text{ N/m}^2 = 20 \text{ Pa (as } r = d/2)$$
4. (c)
5. (c)
6. (c) $hdg = \frac{2T}{r} \Rightarrow h = \frac{2T}{rdg}$
7. (b) $\Delta P = \frac{4T}{r} = 40 \text{ N/m}^2$
8. (b)
9. (b) $\Delta P = \frac{4T}{r} = hdg \Rightarrow T = \frac{rhdg}{4} = \frac{0.35 \times 0.8 \times 1 \times 10^3}{4}$

$$= 70 \text{ dyne/cm} \approx 68.66 \text{ dyne/cm}$$
10. (c) Outside pressure = 1 atm
 Pressure inside first bubble = 1.01 atm
 Pressure inside second bubble = 1.02 atm
 Excess pressure $\Delta P_1 = 1.01 - 1 = 0.01 \text{ atm}$
 Excess pressure $\Delta P_2 = 1.02 - 1 = 0.02 \text{ atm}$

$$\Delta P \propto \frac{1}{r} \Rightarrow r \propto \frac{1}{\Delta P} \Rightarrow \frac{r_1}{r_2} = \frac{\Delta P_2}{\Delta P_1} = \frac{0.02}{0.01} = \frac{2}{1}$$

$$\text{Since } V = \frac{4}{3}\pi r^3 \therefore \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{2}{1}\right)^3 = \frac{8}{1}$$

11. (b) $S = \frac{rhdg}{2 \cos \theta} \Rightarrow \text{Pressure difference} = hdg = \frac{2S}{r} \cos \theta$
12. (c)
13. (c) Excess pressure inside soap bubble is inversely proportional to the radius of bubble i.e. $\Delta P \propto \frac{1}{r}$
 This means that bubbles A and C possess greater pressure inside it than B. So the air will move from A and C towards B.
14. (c) $P_1 V_1 = P_2 V_2 \Rightarrow (H + h)\rho g \times \frac{4}{3}\pi r^3 = H \times \frac{4}{3}\pi (2r)^3$

$$\Rightarrow H + h = 8H \therefore h = 7H$$
15. (c) $r = \sqrt{r_1^2 + r_2^2} = \sqrt{9 + 16} = 5 \text{ cm}$
16. (c) $P_1 V_1 = P_2 V_2 \Rightarrow (H_{Hg} \rho_{Hg} + H_W \rho_W)V = H_{Hg} \rho_{Hg} \times 3V$

$$\Rightarrow H_{Hg} \rho_{Hg} + H_W \frac{\rho_{Hg}}{10} = 3H_{Hg} \rho_{Hg}$$

$$\Rightarrow H_W = 2H_{Hg} \times 10 = \frac{2 \times 75 \times 10}{100} = 15 \text{ m}$$
17. (a) $\Delta P = \frac{4T}{r} \Rightarrow \frac{\Delta P_1}{\Delta P_2} = 4 \therefore \frac{r_2}{r_1} = 4 \text{ and } \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \frac{1}{64}$
18. (b) $\Delta P \propto \frac{1}{r}$
19. (b) Pressure at half the depth = $P_0 + \frac{h}{2}dg$
 Pressure at the bottom = $P_0 + hdg$
 According to given condition

$$P_0 + \frac{h}{2}dg = \frac{2}{3}(P_0 + hdg)$$

$$\Rightarrow 3P_0 + \frac{3h}{2}dg = 2P_0 + 2hdg$$

$$\Rightarrow h = \frac{2P_0}{dg} = \frac{2 \times 10^5}{10^3 \times 10} = 20 \text{ m}$$
20. (a) $\Delta P \propto \frac{1}{r} \Rightarrow \frac{\Delta P_1}{\Delta P_2} = \frac{r_2}{r_1} = \frac{r}{4r} = \frac{1}{4}$
21. (c) $\Delta P = \frac{2T}{R} = \frac{2 \times 70 \times 10^{-3}}{1 \times 10^{-3}} = 140 \text{ N/m}^2$
22. (c) $P = h\rho g$
23. (d)

24. (b) $r = \frac{r_1 r_2}{r_1 - r_2} = \frac{5 \times 4}{5 - 4} = 20 \text{ cm}$

25. (c) Excess pressure inside the air bubble $= \frac{2T}{r}$

$$\Rightarrow P_{in} - P_{out} = \frac{2T}{r} = \frac{2 \times 70 \times 10^{-3}}{0.1 \times 10^{-3}} = 1400 \text{ Pa}$$

$$\Rightarrow P_{in} = 1400 + 1.013 \times 10^5 = 1.027 \times 10^5 \text{ Pa}$$

26. (a) $r_A > r_B$ and $P \propto \frac{1}{r}$ so $P_A < P_B$

So air will flow from B to A i.e. size of A will increase.

27. (a) $\Delta P = \frac{4T}{R} \therefore \Delta P \propto \frac{1}{R}$ ($T = \text{constant}$)

Hence, the internal pressure of smaller bubble is larger than that of larger bubble.

28. (b) $\frac{4T}{R} = h \rho g \therefore T = \frac{R h \rho g}{4}$

$$T = \frac{10^{-2} \times 2 \times 10^{-3} \times 0.8 \times 10^3 \times 9.8}{4} = 3.9 \times 10^{-2} \text{ N/m}$$

29. (a)

30. (d) $\Delta P \propto \frac{1}{r} \Rightarrow \frac{r_1}{r_2} = \frac{\Delta P_2}{\Delta P_1} = \frac{1}{3} \Rightarrow \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \frac{1}{27}$

Equating (i) and (ii) we get, $2T \cos \theta = b x h \rho g$

$$\therefore h = \frac{2T \cos \theta}{x \rho g}$$

5. (d) $6 \times 10^{-2} \times \text{Circumference} = \text{Force}$

$$\therefore \text{Circumference} = \frac{75 \times 10^{-4}}{6 \times 10^{-2}} = 12.5 \times 10^{-2} \text{ m}$$

6. (d) Due to capillarity it absorbs the ink.

7. (b) $r \propto \frac{1}{h} \Rightarrow \frac{r_p}{r_Q} = \frac{h_Q}{h_p} = \frac{h}{\frac{2}{3}h} = \frac{2}{3}$

8. (c) $r \propto \frac{1}{h} \Rightarrow \frac{r_1}{r_2} = \frac{h_2}{h_1} = \frac{6.6}{2.2} = \frac{3}{1}$

9. (c) $\frac{h_2}{h_1} = \frac{r_1}{r_2} = \frac{1}{2} \Rightarrow h_2 = \frac{30}{2} = 15 \text{ cm}$

10. (b)

11. (b)

12. (d) $h = \frac{2T}{r \rho g} = \frac{2 \times 75}{0.005 \times 1 \times 10^3} = 30 \text{ cm}$

13. (c) $T = \frac{r h \rho g}{2} \Rightarrow 75 \times 10^{-3} = \frac{3 \times 10^{-2} \times r \times 10^3 \times 9.8}{2}$
 $\Rightarrow r = \frac{1}{2} \text{ mm} \therefore D = 2r = 1 \text{ mm}$

14. (c) The angle of contact of mercury with glass is obtuse. So it gets depressed below the liquid level outside.

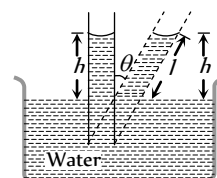
15. (d) The water rises to height h due to capillarity.

16. (b) $h \propto \frac{1}{r}$

17. (b) $h = \frac{2T}{r \rho g} = \frac{2 \times 6 \times 10^{-2}}{5 \times 10^{-4} \times 10^3 \times 10} = 2.4 \times 10^{-2} \text{ m} = 2.4 \text{ cm}$

18. (b) $h \propto \frac{1}{r} \therefore r_1 h_1 = r_2 h_2 \Rightarrow \frac{h_1}{h_2} = \frac{r_2}{r_1} = \frac{0.4}{0.2} = 2 : 1$

19. (b)



Vertical height of the water in the tube remains constant

$$\text{So, } l = \frac{h}{\cos \theta} = \frac{3}{\cos 60^\circ} = 6 \text{ cm}$$

20. (c)

21. (a)

Capillarity

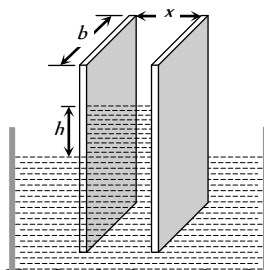
1. (d) $h = \frac{2T \cos \theta}{r \rho g} \therefore h \propto \frac{1}{r}$ (T, θ, d and g are constant)

If r is less then h will be more.

2. (a) $h = \frac{2T \cos \theta}{r \rho g}$. If θ is less than 90° then h will be positive

3. (d) In the state of weightlessness or in gravity free space, water will rise to the upper end of the tube of any length.

4. (b)



Let the width of each plate is b and due to surface tension liquid will rise upto height h then upward force due to surface tension

$$= 2Tb \cos \theta \quad \dots(i)$$

Weight of the liquid rises in between the plates

$$= V \rho g = (bxh) \rho g \quad \dots(ii)$$

22. (d) If lift moves downward with some acceleration then effective g decreases, so h increases.

$$\text{As } h = \frac{2T \cos \theta}{rdg} \therefore h \propto \frac{1}{g}$$

23. (c)

24. (a) $\frac{2T}{r} = hdg \Rightarrow r = \frac{2T}{hdg}$

25. (b) $h \propto \frac{1}{r} \therefore r_1 h_1 = r_2 h_2 \Rightarrow h_2 = \frac{r_1 h_1}{r_2} = 2.4 \text{ mm}$

26. (a)

27. (a) $h \propto \frac{1}{r} \therefore rh = \text{constant}$

28. (a) $h = \frac{2T \cos \theta}{rdg} \therefore h \propto \frac{1}{g}$

$$\text{As } g_m = \frac{g_e}{6} \therefore h_m = 6h_e$$

29. (d) Ascent formula $h = \frac{2T \cos \theta}{rdg}$

$$\Rightarrow \frac{h_1}{h_2} = \frac{T_1}{T_2} \times \frac{d_2}{d_1} \quad (r, \theta \text{ and } g \text{ are constants})$$

$$= \frac{60}{50} \times \frac{0.6}{0.8} = \frac{9}{10}$$

30. (b) $l = \frac{h}{\cos \theta} = \frac{2}{\cos 60^\circ} = 4.0 \text{ cm}$

31. (d) $T = \frac{rhdg}{2 \cos \theta}$. For pure water $\theta = 0^\circ$ so $T = \frac{rhdg}{2}$

32. (c) The length of the water column will be equal to full length of capillary tube.

33. (b) $T = \frac{F}{2\pi r} = \frac{6.28 \times 10^{-4}}{2 \times 3.14 \times 2 \times 10^{-3}} = 5 \times 10^{-2} \text{ N/m}$

34. (a) $h \propto \frac{1}{R}$

35. (a) $h = \frac{2T \cos \theta}{rdg}$, for water $\theta = 0^\circ$

$$\Rightarrow r = \frac{2T}{hdg} = \frac{2 \times 7.2 \times 10^{-2}}{3 \times 10^{-2} \times 10^3 \times 10} = 4.8 \times 10^{-4}$$

$$\therefore d = 2r = 9.6 \times 10^{-4} \text{ m}$$

36. (c) $h = \frac{2T}{rdg} \Rightarrow r = \frac{2T}{hdg} = \frac{2 \times 75 \times 10^{-3}}{15 \times 10^{-3} \times 10^3 \times 10} = 1 \text{ mm}$

37. (d) $h \propto \frac{1}{r}$

38. (a)

39. (a)

40. (b) Mass of liquid in capillary tube

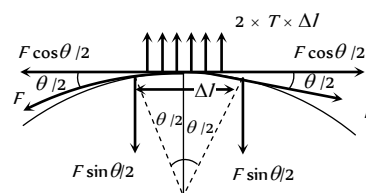
$$M = \pi R^2 H \times \rho \therefore M \propto R^2 \times \left(\frac{1}{R}\right) \quad (\text{As } H \propto 1/R)$$

$\therefore M \propto R$. If radius becomes double then mass will become twice.

41. (c) $h \propto \frac{1}{r} \Rightarrow \frac{h_2}{h_1} = \frac{r_1}{r_2} = \frac{D_1}{D_2} = 2 \Rightarrow h_2 = 2h_1$

Critical Thinking Questions

1. (d) Suppose tension in thread is F , then for small part Δl of thread



$$\Delta l = R\theta \text{ and } 2F \sin \theta / 2 = 2T\Delta l = 2TR\theta$$

$$\Rightarrow F = \frac{TR\theta}{\sin \theta / 2} = \frac{TR\theta}{\theta / 2} = 2TR(\sin \theta / 2 \approx \theta / 2)$$

2. (c) Rise in temperature, $\Delta \theta = \frac{3T}{JSd} \left(\frac{1}{r} - \frac{1}{R} \right)$

$$\therefore \Delta \theta = \frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right) \quad (\text{For water } S = 1 \text{ and } d = 1)$$

3. (b,c) $P_{\text{Bottom}} > P_{\text{Surface}}$. So bubble rises upward.

$$\text{At constant temperature } V \propto \frac{1}{P} \quad (\text{Boyle's law})$$

Since as the bubble rises upward, pressure decreases, then from above law volume of bubble will increase i.e. its size increases.

4. (d) In the satellite, the weight of the liquid column is zero. So the liquid will rise up to the top of the tube.

Graphical Questions

1. (b) $h = \frac{2T \cos \theta}{rdg} \therefore h \propto \frac{1}{r}$. So the graph between h and r will be rectangular hyperbola.

2. (a) $\Delta P = \frac{4T}{r} \therefore \Delta P \propto \frac{1}{r}$

$$\text{As radius of soap bubble increases with time } \therefore \Delta P \propto \frac{1}{t}$$

3. (b) $T_c = T_0(1 - \alpha t)$ i.e. surface tension decreases with increase in temperature.

Assertion and Reason

1. (c) When a liquid is sprayed, the surface area of the liquid increases. Therefore, work has to be done in spraying the liquid, which is directly proportional to the surface tension.

Because on adding soap, surface tension of water decreases, the spraying of water becomes easy.

2. (e) The soap solution, has less surface tension as compared to ordinary water and its surface tension decreases further on heating. The hot soap solution can, therefore spread over large surface area and also it has more wetting power. It is on account of this property that hot soap solution can penetrate and clean the clothes better than the ordinary water.

3. (a) $h = \frac{2T}{Rdg} \Rightarrow hR = \frac{2T}{Rdg} \therefore hR = \text{constant}$

Hence when the tube is of insufficient length, radius of curvature of the liquid meniscus increases, so as to maintain the product hR a finite constant.

i.e. as h decreases, R increases and the liquid meniscus becomes more and more flat, but the liquid does not overflow.

4. (c) Needle floats due to surface tension there is no role of buoyant force in its floating

Buoyant force = $V\sigma g$

Where V = volume of body submerged in liquid

σ = density of liquid.

i.e. the buoyancy of an object depends on the shape of the object.

5. (c) The two glass plates stick together due to surface tension.
6. (e) The presence of impurities either on the liquid surface or dissolved in it, considerably affect the force of surface tension, depending upon the degree of contamination. A highly soluble substance like sodium chloride when dissolved in water increase the surface tension. But the sparingly soluble or substance like phenol when dissolved in water reduces the surface tension of water.
7. (c) With increase in temperature surface tension of the liquid decreases and angle of contact also decreases.
8. (b) We know that the intermolecular distance between the gas molecules is large as compared to that of liquid. Due to it the forces of cohesion in the gas molecules are very small and these are quite large for liquids. Therefore, the concept of surface tension is applicable to liquid but not to gases.
9. (a) Zero surface tension means no opposition to expansion.
10. (a) Since the excess pressure due to surface tension is inversely proportional to its radius, it follows that smaller the bubble, greater is the excess pressure. Thus, when the larger and the smaller bubbles are put in communication, air starts passing from the smaller into the large bubble because excess pressure inside the former is greater than inside the latter. As a result, the smaller bubble shrinks and the larger one swells.
11. (b) When a drop of liquid is poured on a glass plate, the shape of the drop is governed by two forces, the force of gravity. For very small drops, the potential energy due to gravity is

insignificant compared to that due to surface tension. Hence, in this case the shape of the drop is determined by surface tension alone and drop becomes spherical.

12. (a) The height of capillary rise is inversely proportional to radius (or diameter) of capillary tube *i.e.* $h \propto \frac{1}{r}$

So for smaller r the value of h is higher.

13. (c) With increase in temperature of liquid its surface tension decreases so that it tends to acquire larger area. Hence hot soup having low value of surface tension spread properly on our tongue & provides better taste than cold soup.

14. (b) The free surface of liquid tries to acquire a minimum area due to surface tension, hence liquid drop is spherical because sphere has minimum area than other shape.

Surface Tension

Self Evaluation Test -10

- A soap film of surface tension $3 \times 10^{-2} \text{ Nm}^{-1}$ formed in rectangular frame, can support a straw. The length of the film is 10 cm. Mass of the straw the film can support is
 (a) 0.06 gm (b) 0.6 gm
 (c) 6 gm (d) 60 gm
- Energy required to form a soap bubble of diameter 20 cm will be (Surface tension for soap solution is 30 dynes/cm)
 (a) $12000 \pi \text{ ergs}$ (b) $1200 \pi \text{ ergs}$
 (c) $2400 \pi \text{ ergs}$ (d) $24000 \pi \text{ ergs}$
- If the work done in blowing a bubble of volume V is W , then the work done in blowing the bubble of volume $2V$ from the same soap solution will be [MP PET 1989]
 (a) $W/2$ (b) $\sqrt{2} W$
 (c) $\sqrt[3]{2} W$ (d) $\sqrt[3]{4} W$
- Surface tension of soap solution is $2 \times 10^{-2} \text{ N/m}$. The work done in producing a soap bubble of radius 2 cm is
 (a) $64\pi \times 10^{-6} \text{ J}$ (b) $32\pi \times 10^{-6} \text{ J}$
 (c) $16\pi \times 10^{-6} \text{ J}$ (d) $8\pi \times 10^{-6} \text{ J}$
- Excess pressure inside a soap bubble is three times that of the other bubble, then the ratio of their volumes will be
 (a) 1 : 3 (b) 1 : 9
 (c) 1 : 27 (d) 1 : 81
- When a capillary tube is dipped in water it rises upto 8 cm in the tube. What happens when the tube is pushed down such that its end is only 5 cm above the outside water level
 (a) The radius of the meniscus increases and therefore water does not overflow
 (b) The radius of the meniscus decreases and therefore water does not overflow
 (c) The water forms a droplet on top of the tube but does not overflow
 (d) The water start overflowing
- A bubble of 8 mm diameter is formed in the air. The surface tension of soap solution is 30 dynes/cm. The excess pressure inside the bubble is [MP PET 1990]
 (a) 150 dynes/cm (b) 300 dynes/cm
 (c) $3 \times 10^4 \text{ dynes/cm}$ (d) 12 dynes/cm
- The height upto which water will rise in a capillary tube will be
 (a) Maximum when water temperature is 4°C
 (b) Maximum when water temperature is 0°C
 (c) Minimum when water temperature is 4°C
 (d) Same at all temperatures
- Water rises to a height of 10 cm in capillary tube and mercury falls to a depth of 3.112 cm in the same capillary tube. If the density of mercury is 13.6 and the angle of contact for mercury is 135° , the ratio of surface tension of water and mercury is
 (a) 1 : 0.15 (b) 1 : 3
 (c) 1 : 6 (d) 1.5 : 1
- The angle of contact between glass and water is 0° and it rises in a capillary upto 6 cm when its surface tension is 70 dynes/cm. Another liquid of surface tension 140 dynes/cm, angle of contact 60° and relative density 2 will rise in the same capillary by
 (a) 12 cm (b) 24 cm
 (c) 3 cm (d) 6 cm
- A drop of water breaks into two droplets of equal size. In this process, which of the following statement is correct [NCERT 1976]
 (a) The sum of temperature of the two droplets together is equal to the original temperature of the drop
 (b) The sum of masses of the two droplets is equal to the original mass of the drop
 (c) The sum of the radii of two droplets is equal to the radius of the original drop
 (d) The sum of the surface areas of the two droplets is equal to the surface area of the original drop
- A soap bubble of radius R is blown. After heating the solution a second bubble of radius $2R$ is blown. The work required to blow the second bubble in comparison to that required for the first bubble is
 (a) Double
 (b) Slightly less than double
 (c) Slightly less than four times
 (d) Slightly more than four times
- A false statement is
 (a) Angle of contact $\theta < 90^\circ$, if cohesive force < adhesive force
 (b) Angle of contact $\theta > 90^\circ$, if cohesive force > adhesive force
 (c) Angle of contact $\theta = 90^\circ$, if cohesive force = adhesive force
 (d) If the radius of capillary is reduced to half, the rise of liquid column becomes four times
- The diameter of rain-drop is 0.02 cm. If surface tension of water be $72 \times 10^{-3} \text{ newton per metre}$, then the pressure difference of external and internal surfaces of the drop will be
 (a) $1.44 \times 10^4 \text{ dyne} - \text{cm}^{-2}$
 (b) $1.44 \times 10^4 \text{ newton} - \text{m}^{-2}$



- (c) $1.44 \times 10^3 \text{ dyne} - \text{cm}^{-2}$
- (d) $1.44 \times 10^5 \text{ newton} - \text{m}^{-2}$
15. Water rises to a height of 16.3 cm in a capillary of height 18 cm above the water level. If the tube is cut at a height of 12 cm
- (a) Water will come as a fountain from the capillary tube
- (b) Water will stay at a height of 12 cm in the capillary tube
- (c) The height of the water in the capillary will be 10.3 cm
- (d) Water will flow down the sides of the capillary tube [CPMT 1974]

AS Answers and Solutions

(SET - 10)

1. (b) The weight of straw will be balanced by the force of surface tension $\therefore mg = 2Tl \Rightarrow m = \frac{2Tl}{g}$
- $$= \frac{2 \times 3 \times 10^{-2} \times 10 \times 10^{-2}}{9.8} \text{ kg} = 0.6 \text{ gm}$$
2. (d) $E = 8\pi^2 T = 8\pi(10)^2 \times 30 = 24000 \pi \text{ erg}$
3. (d) Work done to form a soap bubble
- $$W = 8\pi R^2 T \quad (\text{As } V \propto R^3 \therefore R \propto V^{1/3})$$
- $\therefore W \propto V^{2/3}$
- $$\frac{W_2}{W_1} = \left(\frac{V_2}{V_1} \right)^{2/3} = (2)^{2/3} \Rightarrow W_2 = (4)^{1/3} W$$
4. (a) $W = 8\pi R^2 T = 8 \times \pi \times (2 \times 10^{-2})^2 \times 2 \times 10^{-2} = 64\pi \times 10^{-6} \text{ J}$
5. (c) $\Delta P \propto \frac{1}{r} \Rightarrow \frac{\Delta P_1}{\Delta P_2} = \frac{r_2}{r_1} \Rightarrow \frac{r_2}{r_1} = \frac{3}{1}$



$$\therefore \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

6. (a) $h = \frac{2T}{Rdg} \Rightarrow hR = \frac{2T}{dg} = \text{constant}$

When h decreases, R increases.

7. (b) $\Delta P = \frac{4T}{r} = \frac{4 \times 30}{0.4} = 300 \text{ dyne/cm}^2$.

8. (c) $h = \frac{2T \cos \theta}{rdg}$. For water, density is maximum at 4°C , so the height is minimum at 4°C .

$$= 1.44 \times 10^4 \text{ dyne/cm}^2$$

15. (b) Because if the length available is less than required, then water will rise upto available height and adjust its radius of curvature.

9. (c) $h = \frac{2T \cos \theta}{rdg} \therefore T = \frac{hrdg}{2 \cos \theta}$
 $\Rightarrow \frac{T_1}{T_2} = \frac{h_1}{h_2} \times \frac{r_1}{r_2} \times \frac{d_1}{d_2} \times \frac{\cos \theta_2}{\cos \theta_1} = \frac{1}{6}$

10. (c) $h = \frac{2T \cos \theta}{rdg} \therefore \frac{h_2}{h_1} = \frac{T_2}{T_1} \times \frac{\cos \theta_2}{\cos \theta_1} \times \frac{d_1}{d_2} \times \frac{r_1}{r_2}$
 $\frac{h_2}{h_1} = \frac{140}{70} \times \frac{\cos 60^\circ}{\cos 0^\circ} \times \frac{1}{2} \times 1 = \frac{1}{2} \Rightarrow h_2 = \frac{h_1}{2} = 3 \text{ cm}$

11. (b)

12. (c) Work done to form a bubble of radius R

$$W_1 = 8\pi R^2 T_1$$

Work done to form a bubble of radius $2R$

$$W_2 = 8\pi (2R)^2 T_2 = 32\pi R^2 T_2 \therefore \frac{W_1}{W_2} = \frac{T_1}{4T_2}$$

If surface tension of soap solution is same then

$$W_2 = 4W_1$$

But in the problem temperature of solution is increased so its surface tension decreases.

$$\therefore W_2 < 4W_1$$

13. (d) If radius of capillary is reduced to half, the rise of liquid column will be two times. as $h \propto 1/r$

14. (a) $\Delta P = \frac{2T}{r} = \frac{2 \times 72 \times 10^{-3}}{0.01 \times 10^{-2}} = 1440 \text{ N/m}^2$



Chapter 11 Fluid Mechanics

Fluid is the name given to a substance which begins to flow when external force is applied on it. Liquids and gases are fluids. Fluids do not have their own shape but take the shape of the containing vessel. The branch of physics which deals with the study of fluids at rest is called hydrostatics and the branch which deals with the study of fluids in motion is called hydrodynamics.

Pressure

The normal force exerted by liquid at rest on a given surface in contact with it is called thrust of liquid on that surface.

The normal force (or thrust) exerted by liquid at rest per unit area of the surface in contact with it, is called pressure of liquid or hydrostatic pressure.

If F be the normal force acting on a surface of area A in contact with liquid, then pressure exerted by liquid on this surface is $P = F / A$

(1) Units : N / m^2 or Pascal (S.I.) and Dyne/cm (C.G.S.)

(2) Dimension : $[P] = \frac{[F]}{[A]} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

(3) At a point pressure acts in all directions and a definite direction is not associated with it. So pressure is a tensor quantity.

(4) Atmospheric pressure : The gaseous envelope surrounding the earth is called the earth's atmosphere and the pressure exerted by the atmosphere is called atmospheric pressure. Its value on the surface of the earth at sea level is nearly $1.013 \times 10^5 N / m^2$ or Pascal in S.I., other practical units of pressure are atmosphere, bar and torr (mm of Hg)

$$1 \text{ atm} = 1.01 \times 10^5 \text{ Pa} = 1.01 \text{ bar} = 760 \text{ torr}$$

The atmospheric pressure is maximum at the surface of earth and goes on decreasing as we move up into the earth's atmosphere.

(5) If P_0 is the atmospheric pressure then for a point at depth h below the surface of a liquid of density ρ , hydrostatic pressure P is given by $P = P_0 + h\rho g$

(6) Hydrostatic pressure depends on the depth of the point below the surface (h), nature of liquid (ρ) and acceleration due to gravity (g) while it is independent of the amount of liquid, shape of the container or cross-sectional area considered. So if a given liquid is filled in vessels of different shapes to same height, the pressure at the base in each vessel's will be the same, though the volume or weight of the liquid in different vessels will be different.

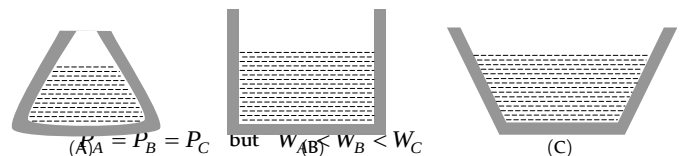
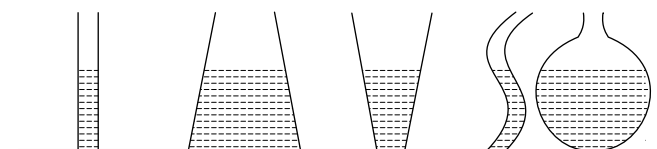


Fig. 11.2

(7) In a liquid at same level, the pressure will be same at all points, if not, due to pressure difference the liquid cannot be at rest. This is why the height of liquid is the same in vessels of different shapes containing different amounts of the same liquid at rest when they are in communication with each other.



(8) Gauge pressure : The pressure difference between hydrostatic pressure P and atmospheric pressure P_0 is called gauge pressure.

Fig. 11.3

$$P - P_0 = h\rho g$$

Density

In a fluid, at a point, density ρ is defined as:

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV}$$

(1) In case of homogenous isotropic substance, it has no directional properties, so is a scalar.

(2) It has dimensions $[ML^{-3}]$ and S.I. unit kg/m^3 while C.G.S. unit g/cc with $1g/cc = 10^3 kg/m^3$

(3) Density of substance means the ratio of mass of substance to the volume occupied by the substance while density of a body means the ratio of mass of a body to the volume of the body. So for a solid body,

Density of body = Density of substance

While for a hollow body, density of body is lesser than that of substance [As $V_{body} > V_{sub.}$]

(4) When immiscible liquids of different densities are poured in a container, the liquid of highest density will be at the bottom while that of lowest density at the top and interfaces will be plane.

(5) Sometimes instead of density we use the term relative density or specific gravity which is defined as :

$$RD = \frac{\text{Density of body}}{\text{Density of water}}$$

(6) If m_1 mass of liquid of density ρ_1 and m_2 mass of density ρ_2 are mixed, then as

$$m = m_1 + m_2 \text{ and } V = (m_1 / \rho_1) + (m_2 / \rho_2)$$

$$[\text{As } V = m / \rho]$$

$$\rho = \frac{m}{V} = \frac{m_1 + m_2}{(m_1 / \rho_1) + (m_2 / \rho_2)} = \frac{\sum m_i}{\sum (m_i / \rho_i)}$$

$$\text{If } m_1 = m_2 \quad \rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = \text{Harmonic mean}$$

(7) If V_1 volume of liquid of density ρ_1 and V_2 volume of liquid of density ρ_2 are mixed, then as:

$$m = \rho_1 V_1 + \rho_2 V_2 \text{ and } V = V_1 + V_2 \quad [\text{As } \rho = m / V]$$

$$\text{If } V_1 = V_2 = V \quad \rho = (\rho_1 + \rho_2) / 2 = \text{Arithmetic Mean}$$

(8) With rise in temperature due to thermal expansion of a given body, volume will increase while mass will remain unchanged, so density will decrease, i.e.,

$$\frac{\rho}{\rho_0} = \frac{(m/V)}{(m/V_0)} = \frac{V_0}{V} = \frac{V_0}{V_0(1 + \gamma\Delta\theta)} \quad [\text{As } V = V_0(1 + \gamma\Delta\theta)]$$

$$\text{or } \rho = \frac{\rho_0}{(1 + \gamma\Delta\theta)} \approx \rho_0(1 - \gamma\Delta\theta)$$

(9) With increase in pressure due to decrease in volume, density will increase, i.e.,

$$\frac{\rho}{\rho_0} = \frac{(m/V)}{(m/V_0)} = \frac{V_0}{V} \quad [\text{As } \rho = \frac{m}{V}]$$

But as by definition of bulk-modulus

$$B = -V_0 \frac{\Delta p}{\Delta V} \text{ i.e., } V = V_0 \left[1 - \frac{\Delta p}{B} \right]$$

$$\text{So } \rho = \rho_0 \left(1 - \frac{\Delta p}{B} \right)^{-1} \approx \rho_0 \left(1 + \frac{\Delta p}{B} \right)$$

Pascal's Law

It states that if gravity effect is neglected, the pressure at every point of liquid in equilibrium of rest is same.

or

The increase in pressure at one point of the enclosed liquid in equilibrium of rest is transmitted equally to all other points of the liquid

and also to the walls of the container, provided the effect of gravity is neglected.

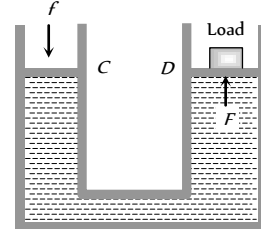
Example : Hydraulic lift, hydraulic press and hydraulic brakes

Working of hydraulic lift : It is used to lift the heavy loads. If a small force f is applied on piston of C then the pressure exerted on the liquid

$$P = f / a \quad [a = \text{Area of cross section of the piston in } C]$$

This pressure is transmitted equally to piston of cylinder D .

Hence the upward force acting on piston of cylinder D .



$$F = P A = \frac{f}{a} A = f \left(\frac{A}{a} \right) \quad \text{Fig. 11.4}$$

As $A \gg a$, therefore $F \gg f$. So heavy load placed on the larger piston is easily lifted upwards by applying a small force.

Archimedes Principle

Accidentally Archimedes discovered that when a body is immersed partly or wholly in a fluid, at rest, it is buoyed up with a force equal to the weight of the fluid displaced by the body. This principle is called Archimedes principle and is a necessary consequence of the laws of fluid statics.

When a body is partly or wholly dipped in a fluid, the fluid exerts force on the body due to hydrostatic pressure. At any small portion of the surface of the body, the force exerted by the fluid is perpendicular to the surface and is equal to the pressure at that point multiplied by the area. The resultant of all these constant forces is called upthrust or buoyancy.

To determine the magnitude and direction of this force consider a body immersed in a fluid of density σ as shown in figure. The forces on the vertical sides of the body will cancel each other. The top surface of the body will experience a downward force.

$$F_1 = AP_1 = A(h_1\sigma g + P_0) \quad [\text{As } P = h\sigma g + P_0]$$

While the lower face of the body will experience an upward force.

$$F_2 = AP_2 = A(h_2\sigma g + P_0)$$

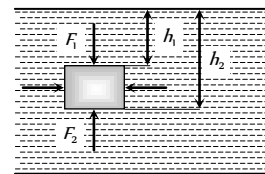


Fig. 11.5

As $h_2 > h_1$, F_2 will be greater than F_1 , so the body will experience a net upward force

$$F = F_2 - F_1 = A\sigma g(h_2 - h_1)$$

If L is the vertical height of the body $F = A\sigma gL = V\sigma g$

$$[\text{As } V = AL = A(h_2 - h_1)]$$

i.e., F = Weight of fluid displaced by the body.

This force is called upthrust or buoyancy and acts vertically upwards (opposite to the weight of the body) through the centre of gravity of

displaced fluid (called centre of buoyancy). Though we have derived this result for a body fully submerged in a fluid, it can be shown to hold good for partly submerged bodies or a body in more than one fluid also.

(1) Upthrust is independent of all factors of the body such as its mass, size, density etc. except the volume of the body inside the fluid.

(2) Upthrust depends upon the nature of displaced fluid. This is why upthrust on a fully submerged body is more in sea water than in fresh water because its density is more than fresh water.

(3) Apparent weight of the body of density (ρ) when immersed in a liquid of density (σ).

$$\begin{aligned}\text{Apparent weight} &= \text{Actual weight} - \text{Upthrust} = W - F_{up} \\ &= V\rho g - V\sigma g = V(\rho - \sigma)g = V\rho g \left(1 - \frac{\sigma}{\rho}\right)\end{aligned}$$

$$\therefore W_{APP} = W \left(1 - \frac{\sigma}{\rho}\right)$$

(4) If a body of volume V is immersed in a liquid of density σ then its weight reduces.

W_1 = Weight of the body in air, W_2 = Weight of the body in water

Then apparent (loss of weight) weight $W_1 - W_2 = V\sigma g$

$$\therefore V = \frac{W_1 - W_2}{\sigma g}$$

(5) Relative density of a body

$$\begin{aligned}(\text{R.D.}) &= \frac{\text{density of body}}{\text{density of water}} \\ &= \frac{\text{Weight of body}}{\text{Weight of equal volume of water}} = \frac{\text{Weight of body}}{\text{Water thrust}} \\ &= \frac{\text{Weight of body}}{\text{Loss of weight in water}} \\ &= \frac{\text{Weight of body in air}}{\text{Weight in air} - \text{weight in water}} = \frac{W_1}{W_1 - W_2}\end{aligned}$$

(6) If the loss of weight of a body in water is 'a' while in liquid is 'b'

$$\begin{aligned}\therefore \frac{\sigma_L}{\sigma_w} &= \frac{\text{Upthrust on body in liquid}}{\text{Upthrust on body in water}} \\ &= \frac{\text{Loss of weight in liquid}}{\text{Loss of weight in water}} = \frac{a}{b} = \frac{W_{\text{air}} - W_{\text{liquid}}}{W_{\text{air}} - W_{\text{water}}}\end{aligned}$$

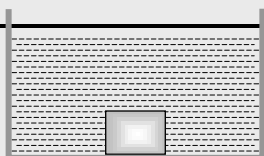
Floatation

(1) **Translatory equilibrium** : When a body of density ρ and volume V is immersed in a liquid of density σ , the forces acting on the body are

Weight of body $W = mg = V\rho g$, acting vertically downwards through centre of gravity of the body.

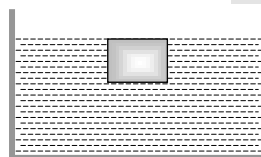
Upthrust force = $V\sigma g$ acting vertically upwards through the centre of gravity of the displaced liquid i.e., centre of buoyancy.

If density of body is greater than that of liquid $\rho > \sigma$



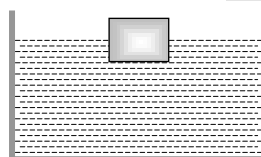
Weight will be more than upthrust so the body will sink

If density of body is equal to that of liquid $\rho = \sigma$



Weight will be equal to upthrust so the body will float fully submerged in neutral equilibrium with its top surface in it just at the top of liquid

If density of body is lesser than that of liquid $\rho < \sigma$



Weight will be less than upthrust so the body will, move upwards and in equilibrium will float and partially immersed in the liquid Such that,

$$W = V_{in}\sigma g \Rightarrow V\rho g = V_{in}\sigma g$$

$$V\rho = V_{in}\sigma \text{ Where } V_{in} \text{ is the volume of body in the liquid}$$

(i) A body will float in liquid only and only if $\rho < \sigma$

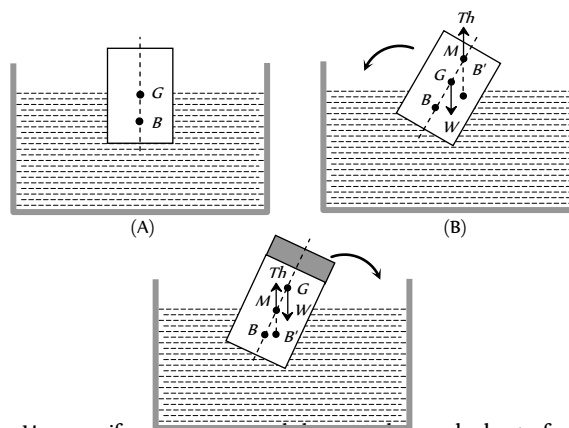
(ii) In case of floating as weight of body = upthrust

So $W_{App} = \text{Actual weight} - \text{upthrust} = 0$

(iii) In case of floating $V\rho g = V_{in}\sigma g$

So the equilibrium of floating bodies is unaffected by variations in g though both thrust and weight depend on g .

(2) **Rotatory Equilibrium** : When a floating body is slightly tilted from equilibrium position, the centre of buoyancy B shifts. The vertical line passing through the new centre of buoyancy B' and initial vertical line meet at a point M called meta-centre. If the meta-centre M is above the centre of gravity the couple due to forces at G (weight of body W) and at B' (upthrust) tends to bring the body back to its original position. So for rotational equilibrium of floating body the meta-centre must always be higher than the centre of gravity of the body.



However, if meta-centre goes below CG, the couple due to forces at G and B' tends to topple the floating body.

That is why a wooden log cannot be made to float vertical in water or a boat is likely to capsize if the sitting passengers stand on it. In these situations CG becomes higher than MG and so the body will topple if slightly tilted.

(3) Application of floatation

(i) When a body floats then the weight of body = Upthrust

$$V\rho g = V_{in}\sigma g \Rightarrow V_{in} = \left(\frac{\rho}{\sigma}\right)V$$

$$\therefore V_{out} = V - V_{in} = \left(1 - \frac{\rho}{\sigma}\right)V$$

$$\text{i.e., Fraction of volume outside the liquid } f_{out} = \frac{V_{out}}{V} = \left[1 - \frac{\rho}{\sigma}\right]$$

$$(ii) \text{ For floatation } V\rho = V_{in}\sigma \Rightarrow \rho = \frac{V_{in}}{V}\sigma = f_{in}\sigma$$

If two different bodies A and B are floating in the same liquid then

$$\frac{\rho_A}{\rho_B} = \frac{(f_{in})_A}{(f_{in})_B}$$

(iii) If the same body is made to float in different liquids of densities σ_A and σ_B respectively.

$$V\rho = (V_{in})_A\sigma_A = (V_{in})_B\sigma_B \quad \therefore \quad \frac{\sigma_A}{\sigma_B} = \frac{(V_{in})_B}{(V_{in})_A}$$

(iv) If a platform of mass M and cross-section A is floating in a liquid of density σ with its height h inside the liquid

$$Mg = hA\sigma g \quad \dots(i)$$

Now if a body of mass m is placed on it and the platform sinks by y then

$$(M + m)g = (y + h)A\sigma g \quad \dots(ii)$$

Subtracting equation (i) from (ii),

$$mg = A\sigma y g, \text{ i.e., } W \propto y \quad \dots(iii)$$

So we can determine the weight of a body by placing it on a floating platform and noting the depression of the platform in the liquid by it.

Streamline, Laminar and Turbulent Flow

(i) **Stream line flow** : Stream line flow of a liquid is that flow in which each element of the liquid passing through a point travels along the same path and with the same velocity as the preceding element passes through that point.

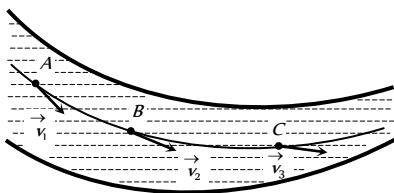


Fig. 11.7

A streamline may be defined as the path, straight or curved, the tangent to which at any point gives the direction of the flow of liquid at that point.

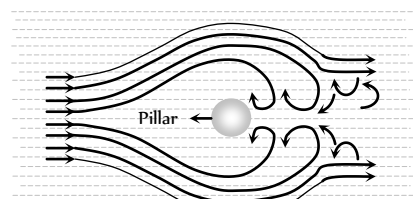
The two streamlines cannot cross each other and the greater is the crowding of streamlines at a place, the greater is the velocity of liquid particles at that place.

Path ABC is streamline as shown in the figure and v_1 , v_2 and v_3 are the velocities of the liquid particles at A , B and C point respectively.

(2) **Laminar flow** : If a liquid is flowing over a horizontal surface with a steady flow and moves in the form of layers of different velocities which do not mix with each other, then the flow of liquid is called laminar flow.

In this flow, the velocity of liquid flow is always less than the critical velocity of the liquid. The laminar flow is generally used synonymously with streamlined flow.

(3) **Turbulent flow** : When a liquid moves with a velocity greater than its critical velocity, the motion of the particles of liquid becomes disordered or irregular. Such a flow is called a turbulent flow.



In a turbulent flow, the path and the velocity of the particles of the liquid change continuously and haphazardly with time from point to point. In a turbulent flow, most of the external energy maintaining the flow is spent in producing eddies in the liquid and only a small fraction of energy is available for forward flow. For example, eddies are seen by the sides of the pillars of a river bridge.

Critical Velocity and Reynold's Number

The critical velocity is that velocity of liquid flow upto which its flow is streamlined and above which its flow becomes turbulent.

Reynold's number is a pure number which determines the nature of flow of liquid through a pipe.

It is defined as the ratio of the inertial force per unit area to the viscous force per unit area for a flowing fluid.

$$N_R = \frac{\text{Inertial force per unit area}}{\text{Viscous force per unit area}}$$

If a liquid of density ρ is flowing through a tube of radius r and cross section A then mass of liquid flowing through the tube per second

$$\frac{dm}{dt} = \text{volume flowing per second} \times \text{density} = Av \times \rho$$

$$\therefore \text{Inertial force per unit area} = \frac{dp/dt}{A} = \frac{v(dm/dt)}{A} = \frac{vAv\rho}{A} =$$

$$v^2\rho$$

$$\text{Viscous force per unit area } F/A = \frac{\eta v}{r}$$

So by the definition of Reynolds number

$$N_R = \frac{\text{Inertial force per unit area}}{\text{Viscous force per unit area}} = \frac{v^2\rho}{\eta v/r} = \frac{v\rho r}{\eta}$$

If the value of Reynold's number

(i) Lies between 0 to 2000, the flow of liquid is streamline or laminar.

(ii) Lies between 2000 to 3000, the flow of liquid is unstable and changing from streamline to turbulent flow.

(iii) Above 3000, the flow of liquid is definitely turbulent.

Equation of Continuity

The equation of continuity is derived from the principle of conservation of mass.

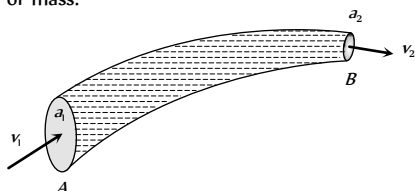


Fig. 11.9

A non-viscous liquid in streamline flow passes through a tube AB of varying cross section. Let the cross sectional area of the pipe at points A and B be a_1 and a_2 respectively. Let the liquid enter with normal velocity v_1 at A and leave with velocity v_2 at B . Let ρ_1 and ρ_2 be the densities of the liquid at point A and B respectively.

Mass of the liquid entering per second at A = Mass of the liquid leaving per second at B

$$a_1 v_1 \rho_1 = a_2 v_2 \rho_2 \text{ and } a_1 v_1 = a_2 v_2$$

[If the liquid is incompressible $\rho_2 = \rho_1$]

$$\text{or } a v = \text{constant or } a \propto \frac{1}{v}$$

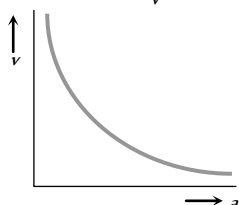


Fig. 11.10

This expression is called the equation of continuity for the steady flow of an incompressible and non-viscous liquid.

(1) The velocity of flow is independent of the liquid (assuming the liquid to be non-viscous)

(2) The velocity of flow will increase if cross-section decreases and vice-versa. That is why :

(a) In hilly region, where the river is narrow and shallow (*i.e.*, small cross-section) the water current will be faster, while in plains where the river is wide and deep (*i.e.*, large cross-section) the current will be slower, and so deep water will appear to be still.

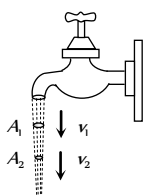


Fig. 11.11

(b) When water falls from a tap, the velocity of falling water under the action of gravity will increase with distance from the tap (*i.e.*, $v_2 > v_1$). So in accordance with continuity equation the cross section of the water stream will decrease (*i.e.*, $A_2 < A_1$), *i.e.*, the falling stream of water becomes narrower.

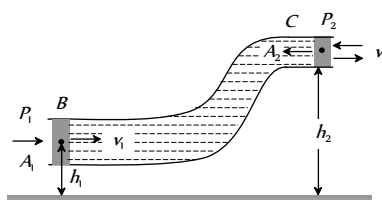
Energy of a Flowing Fluid

A flowing fluid in motion possesses the following three types of energy

Pressure Energy	Potential energy	Kinetic energy
It is the energy possessed by a liquid by virtue of its pressure. It is the measure of work done in pushing the liquid against pressure without imparting any velocity to it.	It is the energy possessed by liquid by virtue of its height or position above the surface of earth or any reference level taken as zero level.	It is the energy possessed by a liquid by virtue of its motion or velocity.
Pressure energy of the liquid PV	Potential energy of the liquid mgh	Kinetic energy of the liquid $\frac{1}{2}mv^2$
Pressure energy per unit mass of the liquid $\frac{P}{\rho}$	Potential energy per unit mass of the liquid gh	Kinetic energy per unit mass of the liquid $\frac{1}{2}v^2$
Pressure energy per unit volume of the liquid P	Potential energy per unit volume of the liquid ρgh	Kinetic energy per unit volume of the liquid $\frac{1}{2}\rho v^2$

Bernoulli's Theorem

According to this theorem the total energy (pressure energy, potential energy and kinetic energy) per unit volume or mass of an incompressible and non-viscous fluid in steady flow through a pipe remains constant throughout the flow, provided there is no source or sink of the fluid along the length of the pipe.



Mathematically for unit volume of liquid flowing through a pipe.

Fig. 11.12

$$P + \rho gh + \frac{1}{2} \rho v^2 = \text{constant}$$

To prove it, consider a liquid flowing steadily through a tube of non-uniform area of cross-section as shown in fig. If P_1 and P_2 are the pressures at the two ends of the tube respectively, work done in pushing the volume V of incompressible fluid from point B to C through the tube will be

$$W = P_1 V - P_2 V = (P_1 - P_2)V \quad \dots(i)$$

This work is used by the fluid in two ways.

(a) In changing the potential energy of mass m (in the volume V) from mgh to mgh ,

$$\text{i.e., } \Delta U = mg(h_2 - h_1) \quad \dots(ii)$$

(b) In changing the kinetic energy from $\frac{1}{2}mv_1^2$ to $\frac{1}{2}mv_2^2$,

$$\text{i.e., } \Delta K = \frac{1}{2}m(v_2^2 - v_1^2) \quad \dots(\text{iii})$$

Now as the fluid is non-viscous, by conservation of mechanical energy

$$W = \Delta U + \Delta K$$

$$\text{i.e., } (P_1 - P_2)V = mg(h_2 - h_1) + \frac{1}{2}m(v_2^2 - v_1^2)$$

$$\text{or } P_1 - P_2 = \rho g(h_2 - h_1) + \frac{1}{2}\rho(v_2^2 - v_1^2) \quad [\text{As } \rho = m/V]$$

$$\text{or } P_1 + \rho gh_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2}\rho v_2^2$$

$$\text{or } P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$$

This equation is the so called Bernoulli's equation and represents conservation of mechanical energy in case of moving fluids.

(i) Bernoulli's theorem for unit mass of liquid flowing through a pipe can also be written as:

$$\frac{P}{\rho} + gh + \frac{1}{2}v^2 = \text{constant}$$

$$\text{(ii) Dividing above equation by } g \text{ we get } \frac{P}{\rho g} + h + \frac{v^2}{2g} = \text{constant}$$

Here $\frac{P}{\rho g}$ is called pressure head, h is called gravitational head and

$\frac{v^2}{2g}$ is called velocity head. From this equation Bernoulli's theorem can be stated as.

"In stream line flow of an ideal liquid, the sum of pressure head, gravitational head and velocity head of every cross section of the liquid is constant."

Applications of Bernoulli's Theorem

(i) Attraction between two closely parallel moving boats (or buses)

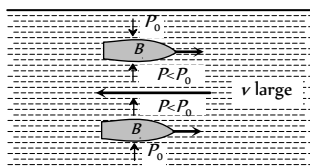


Fig. 11.13

When two boats or buses move side by side in the same direction, the water (or air) in the region between them moves faster than that on the remote sides. Consequently in accordance with *Bernoulli's principle* the pressure between them is reduced and hence due to pressure difference they are pulled towards each other creating the so called attraction.

(ii) Working of an aeroplane

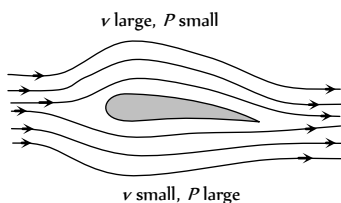


Fig. 11.14

This is also based on Bernoulli's principle. The wings of the aeroplane are of the shape as shown in fig. Due to this specific shape of wings when the aeroplane runs, air passes at higher speed over it as compared to its lower surface. This difference of air speeds above and below the wings, in accordance with Bernoulli's principle, creates a pressure difference, due to which an upward force called 'dynamic lift' (= pressure difference $\times$ area of wing) acts on the plane. If this force becomes greater than the weight of the plane, the plane will rise up.

(iii) Action of atomiser

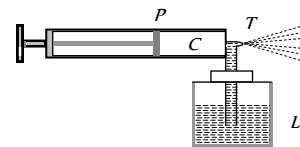


Fig. 11.15

The action of carburetor, paint-gun, scent-spray or insect-sprayer is based on Bernoulli's principle. In all these, by means of motion of a piston P in a cylinder C , high speed air is passed over a tube T dipped in liquid L to be sprayed. High speed air creates low pressure over the tube due to which liquid (paint, scent, insecticide or petrol) rises in it and is then blown off in very small droplets with expelled air.

(iv) Blowing off roofs by wind storms

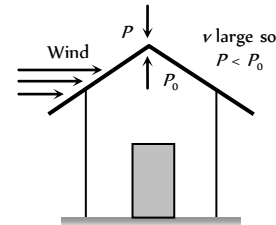


Fig. 11.16

During a tornado or hurricane, when a high speed wind blows over a straw or tin roof, it creates a low pressure (P) in accordance with Bernoulli's principle.

However, the pressure below the roof (i.e., inside the room) is still atmospheric ($= P_0$). So due to this difference of pressure, the roof is lifted up and is then blown off by the wind.

(v) **Magnus effect** : When a spinning ball is thrown, it deviates from its usual path in flight. This effect is called Magnus effect and plays an important role in tennis, cricket and soccer, etc. as by applying appropriate spin the moving ball can be made to curve in any desired direction.

If a ball is moving from left to right and also spinning about a horizontal axis perpendicular to the direction of motion as shown in fig. then relative to the ball, air will be moving from right to left.

The resultant velocity of air above the ball will be $(v + r\omega)$ while below it $(v - r\omega)$. So in accordance with Bernoulli's principle pressure above the ball will be less than below it. Due to this difference of pressure an upward force will act on the ball and hence the ball will deviate from its usual path OA_0 and will hit the ground at A_1 following the path OA_1 i.e., if a ball is thrown with back-spin, the pitch will curve less sharply prolonging the flight.

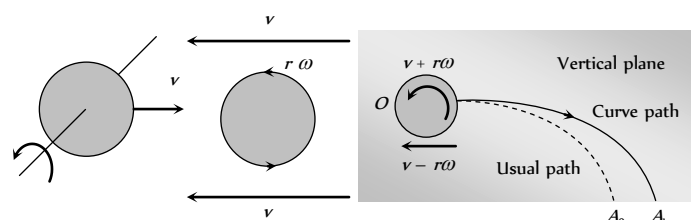


Fig. 11.17

Similarly if the spin is clockwise *i.e.*, the ball is thrown with top-spin, the force due to pressure difference will act in the direction of gravity and so the pitch will curve more sharply shortening the flight.

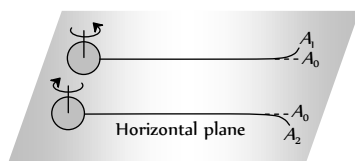


Fig. 11.18

Furthermore, if the ball is spinning about a vertical axis, the curving will be sideways as shown in producing the so called out swing or in swing.

(vi) **Venturimeter** : It is a device based on Bernoulli's theorem used for measuring the rate of flow of liquid through pipes.

It consists of two identical coaxial tubes *A* and *C* connected by a narrow co-axial tube *B*. Two vertical tubes *D* and *E* are mounted on the tubes *A* and *B* to measure the pressure of the flowing liquid.

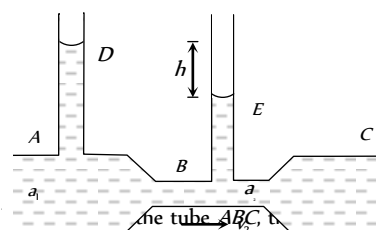


Fig. 11.19

When the liquid flows through the tube *ABC*, the flow in part *B* will be larger than in the tube *A* or *C*. So the pressure in part *B* will be less than that in tube *A* or *C*. By measuring the pressure difference between *A* and *B*, the rate of flow of the liquid in the tube can be calculated.

Let a_1 and a_2 are area of cross section of tube *A* and *B* respectively

v_1, v_2 = Velocity of flow of liquid through *A* and *B* respectively

P_1, P_2 = Liquid pressure at *A* and *B* respectively

$$\therefore P_1 - P_2 = h\rho g \quad \dots(i)$$

[ρ = density of flowing liquid]

From Bernoulli's theorem for horizontal flow of liquid

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) \quad \dots(ii)$$

$$\text{From (i) and (ii) } h\rho g = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} \rho \left[\frac{V^2}{a_2^2} - \frac{V^2}{a_1^2} \right]$$

[As $V = a_1 v_1 = a_2 v_2$]

$$\therefore V^2 = \frac{2a_1^2 a_2^2 h g}{a_1^2 - a_2^2} \text{ or } V = a_1 a_2 \sqrt{\frac{2hg}{a_1^2 - a_2^2}}$$

Velocity of Efflux

If a liquid is filled in a vessel up to height H and a hole is made at a depth h below the free surface of the liquid as shown in fig. then taking the

level of hole as reference level (*i.e.*, zero point of potential energy) and applying Bernoulli's principle to the liquid just inside and outside the hole (assuming the liquid to be at rest inside) we get

$$\therefore (P_0 + h\rho g) + 0 = P_0 + \frac{1}{2} \rho v^2 \text{ or } v = \sqrt{2gh}$$

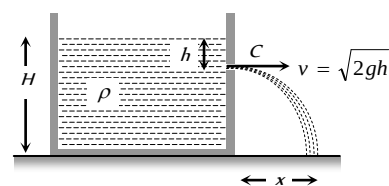


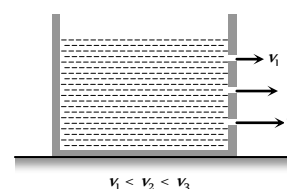
Fig. 11.20

Which is same as the speed that an object would acquire in falling from rest through a distance h and is called velocity of efflux or velocity of flow.

This result was first given by Torricelli, so this is known as Torricelli's theorem.

(i) The velocity of efflux is independent of the nature of liquid, quantity of liquid in the vessel and the area of orifice.

(ii) Greater is the distance of the hole from the free surface of liquid, greater will be the velocity of efflux [*i.e.*, $v \propto \sqrt{h}$]



$$v_1 < v_2 < v_3$$

Fig. 11.21

(iii) As the vertical velocity of liquid at the orifice is zero and it is at a height $(H - h)$ from the base, the time taken by the liquid to reach the base-level

$$t = \sqrt{\frac{2(H - h)}{g}}$$

(iv) Now during time t liquid is moving horizontally with constant velocity v , so it will hit the base level at a horizontal distance x (called range) as shown in figure.

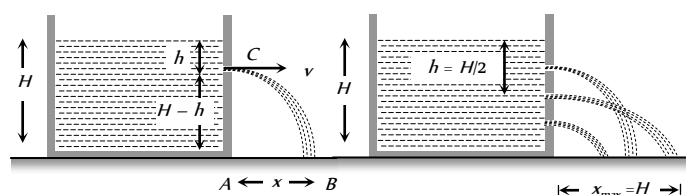


Fig. 11.22

$$\text{Such that } x = vt = \sqrt{2gh} \times \sqrt{\frac{2(H - h)}{g}} = 2\sqrt{h(H - h)}$$

$$\text{For maximum range } \frac{dx}{dh} = 0$$

$$\therefore h = \frac{H}{2}$$

i.e., range x will be maximum when

$$h = \frac{H}{2}$$

$$\therefore \text{Maximum range } x_{\max} = 2\sqrt{\frac{H}{2}\left[H - \frac{H}{2}\right]} = H$$

(v)

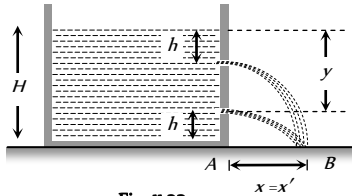


Fig. 11.23

If the level of free surface in a container is at height H from the base and there are two holes at depth h and y below the free surface, then

$$x = 2\sqrt{h(H-h)} \quad \text{and} \quad x' = 2\sqrt{y(H-y)}$$

Now if $x = x'$, i.e., $h(H-h) = y(H-y)$

$$\text{i.e., } y^2 - Hy + h(H-h) = 0$$

$$\text{or } y = \frac{1}{2}[H \pm (H-2h)],$$

$$\text{i.e., } y = h \text{ or } (H-h)$$

i.e., the range will be same if the orifice is at a depth h or $(H-h)$ below the free surface. Now as the distance $(H-h)$ from top means $H - (H-h) = h$ from the bottom, so the range is same for liquid coming out of holes at same distance below the top and above the bottom.

(vi)

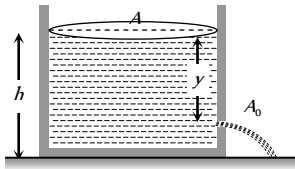


Fig. 11.24

If A_0 is the area of orifice at a depth y below the free surface and A is that of container, the volume of liquid coming out of the orifice per second will be $(dV/dt) = vA_0 = A_0\sqrt{2gy}$ [As $v = \sqrt{2gy}$]

Due to this, the level of liquid in the container will decrease and so if the level of liquid in the container above the hole changes from y to $y - dy$ in time t to $t + dt$ then $-dV = A dy$

So substituting this value of dV in the above equation

$$-A \frac{dy}{dt} = A_0\sqrt{2gy}$$

$$\text{i.e., } \int dt = -\frac{A}{A_0} \frac{1}{\sqrt{2g}} \int y^{-1/2} dy$$

So the time taken for the level to fall from H to H'

$$t = -\frac{A}{A_0} \frac{1}{\sqrt{2g}} \int_H^{H'} y^{-1/2} dy = \frac{A}{A_0} \sqrt{\frac{2}{g}} [\sqrt{H} - \sqrt{H'}]$$

If the hole is at the bottom of the tank, time t to make the tank empty :

$$t = \frac{A}{A_0} \sqrt{\frac{2H}{g}}$$

[As here $H' = 0$]

Viscosity and Newton's law of Viscous Force.

In case of steady flow of a fluid when a layer of fluid slips or tends to slip on adjacent layers in contact, the two layers exert tangential force on each other which tries to destroy the relative motion between them. The property of a fluid due to which it opposes the relative motion between its different layers is called viscosity (or fluid friction or internal friction) and the force between the layers opposing the relative motion is called viscous force.

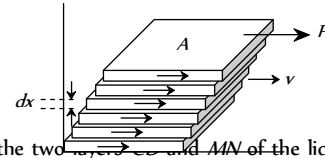


Fig. 11.25

Consider the two layers of the liquid at distances x and $x + dx$ from the fixed surface AB , having the velocities v and $v + dv$ respectively. Then $\frac{dv}{dx}$ denotes the rate of change of velocity with distance and is known as velocity gradient.

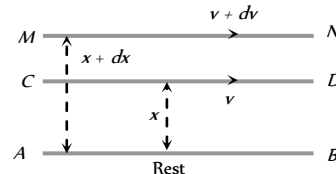


Fig. 11.26

According to Newton's hypothesis, the tangential force F acting on a plane parallel layer is proportional to the area of the plane A and the velocity gradient $\frac{dv}{dx}$ in a direction normal to the layer, i.e.,

$$F \propto A \quad \text{and} \quad F \propto \frac{dv}{dx}$$

$$\therefore F \propto A \frac{dv}{dx}$$

$$\text{or } F = -\eta A \frac{dv}{dx}$$

Where η is a constant called the coefficient of viscosity. Negative sign is employed because viscous force acts in a direction opposite to the flow of liquid.

$$\text{If } A = 1, \frac{dv}{dx} = 1 \text{ then } \eta = F$$

Hence the coefficient of viscosity is defined as the viscous force acting per unit area between two layers moving with unit velocity gradient.

(1) Units : dyne-s-cm or Poise (C.G.S. system); Newton-s-m or Poiseuille or decapoise (S.I. system)

1 Poiseuille = 1 decapoise = 10 Poise

(2) Dimension : $[ML^{-1}T^{-1}]$

(3) Viscosity of liquid is much greater (about 100 times more) than that of gases i.e. $\eta_L > \eta_G$

Example : Viscosity of water = 0.01 Poise.

While of air = 200 μ Poise

(4) With increase in pressure, the viscosity of liquids (except water) increases while that of gases is practically independent of pressure. The viscosity of water decreases with increase in pressure.

(5) Difference between viscosity and solid friction : Viscosity differs from the solid friction in the respect that the viscous force acting between two layers of the liquid depends upon the area of the layers, the relative velocity of two layers and distance between two layers, but the friction between two solid surfaces is independent of the area of surfaces in contact and the relative velocity between them.

(6) From kinetic theory point of view viscosity represents transport of momentum, while diffusion and conduction represents transport of mass and energy respectively.

(7) The viscosity of thick liquids like honey, glycerin, coaltar etc. is more than that of thin liquids like water.

(8) The cause of viscosity in liquids is cohesive forces among molecules where as in gases, it is due to diffusion.

(9) The viscosity of gases increases with increase of temperature, because on increasing temperature the rate of diffusion increases.

(10) The viscosity of liquid decreases with increase of temperature, because the cohesive force between the liquid molecules decreases with increase of temperature

Relation between coefficient of viscosity and temperature; Andrade

$$\text{formula } \eta = \frac{A e^{C\rho/T}}{\rho^{-1/3}}$$

Where T = Absolute temperature of liquid, ρ = density of liquid, A and C are constants.

Stoke's Law and Terminal Velocity

When a body moves through a fluid, the fluid in contact with the body is dragged with it. This establishes relative motion in fluid layers near the body, due to which viscous force starts operating. The fluid exerts viscous force on the body to oppose its motion. The magnitude of the viscous force depends on the shape and size of the body, its speed and the viscosity of the fluid. Stokes established that if a sphere of radius r moves with velocity v through a fluid of viscosity η , the viscous force opposing the motion of the sphere is

$$F = 6\pi\eta rv$$

This law is called Stokes law.

If a spherical body of radius r is dropped in a viscous fluid, it is first accelerated and then its acceleration becomes zero and it attains a constant velocity called terminal velocity.

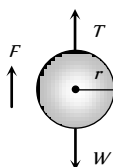


Fig. 11.27

Force on the body

(i) Weight of the body (W) = mg

$$= (\text{volume} \times \text{density}) \times g = \frac{4}{3}\pi r^3 \rho g$$

(ii) Upward thrust (T) = weight of the fluid displaced

$$= (\text{volume} \times \text{density}) \text{ of the fluid} \times g = \frac{4}{3}\pi r^3 \sigma g$$

(iii) Viscous force (F) = $6\pi\eta rv$

When the body attains terminal velocity the net force acting on the body is zero. $\therefore W - T - F = 0$ or $F = W - T$

$$\Rightarrow 6\pi\eta rv = \frac{4}{3}\pi r^3 \rho g - \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 (\rho - \sigma) g$$

$$\therefore \text{Terminal velocity } v = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$$

(i) Terminal velocity depend on the radius of the sphere so if radius is made n -fold, terminal velocity will become n times.

(ii) Greater the density of solid greater will be the terminal velocity

(iii) Greater the density and viscosity of the fluid lesser will be the terminal velocity.

(iv) If $\rho > \sigma$ then terminal velocity will be positive and hence the spherical body will attain constant velocity in downward direction.

(v) If $\rho < \sigma$ then terminal velocity will be negative and hence the spherical body will attain constant velocity in upward direction. Example : Air bubble in a liquid and clouds in sky.

(vi) Terminal velocity graph :

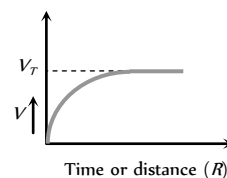


Fig. 11.28

Poiseuille's Formula

Poiseuille studied the stream-line flow of liquid in capillary tubes. He found that if a pressure difference (P) is maintained across the two ends of a capillary tube of length ' l ' and radius r , then the volume of liquid coming out of the tube per second is

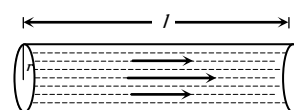


Fig. 11.29

(i) Directly proportional to the pressure difference (P).

(ii) Directly proportional to the fourth power of radius (r) of the capillary tube

(iii) Inversely proportional to the coefficient of viscosity (η) of the liquid.

(iv) Inversely proportional to the length (l) of the capillary tube.

$$\text{i.e. } V \propto \frac{P r^4}{\eta l} \text{ or } V = \frac{K P r^4}{\eta l}$$

$$\therefore V = \frac{\pi P r^4}{8 \eta l}$$

[Where $K = \frac{\pi}{8}$ is the constant of proportionality]

This is known as Poiseuille's equation.

This equation also can be written as,

$$V = \frac{P}{R} \text{ where } R = \frac{8 \eta l}{\pi r^4}$$

R is called as liquid resistance.

(1) Series combination of tubes

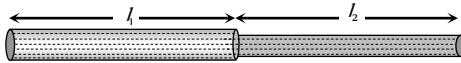


Fig. 11.30

(i) When two tubes of length l_1, l_2 and radii r_1, r_2 are connected in series across a pressure difference P ,

$$\text{Then } P = P_1 + P_2 \quad \dots(i)$$

Where P_1 and P_2 are the pressure difference across the first and second tube respectively

(ii) The volume of liquid flowing through both the tubes i.e. rate of flow of liquid is same.

$$\text{Therefore } V = V_1 = V_2$$

$$\text{i.e., } V = \frac{\pi P_1 r_1^4}{8\eta l_1} = \frac{\pi P_2 r_2^4}{8\eta l_2} \quad \dots(ii)$$

Substituting the value of P_1 and P_2 from equation (ii) to equation (i) we get

$$P = P_1 + P_2 = V \left[\frac{8\eta l_1}{\pi r_1^4} + \frac{8\eta l_2}{\pi r_2^4} \right]$$

$$\therefore V = \frac{P}{\left[\frac{8\eta l_1}{\pi r_1^4} + \frac{8\eta l_2}{\pi r_2^4} \right]} = \frac{P}{R_1 + R_2} = \frac{P}{R_{eff}}$$

Where R_1 and R_2 are the liquid resistance in tubes

(iii) Effective liquid resistance in series combination $R_{eff} = R_1 + R_2$

(2) Parallel combination of tubes

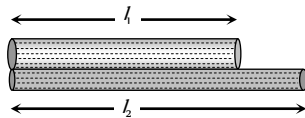


Fig. 11.31

(i) $P = P_1 = P_2$

$$(ii) V = V_1 + V_2 = \frac{P \pi r_1^4}{8\eta l_1} + \frac{P \pi r_2^4}{8\eta l_2}$$

$$= P \left[\frac{\pi r_1^4}{8\eta l_1} + \frac{\pi r_2^4}{8\eta l_2} \right]$$

$$\therefore V = P \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = \frac{P}{R_{eff}}$$

(iii) Effective liquid resistance in parallel combination

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} \text{ or } R_{eff} = \frac{R_1 R_2}{R_1 + R_2}$$

Tips & Tricks

✍ When a liquid is in equilibrium, the force acting on its surface is perpendicular everywhere.

✍ In a liquid, the pressure is same at the same horizontal level.

✍ Pressure at any point is same in all directions.

✍ The pressure is perpendicular to the surface of the fluid.

✍ The pressure at any point in the liquid depends on depth (h) below the surface, density of liquid and acceleration due to gravity.

It is independent of the shape of the containing vessel, or total mass of the liquid.

✍ Force is a vector quantity but pressure is a tensor quantity

✍ Pressure and density play the same role in case of fluids as force and mass play in case of solids.

✍ Bar and millibar are commonly used units for pressure in meteorology.

✍ Sudden fall in atmospheric pressure predicts possibility of a storm.

✍ Water barometer was constructed in 17th century by Von Guericke and fixed on the outside wall of his house. With the help of this barometer Von Guericke made the first recorded scientific weather forecast. He correctly predicted the severe storm after noting a sudden fall in the height of the water column.

✍ The specific gravity is also known as relative density. Thus, S.G. of a substance = $\frac{\text{density of the substance}}{\text{density of water (at } 4^\circ\text{C)}}$

✍ If the specific gravity of the material of a body is x , then its density is

(i) $x \text{ g cm}^3$ in C.G.S.

(ii) $x \times 10^3 \text{ kg m}^3$ in SI.

✍ The number of moles in a sample of any substance containing N molecules is given by

$$\mu = \frac{N}{N_A}$$

✍ The force between atoms and molecules is electrical in nature. However, it does not obey inverse square law.

✍ If two liquids of masses m_1, m_2 and densities ρ_1, ρ_2 are mixed together, then the density of the mixture is given by

$$\rho = \frac{\frac{m_1 + m_2}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}}}{\rho_1 + \rho_2}$$

✍ If two liquids of same mass but different densities are mixed together, then the density of the mixture is harmonic mean of the densities. That is

$$\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} \text{ or } \frac{1}{\rho} = \frac{1}{2} \left[\frac{1}{\rho_1} + \frac{1}{\rho_2} \right]$$

✍ If two drops of same volume but different densities are mixed together, then the density of the mixture is arithmetic mean of the

densities. That is $\rho = \frac{\rho_1 + \rho_2}{2}$

✍ The density of the liquid changes with pressure as follows :

$\rho = \rho_0 \left[1 + \frac{\Delta p}{B} \right]$ where Δp = change in pressure and B is the bulk modulus.

✍ The density of a liquid of bulk modulus B at depth h is given

$$\rho_d = \rho_0 \left[1 + \frac{h \rho g}{B} \right]$$

Where ρ is the average density of the liquid.

✍ The hydrometer can be used to measure density of the liquid or fluid.

✍ If a vessel contains liquid upto a height H and it has a hole in the side at a height h , then the velocity of efflux is $v = \sqrt{2g(H-h)}$. The

time taken by the liquid to reach the ground level is $t = \sqrt{2h/g}$.

Horizontal range of the liquid $R = 2[h(H-h)]^{1/2}$. The range is same for the hole at a height h above the bottom or at the depth h below the surface of the liquid.

The range is maximum for $h = H/2$. It is given by :

$$R_{\max} = 2 \left[\left(\frac{H}{2} \right) \left(H - \frac{H}{2} \right) \right]^{1/2} = H$$

✍ The cross-section of the water stream from a tap decreases as it goes down in accordance with the equation of continuity.

✍ The upthrust on body immersed in a liquid does not depend on the mass, density or shape of the body. It only depends on the volume of the body.

✍ The weight of the plastic bag full of air is same as that of the empty bag because the upthrust is equal to the weight of the air enclosed.

✍ Upthrust depends on the density of the fluid, not the density of the body.

✍ If two bodies have equal upthrust in a liquid, both have the same volume.

✍ When air blows over a roof, the force on the roof is upwards.

✍ If one floats one's back on the surface of water, the apparent weight is zero.

✍ If a body just floats in liquid (density of the body is equal to the density of liquid) then the body sinks if it is pushed downwards.

✍ The line joining the centre of gravity and centre of buoyancy is called central line.

✍ The point where the vertical line through centre of buoyancy intersects the central line is called metacentre.

✍ The floating body is in stable equilibrium where the metacentre is above the centre of gravity. (Centre of gravity is below the centre of

buoyancy)

✍ The floating body is in unstable equilibrium when the metacentre lies below the centre of gravity. (Centre of gravity is above the centre of buoyancy).

✍ The floating body is in the neutral equilibrium when centre of gravity coincides with the metacentre. (Centre of gravity coincides with the centre of buoyancy).

✍ The wooden rod cannot float vertically in a pond of water because centre of gravity lies above the metacentre.

✍ Air bubble in water always goes up. It is because density of air (ρ) is less than the density of water (σ). So the terminal velocity for air bubble is negative, which implies that the air bubble will go up. Positive terminal velocity means the body will fall down.

✍ The faster the air, the lower the pressure.

✍ Wings of an aeroplane are shaped to make air travel further and faster over their top surfaces.

✍ The lift force on a wing or aerofoil is proportional to the square of the speed of flow.

✍ Viscous force between the layers of a liquid is analogous to friction between two solid surfaces.

✍ With increase in temperature, the coefficient of viscosity of liquids decreases but that of gases increases. The reason is that as temperature rises, the atoms of the liquid become more mobile, whereas in case of a gas, the collision frequency of atoms increases as their motion becomes more random.

✍ We cannot sip a drink with a straw on the moon because there is no atmosphere on the moon.

Ordinary Thinking

Objective Questions

Pressure and Density

1. If pressure at half the depth of a lake is equal to $2/3$ pressure at the bottom of the lake then what is the depth of the lake

[RPET 2000]

- (a) 10 m (b) 20 m
(c) 60 m (d) 30 m

2. Two bodies are in equilibrium when suspended in water from the arms of a balance. The mass of one body is 36 g and its density is 9 g/cm. If the mass of the other is 48 g, its density in g/cm is

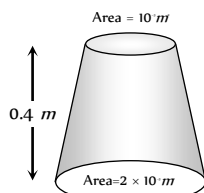
- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) 3 (d) 5

3. An inverted bell lying at the bottom of a lake 47.6 m deep has 50 cm of air trapped in it. The bell is brought to the surface of the lake. The volume of the trapped air will be (atmospheric pressure = 70 cm of Hg and density of Hg = 13.6 g/cm)

- (a) 350 cm (b) 300 cm
(c) 250 cm (d) 22 cm

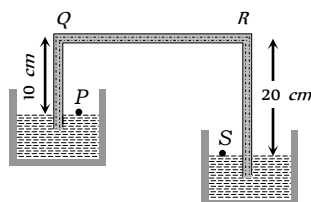
4. A uniformly tapering vessel is filled with a liquid of density 900 kg/m. The force that acts on the base of the vessel due to the liquid is ($g = 10 \text{ ms}^{-2}$)

- (a) 3.6 N
(b) 7.2 N
(c) 9.0 N
(d) 14.4 N



5. A siphon in use is demonstrated in the following figure. The density of the liquid flowing in siphon is 1.5 gm/cc. The pressure difference between the point P and S will be

- (a) 10 N/m
(b) $2 \times 10 \text{ N/m}$
(c) Zero
(d) Infinity



6. The height of a mercury barometer is 75 cm at sea level and 50 cm at the top of a hill. Ratio of density of mercury to that of air is 10. The height of the hill is

- (a) 250 m (b) 2.5 km
(c) 1.25 km (d) 750 m

7. Density of ice is ρ and that of water is σ . What will be the decrease in volume when a mass M of ice melts

- (a) $\frac{M}{\sigma - \rho}$ (b) $\frac{\sigma - \rho}{M}$
(c) $M \left[\frac{1}{\rho} - \frac{1}{\sigma} \right]$ (d) $\frac{1}{M} \left[\frac{1}{\rho} - \frac{1}{\sigma} \right]$

8. Equal masses of water and a liquid of density 2 are mixed together, then the mixture has a density of

- (a) $2/3$ (b) $4/3$
(c) $3/2$ (d) 3

9. A body of density d_1 is counterpoised by Mg of weights of density d_2 in air of density d . Then the true mass of the body is

- (a) M (b) $M \left(1 - \frac{d}{d_2} \right)$
(c) $M \left(1 - \frac{d}{d_1} \right)$ (d) $\frac{M(1 - d/d_2)}{(1 - d/d_1)}$

10. The pressure at the bottom of a tank containing a liquid does not depend on

[Kerala (Engg.) 2002]

- (a) Acceleration due to gravity
(b) Height of the liquid column
(c) Area of the bottom surface
(d) Nature of the liquid

11. When a large bubble rises from the bottom of a lake to the surface. Its radius doubles. If atmospheric pressure is equal to that of column of water height H , then the depth of lake is

[CPMT 1989]

[AIIMS 1995; AFMC 1997]

- (a) H (b) $2H$
(c) $7H$ (d) $8H$

12. The volume of an air bubble becomes three times as it rises from the bottom of a lake to its surface. Assuming atmospheric pressure to be 75 cm of Hg and the density of water to be $1/10$ of the density of mercury, the depth of the lake is

- (a) 5 m (b) 10 m
(c) 15 m (d) 20 m

13. The value of g at a place decreases by 2%. The barometric height of mercury

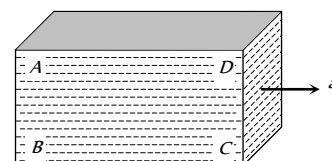
- (a) Increases by 2%
(b) Decreases by 2%
(c) Remains unchanged
(d) Sometimes increases and sometimes decreases

14. A barometer kept in a stationary elevator reads 76 cm. If the elevator starts accelerating up the reading will be

- (a) Zero (b) Equal to 76 cm
(c) More than 76 cm (d) Less than 76 cm

15. A closed rectangular tank is completely filled with water and is accelerated horizontally with an acceleration a towards right. Pressure is (i) maximum at, and (ii) minimum at

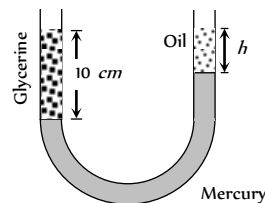
- (a) (i) B (ii) D
(b) (i) C (ii) D
(c) (i) B (ii) C
(d) (i) B (ii) A



16. A beaker containing a liquid is kept inside a big closed jar. If the air inside the jar is continuously pumped out, the pressure in the liquid near the bottom of the liquid will

- (a) Increases
(b) Decreases
(c) Remain constant

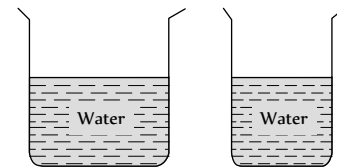
- (d) First decrease and then increase
17. A barometer tube reads 76 cm of mercury. If the tube is gradually inclined at an angle of 60° with vertical, keeping the open end immersed in the mercury reservoir, the length of the mercury column will be
- (a) 152 cm (b) 76 cm
(c) 38 cm (d) $38\sqrt{3}$ cm
18. The height to which a cylindrical vessel be filled with a homogeneous liquid, to make the average force with which the liquid presses the side of the vessel equal to the force exerted by the liquid on the bottom of the vessel, is equal to
- (a) Half of the radius of the vessel
(b) Radius of the vessel
(c) One-fourth of the radius of the vessel
(d) Three-fourth of the radius of the vessel
19. A vertical U-tube of uniform inner cross section contains mercury in both sides of its arms. A glycerin (density = 1.3 g/cm) column of length 10 cm is introduced into one of its arms. Oil of density 0.8 gm/cm is poured into the other arm until the upper surfaces of the oil and glycerin are in the same horizontal level. Find the length of the oil column, Density of mercury = 13.6 g/cm



- (a) 10.4 cm
(b) 8.2 cm
(c) 7.2 cm
(d) 9.6 cm
20. A triangular lamina of area A and height h is immersed in a liquid of density ρ in a vertical plane with its base on the surface of the liquid. The thrust on the lamina is
- (a) $\frac{1}{2} A \rho g h$ (b) $\frac{1}{3} A \rho g h$
(c) $\frac{1}{6} A \rho g h$ (d) $\frac{2}{3} A \rho g h$
21. If two liquids of same masses but densities ρ_1 and ρ_2 respectively are mixed, then density of mixture is given by
- (a) $\rho = \frac{\rho_1 + \rho_2}{2}$ (b) $\rho = \frac{\rho_1 + \rho_2}{2\rho_1\rho_2}$
(c) $\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$ (d) $\rho = \frac{\rho_1\rho_2}{\rho_1 + \rho_2}$
22. If two liquids of same volume but different densities ρ_1 and ρ_2 are mixed, then density of mixture is given by
- (a) $\rho = \frac{\rho_1 + \rho_2}{2}$ (b) $\rho = \frac{\rho_1 + \rho_2}{2\rho_1\rho_2}$
(c) $\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$ (d) $\rho = \frac{\rho_1\rho_2}{\rho_1 + \rho_2}$
23. The density ρ of water of bulk modulus B at a depth y in the ocean is related to the density at surface ρ_0 by the relation
- (a) $\rho = \rho_0 \left[1 - \frac{\rho_0 g y}{B} \right]$ (b) $\rho = \rho_0 \left[1 + \frac{\rho_0 g y}{B} \right]$

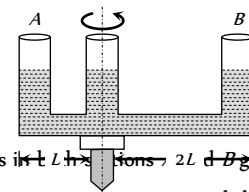
(c) $\rho = \rho_0 \left[1 + \frac{B}{\rho_0 h g y} \right]$ (d) $\rho = \rho_0 \left[1 - \frac{B}{\rho_0 g y} \right]$

24. With rise in temperature, density of a given body changes according to one of the following relations
- (a) $\rho = \rho_0 [1 + \gamma d \theta]$ (b) $\rho = \rho_0 [1 - \gamma d \theta]$
(c) $\rho = \rho_0 \gamma d \theta$ (d) $\rho = \rho_0 / \gamma d \theta$
25. Three liquids of densities $d, 2d$ and $3d$ are mixed in equal volumes. Then the density of the mixture is
- (a) d (b) $2d$
(c) $3d$ (d) $5d$
26. Three liquids of densities $d, 2d$ and $3d$ are mixed in equal proportions of weights. The relative density of the mixture is
- (a) $\frac{11d}{7}$ (b) $\frac{18d}{11}$
(c) $\frac{13d}{9}$ (d) $\frac{23d}{18}$
27. From the adjacent figure, the correct observation is



[KCET 2005]

- (a) The pressure on the bottom of tank (a) is greater than at the bottom of (b).
(b) The pressure on the bottom of the tank (a) is smaller than at the bottom of (b).
(c) The pressure depend on the shape of the container
(d) The pressure on the bottom of (a) and (b) is the same
28. A given shaped glass tube having uniform cross section is filled with water and is mounted on a rotatable shaft as shown in figure. If the tube is rotated with a constant angular velocity ω then



- (a) Water levels in A and B go up
(b) Water level in Section A goes up and that in B comes down
(c) Water level in Section A comes down and that in B it goes up
(d) Water levels remains same in both sections
29. Why the dam of water reservoir is thick at the bottom
- (a) Quantity of water increases with depth
(b) Density of water increases with depth
(c) Pressure of water increases with depth
(d) Temperature of water increases with depth
30. Air is blown through a hole on a closed pipe containing liquid. Then the pressure will
- (a) Increase on sides

[AFMC 2005]

[AFMC 2005]

- (b) Increase downwards
- (c) Increase in all directions
- (d) Never increases

31. Radius of an air bubble at the bottom of the lake is r and it becomes $2r$ when the air bubbles rises to the top surface of the lake. If P cm of water be the atmospheric pressure, then the depth of the lake is

- (a) $2p$
- (b) $8p$
- (c) $4p$
- (d) $7p$

Pascal's Law and Archimides Principle

1. An ice berg of density 900 Kg/m^3 is floating in water of density 1000 Kg/m^3 . The percentage of volume of ice-cube outside the water is

- (a) 20%
- (b) 35%
- (c) 10%
- (d) 25%

2. A log of wood of mass 120 Kg floats in water. The weight that can be put on the raft to make it just sink, should be (density of wood = 600 Kg/m^3) [CPMT 2004]

- (a) 80 Kg
- (b) 50 Kg
- (c) 60 Kg
- (d) 30 Kg

3. A hemispherical bowl just floats without sinking in a liquid of density $1.2 \times 10^3 \text{ kg/m}^3$. If outer diameter and the density of the bowl are 1 m and $2 \times 10^3 \text{ kg/m}^3$ respectively, then the inner diameter of the bowl will be [SCRA 1998]

- (a) 0.94 m
- (b) 0.97 m
- (c) 0.98 m
- (d) 0.99 m

4. In making an alloy, a substance of specific gravity S_1 and mass m_1 is mixed with another substance of specific gravity S_2 and mass m_2 ; then the specific gravity of the alloy is

[CPMT 1995]

- (a) $\left(\frac{m_1 + m_2}{S_1 + S_2} \right)$
- (b) $\left(\frac{S_1 S_2}{m_1 + m_2} \right)$
- (c) $\frac{m_1 + m_2}{\left(\frac{m_1}{S_1} + \frac{m_2}{S_2} \right)}$
- (d) $\frac{\left(\frac{m_1}{S_1} + \frac{m_2}{S_2} \right)}{m_1 + m_2}$

5. A concrete sphere of radius R has a cavity of radius r which is packed with sawdust. The specific gravities of concrete and sawdust are respectively 2.4 and 0.3 for this sphere to float with its entire volume submerged under water. Ratio of mass of concrete to mass of sawdust will be [AIIMS 1995]

- (a) 8
- (b) 4
- (c) 3
- (d) Zero

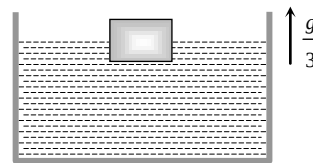
6. A metallic block of density 5 gm cm^{-3} and having dimensions $5 \text{ cm} \times 5 \text{ cm} \times 5 \text{ cm}$ is weighed in water. Its apparent weight will be

- (a) $5 \times 5 \times 5 \times 5 \text{ gf}$
- (b) $4 \times 4 \times 4 \times 4 \text{ gf}$
- (c) $5 \times 4 \times 4 \times 4 \text{ gf}$
- (d) $4 \times 5 \times 5 \times 5 \text{ gf}$

7. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards

with acceleration of $g/3$, the fraction of volume immersed in the liquid will be

- (a) $\frac{1}{2}$
- (b) $\frac{3}{8}$ [Kerla PET 2005]
- (c) $\frac{2}{3}$
- (d) $\frac{3}{4}$



8. A silver ingot weighing 2.1 kg is held by a string so as to be completely immersed in a liquid of relative density 0.8. The relative density of silver is 10.5. The tension in the string in kg-wt is [CPMT 2004]

- (a) 1.6
- (b) 1.94
- (c) 3.1
- (d) 5.25

9. A sample of metal weighs 210 gm in air, 180 gm in water and 120 gm in liquid. Then relative density (RD) of

- (a) Metal is 3
- (b) Metal is 7
- (c) Liquid is 3
- (d) Liquid is $\frac{1}{3}$

10. Two solids A and B float in water. It is observed that A floats with half its volume immersed and B floats with $2/3$ of its volume immersed. Compare the densities of A and B

- (a) 4 : 3
- (b) 2 : 3
- (c) 3 : 4
- (d) 1 : 3

11. The fraction of a floating object of volume V_0 and density d_0 above the surface of a liquid of density d will be

- (a) $\frac{d_0}{d}$
- (b) $\frac{dd_0}{d + d_0}$
- (c) $\frac{d - d_0}{d}$
- (d) $\frac{dd_0}{d - d_0}$

12. Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel. This law was first formulated by

- (a) Bernoulli
- (b) Archimedes
- (c) Boyle
- (d) Pascal

13. A block of steel of size $5 \text{ cm} \times 5 \text{ cm} \times 5 \text{ cm}$ is weighed in water. If the relative density of steel is 7, its apparent weight is

- (a) $6 \times 5 \times 5 \times 5 \text{ gf}$
- (b) $4 \times 4 \times 4 \times 7 \text{ gf}$
- (c) $5 \times 5 \times 5 \times 7 \text{ gf}$
- (d) $4 \times 4 \times 4 \times 6 \text{ gf}$

14. A body is just floating on the surface of a liquid. The density of the body is same as that of the liquid. The body is slightly pushed down. What will happen to the body [AIIMS 1980]

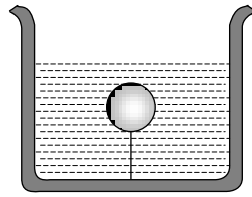
- (a) It will slowly come back to its earlier position
- (b) It will remain submerged, where it is left
- (c) It will sink
- (d) It will come out violently

15. A cork is submerged in water by a spring attached to the bottom of a bowl. When the bowl is kept in an elevator moving with acceleration downwards, the length of spring

- (a) Increases
- (b) Decreases
- (c) Remains unchanged
- (d) None of these

16. A solid sphere of density η (> 1) times lighter than water is suspended in a water tank by a string tied to its base as shown in fig. If the mass of the sphere is m then the tension in the string is given by

- (a) $\left(\frac{\eta-1}{\eta}\right)mg$
 (b) ηmg
 (c) $\frac{mg}{\eta-1}$
 (d) $(\eta-1)mg$



17. A hollow sphere of volume V is floating on water surface with *half* immersed in it. What should be the minimum volume of water poured inside the sphere so that the sphere now sinks into the water

- (a) $V/2$ (b) $V/3$
 (c) $V/4$ (d) V

18. A rectangular block is $5\text{ cm} \times 5\text{ cm} \times 10\text{ cm}$ in size. The block is floating in water with 5 cm side vertical. If it floats with 10 cm side vertical, what change will occur in the level of water?

- (a) No change
 (b) It will rise
 (c) It will fall
 (d) It may rise or fall depending on the density of block

19. A ball whose density is $0.4 \times 10^3\text{ kg/m}^3$ falls into water from a height of 9 cm . To what depth does the ball sink

- (a) 9 cm (b) 6 cm
 (c) 4.5 cm (d) 2.25 cm

20. Two solids A and B float in water. It is observed that A floats with $\frac{1}{2}$ of its body immersed in water and B floats with $\frac{1}{4}$ of its volume above the water level. The ratio of the density of A to that of B is

- (a) $4:3$ (b) $2:3$
 (c) $3:4$ (d) $1:2$

21. A boat carrying steel balls is floating on the surface of water in a tank. If the balls are thrown into the tank one by one, how will it affect the level of water [JK CET 2005]

- (a) It will remain unchanged
 (b) It will rise
 (c) It will fall
 (d) First it will first rise and then fall

22. Two pieces of metal when immersed in a liquid have equal upthrust on them; then

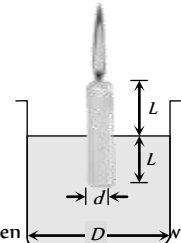
- (a) Both pieces must have equal weights
 (b) Both pieces must have equal densities
 (c) Both pieces must have equal volumes
 (d) Both are floating to the same depth

23. A wooden cylinder floats vertically in water with half of its length immersed. The density of wood is

- (a) Equal of that of water
 (b) Half the density of water
 (c) Double the density of water
 (d) The question is incomplete

24. A candle of diameter d is floating on a liquid in a cylindrical container of diameter D ($D \gg d$) as shown in figure. If it is burning at the rate of 2 cm/hour then the top of the candle will

- (a) Remain at the same height
 (b) Fall at the rate of 1 cm/hour
 (c) Fall at the rate of 2 cm/hour
 (d) Go up the rate of 1 cm/hour



25. An ice block contains a glass ball when within the water containing vessel, the level of water [AFMC 2005]

- (a) Rises (b) Falls
 (c) Unchanged (d) First rises and then falls

26. A large ship can float but a steel needle sinks because of

[AFMC 2005]

- (a) Viscosity (b) Surface tension
 (c) Density (d) None of these

27. Construction of submarines is based on [Kerala PMT 2005]

- (a) Archimedes' principle (b) Bernoulli's theorem
 (c) Pascal's law (d) Newton's laws

Fluid Flow

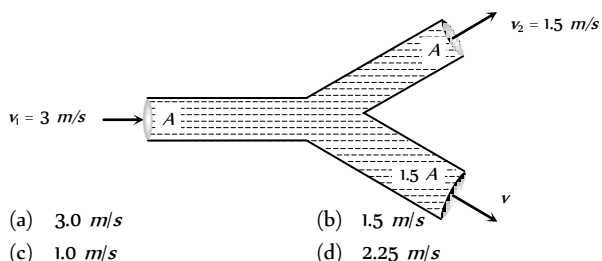
1. In which one of the following cases will the liquid flow in a pipe be most streamlined [Pb. CET 2005]

- (a) Liquid of high viscosity and high density flowing through a pipe of small radius
 (b) Liquid of high viscosity and low density flowing through a pipe of small radius
 (c) Liquid of low viscosity and low density flowing through a pipe of large radius
 (d) Liquid of low viscosity and high density flowing through a pipe of large radius

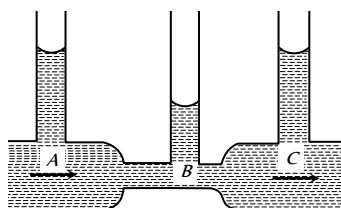
2. Two water pipes of diameters 2 cm and 4 cm are connected with the main supply line. The velocity of flow of water in the pipe of 2 cm diameter is [MNR 1980]

- (a) 4 times that in the other pipe
 (b) $\frac{1}{4}$ times that in the other pipe
 (c) 2 times that in the other pipe
 (d) $\frac{1}{2}$ times that in the other pipe

3. An incompressible liquid flows through a horizontal tube as shown in the following fig. Then the velocity v of the fluid is



- (a) 3.0 m/s (b) 1.5 m/s
(c) 1.0 m/s (d) 2.25 m/s
4. Water enters through end A with speed v_1 and leaves through end B with speed v_2 of a cylindrical tube AB. The tube is always completely filled with water. In case I tube is horizontal and in case II it is vertical with end A upwards and in case III it is vertical with end B upwards. We have $v_1 = v_2$ for
- (a) Case I (b) Case II
(c) Case III (d) Each case
5. Water is moving with a speed of 5.18 m/s through a pipe with a cross-sectional area of 4.20 cm. The water gradually descends 9.66 m as the pipe increase in area to 7.60 cm. The speed of flow at the lower level is
- (a) 3.0 m/s (b) 5.7 m/s
(c) 3.82 m/s (d) 2.86 m/s
6. The velocity of kerosene oil in a horizontal pipe is 5 m/s. If $g = 10 \text{ m/s}^2$ then the velocity head of oil will be
- (a) 1.25 m (b) 12.5 m
(c) 0.125 m (d) 125 m
7. In the following fig. is shown the flow of liquid through a horizontal pipe. Three tubes A, B and C are connected to the pipe. The radii of the tubes A, B and C at the junction are respectively 2 cm, 1 cm and 2 cm. It can be said that the



- (a) Height of the liquid in the tube A is maximum
(b) Height of the liquid in the tubes A and B is the same
(c) Height of the liquid in all the three tubes is the same
(d) Height of the liquid in the tubes A and C is the same
8. A manometer connected to a closed tap reads $3.5 \times 10^4 \text{ N/m}^2$. When the valve is opened, the reading of manometer falls to $3.0 \times 10^4 \text{ N/m}^2$, then velocity of flow of water is
- (a) 100 m/s (b) 10 m/s
(c) 1 m/s (d) $10\sqrt{10}$ m/s
9. Air is streaming past a horizontal air plane wing such that its speed in 120 m/s over the upper surface and 90 m/s at the lower surface. If the density of air is 1.3 kg per metre and the wing is 10 m long and has an average width of 2 m, then the difference of the pressure on the two sides of the wing of
- (a) 4095.0 Pascal (b) 409.50 Pascal
(c) 40.950 Pascal (d) 4.0950 Pascal

10. A large tank filled with water to a height ' h ' is to be emptied through a small hole at the bottom. The ratio of time taken for the level of water to fall from h to $\frac{h}{2}$ and from $\frac{h}{2}$ to zero is

- (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\sqrt{2} - 1$ (d) $\frac{1}{\sqrt{2} - 1}$

11. A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in m/s) through a small hole on the side wall of the cylinder near its bottom is

[AIEEE 2002]

- (a) 10 (b) 20
(c) 25.5 (d) 5

12. There is a hole in the bottom of tank having water. If total pressure at bottom is 3 atm (1 atm = 10 N/m^2) then the velocity of water flowing from hole is

[CPMT 2002]

- (a) $\sqrt{400} \text{ m/s}$ (b) $\sqrt{600} \text{ m/s}$
(c) $\sqrt{60} \text{ m/s}$ (d) None of these

13. There is a hole of area A at the bottom of cylindrical vessel. Water is filled up to a height h and water flows out in t second. If water is filled to a height $4h$, it will flow out in time equal to

- (a) t (b) $4t$
(c) $2t$ (d) $t/4$

14. A cylindrical tank has a hole of 1 cm in its bottom. If the water is allowed to flow into the tank from a tube above it at the rate of 70 cm/sec. then the maximum height up to which water can rise in the tank is

- (a) 2.5 cm (b) 5 cm
(c) 10 cm (d) 0.25 cm

15. A square plate of 0.1 m side moves parallel to a second plate with a velocity of 0.1 m/s, both plates being immersed in water. If the viscous force is 0.002 N and the coefficient of viscosity is 0.01 poise, distance between the plates in m is

[EAMCET (Med.) 2003]

- (a) 0.1 (b) 0.05
(c) 0.005 (d) 0.0005

16. Spherical balls of radius ' r ' are falling in a viscous fluid of viscosity ' η ' with a velocity ' v '. The retarding viscous force acting on the spherical ball is

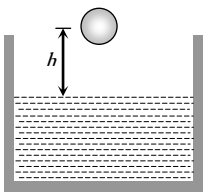
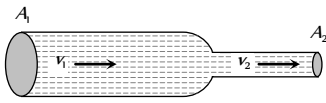
[AIEEE 2004]

- (a) Inversely proportional to ' r ' but directly proportional to velocity ' v '
(b) Directly proportional to both radius ' r ' and velocity ' v '
(c) Inversely proportional to both radius ' r ' and velocity ' v '
(d) Directly proportional to ' r ' but inversely proportional to ' v '

17. A small sphere of mass m is dropped from a great height. After it has fallen 100 m, it has attained its terminal velocity and continues to fall at that speed. The work done by air friction against the sphere during the first 100 m of fall is

[MP PMT 1990]

- (a) Greater than the work done by air friction in the second 100 m

- (b) Less than the work done by air friction in the second 100 m
(c) Equal to 100 mg
(d) Greater than 100 mg
18. Two drops of the same radius are falling through air with a steady velocity of 5 cm per sec. If the two drops coalesce, the terminal velocity would be [MP PMT 1990]
(a) 10 cm per sec (b) 2.5 cm per sec
(c) $5 \times (4)^{1/3}$ cm per sec (d) $5 \times \sqrt{2}$ cm per sec
19. A ball of radius r and density ρ falls freely under gravity through a distance h before entering water. Velocity of ball does not change even on entering water. If viscosity of water is η , the value of h is given by
(a) $\frac{2}{9} r^2 \left(\frac{1-\rho}{\eta} \right) g$
(b) $\frac{2}{81} r^2 \left(\frac{\rho-1}{\eta} \right) g$
(c) $\frac{2}{81} r^4 \left(\frac{\rho-1}{\eta} \right)^2 g$
(d) $\frac{2}{9} r^4 \left(\frac{\rho-1}{\eta} \right)^2 g$
- 
20. The rate of steady volume flow of water through a capillary tube of length l and radius r under a pressure difference of P is V . This tube is connected with another tube of the same length but half the radius in series. Then the rate of steady volume flow through them is (The pressure difference across the combination is P)
(a) $\frac{V}{16}$ (b) $\frac{V}{17}$
(c) $\frac{16V}{17}$ (d) $\frac{17V}{16}$
21. A liquid is flowing in a horizontal uniform capillary tube under a constant pressure difference P . The value of pressure for which the rate of flow of the liquid is doubled when the radius and length both are doubled is [EAMCET 2001]
(a) P (b) $\frac{3P}{4}$
(c) $\frac{P}{2}$ (d) $\frac{P}{4}$
22. We have two (narrow) capillary tubes T_1 and T_2 . Their lengths are l_1 and l_2 and radii of cross-section are r_1 and r_2 respectively. The rate of flow of water under a pressure difference P through tube T_1 is 8 cm/sec. If $l_1 = 2l_2$ and $r_1 = r_2$, what will be the rate of flow when the two tubes are connected in series and pressure difference across the combination is same as before ($= P$)
(a) 4 cm/sec (b) (16/3) cm/sec
(c) (8/17) cm/sec (d) None of these
23. In a laminar flow the velocity of the liquid in contact with the walls of the tube is
(a) Zero
(b) Maximum
(c) In between zero and maximum
(d) Equal to critical velocity
24. In a turbulent flow, the velocity of the liquid molecules in contact with the walls of the tube is
(a) Zero
(b) Maximum
(c) Equal to critical velocity
(d) May have any value
25. The Reynolds number of a flow is the ratio of
(a) Gravity to viscous force
(b) Gravity force to pressure force
(c) Inertia forces to viscous force
(d) Viscous forces to pressure forces
26. Water is flowing through a tube of non-uniform cross-section ratio of the radius at entry and exit end of the pipe is 3 : 2. Then the ratio of velocities at entry and exit of liquid is [RPMT 2001]
(a) 4 : 9 (b) 9 : 4
(c) 8 : 27 (d) 1 : 1
27. Water is flowing through a horizontal pipe of non-uniform cross-section. At the extreme narrow portion of the pipe, the water will have [MP PMT 1992]
(a) Maximum speed and least pressure
(b) Maximum pressure and least speed
(c) Both pressure and speed maximum
(d) Both pressure and speed least
28. A liquid is flowing in a horizontal pipe of non-uniform cross-section from left to right as shown in figure. A_1 and A_2 are the cross-sections of the portions of the tube as shown. Then the ratio of speeds v_1 / v_2 will be [EAMCET (Engg) 2003]
(a) A_1 / A_2
(b) A_2 / A_1
(c) $\sqrt{A_2} / \sqrt{A_1}$
(d) $\sqrt{A_1} / \sqrt{A_2}$
- 
29. In a streamline flow
(a) The speed of a particle always remains same
(b) The velocity of a particle always remains same
(c) The kinetic energies of all the particles arriving at a given point are the same
(d) The moments of all the particles arriving at a given point are the same
30. An application of Bernoulli's equation for fluid flow is found in
(a) Dynamic lift of an aeroplane
(b) Viscosity meter
(c) Capillary rise
(d) Hydraulic press
31. The Working of an atomizer depends upon

[MP PMT 1992; AFMC 2005]

- (a) Bernoulli's theorem (b) Boyle's law
(c) Archimedes principle (d) Newton's law of motion

32. The pans of a physical balance are in equilibrium. Air is blown under the right hand pan; then the right hand pan will

- (a) Move up
(b) Move down
(c) Move erratically
(d) Remain at the same level

33. According to Bernoulli's equation

$$\frac{P}{\rho g} + h + \frac{1}{2} \frac{v^2}{g} = \text{constant}$$

The terms A , B and C are generally called respectively:

- (a) Gravitational head, pressure head and velocity head
(b) Gravity, gravitational head and velocity head
(c) Pressure head, gravitational head and velocity head
(d) Gravity, pressure and velocity head

34. At what speed the velocity head of a stream of water be equal to 40 cm of Hg

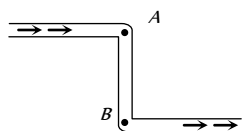
- (a) 282.8 cm/sec (b) 432.6 cm/sec
(c) 632.6 cm/sec (d) 832.6 cm/sec

35. The weight of an aeroplane flying in air is balanced by

- (a) Upthrust of the air which will be equal to the weight of the air having the same volume as the plane
(b) Force due to the pressure difference between the upper and lower surfaces of the wings, created by different air speeds on the surface
(c) Vertical component of the thrust created by air currents striking the lower surface of the wings
(d) Force due to the reaction of gases ejected by the revolving propeller

36. In this figure, an ideal liquid flows through the tube, which is of uniform cross-section. The liquid has velocities v_A and v_B , and pressure P_A and P_B at points A and B respectively

- (a) $v_A = v_B$
(b) $v_B > v_A$
(c) $P_A = P_B$
(d) $P_A > P_B$



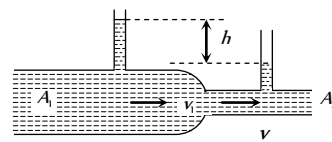
37. A liquid flows through a horizontal tube. The velocities of the liquid in the two sections, which have areas of cross-section A_1 and A_2 , are v_1 and v_2 respectively. The difference in the levels of the liquid in the two vertical tubes is h

(a) The volume of the liquid flowing through the tube in unit time is $A_1 v_1$

(b) $v_2 - v_1 = \sqrt{2gh}$

(c) $v_2^2 - v_1^2 = 2gh$

(d) The energy per unit mass of the liquid is the same in both sections of the tube

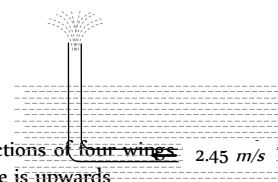


38. A sniper fires a rifle bullet into a gasoline tank making a hole 53.0 m below the surface of gasoline. The tank was sealed at 3.10 atm. The stored gasoline has a density of 660 kg/m³. The velocity with which gasoline begins to shoot out of the hole is

- (a) 27.8 m/s⁻¹ (b) 41.0 m/s⁻¹
(c) 9.6 m/s⁻¹ (d) 19.7 m/s⁻¹

39. An L-shaped tube with a small orifice is held in a water stream as shown in fig. The upper end of the tube is 10.6 cm above the surface of water. What will be the height of the jet of water coming from the orifice? Velocity of water stream is 2.45 m/s

- (a) Zero
(b) 20.0 cm
(c) 10.6 cm
(d) 40.0 cm



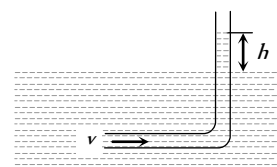
40. Fig. represents vertical sections of four wings horizontally in air. In which case the force is upwards

- (a)  (b) 
(c)  (d) 

41. An L-shaped glass tube is just immersed in flowing water such that its opening is pointing against flowing water. If the speed of water current is v , then

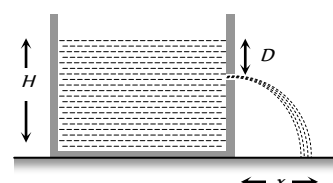
(a) The water in the tube rises to height $\frac{v^2}{2g}$

- (b) The water in the tube rises to height $\frac{g}{2v^2}$
(c) The water in the tube does not rise at all
(d) None of these



42. A tank is filled with water up to a height H . Water is allowed to come out of a hole P in one of the walls at a depth D below the surface of water. Express the horizontal distance x in terms of H and D

- (a) $x = \sqrt{D(H - D)}$
(b) $x = \sqrt{\frac{D(H - D)}{2}}$
(c) $x = 2\sqrt{D(H - D)}$



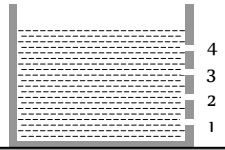


(d) $x = 4\sqrt{D(H-D)}$

43. A cylindrical vessel of 90 cm height is kept filled upto the brim. It has four holes 1, 2, 3, 4 which are respectively at heights of 20 cm, 30 cm, 45 cm and 50 cm from the horizontal floor PQ. The water falling at the maximum horizontal distance from the vessel comes from

[CPMT 1989]

- (a) Hole number 4
(b) Hole number 3
(c) Hole number 2
(d) Hole number 1



44. A rectangular vessel when full of water takes 10 minutes to be emptied through an orifice in its bottom. How much time will it take to be emptied when half filled with water

- (a) 9 minute (b) 7 minute
(c) 5 minute (d) 3 minute

45. A streamlined body falls through air from a height h on the surface of a liquid. If d and D ($D > d$) represents the densities of the material of the body and liquid respectively, then the time after which the body will be instantaneously at rest, is

- (a) $\sqrt{\frac{2h}{g}}$ (b) $\sqrt{\frac{2h}{g} \cdot \frac{D}{d}}$
(c) $\sqrt{\frac{2h}{g} \cdot \frac{d}{D}}$ (d) $\sqrt{\frac{2h}{g} \left(\frac{d}{D-d} \right)}$

46. A large tank is filled with water to a height H . A small hole is made at the base of the tank. It takes T_1 time to decrease the height of water to $\frac{H}{\eta}$ ($\eta > 1$); and it takes T_2 time to take out the rest of water. If $T_1 = T_2$, then the value of η is

- (a) 2 (b) 3
(c) 4 (d) $2\sqrt{2}$

47. Velocity of water in a river is

[CBSE PMT 1988]

- (a) Same everywhere
(b) More in the middle and less near its banks
(c) Less in the middle and more near its banks
(d) Increase from one bank to other bank

48. As the temperature of water increases, its viscosity

- (a) Remains unchanged
(b) Decreases
(c) Increases
(d) Increases or decreases depending on the external pressure

49. The coefficient of viscosity for hot air is

- (a) Greater than the coefficient of viscosity for cold air
(b) Smaller than the coefficient of viscosity for cold air
(c) Same as the coefficient of viscosity for cold air
(d) Increases or decreases depending on the external pressure

50. A good lubricant should have

- (a) High viscosity (b) Low viscosity
(c) Moderate viscosity (d) High density

51. We have three beakers A, B and C containing glycerine, water and kerosene respectively. They are stirred vigorously and placed on a table. The liquid which comes to rest at the earliest is

- (a) Glycerine

- (b) Water
(c) Kerosene
(d) All of them at the same time

52. A small drop of water falls from rest through a large height h in air; the final velocity is

- (a) $\propto \sqrt{h}$
(b) $\propto h$
(c) $\propto (1/h)$
(d) Almost independent of h

53. The rate of flow of liquid in a tube of radius r , length l , whose ends are maintained at a pressure difference P is $V = \frac{\pi Q P r^4}{\eta l}$ where η is coefficient of the viscosity and Q is

[DCE 2002]

- (a) 8 (b) $\frac{1}{8}$
(c) 16 (d) $\frac{1}{16}$

54. In Poiseuille's method of determination of coefficient of viscosity, the physical quantity that requires greater accuracy in measurement is

- (a) Pressure difference
(b) Volume of the liquid collected
(c) Length of the capillary tube
(d) Inner radius of the capillary tube

55. Two capillary tubes of the same length but different radii r_1 and r_2 are fitted in parallel to the bottom of a vessel. The pressure head is P . What should be the radius of a single tube that can replace the two tubes so that the rate of flow is same as before

- (a) $r_1 + r_2$ (b) $r_1^2 + r_2^2$
(c) $r_1^4 + r_2^4$ (d) None of these

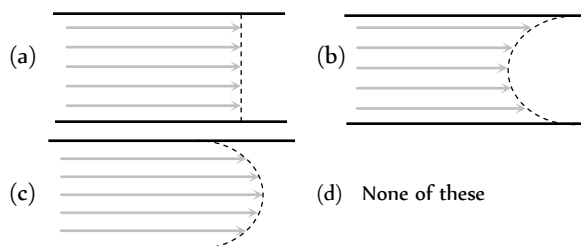
56. Two capillaries of same length and radii in the ratio 1 : 2 are connected in series. A liquid flows through them in streamlined condition. If the pressure across the two extreme ends of the combination is 1 m of water, the pressure difference across first capillary is

- (a) 9.4 m (b) 4.9 m
(c) 0.49 m (d) 0.94 m

57. Water flows in a streamlined manner through a capillary tube of radius a , the pressure difference being P and the rate of flow Q . If the radius is reduced to $a/2$ and the pressure increased to $2P$, the rate of flow becomes

- (a) $4Q$ (b) Q
(c) $\frac{Q}{4}$ (d) $\frac{Q}{8}$

58. A viscous fluid is flowing through a cylindrical tube. The velocity distribution of the fluid is best represented by the diagram



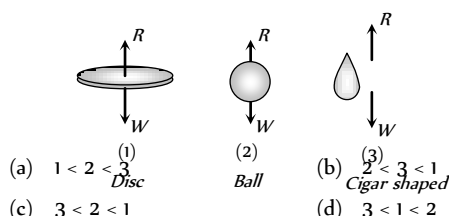
59. Water is flowing in a pipe of diameter 4 cm with a velocity 3 m/s. The water then enters into a tube of diameter 2 cm. The velocity of water in the other pipe is [BCECE 2005]

(a) 3 m/s (b) 6 m/s
(c) 12 m/s (d) 8 m/s

60. Two capillary of length L and $2L$ and of radius R and $2R$ are connected in series. The net rate of flow of fluid through them will be (given rate of the flow through single capillary, $X = \pi PR^4 / 8\eta L$) [DCE 2005]

(a) $\frac{8}{9} X$ (b) $\frac{9}{8} X$
(c) $\frac{5}{7} X$ (d) $\frac{7}{5} X$

61. When a body falls in air, the resistance of air depends to a great extent on the shape of the body, 3 different shapes are given. Identify the combination of air resistances which truly represents the physical situation. (The cross sectional areas are the same).



(a) $1 < 2 < 3$ (b) $3 < 2 < 1$
(c) $3 < 2 < 1$ (d) $3 < 1 < 2$

62. Water falls from a tap, down the streamline

[Orissa JEE 2005]

(a) Area decreases (b) Area increases
(c) Velocity remains same (d) Area remains same

63. A manometer connected to a closed tap reads 4.5×10^5 pascal. When the tap is opened the reading of the manometer falls to 4×10^5 pascal. Then the velocity of flow of water is

(a) 7 ms^{-1} (b) 8 ms^{-1}
(c) 9 ms^{-1} (d) 10 ms^{-1}

64. What is the velocity v of a metallic ball of radius r falling in a tank of liquid at the instant when its acceleration is one-half that of a freely falling body? (The densities of metal and of liquid are ρ and σ respectively, and the viscosity of the liquid is η).

(a) $\frac{r^2 g}{9\eta} (\rho - 2\sigma)$ (b) $\frac{r^2 g}{9\eta} (2\rho - \sigma)$
(c) $\frac{r^2 g}{9\eta} (\rho - \sigma)$ (d) $\frac{2r^2 g}{9\eta} (\rho - \sigma)$

65. Consider the following equation of Bernoulli's theorem.

$$P + \frac{1}{2} \rho V^2 + \rho gh = K \text{ (constant)}$$

The dimensions of K/P are same as that of which of the following [AFMC 2005]

(a) Thrust (b) Pressure
(c) Angle (d) Viscosity

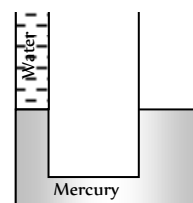
66. An incompressible fluid flows steadily through a cylindrical pipe which has radius $2r$ at point A and radius r at B further along the flow direction. If the velocity at point A is v , its velocity at point B is [Kerala PMT 2005]

(a) $2v$ (b) v
(c) $v/2$ (d) $4v$

Critical Thinking

Objective Questions

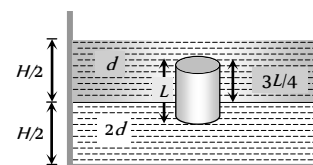
1. A U-tube in which the cross-sectional area of the limb on the left is one quarter, the limb on the right contains mercury (density 13.6 g/cm^3). The level of mercury in the narrow limb is at a distance of 36 cm from the upper end of the tube. What will be the rise in the level of mercury in the right limb if the left limb is filled to the top with water



(a) 1.2 cm
(b) 2.35 cm
(c) 0.36 cm [KCET, 2005]
(d) 0.8 cm

2. A homogeneous solid cylinder of length L ($L < H/2$). Cross-sectional area $A/5$ is immersed such that it floats with its axis vertical at the liquid-liquid interface with length $L/4$ in the denser liquid as shown in the fig. The lower density liquid is open to atmosphere having pressure P_0 . Then density D of solid is given by

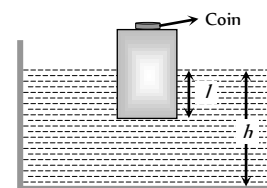
(a) $\frac{5}{4} d$
(b) $\frac{4}{5} d$
(c) d
(d) $\frac{d}{5}$



3. A wooden block, with a coin placed on its top, floats in water as shown in fig. the distance l and h are shown there. After some time the coin falls into the water. Then

[IIT-JEE (Screening) 2002]

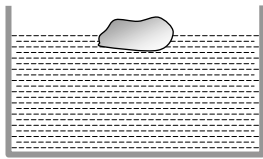
(a) l decreases and h increases
(b) l increases and h decreases
(c) Both l and h increase
(d) Both l and h decrease



4. A vessel contains oil (density = 0.8 gm/cm^3) over mercury (density = 13.6 gm/cm^3). A homogeneous sphere floats with half of its volume immersed in mercury and the other half in oil. The density of the material of the sphere in gm/cm^3 is

- (a) 3.3 (b) 6.4
(c) 7.2 (d) 12.8

5. A body floats in a liquid contained in a beaker. The whole system as shown falls freely under gravity. The upthrust on the body due to the liquid is



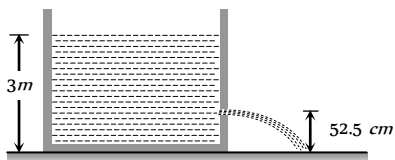
[IIT-JEE 1982]

- (a) Zero
(b) Equal to the weight of the liquid displaced
(c) Equal to the weight of the body in air
(d) Equal to the weight of the immersed position of the body

6. A liquid is kept in a cylindrical vessel which is being rotated about a vertical axis through the centre of the circular base. If the radius of the vessel is r and angular velocity of rotation is ω , then the difference in the heights of the liquid at the centre of the vessel and the edge is

- (a) $\frac{r\omega}{2g}$ (b) $\frac{r^2\omega^2}{2g}$
(c) $\sqrt{2gr\omega}$ (d) $\frac{\omega^2}{2gr^2}$

7. Water is filled in a cylindrical container to a height of $3m$. The ratio of the cross-sectional area of the orifice and the beaker is 0.1 . The square of the speed of the liquid coming out from the orifice is ($g = 10 \text{ m/s}^2$)



[IIT JEE 2004]

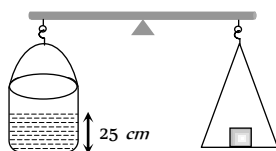
- (a) 50 m/s
(b) 50.5 m/s
(c) 51 m/s
(d) 52 m/s

8. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth $4y$ from the top. When the tank is completely filled with water the quantities of water flowing out per second from both the holes are the same. Then R is equal to

- (a) $2\pi L$ (b) $\frac{L}{\sqrt{2\pi}}$
(c) L (d) $\frac{L}{2\pi}$

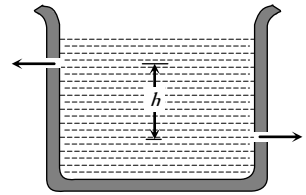
9. A cylinder containing water up to a height of 25 cm has a hole of cross-section $\frac{1}{4} \text{ cm}^2$ in its bottom. It is counterpoised in a balance. What is the initial change in the balancing weight when water begins to flow out

- (a) Increase of 12.5 gm-wt
(b) Increase of 6.25 gm-wt
(c) Decrease of 12.5 gm-wt
(d) Decrease of 6.25 gm-wt

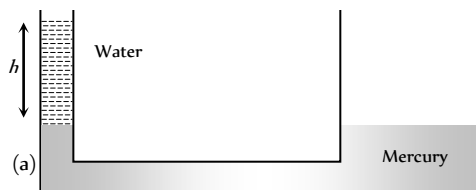


10. There are two identical small holes of area of cross-section a on the opposite sides of a tank containing a liquid of density ρ . The difference in height between the holes is h . Tank is resting on a smooth horizontal surface. Horizontal force which will have to be applied on the tank to keep it in equilibrium is

- (a) $gh\rho a$
(b) $\frac{2gh}{\rho a}$
(c) $2\rho agh$
(d) $\frac{\rho gh}{a}$



11. Two communicating vessels contain mercury. The diameter of one vessel is n times larger than the diameter of the other. A column of water of height h is poured into the left vessel. The mercury level will rise in the right-hand vessel (s = relative density of mercury and ρ = density of water) by



- (c) $\frac{h}{(n+1)^2 s}$ (d) $\frac{h}{n^2 s}$

12. A uniform rod of density ρ is placed in a wide tank containing a liquid of density ρ_0 ($\rho_0 > \rho$). The depth of liquid in the tank is half the length of the rod. The rod is in equilibrium, with its lower end resting on the bottom of the tank. In this position the rod makes an angle θ with the horizontal

- (a) $\sin\theta = \frac{1}{2} \sqrt{\rho_0 / \rho}$ (b) $\sin\theta = \frac{1}{2} \cdot \frac{\rho_0}{\rho}$
(c) $\sin\theta = \sqrt{\rho / \rho_0}$ (d) $\sin\theta = \rho_0 / \rho$

13. A block of ice floats on a liquid of density 1.2 in a beaker then level of liquid when ice completely melt

[IIT-JEE (Screening) 2000]

[IIT-JEE 1994]

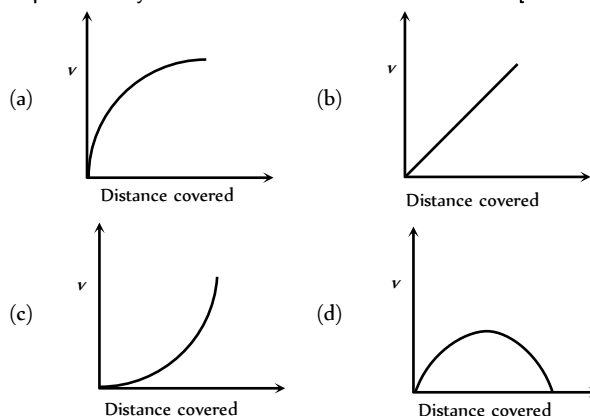
- (a) Remains same (b) Rises
(c) Lowers (d) (a), (b) or (c)

14. A vessel of area of cross-section A has liquid to a height H . There is a hole at the bottom of vessel having area of cross-section a . The time taken to decrease the level from H_1 to H_2 will be

- (a) $\frac{A}{a} \sqrt{\frac{2}{g}} [\sqrt{H_1} - \sqrt{H_2}]$ (b) $\sqrt{2gh}$
(c) $\sqrt{2gh(H_1 - H_2)}$ (d) $\frac{A}{a} \sqrt{\frac{g}{2}} [\sqrt{H_1} - \sqrt{H_2}]$

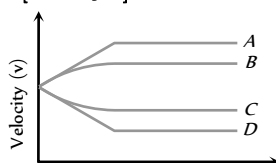
Graphical Questions

1. A lead shot of 1mm diameter falls through a long column of glycerine. The variation of its velocity v with distance covered is represented by [AIIMS 2003]

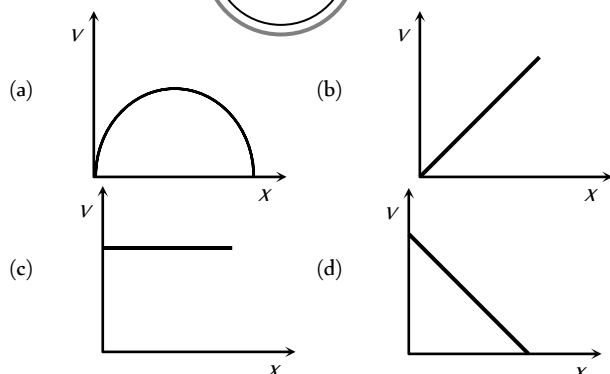
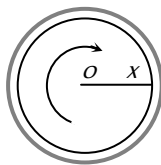


2. A small spherical solid ball is dropped from a great height in a viscous liquid. Its journey in the liquid is best described in the diagram given below by [CPMT 1988]

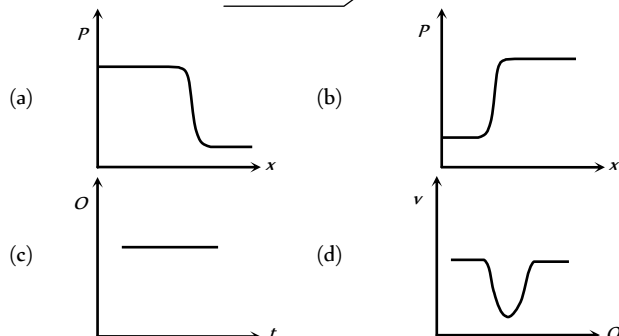
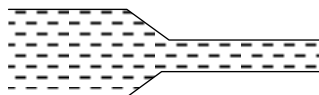
- (a) Curve A
(b) Curve B
(c) Curve C
(d) Curve D



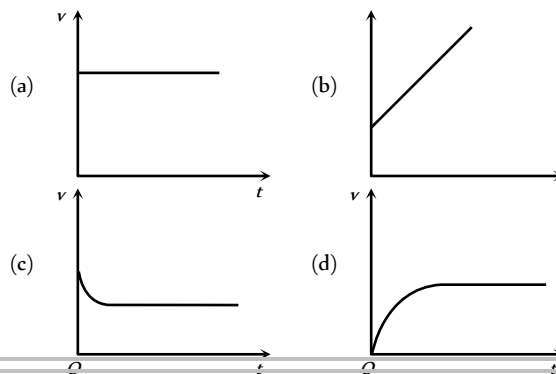
3. The diagram shows a cup of tea seen from above. The tea has been stirred and is now rotating without turbulence. A graph showing the speed v with which the liquid is crossing points at a distance X from O along a radius XO would look like



4. Water flows through a frictionless duct with a cross-section varying as shown in fig. Pressure p at points along the axis is represented by



5. From amongst the following curves, which one shows the variation of the velocity v with time t for a small sized spherical body falling vertically in a long column of a viscous liquid



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

- Assertion : Pascal law is the working principle of a hydraulic lift.
Reason : Pressure is equal to thrust per area.
- Assertion : The blood pressure in humans is greater at the feet than at the brain.
Reason : Pressure of liquid at any point is proportional to height, density of liquid and acceleration due to gravity.
- Assertion : Hydrostatic pressure is a vector quantity.
Reason : Pressure is force divided by area, and force is a vector quantity.
- Assertion : To float, a body must displace liquid whose weight is greater than the actual weight of the body.
Reason : The body will experiences no net downward force, in the case of floating.
- Assertion : A man sitting in a boat which is floating on a pond. If the man drinks some water from the pond, the level of the water in the pond decreases.
Reason : According to Archimede's principle the weight displaced by body is equal to the weight of the body.
- Assertion : A piece of ice floats in water, the level of water remains unchanged when the ice melts completely.
Reason : According to Archimede's principle, the loss in weight of the body in the liquid is equal to the weight of the liquid displaced by the immersed part of the body.
- Assertion : The velocity increases, when water flowing in broader pipe enter a narrow pipe.

- Reason : According to equation of continuity, product of area and velocity is constant.
8. Assertion : The velocity of fall of a man jumping with a parachute first increases and then becomes constant.
- Reason : The constant velocity of fall of man is called terminal velocity.
9. Assertion : The velocity of flow of a liquid is smaller when pressure is larger and viceversa.
- Reason : According to Bernoulli's theorem, for the stream line flow of an ideal liquid, the total energy per unit mass remains constant.
10. Assertion : The shape of an automobile is so designed that its front resembles the stream line pattern of the fluid through which it moves.
- Reason : The resistance offered by the fluid is maximum.
11. Assertion : The size of the needle of a syringe controls flow rate better than the thumb pressure exerted by a doctor while administering an injection.
- Reason : Flow rate is independent of pressure exerted by the thumb of the doctor.
12. Assertion : A fluid flowing out of a small hole in a vessel apply a backward thrust on the vessel.
- Reason : According to equation of continuity, the product of area and velocity remain constant.
13. Assertion : For a floating body to be in stable equilibrium, its centre of buoyancy must be located above the centre of gravity.
- Reason : The torque produced by the weight of the body and the upthrust will restore body back to its normal position, after the body is disturbed.
14. Assertion : Water flows faster than honey.
- Reason : The coefficient of viscosity of water is less than honey.
15. Assertion : The viscosity of liquid increases rapidly with rise of temperature.
- Reason : Viscosity of a liquid is the property of the liquid by virtue of which it opposes the relative motion amongst its different layers.
16. Assertion : Aeroplanes are made to run on the runway before take off, so that they acquire the necessary lift.
- Reason : According to Bernoulli's theorem, as velocity increases pressure decreases and viceversa.
17. Assertion : Sudden fall of pressure at a place indicates storm.
- Reason : Air flows from higher pressure to lower pressure.
18. Assertion : Machine parts are jammed in winter.
- Reason : The viscosity of lubricant used in machine parts increase at low temperature.
19. Assertion : A block of wood is floating in a tank containing water. The apparent weight of the floating block is equal to zero.
- Reason : Because the entire weight of the block is supported by the buoyant force (the upward thrust) due to water.
20. Assertion : A rain drop after falling through some height attains a constant velocity.
- Reason : At constant velocity, the viscous drag is just equal to its weight.
21. Assertion : paper pins are made to have pointed end.

- Reason : Because pointed pins have very small area due to which even for small applied force it exert large pressure on the surface.
22. Assertion : Railways tracks are laid on small sized wooden sleepers.
- Reason : Small sized wooden sleepers are used so that rails exert more pressure on the railway track. Due to which rail does not leave the track
23. Assertion : It is difficult to stop bleeding from a cut in the body at high altitudes.
- Reason : The atmospheric pressure at high altitude is lesser than the blood pressure.
24. Assertion : To empty an oil tank, two holes are made.
- Reason : Oil will come out two holes so it will emptied faster.
25. Assertion : Terminal velocity is same as the critical velocity.
- Reason : The constant velocity of fall of a body through a viscous fluid is called terminal velocity.
26. Assertion : When two boats sails parallel in the same direction and close to each other, they are pulled towards each other.
- Reason : The viscous drag on a spherical body moving with speed v is proportional to v .
27. Assertion : Cars and aeroplanes are streamlined.
- Reason : This is done to reduce the backward drag due to atmosphere.
28. Assertion : Bernoulli's theorem holds for incompressible, non-viscous fluids.
- Reason : The factor $\frac{v^2}{2g}$ is called velocity head.

Answers

Pressure and Density

1	b	2	c	3	b	4	b	5	c
6	b	7	c	8	b	9	d	10	c
11	c	12	c	13	a	14	d	15	a
16	b	17	a	18	b	19	d	20	b
21	c	22	a	23	b	24	b	25	b
26	b	27	d	28	a	29	c	30	c
31	d								

Pascal's Law and Archimides Principle

1	c	2	a	3	c	4	c	5	b
6	d	7	a	8	b	9	bc	10	c
11	c	12	d	13	a	14	b	15	b
16	d	17	a	18	a	19	b	20	b
21	c	22	c	23	b	24	b	25	b
26	d	27	a						



Fluid Flow

1	b	2	a	3	c	4	d	5	d
6	a	7	d	8	b	9	a	10	c
11	b	12	a	13	c	14	a	15	d
16	b	17	b	18	c	19	c	20	b
21	d	22	b	23	a	24	d	25	c
26	a	27	a	28	b	29	a	30	a
31	a	32	b	33	c	34	a	35	b
36	ad	37	acd	38	b	39	b	40	a
41	a	42	c	43	b	44	b	45	d
46	c	47	b	48	b	49	a	50	a
51	a	52	d	53	b	54	d	55	d
56	d	57	d	58	c	59	c	60	a
61	c	62	a	63	d	64	c	65	c
66	d								

Critical Thinking Questions

1	c	2	a	3	d	4	c	5	a
6	b	7	a	8	b	9	c	10	c
11	b	12	a	13	b	14	a		

Graphical Questions

1	a	2	b	3	d	4	a	5	d
---	---	---	---	---	---	---	---	---	---

Assertion and Reason

1	b	2	a	3	e	4	c	5	e
6	a	7	a	8	b	9	a	10	c
11	c	12	a	13	a	14	a	15	e
16	a	17	a	18	a	19	a	20	a
21	a	22	d	23	a	24	c	25	e
26	b	27	a	28	b				

AS Answers and Solutions

Pressure and Density

- (b) Pressure at bottom of the lake = $P_0 + h\rho g$

Pressure at half the depth of a lake = $P_0 + \frac{h}{2}\rho g$

According to given condition

$$P_0 + \frac{1}{2}h\rho g = \frac{2}{3}(P_0 + h\rho g) \Rightarrow \frac{1}{3}P_0 = \frac{1}{6}h\rho g$$

$$\Rightarrow h = \frac{2P_0}{\rho g} = \frac{2 \times 10^5}{10^3 \times 10} = 20 \text{ m}.$$
- (c) Apparent weight = $V(\rho - \sigma)g = \frac{m}{\rho}(\rho - \sigma)g$

where m = mass of the body,
 ρ = density of the body
 σ = density of water

If two bodies are in equilibrium then their apparent weight must be equal.

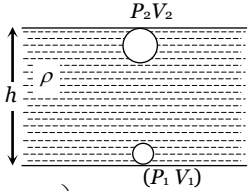
$$\therefore \frac{m_1}{\rho_1}(\rho_1 - \sigma) = \frac{m_2}{\rho_2}(\rho_2 - \sigma)$$

$$\Rightarrow \frac{36}{9}(9 - 1) = \frac{48}{\rho_2}(\rho_2 - 1)$$

By solving we get $\rho_2 = 3$.
- (b) According to Boyle's law, pressure and volume are inversely proportional to each other i.e. $P \propto \frac{1}{V}$

$$\Rightarrow P_1 V_1 = P_2 V_2$$

$$\Rightarrow (P_0 + h\rho_w g)V_1 = P_0 V_2$$

$$\Rightarrow V_2 = \left(1 + \frac{h\rho_w g}{P_0}\right)V_1$$


$$\Rightarrow V_2 = \left(1 + \frac{47.6 \times 10^2 \times 1 \times 1000}{70 \times 13.6 \times 1000}\right)V_1$$

$$\Rightarrow V_2 = (1 + 5)50 \text{ cm}^3 = 300 \text{ cm}^3.$$

[As $P_2 = P_0 = 70 \text{ cm of Hg} = 70 \times 13.6 \times 1000$]
- (b) Force acting on the base

$$F = P \times A = h\rho g A = 0.4 \times 900 \times 10 \times 2 \times 10^{-3} = 7.2 \text{ N}$$
- (c) As the both points are at the surface of liquid and these points are in the open atmosphere. So both point possess similar pressure and equal to 1 atm. Hence the pressure difference will be zero.
- (b) Difference of pressure between sea level and the top of hill

$$\Delta P = (h_1 - h_2) \times \rho_{\text{Hg}} \times g = (75 - 50) \times 10^{-2} \times \rho_{\text{Hg}} \times g \quad \dots(i)$$

and pressure difference due to h meter of air

$$\Delta P = h \times \rho_{\text{air}} \times g \quad \dots(ii)$$

By equating (i) and (ii) we get

$$h \times \rho_{\text{air}} \times g = (75 - 50) \times 10^{-2} \times \rho_{\text{Hg}} \times g$$

$$\therefore h = 25 \times 10^{-2} \left(\frac{\rho_{\text{Hg}}}{\rho_{\text{air}}} \right) = 25 \times 10^{-2} \times 10^4 = 2500 \text{ m}$$

$\therefore$ Height of the hill = 2.5 km.

- (c) Volume of ice = $\frac{M}{\rho}$, volume of water = $\frac{M}{\sigma}$.

$$\therefore \text{Change in volume} = \frac{M}{\rho} - \frac{M}{\sigma} = M \left(\frac{1}{\rho} - \frac{1}{\sigma} \right)$$
- (b) If two liquid of equal masses and different densities are mixed together then density of mixture

$$\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = \frac{2 \times 1 \times 2}{1 + 2} = \frac{4}{3}$$
- (d) Let M_0 = mass of body in vacuum.

Apparent weight of the body in air = Apparent weight of standard weights in air

$\Rightarrow$ Actual weight – upthrust due to displaced air

$=$ Actual weight – upthrust due to displaced air

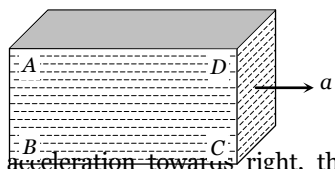
$$\Rightarrow M_0 g - \left(\frac{M_0}{d_1} \right) dg = M g - \left(\frac{M}{d_2} \right) dg \Rightarrow M_0 = \frac{M \left[1 - \frac{d}{d_2} \right]}{\left[1 - \frac{d}{d_1} \right]}$$
- (c) $P = h\rho g$ i.e. pressure does not depend upon the area of bottom surface.
- (c) $P_1 V_1 = P_2 V_2 \Rightarrow (P_0 + h\rho g) \times \frac{4}{3} \pi r^3 = P_0 \times \frac{4}{3} \pi (2r)^3$

Where, h = depth of lake

$$\Rightarrow h\rho g = 7P_0 \Rightarrow h = 7 \times \frac{H\rho g}{\rho g} = 7H.$$
- (c) $P_1 V_1 = P_2 V_2 \Rightarrow (P_0 + h\rho g)V = P_0 \times 3V$

$$\Rightarrow h\rho g = 2P_0 \Rightarrow h = \frac{2 \times 75 \times 13.6 \times g}{13.6 \times 10} = 15 \text{ m}$$
- (a) $h = \frac{P}{\rho g} \therefore h \propto \frac{1}{g}$ (P and ρ are constant)

If value of g decreased by 2% then h will increase by 2%.
- (d) $h = \frac{P}{\rho g} \therefore h \propto \frac{1}{g}$. If lift moves upward with some acceleration then effective g increases. So the value of h decreases i.e. reading will be less than 76 cm.
- (a)



Due to acceleration towards right, there will be a pseudo force in a left direction. So the pressure will

be more on rear side (Points A and B) in comparison with front side (Point D and C).

Also due to height of liquid column pressure will be more at the bottom (points B and C) in comparison with top (point A and D).

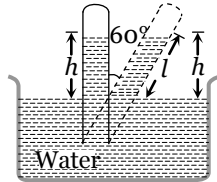
So overall maximum pressure will be at point B and minimum pressure will be at point D.

16. (b) Total pressure at (near) bottom of the liquid

$$P = P_0 + h\rho g$$

As air is continuously pumped out from jar (container), P_0 decreases and hence P decreases.

17. (a) $\cos 60^\circ = \frac{h}{l}$
 $\Rightarrow l = \frac{h}{\cos 60^\circ} = \frac{76}{1/2}$
 $\therefore l = 152 \text{ cm}$



18. (b) Pressure at the bottom = $h\rho g$

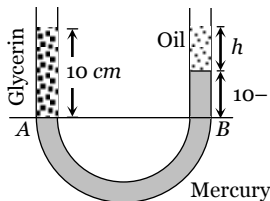
and pressure on the vertical surface = $\frac{1}{2}h\rho g$

Now, according to problem

Force at the bottom = Force on the vertical surface

$$\Rightarrow h\rho g \times \pi r^2 = \frac{1}{2}h\rho g \times 2\pi r h \Rightarrow h = r$$

19. (d)



At the condition of equilibrium

Pressure at point A = Pressure at point B

$$P_A = P_B \Rightarrow 10 \times 1.3 \times g = h \times 0.8 \times g + (10 - h) \times 13.6 \times g$$

By solving we get $h = 9.7 \text{ cm}$

20. (b) Thrust on lamina = pressure at centroid $\times$ Area

$$= \frac{h\rho g}{3} \times A = \frac{1}{3}A\rho gh$$

21. (c) $\rho = \frac{\text{Total mass}}{\text{Total volume}} = \frac{2m}{V_1 + V_2} = \frac{2m}{m\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)}$

$$\therefore \rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$$

22. (a) $\rho = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{2V} = \frac{V(\rho_1 + \rho_2)}{2V} = \frac{\rho_1 + \rho_2}{2}$

23. (b) Bulk modulus, $B = -V_0 \frac{\Delta p}{\Delta V} \Rightarrow \Delta V = -V_0 \frac{\Delta p}{B}$

$$\Rightarrow V = V_0 \left[1 - \frac{\Delta p}{B} \right]$$

$$\therefore \text{Density, } \rho = \rho_0 \left[1 - \frac{\Delta p}{B} \right]^{-1} = \rho_0 \left[1 + \frac{\Delta p}{B} \right]$$

where, $\Delta p = p - p_0 = h\rho_0 g$

= pressure difference between depth and surface of ocean

$$\therefore \rho = \rho_0 \left[1 + \frac{\rho_0 g y}{B} \right] \quad (\text{As } h = y)$$

24. (b) Since, with increase in temperature, volume of given body increases, while mass remains constant so that density will decrease.

$$\text{i.e. } \frac{\rho}{\rho_0} = \frac{m/V}{m/V_0} = \frac{V_0}{V} = \frac{V_0}{V_0(1 + r\Delta\theta)} = (1 - r\Delta\theta)$$

$$\therefore \rho = \rho_0(1 - r\Delta\theta)$$

25. (b) $\rho_{\text{mix}} = \frac{m_1 + m_2 + m_3}{3V} = \frac{V(d + 2d + 3d)}{3V} = 2d$

26. (b) $\rho_{\text{mix}} = \frac{3m}{V_1 + V_2 + V_3} = \frac{3m}{\frac{m}{d} + \frac{m}{2d} + \frac{m}{3d}} = \frac{3 \times 6}{11}d = \frac{18}{11}d$

27. (d) Pressure = $h\rho g$ i.e. pressure at the bottom is independent of the area of the bottom of the tank. It depends on the height of water upto which the tank is filled with water. As in both the tanks, the levels of water are the same, pressure at the bottom is also the same.

28. (a)

29. (c) A torque is acting on the wall of the dam trying to make it topple. The bottom is made very broad so that the dam will be stable.

30. (c)

31. (d)

Pascal's Law and Archimides Principle

1. (c) Let the total volume of ice-berg is V and its density is ρ . If this ice-berg floats in water with volume V_{in}

$$\text{inside it then } V_{\text{in}}\sigma g = V\rho g \Rightarrow V_{\text{in}} = \left(\frac{\rho}{\sigma} \right) V$$

$$\text{or } V_{\text{out}} = V - V_{\text{in}} = \left(\frac{\sigma - \rho}{\sigma} \right) V$$

$$\Rightarrow \frac{V_{\text{out}}}{V} = \left(\frac{\sigma - \rho}{\sigma} \right) = \frac{1000 - 900}{1000} = \frac{1}{10}$$

$$\therefore V_{\text{out}} = 10\% \text{ of } V$$

2. (a) Volume of log of wood $V = \frac{\text{mass}}{\text{density}} = \frac{120}{600} = 0.2 \text{ m}^3$

Let x weight that can be put on the log of wood.

So weight of the body = $(120 + x) \times 10 \text{ N}$

Weight of displaced liquid = $V\sigma g = 0.2 \times 10^3 \times 10 \text{ N}$

The body will just sink in liquid if the weight of the body will be equal to the weight of displaced liquid.

$$\therefore (120 + x) \times 10 = 0.2 \times 10^3 \times 10$$

$$\Rightarrow 120 + x = 200 \therefore x = 80 \text{ kg}$$

3. (c) Weight of the bowl = mg

$$= V\rho g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]\rho g$$

where D = Outer diameter ,

d = Inner diameter

ρ = Density of bowl

Weight of the liquid displaced by the bowl

$$= V\sigma g = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3 \sigma g$$

where σ is the density of the liquid.

$$\text{For the flotation } \frac{4}{3}\pi\left(\frac{D}{2}\right)^3 \sigma g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]\rho g$$

$$\Rightarrow \left(\frac{1}{2}\right)^3 \times 1.2 \times 10^3 = \left[\left(\frac{1}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right] 2 \times 10^4$$

By solving we get $d = 0.98 \text{ m}$.

4. (c) Specific gravity of alloy = $\frac{\text{Density of alloy}}{\text{Density of water}}$

$$= \frac{\text{Mass of alloy}}{\text{Volume of alloy} \times \text{density of water}}$$

$$= \frac{m_1 + m_2}{\left(\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}\right) \times \rho_w} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1 / \rho_w} + \frac{m_2}{\rho_2 / \rho_w}} = \frac{m_1 + m_2}{s_1 + s_2}$$

$$\left[\text{Ass specific gravity of substance} = \frac{\text{density of substance}}{\text{density of water}} \right]$$

5. (b) Let specific gravities of concrete and saw dust are ρ_1 and ρ_2 respectively.

According to principle of floatation weight of whole sphere = upthrust on the sphere

$$\frac{4}{3}\pi(R^3 - r^3)\rho_1 g + \frac{4}{3}\pi r^3 \rho_2 g = \frac{4}{3}\pi R^3 \times 1 \times g$$

$$\Rightarrow R^3 \rho_1 - r^3 \rho_1 + r^3 \rho_2 = R^3$$

$$\Rightarrow R^3(\rho_1 - 1) = r^3(\rho_1 - \rho_2) \Rightarrow \frac{R^3}{r^3} = \frac{\rho_1 - \rho_2}{\rho_1 - 1}$$

$$\Rightarrow \frac{R^3 - r^3}{r^3} = \frac{\rho_1 - \rho_2 - \rho_1 + 1}{\rho_1 - 1}$$

$$\Rightarrow \frac{(R^3 - r^3)\rho_1}{r^3 \rho_2} = \left(\frac{1 - \rho_2}{\rho_1 - 1}\right) \frac{\rho_1}{\rho_2}$$

$$\Rightarrow \frac{\text{Mass of concrete}}{\text{Mass of saw dust}} = \left(\frac{1 - 0.3}{2.4 - 1}\right) \times \frac{2.4}{0.3} = 4$$

6. (d) Apparent weight

$$= V(\rho - \sigma)g = l \times b \times h \times (5 - 1) \times g$$

$$= 5 \times 5 \times 5 \times 4 \times g \text{ Dyne} = 4 \times 5 \times 5 \times 5 \text{ gf.}$$

7. (a) Fraction of volume immersed in the liquid $V_{in} = \left(\frac{\rho}{\sigma}\right)V$

i.e. it depends upon the densities of the block and liquid.

So there will be no change in it if system moves upward or downward with constant velocity or some acceleration.

8. (b) Apparent weight = $V(\rho - \sigma)g = \frac{M}{\rho}(\rho - \sigma)g$

$$= M\left(1 - \frac{\sigma}{\rho}\right)g = 2.1\left(1 - \frac{0.8}{10.5}\right)g = 1.94 \text{ g N}$$

$$= 1.94 \text{ Kg-wt}$$

9. (b, c) Density of metal = ρ , Density of liquid = σ

If V is the volume of sample then according to problem

$$210 = V\rho g \quad \dots(i)$$

$$180 = V(\rho - 1)g \quad \dots(ii)$$

$$120 = V(\rho - \sigma)g \quad \dots(iii)$$

By solving (i), (ii) and (iii) we get $\rho = 7$ and $\sigma = 3$.

10. (c) If two different bodies A and B are floating in the same liquid then $\frac{\rho_A}{\rho_B} = \frac{(f_{in})_A}{(f_{in})_B} = \frac{1/2}{2/3} = \frac{3}{4}$

11. (c) For the floatation $V_0 d_0 g = V_{in} d g \Rightarrow V_{in} = V_0 \frac{d_0}{d}$

$$\therefore V_{out} = V_0 - V_{in} = V_0 - V_0 \frac{d_0}{d} = V_0 \left[\frac{d - d_0}{d} \right]$$

$$\Rightarrow \frac{V_{out}}{V_0} = \frac{d - d_0}{d}.$$

12. (d)

13. (a) Apparent weight = $V(\rho - \sigma)g$

$$= 5 \times 5 \times 5(7 - 1)g = 6 \times 5 \times 5 \times 5 \text{ gf}$$

14. (b)

15. (b) Effective weight $W' = m(g - a)$ which is less than actual weight mg , so the length of spring decreases.

16. (d) Tension in spring T = upthrust - weight of sphere

$$= V\sigma g - V\rho g = V\eta\rho g - V\rho g \quad (\text{As } \sigma = \eta\rho)$$

$$= (\eta - 1)V\rho g = (\eta - 1)mg.$$

17. (a) When body (sphere) is half immersed, then upthrust = weight of sphere

$$\Rightarrow \frac{V}{2} \times \rho_{liq} \times g = V \times \rho \times g \therefore \rho = \frac{\rho_{liq}}{2}$$

When body (sphere) is fully immersed then,

Upthrust = wt. of sphere + wt. of water poured in sphere

$$\Rightarrow V \times \rho_{liq} \times g = V \times \rho \times g + V' \times \rho_{liq} \times g$$

$$\Rightarrow V \times \rho_{liq} = \frac{V \times \rho_{liq}}{2} + V' \times \rho_{liq} \Rightarrow V' = \frac{V}{2}$$

18. (a) Since no change in volume of displaced water takes place, hence level of water remains same.

19. (b) The velocity of ball before entering the water surface

$$v = \sqrt{2gh} = \sqrt{2g \times 9}$$

When ball enters into water, due to upthrust of water the velocity of ball decreases (or retarded)

The retardation, $a = \frac{\text{apparent weight}}{\text{mass of ball}}$

$$= \frac{V(\rho - \sigma)g}{V\rho} = \left(\frac{\rho - \sigma}{\rho} \right) g = \left(\frac{0.4 - 1}{0.4} \right) \times g = -\frac{3}{2}g$$

If h be the depth upto which ball sink, then,

$$0 - v^2 = 2 \times \left(-\frac{3}{2}g \right) \times h \Rightarrow 2g \times 9 = 3gh \therefore h = 6 \text{ cm.}$$

20. (b) Upthrust = weight of body

$$\text{For } A, \frac{V_A}{2} \times \rho_W \times g = V_A \times \rho_A \times g \Rightarrow \rho_A = \frac{\rho_W}{2}$$

$$\text{For } B, \frac{3}{4} V_B \times \rho_W \times g = V_B \times \rho_B \times g \Rightarrow \rho_B = \frac{3}{4} \rho_W$$

(Since $1/4$ of volume of B is above the water surface)

$$\therefore \frac{\rho_A}{\rho_B} = \frac{\rho_W/2}{3/4 \rho_W} = \frac{2}{3}$$

21. (c)

22. (c) Since, up thrust (F) = $V\sigma g$ i.e. $F \propto V$

23. (b) $V\rho g = \frac{V}{2} \sigma g \therefore \rho = \frac{\sigma}{2}$ (σ = density of water)

24. (b)

25. (b)

26. (d)

27. (a)

Fluid Flow

1. (b) For streamline flow, Reynold's number $N_R \propto \frac{r \rho}{\eta}$

should be less. For less value of N_R , radius and density should be small and viscosity should be high.

2. (a) $d_A = 2 \text{ cm}$ and $d_B = 4 \text{ cm} \therefore r_A = 1 \text{ cm}$ and $r_B = 2 \text{ cm}$

From equation of continuity, $av = \text{constant}$

$$\therefore \frac{v_A}{v_B} = \frac{a_B}{a_A} = \frac{\pi(r_B)^2}{\pi(r_A)^2} = \left(\frac{2}{1} \right)^2 \Rightarrow v_A = 4v_B$$

3. (c) If the liquid is incompressible then mass of liquid entering through left end, should be equal to mass of liquid coming out from the right end.

$$\therefore M = m_1 + m_2 \Rightarrow Av_1 = Av_2 + 1.5A \cdot v$$

$$\Rightarrow A \times 3 = A \times 1.5 + 1.5A \cdot v \Rightarrow v = 1 \text{ m/s}$$

4. (d) This happens in accordance with equation of continuity and this equation was derived on the principle of conservation of mass and it is true in every case, either tube remain horizontal or vertical.

5. (d) $a_1 v_1 = a_2 v_2$

$$\Rightarrow 4.20 \times 5.18 = 7.60 \times v_2 \Rightarrow v_2 = 2.86 \text{ m/s}$$

6. (a) Velocity head $h = \frac{v^2}{2g} = \frac{(5)^2}{2 \times 10} = 1.25 \text{ m}$

7. (d) As cross-section areas of both the tubes A and C are same and tube is horizontal. Hence according to equation of continuity $v_A = v_C$ and therefore according to Bernoulli's theorem $P_A = P_C$ i.e. height of liquid is same in both the tubes A and C .

8. (b) Bernoulli's theorem for unit mass of liquid

$$\frac{P}{\rho} + \frac{1}{2}v^2 = \text{constant}$$

As the liquid starts flowing, its pressure energy decreases

$$\frac{1}{2}v^2 = \frac{P_1 - P_2}{\rho} \Rightarrow \frac{1}{2}v^2 = \frac{3.5 \times 10^5 - 3 \times 10^5}{10^3} \Rightarrow v^2$$

$$= \frac{2 \times 0.5 \times 10^5}{10^3} \Rightarrow v^2 = 100 \Rightarrow v = 10 \text{ m/s}$$

9. (a) From the Bernoulli's theorem

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} \times 1.3 \times [(120)^2 - (90)^2]$$

$$= 4095 \text{ N/m}^2 \text{ or Pascal}$$

10. (c) Time taken for the level to fall from H to H'

$$t = \frac{A}{A_0} \sqrt{\frac{2}{g}} [\sqrt{H} - \sqrt{H'}]$$

According to problem- the time taken for the level to

$$\text{fall from } h \text{ to } \frac{h}{2} \quad t_1 = \frac{A}{A_0} \sqrt{\frac{2}{g}} \left[\sqrt{h} - \sqrt{\frac{h}{2}} \right]$$

and similarly time taken for the level to fall from $\frac{h}{2}$

$$\text{to zero} \quad t_2 = \frac{A}{A_0} \sqrt{\frac{2}{g}} \left[\sqrt{\frac{h}{2}} - 0 \right]$$

$$\therefore \frac{t_1}{t_2} = \frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}} - 0} = \sqrt{2} - 1.$$

11. (b) $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = 20 \text{ m/s}$

12. (a) Pressure at the bottom of tank $P = h\rho g = 3 \times 10^5 \frac{\text{N}}{\text{m}^2}$

Pressure due to liquid column

$$P_l = 3 \times 10^5 - 1 \times 10^5 = 2 \times 10^5$$

and velocity of water $v = \sqrt{2gh}$

$$\therefore v = \sqrt{\frac{2P_l}{\rho}} = \sqrt{\frac{2 \times 2 \times 10^5}{10^3}} = \sqrt{400} \text{ m/s}$$

13. (c) Time required to empty the tank $t = \frac{A}{A_0} \sqrt{\frac{2H}{g}}$

$$\therefore \frac{t_2}{t_1} = \sqrt{\frac{H_2}{H_1}} = \sqrt{\frac{4h}{h}} = 2 \therefore t_2 = 2t$$

14. (a) The height of water in the tank becomes maximum when the volume of water flowing into the tank per second becomes equal to the volume flowing out per second.

Volume of water flowing out per second

$$= Av = A\sqrt{2gh} \quad \dots(i)$$

Volume of water flowing in per second

$$= 70 \text{ cm}^3/\text{sec} \quad \dots(ii)$$

From (i) and (ii) we get

$$A\sqrt{2gh} = 70 \Rightarrow 1 \times \sqrt{2gh} = 70$$

$$\Rightarrow 1 \times \sqrt{2 \times 980 \times h} = 70$$

$$\therefore h = \frac{4900}{1960} = 2.5 \text{ cm.}$$

15. (d) $A = (0.1)^2 = 0.01 \text{ m}^2$,
 $\eta = 0.01 \text{ Poise} = 0.001 \text{ decapoise (M.K.S. unit)}$,

$dv = 0.1 \text{ m/s}$ and $F = 0.002 \text{ N}$

$$F = \eta A \frac{dv}{dx}$$

$$\therefore dx = \frac{\eta A dv}{F} = \frac{0.001 \times 0.01 \times 0.1}{0.002} = 0.0005 \text{ m.}$$

16. (b) $F = 6\pi\eta rv$

17. (b) In the first 100 m body starts from rest and its velocity goes on increasing and after 100 m it acquire maximum velocity (terminal velocity). Further, air friction i.e. viscous force which is proportional to velocity is low in the beginning and maximum at $v = v_T$.

Hence work done against air friction in the first 100 m is less than the work done in next 100 m.

18. (c) If two drops of same radius r coalesce then radius of new drop is given by R

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r^3 + \frac{4}{3}\pi r^3 \Rightarrow R^3 = 2r^3 \Rightarrow R = 2^{1/3}r$$

If drop of radius r is falling in viscous medium then it acquire a critical velocity v and $v \propto r^2$

$$\frac{v_2}{v_1} = \left(\frac{R}{r}\right)^2 = \left(\frac{2^{1/3}r}{r}\right)^2$$

$$\Rightarrow v_2 = 2^{2/3} \times v_1 = 2^{2/3} \times (5) = 5 \times (4)^{1/3} \text{ m/s}$$

19. (c) Velocity of ball when it strikes the water surface

$$v = \sqrt{2gh} \quad \dots(i)$$

Terminal velocity of ball inside the water

$$v = \frac{2}{9}r^2g \frac{(\rho-1)}{\eta} \quad \dots(ii)$$

$$\text{Equating (i) and (ii) we get } \sqrt{2gh} = \frac{2}{9} \frac{r^2g}{\eta} (\rho-1)$$

$$\Rightarrow h = \frac{2}{81}r^4 \left(\frac{\rho-1}{\eta}\right)^2 g$$

20. (b) Rate of flow of liquid $V = \frac{P}{R}$

where liquid resistance $R = \frac{8\eta l}{\pi r^4}$

For another tube liquid resistance

$$R' = \frac{8\eta l}{\pi \left(\frac{r}{2}\right)^4} = \frac{8\eta l}{\pi r^4} \cdot 16 = 16R$$

For the series combination

$$V_{\text{New}} = \frac{P}{R+R'} = \frac{P}{R+16R} = \frac{P}{17R} = \frac{V}{17}.$$

21. (d) From $V = \frac{P\pi r^4}{8\eta l} \Rightarrow P = \frac{V8\eta l}{\pi r^4}$

$$\Rightarrow \frac{P_2}{P_1} = \frac{V_2}{V_1} \times \frac{l_2}{l_1} \times \left(\frac{r_1}{r_2}\right)^4 = 2 \times 2 \times \left(\frac{1}{2}\right)^4 = \frac{1}{4}$$

$$\Rightarrow P_2 = \frac{P_1}{4} = \frac{P}{4}.$$

22. (b) $V = \frac{\pi P r^4}{8\eta l} = \frac{8 \text{ cm}^3}{\text{sec}}$

For composite tube

$$V_1 = \frac{P\pi r^4}{8\eta \left(l + \frac{l}{2}\right)} = \frac{2}{3} \frac{\pi P r^4}{8\eta l} = \frac{2}{3} \times 8 = \frac{16}{3} \frac{\text{cm}^3}{\text{sec}}$$

$$\left[\because l_1 = l = 2l_2 \text{ or } l_2 = \frac{l}{2} \right]$$

23. (a)

24. (d)

25. (c)

26. (a) If velocities of water at entry and exit points are v_1 and v_2 , then according to equation of continuity,

$$A_1 v_1 = A_2 v_2 \Rightarrow \frac{v_1}{v_2} = \frac{A_2}{A_1} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

27. (a)

28. (b)

29. (a)

30. (a)

31. (a)

32. (b) According to Bernoulli's theorem.

33. (c)

34. (a) $\frac{v^2}{2g} = h \Rightarrow v = \sqrt{2gh}$

$$= \sqrt{2 \times 10^3 \times 40} = 2\sqrt{2} \times 10^2 = 282.8 \text{ cm/s}$$

35. (b)

36. (a,d)

37. (a,c,d) According to equation of continuity the volume of liquid flowing through the tube in unit time remains constant i.e. $A_1 v_1 = A_2 v_2$, hence option (a) is correct

According to Bernoulli's theorem,

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

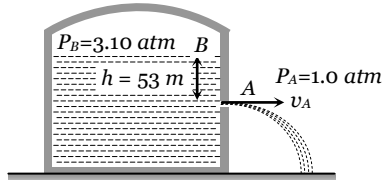
$$\Rightarrow P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) \Rightarrow h \rho g = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

$$\therefore v_2^2 - v_1^2 = 2gh$$

Hence option (c) is correct.

Also, according to Bernoulli's theorem option (d) is correct

38. (b)



According to Bernoulli's theorem,

$$P_B + h \rho g = P_A + \frac{1}{2} \rho v_A^2 \quad (A v_A \gg v_B)$$

$$3.10P + 53 \times 660 \times 10 = P + \frac{1}{2} \times 660 v_A^2$$

$$\Rightarrow 2.1 \times 1.01 \times 10^5 + 3.498 \times 10^5 = \frac{1}{2} \times 660 \times v_A^2$$

$$\Rightarrow 5.619 \times 10^5 = \frac{1}{2} \times 660 \times v_A^2$$

$$\therefore v_A = \sqrt{\frac{2 \times 5.619 \times 10^5}{660}} = 41 \text{ m/s}$$

39. (b) According to Bernoulli's theorem, $h = \frac{v^2}{2g}$

$$\Rightarrow h = \frac{(2.45)^2}{2 \times 10} = 0.314 = 31.4 \text{ cm}$$

$\therefore$ Height of jet coming from orifice

$$= 31.4 - 10.6 = 20.8 \text{ cm}$$

40. (a)

41. (a)

42. (c) Time taken by water to reach the bottom

$$= t = \sqrt{\frac{2(H-D)}{g}}$$

and velocity of water coming out of hole, $v = \sqrt{2gD}$

$\therefore$ Horizontal distance covered $x = v \times t$

$$= \sqrt{2gD} \times \sqrt{\frac{2(H-D)}{g}} = 2\sqrt{D(H-D)}$$

43. (b) Horizontal range will be maximum when $h = \frac{H}{2} = \frac{90}{2}$

$$= 45 \text{ cm i.e. hole 3.}$$

44. (b) Time taken to be emptied for h height, $t = \sqrt{\frac{2h}{g}}$

$$\text{and for } \frac{h}{2} \text{ height, } t' = \sqrt{\frac{2(h/2)}{g}} = \sqrt{\frac{h}{g}}$$

$$\therefore \frac{t'}{t} = \frac{1}{\sqrt{2}} \Rightarrow t' = \frac{t}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 7 \text{ minute}$$

45. (d) Upthrust – weight of body = apparent weight

$$VDg - Vdg = Vda,$$

Where a = retardation of body $\therefore a = \left(\frac{D-d}{d}\right)g$

The velocity gained after fall from h height in air,

$$v = \sqrt{2gh}$$

Hence, time to come in rest,

$$t = \frac{v}{a} = \frac{\sqrt{2gh} \times d}{(D-d)g} = \sqrt{\frac{2h}{g}} \times \frac{d}{(D-d)}$$

$$46. (c) t = \frac{A}{a} \sqrt{\frac{2}{g}} [\sqrt{H_1} - \sqrt{H_2}]$$

$$\text{Now, } T_1 = \frac{A}{a} \sqrt{\frac{2}{g}} \left[\sqrt{H} - \sqrt{\frac{H}{\eta}} \right]$$

$$\text{and } T_2 = \frac{A}{a} \sqrt{\frac{2}{g}} \left[\sqrt{\frac{H}{\eta}} - \sqrt{0} \right]$$

According to problem $T_1 = T_2$

$$\therefore \sqrt{H} - \sqrt{\frac{H}{\eta}} = \sqrt{\frac{H}{\eta}} - 0 \Rightarrow \sqrt{H} = 2\sqrt{\frac{H}{\eta}} \Rightarrow \eta = 4$$

47. (b)

48. (b)

49. (a)

50. (a)

51. (a)

52. (d)

53. (b)

54. (d)

55. (d) $V = V_1 + V_2$

$$\Rightarrow \frac{\pi P r^4}{8\eta l} = \frac{\pi P r_1^4}{8\eta l} + \frac{\pi P r_2^4}{8\eta l} \Rightarrow r^4 = r_1^4 + r_2^4$$

$$\therefore r = (r_1^4 + r_2^4)^{1/4}$$

56. (d) Given, $l_1 = l_2 = 1$, and $\frac{r_1}{r_2} = \frac{1}{2}$

$$V = \frac{\pi P_1 r_1^4}{8\eta l} = \frac{\pi P_2 r_2^4}{8\eta l} \Rightarrow \frac{P_1}{P_2} = \left(\frac{r_2}{r_1}\right)^4 = 16$$

$$\Rightarrow P_1 = 16P_2$$

Since both tubes are connected in series, hence pressure difference across combination,

$$P = P_1 + P_2 \Rightarrow 1 = P_1 + \frac{P_1}{16} \Rightarrow P_1 = \frac{16}{17} = 0.94m$$

57. (d) $V = \frac{\pi P r^4}{8\eta l} \therefore V \propto P r^4$ (η and l are constants)

$$\therefore \frac{V_2}{V_1} = \left(\frac{P_2}{P_1}\right) \left(\frac{r_2}{r_1}\right)^4 = 2 \times \left(\frac{1}{2}\right)^4 = \frac{1}{8} \therefore V_2 = \frac{Q}{8}$$

58. (c)

59. (c) $a_1 v_1 = a_2 v_2 \Rightarrow \frac{v_2}{v_1} = \frac{a_1}{a_2} = \left(\frac{r_1}{r_2}\right)^2$

$\Rightarrow v_2 = 3 \times (2)^2 = 12 \text{ m/s}$

60. (a) Fluid resistance is given by $R = \frac{8\eta l}{\pi^4}$.

When two capillary tubes of same size are joined in parallel, then equivalent fluid resistance is

$R_e = R_1 + R_2 = \frac{8\eta L}{\pi^4} + \frac{8\eta \times 2L}{\pi(2R)^4} = \left(\frac{8\eta L}{\pi^4}\right) \times \frac{9}{8}$

Equivalent resistance becomes $\frac{9}{8}$ times so rate of

flow will be $\frac{8}{9} X$

61. (c) A stream lined body has less resistance due to air.

62. (a)

63. (d) $\frac{P_1 - P_2}{\rho g} = \frac{v^2}{2g} \Rightarrow \frac{4.5 \times 10^5 - 4 \times 10^5}{10^3 \times g} = \frac{v^2}{2g} \therefore v = 10 \text{ m/s}$

64. (c)

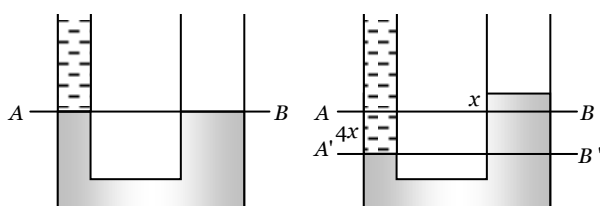
65. (c)

66. (d)

Critical Thinking Questions

1. (c) If the rise of level in the right limb be $x \text{ cm}$, the fall of level of mercury in left limb be $4x \text{ cm}$ because the area of cross section of right limb is 4 times as that of left limb.

$\therefore$ Level of water in left limb is $(36 + 4x) \text{ cm}$.



Now equating pressure at interface of Hg and water (at A' B')

$(36 + 4x) \times 1 \times g = 5x \times 13.6 \times g$

By solving we get $x = 0.56 \text{ cm}$.

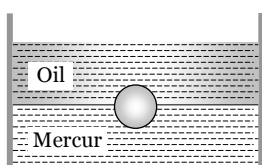
2. (a) Weight of cylinder = upthrust due to both liquids

$V \times D \times g = \left(\frac{A}{5} \times \frac{3}{4} L\right) \times d \times g + \left(\frac{A}{5} \times \frac{L}{4}\right) \times 2d \times g$

$\Rightarrow \left(\frac{A}{5} \times L\right) \times D \times g = \frac{A \times L \times d \times g}{4} \Rightarrow \frac{D}{5} = \frac{d}{4} \therefore D = \frac{5}{4} d$

3. (d) As the block moves up with the fall of coin, l decreases, similarly h will also decrease because when the coin is in water, it displaces water equal to its own volume only.

4. (c)



As the sphere floats in the liquid. Therefore its weight will be equal to the upthrust force on it

Weight of sphere

$= \frac{4}{3} \pi R^3 \rho g \quad \dots(i)$

Upthrust due to oil and mercury

$= \frac{2}{3} \pi R^3 \times \sigma_{oil} g + \frac{2}{3} \pi R^3 \sigma_{Hg} g \quad \dots(ii)$

Equating (i) and (ii)

$\frac{4}{3} \pi R^3 \rho g = \frac{2}{3} \pi R^3 0.8 g + \frac{2}{3} \pi R^3 \times 13.6 g$

$\Rightarrow 2\rho = 0.8 + 13.6 = 14.4 \Rightarrow \rho = 7.2$

5. (a) Upthrust = $V\rho_{liquid}(g - a)$

where, a = downward acceleration,

V = volume of liquid displaced

But for free fall $a = g \therefore$ Upthrust = 0

6. (b) From Bernoulli's theorem,

$P_A + \frac{1}{2} dv_A^2 + dgh_A = P_B + \frac{1}{2} dv_B^2 + dgh_B$

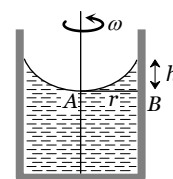
Here, $h_A = h_B$

$\therefore P_A + \frac{1}{2} dv_A^2 = P_B + \frac{1}{2} dv_B^2$

$\Rightarrow P_A - P_B = \frac{1}{2} d[v_B^2 - v_A^2]$

Now, $v_A = 0, v_B = r\omega$ and $P_A - P_B = hdg$

$\therefore hdg = \frac{1}{2} dr^2 \omega^2$ or $h = \frac{r^2 \omega^2}{2g}$

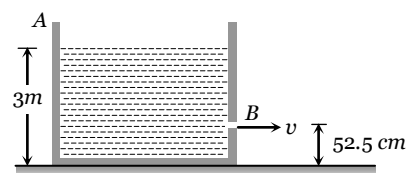


7. (a) Let A = cross-section of tank

a = cross-section hole

V = velocity with which level decreases

v = velocity of efflux



From equation of continuity $av = AV \Rightarrow V = \frac{av}{A}$

By using Bernoulli's theorem for energy per unit volume

Energy per unit volume at point A

= Energy per unit volume at point B

$P + \rho gh + \frac{1}{2} \rho V^2 = P + 0 + \frac{1}{2} \rho v^2$

$\Rightarrow v^2 = \frac{2gh}{1 - \left(\frac{a}{A}\right)^2} = \frac{2 \times 10 \times (3 - 0.525)}{1 - (0.1)^2} = 50 (\text{m/sec})^2$

8. (b) Velocity of efflux when the hole is at depth h ,

$v = \sqrt{2gh}$

Rate of flow of water from square hole

$$Q_1 = a_1 v_1 = L^2 \sqrt{2gy}$$

Rate of flow of water from circular hole

$$Q_2 = a_2 v_2 = \pi R^2 \sqrt{2g(4y)}$$

According to problem $Q_1 = Q_2$

$$\Rightarrow L^2 \sqrt{2gy} = \pi R^2 \sqrt{2g(4y)} \Rightarrow R = \frac{L}{\sqrt{2\pi}}$$

9. (c) Let A = The area of cross section of the hole

v = Initial velocity of efflux

d = Density of water,

Initial volume of water flowing out per second = Av

Initial mass of water flowing out per second = Adv

Rate of change of momentum = Adv^2

Initial downward force on the flowing out water = Adv^2

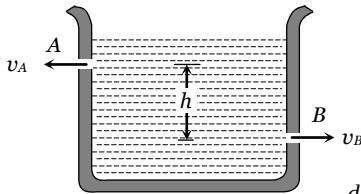
So equal amount of reaction acts upwards on the cylinder.

$$\therefore \text{Initial upward reaction} = Adv^2 \quad [\text{As } v = \sqrt{2gh}]$$

$$\therefore \text{Initial decrease in weight} = Ad(2gh)$$

$$= 2Adgh = 2 \times \left(\frac{1}{4}\right) \times 1 \times 980 \times 25 = 12.5 \text{ gm-wt.}$$

10. (c)



$$\text{Net force (reaction)} = F = F_B - F_A = \frac{dp_B}{dt} - \frac{dp_A}{dt}$$

$$= av_B \rho \times v_B - av_A \rho \times v_A$$

$$\therefore F = a\rho(v_B^2 - v_A^2) \quad \dots(i)$$

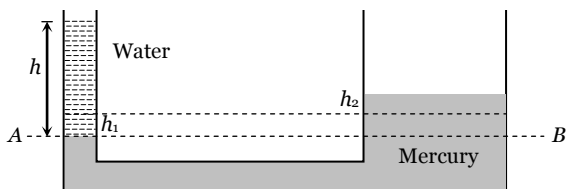
According to Bernoulli's theorem

$$p_A + \frac{1}{2} \rho v_A^2 + \rho gh = p_B + \frac{1}{2} \rho v_B^2 + 0$$

$$\Rightarrow \frac{1}{2} \rho (v_B^2 - v_A^2) = \rho gh \Rightarrow v_B^2 - v_A^2 = 2gh$$

From equation (i), $F = 2a\rho gh$.

11. (b)



If the level in narrow tube goes down by h_1 then in wider tube goes up to h_2 ,

$$\text{Now, } \pi r^2 h_1 = \pi (nr)^2 h_2 \Rightarrow h_1 = n^2 h_2$$

Now, pressure at point A = pressure at point B

$$h\rho g = (h_1 + h_2)\rho' g$$

$$\Rightarrow h = (n^2 h_2 + h_2) s g \left(\text{As } s = \frac{\rho'}{\rho} \right) \Rightarrow h_2 = \frac{h}{(n^2 + 1)s}$$

12. (a) Let $L = PQ$ = length of rod

$$\therefore SP = SQ = \frac{L}{2}$$

Weight of rod, $W = Al\rho g$, acting

At point S

And force of buoyancy,

$$F_B = Al\rho_0 g, [l = PR]$$

which acts at mid-point of PR.

For rotational equilibrium,

$$Al\rho_0 g \times \frac{l}{2} \cos \theta = AL\rho g \times \frac{L}{2} \cos \theta$$

$$\Rightarrow \frac{l^2}{L^2} = \frac{\rho}{\rho_0} \Rightarrow \frac{l}{L} = \sqrt{\frac{\rho}{\rho_0}}$$

$$\text{From figure, } \sin \theta = \frac{h}{l} = \frac{L}{2l} = \frac{1}{2} \sqrt{\frac{\rho_0}{\rho}}$$

13. (b) The volume of liquid displaced by floating ice

$$V_D = \frac{M}{\sigma_L}$$

$$\text{Volume of water formed by melting ice, } V_F = \frac{M}{\sigma_W}$$

$$\text{If } \sigma_1 > \sigma_W, \text{ then, } \frac{M}{\sigma_L} < \frac{M}{\sigma_W} \text{ i.e. } V_D < V_F$$

i.e. volume of liquid displaced by floating ice will be lesser than water formed and so the level of liquid will rise.

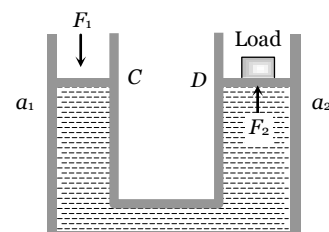
14. (a)

Graphical Questions

- (a)
- (b)
- (d) When we move from centre to circumference, the velocity of liquid goes on decreasing and finally becomes zero.
- (a) When cross-section of duct is decreased, the velocity of water increased and in accordance with Bernoulli's theorem, the pressure P decreased at that place.
- (d)

Assertion and Reason

- 1 (b) According to Pascal's law, if gravity effect is neglected, the pressure at every point of liquid in equilibrium of rest is same.



$$P_1 = P_2 \text{ i.e. } \frac{F_1}{a_1} = \frac{F_2}{a_2} \text{ or } F_2 = \frac{a_2}{a_1} F_1$$

$$\text{As } a_2 \gg a_1 \therefore F_2 \gg F_1$$

This shows that small force (F_1) applied on the smaller piston (of area a_1) will be appearing as a very large force on the larger piston.

- 2 (a) Height of the blood column in the human body is more at feet than at the brain. As $P = h\rho g$, therefore the blood exerts more pressure at the feet than at the brain.
- 3 (e) Since due to applied force on liquid, the pressure is transmitted equally in all directions inside the liquid. That is why there is no fixed direction for the pressure due to liquid. Hence hydrostatic pressure is a scalar quantity.
- 4 (c) Net force = actual weight – upthrust force
= Actual weight – Weight of liquid displaced.
The body will rise above the surface of liquid to such an extent that the weight of the liquid displaced by the immersed part of the body (i.e. upward thrust) becomes equal to the weight of the body. Thus the body will float when upward thrust is more than its actual weight. In this special case the density of solid body is less than the density of liquid.
- 5 (e) The level of water does not change. The reason is that on drinking the water (say $m \text{ gm}$), the weight of man increases by $m \text{ gm}$ and hence water displaced by man increases by $m \text{ gm}$, tending to raise the level. However, this much amount of water has already been consumed by the man. Therefore the level of pond remain same.
- 6 (a)
- 7 (a) In a stream line flow of a liquid, according to equation of continuity $av = \text{constant}$.
Where a is the area of cross section and v is the velocity of liquid flow. When water flowing in a broader pipe enters a narrow pipe, the area of cross-section of water decreases therefore the velocity of water increases.
- 8 (b) As a man jumps-out from a height in air with a parachute, its velocity increases first, because the gravity pull dominates the viscous drag and buoyancy of air which opposes the motion. As the velocity increases, the viscous drag of air also increases and soon a stage is reached where viscous drag and buoyancy of air balances the gravity pull. Then the man with a parachute falls with a constant velocity, called terminal velocity.
- 9 (a) According to Bernoulli's theorem, $P + \frac{1}{2}\rho v^2 = \text{a constant}$
i.e. when velocity is large, the pressure is less in a stream line flow of an ideal liquid through a horizontal tube.
- 10 (c) When a body moves through a fluid, its motion is opposed by the force of fluid friction, which increases with the speed of the body. When cars and planes move through air, their motion is opposed by the air friction, which in turn, depend upon the shape of the body. It is due to this reason that the cars or planes are

given such shape (known as stream lined shaped) so that air friction is minimum. Rather the movement of air layers on the upper and lower side of stream line shape provides a lift which helps in increasing the speed of the car.

- 11 (c) According to Bernoulli's equation,

$$\frac{P}{\rho} + hg + \frac{1}{2}v^2 = \text{constant}$$

Thus, total energy of the injectable medicine depends upon second power of the velocity and first power of the pressure. It implies that total energy of the injectable medicine has greater dependence on its velocity. Therefore, a doctor adjust the flow of the medicine with the help of the size of the needle of the syringe ($a_1 v_1 = a_2 v_2$) rather than the thumb pressure.

- 12 (a) Due to small area of cross-section of the hole, fluid flows out of the vessel with a large speed and thus the fluid possesses a large linear momentum. As no external forces acts on the system, in order to conserve linear momentum, the vessel acquires a velocity in backward direction or in other words a backward thrust results on the vessel.
- 13 (a) The stability of a floating body depends on the relative position of centre of gravity of a body, through which its weight acts and centre of gravity of the displaced water called centre of buoyancy through which the upthrust act.
- 14 (a)
- 15 (e) The viscosity of liquid decreases rapidly with rise of temperature. The variation of viscosity of liquid with temperature is given by $\eta_t = \eta_0(1 + \alpha t + \beta t^2)$
Where η_t and η_0 are the coefficient of viscosities at $t^\circ\text{C}$ and 0°C respectively and α and β are constant.
- 16 (a) According to Bernoulli's theorem, when wind velocity over the wings is larger than the wind velocity under the wings, pressure of wind over the wings becomes less than the pressure of wind under the wing's. This provides the necessary lift to the aeroplane.
- 17 (a)
- 18 (a) Viscosities of fluids are markedly dependent on temperature, increasing for gases and decreasing for liquids as the temperature is increased. Thus important consideration in the design of oils for engine lubrication is to reduce the temperature variation of viscosity as much as possible.
- 19 (a)
- 20 (a) When a body falls through a viscous medium, finally, it attains terminal velocity. At this velocity, viscous force on rain drop balances the weight of the body.
- 21 (a) Smaller the area, larger the pressure exerted by a force
- 22 (d) Railways tracks are laid on large sized wooden sleepers. Due to large sized sleepers the weight of rail act on the large area. Hence, the pressure exerted is reduced appreciably.
- 23 (a)
- 24 (c) When two holes are made in the tin, air keeps on entering through the other hole. Due to this the pressure inside the tin does not become less than atmospheric pressure which happen only one hole is made.

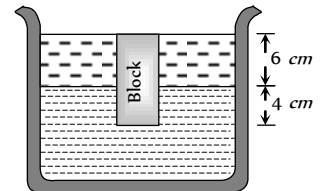


- 25** (e) Terminal velocity and critical velocity are not same.
Critical velocity is the velocity below which the flow of liquid is streamline.
- 26** (b)
- 27** (a)
- 28** (b)

Fluid Mechanics

Self Evaluation Test -11

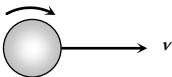
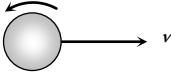
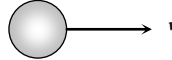
- A tank 5 m high is half filled with water and then is filled to the top with oil of density 0.85 g/cm . The pressure at the bottom of the tank, due to these liquids is
 - 1.85 g/cm
 - 89.25 g/cm
 - 462.5 g/cm
 - 500 g/cm
- Two substances of densities ρ_1 and ρ_2 are mixed in equal volume and the relative density of mixture is 4. When they are mixed in equal masses, the relative density of the mixture is 3. The values of ρ_1 and ρ_2 are
 - $\rho_1 = 6$ and $\rho_2 = 2$
 - $\rho_1 = 3$ and $\rho_2 = 5$
 - $\rho_1 = 12$ and $\rho_2 = 4$
 - None of these
- A wooden block of volume 1000 cm is suspended from a spring balance. It weighs 12 N in air. It is suspended in water such that half of the block is below the surface of water. The reading of the spring balance is
 - 10 N
 - 9 N
 - 8 N
 - 7 N
- Two different liquids are flowing in two tubes of equal radius. The ratio of coefficients of viscosity of liquids is $52:49$ and the ratio of their densities is $13:1$, then the ratio of their critical velocities will be
 - $4 : 49$
 - $49 : 4$
 - $2 : 7$
 - $7 : 2$
- Two capillary tubes of same radius r but of lengths l and l_1 are fitted in parallel to the bottom of a vessel. The pressure head is P . What should be the length of a single tube that can replace the two tubes so that the rate of flow is same as before
 - $l_1 + l_2$
 - $\frac{1}{l_1} + \frac{1}{l_2}$
 - $\frac{l_1 l_2}{l_1 + l_2}$
 - $\frac{1}{l_1 + l_2}$
- A capillary tube is attached horizontally to a constant head arrangement. If the radius of the capillary tube is increased by 10% then the rate of flow of liquid will change nearly by
 - + 10%
 - + 46%
 - 10%
 - 40%
- Two stretched membranes of area 2 cm and 3 cm are placed in a liquid at the same depth. The ratio of pressures on them is
 - $1 : 1$
 - $2 : 3$
 - $3 : 2$
 - $2 : 3$
- Three identical vessels are filled to the same height with three different liquids A, B and C ($\rho_A > \rho_B > \rho_C$). The pressure at the base will be
 - Equal in all vessels
 - Maximum in vessel A
 - Maximum in vessel B
 - Maximum in vessel C
- Three identical vessels are filled with equal masses of three different liquids A, B and C ($\rho_A > \rho_B > \rho_C$). The pressure at the base will be
 - Equal in all vessels
 - Maximum in vessel A
 - Maximum in vessel B
 - Maximum in vessel C
- A piston of cross-section area 100 cm is used in a hydraulic press to exert a force of 10^5 dynes on the water. The cross-sectional area of the other piston which supports an object having a mass 2000 kg . is
 - 100 cm
 - 10 cm
 - $2 \times 10 \text{ cm}$
 - $2 \times 10^5 \text{ cm}$
- A cubical block of wood 10 cm on a side floats at the interface between oil and water with its lower surface horizontal and 4 cm below the interface. The density of oil is 0.6 g/cm^{-3} . The mass of block is
 - 706 g
 - 607 g
 - 760 g
 - 670 g
- A spherical ball of radius r and relative density 0.5 is floating in equilibrium in water with half of it immersed in water. The work done in pushing the ball down so that whole of it is just immersed in water is : (where ρ is the density of water)
 - $\frac{5}{12} \pi r^4 \rho g$
 - $0.5 \pi r g$
 - $\frac{4}{3} \pi r^3 \rho g$
 - $\frac{2}{3} \pi r^4 \rho g$
- If W be the weight of a body of density ρ in vacuum then its apparent weight in air of density σ is
 - $\frac{W\rho}{\sigma}$
 - $W\left(\frac{\rho}{\sigma} - 1\right)$
 - $\frac{W}{\rho} \sigma$
 - $W\left(1 - \frac{\sigma}{\rho}\right)$
- Which of the following is not the characteristic of turbulent flow
 - Velocity more than the critical velocity
 - Velocity less than the critical velocity
 - Irregular flow
 - Molecules crossing from one layer to another
- Water coming out of the mouth of a tap and falling vertically in streamline flow forms a tapering column, i.e., the area of cross-section of the liquid column decreases as it moves



down. Which of the following is the most accurate explanation for this

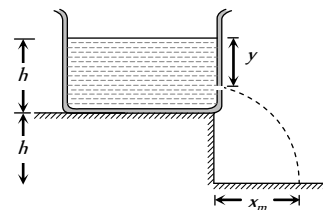
- (a) As the water moves down, its speed increases and hence its pressure decreases. It is then compressed by the atmosphere
- (b) Falling water tries to reach a terminal velocity and hence reduces the area of cross-section to balance upward and downward forces
- (c) The mass of water flowing past any cross-section must remain constant. Also, water is almost incompressible. Hence, the rate of volume flow must remain constant. As this is equal to velocity $\times$ area, the area decreases as velocity increases
- (d) The surface tension causes the exposed surface area of the liquid to decrease continuously

16. To get the maximum flight, a ball must be thrown as

- (a) 
- (b) 
- (c) 
- (d) Any of (a), (b) and (c)

17. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then

- (a) $x_m = 2h$
- (b) $x_m = 1.5h$
- (c) $y = h$
- (d) $y = 0.75h$



18. The relative velocity of two consecutive layers is 8 cm/s . If the perpendicular distance between the layers is 0.1 cm , then the velocity gradient will be

- (a) 8 sec
- (b) 80 sec
- (c) 0.8 sec
- (d) 0.08 sec

19. Under a constant pressure head, the rate of flow of liquid through a capillary tube is V . If the length of the capillary is doubled and the diameter of the bore is halved, the rate of flow would become

- (a) $V/4$
- (b) $16V$
- (c) $V/8$
- (d) $V/32$

AS Answers and Solutions

(SET - 11)

1. (c) Pressure at the bottom $P = (h_1 d_1 + h_2 d_2) \frac{g}{\text{cm}^2}$

$$= [250 \times 1 + 250 \times 0.85] = 250 [1.85] \frac{g}{\text{cm}^2}$$

$$= 462.5 \frac{g}{\text{cm}^2}$$

2. (a) When substances are mixed in equal volume then density

$$= \frac{\rho_1 + \rho_2}{2} = 4 \Rightarrow \rho_1 + \rho_2 = 8 \quad \dots\dots(i)$$

When substances are mixed in equal masses then density

$$= \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = 3$$

$$\Rightarrow 2\rho_1\rho_2 = 3(\rho_1 + \rho_2) \dots\dots(ii)$$

By solving (i) and (ii) we get $\rho_1 = 6$ and $\rho_2 = 2$.

3. (d) Reading of the spring balance

= Apparent weight of the block

= Actual weight – upthrust

$$= 12 - V_{in}\sigma g$$

$$= 12 - 500 \times 10^{-6} \times 10^3 \times 10 = 12 - 5 = 7N.$$

4. (a) Critical velocity $v = N_R \frac{\eta}{\rho r}$

$$\Rightarrow \frac{v_1}{v_2} = \frac{\eta_1}{\eta_2} \times \frac{\rho_2}{\rho_1} = \frac{52}{49} \times \frac{1}{13} = \frac{4}{49}.$$

5. (c) For parallel combination $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}$

$$\Rightarrow \frac{\pi r^4}{8\eta l} = \frac{\pi r^4}{8\eta l_1} + \frac{\pi r^4}{8\eta l_2} \Rightarrow \frac{1}{l} = \frac{1}{l_1} + \frac{1}{l_2} \quad \therefore l = \frac{l_1 l_2}{l_1 + l_2}$$

6. (b) $V = \frac{P\pi r^4}{8\eta l} \Rightarrow \frac{V_2}{V_1} = \left(\frac{r_2}{r_1}\right)^4$

$$\Rightarrow V_2 = V_1 \left(\frac{110}{100}\right)^4 = V_1 (1.1)^4 = 1.4641V$$

$$\frac{\Delta V}{V} = \frac{V_2 - V_1}{V} = \frac{1.4641V - V}{V} = 0.46 \quad \text{or} \quad 46\%.$$

7. (a) Pressure is independent of area of cross section

8. (b) $P \propto \rho$

9. (a) $P = \frac{F}{A} = \frac{mg}{A}$

10. (c) $P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{10^7}{10^2} = \frac{2000 \times 10^3 \times 10^3}{A_2}$

$$\therefore A_2 = 2 \times 10^4 \text{ cm}^2 \quad (g = 980 \approx 10^3 \text{ cm/s}^2)$$

11. (c) Weight of block

= Weight of displaced oil + Weight of displaced water

$$\Rightarrow mg = V_1 \rho_o g + V_2 \rho_w g$$

$$\Rightarrow m = (10 \times 10 \times 6) \times 0.6 + (10 \times 10 \times 4) \times 1 = 760 \text{ gm}.$$

12. (a)

13. (d) Apparent weight in air = $W - \text{upthrust} = V\rho g - V\sigma g$

$$= V\rho g \left(1 - \frac{\sigma}{\rho}\right) = W \left(1 - \frac{\sigma}{\rho}\right)$$

14. (b)

15. (c)

16. (b)

17. (a,c) Velocity of liquid through orifice, $v = \sqrt{2gy}$

and time taken by liquid to reach the ground

$$t = \sqrt{\frac{2(h+h-y)}{g}} = \sqrt{\frac{2(2h-y)}{g}}$$

$\therefore$ Horizontal distance covered by liquid

$$x = v.t. = \sqrt{2gy} \times \sqrt{\frac{2(2h-y)}{g}} = \sqrt{4y(2h-y)}$$

$$\Rightarrow x^2 = 4y(2h-y)$$

$$\Rightarrow \frac{d(x)^2}{dy} = 8h - 8y$$

for x to be maximum, $\frac{d}{dy}(x^2) = 0$

$$\therefore 8h - 8y = 0 \quad \text{or} \quad h = y$$

$$\text{So } x_m = \sqrt{4h(2h-h)} = 2h$$

18. (b) $\frac{dv}{dx} = \frac{8}{0.1} = 80 \text{ s}^{-1}$

19. (d) Rate of flow under a constant pressure head,

$$V = \frac{\pi p r^4}{8\eta l} \Rightarrow V \propto \frac{r^4}{l} \Rightarrow \frac{V_2}{V_1} = \left(\frac{r_2}{r_1}\right)^4 \times \frac{l_1}{l_2} = \left(\frac{1}{2}\right)^4 \times \frac{1}{2}$$

$$\Rightarrow V_2 = \frac{V_1}{32} = \frac{V}{32}$$



Chapter 12

Thermometry, Thermal Expansion and Calorimetry

Temperature



Temperature is defined as the degree of hotness or coldness of a body. The natural flow of heat is from higher temperature to lower temperature.

Two bodies are said to be in thermal equilibrium with each other, when no heat flows from one body to the other. That is when both the bodies are at the same temperature.

(1) Temperature is one of the seven fundamental quantities with dimension $[\theta]$. It is a scalar physical quantity with S.I. unit kelvin.

(2) When heat is given to a body and its state does not change, the temperature of the body rises and if heat is taken from a body its temperature falls *i.e.* temperature can be regarded as the effect of cause "heat".

(3) According to kinetic theory of gases, temperature (macroscopic physical quantity) is a measure of average translational kinetic energy of a molecule (microscopic physical quantity).

(4) Although the temperature of a body can be raised without limit, it cannot be lowered without limit and theoretically limiting low temperature is taken to be zero of the kelvin scale.

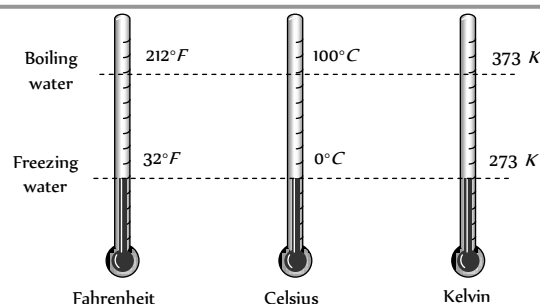
(5) Highest possible temperature achieved in laboratory is about $10^8 K$ while lowest possible temperature attained is $10^{-8} K$.

(6) Temperature of the core of the sun is $10^7 K$ while that of its surface is $6000 K$.

(7) Normal temperature of human body is $310.15 K$ ($37^\circ C = 98.6^\circ F$).

(8) NTP or STP implies $273.15 K$ ($0^\circ C = 32^\circ F$)

Scales of Temperature



The centigrade ($^\circ C$), Fahrenheit ($^\circ F$), Kelvin (K), Reaumur (R), Rankine (Ra) are commonly used temperature scales.

(1) To construct a scale of temperature, two fixed points are taken. First fixed point is the freezing point (ice point) of water, it is called lower fixed point (LFP). The second fixed point is the boiling point (steam point) of water, it is called upper fixed point (UFP).

(2) **Celsius scale** : In this scale LFP (ice point) is taken 0° and UFP (steam point) is taken 100° . The temperature measured on this scale all in degree Celsius ($^\circ C$).

(3) **Farenheite scale** : This scale of temperature has LFP as $32^\circ F$ and UFP as $212^\circ F$. The change in temperature of $1^\circ F$ corresponds to a change of less than 1° on Celsius scale.

(4) **Kelvin scale** : The Kelvin temperature scale is also known as thermodynamic scale. The triple point of water is also selected to be the zero of scale of temperature. The temperature measured on this scale are in Kelvin (K).

The triple point of water is that point on a $P-T$ diagram where the three phases of water, the solid, the liquid and the gas, can coexist in equilibrium.

Table 12.1 : Different measuring scales

Scale	Symbol for each degree	LFP	UFP	Number of divisions on the scale
Celsius	$^{\circ}C$	$0^{\circ}C$	$100^{\circ}C$	100
Fahrenheit	$^{\circ}F$	$32^{\circ}F$	$212^{\circ}F$	180
Reaumer	$^{\circ}R$	$0^{\circ}R$	$80^{\circ}R$	80
Rankine	$^{\circ}Ra$	$460^{\circ}Ra$	$672^{\circ}Ra$	212
Kelvin	K	$273.15 K$	$373.15 K$	100

(5) Temperature on one scale can be converted into other scale by using the following identity.

$$\frac{\text{Reading on any scale} - \text{LFP}}{\text{UFP} - \text{LFP}} = \text{Constant for all scales}$$

(6) All these temperatures are related to each other by the following relationship

$$\frac{C - 0}{100} = \frac{F - 32}{212 - 32} = \frac{K - 273.15}{373.15 - 273.15} = \frac{R - 0}{80 - 0} = \frac{Ra - 460}{672 - 460}$$

$$\text{or } \frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273}{5} = \frac{R}{4} = \frac{Ra - 460}{10.6}$$

(7) The Celsius and Kelvin scales have different zero points but the same size degrees. Therefore any temperature difference is the same on the Celsius and Kelvin scales ($T_2 - T_1$) $^{\circ}C = (T_2 - T_1) K$.

Thermometry

A branch of science which deals with the measurement of temperature of a substance is known as thermometry.

(i) The linear variation in some physical properties of a substance with change of temperature is the basic principle of thermometry and these properties are defined as thermometric property (x) of the substance.

(ii) Thermometric properties (x) may be as follows

(i) Length of liquid in capillary

(ii) Pressure of gas at constant volume.

(iii) Volume of gas at constant pressure.

(iv) Resistance of a given platinum wire.

(3) In old thermometry, freezing point ($0^{\circ}C$) and steam point ($100^{\circ}C$) are taken to define the temperature scale. So if the thermometric property at temperature $0^{\circ}C$, $100^{\circ}C$ and $t^{\circ}C$ are x_0 , x_{100} and x respectively then

$$\frac{t - 0}{100 - 0} = \frac{x - x_0}{x_{100} - x_0} \Rightarrow t^{\circ}C = \frac{x - x_0}{x_{100} - x_0} \times 100^{\circ}C$$

(4) In modern thermometry instead of two fixed points only one reference point is chosen (triple point of water $273.16 K$) the other is itself $0 K$ where the value of thermometric property is assumed to be zero.

So if the value of thermometric property at $0 K$, $273.16 K$ and $T K$ are 0 , x_0 and x respectively then

$$\frac{T}{273.16} = \frac{x}{x_0} \Rightarrow T = 273.16 \left[\frac{x}{x_0} \right] K$$

Thermometers



An instrument used to measure the temperature of a body is called a thermometer

It works by absorbing some heat from the body, so the temperature recorded by it is lesser than the actual value unless the body is at constant temperature. Some common types of thermometers are as follows

(i) **Liquid (mercury) thermometers** : In liquid thermometers mercury is preferred over other liquids as its expansion is large and uniform and it has high thermal conductivity and low specific heat.

(i) Range of temperature : $-50^{\circ}C$ to $350^{\circ}C$
(freezing point) (boiling point)

(ii) Upper limit of range of mercury thermometer can be raised upto $550^{\circ}C$ by filling nitrogen in space over mercury under pressure (which elevates boiling point of mercury).

(iii) Mercury thermometer with cylindrical bulbs are more sensitive than those with spherical bulbs.

(iv) If alcohol is used instead of mercury then range of temperature measurement becomes $-80^{\circ}C$ to $350^{\circ}C$

(v) Formula : $t = \frac{l - l_0}{l_{100} - l_0} \times 100^{\circ}C$

(2) **Gas thermometers** : These are more sensitive and accurate than liquid thermometers as expansion of gases is more than that of liquids. The thermometers using a gas as thermoelectric substance are called ideal gas thermometers. These are of two types

(i) **Constant pressure gas thermometers**

(a) Principle $V \propto T$ (if $P = \text{constant}$)

(b) Formula : $t = \frac{V - V_0}{V_{100} - V_0} \times 100^{\circ}C$ or $T = 273.16 \frac{V}{V_{Tr}} K$

(ii) **Constant volume gas thermometers**

(a) Principle $P \propto T$ (if $V = \text{constant}$)

(b) Formula : $t = \frac{P - P_0}{P_{100} - P_0} \times 100^{\circ}C$ or $T = 273.16 \frac{P}{P_{Tr}} K$

(c) Range of temperature :

Hydrogen gas thermometer : $-200^{\circ}C$ to $500^{\circ}C$

Nitrogen gas thermometer : $-200^{\circ}C$ to $1600^{\circ}C$

Helium gas thermometer : $-268^{\circ}C$ to $500^{\circ}C$

(3) **Resistance thermometers** : Usually platinum is used in resistance thermometers due to high melting point and large value of temperature coefficient of resistance.

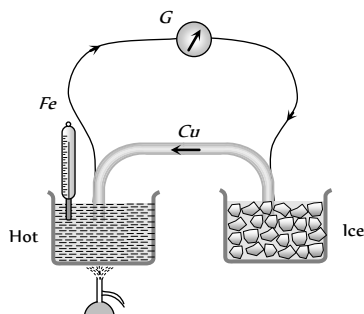
Resistance of metals varies with temperature according to relation.
 $R = R_0(1 + \alpha t)$ where α is the temperature coefficient of resistance and t is change in temperature.

(i) Formula : $t = \frac{R - R_0}{R_{100} - R_0} \times 100^{\circ}C$ or $T = 273.16 \frac{R}{R_{Tr}} K$

(ii) Temperature range : For Platinum resistance thermometer it is $-200^{\circ}C$ to $1200^{\circ}C$

For Germanium resistance thermometer it is 4 to $77 K$.

(4) **Thermoelectric thermometers** : These are based on “Seebeck effect” according to which when two distinct metals are joined to form a closed circuit called thermocouple and the difference in temperature is maintained between their junctions, an emf is developed. The emf is called thermo-emf and if one junction is at 0°C , thermoelectric emf varies with temperature of hot junction (t) according to $e = at + bt^2$; where a and b are constants.



Thermoelectric thermometers have low thermal capacity and high thermal conductivity, so can be used to measure quickly changing temperature

Table 12.2 : Different temperature range

Thermo couple	Temperature range
Copper-iron thermocouple	0°C to 260°C
Iron-constantan thermocouple	0°C to 800°C
Tungsten-molybdenum thermocouple	2000°C to 3000°C

(5) **Pyrometers** : These are the devices used to measure the temperature by measuring the intensity of radiations received from the body. They are based on the fact that the amount of radiations emitted from a body per unit area per second is directly proportional to the fourth power of temperature (Stefan’s law).

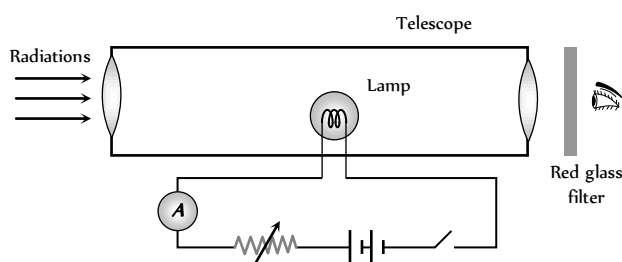


Fig. 12.3

(i) These can be used to measure temperatures ranging from 800°C to 6000°C .

(ii) They cannot measure temperature below 800°C because the amount of radiations is too small to be measured.

(6) **Vapour pressure thermometer** : These are used to measure very low temperatures. They are based on the fact that saturated vapour pressure P of a liquid depends on the temperature according to the relation

$$\log P = a + bT + \frac{c}{T}$$

The range of these thermometers varies from 0.71 K to 120 K for different liquid vapours.

Thermal Expansion



expands. According to atomic theory of matter, a symmetry in potential energy curve is responsible for thermal expansion. As with rise in temperature the amplitude of vibration and hence energy of atoms increases, hence the average distance between the atoms increases. So the matter as a whole expands.

(1) Thermal expansion is minimum in case of solids but maximum in case of gases because intermolecular force is maximum in solids but minimum in gases.

(2) Solids can expand in one dimension (linear expansion), two dimension (superficial expansion) and three dimension (volume expansion) while liquids and gases usually suffers change in volume only.

(3) **Linear expansion** : When a solid is heated and its length increases, then the expansion is called linear expansion.

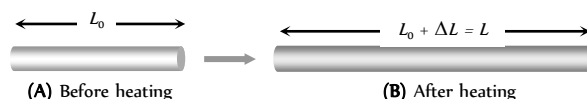


Fig. 12.4

(i) Change in length $\Delta L = L_0 \alpha \Delta T$

(L_0 = Original length, ΔT = Temperature change)

(ii) Final length $L = L_0 (1 + \alpha \Delta T)$

(iii) Co-efficient of linear expansion $\alpha = \frac{\Delta L}{L_0 \Delta T}$

(iv) Unit of α is $^\circ\text{C}^{-1}$ or K^{-1} . Its dimension is $[\theta^{-1}]$

(4) **Superficial (areal) expansion** : When the temperature of a 2D object is changed, its area changes, then the expansion is called superficial expansion.

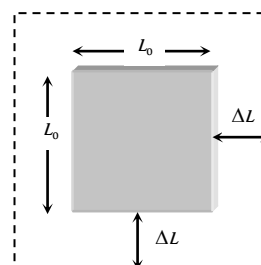


Fig. 12.5

(i) Change in area is $\Delta A = A_0 \beta \Delta T$

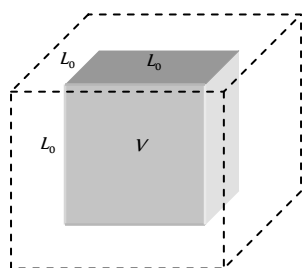
(A_0 = Original area, ΔT = Temperature change)

(ii) Final area $A = A_0 (1 + \beta \Delta T)$

(iii) Co-efficient of superficial expansion $\beta = \frac{\Delta A}{A_0 \Delta T}$

(iv) Unit of β is $^\circ\text{C}$ or K .

(5) **Volume or cubical expansion** : When a solid is heated and its volume increases, then the expansion is called volume or cubical expansion.



- (i) Change in volume is $\Delta V = V_0 \gamma \Delta T$
 (ii) Final volume $V = V_0(1 + \gamma \Delta T)$

Fig. 12.6

(iii) Volume co-efficient of expansion $\gamma = \frac{\Delta V}{V_0 \Delta T}$

(iv) Unit of γ is $^{\circ}\text{C}^{-1}$ or K^{-1} .

(6) **More about α , β and γ** : The co-efficient α , β and γ for a solid are related to each other as follows

$$\alpha = \frac{\beta}{2} = \frac{\gamma}{3} \Rightarrow \alpha : \beta : \gamma = 1 : 2 : 3$$

(i) Hence for the same rise in temperature

Percentage change in area = 2 × percentage change in length.

Percentage change in volume = 3 × percentage change in length.

(ii) The three coefficients of expansion are not constant for a given solid. Their values depend on the temperature range in which they are measured.

(iii) The values of α , β , γ are independent of the units of length, area and volume respectively.

(iv) For anisotropic solids $\gamma = \alpha_x + \alpha_y + \alpha_z$ where α_x , α_y and α_z represent the mean coefficients of linear expansion along three mutually perpendicular directions.

(7) **Contraction on heating** : Some rubber like substances contract with rising temperature, because transverse vibration of atoms of substance dominate over longitudinal vibration which is responsible for expansion.

Table 12.3 : α and γ for some materials

Material	$\alpha [^{\circ}\text{C}^{-1} \text{ or } (^{\circ}\text{C})^{-1}]$	$\gamma [^{\circ}\text{C}^{-1} \text{ or } (^{\circ}\text{C})^{-1}]$
Steel	1.2×10^{-5}	3.6×10^{-5}
Copper	1.7×10^{-5}	5.1×10^{-5}
Brass	2.0×10^{-5}	6.0×10^{-5}
Aluminium	2.4×10^{-5}	7.2×10^{-5}

Application of Thermal Expansion in Solids

(i) **Bi-metallic strip** : Two strips of equal lengths but of different materials (different coefficient of linear expansion) when join together, it is called "bi-metallic strip", and can be used in thermostat to break or make electrical contact. This strip has the characteristic property of bending on

heating due to unequal linear expansion of the two metal. The strip will bend with metal of greater α on outer side i.e. convex side.

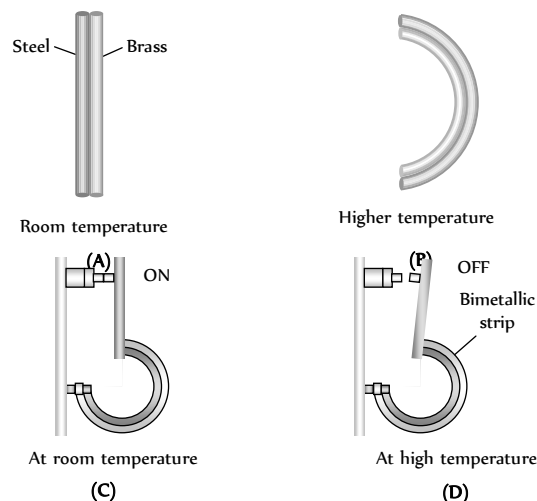


Fig. 12.7

(2) **Effect of temperature on the time period of a simple pendulum** : A pendulum clock keeps proper time at temperature θ . If temperature is increased to $\theta' (> \theta)$ then due to linear expansion, length of pendulum and hence its time period will increase.

$$\text{Fractional change in time period} = \frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta$$

(i) Due to increment in its time period, a pendulum clock becomes slow in summer and will lose time.

$$\text{Loss of time in a time period } \Delta T = \frac{1}{2} \alpha \Delta \theta T$$

(ii) Time lost by the clock in a day ($t = 86400 \text{ sec}$)

$$\Delta t = \frac{1}{2} \alpha \Delta \theta t = \frac{1}{2} \alpha \Delta \theta (86400) = 43200 \alpha \Delta \theta \text{ sec}$$

(iii) The clock will lose time i.e. will become slow if $\theta' > \theta$ (in summer)

and will gain time i.e. will become fast if $\theta' < \theta$ (in winter).

(iv) The gain or loss in time is independent of time period T and depends on the time interval t .

(v) Since coefficient of linear expansion (α) is very small for invar, hence pendulums are made of invar to show the correct time in all seasons.

(3) **Thermal stress in a rigidly fixed rod** : When a rod whose ends are rigidly fixed such as to prevent expansion or contraction, undergoes a change in temperature, due to thermal expansion or contraction, a compressive or tensile stress is developed in it. Due to this thermal stress the rod will exert a large force on the supports. If the change in temperature of a rod of length L is $\Delta \theta$ then

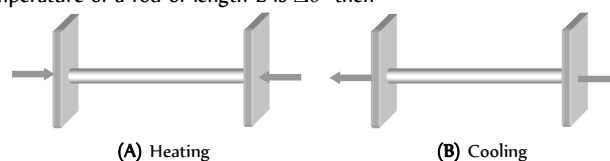


Fig. 12.8

$$\text{Thermal strain} = \frac{\Delta L}{L} = \alpha \Delta \theta$$

$$\left[\text{As } \alpha = \frac{\Delta L}{L} \times \frac{1}{\Delta \theta} \right]$$

So Thermal stress = $Y\alpha\Delta\theta$

$$\left[\text{As } Y = \frac{\text{stress}}{\text{strain}} \right]$$

or Force on the supports $F = YA\alpha\Delta\theta$

(4) **Error in scale reading due to expansion or contraction** : If a scale gives correct reading at temperature θ , at temperature $\theta' (> \theta)$ due to linear expansion of scale, the scale will expand and scale reading will be lesser than true value so that,

$$\text{True value} = \text{Scale reading} [1 + \alpha(\theta' - \theta)]$$

$$\text{i.e. } TV = SR [1 + \alpha\Delta\theta] \text{ with } \Delta\theta = (\theta' - \theta)$$

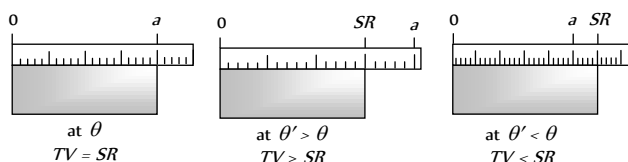


Fig. 12.9

However, if $\theta' < \theta$, due to contraction of scale, scale reading will be more than true value, so true value will be lesser than scale reading and will still be given by above equation with $\Delta\theta = (\theta' - \theta)$ negative.

(5) **Expansion of cavity** : Thermal expansion of an isotropic object may be imagined as a photographic enlargement. So if there is a hole A in a plate C (or cavity A inside a body C), the area of hole (or volume of cavity) will increase when body expands on heating, just as if the hole (or cavity) were solid B of the same material. Also the expansion of area (or volume) of the body C will be independent of shape and size of hole (or cavity), i.e., will be equal to that of D .

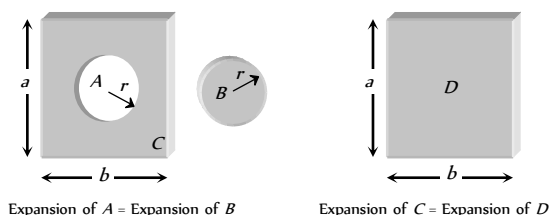


Fig. 12.10

(6) Some other application

(i) When rails are laid down on the ground, space is left between the ends of two rails.

(ii) The transmission cable are not tightly fixed to the poles.

(iii) Test tubes, beakers and crucibles are made of pyrex-glass or silica because they have very low value of coefficient of linear expansion.

(iv) The iron rim to be put on a cart wheel is always of slightly smaller diameter than that of wheel.

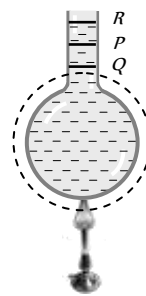
(v) A glass stopper jammed in the neck of a glass bottle can be taken out by warming the neck of the bottle

Thermal Expansion in Liquids

(1) Liquids do not have linear and superficial expansion but these only have volume expansion.

(2) Since liquids are always to be heated along with a vessel which contains them so initially on heating the system (liquid + vessel), the level of liquid in vessel falls (as vessel expands more since it absorbs heat and liquid

expands less) but later on, it starts rising due to faster expansion of the liquid.



$PQ \rightarrow$ represents expansion of vessel

$QR \rightarrow$ represents the real expansion of liquid

$PR \rightarrow$ Represent the apparent expansion of liquid

Fig. 12.11

(3) The actual increase in the volume of the liquid = The apparent increase in the volume of liquid + the increase in the volume of the vessel.

(4) Liquids have two coefficients of volume expansion.

(i) **Co-efficient of apparent expansion (γ)** : It is due to apparent (that appears to be, but is not) increase in the volume of liquid if expansion of vessel containing the liquid is not taken into account.

$$\gamma_a = \frac{\text{Apparent expansion in volume}}{\text{Initial volume} \times \Delta\theta} = \frac{(\Delta V)_a}{V \times \Delta\theta}$$

(ii) **Co-efficient of real expansion (γ_r)** : It is due to the actual increase in volume of liquid due to heating.

$$\gamma_r = \frac{\text{Real increase in volume}}{\text{Initial volume} \times \Delta\theta} = \frac{(\Delta V)_r}{V \times \Delta\theta}$$

(iii) Also coefficient of expansion of flask $\gamma_{\text{vessel}} = \frac{(\Delta V)_{\text{vessel}}}{V \times \Delta\theta}$

(iv) $\gamma_{\text{Real}} = \gamma_{\text{Apparent}} + \gamma_{\text{Vessel}}$

(v) Change (apparent change) in volume in liquid relative to vessel $\Delta V_{\text{app}} = V \gamma_{\text{app}} \Delta\theta = V(\gamma_{\text{Real}} - \gamma_{\text{Vessel}})\Delta\theta = V(\gamma_r - 3\alpha)\Delta\theta$

α = Coefficient of linear expansion of the vessel.

Table 12.4 : Different level of liquid in vessel

γ	ΔV	Level
$\gamma_{\text{Real}} > \gamma_{\text{Vessel}} (=3\alpha) \Rightarrow \gamma_{\text{app}} > 0$	ΔV_{app} is positive	Level of liquid in vessel will rise on heating.
$\gamma_{\text{Real}} < \gamma_{\text{Vessel}} (=3\alpha) \Rightarrow \gamma_{\text{app}} < 0$	ΔV_{app} is negative	Level of liquid in vessel will fall on heating.
$\gamma_{\text{Real}} = \gamma_{\text{Vessel}} (=3\alpha) \Rightarrow \gamma_{\text{app}} = 0$	$\Delta V_{\text{app}} = 0$	level of liquid in vessel will remain same.

(5) **Anomalous expansion of water** : Generally matter expands on heating and contracts on cooling. In case of water, it expands on heating if its temperature is greater than 4°C . In the range 0°C to 4°C , water contracts on heating and expands on cooling, i.e. γ is negative. This behaviour of water in the range from 0°C to 4°C is called anomalous expansion.

This anomalous behaviour of water causes ice to form first at the surface of a lake in cold weather. As winter approaches, the water temperature increases initially at the surface. The water there sinks because of its increased density. Consequently, the surface reaches 0°C first and the lake becomes covered with ice. Aquatic life is able to survive the cold winter as the lake bottom remains unfrozen at a temperature of about 4°C .

At 4°C , density of water is maximum while its specific volume is minimum.

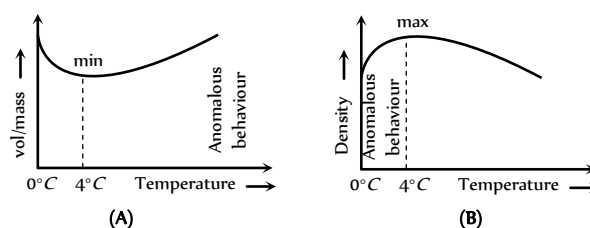


Fig. 12.12

(6) **Effect of temperature on upthrust** : The thrust on V volume of a body in a liquid of density σ is given by $Th = V\sigma g$

Now with rise in temperature by $\Delta\theta^\circ C$, due to expansion, volume of the body will increase while density of liquid will decrease according to the relations $V' = V(1 + \gamma_S \Delta\theta)$ and $\sigma' = \sigma / (1 + \gamma_L \Delta\theta)$

$$\text{So the thrust } Th' = V'\sigma'g \Rightarrow \frac{Th'}{Th} = \frac{V'\sigma'g}{V\sigma g} = \frac{(1 + \gamma_S \Delta\theta)}{(1 + \gamma_L \Delta\theta)}$$

and apparent weight of the body $W_{\text{app}} = \text{Actual weight} - \text{Thrust}$

As $\gamma_S < \gamma_L \therefore Th' < Th$ with rise in temperature thrust also decreases and apparent weight of body increases.

Variation of Density with Temperature

Most substances (solid and liquid) expand when they are heated, i.e., volume of a given mass of a substance increases on heating, so the density should decrease $\left(\text{as } \rho \propto \frac{1}{V} \right)$. For a given mass $\rho \propto \frac{1}{V} \Rightarrow$

$$\frac{\rho'}{\rho} = \frac{V}{V'} = \frac{V}{V + \Delta V} = \frac{V}{V + \gamma V \Delta\theta} = \frac{1}{1 + \gamma \Delta\theta}$$

$$\Rightarrow \rho' = \frac{\rho}{1 + \gamma \Delta\theta} = \rho(1 + \gamma \Delta\theta)^{-1} = \rho(1 - \gamma \Delta\theta)$$

Expansion of Gases

Gases have no definite shape, therefore gases have only volume expansion. Since the expansion of container is negligible in comparison to the gases, therefore gases have only real expansion.

(1) **Coefficient of volume expansion** : At constant pressure, the unit volume of a given mass of a gas, increases with $1^\circ C$ rise of temperature, is called coefficient of volume expansion.

$$\alpha = \frac{\Delta V}{V_0} \times \frac{1}{\Delta\theta} \Rightarrow \text{Final volume } V' = V(1 + \alpha \Delta\theta)$$

$$(2) \text{ Coefficient of pressure expansion : } \beta = \frac{\Delta P}{P} \times \frac{1}{\Delta\theta}$$

$$\therefore \text{ Final pressure } P' = P(1 + \beta \Delta\theta)$$

For an ideal gas, coefficient of volume expansion is equal to the coefficient of pressure expansion i.e. $\alpha = \beta = \frac{1}{273}^\circ C^{-1}$

Heat

(1) The form of energy which is exchanged among various bodies or system on account of temperature difference is defined as heat.

(2) We can change the temperature of a body by giving heat (temperature rises) or by removing heat (temperature falls) from body.

(3) The amount of heat (Q) is given to a body depends upon its mass (m), change in its temperature ($\Delta\theta^\circ = \Delta\theta$) and nature of material i.e. $Q = m.c.\Delta\theta$; where c = specific heat of material.

(4) Heat is a scalar quantity. Its units are *joule, erg, cal, kcal* etc.

(5) The calorie (*cal*) is defined as the amount of heat required to raise the temperature of 1 gm of water from $14.5^\circ C$ to $15.5^\circ C$.

$$\text{Also } 1 \text{ kcal} = 1000 \text{ cal} = 4186 \text{ J and } 1 \text{ cal} = 4.18 \text{ J}$$

(6) **British Thermal Unit (BTU)** : One BTU is the quantity of heat required to raise the temperature of one pound (1 lb) of water from $63^\circ F$ to $64^\circ F$

$$1 \text{ BTU} = 778 \text{ ft. lb} = 252 \text{ cal} = 1055 \text{ J}$$

(7) In solids thermal energy is present in the form of kinetic energy, in liquids, in the form of translatory energy of molecules. In gas it is due to the random motion of molecules.

(8) Heat always flows from a body of higher temperature to lower temperature till their temperature becomes equal (Thermal equilibrium).

(9) The heat required for a given temperature increase depends only on how many atoms the sample contains, not on the mass of an individual atom.

Specific Heat

When a body is heated its temperature rises (except during a change in phase).

(i) **Gram specific heat** : The amount of heat energy required to raise the temperature of unit mass of a body through $1^\circ C$ (or K) is called specific heat of the material of the body.

If Q heat changes the temperature of mass m by $\Delta\theta$ then specific heat

$$c = \frac{Q}{m\Delta\theta}$$

(i) Units : *Calorie/gm* $\times^\circ C$ (practical), *J/kg* $\times K$ (S.I.) Dimension : $[L^2 T^{-2} \theta^{-1}]$

(ii) For an infinitesimal temperature change $d\theta$ and corresponding quantity of heat dQ .

$$\text{Specific heat } c = \frac{1}{m} \cdot \frac{dQ}{d\theta}$$

(2) **Molar specific heat** : Molar specific heat of a substance is defined as the amount of heat required to raise the temperature of one gram mole of the substance through a unit degree it is represented by (capital) C .

$$\text{Molar specific heat } (C) = M \times \text{Gram specific heat } (c)$$

(M = Molecular mass of substance)

$$C = M \frac{Q}{m\Delta\theta} = \frac{1}{\mu} \frac{Q}{\Delta\theta} \quad \left(\text{where, Number of moles } \mu = \frac{m}{M} \right)$$

Units : *calorie/mole* $\times^\circ C$ (practical); *J/mole* $\times \text{kelvin}$ (S.I.)
Dimension : $[ML^2 T^{-2} \theta^{-1}]$

Specific Heat of Solids

When a solid is heated through a small range of temperature, its volume remains more or less constant. Therefore specific heat of a solid may be called its specific heat at constant volume C_v .

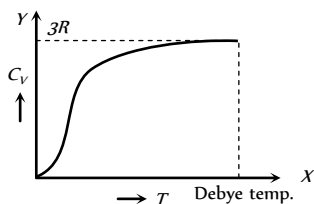


Fig. 12.13

- (1) From the graph it is clear that at $T = 0$, C_v tends to zero.
- (2) With rise in temperature, C_v increases and at a particular temperature (called Debye's temperature) it becomes constant $= 3R = 6 \text{ cal/mole} \times \text{kelvin} = 25 \text{ J/mole} \times \text{kelvin}$.
- (3) For most of the solids, Debye temperature is close to room temperature.
- (4) **Dulong and Petit law** : Average molar specific heat of all metals at room temperature is constant, being nearly equal to $3R = 6 \text{ cal/mole} \times \text{K} = 25 \text{ J/mole} \times \text{K}$, where R is gas constant for one mole of the gas. This statement is known as Dulong and Petit law.
- (5) **Debye's law** : It was observed that at very low temperature molar specific heat $\propto T^3$ (exception are Sn , Pb and Pt).

(6) **Specific heat of ice** : In C.G.S. $c_{\text{ice}} = 0.5 \frac{\text{cal}}{\text{gm} \times ^\circ\text{C}}$

In S.I. $c = 500 \frac{\text{cal}}{\text{kg} \times ^\circ\text{C}} = 2100 \frac{\text{Joule}}{\text{kg} \times ^\circ\text{C}}$.

Table 12.5 : Specific heat of some solids at room temperature and atmospheric pressure

Substance	Specific heat ($\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$)	Molar specific heat ($\text{J} \cdot \text{g mole}^{-1} \cdot \text{K}^{-1}$)
Aluminium	900.0	24.4
Copper	386.4	24.5
Silver	236.1	25.5
Lead	127.7	26.5
Tungsten	134.4	24.9

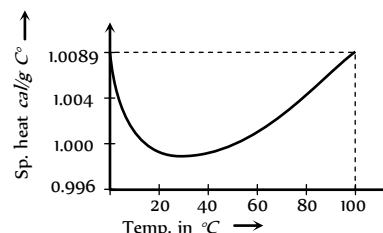
Specific Heat of Liquid (Water)

- (1) Among all known solids and liquids specific heat of water is maximum *i.e.* water takes more time to heat and more time to cool *w.r.t.* other solids and liquids.
- (2) It is observed that by increasing temperature, initially specific heat of water goes on decreasing, becomes minimum at 37°C and then it start increasing. Specific heat of water is –

$$\frac{1 \text{ cal}}{\text{gm} \times ^\circ\text{C}} = 1000 \frac{\text{cal}}{\text{kg} \times ^\circ\text{C}} = 4200 \frac{\text{J}}{\text{kg} \times ^\circ\text{C}}$$

(This value is obtained between the temperature 14.5°C to 15.5°C)

- (3) The variation of specific heat with temperature for water is shown in the figure. Usually this temperature dependence of specific heat is neglected.



- (4) As specific heat of water is large; by absorbing or releasing large amount of heat its temperature changes by small amount. This is why, it is used in hot water bottles or as coolant in radiators.

Specific Heat of Gases

- (1) In case of gases, heat energy supplied to a gas is spent not only in raising the temperature of the gas but also in expansion of gas against atmospheric pressure.
- (2) Hence specific heat of a gas, which is the amount of heat energy required to raise the temperature of one gram of gas through a unit degree shall not have a single or unique value.
- (3) If the gas is compressed suddenly and no heat is supplied from outside *i.e.* $\Delta Q = 0$, but the temperature of the gas raises on the account of compression.

$$\Rightarrow c = \frac{Q}{m(\Delta\theta)} = \frac{0}{m\Delta\theta} = 0$$

- (4) If the gas is heated and allowed to expand at such a rate that rise in temperature due to heat supplied is exactly equal to fall in temperature due to expansion of the gas. *i.e.* $\Delta\theta = 0$

$$\Rightarrow c = \frac{Q}{m(\Delta\theta)} = \frac{Q}{0} = \infty$$

- (5) If rate of expansion of the gas were slow, the fall in temperature of the gas due to expansion would be smaller than the rise in temperature of the gas due to heat supplied. Therefore, there will be some net rise in temperature of the gas *i.e.* ΔT will be positive.

$$\Rightarrow c = \frac{Q}{m(\Delta\theta)} = \text{Positive}$$

- (6) If the gas were to expand very fast, fall of temperature of gas due to expansion would be greater than rise in temperature due to heat supplied. Therefore, there will be some net fall in temperature of the gas *i.e.* $\Delta\theta$ will be negative.

$$\Rightarrow c = \frac{Q}{m(-\Delta\theta)} = \text{Negative}$$

Hence the specific heat of gas can have any positive value ranging from zero to infinity. Further it can even be negative. The exact value depends upon the mode of heating the gas. Out of many values of specific heat of a gas, two are of special significance, namely C_p and C_v , in the chapter "Kinetic theory of gases" we will discuss this topic in detail.

Specific heat of steam : $c_{\text{steam}} = 0.47 \text{ cal/gm} \times ^\circ\text{C}$

Phase Change and Latent Heat

(i) **Phase** : We use the term phase to describe a specific state of matter, such as solid, liquid or gas. A transition from one phase to another is called a phase change.

(i) For any given pressure a phase change takes place at a definite temperature, usually accompanied by absorption or emission of heat and a change of volume and density.

(ii) In phase change ice at 0°C melts into water at 0°C . Water at 100°C boils to form steam at 100°C .

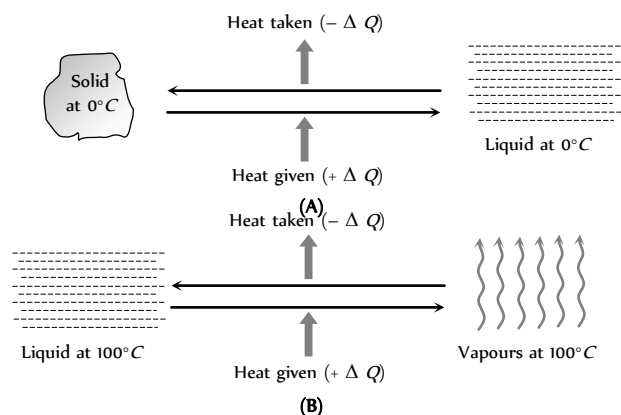


Fig. 12.15

(iii) In solids, the forces between the molecules are large and the molecules are almost fixed in their positions inside the solid. In a liquid, the forces between the molecules are weaker and the molecules may move freely inside the volume of the liquid. However, they are not able to come out of the surface. In vapours or gases, the intermolecular forces are almost negligible and the molecules may move freely anywhere in the container. When a solid melts, its molecules move apart against the strong molecular attraction. This needs energy which must be supplied from outside. Thus, the internal energy of a given body is larger in liquid phase than in solid phase. Similarly, the internal energy of a given body in vapour phase is larger than that in liquid phase.

(iv) In case of change of state if the molecules come closer, energy is released and if the molecules move apart, energy is absorbed.

(2) **Latent heat** : The amount of heat required to change the state of the mass m of the substance is written as : $Q = mL$, where L is the latent heat. Latent heat is also called as Heat of Transformation. Its unit is cal/gm or J/kg and Dimension: $[L^2T^{-2}]$

(i) **Latent heat of fusion** : The latent heat of fusion is the heat energy required to change 1 kg of the material in its solid state at its melting point to 1 kg of the material in its liquid state. It is also the amount of heat energy released when at melting point 1 kg of liquid changes to 1 kg of solid. For water at its normal freezing temperature or melting point (0°C), the latent heat of fusion (or latent heat of ice) is

$$L_F = L_{\text{ice}} \approx 80 \text{ cal/gm} \approx 60 \text{ kJ/mol} \approx 336 \text{ kilo joule/kg}$$

(ii) **Latent heat of vaporisation** : The latent heat of vaporisation is the heat energy required to change 1 kg of the material in its liquid state at its boiling point to 1 kg of the material in its gaseous state. It is also the amount of heat energy released when 1 kg of vapour changes into 1 kg of liquid. For water at its normal boiling point or condensation temperature (100°C), the latent heat of vaporisation (latent heat of steam) is

$$L_V = L_{\text{steam}} \approx 540 \text{ cal/gm} \approx 40.8 \text{ kJ/mol} \approx 2260 \text{ kilo joule/kg}$$

(iii) Latent heat of vaporisation is more than the latent heat of fusion. This is because when a substance gets converted from liquid to vapour, there is a large increase in volume. Hence more amount of heat is required.

But when a solid gets converted to a liquid, then the increase in volume is negligible. Hence very less amount of heat is required. So, latent heat of vaporisation is more than the latent heat of fusion.

Thermal Capacity and Water Equivalent

(i) **Thermal capacity** : It is defined as the amount of heat required to raise the temperature of the whole body (mass m) through 0°C or 1K .

$$\text{Thermal capacity} = mc = \mu C = \frac{Q}{\Delta\theta}$$

The value of thermal capacity of a body depends upon the nature of the body and its mass.

$$\text{Dimension : } [ML^2T^{-2}\theta^{-1}], \text{ Unit : } \text{cal}^{\circ}\text{C (practical) Joule/k (S.I.)}$$

(2) **Water Equivalent** : Water equivalent of a body is defined as the mass of water which would absorb or evolve the same amount of heat as is done by the body in rising or falling through the same range of temperature. It is represented by W .

If m = Mass of the body, c = Specific heat of body, $\Delta\theta$ = Rise in temperature.

$$\text{Then heat given to body } \Delta Q = mc\Delta\theta \quad \dots (i)$$

If same amount of heat is given to $W\text{ gm}$ of water and its temperature also rises by $\Delta\theta$. Then

$$\text{heat given to water } Q = W \times 1 \times \Delta\theta \dots (ii) \quad [\text{As } c_{\text{water}} = 1]$$

$$\text{From equation (i) and (ii) } \Delta Q = mc\Delta\theta = W \times 1 \times \Delta\theta$$

$$\Rightarrow \text{Water equivalent (W)} = mc\text{ gm}$$

$$(i) \text{ Unit : Kg (S.I.)} \quad \text{Dimension : } [ML^0T^0]$$

(ii) Unit of thermal capacity is J/kg while unit of water equivalent is kg .

(iii) Thermal capacity of the body and its water equivalent are numerically equal.

(iv) If thermal capacity of a body is expressed in terms of mass of water it is called water-equivalent of the body.

Some Important Terms

(i) **Evaporation** : Vaporisation occurring from the free surface of a liquid is called evaporation. Evaporation is the escape of molecules from the surface of a liquid. This process takes place at all temperatures and increases with the increase of temperature. Evaporation leads to cooling because the faster molecules escape and, therefore, the average kinetic energy of the molecules of the liquid (and hence the temperature) decreases.



Evaporation cools hot water produced by power plants

(2) **Melting (or fusion)/freezing (or solidification)** : The phase change of solid to liquid is called melting or fusion. The reverse phenomenon is called freezing or solidification.

When pressure is applied on ice, it melts. As soon as the pressure is



Fig. 12.17

removed, it freezes again. This phenomenon is called **regelation**.

(3) **Vaporisation/liquefaction (condensation)** : The phase change from liquid to vapour is called vaporisation. The reverse transition is called liquefaction or condensation.

(4) **Sublimation** : Sublimation is the conversion of a solid directly into vapours. Sublimation takes place when boiling point is less than the melting point. A block of ice sublimates into vapours on the surface of moon because of very very low pressure on its surface. Heat required to change unit mass of solid directly into vapours at a given temperature is called heat of sublimation at that temperature.

(5) **Hoar frost** : Direct conversion of vapours into solid is called hoar frost. This process is just reverse of the process of sublimation, *e.g.*, formation of snow by freezing of clouds.



Fig. 12.18

(6) **Vapour pressure** : When the space above a liquid is closed, it soon becomes saturated with vapour and a dynamic equilibrium is established. The pressure exerted by this vapour is called Saturated Vapour Pressure (S.V.P.) whose value depends only on the temperature – it is independent of any external pressure. If the volume of the space is reduced, some vapour liquefies, but the pressure is unchanged.

A saturated vapour does not obey the gas law whereas the unsaturated vapour obeys them fairly well. However, a vapour differs from a gas in that the former can be liquefied by pressure alone, whereas the latter cannot be liquefied unless it is first cooled.

(7) **Boiling** : As the temperature of a liquid is increased, the rate of evaporation also increases. A stage is reached when bubbles of vapour start forming in the body of the liquid which rise to the surface and escape. A liquid boils at a temperature at which the S.V.P. is equal to the external pressure.



Fig. 12.19

It is a fast process. The boiling point changes on mixing impurities.

(8) **Dew point** : It is that temperature at which the mass of water vapour present in a given volume of air is just sufficient to saturate it, *i.e.* the temperature at which the actual vapour pressure becomes equal to the saturated vapour pressure.

(9) **Humidity** : Atmospheric air always contains some water vapour. The mass of water vapour per unit volume is called absolute humidity.

The ratio of the mass of water vapour (m) actually present in a given volume of air to the mass of water vapour (M) required to saturate the same volume at the same temperature is called the relative humidity (R.H.).

Generally, it is expressed as a percentage, *i.e.*, $\text{R.H.}(\%) = \frac{m}{M} \times 100(\%)$

R.H. May also be defined as the ratio of the actual vapour pressure (p) of water at the same temperature, *i.e.* $\text{R.H.}(\%) = \frac{p}{P} \times 100(\%)$

Thus R.H. may also be defined as

$$\text{R.H.}(\%) = \frac{\text{S.V.P. at dew point}}{\text{S.V. P. at given temperature}} \times 100$$

(10) **Variation of melting point with pressure** : For those substances with contract on melting (*e.g.* water and rubber), the melting point decreases with pressure. The reason is the pressure helps shrinking and hence melting. Most substances expand on melting. (*e.g.* max, sulphur *etc.*)

An increase of pressure opposes the melting of such substances and their melting point is raised.

(11) **Variation of latent heat with temperature and pressure** : The latent heat of vapourization of a substance varies with temperature and hence pressure because the boiling point depends on pressure. It increases as the temperature is decreased. For example, water at 1 atm boils at 100°C and has latent heat 2259 J/g but at 0.5 atm it boils at 82°C and has latent heat 2310 J/g

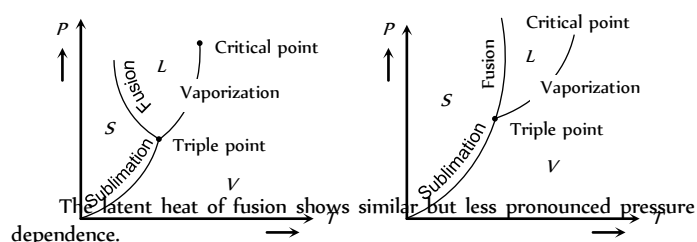


Fig. 12.20

The figures show the P - T graphs for (a) a substance (*e.g.*, water) which contracts on melting and (b) a substance (*e.g.* wax) which expands on melting. The P - T graph consists of three curves.

(i) Sublimation curve which connects points at which vapour (V) and solid (S) exist in equilibrium.

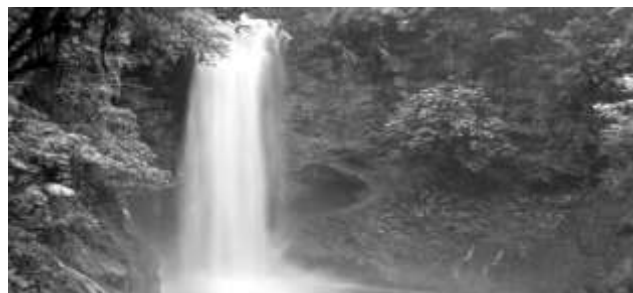
(ii) Vapourization curve which shows vapour and liquid (L) existing in equilibrium.

(iii) Fusion curve which shows liquid and solid existing in equilibrium.

The three curves meet at a single point which is called the triple point. It is that unique temperature-pressure point for a substance at which all the three phases exist in equilibrium.

(12) **Freezing mixture** : If salt is added to ice, then the temperature of mixture drops down to less than 0°C. This is so because, some ice melts down to cool the salt to 0°C. As a result, salt gets dissolved in the water formed and saturated solution of salt is obtained; but the ice point (freezing point) of the solution formed is always less than that of pure water. So, ice cannot be in the solid state with the salt solution at 0°C. The ice which is in contact with the solution, starts melting and it absorbs the required latent heat from the mixture, so the temperature of mixture falls down.

Joule's Law (Heat and Mechanical Work)



Whenever heat is converted into mechanical work or mechanical work is converted into heat, then the ratio of work done to heat produced always

remains constant. *i.e.* $W \propto Q$ or $\frac{W}{Q} = J$

This is Joule's law and J is called mechanical equivalent of heat.

(1) From $W = JQ$ if $Q = 1$ then $J = W$. Hence the amount of work done necessary to produce unit amount of heat is defined as the mechanical equivalent of heat.

(2) J is neither a constant, nor a physical quantity rather it is a conversion factor which used to convert *Joule* or *erg* into *calorie* or *kilo calories* vice-versa.

$$(3) \text{ Value of } J = 4.2 \frac{\text{Joule}}{\text{cal}} = 4.2 \times 10^7 \frac{\text{erg}}{\text{cal}}$$

$$= 4.2 \times 10^3 \frac{\text{Joule}}{\text{kcal}}.$$

(4) When water in a stream falls from height h , then its potential energy is converted into heat and temperature of water rises slightly.

$$\text{From } W = JQ \Rightarrow mgh = J(mc \Delta\theta)$$

[where m = Mass of water, c = Specific heat of water, $\Delta\theta$ = temperature rise]

$$\Rightarrow \text{Rise in temperature } \Delta\theta = \frac{gh}{Jc} ^\circ\text{C}$$

(5) The kinetic energy of a bullet fired from a gun gets converted into heat on striking the target. By this heat the temperature of bullet increases by $\Delta\theta$.

$$\text{From } W = JQ \Rightarrow \frac{1}{2}mv^2 = J(ms \Delta\theta)$$

[where m = Mass of the bullet, v = Velocity of the bullet, c = Specific heat of the bullet]

$$\Rightarrow \text{Rise in temperature } \Delta t = \frac{v^2}{2Jc} ^\circ\text{C}$$

If the temperature of bullet rises upto the melting point of the bullet and bullet melts then.

$$\text{From } W = J(Q_{\text{temperature change}} + Q_{\text{heat of fusion}})$$

$$\Rightarrow \frac{1}{2}mv^2 = J(mc \Delta\theta + mL); \quad L = \text{Latent heat of bullet}$$

$$\Rightarrow \text{Rise in temperature } \Delta\theta = \left[\frac{\left(\frac{v^2}{2J} - L \right)}{c} \right] ^\circ\text{C}$$

(6) If m kg ice-block falls down through some height (h) and melts partially (m' kg) then its potential energy gets converted into heat of melting.

$$\text{From } W = JQ \Rightarrow mgh = Jm'L \Rightarrow h = \frac{m'}{m} \left(\frac{JL}{g} \right)$$

$$\text{If ice-block melts completely then } m' = m \Rightarrow h = \frac{JL}{g} \text{ meter}$$

Principle of Calorimetry

Calorimetry means 'measuring heat'.

When two bodies (one being solid and other liquid or both being liquid) at different temperatures are mixed, heat will be transferred from body at higher temperature to a body at lower temperature till both acquire same temperature. The body at higher temperature releases heat while body at lower temperature absorbs it, so that

Heat lost = Heat gained

i.e. principle of calorimetry represents the law of conservation of heat energy.

(1) Temperature of mixture (θ) is always $\geq$ lower temperature (θ) and $\leq$ higher temperature (θ), i.e., $\theta_L \leq \theta_{\text{mix}} \leq \theta_H$.

It means the temperature of mixture can never be lesser than lower temperatures (as a body cannot be cooled below the temperature of cooling body) and greater than higher temperature (as a body cannot be heated above the temperature of heating body). Furthermore usually rise in temperature of one body is not equal to the fall in temperature of the other body though heat gained by one body is equal to the heat lost by the other.

(2) **Mixing of two substances when temperature changes only** : It means no phase change. Suppose two substances having masses m_1 and m_2 , gram specific heat c_1 and c_2 , temperatures θ_1 and θ_2 ($\theta_1 > \theta_2$) are mixed together such that temperature of mixture at equilibrium is θ_{mix} .

Hence, Heat lost = Heat gained

$$\Rightarrow m_1c_1(\theta_1 - \theta_{\text{mix}}) = m_2c_2(\theta_{\text{mix}} - \theta_2) \Rightarrow$$

$$\theta_{\text{mix}} = \frac{m_1c_1\theta_1 + m_2c_2\theta_2}{m_1c_1 + m_2c_2}$$

Table 12.6 : Temperature of mixture in different cases

Condition	Temperature of mixture
If bodies are of same material i.e. $c_1 = c_2$	$\theta_{\text{mix}} = \frac{m_1\theta_1 + m_2\theta_2}{m_1 + m_2}$
If bodies are of same mass $m_1 = m_2$	$\theta_{\text{mix}} = \frac{\theta_1c_1 + \theta_2c_2}{c_1 + c_2}$
If $m_1 = m_2$ and $c_1 = c_2$	$\theta_{\text{mix}} = \frac{\theta_1 + \theta_2}{2}$

(3) **Mixing of two substances when temperature and phase both changes or only phase changes**: A very common example for this category is ice-water mixing.

Suppose water at temperature $\theta^\circ\text{C}$ is mixed with ice at 0°C , first ice will melt and then its temperature rises to attain thermal equilibrium. Hence; Heat given = Heat taken

$$\Rightarrow m_W C_W (\theta_W - \theta_{\text{mix}}) = m_i L_i + m_i C_W (\theta_{\text{mix}} - 0^\circ)$$

$$\Rightarrow \theta_{\text{mix}} = \frac{m_W \theta_W - \frac{m_i L_i}{C_W}}{m_W + m_i}$$

$$(i) \text{ If } m_W = m_i \text{ then } \theta_{\text{mix}} = \frac{\theta_W - \frac{L_i}{C_W}}{2}$$

(ii) By using this formulae if $\theta_{\text{mix}} < 0^\circ$ then take $\theta_{\text{mix}} = 0^\circ\text{C}$

Heating Curve

If to a given mass (m) of a solid, heat is supplied at constant rate P and a graph is plotted between temperature and time, the graph is as shown in figure and is called heating curve. From this curve it is clear that

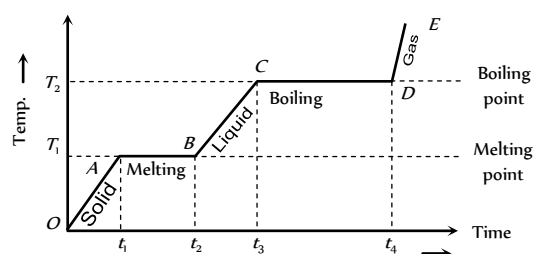


Fig. 12.21

(1) In the region OA temperature of solid is changing with time so,
 $Q = mc_s \Delta T \Rightarrow P \Delta t = mc_s \Delta T$ [as $Q = P \Delta t$]

But as $(\Delta T / \Delta t)$ is the slope of temperature-time curve

$$c_s \propto \frac{1}{\text{Slope of line } OA}$$

i.e. specific heat (or thermal capacity) is inversely proportional to the slope of temperature-time curve.

(2) In the region AB temperature is constant, so it represents change of state, *i.e.*, melting of solid with melting point T . At A melting starts and at B all solid is converted into liquid. So between A and B substance is partly solid and partly liquid. If L is the latent heat of fusion. $Q = mL_F$ or

$$L_F = \frac{P(t_2 - t_1)}{m} \quad [\text{as } Q = P(t_2 - t_1)]$$

or $L \propto \text{length of line } AB$

i.e. Latent heat of fusion is proportional to the length of line of zero slope. [In this region specific heat $\propto \frac{1}{\tan 0} = \infty$]

(3) In the region BC temperature of liquid increases so specific heat (or thermal capacity) of liquid will be inversely proportional to the slope of line BC

$$\text{i.e., } c_L \propto \frac{1}{\text{Slope of line } BC}$$

(4) In the region CD temperature is constant, so it represents the change of state, *i.e.*, boiling with boiling point T . At C all substance is in liquid state while at D in vapour state and between C and D partly liquid and partly gas. The length of line CD is proportional to latent heat of vaporisation

$$\text{i.e., } L \propto \text{Length of line } CD \quad [\text{In this region specific heat } \propto \frac{1}{\tan 0} = \infty]$$

(5) The line DE represents gaseous state of substance with its temperature increasing linearly with time. The reciprocal of slope of line will be proportional to specific heat or thermal capacity of substance in vapour state.

Tips & Tricks

✍ After snow falls, the temperature of the atmosphere becomes very low. This is because the snow absorbs the heat from



the atmosphere to melt down.

So, in the mountains, when

snow falls, one does not feel

too cold, but when ice melts, he feels too cold.

✍ There is more shivering effect of **ice-cream** on teeth as compared to that of water

(obtained from ice).

This is because, when ice-cream

melts down, it absorbs large

amount of heat from teeth.



✍ Branch of physics dealing with production and measurement of temperatures close to $0K$ is known as cryogenics while that dealing with the measurement of very high temperature is called as pyrometry.

✍ It is more painful to get burnt by steam rather than by boiling water at same temperature. This is so because when steam at $100^\circ C$ gets converted to water at $100^\circ C$, then it gives out 536 *calories* of heat. So, it is clear that steam at $100^\circ C$ has more heat than water at $100^\circ C$ (*i.e.*, boiling of water).

✍ A solid and hollow sphere of same radius and material, heated to the same temperature then expansion of both will be equal because thermal expansion of isotropic solids is similar to true photographic enlargement. It means the expansion of cavity is same as if it has been a solid body of the same material. But if same heat is given to the two spheres, due to lesser mass, rise in temperature of hollow sphere will be

$$\text{more } \left\{ \text{As } \Delta \theta = \frac{\Delta Q}{mc} \right\}.$$

Hence its expansion will be more.

✍ Specific heat of a substance can also be negative. Negative specific heat means that in order to raise the temperature, a certain quantity of heat is to be withdrawn from the body.

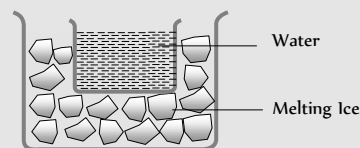
e.g. Specific heat of saturated vapours.

✍ Specific heat for hydrogen is maximum ($3.5 \text{ cal/gm}^\circ C$) and it is minimum for radon and actinium ($\approx 0.022 \text{ cal/gm}^\circ C$).

✍ The minimum possible temperature is $0 K$.

✍ Amount of steam at $100^\circ C$ required to just melt $m \text{ gm}$ of ice at $0^\circ C$ is $m/8 \text{ gm}$.

✍ If we put the beaker containing water in melting ice, the water in the beaker will cool to $0^\circ C$ but will never freeze.



✍ A pressure in excess of 25 atm is required to make helium solidify. At 1 atm pressure, helium remains a liquid down to absolute zero.

✍ Boiling temperature of water, if pressure is different from normal pressure is $t_{\text{boil}} = [100^\circ C - (760 - P \text{ in } mm) \times 0.037]^\circ C$

✍ Confusing S.I. and C.G.S. units

It is advised to do questions on calorimetry in C.G.S. as calculations becomes simple. If the final answer is in joules, then convert cal into joules.

✍ Invar and quartz have very small values of co-efficient of linear expansion.

✍ In S.I. nomenclature " degree" is not used with the kelvin scale; *e.g.* $273^{\circ}K$ is wrong while $273 K$ is correct to write.

✍ Magnetic thermometer is recommended for measuring very low temperature ($2K$).

✍ The most sensitive thermometer is gas thermometer.

✍ Dew formation is more probable on a cloudiness calm night.

✍ In winters, generally fog disappear before noon. Because, the atmosphere warms up and tends to be unsaturated. The condensed vapours reevaporates and the fog disappears.

✍ Standardisation of thermometer is obtained with gas thermometer. Because coefficient of expansion of gas is very large.

✍ Dogs hang their tongues in order to expose a surface to the air for evaporation and hence, cooling. They do not sweat.



Ordinary Thinking

Objective Questions

Thermometry

- On the Celsius scale the absolute zero of temperature is at
[CBSE PMT 1994]
(a) $0^{\circ}C$ (b) $-32^{\circ}C$
(c) $100^{\circ}C$ (d) $-273.15^{\circ}C$
- Oxygen boils at $-183^{\circ}C$. This temperature is approximately
[CPMT 1992]
(a) $215^{\circ}F$ (b) $-297^{\circ}F$
(c) $329^{\circ}F$ (d) $361^{\circ}F$
- Recently, the phenomenon of superconductivity has been observed at $95 K$. This temperature is nearly equal to
[CPMT 1990]
(a) $-288^{\circ}F$ (b) $-146^{\circ}F$
(c) $-368^{\circ}F$ (d) $+178^{\circ}F$
- The temperature of a substance increases by $27^{\circ}C$. On the Kelvin scale this increase is equal to
[CPMT 1993]
(a) $300 K$ (b) $2.46 K$
(c) $27 K$ (d) $7 K$

- The resistance of a resistance thermometer has values 2.71 and 3.70 ohm at $10^{\circ}C$ and $100^{\circ}C$. The temperature at which the resistance is 3.26 ohm is
[CPMT 1994]
(a) $40^{\circ}C$ (b) $50^{\circ}C$
(c) $60^{\circ}C$ (d) $70^{\circ}C$
- No other thermometer is as suitable as a platinum resistance thermometer to measure temperature in the entire range of
[MNR 1993]
(a) $0^{\circ}C$ to $100^{\circ}C$ (b) $100^{\circ}C$ to $1500^{\circ}C$
(c) $-50^{\circ}C$ to $+350^{\circ}C$ (d) $-200^{\circ}C$ to $600^{\circ}C$



7. The temperature of the sun is measured with
[Pb. PMT 1998; CPMT 1998; Pb. PET 1997, 2001]
(a) Platinum thermometer
(b) Gas thermometer
(c) Pyrometer
(d) Vapour pressure thermometer
8. Absolute temperature can be calculated by [AFMC 1994]
(a) Mean square velocity (b) Motion of the molecule
(c) Both (a) and (b) (d) None of the above
9. Thermoelectric thermometer is based on [CPMT 1993, 95; AFMC 1998]
(a) Photoelectric effect (b) Seeback effect
(c) Compton effect (d) Joule effect
10. Maximum density of H_2O is at the temperature [CPMT 1996; Pb. PMT 1996]
(a) $32^\circ F$ (b) $39.2^\circ F$
(c) $42^\circ F$ (d) $4^\circ F$
11. The study of physical phenomenon at low temperatures (below liquid nitrogen temperature) is called [CPMT 1992]
(a) Refrigeration (b) Radiation
(c) Cryogenics (d) Pyrometry
12. 'Stem Correction' in platinum resistance thermometers are eliminated by the use of [AIIMS 1998]
(a) Cells (b) Electrodes
(c) Compensating leads (d) None of the above
13. The absolute zero is the temperature at which [AIIMS 1998]
(a) Water freezes
(b) All substances exist in solid state
(c) Molecular motion ceases
(d) None of the above
14. Absolute scale of temperature is reproduced in the laboratory by making use of a [SCRA 1998]
(a) Radiation pyrometer
(b) Platinum resistance thermometer
(c) Constant volume helium gas thermometer
(d) Constant pressure ideal gas thermometer
15. Absolute zero (0 K) is that temperature at which [AFMC 1993]
(a) Matter ceases to exist
(b) Ice melts and water freezes
(c) Volume and pressure of a gas becomes zero
(d) None of these
16. On which of the following scales of temperature, the temperature is never negative [EAMCET 1997]
(a) Celsius (b) Fahrenheit
(c) Reaumur (d) Kelvin
17. The temperature on Celsius scale is $25^\circ C$. What is the corresponding temperature on the Fahrenheit scale [AFMC 2001]
(a) $40^\circ F$ (b) $77^\circ F$
(c) $50^\circ F$ (d) $45^\circ F$
18. One quality of a thermometer is that its heat capacity should be small. If P is a mercury thermometer, Q is a resistance thermometer and R thermocouple type then [CPMT 1997]
(a) P is best, R worst (b) R is best, P worst
(c) R is best, Q worst (d) P is best, Q worst
19. Two thermometers are used to record the temperature of a room. If the bulb of one is wrapped in wet hanky [AFMC 1997]
(a) The temperature recorded by both will be same
(b) The temperature recorded by wet-bulb thermometer will be greater than that recorded by the other
(c) The temperature recorded by dry-bulb thermometer will be greater than that recorded by the other
(d) None of the above
20. The temperature of a body on Kelvin scale is found to be x K. When it is measured by Fahrenheit thermometer, it is found to be $x^\circ F$, then the value of x is [UPSEAT 2000; Pb. CET 2004]
(a) 40 (b) 313
(c) 574.25 (d) 301.25
21. A centigrade and a Fahrenheit thermometer are dipped in boiling water. The water temperature is lowered until the Fahrenheit thermometer registers 140° . What is the fall in temperature as registered by the Centigrade thermometer [CBSE PMT 1992; AIIMS 1998]
(a) 30° (b) 40°
(c) 60° (d) 80°
22. At what temperature the centigrade (Celsius) and Fahrenheit, readings are the same [RPMT 1997, 99, 2003; BHU 1997; MNR 1992; DPMT 1998; CPMT 1995; UPSEAT 1999; KCET 2000]
(a) -40° (b) $+40^\circ$
(c) 36.6° (d) -37°
23. Standardisation of thermometers is obtained with [CPMT 1996]
(a) Jolly's thermometer
(b) Platinum resistance thermometer
(c) Thermocouple thermometer
(d) Gas thermometer
24. The gas thermometers are more sensitive than liquid thermometers because [CPMT 1993]
(a) Gases expand more than liquids
(b) Gases are easily obtained
(c) Gases are much lighter
(d) Gases do not easily change their states
25. Mercury thermometers can be used to measure temperatures upto [CBSE PMT 1992, 96; BHU 1998; UPSEAT 1998]
(a) $100^\circ C$ (b) $212^\circ C$
(c) $360^\circ C$ (d) $500^\circ C$
26. A constant volume gas thermometer shows pressure reading of 50 cm and 90 cm of mercury at $0^\circ C$ and $100^\circ C$ respectively. When the pressure reading is 60 cm of mercury, the temperature is [MNR 1991; UPSEAT 2000; Pb. CET 2004]
(a) $25^\circ C$ (b) $40^\circ C$

- (c) 15°C (d) 12.5°C
27. Mercury boils at 367°C . However, mercury thermometers are made such that they can measure temperature up to 500°C . This is done by [CPMT 2004]
- (a) Maintaining vacuum above mercury column in the stem of the thermometer
(b) Filling nitrogen gas at high pressure above the mercury column
(c) Filling nitrogen gas at low pressure above the mercury level
(d) Filling oxygen gas at high pressure above the mercury column
28. A device used to measure very high temperature is [KCET 1998]
- (a) Pyrometer (b) Thermometer
(c) Bolometer (d) Calorimeter
29. The absolute zero temperature in Fahrenheit scale is [DCE 1996]
- (a) -273°F (b) -32°F
(c) -460°F (d) -132°F
30. A constant pressure air thermometer gave a reading of 47.5 units of volume when immersed in ice cold water, and 67 units in a boiling liquids. The boiling point of the liquid will be
- (a) 135°C (b) 125°C
(c) 112°C (d) 100°C
31. If a thermometer reads freezing point of water as 20°C and boiling point as 150°C , how much thermometer read when the actual temperature is 60°C [AFMC 2004]
- (a) 98°C (b) 110°C
(c) 40°C (d) 60°C
32. If temperature of an object is 140°F , then its temperature in centigrade is [RPMT 1999]
- (a) 105°C (b) 32°C
(c) 140°C (d) 60°C
33. Of the following thermometers, the one which can be used for measuring a rapidly changing temperature is a [CPMT 1992]
- (a) Thermocouple thermometer
(b) Gas thermometer
(c) Maximum resistance thermometer
(d) Vapour pressure thermometer
34. On centigrade scale the temperature of a body increases by 30 degrees. The increase in temperature on Fahrenheit scale is
- (a) 50° (b) 40°
(c) 30° (d) 54°
35. The correct value of 0°C on Kelvin scale will be [RPMT 1999]
- (a) 273.15 K (b) 273.00 K
(c) 273.05 K (d) 273.63 K
2. A vertical column 50 cm long at 50°C balances another column of same liquid 60 cm long at 100°C . The coefficient of absolute expansion of the liquid is [EAMCET 1990]
- (a) $0.005/^{\circ}\text{C}$ (b) $0.0005/^{\circ}\text{C}$
(c) $0.002/^{\circ}\text{C}$ (d) $0.0002/^{\circ}\text{C}$
3. The apparent coefficient of expansion of a liquid when heated in a copper vessel is C and when heated in a silver vessel is S . If A is the linear coefficient of expansion of copper, then the linear coefficient of expansion of silver is [EAMCET 1991]
- (a) $\frac{C+S-3A}{3}$ (b) $\frac{C+3A-S}{3}$
(c) $\frac{S+3A-C}{3}$ (d) $\frac{C+S+3A}{3}$
4. A uniform metal rod is used as a bar pendulum. If the room temperature rises by 10°C , and the coefficient of linear expansion of the metal of the rod is 2×10^{-5} per $^{\circ}\text{C}$, the period of the pendulum will have percentage increase of [NSEP 1992]
- (a) -2×10^{-5} [AIIMS 1994] (b) -1×10^{-5}
(c) 2×10^{-5} (d) 1×10^{-5}
5. A bar of iron is 10 cm at 20°C . At 19°C it will be (α of iron = $11 \times 10^{-6}/^{\circ}\text{C}$) [EAMCET 1997]
- (a) 11×10^{-6} cm longer (b) 11×10^{-6} cm shorter
(c) 11×10^{-6} cm shorter (d) 11×10^{-6} cm longer
6. When a rod is heated but prevented from expanding, the stress developed is independent of [EAMCET 1997]
- (a) Material of the rod (b) Rise in temperature
(c) Length of rod (d) None of above
7. Expansion during heating [CBSE PMT 1994]
- (a) Occurs only in solids
(b) Increases the weight of a material
(c) Decreases the density of a material
(d) Occurs at the same rate for all liquids and solids
8. On heating a liquid of coefficient of cubical expansion γ in a container having coefficient of linear expansion $\gamma/3$, the level of liquid in the container will [EAMCET 1993]
- (a) Rise [UPSEAT 2005]
(b) Fall
(c) Will remain almost stationary
(d) It is difficult to say
9. A pendulum clock keeps correct time at 0°C . Its mean coefficient of linear expansions is $\alpha/^{\circ}\text{C}$, then the loss in seconds per day by the clock if the temperature rises by $t^{\circ}\text{C}$ is
- (a) $\frac{1}{2} \alpha t \times 864000$ (b) $\frac{1}{2} \alpha t \times 86400$
 $1 - \frac{\alpha t}{2}$

Thermal Expansion

1. When a copper ball is heated, the largest percentage increase will occur in its [EAMCET 1992]
- (a) Diameter (b) Area
(c) Volume (d) Density

$$(c) \frac{\frac{1}{2} \alpha t \times 86400}{\left(1 - \frac{\alpha t}{2}\right)^2} \quad (d) \frac{\frac{1}{2} \alpha t \times 86400}{1 + \frac{\alpha t}{2}}$$

10. When a bimetallic strip is heated, it [CBSE PMT 1990]

- (a) Does not bend at all
(b) Gets twisted in the form of an helix
(c) Bend in the form of an arc with the more expandable metal outside
(d) Bends in the form of an arc with the more expandable metal inside

11. A solid ball of metal has a concentric spherical cavity within it. If the ball is heated, the volume of the cavity will

[AFMC 1997; Orissa PMT 2004]

- (a) Increase (b) Decrease
(c) Remain unaffected (d) None of these

12. A litre of alcohol weighs [AFMC 1994]

- (a) Less in winter than in summer
(b) Less in summer than in winter
(c) Some both in summer and winter
(d) None of the above

13. 5 litre of benzene weighs [MNR 1996]

- (a) More in summer than in winter
(b) More in winter than in summer
(c) Equal in winter and summer
(d) None of the above

14. Water has maximum density at [Pb. PMT 1997]

- (a) 0°C (b) 32°F
(c) -4°C (d) 4°C

15. At some temperature T , a bronze pin is a little large to fit into a hole drilled in a steel block. The change in temperature required for an exact fit is minimum when

[SCRA 1998]

- (a) Only the block is heated
(b) Both block and pin are heated together
(c) Both block and pin are cooled together
(d) Only the pin is cooled

16. If the length of a cylinder on heating increases by 2%, the area of its base will increase by [CPMT 1993; BHU 1997]

- (a) 0.5% (b) 2%
(c) 1% (d) 4%

17. The volume of a gas at 20°C is 100 cm at normal pressure. If it is heated to 100°C , its volume becomes 125 cm at the same pressure, then volume coefficient of the gas at normal pressure is

- (a) $0.0015/^\circ\text{C}$ (b) $0.0045/^\circ\text{C}$
(c) $0.0025/^\circ\text{C}$ (d) $0.0033/^\circ\text{C}$

18. The coefficient of superficial expansion of a solid is $2 \times 10^{-4}/^\circ\text{C}$. Its coefficient of linear expansion is [KCET 1999]

- (a) $4 \times 10^{-4}/^\circ\text{C}$ (b) $3 \times 10^{-4}/^\circ\text{C}$
(c) $2 \times 10^{-4}/^\circ\text{C}$ (d) $1 \times 10^{-4}/^\circ\text{C}$

19. Density of substance at 0°C is 10 gm/cc and at 100°C , its density is 9.7 gm/cc. The coefficient of linear expansion of the substance will be

[BHU 1996; Pb. PMT 1999; DPMT 1998, 2003]

- (a) 10^{-4} (b) 10^{-5}
(c) 10^{-6} (d) 10^{-7}

20. Coefficient of real expansion of mercury is $0.18 \times 10^{-4}/^\circ\text{C}$. If the density of mercury at 0°C is 13.6 gm/cc, its density at 473K is

- (a) 13.11 gm/cc (b) 26.22 gm/cc
(c) 52.11 gm/cc (d) None of these

21. The real coefficient of volume expansion of glycerine is $0.000597/^\circ\text{C}$ and linear coefficient of expansion of glass is $0.000009/^\circ\text{C}$. Then the apparent volume coefficient of expansion of glycerine is

- (a) $0.000558/^\circ\text{C}$ (b) $0.00057/^\circ\text{C}$
(c) $0.00027/^\circ\text{C}$ (d) $0.00066/^\circ\text{C}$

22. A beaker is completely filled with water at 4°C . It will overflow if [EAMCET 1999]

- (a) Heated above 4°C
(b) Cooled below 4°C
(c) Both heated and cooled above and below 4°C respectively
(d) None of the above

23. The volume of a metal sphere increases by 0.24% when its temperature is raised by 40°C . The coefficient of linear expansion of the metal is $^\circ\text{C}$ [Kerala PMT 2005]

- (a) 2×10^{-4} (b) 6×10^{-4}
(c) 2.1×10^{-4} (d) 1.2×10^{-4}

24. Ratio among linear expansion coefficient (α), areal expansion coefficient (β) and volume expansion coefficient (γ) is

- (a) 1 : 2 : 3 (b) 3 : 2 : 1
(c) 4 : 3 : 2 (d) None of these

25. If on heating liquid through 80°C , the mass expelled is $(1/100)$ of mass still remaining, the coefficient of apparent expansion of liquid is [RPMT 2004]

- (a) $1.25 \times 10^{-4}/^\circ\text{C}$ (b) $12.5 \times 10^{-4}/^\circ\text{C}$
(c) $1.25 \times 10^{-3}/^\circ\text{C}$ (b) None of these

26. In cold countries, water pipes sometimes burst, because

- (a) Pipe contracts
(b) Water expands on freezing
(c) When water freezes, pressure increases
(d) When water freezes, it takes heat from pipes

27. A cylindrical metal rod of length L is shaped into a ring with a small gap as shown. On heating the system



[Pb. PET 2002; DPMT 2001]

- (a) x decreases, r and d increase
(b) x and r increase, d decreases
(c) x , r and d all increase
(d) Data insufficient to arrive at a conclusion

28. The length of a metallic rod is 5 m at 0°C and becomes 5.01 m, on heating upto 100°C . The linear expansion of the metal will be

- (a) $2.33 \times 10^{-5} / ^\circ C$ (b) $6.0 \times 10^{-5} / ^\circ C$
(c) $4.0 \times 10^{-5} / ^\circ C$ (d) $2.0 \times 10^{-5} / ^\circ C$
29. A metal rod of silver at $0^\circ C$ is heated to $100^\circ C$. Its length is increased by 0.19 cm . Coefficient of cubical expansion of the silver rod is [UPSEAT 2001]
(a) $5.7 \times 10^{-5} / ^\circ C$ (b) $0.63 \times 10^{-5} / ^\circ C$
(c) $1.9 \times 10^{-5} / ^\circ C$ (d) $16.1 \times 10^{-5} / ^\circ C$
30. A brass disc fits simply in a hole of a steel plate. The disc from the hole can be loosened if the system [UPSEAT 2001]
(a) First heated then cooled (b) First cooled then heated
(c) Is heated (d) Is cooled
31. An iron bar of length 10 m is heated from $0^\circ C$ to $100^\circ C$. If the coefficient of linear thermal expansion of iron is $10 \times 10^{-6} / ^\circ C$, the increase in the length of bar is [UPSEAT 2005]
(a) 0.5 cm (b) 1.0 cm
(c) 1.5 cm (d) 2.0 cm
32. If a cylinder of diameter 1.0 cm at $30^\circ C$ is to be solid into a hole of diameter 0.9997 cm in a steel plate at the same temperature, then minimum required rise in the temperature of the plate is : (Coefficient of linear expansion of steel $= 12 \times 10^{-6} / ^\circ C$)
(a) $25^\circ C$ (b) $35^\circ C$
(c) $45^\circ C$ (d) $55^\circ C$
33. Surface of the lake is at $2^\circ C$. Find the temperature of the bottom of the lake [Orissa JEE 2002]
(a) $2^\circ C$ (b) $3^\circ C$
(c) $4^\circ C$ (d) $1^\circ C$
34. Two rods, one of aluminum and the other made of steel, having initial length l_1 and l_2 are connected together to form a single rod of length $l_1 + l_2$. The coefficients of linear expansion for aluminum and steel are α_a and α_s respectively. If the length of each rod increases by the same amount when their temperature are raised by $t^\circ C$, then find the ratio $\frac{l_1}{(l_1 + l_2)}$
(a) $\frac{\alpha_s}{\alpha_a}$ (b) $\frac{\alpha_a}{\alpha_s}$
(c) $\frac{\alpha_s}{(\alpha_a + \alpha_s)}$ (d) $\frac{\alpha_a}{(\alpha_a + \alpha_s)}$
- (c) No change in temperature takes place whether heat is taken in or given out
(d) All of the above
4. A gas in an airtight container is heated from $25^\circ C$ to $90^\circ C$. The density of the gas will [BCECE 1997]
(a) Increase slightly (b) Increase considerably
(c) Remain the same (d) Decrease slightly
5. A quantity of heat required to change the unit mass of a solid substance, from solid state to liquid state, while the temperature remains constant, is known as [AIIMS 1998]
(a) Latent heat (b) Sublimation
(c) Hoar frost (d) Latent heat of fusion
6. The latent heat of vaporization of a substance is always [SCRA 1998]
(a) Greater than its latent heat of fusion
(b) Greater than its latent heat of sublimation
(c) Equal to its latent heat of sublimation
(d) Less than its latent heat of fusion
7. The factor not needed to calculate heat lost or gained when there is no change of state is [AFMC 1997; BHU 1997]
(a) Weight (b) Specific heat
(c) Resistance [EAMCET 2001] (d) Temperature change
8. 540 g of ice at $0^\circ C$ is mixed with 540 g of water at $80^\circ C$. The final temperature of the mixture is [AFMC 1994]
(a) $0^\circ C$ (b) $40^\circ C$
(c) $80^\circ C$ (d) Less than $0^\circ C$
9. Water is used to cool radiators of engines, because [AFMC 2001]
(a) Of its lower density (b) It is easily available
(c) It is cheap (d) It has high specific heat
10. How much heat energy is gained when 5 kg of water at $20^\circ C$ is brought to its boiling point (Specific heat of water $= 4.2 \text{ kJ kg}^{-1} ^\circ C^{-1}$) [BHU 2001]
(a) 1680 kJ (b) 1700 kJ
(c) 1720 kJ (d) 1740 kJ
- [IIT-JEE Screening 2003]
11. Melting point of ice [CBSE PMT 1993]
(a) Increases with increasing pressure
(b) Decreases with increasing pressure
(c) Is independent of pressure
(d) Is proportional to pressure
12. Heat required to convert one gram of ice at $0^\circ C$ into steam at $100^\circ C$ is (given $L_v = 536 \text{ cal/gm}$) [Pb. PMT 1990]
(a) 100 calorie (b) 0.01 kilocalorie
(c) 716 calorie (d) 1 kilocalorie
13. 80 gm of water at $30^\circ C$ are poured on a large block of ice at $0^\circ C$. The mass of ice that melts is [CBSE PMT 1989]
(a) 30 gm (b) 80 gm
(c) 1600 gm (d) 150 gm
14. The saturation vapour pressure of water at $100^\circ C$ is [EAMCET 1997]
(a) 739 mm of mercury (b) 750 mm of mercury
(c) 760 mm of mercury (d) 712 mm of mercury
15. Two spheres made of same substance have diameters in the ratio $1 : 2$. Their thermal capacities are in the ratio of [JIPMER 1999]

Calorimetry

1. When vapour condenses into liquid [CPMT 1990]
(a) It absorbs heat (b) It liberates heat
(c) Its temperature increases (d) Its temperature decreases
2. At NTP water boils at $100^\circ C$. Deep down the mine, water will boil at a temperature [CPMT 1996]
(a) $100^\circ C$ (b) $> 100^\circ C$
(c) $< 100^\circ C$ (d) Will not boil at all
3. If specific heat of a substance is infinite, it means [AIIMS 1997]
(a) Heat is given out
(b) Heat is taken in
- (a) 739 mm of mercury (b) 750 mm of mercury
(c) 760 mm of mercury (d) 712 mm of mercury
15. Two spheres made of same substance have diameters in the ratio $1 : 2$. Their thermal capacities are in the ratio of [JIPMER 1999]

- (a) 1 : 2 (b) 1 : 8
(c) 1 : 4 (d) 2 : 1
16. Work done in converting one gram of ice at -10°C into steam at 100°C is
[MP PET/PMT 1988; EAMCET (Med.) 1995; MP PMT 2003]
(a) 3045 J (b) 6056 J
(c) 721 J (d) 616 J
17. If mass energy equivalence is taken into account, when water is cooled to form ice, the mass of water should
[AIEEE 2002]
(a) Increase (b) Remain unchanged
(c) Decrease (d) First increase then decrease
18. Compared to a burn due to water at 100°C , a burn due to steam at 100°C is
[KCET 1999; UPSEAT 1999]
(a) More dangerous (b) Less dangerous
(c) Equally dangerous (d) None of these
19. 50 gm of copper is heated to increase its temperature by 10°C . If the same quantity of heat is given to 10 gm of water, the rise in its temperature is (Specific heat of copper = $420 \text{ Joule-kg}^{-1}\text{C}^{-1}$)
(a) 5°C (b) 6°C
(c) 7°C (d) 8°C
20. Two liquids A and B are at 32°C and 24°C . When mixed in equal masses the temperature of the mixture is found to be 28°C . Their specific heats are in the ratio of [DPMT 1996]
(a) 3 : 2 (b) 2 : 3
(c) 1 : 1 (d) 4 : 3
21. A beaker contains 200 gm of water. The heat capacity of the beaker is equal to that of 20 gm of water. The initial temperature of water in the beaker is 20°C . If 440 gm of hot water at 92°C is poured in it, the final temperature (neglecting radiation loss) will be nearest to [NSEP 1994]
(a) 58°C (b) 68°C
(c) 73°C (d) 78°C
22. Amount of heat required to raise the temperature of a body through 1K is called its
[KCET 1996; MH CET 2001; AIEEE 2002]
(a) Water equivalent (b) Thermal capacity
(c) Entropy (d) Specific heat
23. A metallic ball and highly stretched spring are made of the same material and have the same mass. They are heated so that they melt, the latent heat required [AIIMS 2002]
(a) Are the same for both
(b) Is greater for the ball
(c) Is greater for the spring
(d) For the two may or may not be the same depending upon the metal
24. A liquid of mass m and specific heat c is heated to a temperature $2T$. Another liquid of mass $m/2$ and specific heat $2c$ is heated to a temperature T . If these two liquids are mixed, the resulting temperature of the mixture is
[EAMCET 1992]
(a) $(2/3)T$ (b) $(8/5)T$
(c) $(3/5)T$ (d) $(3/2)T$
25. Calorie is defined as the amount of heat required to raise temperature of 1g of water by 1°C and it is defined under which of the following conditions
[IIT-JEE (Screening) 2005]
(a) From 14.5°C to 15.5°C at 760 mm of Hg
(b) From 98.5°C to 99.5°C at 760 mm of Hg
(c) From 13.5°C to 14.5°C at 76 mm of Hg
(d) From 3.5°C to 4.5°C at 76 mm of Hg
26. 100 gm of ice at 0°C is mixed with 100 g of water at 100°C . What will be the final temperature of the mixture
[SCRA 1996; AMU 1999]
(a) 10°C (b) 20°C
(c) 30°C (d) 40°C
27. At atmospheric pressure, the water boils at 100°C . If pressure is reduced, it will boil at [MP PMT 1984]
(a) Higher temperature (b) Lower temperature
(c) At the same temperature (d) At critical temperature
28. A closed bottle containing water at 30°C is carried to the moon in a space-ship. If it is placed on the surface of the moon, what will happen to the water as soon as the lid is opened
(a) Water will boil [EAMCET (Med.) 2000]
(b) Water will freeze
(c) Nothing will happen on it
(d) It will decompose into H_2 and O_2
29. The thermal capacity of 40 gm of aluminium (specific heat = $0.2 \text{ cal/gm}^{\circ}\text{C}$) is [CBSE PMT 1990]
(a) $40 \text{ cal}^{\circ}\text{C}$ (b) $160 \text{ cal}^{\circ}\text{C}$
(c) $200 \text{ cal}^{\circ}\text{C}$ (d) $8 \text{ cal}^{\circ}\text{C}$
30. If temperature scale is changed from $^{\circ}\text{C}$ to $^{\circ}\text{F}$, the numerical value of specific heat will [CPMT 1984]
(a) Increases (b) Decreased
(c) Remains unchanged (d) None of the above
31. By exerting a certain amount of pressure on an ice block, you
(a) Lower its melting point
(b) Make it melt at 0°C only
(c) Make it melt at a faster rate
(d) Raise its melting point
32. When we rub our palms they gets heated but to a maximum temperature because
(a) Heat is absorbed by our palm
(b) Heat is lost in the environment
(c) Produced of heat is stopped
(d) None of the above
33. A bullet moving with a uniform velocity v , stops suddenly after hitting the target and the whole mass melts be m , specific heat S , initial temperature 25°C , melting point 475°C and the latent heat L . Then v is given by [NCERT 1972]
(a) $mL = mS(475 - 25) + \frac{1}{2} \cdot \frac{mv^2}{J}$
(b) $mS(475 - 25) + mL = \frac{mv^2}{2J}$
(c) $mS(475 - 25) + mL = \frac{mv^2}{J}$

$$(d) \quad mS(475 - 25) - mL = \frac{mv^2}{2J}$$

34. A water fall is 84 metres high. If half of the potential energy of the falling water gets converted to heat, the rise in temperature of water will be [JIPMER 2002]

(a) 0.098°C (b) 0.98°C
(c) 9.8°C (d) 0.0098°C

35. A body of mass 5 kg falls from a height of 30 metre. If its all mechanical energy is changed into heat, then heat produced will be

(a) 350 cal (b) 150 cal
(c) 60 cal (d) 6 cal

36. In supplying 400 calories of heat to a system, the work done will be

(a) 400 joules (b) 1672 joules
(c) 1672 watts (d) 1672 ergs

37. 0.93 watt-hour of energy is supplied to a block of ice weighing 10 gm. It is found that

[NCERT 1973; DPMT 1999]

(a) Half of the block melts
(b) The entire block melts and the water attains a temperature of 4°C
(c) The entire block just melts
(d) The block remains unchanged

38. The weight of a person is 60 kg. If he gets 10 calories heat through food and the efficiency of his body is 28%, then upto how much height he can climb (approximately)

[AFMC 1997]

(a) 100 m (b) 200 m
(c) 400 m (d) 1000 m

39. The temperature of Bhakhra dam water at the ground level with respect to the temperature at high level should be

(a) Greater (b) Less
(c) Equal (d) 0°C

40. The height of a waterfall is 84 metre. Assuming that the entire kinetic energy of falling water is converted into heat, the rise in temperature of the water will be

$$(g = 9.8 \text{ m/s}^2, J = 4.2 \text{ joule/cal}) \quad [\text{MP PET 1994}]$$

(a) 0.196°C (b) 1.960°C
(c) 0.96°C (d) 0.0196°C

41. Hailstone at 0°C falls from a height of 1 km on an insulating surface converting whole of its kinetic energy into heat. What part of it will melt ($g = 10 \text{ m/s}^2$) [MP PMT 1994]

(a) $\frac{1}{33}$ (b) $\frac{1}{8}$
(c) $\frac{1}{33} \times 10^{-4}$ (d) All of it will melt

42. The SI unit of mechanical equivalent of heat is

[MP PMT/PET 1998]

(a) Joule $\times$ Calorie (b) Joule/Calorie

(c) Calorie $\times$ Erg (d) Erg/Calorie

43. Of two masses of 5 kg each falling from height of 10 m, by which 2 kg water is stirred. The rise in temperature of water will be

(a) 2.6°C (b) 1.2°C
(c) 0.32°C (d) 0.12°C

44. A lead ball moving with a velocity V strikes a wall and stops. If 50% of its energy is converted into heat, then what will be the increase in temperature (Specific heat of lead is S)

[CPMT 1975]

[RPMT 1996]

(a) $\frac{2V^2}{JS}$ (b) $\frac{V^2}{4JS}$

[MP PMT 1989]

(c) $\frac{V^2}{J}$ (d) $\frac{V^2S}{2J}$

45. The mechanical equivalent of heat J is [MP PET 2000]

(a) A constant (b) A physical quantity
(c) A conversion factor (d) None of the above

46. Water falls from a height of 210 m. Assuming whole of energy due to fall is converted into heat the rise in temperature of water would be ($J = 4.3 \text{ Joule/cal}$)

[Pb. PMT 2002]

(a) 42°C (b) 49°C
(c) 0.49°C (d) 4.9°C

47. A block of mass 100 gm slides on a rough horizontal surface. If the speed of the block decreases from 10 m/s to 5 m/s, the thermal energy developed in the process is

[UPSEAT 2002]

(a) 3.75 J (b) 37.5 J
(c) 0.375 J (d) 0.75 J

48. 4200 J of work is required for [MP PMT 1986]

(a) Increasing the temperature of 10 gm of water through 10°C
(b) Increasing the temperature of 100 gm of water through 10°C
(c) Increasing the temperature of 1 kg of water through 10°C
(d) Increasing the temperature of 10 kg of water through 10°C

49. At 100°C , the substance that causes the most severe burn, is

[KCET 1999; UPSEAT 1999]

(a) Oil (b) Steam
(c) Water (d) Hot air

50. In a water-fall the water falls from a height of 100 m. If the entire K.E. of water is converted into heat, the rise in temperature of water will be [MP PMT 2001]

(a) 0.23°C (b) 0.46°C
(c) 2.3°C (d) 0.023°C

51. A lead bullet of 10 g travelling at 300 m/s strikes against a block of wood and comes to rest. Assuming 50% of heat is absorbed by the bullet, the increase in its temperature is

(Specific heat of lead = $150 \text{ J/kg} \cdot \text{K}$) [EAMCET 2001]

(a) 100°C (b) 125°C
(c) 150°C (d) 200°C

52. The temperature at which the vapour pressure of a liquid becomes equals to the external (atmospheric) pressure is its
[Kerala (Engg.) 2001]
- (a) Melting point (b) Sublimation point
(c) Critical temperature (d) Boiling point
53. When the pressure on water is increased the boiling temperature of water as compared to 100°C will be
[RPET 1999]
- (a) Lower
(b) The same
(c) Higher
(d) On the critical temperature
54. Calorimeters are made of which of the following
[AFMC 2000]
- (a) Glass (b) Metal
(c) Wood (d) Either (a) or (c)
55. Triple point of water is
[CPMT 2002]
- (a) 273.16°F (b) 273.16 K
(c) 273.16°C (d) 273.16 R
56. A liquid boils when its vapour pressure equals
[MP PET 2002]
- (a) The atmospheric pressure
(b) Pressure of 76.0 cm column of mercury
(c) The critical pressure
(d) The dew point of the surroundings
57. The amount of work, which can be obtained by supplying 200 cal of heat, is
[Pb. PET 2001, 03; BHU 2004]
- (a) 840 dyne (b) 840 W
(c) 840 erg (d) 840 J
58. How many grams of a liquid of specific heat 0.2 at a temperature 40°C must be mixed with 100 gm of a liquid of specific heat of 0.5 at a temperature 20°C , so that the final temperature of the mixture becomes 32°C
[Pb. PET 1999]
- (a) 175 gm (b) 300 g
(c) 295 gm (d) 375 g
59. 1 g of a steam at 100°C melt how much ice at 0°C ? (Latent heat of ice = 80 cal/gm and latent heat of steam = 540 cal/gm)
[Pb. PET 2003]
- (a) 1 gm (b) 2 gm
(c) 4 gm (d) 8 gm
60. 5 g of ice at 0°C is dropped in a beaker containing 20 g of water at 40°C . The final temperature will be
[Pb. PET 2003]
- (a) 32°C (b) 16°C
(c) 8°C (d) 24°C
61. One kilogram of ice at 0°C is mixed with one kilogram of water at 80°C . The final temperature of the mixture is
(Take : specific heat of water = $4200\text{ J kg}^{-1}\text{ K}^{-1}$, latent heat of ice = 336 kJ kg^{-1})
[KCET 2002]
- (a) 40°C (b) 60°C
(c) 0°C (d) 50°C
62. During constant temperature, we feel colder on a day when the relative humidity will be
[Pb. PMT 1996]
- (a) 25% (b) 12.5%
(c) 50% (d) 75%
63. Which of the following is the unit of specific heat
[MH CET 2004]
- (a) $\text{J kg }^{\circ}\text{C}^{-1}$ (b) $\text{J / kg }^{\circ}\text{C}$
(c) $\text{kg }^{\circ}\text{C / J}$ (d) $\text{J / kg }^{\circ}\text{C}^{-2}$
64. 50 gm of ice at 0°C is mixed with 50 gm of water at 80°C , final temperature of mixture will be
[DCE 2002]
- (a) 0°C (b) 40°C
(c) 40°C (d) 4°C
65. The freezing point of the liquid decreases when pressure is increased, if the liquid
[DCE 1995]
- (a) Expands while freezing
(b) Contracts while freezing
(c) Does not change in volume while freezing
(d) None of these
66. The relative humidity on a day, when partial pressure of water vapour is $0.012 \times 10^5\text{ Pa}$ at 12°C is (take vapour pressure of water at this temperature as $0.016 \times 10^5\text{ Pa}$)
[AIIMS 1998]
- (a) 70% (b) 40%
(c) 75% (d) 25%
67. A hammer of mass 1 kg having speed of 50 m/s, hit a iron nail of mass 200 gm. If specific heat of iron is $0.105\text{ cal/gm }^{\circ}\text{C}$ and half the energy is converted into heat, the raise in temperature of nail is
[RPMT 1999]
- (a) 7.1°C (b) 9.2°C
(c) 10.5°C (d) 12.1°C
68. Latent heat of 1 gm of steam is 536 cal/gm, then its value in joule/kg is
[RPMT 1999]
- (a) 2.25×10^6 (b) 2.25×10^3
(c) 2.25 (d) None
69. Which of the following has maximum specific heat
[RPMT 1999]
- (a) Water (b) Alcohol
(c) Glycerine (d) Oil
70. 50 gm ice at 0°C in insulator vessel, 50g water of 100°C is mixed in it, then final temperature of the mixture is (neglect the heat loss)
[Pb. PET 2000]
- (a) 10°C (b) $0^{\circ} < T_m < 20^{\circ}\text{C}$
(c) 20°C (d) Above 20°C
71. A stationary object at 4°C and weighing 3.5 kg falls from a height of 2000 m on a snow mountain at 0°C . If the temperature of the object just before hitting the snow is 0°C and the object comes to rest immediately ($g = 10\text{ m / s}^2$) and (latent heat of ice = $3.5 \times 10^5\text{ joule/ sec}$), then the object will melt
[Pb. PET 2000]
- (a) 2 kg of ice (b) 200 gm of ice
(c) 20 gm ice (d) 2 gm of ice

72. 300 gm of water at 25°C is added to 100 gm of ice at 0°C . The final temperature of the mixture is [MP PET 2004]
- (a) $-\frac{5}{3}^{\circ}\text{C}$ (b) $-\frac{5}{2}^{\circ}\text{C}$
(c) -5°C (d) 0°C
73. Calculate the amount of heat (in calories) required to convert 5 gm of ice at 0°C to steam at 100°C [DPMT 2005]
- (a) 3100 (b) 3200
(c) 3600 (d) 4200
74. 2gm of steam condenses when passed through 40gm of water initially at 25°C . The condensation of steam raises the temperature of water to 54.3°C . What is the latent heat of steam
- (a) 540 cal/g (b) 536 cal/g
(c) 270 cal/g (d) 480 cal/g
75. 10 gm of ice at 0°C is mixed with 100 gm of water at 50°C . What is the resultant temperature of mixture [AFMC 2005]
- (a) 31.2°C (b) 32.8°C
(c) 36.7°C (d) 38.2°C
76. Three liquids with masses m_1, m_2, m_3 are thoroughly mixed. If their specific heats are C_1, C_2, C_3 and their temperatures T_1, T_2, T_3 respectively, then the temperature of the mixture is
- (a) $\frac{C_1 T_1 + C_2 T_2 + C_3 T_3}{m_1 C_1 + m_2 C_2 + m_3 C_3}$
(b) $\frac{m_1 C_1 T_1 + m_2 C_2 T_2 + m_3 C_3 T_3}{m_1 C_1 + m_2 C_2 + m_3 C_3}$
(c) $\frac{m_1 C_1 T_1 + m_2 C_2 T_2 + m_3 C_3 T_3}{m_1 T_1 + m_2 T_2 + m_3 T_3}$
(d) $\frac{m_1 T_1 + m_2 T_2 + m_3 T_3}{C_1 T_1 + C_2 T_2 + C_3 T_3}$
77. The point on the pressure temperature phase diagram where all the phases co-exist is called [MH CET 2005]
- (a) Sublimation (b) Fusion point
(c) Triple point (d) Vaporisation point
78. Boiling water is changing into steam. At this stage the specific heat of water is [UPSEAT 1998]
- (a) < 1 (b) ∞
(c) 1 (d) 0
79. A vessel contains 110 g of water. The heat capacity of the vessel is equal to 10 g of water. The initial temperature of water in vessel is 10°C . If 220 g of hot water at 70°C is poured in the vessel, the final temperature neglecting radiation loss, will be
- (a) 70°C (b) 80°C
(c) 60°C (d) 50°C
80. The thermal capacity of a body is 80 cal, then its water equivalent is [UPSEAT 2001]

- (a) 80 cal/gm (b) 8 gm
(c) 80 gm (d) 80 kg

81. A liquid of mass M and specific heat S is at a temperature $2t$. If another liquid of thermal capacity 1.5 times, at a temperature of $\frac{t}{3}$ is added to it, the resultant temperature will be
- (a) $\frac{4}{3}t$ (b) t
(c) $\frac{t}{2}$ (d) $\frac{2}{3}t$
82. Dry ice is [CPMT 2000]
- (a) Ice cube (b) Sodium chloride
(c) Liquid nitrogen [JEE 2005] (d) Solid carbon dioxide

Critical Thinking

Objective Questions

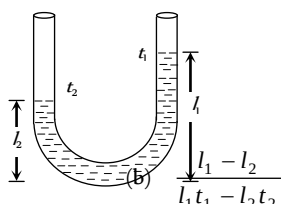
1. A glass flask is filled up to a mark with 50 cc of mercury at 18°C . If the flask and contents are heated to 38°C , how much mercury will be above the mark? (α for glass is $9 \times 10^{-6}/^{\circ}\text{C}$ and coefficient of real expansion of mercury is $180 \times 10^{-6}/^{\circ}\text{C}$)
- (a) 0.85 cc (b) 0.46 cc
(c) 0.153 cc (d) 0.05 cc
2. The coefficient of apparent expansion of mercury in a glass vessel is $153 \times 10^{-6}/^{\circ}\text{C}$ and in a steel vessel is $144 \times 10^{-6}/^{\circ}\text{C}$. If α for steel is $12 \times 10^{-6}/^{\circ}\text{C}$, then that of glass is [EAMCET 1997]
- (a) $9 \times 10^{-6}/^{\circ}\text{C}$ (b) $6 \times 10^{-6}/^{\circ}\text{C}$
(c) $36 \times 10^{-6}/^{\circ}\text{C}$ (d) $27 \times 10^{-6}/^{\circ}\text{C}$
3. Solids expand on heating because [CPMT 1990]
- (a) Kinetic energy of the atoms increases
(b) Potential energy of the atoms increases
(c) Total energy of the atoms increases
(d) The potential energy curve is asymmetric about the equilibrium distance between neighbouring atoms
4. An iron tyre is to be fitted on to a wooden wheel 1m in diameter. The diameter of tyre is 6 mm smaller than that of wheel. The tyre should be heated so that its temperature increases by a minimum of (the coefficient of cubical expansion of iron is $3.6 \times 10^{-6}/^{\circ}\text{C}$) [CPMT 1989]
- (a) 167°C (b) 334°C
(c) 500°C (d) 1000°C
5. A glass flask of volume one litre at 0°C is filled, level full of mercury at this temperature. The flask and mercury are now heated to 100°C . How much mercury will spill out, if coefficient of volume expansion of mercury is $1.82 \times 10^{-4}/^{\circ}\text{C}$ and linear expansion of glass is $0.1 \times 10^{-4}/^{\circ}\text{C}$ respectively [UPSEAT 2000] [MNR 1994]
- (a) 21.2 cc (b) 15.2 cc
(c) 1.52 cc (d) 2.12 cc

6. A steel scale measures the length of a copper wire as 80.0 cm , when both are at 20°C (the calibration temperature for scale). What would be the scale read for the length of the wire when both are at 40°C ? (Given $\alpha_{\text{steel}} = 11 \times 10^{-6} \text{ per } ^\circ\text{C}$ and $\alpha_{\text{copper}} = 17 \times 10^{-6} \text{ per } ^\circ\text{C}$) [CPMT 2004]
- (a) 80.0096 cm (b) 80.0272 cm
(c) 1 cm (d) 25.2 cm
7. A bimetallic strip is formed out of two identical strips, one of copper and other of brass. The coefficients of linear expansion of the two metals are α_C and α_B . On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R . Then R is [IIT-JEE (Screening) 1999]
- (a) Proportional to ΔT
(b) Inversely proportional to ΔT
(c) Proportional to $|\alpha_B - \alpha_C|$
(d) Inversely proportional to $|\alpha_B - \alpha_C|$
8. Two metal strips that constitute a thermostat must necessarily differ in their [IIT-JEE 1992]
- (a) Mass
(b) Length
(c) Resistivity
(d) Coefficient of linear expansion
9. A metal ball immersed in alcohol weighs W_1 at 0°C and W_2 at 59°C . The coefficient of cubical expansion of the metal is less than that of alcohol. Assuming that the density of metal is large compared to that of alcohol, it can be shown that [CPMT 1998]
- (a) $W_1 > W_2$ (b) $W_1 = W_2$
(c) $W_1 < W_2$ (d) $W_2 = (W_1 / 2)$
10. The coefficient of volumetric expansion of mercury is $18 \times 10^{-6} / ^\circ\text{C}$. A thermometer bulb has a volume 10^{-7} m^3 and cross section of stem is 0.004 cm^2 . Assuming that bulb is filled with mercury at 0°C then the length of the mercury column at 100°C is [Pb. PMT 1998, CPMT 1997, 2001]
- (a) 18.8 mm (b) 9.2 mm
(c) 7.4 cm (d) 4.5 cm
11. A piece of metal weight 46 gm in air, when it is immersed in the liquid of specific gravity 1.24 at 27°C it weighs 30 gm . When the temperature of liquid is raised to 42°C the metal piece weight 30.5 gm , specific gravity of the liquid at 42°C is 1.20, then the linear expansion of the metal will be [BHU 1995]
- (a) $3.316 \times 10^{-6} / ^\circ\text{C}$ (b) $2.316 \times 10^{-6} / ^\circ\text{C}$
(c) $4.316 \times 10^{-6} / ^\circ\text{C}$ (d) None of these
12. It is known that wax contracts on solidification. If molten wax is taken in a large vessel and it is allowed to cool slowly, then
- (a) It will start solidifying from the top downward
(b) It will start solidifying from the bottom upward
(c) It will start solidifying from the middle, upward and downward at equal rates
(d) The whole mass will solidify simultaneously
13. A substance of mass $m \text{ kg}$ requires a power input of P watts to remain in the molten state at its melting point. When the power is turned off, the sample completely solidifies in time $t \text{ sec}$. What is the latent heat of fusion of the substance [IIT JEE 1992]
- (a) $\frac{Pm}{t}$ (b) $\frac{Pt}{m}$
(c) $\frac{m}{Pt}$ (d) $\frac{t}{Pm}$
14. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C . The mass of the steam condensed in kg is [IIT 1995]
- (a) 0.130 (b) 0.065
(c) 0.260 (d) 0.135
15. 2 kg of ice at -20°C is mixed with 5 kg of water at 20°C in an insulating vessel having a negligible heat capacity. Calculate the final mass of water remaining in the container. It is given that the specific heats of water and ice are $1 \text{ kcal/kg per } ^\circ\text{C}$ and $0.5 \text{ kcal/kg per } ^\circ\text{C}$ while the latent heat of fusion of ice is 80 kcal/kg [IIT-JEE (Screening) 2003]
- (a) 7 kg (b) 6 kg
(c) 4 kg (d) 2 kg
16. Water of volume 2 litre in a container is heated with a coil of 1 kW at 27°C . The lid of the container is open and energy dissipates at rate of 160 J/s . In how much time temperature will rise from 27°C to 77°C [Given specific heat of water is 4.2 kJ/kg]
- (a) $8 \text{ min } 20 \text{ s}$ (b) $6 \text{ min } 2 \text{ s}$
(c) 7 min (d) 14 min
17. A lead bullet at 27°C just melts when stopped by an obstacle. Assuming that 25% of heat is absorbed by the obstacle, then the velocity of the bullet at the time of striking (M.P. of lead = 327°C , specific heat of lead = $0.03 \text{ cal/gm per } ^\circ\text{C}$, latent heat of fusion of lead = 6 cal/gm and $J = 4.2 \text{ joule/cal}$) [IIT 1981]
- (a) 410 m/sec (b) 1230 m/sec
(c) 307.5 m/sec (d) None of the above
18. If two balls of same metal weighing 5 gm and 10 gm strike with a target with the same velocity. The heat energy so developed is used for raising their temperature alone, then the temperature will be higher
- (a) For bigger ball
(b) For smaller ball
(c) Equal for both the balls
(d) None is correct from the above three
19. The temperature of equal masses of three different liquids A , B and C are 12°C , 19°C and 28°C respectively. The temperature when A and B are mixed is 16°C and when B and C are mixed is 23°C . The temperature when A and C are mixed is [CBSE PMT 1994]

- (a) 18.2°C (b) 22°C
(c) 20.2°C (d) 25.2°C
20. In an industrial process 10 kg of water per hour is to be heated from 20°C to 80°C . To do this steam at 150°C is passed from a boiler into a copper coil immersed in water. The steam condenses in the coil and is returned to the boiler as water at 90°C . how many kg of steam is required per hour.

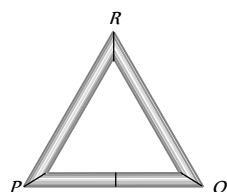
(Specific heat of steam = 1 calorie per gm $^{\circ}\text{C}$, Latent heat of vaporisation = 540 cal/gm)

- (a) 1 gm (b) 1 kg
(c) 10 gm (d) 10 kg
21. In a vertical U-tube containing a liquid, the two arms are maintained at different temperatures t_1 and t_2 . The liquid columns in the two arms have heights l_1 and l_2 respectively. The coefficient of volume expansion of the liquid is equal to



- (a) $\frac{l_1 - l_2}{l_2 t_1 - l_1 t_2}$ (b) $\frac{l_1 - l_2}{l_1 t_1 - l_2 t_2}$
(c) $\frac{l_1 + l_2}{l_2 t_1 + l_1 t_2}$ (d) $\frac{l_1 + l_2}{l_1 t_1 + l_2 t_2}$
22. The coefficient of linear expansion of crystal in one direction is α_1 and that in every direction perpendicular to it is α_2 . The coefficient of cubical expansion is
- (a) $\alpha_1 + \alpha_2$ (b) $2\alpha_1 + \alpha_2$
(c) $\alpha_1 + 2\alpha_2$ (d) None of these

23. Three rods of equal length l are joined to form an equilateral triangle PQR . O is the mid point of PQ . Distance OR remains same for small change in temperature. Coefficient of linear expansion for PR and RQ is same i.e. α_2 but that for PQ is α_1 . Then



- (a) $\alpha_2 = 3\alpha_1$
(b) $\alpha_2 = 4\alpha_1$
(c) $\alpha_1 = 3\alpha_2$
(d) $\alpha_1 = 4\alpha_2$
24. A one litre glass flask contains some mercury. It is found that at different temperatures the volume of air inside the flask remains the same. What is the volume of mercury in this flask if coefficient of linear expansion of glass is $9 \times 10^{-6}/^{\circ}\text{C}$ while of volume expansion of mercury is $1.8 \times 10^{-4}/^{\circ}\text{C}$

- (a) 50 cc (b) 100 cc
(c) 150 cc (d) 200 cc
25. 10 gm of ice at -20°C is dropped into a calorimeter containing 10 gm of water at 10°C ; the specific heat of water is twice that of ice. When equilibrium is reached, the calorimeter will contain

- (b) 20 gm of ice
(c) 10 gm ice and 10 gm water
(d) 5 gm ice and 15 gm water

26. A rod of length 20 cm is made of metal. It expands by 0.075 cm when its temperature is raised from 0°C to 100°C . Another rod of a different metal B having the same length expands by 0.045 cm for the same change in temperature. A third rod of the same length is composed of two parts, one of metal A and the other of metal B . This rod expands by 0.060 cm for the same change in temperature. The portion made of metal A has the length

- (a) 20 cm (b) 10 cm
(c) 15 cm (d) 18 cm

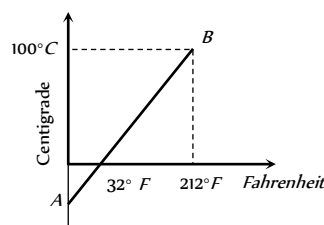
27. Steam is passed into 22 gm of water at 20°C . The mass of water that will be present when the water acquires a temperature of 90°C (Latent heat of steam is 540 cal/gm) is

[SCRA 1994]

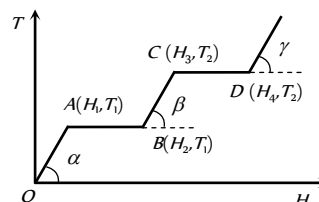
- (a) 24.8 gm (b) 24 gm
(c) 36.6 gm (d) 30 gm

Graphical Questions

1. The graph AB shown in figure is a plot of temperature of a body in degree celsius and degree Fahrenheit. Then

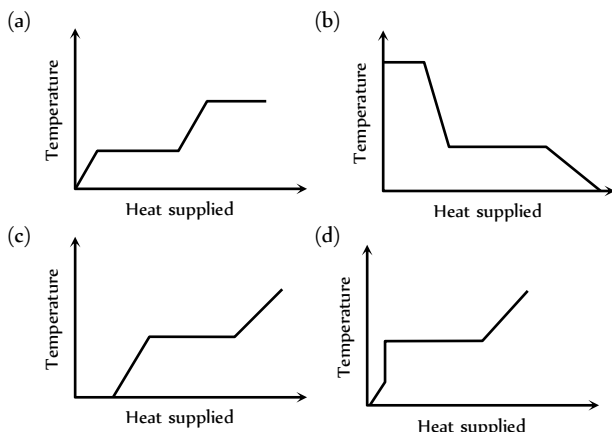


- (a) Slope of line AB is 9/5 (b) Slope of line AB is 5/9
(c) Slope of line AB is 1/9 (d) Slope of line AB is 3/9
2. The graph shows the variation of temperature (T) of one kilogram of a material with the heat (H) supplied to it. At O , the substance is in the solid state. From the graph, we can conclude that

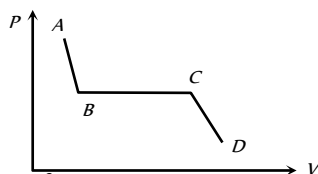


- (a) T_2 is the melting point of the solid
(b) BC represents the change of state from solid to liquid
(c) $(H_2 - H_1)$ represents the latent heat of fusion of the substance
(d) $(H_3 - H_1)$ represents the latent heat of vaporization of the liquid

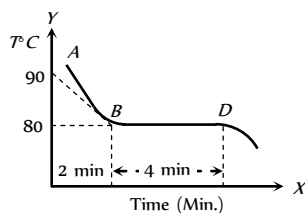
3. A block of ice at -10°C is slowly heated and converted to steam at 100°C . Which of the following curves represents the phenomenon qualitatively [IIT-JEE (Screening) 2000]



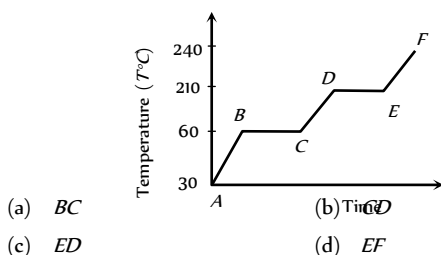
4. The portion AB of the indicator diagram representing the state of matter denotes



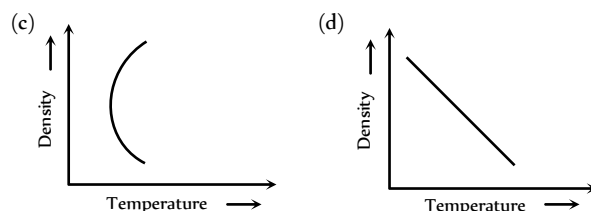
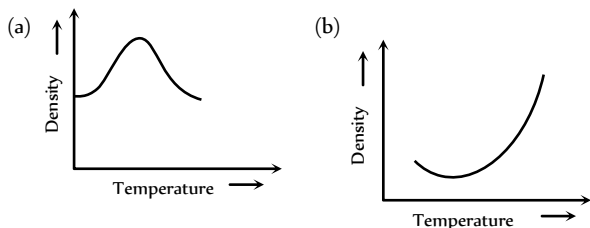
- (a) The liquid state of matter
(b) Gaseous state of matter
(c) Change from liquid to gaseous state
(d) Change from gaseous state to liquid state
5. The figure given below shows the cooling curve of pure wax material after heating. It cools from A to B and solidifies along BD . If L and C are respective values of latent heat and the specific heat of the liquid wax, the ratio L/C is



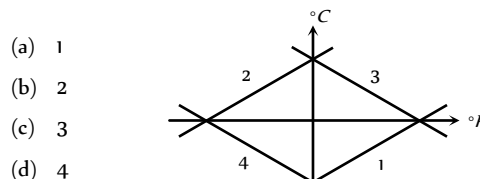
- (a) 40
(b) 80
(c) 100
(d) 20
6. A solid substance is at 30°C . To this substance heat energy is supplied at a constant rate. Then temperature versus time graph is as shown in the figure. The substance is in liquid state for the portion (of the graph) [RPET 1990, 94]



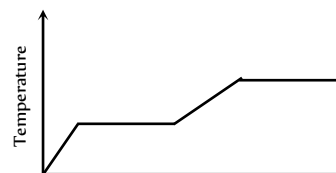
- (a) BC
(b) CD
(c) ED
(d) EF
7. The variation of density of water with temperature is represented by the



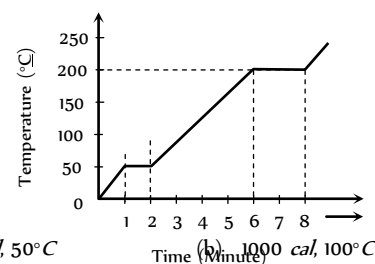
8. If a graph is plotted taking the temperature in Fahrenheit along Y -axis and the corresponding temperature in Celsius along the X -axis, it will be a straight line [AIIMS 1997]
- (a) Having a $+ve$ intercept on Y -axis
(b) Having a $+ve$ intercept on X -axis
(c) Passing through the origin
(d) Having a $-ve$ intercepts on both the axis
9. Which of the curves in figure represents the relation between Celsius and Fahrenheit temperatures



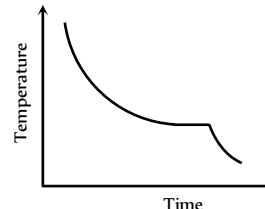
10. Heat is supplied to a certain homogenous sample of matter, at a uniform rate. Its temperature is plotted against time, as shown. Which of the following conclusions can be drawn



- (a) Its specific heat capacity is greater in the solid state than in the liquid state
(b) Its specific heat capacity is greater in the liquid state than in the solid state
(c) Its latent heat of vaporization is greater than its latent heat of fusion
(d) Its latent heat of vaporization is smaller than its latent of fusion
11. A student takes 50gm wax (specific heat = $0.6\text{ kcal/kg}^{\circ}\text{C}$) and heats it till it boils. The graph between temperature and time is as follows. Heat supplied to the wax per minute and boiling point are respectively [BHU 1994]

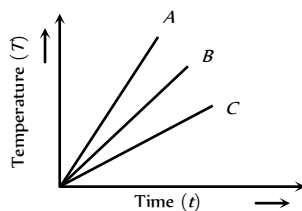


- (a) $500\text{ cal}, 50^{\circ}\text{C}$
(b) $1000\text{ cal}, 100^{\circ}\text{C}$
(c) $1500\text{ cal}, 200^{\circ}\text{C}$
(d) 200°C
12. The graph signifies [JIPMER 1999]

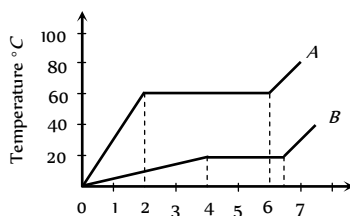


- (a) Adiabatic expansion of a gas
- (b) Isothermal expansion of a gas
- (c) Change of state from liquid to solid
- (d) Cooling of a heated solid

13. Which of the substances A , B or C has the highest specific heat ?
The temperature vs time graph is shown



- (a) A
 - (b) B
 - (c) C
 - (d) All have equal specific heat
14. Two substances A and B of equal mass m are heated at uniform rate of 6 cal/s under similar conditions. A graph between temperature and time is shown in figure. Ratio of heat absorbed H_A / H_B by them for complete fusion is



- (a) $\frac{9}{4}$
- (b) $\frac{4}{9}$
- (c) $\frac{8}{5}$
- (d) $\frac{5}{8}$

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below :

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (c) If assertion is true but reason is false.
- (d) If the assertion and reason both are false.
- (e) If assertion is false but reason is true.

1. Assertion : The melting point of ice decreases with increase of pressure.
Reason : Ice contracts on melting. [AIIMS 2004]
2. Assertion : Fahrenheit is the smallest unit measuring temperature.
Reason : Fahrenheit was the first temperature scale used for measuring temperature. [AIIMS 1999]
3. Assertion : Melting of solid causes no change in internal energy.

Reason : Latent heat is the heat required to melt a unit mass of solid. [AIIMS 1998]

4. Assertion : Specific heat capacity is the cause of formation of land and sea breeze.

Reason : The specific heat of water is more than land.

[AIIMS 1995]

5. Assertion : A brass disc is just fitted in a hole in a steel plate. The system must be cooled to loosen the disc from the hole.

Reason : The coefficient of linear expansion for brass is greater than the coefficient of linear expansion for steel.

6. Assertion : The coefficient of volume expansion has dimension K .

Reason : The coefficient of volume expansion is defined as the change in volume per unit volume per unit change in temperature.

7. Assertion : The temperature at which Centigrade and Fahrenheit thermometers read the same is -40° .

Reason : There is no relation between Fahrenheit and Centigrade temperature.

8. Assertion : When a solid iron ball is heated, percentage increase in its volume is largest.

Reason : Coefficient of superficial expansion is twice that of linear expansion where as coefficient of volume expansion is three time of linear expansion.

9. Assertion : A beaker is completely filled with water at 4°C . It will overflow, both when heated or cooled.

Reason : There is expansion of water below and above 4°C .

10. Assertion : Latent heat of fusion of ice is 336000 J/kg .

Reason : Latent heat refers to change of state without any change in temperature

11. Assertion : Two bodies at different temperatures, if brought in thermal contact do not necessary settle to the mean temperature.

Reason : The two bodies may have different thermal capacities.

12. Assertion : Specific heat of a body is always greater than its thermal capacity.

Reason : Thermal capacity is the required for raising temperature of unit mass of the body through unit degree.

13. Assertion : Water kept in an open vessel will quickly evaporate on the surface of the moon.

Reason : The temperature at the surface of the moon is much higher than boiling point of the water.

14. Assertion : The molecules at 0°C ice and 0°C water will have same potential energy.

Reason : Potential energy depends only on temperature of the system.

Answers

Thermometry

1	d	2	b	3	a	4	c	5	b
6	d	7	c	8	a	9	b	10	b
11	c	12	c	13	c	14	c	15	c
16	d	17	b	18	c	19	c	20	c
21	c	22	a	23	d	24	a	25	c
26	a	27	b	28	a	29	c	30	c
31	a	32	d	33	a	34	d	35	a

Thermal Expansion

1	c	2	a	3	b	4	d	5	c
6	c	7	c	8	c	9	b	10	c
11	a	12	b	13	b	14	d	15	a
16	d	17	d	18	d	19	d	20	a
21	b	22	c	23	a	24	a	25	a
26	b	27	c	28	d	29	a	30	d
31	b	32	a	33	a	34	c		

Calorimetry

1	b	2	b	3	c	4	c	5	d
6	a	7	c	8	a	9	d	10	a
11	b	12	c	13	a	14	c	15	b
16	a	17	b	18	a	19	a	20	c
21	b	22	b	23	a	24	d	25	a
26	a	27	b	28	a	29	d	30	b
31	a	32	b	33	b	34	a	35	a
36	b	37	c	38	b	39	a	40	a
41	a	42	b	43	d	44	b	45	c
46	c	47	a	48	b	49	b	50	a
51	c	52	d	53	c	54	b	55	b
56	a	57	d	58	d	59	d	60	b
61	c	62	a	63	a	64	a	65	a
66	c	67	a	68	a	69	a	70	a
71	b	72	d	73	c	74	a	75	d
76	b	77	c	78	b	79	d	80	c
81	b	82	d						

Critical Thinking Questions

1	c	2	a	3	d	4	c	5	b
6	a	7	bd	8	d	9	c	10	d
11	b	12	b	13	b	14	a	15	b
16	a	17	a	18	c	19	c	20	b
21	a	22	c	23	d	24	c	25	c
26	b	27	a						

Graphical Questions

1	b	2	c	3	a	4	a	5	d
6	b	7	a	8	a	9	a	10	bc
11	c	12	c	13	c	14	c		

Assertion and Reason

1	a	2	c	3	e	4	a	5	a
6	a	7	c	8	a	9	a	10	b
11	a	12	d	13	a	14	d		

Answers and Solutions

Thermometry

- (d) $T = 273.15 + t^{\circ}\text{C} \Rightarrow 0 = 273.15 + t^{\circ}\text{C} \Rightarrow t = -273.15^{\circ}\text{C}$
- (b) $\frac{C}{5} = \frac{F - 32}{9} \Rightarrow \frac{-183}{5} = \frac{F - 32}{9} \Rightarrow F = -297^{\circ}\text{F}$
- (a) $\frac{F - 32}{9} = \frac{K - 273}{5} \Rightarrow \frac{F - 32}{9} = \frac{95 - 273}{5} \Rightarrow F = -288^{\circ}\text{F}$
- (c) Temperature change in Celsius scale = Temperature change in Kelvin scale = 27 K
- (b) Change in resistance $3.70 - 2.71 = 0.99\ \Omega$ corresponds to interval of temperature 90°C .
So change in resistance $3.26 - 2.71 = 0.55\ \Omega$ Corresponds to change in temperature
 $= \frac{90}{0.99} \times 0.55 = 50^{\circ}\text{C}$
- (d) -200°C to 600°C can be measured by platinum resistance thermometer.
- (c) Pyrometer can measure temperature from 800°C to 6000°C . Hence temperature of sun is measured with pyrometer.
- (a) $\bar{v}^2 \propto T$
- (b) Thermoelectric thermometer is based on Seebeck Effect.
- (b) Maximum density of water is at 4°C
Also $\frac{C}{5} = \frac{F - 32}{9} \Rightarrow \frac{4}{5} = \frac{F - 32}{9} \Rightarrow F = 39.2^{\circ}\text{F}$
- (c) Production and measurement of temperature close to 0 K is done in cryogenics
- (c)
- (c) At absolute zero (*i.e.* 0 K) v_{rms} becomes zero.
- (c)
- (c) We know that $P = P_0(1 + \gamma t)$ and $V = V_0(1 + \gamma t)$

and $\gamma = (1/273)/^{\circ}\text{C}$ for $t = -273^{\circ}\text{C}$, we have $P = 0$ and $V = 0$

Hence, at absolute zero, the volume and pressure of the gas become zero.

16. (d) Zero kelvin $= -273^{\circ}\text{C}$ (absolute temperature). As no matter can attain this temperature, hence temperature can never be negative on Kelvin scale.
17. (b) $\frac{C}{5} = \frac{F-32}{9} \Rightarrow \frac{25}{5} = \frac{F-32}{9} \Rightarrow F = 77^{\circ}\text{F}$.
18. (c) Thermoelectric thermometer is used for finding rapidly varying temperature.
19. (c) Due to evaporation cooling is caused which lowers the temperature of bulb wrapped in wet hanky.
20. (c) $\frac{F-32}{9} = \frac{K-273}{5} \Rightarrow \frac{x-32}{9} = \frac{x-273}{5} \Rightarrow x = 574.25$
21. (c) $\frac{C}{5} = \frac{F-32}{9} \Rightarrow \frac{C}{5} = \frac{(140-32)}{9} \Rightarrow C = 60^{\circ}$
22. (a) $\frac{C}{5} = \frac{F-32}{9} \Rightarrow \frac{t}{5} = \frac{t-32}{9} \Rightarrow t = -40^{\circ}$
23. (d) Standardisation of thermometers is done with gas thermometer.
24. (a) For gases γ is more.
25. (c) The boiling point of mercury is 400°C . Therefore, the mercury thermometer can be used to measure the temperature upto 360°C .
26. (a) $t = \frac{(P_t - P_0)}{(P_{100} - P_0)} \times 100^{\circ}\text{C} = \frac{(60 - 50)}{(90 - 50)} \times 100 = 25^{\circ}\text{C}$
27. (b) By filling nitrogen gas at high pressure, the boiling point of mercury is increased which extend the range upto 500°C .
28. (a) Pyrometer is used to measure very high temperature.
29. (c) $\frac{F-32}{9} = \frac{K-273}{5} \Rightarrow \frac{F-32}{9} = \frac{0-273}{5}$
 $\Rightarrow F = -459.4^{\circ}\text{F} \approx -460^{\circ}\text{F}$

30. (c) Initial volume $V_1 = 47.5$ units
Temperature of ice cold water $T_1 = 0^\circ\text{C} = 273\text{ K}$
Final volume of $V_2 = 67$ units

Applying Charles's law, we have $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

(where temperature T_2 is the boiling point)

$$\text{or } T_2 = \frac{V_2}{V_1} \times T_1 = \frac{67 \times 273}{47.5} = 385\text{ K} = 112^\circ\text{C}$$

31. (a) Temperature on any scale can be converted into other scale by
 $\frac{x - LFP}{UFP - LFP} = \text{Constant for all scales}$ $\frac{x - 20}{150 - 20} = \frac{60}{100} \Rightarrow$
 $x = 98^\circ\text{C}$

32. (d) $\frac{C}{5} = \frac{F - 32}{9} \Rightarrow \frac{C}{5} = \frac{140 - 32}{9} \Rightarrow C = 60^\circ\text{C}$

33. (a) Rapidly changing temperature is measured by thermocouple thermometers.

34. (d) Difference of $100^\circ\text{C} = \text{difference of } 180^\circ\text{F}$

$$\therefore \text{Difference of } 30^\circ = \frac{180}{100} \times 30 = 54^\circ$$

35. (a)

Thermal Expansion

1. (c) When a copper ball is heated, it's size increases. As Volume $\propto (\text{radius})^3$ and Area $\propto (\text{radius})^2$, so percentage increase will be largest in it's volume. Density will decrease with rise in temperature.

2. (a) $\frac{h_1}{h_2} = \frac{\rho_1}{\rho_2} = \frac{(1 + \gamma \theta_1)}{(1 + \gamma \theta_2)} \quad \left[\because \rho = \frac{\rho_0}{(1 + \gamma \theta)} \right]$

$$\Rightarrow \frac{50}{60} = \frac{1 + \gamma \times 50}{1 + \gamma \times 100} \Rightarrow \gamma = 0.005 / ^\circ\text{C}$$

3. (b) $\gamma_r = \gamma_a + \gamma_v$; where γ_r = coefficient of real expansion,
 γ_a = coefficient of apparent expansion and γ_v = coefficient of expansion of vessel.

For copper $\gamma_r = C + 3\alpha_{Cu} = C + 3A$

For silver $\gamma_r = S + 3\alpha_{Ag}$

$$\Rightarrow C + 3A = S + 3\alpha_{Ag} \Rightarrow \alpha_{Ag} = \frac{C - S + 3A}{3}$$

4. (d) Fractional change in period

$$\frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta = \frac{1}{2} \times 2 \times 10^{-6} \times 10 = 10^{-5}$$

$$\% \text{ change} = \frac{\Delta T}{T} \times 100 = 10^{-5} \times 100 = 10^{-3} \%$$

5. (c) $L = L_0(1 + \alpha \Delta \theta) \Rightarrow \frac{L_1}{L_2} = \frac{1 + \alpha(\Delta \theta)_1}{1 + \alpha(\Delta \theta)_2}$
 $\Rightarrow \frac{10}{L_2} = \frac{1 + 11 \times 10^{-6} \times 20}{1 + 11 \times 10^{-6} \times 19} \Rightarrow L_2 = 9.99989$

$\Rightarrow$ Length is shorten by

$$10 - 9.99989 = 0.00011 = 11 \times 10^{-5} \text{ cm}$$

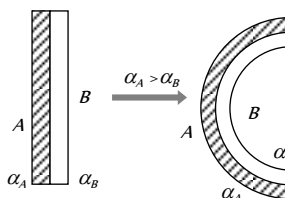
6. (c) Stress $= Y \alpha \Delta \theta$; hence it is independent of length.
7. (c) Solids, liquids and gases all expand on being heated as result density (= mass/volume) decreases.
8. (c) As coefficient of cubical expansion of liquid equals coefficient of cubical expansion of vessel, the level of liquid will not change on heating.

9. (b) Loss in time per second $\frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta = \frac{1}{2} \alpha (t - 0)$

$\Rightarrow$ loss in time per day

$$\Delta t = \left(\frac{1}{2} \alpha t \right) t = \frac{1}{2} \alpha t \times (24 \times 60 \times 60) = \frac{1}{2} \alpha t \times 86400$$

10. (c) A bimetallic strip on being heated bends in the form of an arc with more expandable metal (A) outside (as shown) correct.



11. (a) When the ball is heated, expansion of ball and cavity both occurs, hence volume of cavity increases.
12. (b) In summer alcohol expands, density decreases, so 1 litre of alcohol will weigh less in summer than in winter.
13. (b) Similar to previous question, benzene contracts in winter. So 5 litre of benzene will weigh more in winter than in summer.
14. (d) Water has maximum density at 4°C .
15. (a) Since coefficient of expansion of steel is greater than that of bronze. Hence with small increase in it's temperature the hole expand sufficiently.

16. (d) $A \propto L^2 \Rightarrow \frac{\Delta A}{A} = 2 \cdot \frac{\Delta L}{L} \Rightarrow \frac{\Delta A}{A} = 2 \times 2 = 4\%$

17. (d) $\frac{V_1}{V_2} = \frac{1 + \gamma t_1}{1 + \gamma t_2} \Rightarrow \frac{100}{125} = \frac{1 + \gamma \times 20}{1 + \gamma \times 100} \Rightarrow \gamma = 0.0033 / ^\circ\text{C}$

18. (d) $\alpha = \frac{\beta}{2} = \frac{2 \times 10^{-5}}{2} = 10^{-5} / ^\circ\text{C}$

19. (d) Coefficient of volume expansion

$$\gamma = \frac{\Delta \rho}{\rho \cdot \Delta T} = \frac{(\rho_1 - \rho_2)}{\rho \cdot (\Delta \theta)} = \frac{(10 - 9.7)}{10 \times (100 - 0)} = 3 \times 10^{-4}$$

Hence, coefficient of linear expansion

$$\alpha = \frac{\gamma}{3} = 10^{-4} / ^\circ\text{C}$$

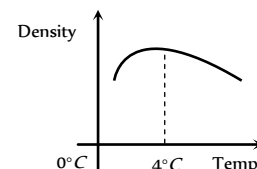
20. (a) $\rho = \rho_0(1 - \gamma \cdot \Delta \theta) = 13.6[1 - 0.18 \times 10^{-3}(473 - 273)]$
 $= 13.6[1 - 0.036] = 13.11 \text{ gm/cc}$

21. (b) As we know $\gamma_{\text{real}} = \gamma_{\text{app.}} + \gamma_{\text{vessel}}$

$$\Rightarrow \gamma_{\text{app.}} = \gamma_{\text{glycerine}} - \gamma_{\text{glass}}$$

$$= 0.000597 - 0.000027 = 0.00057 / ^\circ\text{C}$$

22. (c) Water has maximum density at 4°C , so if the water is heated above 4°C or cooled below 4°C density decreases i.e. volume increases. In other words, it expands so it overflows in both the cases.



23. (a)

$$\gamma = \frac{\Delta V}{V \cdot \Delta T} = \frac{0.24}{100 \times 40} = 6 \times 10^{-5} / ^\circ\text{C}$$

$$\Rightarrow \alpha = \frac{\gamma}{3} = 2 \times 10^{-5} / ^\circ\text{C}$$

24. (a) As $\alpha = \frac{\beta}{2} = \frac{\gamma}{3} \Rightarrow \alpha : \beta : \gamma = 1 : 2 : 3$

25. (a) $\gamma_{\text{app.}} = \frac{\text{Mass expelled}}{\text{Mass remained} \times \Delta T}$
 $= \frac{x/100}{x \times 80} = \frac{1}{8000} = 1.25 \times 10^{-4} / ^\circ\text{C}$

26. (b) In anomalous expansion, water contracts on heating and expands on cooling in the range 0°C to 4°C . Therefore water pipes sometimes burst, in cold countries.

27. (c) On heating the system; x, r, d all increases, since the expansion of isotropic solids is similar to true photographic enlargement.

28. (d) $\alpha = \frac{\Delta L}{L_0 \times \Delta \theta} = \frac{0.01}{5 \times 100} = 2 \times 10^{-5} / ^\circ\text{C}$

29. (a) $\alpha = \frac{\Delta L}{L_0(\Delta \theta)} = \frac{0.19}{100(100-0)} = 1.9 \times 10^{-5} / ^\circ\text{C}$

$$\text{Now } \gamma = 3\alpha = 3 \times 1.9 \times 10^{-5} / ^\circ\text{C} = 5.7 \times 10^{-5} / ^\circ\text{C}$$

30. (d) Since, the coefficient of linear expansion of brass is greater than that of steel. On cooling, the brass contracts more, so, it get loosened.

31. (b) Increase in length $\Delta L = L_0 \alpha \Delta \theta$
 $= 10 \times 10 \times 10^{-5} \times (100 - 0) = 10^{-3} \text{ m} = 1 \text{ mm}$

32. (a) $\alpha = \frac{\Delta L}{L_0 \Delta \theta} = \frac{(1 - 0.9997)}{0.9997 \times 12 \times 10^{-6}} = 25^\circ\text{C}$

33. (c) The densest layer of water will be at bottom. The density of water is maximum at 4°C . So the temperature of bottom of lake will be 4°C .

34. (c) Given $\Delta l_1 = \Delta l_2$ or $l_1 \alpha_a t = l_2 \alpha_s t$
 $\therefore \frac{l_1}{l_2} = \frac{\alpha_s}{\alpha_a}$ or $\frac{l_1}{l_1 + l_2} = \frac{\alpha_s}{\alpha_a + \alpha_s}$

Calorimetry

- (b) In vapor to liquid phase transition, heat liberates.
- (b) Pressure inside the mines is greater than that of normal. Pressure. Also we know that boiling point increases with increase in pressure.

3. (c) $Q = m.c.\Delta\theta \Rightarrow c = \frac{Q}{m.\Delta\theta}$; when $\Delta\theta = 0 \Rightarrow c = \infty$

4. (c) Mass and volume of the gas will remain same, so density will also remain same.

5. (d)

6. (a) The latent heat of vaporization is always greater than latent heat of fusion because in liquid to vapour phase change there is a large increase in volume. Hence more heat is required as compared to solid to liquid phase change.

7. (c) When state is not changing $\Delta Q = mc\Delta\theta$.

8. (a) Heat taken by ice to melt at 0°C is

$$Q_1 = mL = 540 \times 80 = 43200 \text{ cal}$$

Heat given by water to cool upto 0°C is

$$Q_2 = ms\Delta\theta = 540 \times 1 \times (80 - 0) = 43200 \text{ cal}$$

Hence heat given by water is just sufficient to melt the whole ice and final temperature of mixture is 0°C .

Short trick : For these type of frequently asked questions you can remember the following formula

$$\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{c_w}}{m_i + m_w} \quad (\text{See theory for more details})$$

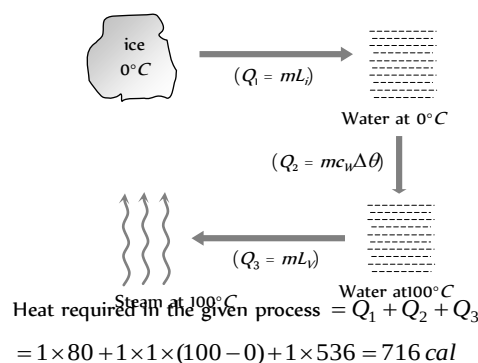
If $m_w = m_i$ then $\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{80 - \frac{80}{1}}{2} = 0^\circ\text{C}$

9. (d) Due to large specific heat of water, it releases large heat with very small temperature change.

10. (a) $Q = m.c.\Delta\theta = 5 \times (1000 \times 4.2) \times (100 - 20)$
 $= 1680 \times 10^3 \text{ J} = 1680 \text{ kJ}$

11. (b) Melting point of ice decreases with increase in pressure (as ice expands on solidification).

12. (c) Conversion of ice (0°C) into steam (100°C) is as follows



13. (a) If $m \text{ gm}$ ice melts then

Heat lost = Heat gain

$$80 \times 1 \times (30 - 0) = m \times 80 \Rightarrow m = 30 \text{ gm}$$

14. (c) At boiling point saturation vapour pressure becomes equal to atmospheric pressure. Therefore, at 100°C for water. S.V.P. = 760 mm of Hg (atm pressure).

15. (b) Thermal capacity = Mass $\times$ Specific heat

Due to same material both spheres will have same specific heat.

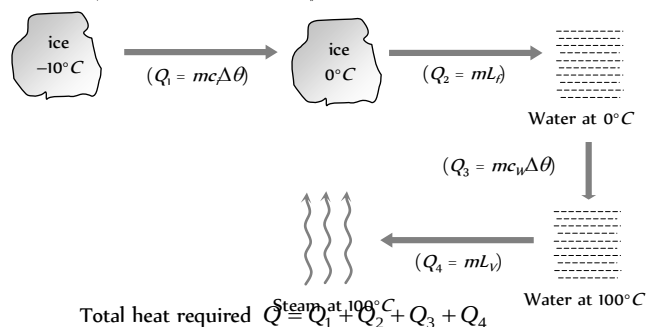
Also mass = Volume (V) $\times$ Density (ρ)

$\therefore$ Ratio of thermal capacity

$$= \frac{m_1}{m_2} = \frac{V_1 \rho}{V_2 \rho} = \frac{\frac{4}{3} \pi r_1^3}{\frac{4}{3} \pi r_2^3} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1}{2}\right)^3 = 1 : 8$$

16. (a) Ice (-10°C) converts into steam as follows

($c =$ Specific heat of ice, $c_s =$ Specific heat of water)



$$\Rightarrow Q = 1 \times 0.5(10) + 1 \times 80 + 1 \times 1 \times (100 - 0) + 1 \times 540$$

$$= 725 \text{ cal}$$

Hence work done $W = JQ = 4.2 \times 725 = 3045 \text{ J}$

17. (b) When water is cooled at 0°C to form ice then 80 cal/gm (latent heat) energy is released. Because potential energy of the molecules decreases. Mass will remain constant in the process of freezing of water.

18. (a) Steam at 100°C contains extra 540 cal/gm energy as compare to water at 100°C . So it's more dangerous to burn with steam than water.

19. (a) Same amount of heat is supplied to copper and water so $m_c c_c \Delta\theta_c = m_w c_w \Delta\theta_w$

$$\Rightarrow \Delta\theta_w = \frac{m_c c_c (\Delta\theta)_c}{m_w c_w} = \frac{50 \times 10^{-3} \times 420 \times 10}{10 \times 10^{-3} \times 4200} = 5^\circ\text{C}$$

20. (c) Temperature of mixture $\theta_{\text{mix}} = \frac{\theta_A c_A + \theta_B c_B}{c_A + c_B}$

$$\Rightarrow 28 = \frac{32 \times c_A + 24 \times c_B}{c_A + c_B}$$

$$\Rightarrow 28 c_A + 28 c_B = 32 c_A + 24 c_B \Rightarrow \frac{c_A}{c_B} = \frac{1}{1}$$

21. (b) Heat lost by hot water = Heat gained by cold water in beaker + Heat absorbed by beaker

$$\Rightarrow 440(92 - \theta) = 200 \times (\theta - 20) + 20 \times (\theta - 20)$$

$$\Rightarrow \theta = 68^\circ\text{C}$$

22. (b) $Q = m.c.\Delta\theta$; if $\Delta\theta = 1 \text{ K}$ then $Q = mc = \text{Thermal capacity}$.

23. (a) Latent heat is independent of configuration. Ordered energy spent in stretching the spring will not contribute to heat which is disordered kinetic energy of molecules of substance.

24. (d) Temperature of mixture

$$\theta_{\text{mix}} = \frac{m_1 c_1 \theta_1 + m_2 c_2 \theta_2}{m_1 c_1 + m_2 c_2} = \frac{m \times c \times 2T + \frac{m}{2} (2c)T}{m.c + \frac{m}{2} (2c)} = \frac{3}{2} T$$

25. (a)

$$\theta_w - \frac{L_i}{c_w} = \frac{100 - 80}{2}$$

26. (a) $\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{100 - 80}{2} = 10^\circ\text{C}$

27. (b) When pressure decreases, boiling point also decreases.

28. (a) Boiling occurs when the vapour pressure of liquid becomes equal to the atmospheric pressure. At the surface of moon, atmospheric pressure is zero, hence boiling point decreases and water begins to boil at 30°C .

29. (d) Thermal capacity $= mc = 40 \times 0.2 = 8 \text{ cal}/^\circ\text{C}$.

30. (b) $Q = m.c.\Delta\theta \Rightarrow c = \frac{Q}{m.\Delta\theta}$

In temperature measurement scale $\Delta\theta^\circ\text{F} > \Delta\theta^\circ\text{C}$ so $(c)_F < (c)_C$.

31. (a) Increasing pressure lowers melting point of ice.

32. (b) Work done changes into heat energy, when the temperature of palm becomes above the atmosphere so it starts losing heat to the surroundings.

33. (b) Firstly the temperature of bullet rises up to melting point, then it melts. Hence according to $W = JQ$.

$$\Rightarrow \frac{1}{2} mv^2 = J[m.c.\Delta\theta + mL] = J[m S (475 - 25) + mL]$$

$$\Rightarrow mS(475 - 25) + mL = \frac{mv^2}{2J}$$

34. (a) As $W = JQ \Rightarrow \frac{1}{2} (mgh) = J \times mc\Delta\theta \Rightarrow \Delta\theta = \frac{gh}{2Jc}$

$$\Delta\theta = \frac{9.8 \times 84}{2 \times 4.2 \times 1000} = 0.098^\circ\text{C}$$

$$(\because c_{\text{water}} = 1000 \frac{\text{cal}}{\text{kg} \times ^\circ\text{C}})$$

Short trick : Remember the value of $\frac{g}{Jc_w} = 0.0023$, here

$$\Delta\theta = \frac{1}{2} \times (0.0023)h = \frac{1}{2} \times 0.0023 \times 84 = 0.098^\circ\text{C}$$

35. (a) $W = JQ \Rightarrow mgh = J \times Q$

$$\Rightarrow Q = \frac{mgh}{J} = \frac{5 \times 9.8 \times 30}{4.2} = 350 \text{ cal}$$

36. (b) $W = JQ = 4.18 \times 400 = 1672 \text{ joule}$

37. (c) Energy supplied $= 0.93 \times 3600 \text{ joules} = 3348 \text{ joules}$

Heat required to melt 10 gms of ice

$$= 10 \times 80 \times 4.18 = 3344 \text{ joules}$$

Hence block of ice just melts.

38. (b) Suppose person climbs upto height h , then by using

$$W = JQ \Rightarrow mgh = JQ$$

$$\Rightarrow 60 \times 9.8 \times h = 4.2 \times \left(10^5 \times \frac{28}{100}\right) \Rightarrow h = 200 \text{ m}$$

39. (a) When water falls from a height, loss of potential energy causes rise in temperature.

40. (a) $W = JQ \Rightarrow mgh = J(m.c.\Delta\theta)$

$$\Rightarrow \Delta\theta = \frac{gh}{Jc} = 0.0023 h = 0.0023 \times 84 = 0.196^\circ\text{C}$$

41. (a) Suppose m' kg ice melts out of m kg then by using

$$W = JQ \Rightarrow mgh = J(m'L). \text{ Hence fraction of ice melts}$$

$$= \frac{m'}{m} = \frac{gh}{JL} = \frac{9.8 \times 1000}{4.18 \times 80} = \frac{1}{33}$$

42. (b) $J = \frac{W}{Q} = \frac{\text{Joule}}{\text{cal}}$

43. (d) $W = JQ \Rightarrow (2m)gh = J \times m'c\Delta\theta$

$$\Rightarrow 2 \times 5 \times 10 \times 10 = 4.2(2 \times 1000 \times \Delta\theta)$$

$$\Rightarrow \Delta\theta = 0.1190^\circ\text{C} = 0.12^\circ\text{C}$$

44. (b) $W = JQ \Rightarrow \frac{1}{2} \left(\frac{1}{2} mV^2\right) = J \times mS\Delta\theta \Rightarrow \Delta\theta = \frac{V^2}{4JS}$

45. (c) J is a conversion

46. (c) $\Delta\theta = 0.0023 h = 0.0023 \times 210 = 0.483^\circ\text{C} \approx 0.49^\circ\text{C}$.

47. (a) According to energy conservation, change in kinetic energy appears in the form of heat (thermal energy).

$$\Rightarrow \text{i.e. Thermal energy} = \frac{1}{2} m(v_1^2 - v_2^2) \quad \left[\because \underset{\text{(Joule)}}{W} = \underset{\text{(Joule)}}{Q} \right]$$

$$= \frac{1}{2} (100 \times 10^{-3}) (10^2 - 5^2) = 3.75 \text{ J}$$

48. (b) Work done to raise the temperature of 100 gm water through 10°C is

$$W = JQ = 4.2 \times (100 \times 10^{-3} \times 1000 \times 10) = 4200 \text{ J}$$

49. (b) Among all the option, latent heat of steam is highest.

50. (a) $\Delta\theta = 0.0023 h = 0.0023 \times 100 = 0.23^\circ\text{C}$

51. (c) Since specific heat of lead is given in *Joules*, hence use $W = Q$ instead of $W = JQ$.

$$\Rightarrow \frac{1}{2} \times \left(\frac{1}{2} m v^2 \right) = m.c.\Delta\theta \Rightarrow \Delta\theta = \frac{v^2}{4c} = \frac{(300)^2}{4 \times 150} = 150^\circ\text{C}.$$

52. (d) At boiling point, vapour pressure becomes equal to the external pressure.

53. (c) When pressure increases boiling point also increases.

54. (b) Calorimeters are made by conducting materials.

55. (b) Triple point of water is 273.16 K .

56. (a)

57. (d) $W = JQ \Rightarrow W = 4.2 \times 200 = 840 \text{ J}.$

58. (d) Temperature of mixture $\theta = \frac{m_1 c_1 \theta_1 + m_2 c_2 \theta_2}{m_1 c_1 + m_2 c_2}$

$$\Rightarrow 32 = \frac{m_1 \times 0.2 \times 40 + 100 \times 0.5 \times 20}{m_1 \times 0.2 + 100 \times 0.5} \Rightarrow m_1 = 375 \text{ gm}$$

59. (d) Suppose $m \text{ gm}$ ice melted, then heat required for its melting $= mL = m \times 80 \text{ cal}$

Heat available with steam for being condensed and then brought to 0°C

$$= 1 \times 540 + 1 \times 1 \times (100 - 0) = 640 \text{ cal}$$

$$\Rightarrow \text{Heat lost} = \text{Heat taken}$$

$$\Rightarrow 640 = m \times 80 \Rightarrow m = 8 \text{ gm}$$

Short trick: You can remember that amount of steam (m') at

100°C required to melt $m \text{ gm}$ ice at 0°C is $m' = \frac{m}{8}$.

$$\text{Here, } m = 8 \times m' = 8 \times 1 = 8 \text{ gm}$$

60. (b) For water and ice mixing $\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{c_w}}{m_i + m_w}$

$$= \frac{20 \times 40 - \frac{5 \times 80}{1}}{5 + 20} = 16^\circ\text{C}$$

61. (c) $\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{c_w}}{m_i + m_w}$

$$\therefore m_i = m_w \Rightarrow \theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{80 - \frac{336}{4.2}}{2} = 0^\circ\text{C}$$

62. (a) When the relative humidity is low (approx. 25%), the evaporation from our body is faster. Thus we feel colder.

63. (a) $c = \frac{Q}{m.\Delta\theta} \rightarrow \frac{J}{\text{kg} \times ^\circ\text{C}}$

64. (a) $\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{80 - \frac{80}{1}}{2} = 0$

65. (a) Freezing point of water decreases when pressure increases, because water expands on solidification while "except water" for other liquid freezing point increases with increase in pressure.

Since the liquid in question is water. Hence, it expands on freezing.

66. (c) Partial pressure of water vapour $P_w = 0.012 \times 10^5 \text{ Pa}$,

$$\text{Vapour pressure of water } P_v = 0.016 \times 10^5 \text{ Pa}.$$

The relative humidity at a given temperature is given by

$$= \frac{\text{Partial pressure of water vapour}}{\text{Vapour pressure of water}}$$

$$= \frac{0.012 \times 10^5}{0.016 \times 10^5} = 0.75 = 75\%$$

67. (a) $W = JQ \Rightarrow \frac{1}{2} \left(\frac{1}{2} M v^2 \right) = J(m.c.\Delta\theta)$

$$\Rightarrow \frac{1}{4} \times 1 \times (50)^2 = 4.2[200 \times 0.105 \times \Delta\theta] \Rightarrow \Delta\theta = 7.1^\circ\text{C}$$

68. (a) $536 \frac{\text{cal}}{\text{gm}} = \frac{536 \times 4.2 \text{ J}}{10^{-3} \text{ kg}} = 2.25 \times 10^6 \text{ J/kg}$

69. (a) Water has maximum specific heat.

70. (a) $\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{100 - \frac{80}{1}}{2} = 10^\circ\text{C}$

71. (b) Suppose $m \text{ kg}$ of ice melts then by using $\underset{\text{(Joules)}}{W} = \underset{\text{(Joules)}}{H}$

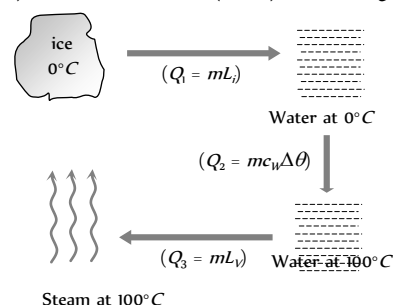
$$\Rightarrow Mgh = mL \Rightarrow 3.5 \times 10 \times 2000 = m \times 3.5 \times 10^5$$

$$\Rightarrow m = 0.2 \text{ kg} = 200 \text{ gm}$$

72. (d) $\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{S_w}}{m_i + m_w} = \frac{300 \times 25 - \frac{100 \times 80}{1}}{100 + 300} = -1.25^\circ\text{C}$

Which is not possible. Hence $\theta_{\text{mix}} = 0^\circ\text{C}$

73. (c) Ice (0°C) converts into steam (100°C) in following three steps.



Total heat required $Q = Q_1 + Q_2 + Q_3$

$$= 5 \times 80 + 5 \times 1 \times (100 - 0) + 5 \times 540 = 3600 \text{ cal}$$

74. (a) Let L be the latent heat and using principle of calorimetry.

$$2L + 2(100 - 54.3) = 40 \times (54.3 - 25.3)$$

$$\Rightarrow L = 540.3 \text{ cal/gm.}$$

75. (d) $\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{c_w}}{m_i + m_w} = \frac{100 \times 50 - 10 \times \frac{80}{1}}{10 + 100} \approx 38.2^\circ \text{C}$

76. (b) Let the final temperature be $T^\circ \text{C}$.

Total heat supplied by the three liquids in coming down to 0°C =

$$m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3 \quad \dots (i)$$

Total heat used by three liquids in raising temperature from 0°C to $T^\circ \text{C}$

$$= m_1 c_1 T + m_2 c_2 T + m_3 c_3 T \quad \dots (ii)$$

By equating (i) and (ii) we get

$$(m_1 c_1 + m_2 c_2 + m_3 c_3) T$$

$$= m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3$$

$$\Rightarrow T = \frac{m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3}{m_1 c_1 + m_2 c_2 + m_3 c_3}$$

77. (c) At triple point all the phases co-exist

78. (b) $c = \frac{Q}{m \Delta \theta}$; as $\Delta \theta = 0$, hence c becomes ∞ .

79. (d) Let final temperature of water be θ

Heat taken = Heat given

$$110 \times 1 (\theta - 10) + 10 (\theta - 10) = 220 \times 1 (70 - \theta)$$

$$\Rightarrow \theta = 48.8^\circ \text{C} \approx 50^\circ \text{C}$$

80. (c) We know that thermal capacity of a body expressed in calories is equal to water equivalent of the body expressed in grams.

81. (b) $\theta_{\text{mix}} = \frac{m_1 c_1 \theta_1 + m_2 c_2 \theta_2}{m_1 c_1 + m_2 c_2} = \frac{m s (2t) + 1.5 (m s) \times \frac{t}{3}}{m s + 1.5 (m s)} = t$

82. (d) We know that when solid carbon dioxide is heated, it becomes vapour directly without passing through its liquid phase. Therefore it is called dry ice.

Critical Thinking Questions

1. (c) Due to volume expansion of both mercury and flask, the change in volume of mercury relative to flask is given by

$$\Delta V = V_0 [\gamma_L - \gamma_g] \Delta \theta = V [\gamma_m - 3\alpha_g] \Delta \theta$$

$$= 50 [180 \times 10^{-6} - 3 \times 9 \times 10^{-6}] (38 - 18) = 0.153 \text{ cc}$$

2. (a) $\gamma_{\text{net}} = \gamma_{\text{net}} + \gamma_{\text{net}}$

$$\text{So } (\gamma_{\text{net}} + \gamma_{\text{net}})_{\text{net}} = (\gamma_{\text{net}} + \gamma_{\text{net}})_{\text{net}}$$

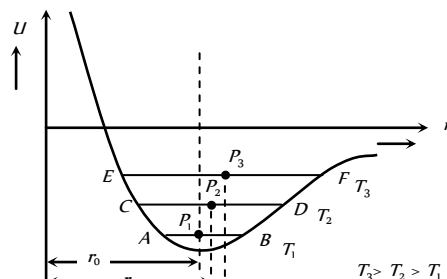
$$\Rightarrow 153 \times 10^{-6} + (\gamma_{\text{net}})_{\text{net}} = (144 \times 10^{-6} + \gamma_{\text{net}})_{\text{net}}$$

$$\text{Further, } (\gamma_{\text{net}})_{\text{net}} = 3\alpha = 3 \times (12 \times 10^{-6}) = 36 \times 10^{-6} \text{ } ^\circ \text{C}$$

$$\Rightarrow 153 \times 10^{-6} + (\gamma_{\text{net}})_{\text{net}} = 144 \times 10^{-6} + 36 \times 10^{-6}$$

$$\Rightarrow (\gamma_{\text{net}})_{\text{net}} = 3\alpha = 27 \times 10^{-6} \text{ } ^\circ \text{C} \Rightarrow \alpha = 9 \times 10^{-6} \text{ } ^\circ \text{C}$$

3. (d) The expansion of solids can be well understood by potential energy curve for two adjacent atoms in a crystalline solid as a function of their internuclear separation (r).



At ordinary temperature : Each molecule of the solid vibrates about its equilibrium position P between A and B (r_1 is the equilibrium distance of it from some other molecule).

At high temperature : Amplitude of vibration increases ($C \leftrightarrow D$ and $E \leftrightarrow F$). Due to asymmetry of the curve, the equilibrium positions (P_1 and P_2) of molecule displaced. Hence its distance from other molecule increases ($r_2 > r_1 > r_0$).

Thus, on raising the temperature, the average equilibrium distance between the molecules increases and the solid as a whole expands.

4. (c) Initial diameter of tyre = $(1000 - 6) \text{ mm} = 994 \text{ mm}$, so initial radius of tyre $R = \frac{994}{2} = 497 \text{ mm}$

$$\text{and change in diameter } \Delta D = 6 \text{ mm so } \Delta R = \frac{6}{2} = 3 \text{ mm}$$

After increasing temperature by $\Delta \theta$ tyre will fit onto wheel
Increment in the length (circumference) of the iron tyre

$$\Delta L = L \times \alpha \times \Delta \theta = L \times \frac{\gamma}{3} \times \Delta \theta \quad [\text{As } \alpha = \frac{\gamma}{3}]$$

$$2\pi \Delta R = 2\pi R \left(\frac{\gamma}{3} \right) \Delta \theta \Rightarrow \Delta \theta = \frac{3}{\gamma} \frac{\Delta R}{R} = \frac{3 \times 3}{3.6 \times 10^{-5} \times 497}$$

$$\Rightarrow \Delta \theta = 500^\circ \text{C}$$

5. (b) Due to volume expansion of both liquid and vessel, the change in volume of liquid relative to container is given by $\Delta V = V_0 [\gamma_L - \gamma_g] \Delta \theta$

$$\text{Given } V_0 = 1000 \text{ cc, } \alpha_g = 0.1 \times 10^{-4} \text{ } ^\circ \text{C}$$

$$\therefore \gamma_g = 3\alpha_g = 3 \times 0.1 \times 10^{-4} \text{ } ^\circ \text{C} = 0.3 \times 10^{-4} \text{ } ^\circ \text{C}$$

$$\therefore \Delta V = 1000 [1.82 \times 10^{-4} - 0.3 \times 10^{-4}] \times 100 = 15.2 \text{ cc}$$

6. (a) With temperature rise (same 25°C for both), steel scale and copper wire both expand. Hence length of copper wire w.r.t. steel scale or apparent length of copper wire after rise in temperature

$$L_{\text{app}} = L'_{\text{cu}} - L'_{\text{steel}} = [L_0 (1 + \alpha_{\text{Cu}} \Delta \theta) - L_0 (1 + \alpha_{\text{S}} \Delta \theta)]$$

$$\Rightarrow L_{\text{app}} = L_0 (\alpha_{\text{Cu}} - \alpha_{\text{S}}) \Delta \theta$$

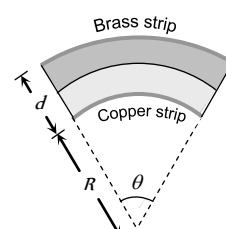
$$= 80 (17 \times 10^{-6} - 11 \times 10^{-6}) \times 20 = 80.0096 \text{ cm}$$

7. (b, d) Let L be the initial length of each strip before heating.

Length after heating will be

$$L_B = L_0 (1 + \alpha_B \Delta T) = (R + d) \theta$$

$$L_C = L_0 (1 + \alpha_C \Delta T) = R \theta$$



$$\Rightarrow \frac{R+d}{R} = \frac{1+\alpha_B \Delta T}{1+\alpha_C \Delta T}$$

$$\Rightarrow 1 + \frac{d}{R} = 1 + (\alpha_B - \alpha_C) \Delta T$$

$$\Rightarrow R = \frac{d}{(\alpha_B - \alpha_C) \Delta T} \Rightarrow R \propto \frac{1}{\Delta T} \text{ and } R \propto \frac{1}{(\alpha_B - \alpha_C)}$$

8. (d) Thermostat is used in electric apparatus like refrigerator, Iron etc for automatic cut off. Therefore for metallic strips to bend on heating their coefficient of linear expansion should be different.

9. (c) As the coefficient of cubical expansion of metal is less as compared to the coefficient of cubical expansion of liquid, we may neglect the expansion of metal ball. So when the ball is immersed in alcohol at 0°C , it displaces some volume V of alcohol at 0°C and has weight W .

$$\therefore W_i = W_a - V\rho_g$$

where W_i = weight of ball in air

$$\text{Similarly, } W_s = W_a - V\rho_s$$

where ρ_s = density of alcohol at 0°C

and ρ_g = density of alcohol at 50°C

As $\rho_s < \rho_g$, $\Rightarrow W_i > W_s$ or $W_i < W_s$

10. (d) $V = V_0(1 + \gamma \Delta\theta) \Rightarrow$ Change in volume

$$V - V_0 = \Delta V = A \Delta l = V_0 \gamma \Delta\theta$$

$$\Rightarrow \Delta l = \frac{V_0 \Delta\theta}{A} = \frac{10^{-6} \times 18 \times 10^{-5} \times (100 - 0)}{0.004 \times 10^{-4}}$$

$$= 45 \times 10^{-7} \text{ m} = 4.5 \text{ cm}$$

11. (b) Loss of weight at 27°C is

$$= 46 - 30 = 16 = V_1 \times 1.24 \rho_s \times g \quad \dots(i)$$

Loss of weight at 42°C is

$$= 46 - 30.5 = 15.5 = V_2 \times 1.2 \rho_s \times g \quad \dots(ii)$$

$$\text{Now dividing (i) by (ii), we get } \frac{16}{15.5} = \frac{V_1}{V_2} \times \frac{1.24}{1.2}$$

$$\text{But } \frac{V_2}{V_1} = 1 + 3\alpha(t - t_0) = \frac{15.5 \times 1.24}{16 \times 1.2} = 1.001042$$

$$\Rightarrow 3\alpha(42^\circ - 27^\circ) = 0.001042 \Rightarrow \alpha = 2.316 \times 10^{-6} / ^\circ\text{C}$$

12. (b) Substances are classified into two categories

(i) water like substances which expand on solidification.

(ii) CO_2 like (Wax, Ghee etc.) which contract on solidification.

Their behaviour regarding solidification is opposite.

Melting point of ice decreases with rise of temp but that of wax etc increases with increase in temperature. Similarly ice starts forming from top downwards whereas wax starts its formation from bottom.

13. (b) Heat lost in t sec = mL or heat lost per sec = $\frac{mL}{t}$. This must be the heat supplied for keeping the substance in molten state per sec.

$$\therefore \frac{mL}{t} = P \quad \text{or} \quad L = \frac{Pt}{m}$$

14. (a) Heat is lost by steam in two stages (i) for change of state from steam at 100°C to water at 100°C is $m \times 540$ (ii) to change water at 100°C to water at 80°C is $m \times 1 \times (100 - 80)$, where m is the mass of steam condensed.

Total heat lost by steam is $m \times 540 + m \times 20 = 560 \text{ m (cal)}$

Heat gained by calorimeter and its contents is

$$= (1.1 + 0.02) \times (80 - 15) = 1.12 \times 65 \text{ cal}$$

using Principle of calorimetry, Heat gained = heat lost

$$\therefore 560 \text{ m} = 1.12 \times 65, \text{ m} = 0.130 \text{ gm}$$

15. (b) Initially ice will absorb heat to raise its temperature to 0°C then its melting takes place

If m = Initial mass of ice, m_i = Mass of ice that melts and m_w = Initial mass of water

By Law of mixture Heat gained by ice = Heat lost by water $\Rightarrow$

$$m_i \times c \times (20) + m_i \times L = m_w c_w [20]$$

$$\Rightarrow 2 \times 0.5(20) + m_i \times 80 = 5 \times 1 \times 20 \Rightarrow m_i = 1 \text{ kg}$$

So final mass of water = Initial mass of water + Mass of ice that melts = $5 + 1 = 6 \text{ kg}$.

16. (a) Heat gained by the water = (Heat supplied by the coil) - (Heat dissipated to environment)

$$\Rightarrow mc \Delta\theta = P_{\text{Coil}} t - P_{\text{Loss}} t$$

$$\Rightarrow 2 \times 4.2 \times 10^3 \times (77 - 27) = 1000 t - 160 t$$

$$\Rightarrow t = \frac{4.2 \times 10^5}{840} = 500 \text{ sec} = 8 \text{ min } 20 \text{ sec}$$

17. (a) If mass of the bullet is $m \text{ gm}$,

then total heat required for bullet to just melt down

$$Q = m c \Delta\theta + m L = m \times 0.03 (327 - 27) + m \times 6$$

$$= 15 \text{ m cal} = (15m \times 4.2) \text{ J}$$

Now when bullet is stopped by the obstacle, the loss in its

$$\text{mechanical energy} = \frac{1}{2} (m \times 10^{-3}) v^2 \text{ J}$$

$$(\text{As } m \text{ gm} = m \times 10^{-3} \text{ kg})$$

As 25% of this energy is absorbed by the obstacle,

The energy absorbed by the bullet

$$Q_2 = \frac{75}{100} \times \frac{1}{2} m v^2 \times 10^{-3} = \frac{3}{8} m v^2 \times 10^{-3} \text{ J}$$

Now the bullet will melt if $Q_2 \geq Q_1$

$$\text{i.e. } \frac{3}{8} m v^2 \times 10^{-3} \geq 15m \times 4.2 \Rightarrow v_{\min} = 410 \text{ m/s}$$

18. (c) Energy = $\frac{1}{2} m v^2 = mc \Delta\theta \Rightarrow \Delta\theta \propto v^2$

Temperature does not depend upon the mass of the balls.

19. (c) Heat gain = heat lost

$$C_A(16 - 12) = C_B(19 - 16) \Rightarrow \frac{C_A}{C_B} = \frac{3}{4}$$

$$\text{and } C_A(23 - 19) = C_B(28 - 23) \Rightarrow \frac{C_B}{C_C} = \frac{5}{4}$$

$$\Rightarrow \frac{C_A}{C_C} = \frac{15}{16} \quad \dots(i)$$

If θ is the temperature when A and C are mixed then,

$$C_A(\theta - 12) = C_C(28 - \theta) \Rightarrow \frac{C_A}{C_C} = \frac{28 - \theta}{\theta - 12} \quad \dots(ii)$$

On solving equation (i) and (ii) $\theta = 20.2^\circ\text{C}$.

20. (b) Suppose m kg steam required per hour
Heat released by steam in following three steps

(i) When 150°C steam $\xrightarrow{Q_1}$ 100°C steam

$$Q_1 = mc_{\text{steam}} \Delta\theta = m \times 1 (150 - 100) = 50 m \text{ cal}$$

(ii) When 150°C steam $\xrightarrow{Q_2}$ 100°C water

$$Q_2 = mL_v = m \times 540 = 540 m \text{ cal}$$

(iii) When 100°C water $\xrightarrow{Q_3}$ 90°C water

$$Q_3 = mc_w \Delta\theta = m \times 1 \times (100 - 90) = 10 m \text{ cal}$$

Hence total heat given by the steam $Q = Q_1 + Q_2 + Q_3 = 600 m \text{ cal}$
... (i)

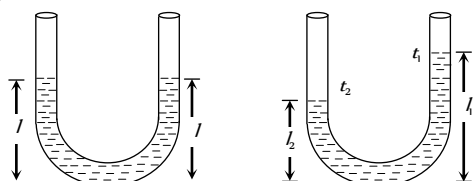
Heat taken by 10 kg water

$$Q' = mc_w \Delta\theta = 10 \times 10^3 \times 1 \times (80 - 20) = 600 \times 10^3 \text{ cal}$$

Hence $Q = Q' \Rightarrow 600 m = 600 \times 10$

$$\Rightarrow m = 10 \text{ gm} = 1 \text{ kg.}$$

21. (a) Suppose, height of liquid in each arm before rising the temperature is l .



With temperature rise height of liquid in each arm increases i.e. $l_1 > l$ and $l_2 > l$

$$\text{Also } l = \frac{l_1}{1 + \gamma t_1} = \frac{l_2}{1 + \gamma t_2}$$

$$\Rightarrow l_1 + \gamma l_1 t_2 = l_2 + \gamma l_2 t_1 \Rightarrow \gamma = \frac{l_1 - l_2}{l_2 t_1 - l_1 t_2}$$

22. (c) $V = V_0(1 + \gamma \Delta\theta)$

$$L^3 = L_0(1 + \alpha_1 \Delta\theta)L_0^2(1 + \alpha_2 \Delta\theta)^2 = L_0^3(1 + \alpha_1 \Delta\theta)(1 + \alpha_2 \Delta\theta)^2$$

Since $L_0^3 = V_0$ and $L^3 = V$

$$\text{Hence } 1 + \gamma \Delta\theta = (1 + \alpha_1 \Delta\theta)(1 + \alpha_2 \Delta\theta)^2$$

$$\cong (1 + \alpha_1 \Delta\theta)(1 + 2\alpha_2 \Delta\theta) \cong (1 + \alpha_1 \Delta\theta + 2\alpha_2 \Delta\theta)$$

$$\Rightarrow \gamma = \alpha_1 + 2\alpha_2$$

23. (d) $(OR)^2 = (PR)^2 - (PO)^2 = l^2 - \left(\frac{l}{2}\right)^2$
 $= [l(1 + \alpha_2 t)]^2 - \left[\frac{l}{2}(1 + \alpha_1 t)\right]^2$

$$l^2 - \frac{l^2}{4} = l^2(1 + \alpha_2^2 t^2 + 2\alpha_2 t) - \frac{l^2}{4}(1 + \alpha_1^2 t^2 + 2\alpha_1 t)$$

Neglecting $\alpha_2^2 t^2$ and $\alpha_1^2 t^2$

$$0 = l^2(2\alpha_2 t) - \frac{l^2}{4}(2\alpha_1 t) \Rightarrow 2\alpha_2 = \frac{2\alpha_1}{4} \Rightarrow \alpha_1 = 4\alpha_2$$

24. (c) It is given that the volume of air in the flask remains the same. This means that the expansion in volume of the vessel is exactly equal to the volume expansion of mercury.

$$\text{i.e., } \Delta V_g = \Delta V_L \text{ or } V_g \gamma_g \Delta\theta = V_L \gamma_L \Delta\theta$$

$$\therefore V_L = \frac{V_g \gamma_g}{\gamma_L} = \frac{1000 \times (3 \times 9 \times 10^{-6})}{1.8 \times 10^{-4}} = 150 \text{ cc}$$

25. (c) Heat given by water $Q_1 = 10 \times 10 = 100 \text{ cal}$.

Heat taken by ice to melt

$$Q_2 = 10 \times 0.5 \times [0 - (-20)] + 10 \times 80 = 900 \text{ cal}$$

As $Q_1 < Q_2$, so ice will not completely melt and final temperature $= 0^\circ\text{C}$.

As heat given by water in cooling up to 0°C is only just sufficient to increase the temperature of ice from -20°C to 0°C , hence mixture in equilibrium will consist of 10 gm ice and 10 gm water at 0°C .

26. (b) $\Delta L = L_0 \alpha \Delta\theta$

$$\text{Rod A : } 0.075 = 20 \times \alpha_A \times 100 \Rightarrow \alpha_A = \frac{75}{2} \times 10^{-6} / ^\circ\text{C}$$

$$\text{rod B : } 0.045 = 20 \times \alpha_B \times 100 \Rightarrow \alpha_B = \frac{45}{2} \times 10^{-6} / ^\circ\text{C}$$

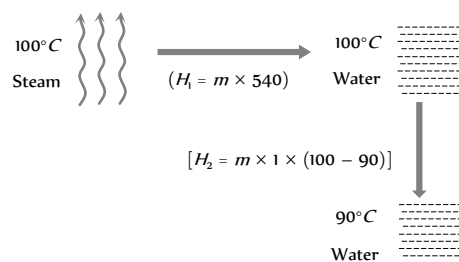
For composite rod : $x \text{ cm}$ of A and $(20 - x) \text{ cm}$ of B we have

$$0.060 = x \alpha_A \times 100 + (20 - x) \alpha_B \times 100$$

$$= x \left[\frac{75}{2} \times 10^{-6} \times 100 + (20 - x) \times \frac{45}{2} \times 10^{-6} \times 100 \right]$$

On solving we get $x = 10 \text{ cm}$.

27. (a) Let $m \text{ gm}$ of steam get condensed into water (By heat loss). This happens in following two steps.



Heat gained by water (20°C) to raise its temperature upto 90°
 $= 22 \times 1 \times (90 - 20)$

Hence, in equilibrium heat lost = Heat gain

$$\Rightarrow m \times 540 + m \times 1 \times (100 - 90) = 22 \times 1 \times (90 - 20)$$

$$\Rightarrow m = 2.8 \text{ gm}$$

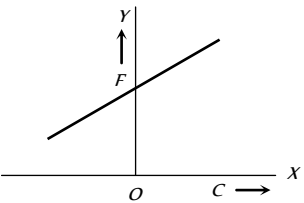
The net mass of the water present in the mixture
 $= 22 + 2.8 = 24.8 \text{ gm}$.

Graphical Questions

- (b) Relation between Celsius and Fahrenheit scale of temperature is

$$\frac{C}{5} = \frac{F - 32}{9} \Rightarrow C = \frac{5}{9}F - \frac{160}{9}$$
 Equating above equation with standard equation of line
 $y = mx + c$ we get slope of the line AB is $m = \frac{5}{9}$
- (c) Since in the region AB temperature is constant therefore at this temperature phase of the material changes from solid to liquid and $(H - H)$ heat will be absorb by the material. This heat is known as the heat of melting of the solid.
 Similarly in the region CD temperature is constant therefore at this temperature phase of the material changes from liquid to gas and $(H - H)$ heat will be absorb by the material. This heat as known as the heat of vaporisation of the liquid.
- (a) Initially, on heating temperature rises from -10°C to 0°C . Then ice melts and temperature does not rise. After the whole ice has melted, temperature begins to rise until it reaches 100°C . Then it becomes constant, as at the boiling point will not rise.
- (a) The volume of matter in portion AB of the curve is almost constant and pressure is decreasing. These are the characteristics of liquid state.
- (d) Let the quantity of heat supplied per minute be Q . Then quantity of heat supplied in $2 \text{ min} = mC(90 - 80)$
 In 4 min , heat supplied $= 2mC(90 - 80)$

$$\therefore 2mC(90 - 80) = mL \Rightarrow \frac{L}{C} = 20$$
- (b) In the given graph CD represents liquid state.
- (a) Density of water is maximum at 4°C and is less on either side of this temperature.
- (a) We know that, $\frac{C}{100} = \frac{F - 32}{180}$ or $F = \frac{9}{5}C + 32$
 Equation of straight line is, $y = mx + c$
 Hence, $m = (9/5)$, positive and $c = 32$ positive. The graph is shown in figure.

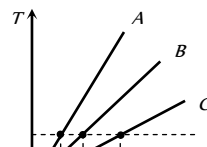

- (a)

$$\frac{C}{5} = \frac{F - 32}{9} \Rightarrow C = \left(\frac{5}{9}\right)F - \frac{20}{3}$$
 Hence graph between $^\circ\text{C}$ and $^\circ\text{F}$ will be a straight line with positive slope and negative intercept.
- (bc) The horizontal parts of the curve, where the system absorbs heat at constant temperature must depict changes of state. Here the latent heats are proportional to lengths of the horizontal parts. In the sloping parts, specific heat capacity is inversely proportional to the slopes.
- (c) Since specific heat $= 0.6 \text{ kcal/gm} \times ^\circ\text{C} = 0.6 \text{ cal/gm} \times ^\circ\text{C}$
 From graph it is clear that in a minute, the temperature is raised from 0°C to 50°C .

$\Rightarrow$ Heat required for a minute $= 50 \times 0.6 \times 50 = 1500 \text{ cal}$

Also from graph, Boiling point of wax is 200°C .

- (c)
- (c) Substances having more specific heat take longer time to get heated to a higher temperature and longer time to get cooled.



If we draw a line parallel to the time axis then it cuts the given graphs at three different points. Corresponding t points on the times axis shows that

$$t_C > t_B > t_A \Rightarrow C_C > C_B > C_A$$

- (c) From given curve,
 Melting point for $A = 60^\circ\text{C}$
 and melting point for $B = 20^\circ\text{C}$
 Time taken by A for fusion $= (6 - 2) = 4 \text{ minute}$
 Time taken by B for fusion $= (6.5 - 4) = 2.5 \text{ minute}$
 Then $\frac{H_A}{H_B} = \frac{6 \times 4 \times 60}{6 \times 2.5 \times 60} = \frac{8}{5}$.

Assertion and Reason

- (a) With rise in pressure melting point of ice decreases. Also ice contracts on melting
- (c) Celsius scale was the first temperature scale and Fahrenheit is the smallest unit measuring temperature.
- (e) Melting is associated with increasing of internal energy without change in temperature. In view of the reason being correct the amount of heat absorbed or given out during change of state is expressed $Q = mL$, where m is the mass of the substance and L is the latent heat of the substance.
- (a) The temperature of land rises rapidly as compared to sea because of specific heat of land is five times less than that of sea water. Thus, the air above the land become hot and light so rises up so because of pressure drops over land. To compensate the drop of pressure, the cooler air starts from sea starts blowing towards lands, so, setting up sea breeze. During night land as well sea radiate heat energy. The temperature of land falls more rapidly as compared to sea water, as sea water consists of higher specific heat capacity. The air above sea water being warm and light rises up and to take its place the cold air from land starts blowing towards sea and so set up breeze.
- (a) Linear expansion for brass $(19 \times 10^{-4}) >$ linear expansion for steel (11×10^{-4}) . On cooling the disk shrinks to a greater extent than the hole and hence it will get loose.
- (a) As, $\gamma = \frac{\Delta V}{V \Delta T}$ i.e., units of coefficient of volume expansion is K .
- (c) The relation between F and C scale is, $\frac{C}{5} = \frac{F - 32}{9}$. If $F = C$
 $\Rightarrow C = -40^\circ\text{C}$ i.e., at -40° the Centigrade and Fahrenheit thermometers reads the same.

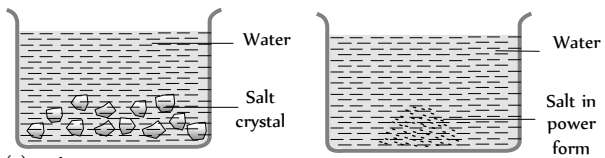
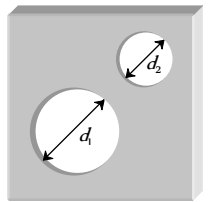
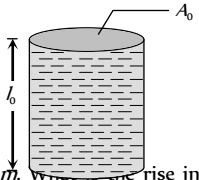
8. (a) As $\beta = 2\alpha$ and $\gamma = 3\alpha$, i.e., coefficient of volume expansion of solid is three times coefficient of linear expansion and 1.5 times the coefficient of superficial expansion, on heating a solid iron ball, percentage increase in its volume is largest.
9. (a) Water has maximum density at 4°C . On heating above 4°C or cooling below 4°C , density of water decreases and its volume increases. Therefore, water overflows in both the cases.
10. (b) The Latent heat of fusion of ice is amount of heat required to convert unit mass of ice at 0°C into water at 0°C . For fusion of ice

$$L = 80 \text{ cal/gm} = 80000 \text{ cal/gm} = 8000 \times 4.2 \text{ J/kg} \\ = 336000 \text{ J/kg}.$$

11. (a) When two bodies at temperature T_1 and T_2 are brought in thermal contact, they do settle to the mean temperature $(T_1 + T_2)/2$. They will do so, in case the two bodies were of same mass and material i.e., same thermal capacities. In other words, the two bodies may be having different thermal capacities, that's why they do not settle to the mean temperature, when brought together.
12. (d) Specific heat of a body is the amount of heat required to raise the temperature of unit mass of the body through unit degree. When mass of a body is less than unity, then its thermal capacity is less than its specific heat and vice-versa.
13. (a) Water would evaporate quickly because there is no atmosphere on moon, due to which surface temperature of moon is much higher than earth (Maximum surface temperature of moon is 123°C).
14. (d) The potential energy of water molecules is more. The heat given to melt the ice at 0°C is used up in increasing the potential energy of water molecules formed at 0°C .

Thermometry, Thermal Expansion
and
Calorimetry

SET Self Evaluation Test -12

- Out of the following, in which vessel will the temperature of the solution be higher after the salt is completely dissolved.
 
 - A
 - B
 - Equal in both
 - Information is not sufficient
- Fire is extinguished more effectively by
 - Hot water
 - Cold water
 - Equally by both
 - Ice
- An ideal thermometer should have
 - Large heat capacity
 - Medium heat capacity
 - Small heat capacity
 - Variable heat capacity
- A steel meter scale is to be ruled so that millimeter intervals are accurate within about $5 \times 10^{-5} \text{ mm}$ at a certain temperature. The maximum temperature variation allowable during the ruling is (Coefficient of linear expansion of steel $= 10 \times 10^{-6} \text{ K}^{-1}$)
 - 2°C
 - 5°C
 - 7°C
 - 10°C
- During illness an 80 kg man ran a fever of 102.2°F instead of normal body temperature of 98.6°F . Assuming that human body is mostly water, how much heat is required to raise his temperature by that amount
 - 100 kcal
 - 160 kcal
 - 50 kcal
 - 92 kcal
- Two holes of unequal diameters d_1 and d_2 ($d_1 > d_2$) are cut in a metal sheet. If the sheet is heated
 
 - Both d_1 and d_2 will decrease
 - Both d_1 and d_2 will increase
 - d_1 will increase, d_2 will decrease
 - d_1 will decrease, d_2 will increase
- If earth suddenly stops rotating about its own axis, the increase in its temperature will be
 - $\frac{R^2 \omega^2}{5Js}$
 - $\frac{R^2 \omega^2}{Js}$
 - $\frac{Rm \omega^2}{5Js}$
 - None of these
- Latent heat of ice is 80 cal/gm . A man melts 60 g of ice by chewing in 1 minute . His power is
 - 4800 W
 - 336 W
 - 1.33 W
 - 0.75 W
- A faulty thermometer has its lower fixed point marked as -10°C and upper fixed point marked as 110° and upper fixed point marked as 110° . If the temperature of the body shown in this scale is 62° , the temperature shown on the Celsius scale is
 - 72°C
 - 82°C
 - 60°C
 - 42°C
- If there are no heat losses, the heat released by the condensation of $x \text{ gm}$ of steam at 100°C into water at 100°C can be used to convert $y \text{ gm}$ of ice at 0°C into water at 100°C . Then the ratio $y : x$ is nearly
 - $1 : 1$
 - $2.5 : 1$
 - $2 : 1$
 - $3 : 1$
- The figure shows a glass tube (linear co-efficient of expansion is α) completely filled with a liquid of volume expansion co-efficient γ . On heating length of the liquid column does not change. Choose the correct relation between γ and α

 - $\gamma = \alpha$
 - $\gamma = 2\alpha$
 - $\gamma = 3\alpha$ [EAMCET 2001]
 - $\gamma = \frac{\alpha}{3}$
- Water falls from a height 500 m . What is the rise in temperature of water at bottom if whole energy remains in the water
 - 0.96°C
 - 1.02°C
 - 1.16°C
 - 0.23°C
- A steel ball of mass 0.1 kg falls freely from a height of 10 m and bounces to a height of 5.4 m from the ground. If the dissipated energy in this process is absorbed by the ball, the rise in its temperature is
(Specific heat of steel $= 460 \text{ Joule} - \text{kg}^{-1} \text{C}^{-1}$, $g = 10 \text{ ms}^{-2}$)
 - 0.01°C
 - 0.1°C
 - 1°C
 - 1.1°C
- 1 gm of ice at 0°C is mixed with 1 gm of water at 100°C the resulting temperature will be [AIIMS 1994]
 - 5°C
 - 0°C
 - 10°C
 - ∞
- The amount of heat required to change 1 gm (0°C) of ice into water of 100°C , is [RPMT 1999]
 - 716 cal
 - 500 cal
 - 180 cal
 - 100 cal

1. (b) When salt crystals dissolves, crystal lattice is destroyed. The process requires a certain amount of energy (latent heat) which is taken from the water.

In vessel (B), a part of intermolecular bonds has already been destroyed in crushing the crystal. Hence less energy is require to dissolve the powder and the water will be at higher temperature.

2. (a) Fire is extinguished by the vaporisation do water which lowers the temperature of the burning body. Further, the water vapour envelops the body, keeping oxygen away. Hot water evaporates more than cold water as
3. (c) The thermometer has to attain the temperature of the body. To do this, it should draw as little heat from the body as possible, so that the existing temperature of the body is not disturbed.

4. (b) As we know $\alpha = \frac{\Delta L}{L_0 \Delta \theta} \Rightarrow \Delta \theta = \frac{\Delta L}{\alpha L_0} = \frac{5 \times 10^{-5}}{10 \times 10^{-6} \times 1} = 5^\circ \text{C}$

5. (b) Since $102.2^\circ \text{F} \rightarrow 39^\circ \text{C}$ and $98.6^\circ \text{F} \rightarrow 37^\circ \text{C}$
Hence $\Delta Q = m \cdot s \cdot \Delta \theta = 80 \times 1000 \times (39 - 37)$
 $= 16 \times 10^5 \text{ cal} = 160 \text{ kcal}$.

6. (b) If the sheet is heated then both d and d' will increase since the thermal expansion of isotropic solid is similar to true photographic enlargement.

7. (a) $W = JQ \Rightarrow \frac{1}{2} I \omega^2 = J(MS \Delta \theta) \Rightarrow \frac{1}{2} \left(\frac{2}{5} MR^2 \right) \omega^2$
 $= J(MS \Delta \theta) \Rightarrow \Delta \theta = \frac{R^2 \omega^2}{5Js}$

8. (b) $W = JQ = J(mL) \Rightarrow P \times t = J(mL) \Rightarrow P = J \left(\frac{m}{t} \right) L$;

where $\frac{m}{t}$ = rate of melting of ice by chewing

$= \frac{60 \text{ gm}}{\text{min}} = \frac{1 \text{ gm}}{\text{sec}} \Rightarrow P = 4.2 \times 1 \times 80 = 336 \text{ W}$.

9. (c) $\frac{X-L}{U-L} = \frac{C}{100} \Rightarrow \frac{62-(10)}{110-(-10)} = \frac{C}{100} \quad (C = 60^\circ \text{C})$

10. (d) Heat released to convert $x \text{ gm}$ of steam at 100°C to water at 100°C is $x \times 540 \text{ cal}$.

If $y \text{ gm}$ of ice is converted from 0°C to water at 100°C it requires heat $y \times 80 + y \times 1 \times 100 = 180 y$

$\therefore x \times 540 = 180 y \text{ or } \frac{y}{x} = \frac{540}{180} = \frac{3}{1}$

11. (b) When length of the liquid column remains constant, then the level of liquid moves down with respect to the container, thus γ must be less than 3α .

Now we can write $V = V_0(1 + \gamma \Delta T)$

Since $V = A l = [A_0(1 + 2\alpha \Delta T)] l = V_0(1 + 2\alpha \Delta T)$

Hence $V_0(1 + \gamma \Delta T) = V_0(1 + 2\alpha \Delta T) \Rightarrow \gamma = 2\alpha$.

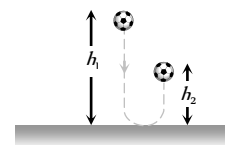
12. (c) By using

$\Delta \theta = 0.0023 h = 0.0023 \times 500 = 1.15^\circ \text{C} \approx 1.16^\circ \text{C}$

13. (b) According to energy conservation, change in potential energy of the ball, appears in the form of heat which raises the temperature of the ball.

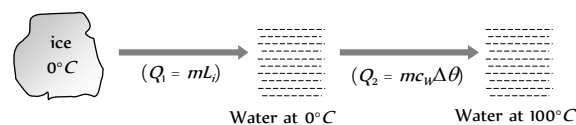
i.e. $mg(h_1 - h_2) = m \cdot c \cdot \Delta \theta$

$\Rightarrow \Delta \theta = \frac{g(h_1 - h_2)}{c}$
 $= \frac{10(10 - 5.4)}{460} = 0.1^\circ \text{C}$



14. (c) $\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{C_w}}{2} = \frac{100 - \frac{80}{1}}{2} = 10^\circ \text{C}$

15. (c) Ice (0°C) converts into water (100°C) in following two steps.



Total heat required

$Q = Q_1 + Q_2 = 1 \times 80 + 1 \times 1 \times (100 - 0) = 180 \text{ cal}$



Chapter 13

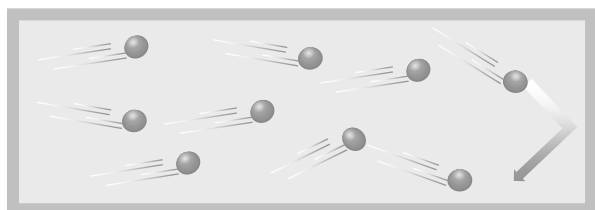
Kinetic Theory of Gases

Gas

In gases the intermolecular forces are very weak and its molecule may fly apart in all directions. So the gas is characterized by the following properties.

- (i) It has no shape and size and can be obtained in a vessel of any shape or size.
- (ii) It expands indefinitely and uniformly to fill the available space.
- (iii) It exerts pressure on its surroundings.
- (iv) Intermolecular forces in a gas are minimum.
- (v) They can easily compressed and expand.

Assumption of Ideal Gases (or Kinetic Theory of Gases)



Kinetic theory of gases relates the macroscopic properties of gases (such as pressure, temperature *etc.*) to the microscopic properties of the gas molecules (such as speed, momentum, kinetic energy of molecule *etc.*)

Actually it attempts to develop a model of the molecular behaviour which should result in the observed behaviour of an ideal gas. It is based on following assumptions :

- (1) Every gas consists of extremely small particles known as molecules. The molecules of a given gas are all identical but are different than those of another gas.
- (2) The molecules of a gas are identical, spherical, rigid and perfectly elastic point masses.
- (3) Their size is negligible in comparison to intermolecular distance (10^{-10} m)

(4) The volume of molecules is negligible in comparison to the volume of gas. (The volume of molecules is only 0.014% of the volume of the gas).

(5) Molecules of a gas keep on moving randomly in all possible direction with all possible velocities.

(6) The speed of gas molecules lie between zero and infinity

(7) The gas molecules keep on colliding among themselves as well as with the walls of containing vessel. These collisions are perfectly elastic.

(8) The time spent in a collision between two molecules is negligible in comparison to time between two successive collisions.

(9) The number of collisions per unit volume in a gas remains constant.

(10) No attractive or repulsive force acts between gas molecules.

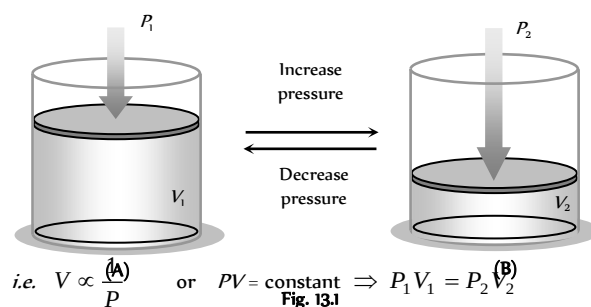
(11) Gravitational attraction among the molecules is ineffective due to extremely small masses and very high speed of molecules.

(12) Molecules constantly collide with the walls of container due to which their momentum changes. The change in momentum is transferred to the walls of the container. Consequently pressure is exerted by gas molecules on the walls of container.

(13) The density of gas is constant at all points of the container.

Gas Laws

(1) **Boyle's law** : For a given mass of an ideal gas at constant temperature, the volume of a gas is inversely proportional to its pressure.



- (i) $PV = P\left(\frac{m}{\rho}\right) = \text{constant} \Rightarrow \frac{P}{\rho} = \text{constant} \text{ or } \frac{P_1}{\rho_1} = \frac{P_2}{\rho_2}$
- (As volume = $\frac{m}{\rho(\text{Density of the gas})}$ and $m = \text{constant}$)
- (ii) $PV = P\left(\frac{N}{n}\right) = \text{constant} \Rightarrow \frac{P}{n} = \text{constant} \text{ or } \frac{P_1}{n_1} = \frac{P_2}{n_2}$
- (iii) As number of molecules per unit volume $n = \frac{N}{V}$
- $\Rightarrow V = \frac{N}{n}$ also $N = \text{constant}$
- (iv) Graphical representation : If m and T are constant

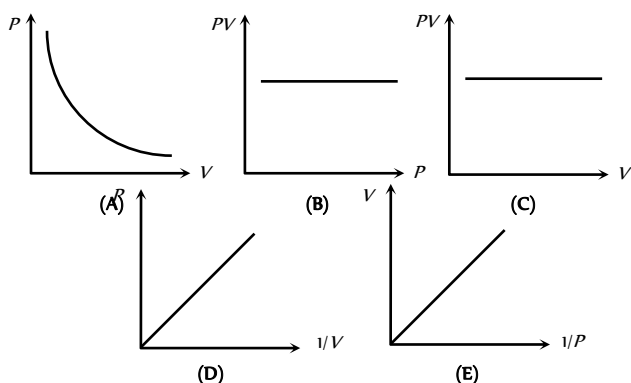
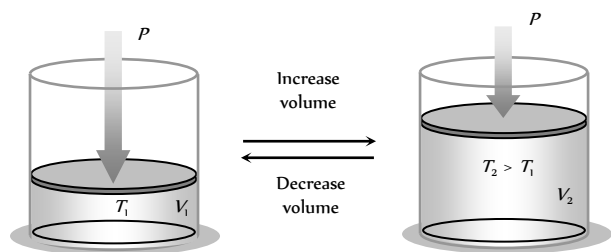


Fig. 13.2

(2) **Charles's law** : If the pressure remaining constant, the volume of the given mass of a gas is directly proportional to its absolute temperature.



i.e., $V \propto T \Rightarrow \frac{V}{T} = \text{constant} \Rightarrow \frac{V_1}{T_1} = \frac{V_2}{T_2}$

Fig. 13.3

- (i) $\frac{V}{T} = \frac{m}{\rho T} = \text{constant}$ (As volume $V = \frac{m}{\rho}$)
- or $\rho T = \text{constant} \Rightarrow \rho_1 T_1 = \rho_2 T_2$

(ii) If the pressure remains constant, the volume of the given mass of a gas increases or decreases by $\frac{1}{273.15}$ of its volume at 0°C for each 1°C rise or fall in temperature.

$$V_t = V_0 \left(1 + \frac{1}{273.15} t \right)$$

This is Charles's law for centigrade scale.

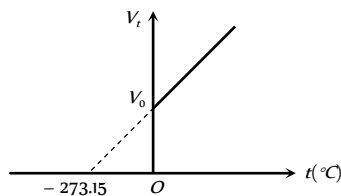


Fig. 13.4

- (v) Graphical representation: If m and P are constant

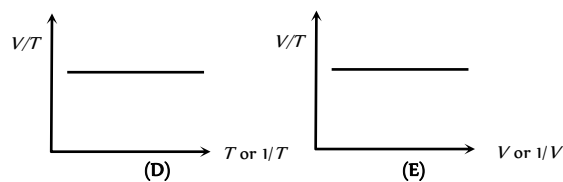
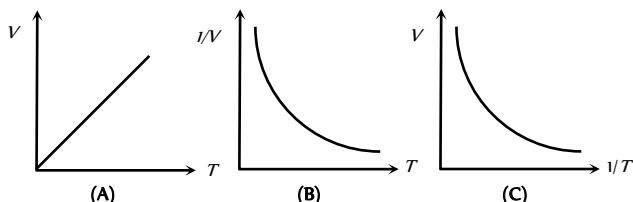


Fig. 13.5

(3) **Gay-Lussac's law or pressure law** : The volume remaining constant, the pressure of a given mass of a gas is directly proportional to its absolute temperature.

$$P \propto T \text{ or } \frac{P}{T} = \text{constant} \Rightarrow \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

(i) The volume remaining constant, the pressure of a given mass of a gas increases or decreases by $\frac{1}{273.15}$ of its pressure at 0°C for each 1°C rise or fall in temperature.

$$P_t = P_0 \left[1 + \frac{1}{273.15} t \right]$$

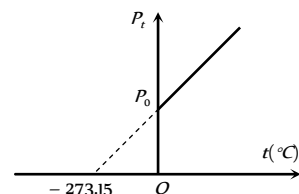


Fig. 13.5

This is pressure law for centigrade scale.

- (ii) **Graphical representation** : If m and V are constants

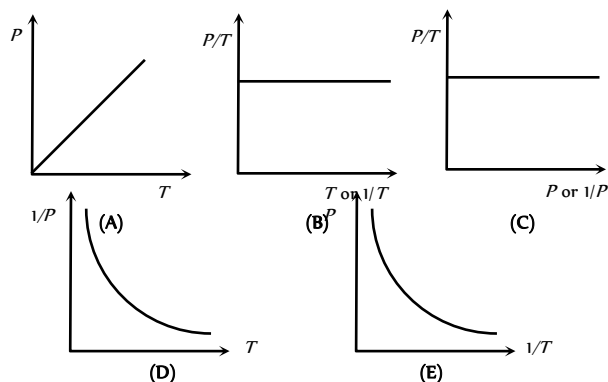


Fig. 13.6

(4) **Avogadro's law** : Equal volume of all the gases under similar conditions of temperature and pressure contain equal number of molecules i.e. $N_1 = N_2$.

(5) **Graham's law of diffusion** : When two gases at the same pressure and temperature are allowed to diffuse into each other, the rate of diffusion of each gas is inversely proportional to the square root of the density of the gas i.e. $r \propto \frac{1}{\sqrt{\rho}} \propto \frac{1}{\sqrt{M}}$ (M is the molecular weight of the gas) $\Rightarrow$

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \sqrt{\frac{M_2}{M_1}}$$

If V is the volume of gas diffused in t sec then

$$r = \frac{V}{t} \Rightarrow \frac{r_1}{r_2} = \frac{V_1}{V_2} \times \frac{t_2}{t_1}$$

(6) **Dalton's law of partial pressure** : The total pressure exerted by a mixture of non-reacting gases occupying a vessel is equal to the sum of the

individual pressures which each gases exert if it alone occupied the same volume at a given temperature.

$$\text{For } n \text{ gases } P = P_1 + P_2 + P_3 + \dots + P_n$$

where P = Pressure exerted by mixture and $P_1, P_2, P_3, \dots, P_n$ = Partial pressure of component gases.

Equation of State or Ideal Gas Equation

The equation which relates the pressure (P) volume (V) and temperature (T) of the given state of an ideal gas is known as ideal gas equation or equation of state.

$$\text{For 1 mole of gas } \frac{PV}{T} = R \text{ (constant)} \Rightarrow PV = RT$$

where R = universal gas constant.

Table 13.1 : Different forms of gas equation

Quantity of gas	Equation	Constant
1 mole gas	$PV = RT$	R = universal gas constant
μ mole gas	$PV = \mu RT$	
1 molecule of gas	$PV = \left(\frac{R}{N_A}\right)T = kT$	k = Boltzmann's constant
N molecules of gas	$PV = NkT$	
1 gm of gas	$PV = \left(\frac{R}{M}\right)T = rT$	r = Specific gas constant
m gm of gas	$PV = mrT$	

(1) **Universal gas constant (R)** : Universal gas constant signifies the work done by (or on) a gas per mole per kelvin.

$$R = \frac{PV}{\mu T} = \frac{\text{Pressure} \times \text{Volume}}{\mu \times \text{Temperature}} = \frac{\text{Work done}}{\mu \times \text{Temperature}}$$

(i) At S.T.P. the value of universal gas constant is same for all gases R

$$= 8.31 \frac{\text{J}}{\text{mole} \times \text{kelvin}} = 1.98 \frac{\text{cal}}{\text{mole} \times \text{kelvin}} \approx 2 \frac{\text{cal}}{\text{mol} \times \text{kelvin}}$$

$$= 0.8221 \frac{\text{litre} \times \text{atm}}{\text{mole} \times \text{kelvin}}$$

(ii) Dimension : $[ML^2T^{-2}\theta^{-1}]$

(2) **Boltzman's constant (k)** : It is represented by per mole gas

$$\text{constant i.e., } k = \frac{R}{N} = \frac{8.31}{6.023 \times 10^{23}} = 1.38 \times 10^{-23} \text{ J/K}$$

It's dimension : $[ML^2T^{-2}\theta^{-1}]$

(3) **Specific gas constant (r)** : It is represented by per gram gas

$$\text{constant i.e., } r = \frac{R}{M}. \text{ It's unit is } \frac{\text{Joule}}{\text{gm} \times \text{kelvin}} \text{ and dimension}$$

$$[L^2T^{-2}\theta^{-1}]$$

Since the value of M is different for different gases. Hence the value of r is different for different gases. e.g. It is maximum for hydrogen $r_{H_2} = \frac{R}{2}$

(1) The gases actually found in nature are called real gases.

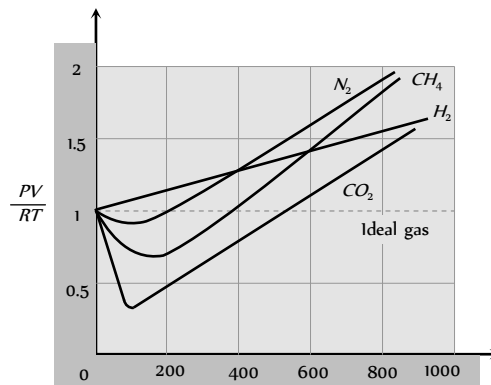
(2) They do not obeys gas Laws.

(3) For exactly one mole of an ideal gas $\frac{PV}{RT} = 1$. Plotting the

experimentally determined value of $\frac{PV}{RT}$ for exactly one mole of various

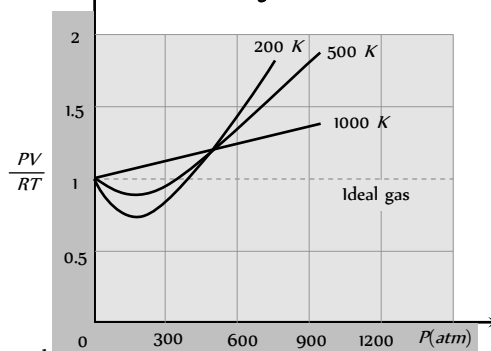
real gases as a function of pressure P , shows a deviation from identity.

(4) The quantity $\frac{PV}{RT}$ is called the compressibility factor and should be unit for an ideal gas.



(5) Deviation from ideal behaviour as a function of pressure (at constant temperature)

Fig. 13.7



(6) A real gas behaves as ideal gas most closely at low pressure and high temperature. Also can actual gas can be liquefied most easily which deviates most from ideal gas behaviour at low temperature and high pressure.

Fig. 13.8

(7) **Equation of state for real gases** : It is given by Vander Waal's with two correction in ideal gas equation. The it know as Vander Waal's gas equation.

(i) **Volume correction** : Due to finite size of molecule, a certain portion of volume of a gas is covered by the molecules themselves. Therefore the space available for the free motion of molecules of gas will be slightly less than the volume V of a gas. Hence the effective volume becomes $(V - b)$.

(ii) **Pressure correction** : Due to intermolecular force in real gases, molecule do not exert that force on the wall which they would have exerted in the absence of intermolecular force. Therefore the observed pressure P of the gas will be less than that present in the absence of intermolecular force. Hence the effective pressure becomes $\left(P + \frac{a}{V^2}\right)$.

(iii) **Vander Waal's gas equations**

$$\text{For 1 mole of gas } \left(P + \frac{a}{V^2}\right)(V - b) = RT$$

Real Gases

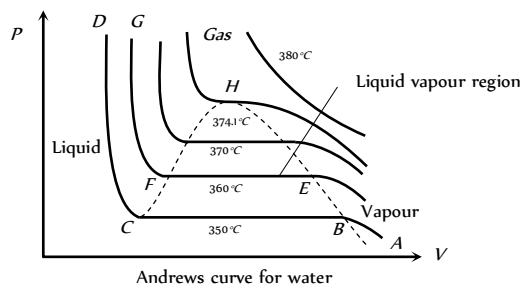
$$\text{For } \mu \text{ moles of gas } \left(P + \frac{a\mu^2}{V^2} \right) (V - \mu b) = \mu RT$$

Here a and b are constant called Vander Waal's constant.

Dimension : $[a] = [ML^5 T^{-2}]$ and $[b] = [L]$

Units : $a = N \times m$ and $b = m$.

(8) **Andrews curves** : The pressure (P) versus volume (V) curves for actual gases are called Andrews curves.



Andrews curve for water

(i) At 350°C , part AB represents vapour phase of water, in this part Boyle's law is obeyed ($P \propto \frac{1}{V}$). Part BC represents the co-existence of vapour and liquid phases. At point C , vapours completely change to liquid phase. Part CD is parallel to pressure axis which shows that compressibility of the water is negligible.

(ii) At 360°C portion representing the co-existence of liquid vapour phase is shorter.

(iii) At 370°C this portion is further decreased.

(iv) At 374.1°C , it reduces to point (H) called critical point and the temperature 374.1°C is called critical temperature (T) of water.

(v) The phase of water (at 380°C) above the critical temperature is called gaseous phase.

(9) **Critical temperature, pressure and volume** : The point on the P - V curve at which the matter gets converted from gaseous state to liquid state is known as critical point. At this point the difference between the liquid and vapour vanishes i.e. the densities of liquid and vapour become equal.

(i) **Critical temperature (T)** : The maximum temperature below which a gas can be liquefied by pressure alone is called critical temperature and is characteristic of the gas. A gas cannot be liquefied if its temperature is more than critical temperature.

CO_2 (31.1°C), O_2 (-118°C), N_2 (-147.1°C) and H_2O (374.1°C)

(ii) **Critical pressure (P)** : The minimum pressure necessary to liquify a gas at critical temperature is defined as critical pressure CO_2 (73.87 bar) and O_2 (49.7 atm)

(iii) **Critical volume (V)** : The volume of 1 mole of gas at critical pressure and critical temperature is defined as critical volume CO_2 ($95 \times 10^{-6} \text{ m}^3$)

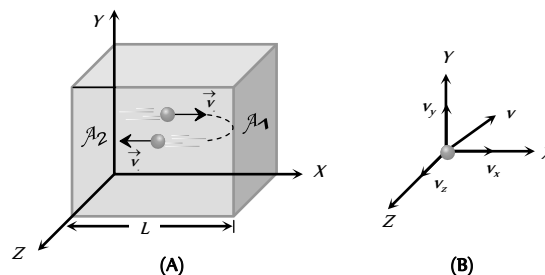
(iv) **Relation between Vander Waal's constants and T , P , V**

$$T_c = \frac{8a}{27Rb}, P_c = \frac{a}{27b^2}, V_c = 3b,$$

$$a = \frac{27R^2 T_c^2}{64 P_c}, b = \frac{R}{8} \left(\frac{T_c}{P_c} \right) \text{ and } \frac{P_c V_c}{T_c} = \frac{3}{8} R$$

Pressure of an Ideal Gas

Consider an ideal gas (consisting of N molecules each of mass m) enclosed in a cubical box of side L .



(1) **Instantaneous velocity** : Any molecule of gas moves with velocity $\vec{v}$ in any direction

where $\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} \Rightarrow v = \sqrt{v_x^2 + v_y^2 + v_z^2}$. Due to random motion of molecule $v_x = v_y = v_z \Rightarrow v^2 = 3v_x^2 = 3v_y^2 = 3v_z^2$

(2) **Time during collision** : Time between two successive collision with the wall A_1 .

$$\Delta t = \frac{\text{Distance travelled by molecule between two successive collision}}{\text{Velocity of molecule}} = \frac{2L}{v_x}$$

(3) **Collision frequency (n)** : It means the number of collision per second. Hence $n = \frac{1}{\Delta t} = \frac{v_x}{2L}$

(4) **Change in momentum** : This molecule collides with the shaded wall (A_1) with velocity v_x and rebounds with velocity $-v_x$.

The change in momentum of the molecule

$$\Delta p = (-mv_x) - (mv_x) = -2mv_x$$

As the momentum remains conserved in a collision, the change in momentum of the wall A_1 is $\Delta p = 2mv_x$

After rebound this molecule travel toward opposite wall A_1 with velocity $-v_x$, collide to it and again rebound with velocity v_x towards wall A_1 .

(5) **Force on wall** : Force exerted by a single molecule on shaded wall is equal to rate at which the momentum is transferred to the wall by this molecule.

$$\text{i.e. } F_{\text{Single molecule}} = \frac{\Delta p}{\Delta t} = \frac{2mv_x}{(2L/v_x)} = \frac{mv_x^2}{L}$$

The total force on the wall A_1 due to all the molecules

$$F_x = \frac{m}{L} \sum v_x^2 = \frac{m}{L} (v_{x1}^2 + v_{x2}^2 + v_{x3}^2 + \dots) = \frac{mN}{L} \overline{v_x^2}$$

$\overline{v_x^2}$ = mean square of x component of the velocity.

(6) **Pressure** : Now pressure is defined as force per unit area, hence pressure on shaded wall $P_x = \frac{F_x}{A} = \frac{mN}{AL} \overline{v_x^2} = \frac{mN}{V} \overline{v_x^2}$

For any molecule, the mean square velocity $\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}$; by symmetry $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} \Rightarrow \overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{\overline{v^2}}{3}$

Total pressure inside the container

$$P = \frac{1}{3} \frac{mN}{V} \overline{v^2} = \frac{1}{3} \frac{mN}{V} v_{rms}^2 \quad (\text{where } v_{rms} = \sqrt{\overline{v^2}})$$

(7) **Relation between pressure and kinetic energy** : As we know

$$P = \frac{1}{3} \frac{mN}{V} v_{rms}^2 = \frac{1}{3} \frac{M}{V} v_{rms}^2 \Rightarrow P = \frac{1}{3} \rho v_{rms}^2 \quad \dots (i)$$

$$[As M = mN = \text{Total mass of the gas and } \rho = \frac{M}{V}]$$

$$\therefore \text{K.E. per unit volume } E = \frac{1}{2} \left(\frac{M}{V} \right) v_{rms}^2 = \frac{1}{2} \rho v_{rms}^2 \quad \dots (ii)$$

$$\text{From (i) and (ii), we get } P = \frac{2}{3} E$$

i.e. the pressure exerted by an ideal gas is numerically equal to the two third of the mean kinetic energy of translation per unit volume of the gas.

(8) **Effect of mass, volume and temperature on pressure** :

$$P = \frac{1}{3} \frac{mN}{V} v_{rms}^2 \quad \text{or} \quad P \propto \frac{(mN)T}{V} \quad [As v_{rms}^2 \propto T]$$

(i) If volume and temperature of a gas are constant $P \propto mN$ i.e. Pressure $\propto$ (Mass of gas).

i.e. if mass of gas is increased, number of molecules and hence number of collision per second increases i.e. pressure will increase.

(ii) If mass and temperature of a gas are constant. $P \propto (1/V)$, i.e., if volume decreases, number of collisions per second will increase due to lesser effective distance between the walls resulting in greater pressure.

(iii) If mass and volume of gas are constant, $P \propto (v_{rms})^2 \propto T$

i.e., if temperature increases, the mean square speed of gas molecules will increase and as gas molecules are moving faster, they will collide with the walls more often with greater momentum resulting in greater pressure.

Various Speeds of Gas Molecules

The motion of molecules in a gas is characterised by any of the following three speeds.

(1) **Root mean square speed** : It is defined as the square root of mean of squares of the speed of different molecules

$$i.e. v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + v_4^2 + \dots}{N}} = \sqrt{\overline{v^2}}$$

(i) From the expression of pressure $P = \frac{1}{3} \rho v_{rms}^2$

$$\Rightarrow v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3PV}{\text{Mass of gas}}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

where $\rho = \frac{\text{Mass of gas}}{V}$ = Density of the gas, $M = \mu \times$ (mass of

gas), $pV = \mu RT$, $R = kN_A$, k = Boltzmann's constant,

$$m = \frac{M}{N_A} = \text{mass of each molecule.}$$

(ii) With rise in temperature rms speed of gas molecules increases as $v_{rms} \propto \sqrt{T}$.

(iii) With increase in molecular weight rms speed of gas molecule decreases as $v_{rms} \propto \frac{1}{\sqrt{M}}$. e.g., rms speed of hydrogen molecules is four times that of oxygen molecules at the same temperature.

(iv) rms speed of gas molecules is of the order of km/s e.g., at NTP for hydrogen gas

$$(v_{rms}) = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.31 \times 273}{2 \times 10^{-3}}} = 1840 \text{ m/s.}$$

(v) rms speed of gas molecules is $\sqrt{\frac{3}{\gamma}}$ times that of speed of sound

$$\text{in gas, as } v_{rms} = \sqrt{\frac{3RT}{M}} \text{ and } v_s = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v_{rms} = \sqrt{\frac{3}{\gamma}} v_s$$

(vi) rms speed of gas molecules does not depend on the pressure of gas (if temperature remains constant) because $P \propto \rho$ (Boyle's law) if pressure is increased n times then density will also increase by n times but v_{rms} remains constant.

(vii) Moon has no atmosphere because v_{rms} of gas molecules is more than escape velocity (v_e).

A planet or satellite will have atmosphere only if $v_{rms} < v_e$

(viii) At $T = 0$; $v_{rms} = 0$ i.e. the rms speed of molecules of a gas is zero at 0 K . This temperature is called absolute zero.

(2) **Most probable speed** : The particles of a gas have a range of speeds. This is defined as the speed which is possessed by maximum fraction of total number of molecules of the gas. e.g., if speeds of 10 molecules of a gas are 1, 2, 2, 3, 3, 3, 4, 5, 6, 6 km/s , then the most probable speed is 3 km/s , as maximum fraction of total molecules possess this speed.

$$\text{Most probable speed } v_{mp} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2kT}{m}}$$

(3) **Average speed** : It is the arithmetic mean of the speeds of molecules in a gas at given temperature.

$$v_{av} = \frac{v_1 + v_2 + v_3 + v_4 + \dots}{N}$$

and according to kinetic theory of gases

$$\text{Average speed } v_{av} = \sqrt{\frac{8P}{\pi\rho}} = \sqrt{\frac{8}{\pi} \frac{RT}{M}} = \sqrt{\frac{8}{\pi} \frac{kT}{m}}$$

Maxwell's Law (or the Distribution of Molecular Speeds)

(1) The v_{rms} gives us a general idea of molecular speeds in a gas at a given temperature. This doesn't mean that the speed of each molecule is v_{rms} . Many of the molecules have speed less than v_{rms} and many have speeds greater than v_{rms} .

(2) Maxwell derived an equation given the distribution of molecules in different speed as follow

$$dN = 4\pi N \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}} dv$$

where dN = Number of molecules with speeds between v and $v + dv$.

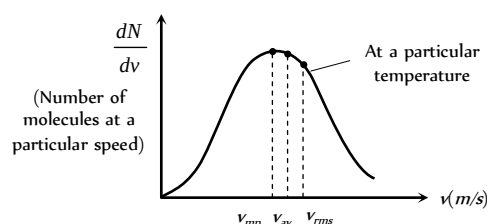


Fig. 13.11

(3) Graph between $\frac{dN}{dv}$ (number of molecules at a particular speed)

and v (speed of these molecules). From the graph it is seen that $\frac{dN}{dv}$ is maximum at most probable speed.

This graph also represent that $v_{rms} > v_{av} > v_{mp}$

(Order remember trick RAM)

$$\Rightarrow \sqrt{\frac{3RT}{M}} > \sqrt{\frac{8RT}{\pi M}} > \sqrt{\frac{2RT}{M}} \Rightarrow$$

$$1.77 \sqrt{\frac{RT}{M}} > 1.6 \sqrt{\frac{RT}{M}} > 1.41 \sqrt{\frac{RT}{M}}$$

Area bonded by this curve with speed axis represents the number of molecules corresponds to that velocity range. This curve is asymmetric curve.

Effect of temperature on velocity distribution : With temperature rise the $\frac{dN}{dv}$ vs v . Curve shift towards right and becomes broader.

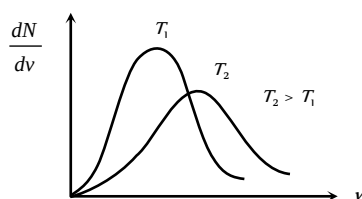


Fig. 13.12

(Because with temperature rise average molecular speed increases).

Mean Free Path.

(1) The distance travelled by a gas molecule between two successive collisions is known as free path.

$$\lambda = \frac{\text{Total distance travelled by a gas molecule between successive collisions}}{\text{Total number of collisions}}$$

During two successive collisions, a molecule of a gas moves in a straight line with constant velocity and

Let $\lambda_1, \lambda_2, \lambda_3, \dots$ be the distance travelled by a gas molecule during n collisions respectively, then the mean free path of a gas molecule is given by

$$\lambda = \frac{\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n}{n}$$

$$(2) \lambda = \frac{1}{\sqrt{2} \pi n d^2}$$

where d = Diameter of the molecule,

n = Number of molecules per unit volume

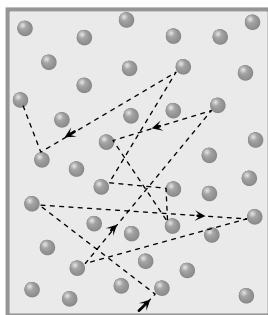


Fig. 13.13

$$(3) \text{ As } PV = \mu RT = \mu NkT \Rightarrow \frac{N}{V} = \frac{P}{kT} = n = \text{Number of molecule}$$

$$\text{per unit volume so } \lambda = \frac{1}{\sqrt{2}} \frac{kT}{\pi d^2 P}.$$

$$(4) \text{ From } \lambda = \frac{1}{\sqrt{2} \pi n d^2} = \frac{m}{\sqrt{2} \pi (mn) d^2} = \frac{m}{\sqrt{2} \pi^2 \rho}$$

[As m = Mass each molecule, mn = Mass per unit volume = Density = ρ]

$$(5) \text{ If average speed of molecule is } v \text{ then } \lambda = v \times \frac{t}{N} = v \times T$$

[As N = Number of collision in time t , T = time interval between two collisions].

$$(i) \text{ As } \lambda \propto \frac{1}{\rho} \text{ and } \lambda \propto m \text{ i.e. the mean free path is inversely}$$

proportional to the density of a gas and directly proportional to the mass of each molecule.

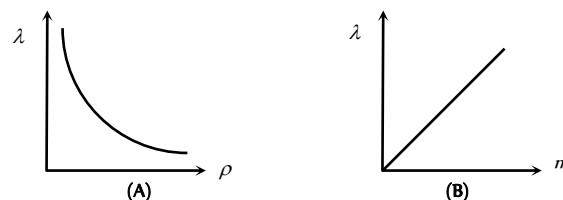


Fig. 13.14

$$(ii) \text{ As } \lambda = \frac{1}{\sqrt{2}} \frac{kT}{\pi d^2 P}.$$

For constant volume and hence constant number density n of gas molecules, $\frac{P}{T}$ is constant so that λ will not depend on P and T . But if volume of given mass of a gas is allowed to change with P or T then $\lambda \propto T$ at constant pressure and $\lambda \propto \frac{1}{P}$ at constant temperature.

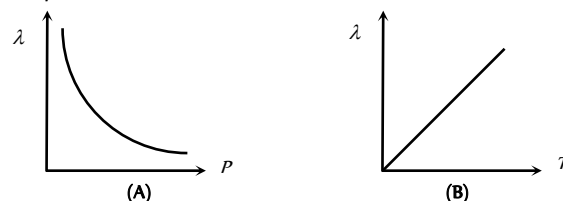


Fig. 13.15

Degree of Freedom

The term degree of freedom of a system refers to the possible independent motions, systems can have. or

The total number of independent modes (ways) in which a system can possess energy is called the degree of freedom (f).

The independent motions can be translational, rotational or vibrational or any combination of these.

So the degree of freedom are of three types :

- (i) Translational degree of freedom
- (ii) Rotational degree of freedom
- (iii) Vibrational degree of freedom

General expression for degree of freedom

$$f = 3A - B; \text{ where } A = \text{Number of independent particles,}$$

$$B = \text{Number of independent restriction}$$

(1) **Monoatomic gas :** Molecule of monoatomic gas can move in any direction in space so it can have three independent motions and hence 3 degrees of freedom (all translational)

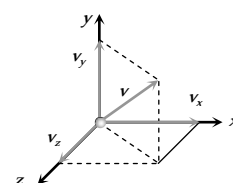


Fig. 13.16

(2) **Diatomic gas** : Molecules of diatomic gas are made up of two atoms joined rigidly to one another through a bond. This cannot only move bodily, but also rotate about one of the three co-ordinate axes. However its moment of inertia about the axis joining the two atoms is negligible compared to that about the other two axes.

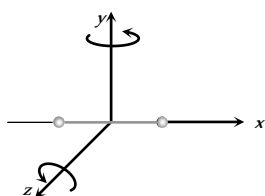


Fig. 13.17

Hence it can have only two rotational motion. Thus a diatomic molecule has 5 degree of freedom : 3 translational and 2 rotational.

(3) **Triatomic gas (Non-linear)** : A non-linear molecule can rotate about any of three co-ordinate axes. Hence it has 6 degrees of freedom : 3 translational and 3 rotational.

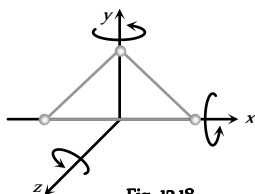


Fig. 13.18

Table 13.2 ; Degree of freedom for different gases

Atomicity of gas	Example	A	B	$f = 3$ $A - B$	Figure
Monoatomic	He, Ne, Ar	1	0	$f = 3$	
Diatomic	H_2 , O_2 , N_2 , Cl_2 etc.	2	1	$f = 5$	
Triatomic non linear	H_2O	3	3	$f = 6$	
Triatomic linear	CO_2 , $BeCl_2$	3	2	$f = 7$	

The above degrees of freedom are shown at room temperature. Further at high temperature, in case of diatomic or polyatomic molecules, the atoms with in the molecule may also vibrate with respect to each other. In such cases, the molecule will have an additional degrees of freedom, due to vibrational motion.

An object which vibrates in one dimension has two additional degree of freedom. One for the potential energy and one for the kinetic energy of vibration.

A diatomic molecule that is free to vibrate (in addition to translation and rotation) will have $7(2 + 3 + 2)$ degrees of freedom.

Kinetic Energy of Ideal Gas

In ideal gases, the molecules are considered as point particles. For point particles, there is no internal excitation, no vibration and no rotation. The point particles can have only translational motion and thus only translational energy. For an ideal gas the internal energy can only be translational kinetic energy.

Hence kinetic energy (or internal energy) of 1 mole ideal gas

$$E = \frac{1}{2} M v_{rms}^2 = \frac{1}{2} M \times \frac{3RT}{M} = \frac{3}{2} RT$$

Table 13.3 : Various Translational kinetic energies

1 mole gas	$\frac{3}{2} RT$; R = Universal gas constant
μ mole gas	$\frac{3}{2} \mu RT$
1 molecule	$\frac{3}{2} k T$; k = Boltzmann's constant
N molecule	$\frac{3}{2} N k T$
1 gm gas	$\frac{3}{2} r T$; r = Specific gas constant
m gm gas	$\frac{3}{2} m r T$

(i) Kinetic energy per molecule of gas does not depends upon the mass of the molecule but only depends upon the temperature of the gas. As

$$E = \frac{3}{2} kT \quad \text{or} \quad E \propto T \quad \text{i.e. molecules of different gases say He, H and O}_2$$

etc. at same temperature will have same translational kinetic energy though their *r.m.s.* speed are different.

$$(2) \text{ For two gases at the same temperature } m_1(v_{rms})_1^2 = m_2(v_{rms})_2^2$$

(3) Kinetic energy per mole of gas depends only upon the temperature of gas.

(4) Kinetic energy per *gram* of gas depend upon the temperature as well as molecular weight (or mass of one molecule) of the gas.

$$E_{gram} = \frac{3}{2} \frac{k}{m} T \Rightarrow E_{gram} \propto \frac{T}{m}$$

(5) From the above expressions it is clear that higher the temperature of the gas, more will be the average kinetic energy possessed by the gas molecules at $T = 0$, $E = 0$ i.e. at absolute zero the molecular motion stops.

Law of Equipartition of Energy

According to this law, for any system in thermal equilibrium, the total energy is equally distributed among its various degree of freedom. And each

degree of freedom is associated with energy $\frac{1}{2} kT$ (where

$$k = 1.38 \times 10^{-23} \text{ J/K, } T = \text{absolute temperature of the system}).$$

(i) At a given temperature T , all ideal gas molecules no matter what their mass have the same average translational kinetic energy; namely,

$$\frac{3}{2} kT. \text{ When measure the temperature of a gas, we are also measuring the}$$

average translational kinetic energy of it's molecules.

(2) At same temperature gases with different degrees of freedom (e.g., He and H) will have different average energy or internal energy namely

$$\frac{f}{2} kT. \text{ (} f \text{ is different for different gases)}$$

(3) Different energies of a system of degree of freedom f are as follows

$$(i) \text{ Total energy associated with each molecule } = \frac{f}{2} kT$$

$$(ii) \text{ Total energy associated with } N \text{ molecules } = \frac{f}{2} NkT$$

$$(iii) \text{ Total energy associated with } \mu \text{ mole } = \frac{f}{2} RT$$

(iv) Total energy associated with μ molen = $\frac{f}{2} \mu RT$

(v) Total energy associated with each gram = $\frac{f}{2} r T$

(iv) Total energy associated with m gram = $\frac{f}{2} m r T$

Specific Heat (C_p and C_v) of a Gas

The specific heat of gas can have many values, but out of them following two values are very important

(1) **Specific heat at constant volume (C_v)** : The specific heat of a gas at constant volume is defined as the quantity of heat required to raise the temperature of unit mass of gas through $1^\circ C$ or $1 K$ when its volume is kept

constant, i.e., $C_v = \frac{(\Delta Q)_v}{m \Delta T}$

If instead of unit mass, 1 mole of gas is considered, the specific heat is called molar specific heat at constant volume and is represented by capital C_v .

$$C_v = M C_v = \frac{M(\Delta Q)_v}{m \Delta T} = \frac{1}{\mu} \frac{(\Delta Q)_v}{\Delta T} \quad \left[\text{As } \mu = \frac{m}{M} \right]$$

(2) **Specific heat at constant from (C_p)** : The specific heat of a gas at constant pressure is defined as the quantity of heat required to raise the temperature of unit mass of gas through $1 K$ when its pressure is kept

constant, i.e., $C_p = \frac{(\Delta Q)_p}{m \Delta T}$

If instead of unit mass, 1 mole of gas is considered, the specific heat is called molar specific heat at constant pressure and is represented by C_p .

$$C_p = M C_p = \frac{M(\Delta Q)_p}{m \Delta T} = \frac{1}{\mu} \frac{(\Delta Q)_p}{\Delta T} \quad \left[\text{As } \mu = \frac{m}{M} \right]$$

Mayer's Formula

(1) Out of two principle specific heats of a gas, C_p is more than C_v because in case of C_v volume of gas is kept constant and heat is required only for raising the temperature of one gram mole of the gas through $1^\circ C$ or $1 K$. Hence no heat, what so ever, is spent in expansion of the gas.

It means that heat supplied to the gas increases its internal energy only i.e. $(\Delta Q)_v = \Delta U = \mu C_v \Delta T$ (i)

(2) While in case of C_p the heat is used in two ways

(i) In increasing the temperature of the gas by ΔT

(ii) In doing work, due to expansion at constant pressure (ΔW)

So $(\Delta Q)_p = \Delta U + \Delta W = \mu C_p \Delta T$ (ii)

From equation (i) and (ii) $\mu C_p \Delta T - \mu C_v \Delta T = \Delta W$

$$\Rightarrow \mu \Delta T (C_p - C_v) = P \Delta V \Rightarrow C_p - C_v = \frac{P \Delta V}{\mu \Delta T} = R$$

[For constant pressure, $\Delta W = P \Delta V$ also from $PV = \mu RT$,

$$P \Delta V = \mu R \Delta T]$$

This relation is called Mayer's formula and shows that $C_p > C_v$ i.e. molar specific heat at constant pressure is greater than that at constant volume.

Specific Heat in Terms of Degree of Freedom

(1) C_v : For a gas at temperature T , the internal energy

$$U = \frac{f}{2} \mu RT \Rightarrow \text{Change in energy } \Delta U = \frac{f}{2} \mu R \Delta T \quad \dots (i)$$

Also, as we know for any gas heat supplied at constant volume $(\Delta Q)_v = \mu C_v \Delta T = \Delta U$ (ii)

From equation (i) and (ii) $C_v = \frac{1}{2} f R$

(2) C_p : From the Mayer's formula $C_p - C_v = R$

$$\Rightarrow C_p = C_v + R = \frac{f}{2} R + R = \left(\frac{f}{2} + 1 \right) R$$

(3) **Ratio of C_p and C_v (γ)** : $\gamma = \frac{C_p}{C_v} = \frac{\left(\frac{f}{2} + 1 \right) R}{\frac{f}{2} R} = 1 + \frac{2}{f}$

(i) Value of γ is different for monoatomic, diatomic and triatomic gases. $\gamma_{mono} = \frac{5}{3} = 1.6, \gamma_{di} = \frac{7}{5} = 1.4, \gamma_{tri} = \frac{4}{3} = 1.33$

(ii) Value of γ is always more than 1. So we can say that always $C_p > C_v$.

Gaseous Mixture

If two non-reactive gases are enclosed in a vessel of volume V . In the mixture μ_1 moles of one gas are mixed with μ_2 moles of another gas. If N_A is Avogadro's number then

Number of molecules of first gas $N_1 = \mu_1 N_A$

and number of molecules of second gas $N_2 = \mu_2 N_A$

(1) Total mole fraction $\mu = (\mu_1 + \mu_2)$.

(2) If M_1 is the molecular weight of first gas and M_2 that of second gas.

Then molecular weight of mixture $M = \frac{\mu_1 M_1 + \mu_2 M_2}{\mu_1 + \mu_2}$

(3) Specific heat of the mixture at constant volume will be

$$C_{v_{mix}} = \frac{\mu_1 C_{v_1} + \mu_2 C_{v_2}}{\mu_1 + \mu_2} = \frac{\frac{m_1}{M_1} C_{v_1} + \frac{m_2}{M_2} C_{v_2}}{\frac{m_1}{M_1} + \frac{m_2}{M_2}}$$

(4) Specific heat of the mixture at constant pressure will be

$$C_{p_{mix}} = \frac{\mu_1 C_{p_1} + \mu_2 C_{p_2}}{\mu_1 + \mu_2} = \frac{\mu_1 \left(\frac{\gamma_1}{\gamma_1 - 1} \right) R + \mu_2 \left(\frac{\gamma_2}{\gamma_2 - 1} \right) R}{\mu_1 + \mu_2}$$

$$= \frac{R}{\mu_1 + \mu_2} \left[\mu_1 \left(\frac{\gamma_1}{\gamma_1 - 1} \right) + \mu_2 \left(\frac{\gamma_2}{\gamma_2 - 1} \right) \right]$$

$$= \frac{R}{\frac{m_1}{M_1} + \frac{m_2}{M_2}} \left[\frac{m_1}{M_1} \left(\frac{\gamma_1}{\gamma_1 - 1} \right) + \frac{m_2}{M_2} \left(\frac{\gamma_2}{\gamma_2 - 1} \right) \right]$$

(5) $\gamma_{mixture} = \frac{C_{p_{mix}}}{C_{v_{mix}}} = \frac{\frac{(\mu_1 C_{p_1} + \mu_2 C_{p_2})}{\mu_1 + \mu_2}}{\frac{(\mu_1 C_{v_1} + \mu_2 C_{v_2})}{\mu_1 + \mu_2}}$

$$= \frac{\mu_1 C_{P_1} + \mu_2 C_{P_2}}{\mu_1 C_{V_1} + \mu_2 C_{V_2}} = \frac{\left\{ \mu_1 \left(\frac{\gamma_1}{\gamma_1 - 1} \right) R + \mu_2 \left(\frac{\gamma_2}{\gamma_2 - 1} \right) R \right\}}{\left\{ \mu_1 \left(\frac{R}{\gamma_1 - 1} \right) + \mu_2 \left(\frac{R}{\gamma_2 - 1} \right) \right\}}$$

$$\therefore \gamma_{\text{mixture}} = \frac{\frac{\mu_1 \gamma_1}{\gamma_1 - 1} + \frac{\mu_2 \gamma_2}{\gamma_2 - 1}}{\frac{\mu_1}{\gamma_1 - 1} + \frac{\mu_2}{\gamma_2 - 1}} = \frac{\mu_1 \gamma_1 (\gamma_2 - 1) + \mu_2 \gamma_2 (\gamma_1 - 1)}{\mu_1 (\gamma_2 - 1) + \mu_2 (\gamma_1 - 1)}$$

Tips & Tricks

✍ The cooking gas cylinder contains L.P.G. (Liquid Petroleum gas) which is saturated. And as pressure

of saturated vapours is independent of volume (at constant temperature), the pressure of gas coming out of the cylinder remains constant till the cylinder becomes empty.



✍ If the number of molecules in a gas increases, then the temperature, kinetic energy and pressure of the gas increases because $P \propto n$, $T \propto n$ and kinetic energy $\propto T \propto n$.

✍ At constant volume if T increases then $\bar{v}$, v_{rms} , P and collision frequency increases.

✍ If two gases are filled in vessel then nothing can be predicted about the pressure of gases. However their mean molecular energies will be same but their rms velocities will be different.

✍ The average distance between two gas molecules at NTP is $10^{-7} m$.

✍ The space available for a single gas molecule at NTP is $37.2 \times 10^{-28} m^3$.

✍ The molecules of gases will escape out from a planet if the temperature of planet $T \leq \frac{M v_e^2}{3R}$; where v_e = escape velocity from the planet, R = universal, gas constant and M = Molecular mass of the gas.

✍ As f (degree of freedom) increases then $C_p \uparrow$, $C_v \uparrow$ and $\gamma \uparrow$.

✍ The number of molecules present in 1 gm mole of a gas is defined as Avogadro number (N).

$$N_A = 6.023 \times 10^{23} \text{ per gm mole} = 6.023 \times 10^{26} \text{ per kg mole.}$$

At S.T.P. or N.T.P. ($T = 273 K$ and $P = 1 atm$) 22.4 litre of each gas has 6.023×10^{23} molecule

✍ One mole of any gas at S.T.P. occupy 22.4 litre of volume

e.g. 32 gm oxygen, 28 gm nitrogen and 2gm hydrogen occupy the same volume at S.T.P.

✍ For any gas 1 mole = M gram = 22.4 litre = 6.023×10^{23} molecule.

$$\text{✍ } v_1 : v_2 : v_3 = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} = \sqrt{3} : \sqrt{2.5} : \sqrt{2}$$

✍ For oxygen gas molecules $v_{rms} = 461 m/s$, $v_{avg} = 424.7 m/s$ and $v_{p} = 376.4 m/s$

✍ An atom in a solid though has no degree of freedom for translational and rotational motion, due to vibration along 3 axes has $3 \times 2 = 6$ degrees of freedom (and not like an ideal gas molecule). When a diatomic or polyatomic gas dissociates into atoms it behaves as monoatomic gas whose degree of freedom are changed accordingly

✍ In General a polyatomic molecule has 3 translational, 3 rotational degree of freedom and a certain number of vibration mode f_{vib} . Hence

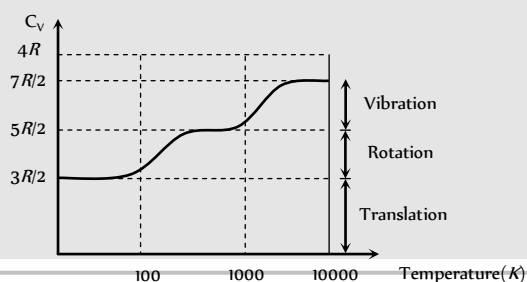
$$\gamma_{poly} = \frac{4 + f_{vib}}{3 + f_{vib}}$$

✍ Only average translational kinetic energy of a gas contributes to its temperature. Two gases with the same average translational kinetic energy have the same temperature even if one has greater rotational energy and thus greater internal energy.

✍ Unsaturated vapours obey gas laws while saturated vapours don't.

✍ For real gases effective volume is considered as $(V - \mu b)$ where $b = 4 N_A \left(\frac{4}{3} \pi r^3 \right)$; r = radius of each molecule and N_A = avogadro number.

✍ Variation of degree of freedom of a diatomic gas (H_2) with temperature. At very low temperature only translation is possible. as the temperature increases rotational motion can begin. At still higher temperatures vibratory motion can begin.



Ordinary Thinking

Objective Questions

Gas Laws

- The temperature of a gas at pressure P and volume V is $27^\circ C$. Keeping its volume constant if its temperature is raised to $927^\circ C$, then its pressure will be [MP PMT 1985]
 - $2 P$
 - $3 P$
 - $4 P$
 - $6 P$
- 4 moles of an ideal gas is at $0^\circ C$. At constant pressure it is heated to double its volume, then its final temperature will be
 - $0^\circ C$
 - $273^\circ C$
 - $546^\circ C$
 - $136.5^\circ C$
- Every gas (real gas) behaves as an ideal gas [CPMT 1997; RPMT 2000; MP PET 2001]

- (a) At high temperature and low pressure
 (b) At low temperature and high pressure
 (c) At normal temperature and pressure
 (d) None of the above
4. Boyle's law holds for an ideal gas during [AFMC 1994; KCET 1999]
 (a) Isobaric changes (b) Isothermal changes
 (c) Isochoric changes (d) Isotonic changes
5. S.I. unit of universal gas constant is [MNR 1988; MP PMT 1994; UPSEAT 1999]
 (a) $\text{cal}/^\circ\text{C}$ (b) J/mol
 (c) $\text{J mol}^{-1} \text{K}^{-1}$ (d) J/kg
6. Molecules of a gas behave like [J & K CET 2000]
 (a) Inelastic rigid sphere
 (b) Perfectly elastic non-rigid sphere
 (c) Perfectly elastic rigid sphere
 (d) Inelastic non-rigid sphere

Ordinary Thinking Objective Questions

Communication

1. In short wave communication waves of which of the following frequencies will be reflected back by the ionospheric layer, having electron density 10^6 per m [AIIMS 2003]
 - (a) 2 MHz (b) 10 MHz
 - (c) 12 MHz (d) 18 MHz
2. In an amplitude modulated wave for audio frequency of 500 cycle/second, the appropriate carrier frequency will be [AMU 1996]
 - (a) 50 cycles/sec (b) 100 cycles/sec
 - (c) 500 cycles/sec (d) 50,000 cycles/sec
3. AM is used for broadcasting because
 - (a) It is more noise immune than other modulation systems
 - (b) It requires less transmitting power compared with other systems
 - (c) Its use avoids receiver complexity
 - (d) No other modulation system can provide the necessary bandwidth faithful transmission
4. Range of frequencies allotted for commercial FM radio broadcast is
 - (a) 88 to 108 MHz (b) 88 to 108 kHz
 - (c) 8 to 88 MHz (d) 88 to 108 GHz
5. The velocity factor of a transmission line x . If dielectric constant of the medium is 2.6, the value of x is [AFMC 1995]
 - (a) 0.26 (b) 0.62
 - (c) 2.6 (d) 6.2
6. The process of superimposing signal frequency (*i.e.* audio wave) on the carrier wave is known as [AIIMS 1987]
 - (a) Transmission (b) Reception
 - (c) Modulation (d) Detection
7. Long distance short-wave radio broadcasting uses [AFMC 1996]
 - (a) Ground wave (b) Ionospheric wave
 - (c) Direct wave (d) Sky wave
8. A step index fibre has a relative refractive index of 0.88%. What is the critical angle at the corecladding interface [Manipal 2003]
 - (a) 60° (b) 75°
 - (c) 45° (d) None of these
9. The characteristic impedance of a coaxial cable is of the order of
 - (a) 50 Ω (b) 200 Ω
 - (c) 270 Ω (d) None of these
10. In which frequency range, space waves are normally propagated
 - (a) HF (b) VHF
 - (c) UHF (d) SHF
11. If μ and μ_c are the refractive indices of the materials of core and cladding of an optical fibre, then the loss of light due to its leakage can be minimised by having [BVP 2003]
 - (a) $\mu > \mu_c$ (b) $\mu < \mu_c$
 - (c) $\mu = \mu_c$ (d) None of these
12. Through which mode of propagation, the radio waves can be sent from one place to another [JIPMER 2003]
 - (a) Ground wave propagation
 - (b) Sky wave propagation
 - (c) Space wave propagation
 - (d) All of them
13. A laser beam of pulse power 10^6 watt is focussed on an object are 10^{-4} cm. The energy flux in watt/cm at the point of focus is
 - (a) 10^6 (b) 10^4
 - (c) 10^8 (d) 10^2
14. The carrier frequency generated by a tank circuit containing 1 nF capacitor and 10 μH inductor is [AFMC 2003]
 - (a) 1592 Hz (b) 1592 MHz
 - (c) 1592 kHz (d) 159.2 Hz
15. Broadcasting antennas are generally [AFMC 2003]
 - (a) Omnidirectional type (b) Vertical type
 - (c) Horizontal type (d) None of these
16. For television broadcasting, the frequency employed is normally
 - (a) 30-300 MHz (b) 30-300 GHz
 - (c) 30-300 KHz (d) 30-300 Hz
17. The radio waves of frequency 300 MHz to 3000 MHz belong to [MNR 1997]
 - (a) High frequency band
 - (b) Very high frequency band
 - (c) Ultra high frequency band
 - (d) Super high frequency band
18. An antenna behaves as resonant circuit only when its length is
 - (a) $\frac{\lambda}{2}$
 - (b) $\frac{\lambda}{4}$
 - (c) λ
 - (d) $\frac{\lambda}{2}$ or integral multiple of $\frac{\lambda}{2}$
19. Maximum useable frequency (MUF) in F-region layer is x , when the critical frequency is 60 MHz and the angle of incidence is 70° . Then x is [Himachal PMT 2003]
 - (a) 150 MHz (b) 170 MHz
 - (c) 175 MHz (d) 190 MHz
20. The electromagnetic waves of frequency 2 MHz to 30 MHz are
 - (a) In ground wave propagation
 - (b) In sky wave propagation
 - (c) In microwave propagation
 - (d) In satellite communication [CPMT 2003]
21. A laser is a coherent source because it contains [JIPMER 2003]
 - (a) Many wavelengths [EAMCET 2002]
 - (b) Uncoordinated wave of a particular wavelength

- (c) Coordinated wave of many wavelengths
(d) Coordinated waves of a particular wavelength
22. The attenuation in optical fibre is mainly due to [AFMC 2003]
(a) Absorption
(b) Scattering
(c) Neither absorption nor scattering
(d) Both (a) and (b)
23. The maximum distance upto which TV transmission from a TV tower of height h can be received is proportional to [AIIMS 2003]
(a) h (b) h
(c) h (d) h
24. A laser beam is used for carrying out surgery because it [AIIMS 2003]
(a) Is highly monochromatic (b) Is highly coherent
(c) Is highly directional (d) Can be sharply focussed
25. Laser beams are used to measure long distances because [DCE 2002, 03]
(a) They are monochromatic
(b) They are highly polarised
(c) They are coherent
(d) They have high degree of parallelism
26. An oscillator is producing FM waves of frequency 2 kHz with a variation of 10 kHz. What is the modulating index [DCE 2004]
(a) 0.20 (b) 5.0
(c) 0.67 (d) 1.5
27. The maximum peak to peak voltage of an AM wave is 24 mV and the minimum peak to peak voltage is 8 mV. The modulation factor is
(a) 10% (b) 20%
(c) 25% (d) 50%
28. Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are
(a) 1490, 1510 (b) 1510, 1490
(c) $\frac{1}{1490}, \frac{1}{1510}$ (d) $\frac{1}{1510}, \frac{1}{1490}$
29. What is the modulation index of an over modulated wave
(a) 1 (b) Zero
(c) < 1 (d) > 1
30. Basically, the product modulator is
(a) An amplifier (b) A mixer
(c) A frequency separator (d) A phase separator
31. If f_c and f_m represent the carrier wave frequencies for amplitude and frequency modulations respectively, then
(a) $f_a > f_f$ (b) $f_a < f_f$
(c) $f_a \approx f_f$ (d) $f_a \geq f_f$
32. Which of the following is the disadvantage of FM over AM
(a) Larger band width requirement
(b) Larger noise
(c) Higher modulation power
(d) Low efficiency
33. If a number of sine waves with modulation indices $n_1, n_2, n_3, \dots$ modulate a carrier wave, then total modulation index (n) of the wave is
(a) $n_1 + n_2 + \dots + 2(n_1 + n_2 + \dots)$
(b) $\sqrt{n_1^2 + n_2^2 + n_3^2 + \dots}$
(c) $\sqrt{n_1^2 + n_2^2 + n_3^2 + \dots}$
(d) None of these
34. An AM wave has 1800 watt of total power content. For 100% modulation the carrier should have power content equal to
(a) 1000 watt (b) 1200 watt
(c) 1500 watt (d) 1600 watt
35. The frequency of a FM transmitter without signal input is called
(a) Lower side band frequency
(b) Upper side band frequency
(c) Resting frequency
(d) None of these
36. What type of modulation is employed in India for radio transmission
(a) Amplitude modulation (b) Frequency modulation
(c) Pulse modulation (d) None of these
37. When the modulating frequency is doubled, the modulation index is halved and the modulating voltage remains constant, the modulation system is
(a) Amplitude modulation (b) Phase modulation
(c) Frequency modulation (d) All of the above
38. An antenna is a device
(a) That converts electromagnetic energy into radio frequency signal
(b) That converts radio frequency signal into electromagnetic energy
(c) That converts guided electromagnetic waves into free space electromagnetic waves and vice-versa
(d) None of these
39. While tuning in a certain broadcast station with a receiver, we are actually
(a) Varying the local oscillator frequency
(b) Varying the frequency of the radio signal to be picked up
(c) Tuning the antenna
(d) None of these
40. Indicate which one of the following system is digital
(a) Pulse position modulation
(b) Pulse code modulation

- (c) Pulse width modulation
(d) Pulse amplitude modulation
41. In a communication system, noise is most likely to affect the signal
(a) At the transmitter
(b) In the channel or in the transmission line
(c) In the information source
(d) At the receiver
42. The waves used in telecommunication are
(a) IR (b) UV
(c) Microwave (d) Cosmic rays
43. In an FM system a 7 kHz signal modulates 108 MHz carrier so that frequency deviation is 50 kHz. The carrier swing is
(a) 7.143 (b) 8
(c) 0.71 (d) 350
44. Consider telecommunication through optical fibres. Which of the following statements is not true [AIEEE 2003]
(a) Optical fibres may have homogeneous core with a suitable cladding
(b) Optical fibres can be of graded refractive index
(c) Optical fibres are subject to electromagnetic interference from outside
(d) Optical fibres have extremely low transmission loss
45. The phenomenon by which light travels in an optical fibres is
(a) Reflection (b) Refraction
(c) Total internal reflection (d) Transmission
46. Television signals on earth cannot be received at distances greater than 100 km from the transmission station. The reason behind this is that [DCE 1995]
(a) The receiver antenna is unable to detect the signal at a distance greater than 100 km
(b) The TV programme consists of both audio and video signals
(c) The TV signals are less powerful than radio signals
(d) The surface of earth is curved like a sphere
47. Advantage of optical fibre [DCE 2005]
(a) High bandwidth and EM interference
(b) Low bandwidth and EM interference
(c) High band width, low transmission capacity and no EM interference
(d) High bandwidth, high data transmission capacity and no EM interference
48. In frequency modulation [Kerala PMT 2005]
(a) The amplitude of modulated wave varies as frequency of carrier wave
(b) The frequency of modulated wave varies as amplitude of modulating wave
(c) The amplitude of modulated wave varies as amplitude of carrier wave
(d) The frequency of modulated wave varies as frequency of modulating wave
(e) The frequency of modulated wave varies as frequency of carrier wave
49. Audio signal cannot be transmitted because [Kerala PMT 2005]
(a) The signal has more noise
(b) The signal cannot be amplified for distance communication
(c) The transmitting antenna length is very small to design
(d) The transmitting antenna length is very large and impracticable
(e) The signal is not a radio signal
50. In which of the following remote sensing technique is not used
(a) Forest density (b) Pollution
(c) Wetland mapping (d) Medical treatment
51. For sky wave propagation of a 10 MHz signal, what should be the minimum electron density in ionosphere [AIIMS 2005]
(a) $\sim 1.2 \times 10^6 \text{ m}^{-3}$ (b) $\sim 10^6 \text{ m}^{-3}$
(c) $\sim 10^7 \text{ m}^{-3}$ (d) $\sim 10^8 \text{ m}^{-3}$
52. What should be the maximum acceptance angle at the aircore interface of an optical fibre if n_1 and n_2 are the refractive indices of the core and the cladding, respectively [AIIMS 2005]
(a) $\sin^{-1}(n_2 / n_1)$ (b) $\sin^{-1} \sqrt{n_1^2 - n_2^2}$
(c) $\left[\tan^{-1} \frac{n_2}{n_1} \right]$ (d) $\left[\tan^{-1} \frac{n_1}{n_2} \right]$



[DCE 2001]

Critical Thinking

Objective Questions

1. A sky wave with a frequency 55 MHz is incident on D-region of earth's atmosphere at 45°. The angle of refraction is (electron density for D-region is 400 electron/cm) [Haryana PMT 2003]
(a) 60° (b) 45°
(c) 30° (d) 15°
2. In a diode AM-detector, the output circuit consist of $R = 1k\Omega$ and $C = 10 \mu F$. A carrier signal of 100 kHz is to be detected. Is it good
(a) Yes
(b) No
(c) Information is not sufficient
(d) None of these
3. Consider an optical communication system operating at $\lambda = 800 \text{ nm}$. Suppose, only 1% of the optical source frequency is the available channel bandwidth for optical communication. How many channels can be accommodated for transmitting audio signals requiring a bandwidth of 8 kHz
(a) 4.8×10^4 (b) 48
(c) 6.2×10^4 (d) 4.8×10^5
4. A photodetector is made from a semiconductor $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ with $E_g = 0.73 \text{ eV}$. What is the maximum wavelength, which it can detect
(a) 1000 nm (b) 1703 nm
(c) 500 nm (d) 173 nm

5. A transmitter supplies 9 kW to the aerial when unmodulated. The power radiated when modulated to 40% is
 (a) 5 kW (b) 9.72 kW
 (c) 10 kW (d) 12 kW
6. The antenna current of an AM transmitter is 8 A when only carrier is sent but increases to 8.96 A when the carrier is sinusoidally modulated. The percentage modulation is
 (a) 50% (b) 60%
 (c) 65% (d) 71%
7. The total power content of an AM wave is 1500 W . For 100% modulation, the power transmitted by the carrier is
 (a) 500 W (b) 700 W
 (c) 750 W (d) 1000 W
8. The total power content of an AM wave is 900 W . For 100% modulation, the power transmitted by each side band is
 (a) 50 W (b) 100 W
 (c) 150 W (d) 200 W
9. The modulation index of an FM carrier having a carrier swing of 200 kHz and a modulating signal 10 kHz is
 (a) 5 (b) 10
 (c) 20 (d) 25
10. A 500 Hz modulating voltage fed into an FM generator produces a frequency deviation of 2.25 kHz . If amplitude of the voltage is kept constant but frequency is raised to 6 kHz then the new deviation will be
 (a) 4.5 kHz (b) 54 kHz
 (c) 27 kHz (d) 15 kHz
11. The audio signal used to modulate $60 \sin(2\pi \times 10^4 t)$ is $15 \sin 300\pi t$. The depth of modulation is
 (a) 50% (b) 40%
 (c) 25% (d) 15%
12. The bit rate for a signal, which has a sampling rate of 8 kHz and where 16 quantisation levels have been used is
 (a) 32000 bits/sec (b) 16000 bits/sec
 (c) 64000 bits/sec (d) 72000 bits/sec
13. An amplitude modulated wave is modulated to 50%. What is the saving in power if carrier as well as one of the side bands are suppressed
 (a) 70% (b) 65.4%
 (c) 94.4% (d) 25.5%
14. In AM, the centpercent modulation is achieved when
 (a) Carrier amplitude = signal amplitude
 (b) Carrier amplitude $\neq$ signal amplitude
 (c) Carrier frequency = signal frequency
 (d) Carrier frequency $\neq$ signal frequency

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : Diode lasers are used as optical sources in optical communication.
 Reason : Diode lasers consume less energy.
 [AIIMS 2005]
2. Assertion : Television signals are received through sky-wave propagation.
 Reason : The ionosphere reflects electromagnetic waves of frequencies greater than a certain critical frequency.
 [AIIMS 2005]
3. Assertion : In high latitude one sees colourful curtains of light hanging down from high altitudes.
 Reason : The high energy charged particles from the sun are deflected to polar regions by the magnetic field of the earth.
 [AIIMS 2003]
4. Assertion : Short wave bands are used for transmission of radio waves to a large distance.
 Reason : Short waves are reflected by ionosphere
 [AIIMS 1994]
5. Assertion : The electrical conductivity of earth's atmosphere decreases with altitude.
 Reason : The high energy particles (*i.e.* γ -rays and cosmic rays) coming from outer space and entering our earth's atmosphere cause ionisation of the atoms of the gases present there and the pressure of gases decreases with increase in altitude.
6. Assertion : The electromagnetic waves of shorter wavelength can travel longer distances on earth's surface than those of longer wavelengths.
 Reason : Shorter the wavelength, the larger is the velocity of wave propagation.
7. Assertion : The surface wave propagation is used for medium wave band and for television broadcasting.
 Reason : The surface waves travel directly from transmitting antenna to receiver antenna through atmosphere.
8. Assertion : The television broadcasting becomes weaker with increasing distance.
 Reason : The power transmitted from TV transmitter varies inversely as the distance of the receiver
9. Assertion : Microwave propagation is better than the sky wave propagation.
 Reason : Microwaves have frequencies 100 to 300 GHz , which have very good directional properties.
10. Assertion : Satellite is an ideal platform for remote sensing.
 Reason : Satellite in polar orbit can provide global coverage or continuous coverage of the fixed area in geostationary configuration.
11. Assertion : Fax is a modulating and demodulating device.



1630 Communication

Reason : It is necessary for exact reproduction of a document.

12. Assertion : A dish antenna is highly directional.

Reason : This is because a dipole antenna is omni directional.

Answers

Communication

1	a	2	d	3	c	4	a	5	b
6	c	7	c	8	d	9	c	10	c
11	a	12	d	13	b	14	c	15	b
16	a	17	c	18	d	19	c	20	b
21	d	22	d	23	a	24	d	25	d
26	b	27	d	28	a	29	d	30	b
31	b	32	a	33	c	34	b	35	c
36	a	37	c	38	c	39	a	40	b
41	b	42	c	43	a	44	c	45	c
46	d	47	d	48	b	49	d	50	d
51	a	52	b						

Critical Thinking Questions

1	b	2	b	3	a	4	b	5	b
6	d	7	d	8	c	9	b	10	b
11	c	12	a	13	c	14	a		

A Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : In pressure-temperature (P - T) phase diagram of water, the slope of the melting curve is found to be negative.
Reason : Ice contracts on melting to water. [AIIMS 2005]
2. Assertion : For gas atom the number of degrees of freedom is 3.
Reason : $\frac{C_p}{C_v} = \gamma$ [AIIMS 2000]
3. Assertion : A gas have a unique value of specific heat.
Reason : Specific heat is defined as the amount of heat required to raise the temperature of unit mass of the substance through unit degree.
4. Assertion : A gas can be liquified at any temperature by increase of pressure alone.
Reason : On increasing pressure the temperature of gas decreases.
5. Assertion : Equal masses of helium and oxygen gases are given equal quantities of heat. There will be a greater rise in the temperature of helium compared to that of oxygen.
Reason : The molecular weight of oxygen is more than the molecular weight of helium.
6. Assertion : Absolute zero is the temperature corresponding to zero energy.
Reason : The temperature at which no molecular motion cease is called absolute zero temperature.
7. Assertion : The ratio of specific heat gas at constant pressure and specific heat at constant volume for a diatomic gas is more than that for a monatomic gas.
Reason : The molecules of a monatomic gas have more degree of freedom than those of a diatomic gas.
8. Assertion : At room temperature, water does not sublime from water to steam.
Reason : The critical point of water is much above the room temperature.
9. Assertion : Specific heat of a gas at constant pressure (C) is greater than its specific heat at constant volume (C_v).
Reason : At constant pressure, some heat is spent in expansion of the gas.
10. Assertion : The internal energy of a real gas is function of both, temperature and volume.
Reason : Internal kinetic energy depends on temperature and internal potential energy depends on volume.

11. Assertion : For an ideal gas, at constant temperature, the product of the pressure and volume is constant.
Reason : The mean square velocity of the molecules is inversely proportional to mass. [AIIMS 1998]
12. Assertion : If a gas container in motion is suddenly stopped, the temperature of the gas rises.
Reason : The kinetic energy of ordered mechanical motion is converted in to the kinetic energy of random motion of gas molecules.
13. Assertion : Internal energy of an ideal gas does not depend upon volume of the gas
Reason : Internal energy of ideal gas depends on temperature of gas.
14. Assertion : At low density, variables of gases P , V and T follows the equation $PV = \mu RT$
Reason : At low density real gases are more closely to ideal gases
15. Assertion : Maxwell speed distribution graph is symmetric about most probable speed
Reason : rms speed of ideal gas, depends upon it's type (monoatomic, diatomic and polyatomic)

Answers

Gas Laws

1	c	2	b	3	a	4	b	5	c
6	c	7	c	8	c	9	a	10	d
11	a	12	d	13	b	14	d	15	c
16	b	17	c	18	a	19	a	20	d
21	c	22	b	23	a	24	a	25	a
26	a	27	c	28	a	29	c	30	c
31	d	32	c	33	a	34	d	35	a
36	a	37	c	38	a	39	a	40	c
41	d	42	b	43	a	44	c	45	a
46	a	47	d	48	d	49	c	50	d
51	c	52	c	53	c	54	b	55	d
56	c	57	c	58	a	59	d	60	a
61	d	62	c	63	d	64	d	65	c
66	b	67	a	68	d	69	d	70	c
71	c	72	c	73	d	74	b	75	c
76	c	77	c	78	a	79	c	80	c
81	b	82	a	83	c	84	b	85	c
86	d	87	c	88	c	89	b	90	a
91	b	92	a	93	a	94	d	95	a
96	c	97	a	98	a	99	c	100	c

Speed of Gas

1	b	2	a	3	c	4	c	5	a
6	a	7	d	8	d	9	a	10	a
11	c	12	d	13	b	14	c	15	a

16	a	17	a	18	d	19	c	20	a
21	d	22	c	23	c	24	a	25	d
26	a	27	b	28	d	29	a	30	b
31	c	32	b	33	d	34	c	35	a
36	b	37	a	38	c	39	d	40	a
41	d	42	a	43	c	44	b	45	b
46	c	47	c	48	b	49	b	50	a
51	b	52	b	53	b	54	d	55	a
56	a	57	c	58	b	59	d	60	a
61	c	62	b	63	b	64	d	65	a
66	b	67	b	68	a				

Degree of Freedom and Specific Heat

1	a	2	c	3	a	4	a	5	c
6	d	7	c	8	b	9	d	10	d
11	c	12	a	13	b	14	d	15	a
16	a	17	a	18	a	19	b	20	a
21	c	22	b	23	c	24	d	25	b
26	d	27	d	28	a	29	b	30	d
31	a	32	c	33	a	34	c	35	a
36	d	37	a	38	a	39	b	40	c
41	b	42	b	43	b	44	d	45	b
46	c	47	c	48	d				

Pressure and Energy

1	c	2	b	3	c	4	d	5	d
6	d	7	d	8	a	9	a	10	b
11	d	12	c	13	c	14	a	15	d
16	d	17	b	18	b	19	c	20	a
21	c	22	b	23	b	24	c	25	a
26	b	27	d	28	d	29	c	30	d
31	a	32	a	33	c	34	c	35	d
36	c	37	a	38	b	39	ac	40	d
41	d	42	b	43	a	44	a	45	a
46	b	47	a	48	a	49	d	50	a
51	c	52	c	53	d	54	c	55	b
56	c	57	d	58	d	59	c	60	c
61	a	62	c	63	c	64	a		

Critical Thinking Questions

1	d	2	d	3	a	4	b	5	a
6	acd	7	b	8	b	9	cd	10	b
11	b	12	bc	13	d	14	d	15	a
16	c	17	d	18	b	19	d	20	d

21	a	22	c	23	b	24	a	25	c
26	c	27	c	28	c	29	d	30	d
31	d	32	b	33	a	34	a	35	d
36	c	37	c	38	d				

Graphical Questions

1	d	2	b	3	c	4	c	5	c
6	c	7	a	8	b	9	a	10	c
11	b	12	c	13	c	14	a	15	b
16	b	17	a	18	a	19	b	20	c
21	a	22	c	23	b	24	b	25	a

Assertion and Reason

1	a	2	b	3	e	4	d	5	b
6	e	7	d	8	a	9	a	10	a
11	b	12	a	13	b	14	a	15	d

AS Answers and Solutions

Gas Laws

1. (c) Using Charles's law $\frac{P_1}{P_2} = \frac{T_1}{T_2}$

$$\text{or } P_2 = \frac{P_1 T_2}{T_1} = \frac{P(273 + 927)}{(273 + 27)} = 4P.$$

2. (b) $\frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow T_2 = 2 \times T_1 = 2 \times (273 + 0) = 546K$

$$\Rightarrow T_2 = 273 \times 2 = 546K \Rightarrow 273^\circ C \Rightarrow 273^\circ C$$

Kinetic Theory of Gases

SET Self Evaluation Test -13

1. An ideal gas has an initial pressure of 3 pressure units and an initial volume of 4 volume units. The table gives the final pressure and volume of the gas (in those same units) in four, processes. Which processes start and end on the same isotherm

(a) A
(b) B
(c) C
(d) D

	A	B	C	D
P	5	4	12	6
V	4	3	1	2

2. Suppose ideal gas equation follows $6VP^3 = \text{constant}$. Initial temperature and volume of the gas are T and V respectively. If gas expand to $27V$ then its temperature will be come

(a) T
(b) $9T$
(c) $27T$
(d) $T/9$

3. One mole of a monoatomic ideal gas is mixed with one mole of a diatomic ideal gas. The molar specific heat of the mixture at constant volume is

(a) 8
(b) $\frac{3}{2}R$
(c) $2R$
(d) $2.5R$

4. When the temperature of a gas is raised from 27°C to 90°C , the percentage increase in the *r.m.s.* velocity of the molecules will be

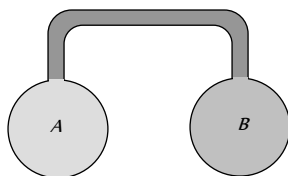
(a) 10%
(b) 15%
(c) 20%
(d) 17.5%

5. A gas is enclosed in a closed pot. On keeping this pot in a train moving with high speed, the temperature of the gas

(a) Will increase
(b) Will decrease
(c) Will remain the same
(d) Will change according to the nature of the gas

6. Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300K . Now if one vessel is immersed in a bath of constant temperature 600K and the other in a bath of constant temperature 300K . Then the common pressure will be

(a) 1 atm
(b) $\frac{4}{5}\text{ atm}$
(c) $\frac{4}{3}\text{ atm}$
(d) $\frac{3}{4}\text{ atm}$



7. The *r.m.s.* velocity of a gas at a certain temperature is $\sqrt{2}$ times than that of the oxygen molecules at that temperature. The gas can be

(a) H_2
(b) He
(c) CH_4
(d) SO_2

8. At what temperature, the mean kinetic energy of O_2 will be the same for H_2 molecules at -73°C

(a) 127°C
(b) 527°C
(c) -73°C
(d) -173°C

9. The volume of a gas at pressure $21 \times 10^4 \text{ N/m}^2$ and temperature 27°C is 83 litres . If $R = 8.3 \text{ J/mol/K}$, then the quantity of gas in *gm-mole* will be

(a) 15
(b) 42
(c) 7
(d) 14

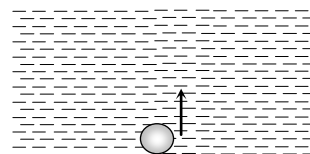
10. The pressure and temperature of an ideal gas in a closed vessel are 720 kPa and 40°C respectively. If $\frac{1}{4}$ th of the gas is released from

the vessel and the temperature of the remaining gas is raised to 353°C , the final pressure of the gas is [EAMCET (Med.) 2000]

(a) 1440 kPa
(b) 1080 kPa
(c) 720 kPa
(d) 540 kPa

11. An air bubble doubles its radius on raising from the bottom of water reservoir to be the surface of water in it. If the atmospheric pressure is equal to 10 m of water, the height of water in the reservoir is [EAMCET Med.1999]

(a) 10 m
(b) 20 m
(c) 70 m
(d) 80 m



12. If the *r.m.s.* velocity of a gas at a given temperature (Kelvin scale) is 300 m/sec . What will be the *r.m.s.* velocity of a gas having twice the molecular weight and half the temperature on Kelvin scale =

(a) 300 m/sec
(b) 600 m/sec
(c) 75 m/sec
(d) 150 m/sec

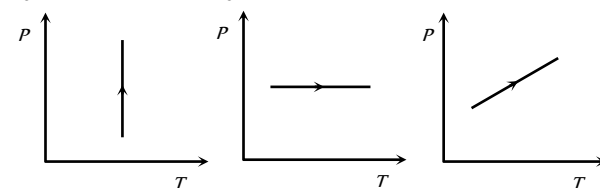
13. The ratio of two specific heats $\frac{C_p}{C_v}$ of CO is

(a) 1.33
(b) 1.40
(c) 1.29
(d) 1.66

14. The energy of a gas/litre is 300 joules , then its pressure will be

(a) $3 \times 10^5 \text{ N/m}^2$
(b) $6 \times 10^5 \text{ N/m}^2$
(c) 10^5 N/m^2
(d) $2 \times 10^5 \text{ N/m}^2$

15. Pressure versus temperature graphs of an ideal gas are as shown in figure. Choose the wrong statement



(a) Density of gas is increasing in graph (i)
(b) Density of gas is decreasing in graph (ii)
(c) Density of gas is constant in graph (iii)
(d) None of these

16. If pressure of CO_2 (real gas) in a container is given by

$$P = \frac{RT}{2V - b} - \frac{a}{4b^2} \text{ then mass of the gas in container is}$$

- (a) 11 gm (b) 22 gm
(c) 33 gm (d) 44 gm
17. A cylinder of fixed capacity 44.8 litre. contains a monatomic gas at standard temperature and pressure. The amount of heat required to cylinder by $10^\circ C$ will be.

(R = universal gas constant)

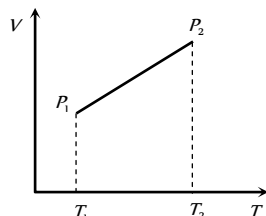
- (a) R (b) $10R$
(c) $20R$ (d) $30R$
18. A pressure cooker contains air at 1 atm and $30^\circ C$. If the safety value of the cooler blows when the inside pressure ≥ 3 atm, then the maximum temperature of the air, inside the cooker can be
- (a) $90^\circ C$ (b) $636^\circ C$
(c) $909^\circ C$ (d) $363^\circ C$

19. One mole of an ideal monatomic gas requires 210 J heat to raise the temperature by 10 K, when heated at constant temperature. If the same gas is heated at constant volume to raise the temperature by 10 K then heat required is

[Pb. PET 2000]

- (a) 238 J (b) 126 J
(c) 210 J (d) 350 J
20. From the following V - T diagram we can conclude

- (a) $P_1 = P_2$
(b) $P_1 > P_2$
(c) $P_1 < P_2$
(d) None of these



21. A cylinder contains 10 kg of gas at pressure of $10^7 N/m^2$. The quantity of gas taken out of the cylinder, if final pressure is $2.5 \times 10^6 N/m^2$, will be (Temperature of gas is constant)

[EAMCET 1998; Pb. PMT 1999; 2003; DPMT 199, 2003]

- (a) 15.2 kg (b) 3.7 kg
(c) Zero (d) 7.5 kg
22. Certain amount of an ideal gas are contained in a closed vessel. The vessel is moving with a constant velocity v . The molecular mass of

gas is M . The rise in temperature of the gas when the vessel is suddenly stopped is ($\gamma = C_p / C_v$)

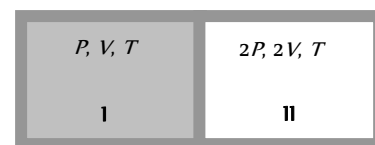
- (a) $\frac{Mv^2}{2R(\gamma + 1)}$ (b) $\frac{Mv^2(\gamma - 1)}{2R}$
(c) $\frac{Mv^2}{2R(\gamma + 1)}$ (d) $\frac{Mv^2}{2R(\gamma - 1)}$

23. Air is filled at $60^\circ C$ in a vessel of open mouth. The vessel is heated to a temperature T so that 1/4th part of air escapes. Assuming the volume of the vessel remaining constant, the value of T is

- (a) $80^\circ C$ (b) $444^\circ C$
(c) $333^\circ C$ (d) $171^\circ C$

24. A partition divides a container having insulated walls into two compartments I and II. the same gas fills the two compartments. The ratio of the number of molecules in compartments I and II is

- (a) 1 : 6
(b) 6 : 1
(c) 4 : 1
(d) 1 : 4



25. Considering the gases to be ideal, the value of $\gamma = \frac{C_p}{C_v}$ for a gaseous mixture consisting of 3 moles of carbon dioxide and 2 moles of oxygen will be ($\gamma_{O_2} = 1.4$, $\gamma_{CO_2} = 1.3$)

[UPSEAT 2000; Pb. PET 2004]

- (a) 1.37 (b) 1.34
(c) 1.55 (d) 1.63

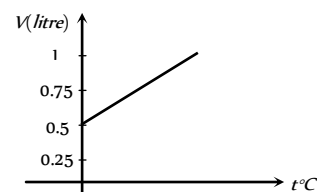
26. A jar has a mixture of hydrogen and oxygen gas in the ratio of 1 : 5. The ratio of mean kinetic energies of hydrogen and oxygen molecules is

[CPMT 1977]

- (a) 1 : 16 (b) 1 : 4
(c) 1 : 5 (d) 1 : 1

27. Graph between volume and temperature for a gas is shown in figure. If α = volume coefficient of gas = $\frac{1}{273}$ per $^\circ C$, then what is the volume of gas at a temperature of $819^\circ C$

- (a) $1 \times 10^{-3} m^3$
(b) $2 \times 10^{-3} m^3$
(c) $3 \times 10^{-3} m^3$
(d) $4 \times 10^{-3} m^3$



AS Answers and Solutions

(SET -13)

1. (c) For same isotherm ; $T \rightarrow$ constant

$$\therefore P \propto \frac{1}{V} \Rightarrow P_1 V_1 = P_2 V_2$$

2. (b) $VP^3 = \text{constant} = k \Rightarrow P = \frac{k}{V^{1/3}}$

$$\text{Also } PV = \mu RT \Rightarrow \frac{k}{V^{1/3}} \cdot V = \mu RT \Rightarrow V^{2/3} = \frac{\mu RT}{k}$$

$$\text{Hence } \left(\frac{V_1}{V_2} \right)^{2/3} = \frac{T_1}{T_2} \Rightarrow \left(\frac{V}{27V} \right)^{2/3} = \frac{T}{T_2} \Rightarrow T_2 = 9T$$

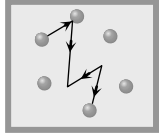
3. (c) $(C_v)_{\text{mix}} = \frac{\mu_1 C_{v1} + \mu_2 C_{v2}}{\mu_1 + \mu_2} = \frac{1 \times \frac{3}{2} R + 1 \times \frac{5}{2} R}{1 + 1} = 2R$

$$\left((C_v)_{\text{mono}} = \frac{3}{2} R, (C_v)_{\text{di}} = \frac{5}{2} R \right)$$

4. (a) $v_{rms} = \sqrt{\frac{3RT}{M}} \Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{(273+90)}{(273+30)}} = 1.1$

% increase $= \left(\frac{v_2}{v_1} - 1 \right) \times 100 = 0.1 \times 100 = 10\%$

5. (c) Temperature of the gas is concerned only with its disordered motion. It is not concerned with its ordered motion.



Motion of train (order of motion)

Motion of molecule (Disordered motion)

6. (c) $\mu = \mu_1 + \mu_2$

$$\frac{P(2V)}{RT_1} = \frac{P'V}{RT_1} + \frac{P'V}{RT_2} \Rightarrow \frac{2P}{RT_1} = \frac{P'}{R} \left[\frac{T_2 + T_1}{T_1 T_2} \right]$$

$$P' = \frac{2PT_2}{(T_1 + T_2)} = \frac{2 \times 1 \times 600}{(300 + 600)} = \frac{4}{3} \text{ atm}$$

7. (c) $v_{rms} \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}}$

$$\therefore \frac{1}{\sqrt{2}} = \sqrt{\frac{M_2}{32}} \Rightarrow M_2 = 16. \text{ Hence the gas is } CH_4.$$

8. (c) Mean kinetic energy of molecule depends upon temperature only. For O_2 it is same as that of H_2 at the same temperature of $-73^\circ C$.

9. (c) $PV = \mu RT \Rightarrow \mu = \frac{PV}{RT} = \frac{21 \times 10^4 \times 83 \times 10^{-3}}{8.3 \times 300} = 7$

10. (b) $P_1 = 720 \text{ kPa}, T_1 = 40^\circ C = 273 + 40 = 313 K$

$$P \propto mT \Rightarrow \frac{P_2}{P_1} = \frac{m_2}{m_1} \frac{T_2}{T_1} = \frac{3}{4} \times \frac{626}{313} = 1.5$$

$$\Rightarrow P_2 = 1.5P_1 = 1.5 \times 720 = 1080 \text{ kPa}$$

11. (c) According to Boyle's law $(P_1 V_1)_{\text{bottom}} = (P_2 V_2)_{\text{top}}$

$$(10 + h) \times \frac{4}{3} \pi r_1^3 = 10 \times \frac{4}{3} \pi r_2^3 \quad \text{but } r_2 = 2r_1$$

$$\therefore (10 + h)r_1^3 = 10 \times 8r_1^3 \Rightarrow 10 + h = 80 \quad \therefore h = 70 \text{ m}$$

12. (d) $v_{rms} = \sqrt{\frac{3RT}{M}} \Rightarrow v_{rms} \propto \sqrt{\frac{T}{M}}$

$$\frac{v_2}{v_1} = \sqrt{\frac{M_1}{M_2} \times \frac{T_2}{T_1}} = \sqrt{\frac{1}{2} \times \frac{1}{2}} \Rightarrow v_2 = \frac{v_1}{2} = \frac{300}{2} = 150 \text{ m/sec}$$

13. (bc) Co is diatomic gas, for diatomic gas

$$C_p = \frac{7}{2} R \text{ and } C_v = \frac{5}{2} R \Rightarrow \gamma_{di} = \frac{C_p}{C_v} = \frac{7R/2}{5R/2} = 1.4$$

14. (d) Energy $= 300 \text{ J/litre} = 300 \times 10^3 \text{ J/m}^3$

$$P = \frac{2}{3} E = \frac{2 \times 300 \times 10^3}{3} = 2 \times 10^5 \text{ N/m}^2$$

15. (c) $\rho = \frac{PM}{RT}$

Density ρ remains constant when P/T or volume remains constant.

In graph (i) Pressure is increasing at constant temperature hence volume is decreasing so density is increasing. Graphs (ii) and (iii) volume is increasing hence, density is decreasing. Note that volume would have been constant in case the straight line in graph (iii) had passed through origin.

16. (b) Vander wall's gas equation for μ mole of real gas

$$\left(P + \frac{\mu^2 a}{V^2} \right) (V - \mu b) = \mu RT \Rightarrow P = \frac{\mu RT}{V - \mu b} - \frac{\mu^2 a}{V^2}$$

on comparing the given equation with this standard equation

we get $\mu = \frac{1}{2}$. Hence $\mu = \frac{m}{M} \Rightarrow$ mass of gas

$$m = \mu M = \frac{1}{2} \times 44 = 22 \text{ gm.}$$

17. (d) As we know 1 mol of any ideal gas at STP occupies a volume of 22.4 litres.

Hence number of moles of gas $\mu = \frac{44.8}{22.4} = 2$

Since the volume of cylinder is fixed,

Hence $(\Delta Q)_V = \mu \omega \Delta T$

$$= 2 \times \frac{3}{2} R \times 10 = 30R \quad \left(\because (C_V)_{\text{mono}} = \frac{3}{2} R \right)$$

18. (b) Since volume is constant,

Hence $\frac{P_1}{P_2} = \frac{T_1}{T_2} \Rightarrow \frac{1}{3} = \frac{(273+30)}{T_2}$

$$\Rightarrow T_2 = 909 K = 636^\circ C$$

19. (b) $(\Delta Q)_P = \mu C_P \Delta T$ and $(\Delta Q)_V = \mu C_V \Delta T$

$$\Rightarrow \frac{(\Delta Q)_V}{(\Delta Q)_P} = \frac{C_V}{C_P} = \frac{\frac{3}{2} R}{\frac{5}{2} R} = \frac{3}{5}$$

$$\left[\because (C_V)_{\text{mono}} = \frac{3}{2} R, (C_P)_{\text{mono}} = \frac{5}{2} R \right]$$

$$\Rightarrow (\Delta Q)_V = \frac{3}{5} \times (\Delta Q)_P = \frac{3}{5} \times 210 = 126 \text{ J}$$

20. (b) In case of given graph, V and T are related as $V = aT - b$, where a and b are constants.

From ideal gas equation, $PV = \mu RT$

We find $P = \frac{\mu RT}{aT - b} = \frac{\mu R}{a - b/T}$

Since $T_1 > T_2$, therefore $P_1 < P_2$.

21. (d) $PV = \mu rT \Rightarrow P \propto m$ [$\because V, r, T \rightarrow \text{constant}$]

$$\Rightarrow \frac{m_1}{m_2} = \frac{P_1}{P_2} \Rightarrow \frac{10}{m_2} = \frac{10^7}{2.5 \times 10^6} \Rightarrow m_2 = 2.5 \text{ kg.}$$

Hence mass of the gas taken out of the cylinder

$$= 10 - 2.5 = 7.5 \text{ kg.}$$

22. (b) If m is the total mass of the gas then its kinetic

$$\text{energy} = \frac{1}{2}mv^2$$

When the vessel is suddenly stopped then total kinetic energy will increase the temperature of the gas. Hence

$$\frac{1}{2}mv^2 = \mu C_v \Delta T = \frac{m}{M} C_v \Delta T \quad \left[\text{As } C_v = \frac{R}{\gamma - 1} \right]$$

$$\Rightarrow \frac{m}{M} \frac{R}{\gamma - 1} \Delta T = \frac{1}{2}mv^2 \Rightarrow \Delta T = \frac{Mv^2(\gamma - 1)}{2R}$$

23. (d) For open mouth vessel, pressure is constant.

Volume is also given constant

$$\text{Hence from } PV = \mu RT = \left(\frac{m}{M} \right) RT \Rightarrow T \propto \frac{1}{m} \Rightarrow \frac{T_1}{T_2} = \frac{m_2}{m_1}$$

$\therefore \frac{1}{4}$ th part escapes, so remaining mass in the vessel

$$m_2 = \frac{3}{4}m_1 \Rightarrow \frac{(273 + 60)}{T} = \frac{3/4 m_1}{m_1}$$

$$\Rightarrow T = 444\text{K} = 171^\circ\text{C}$$

24. (d) $n = \frac{PV}{kT}$ Now, $n' = \frac{(2P)(2V)}{kT} = 4 \frac{PV}{kT} = 4n$ or $\frac{n}{n'} = \frac{1}{4}$.

$$25. (b) \gamma_{\text{mix}} = \frac{\frac{\mu_1 \gamma_1}{\gamma_1 - 1} + \frac{\mu_2 \gamma_2}{\gamma_2 - 1}}{\frac{\mu_1}{\gamma_1 - 1} + \frac{\mu_2}{\gamma_2 - 1}} = \frac{\frac{3 \times 1.3}{(1.3 - 1)} + \frac{2 \times 1.4}{(1.4 - 1)}}{\frac{3}{(1.3 - 1)} + \frac{2}{(1.4 - 1)}} = 1.33$$

26. (d) In mixture gases will acquire thermal equilibrium (*i.e.*, same temperature) so their kinetic energies will also be same.

$$27. (b) V_t = V_0(1 + \alpha t) = 0.5 \left(1 + \frac{1}{273} \times 819 \right) = 2 \text{ litre} = 2 \times 10^{-3} \text{ m}^3$$



Chapter 14 Thermodynamics



Thermodynamics is a branch of science which deals with exchange of heat energy between bodies and conversion of the heat energy into mechanical energy and vice-versa.

Some Definitions

(i) Thermodynamic system

(i) It is a collection of an extremely large number of atoms or molecules

(ii) It is confined with in certain boundaries.

(iii) Anything outside the thermodynamic system to which energy or matter is exchanged is called its surroundings.

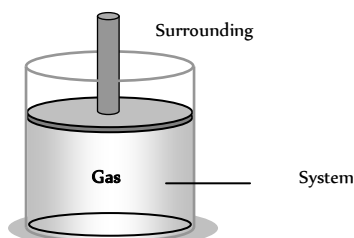


Fig. 14.1

(iv) Thermodynamic system may be of three types

(a) Open system : It exchange both energy and matter with the surrounding.

(b) Closed system : It exchange only energy (not matter) with the surroundings.

(c) Isolated system : It exchange neither energy nor matter with the surrounding.

(2) **Thermodynamic variables and equation of state** : A thermodynamic system can be described by specifying its pressure, volume, temperature, internal energy and the number of moles. These parameters are called thermodynamic variables. The relation between the thermodynamic variables (P , V , T) of the system is called equation of state.

For μ moles of an ideal gas, equation of state is $PV = \mu RT$ and for 1 mole of an it ideal gas is $PV = RT$

(3) **Thermodynamic equilibrium** : In steady state thermodynamic variables are independent of time and the system is said to be in the state of thermodynamic equilibrium. For a system to be in thermodynamic equilibrium, the following conditions must be fulfilled.

(i) Mechanical equilibrium : There is no unbalanced force between the system and its surroundings.

(ii) Thermal equilibrium : There is a uniform temperature in all parts of the system and is same as that of surrounding.

(iii) Chemical equilibrium : There is a uniform chemical composition through out the system and the surrounding.

(4) **Thermodynamic process** : The process of change of state of a system involves change of thermodynamic variables such as pressure P , volume V and temperature T of the system. The process is known as thermodynamic process. Some important processes are

(i) Isothermal process : Temperature remain constant

(ii) Adiabatic process : No transfer of heat

(iii) Isobaric process : Pressure remains constant

(iv) Isochoric (isovolumic process) : Volume remains constant

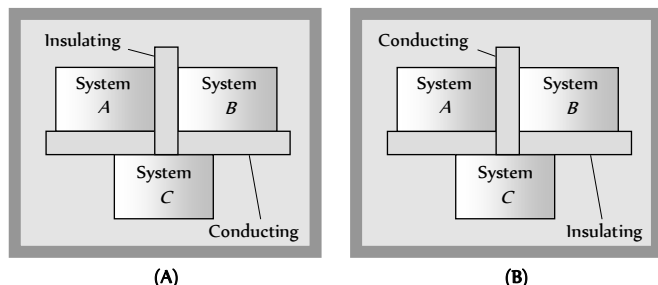
(v) Cyclic and non-cyclic process : Incyclic process Initial and final states are same while in non-cyclic process these states are different.

(vi) Reversible and irreversible process :

(5) **Indicator diagram** : Whenever the state of a gas (P, V, T) is changed, we say the gaseous system is undergone a thermodynamic process. The graphical representation of the change in state of a gas by a thermodynamic process is called indicator diagram. Indicator diagram is plotted generally in pressure and volume of gas.

Zeroth Law of Thermodynamics

If systems A and B are each in thermal equilibrium with a third system C , then A and B are in thermal equilibrium with each other.



(1) The zeroth law leads to the concept of temperature. All bodies in thermal equilibrium must have a common property which has the same value for all of them. This property is called the temperature.

(2) The zeroth law came to light long after the first and second laws of thermodynamics had been discovered and numbered. It is so named because it logically precedes the first and second laws of thermodynamics.

Heat, Internal Energy and Work in Thermodynamics

(1) **Heat (ΔQ)** : It is the energy that is transferred between a system and its environment because of the temperature difference between them.

(i) Heat is a path dependent quantity *e.g.* Heat required to change the temperature of a given gas at a constant pressure is different from that required to change the temperature of same gas through same amount at constant volume.

(ii) For gases when heat is absorbed and temperature changes $\Rightarrow \Delta Q = \mu C \Delta T$

$$\text{At constant pressure } (\Delta Q)_p = \mu C_p \Delta T$$

$$\text{At constant volume } (\Delta Q)_v = \mu C_v \Delta T$$

(2) **Internal energy (U)** : Internal energy of a system is the energy possessed by the system due to molecular motion and molecular configuration.

The energy due to molecular motion is called internal kinetic energy U_K and that due to molecular configuration is called internal potential energy U_P , *i.e.* Total internal energy $U = U_K + U_P$

(i) For an ideal gas, as there is no molecular attraction $U_P = 0$

i.e. internal energy of an ideal gas is totally kinetic and is given by

$$U = U_K = \frac{3}{2} \mu R T \text{ and change in internal energy } \Delta U = \frac{3}{2} \mu R \Delta T$$

(ii) In case of gases whatever be the process

$$\Delta U = \mu \frac{f}{2} R \Delta T = \mu C_v \Delta T = \mu \frac{R}{(\gamma - 1)} \Delta T = \frac{\mu R (T_f - T_i)}{\gamma - 1}$$

$$= \frac{\mu R T_f - \mu R T_i}{\gamma - 1} = \frac{(P_f V_f - P_i V_i)}{\gamma - 1}$$

(iii) Change in internal energy does not depend on the path of the process. So it is called a point function *i.e.* it depends only on the initial and final states of the system, *i.e.* $\Delta U = U_f - U_i$

(3) **Work (ΔW)** : Suppose a gas is confined in a cylinder that has a movable piston at one end. If P be the pressure of the gas in the cylinder, then force exerted by the gas on the piston of the cylinder $F = PA$ (A = Area of cross-section of piston)

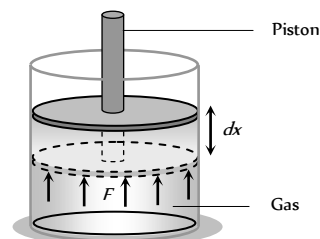


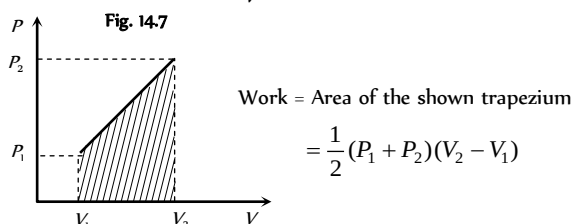
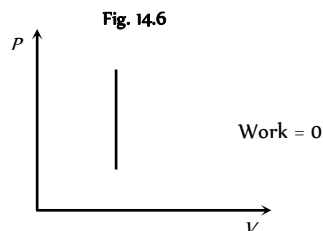
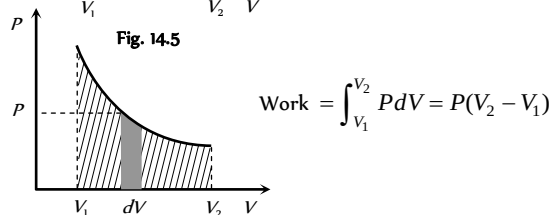
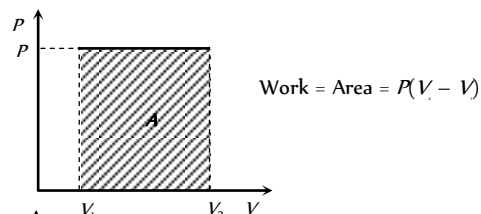
Fig. 14.4

When the piston is pushed outward an infinitesimal distance dx , the work done by the gas $dW = F \cdot dx = P(A \cdot dx) = P dV$

For a finite change in volume from V_i to V_f

$$\text{Total amount of work done } W = \int_{V_i}^{V_f} P dV = P(V_f - V_i)$$

(i) If we draw indicator diagram, the area bounded by PV -graph and volume axis represents the work done



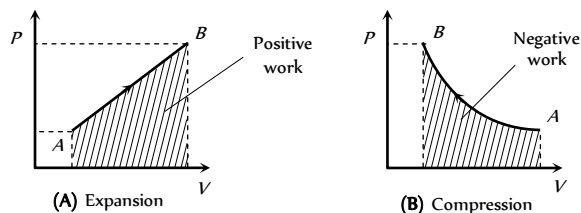
(ii) From $\Delta W = P \Delta V = P(V_f - V_i)$

If system expands against some external force then $V_f > V_i$

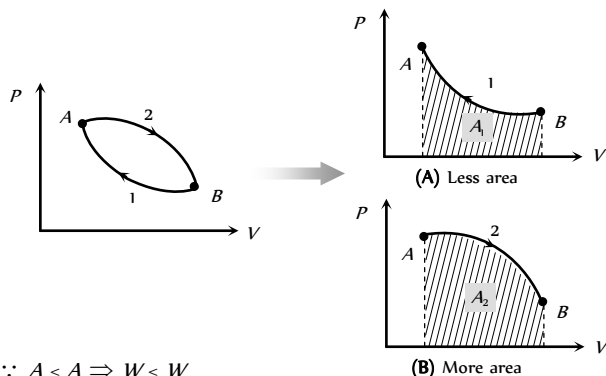
$\Rightarrow \Delta W = \text{positive}$

If system contracts because of external force then $V_f < V_i$

$\Rightarrow \Delta W = \text{negative}$



(iii) Like heat, work done is also depends upon initial and final state of the system and path adopted for the process



$\therefore A_1 < A_2 \Rightarrow W_1 < W_2$

(iv) In cyclic process, work done is equal to the area of closed curve. It is positive if the cycle is clockwise and it is negative if the cycle is anticlockwise.

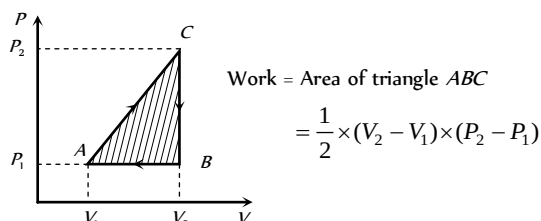


Fig. 14.11

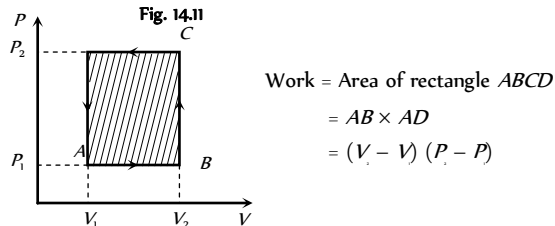


Fig. 14.12

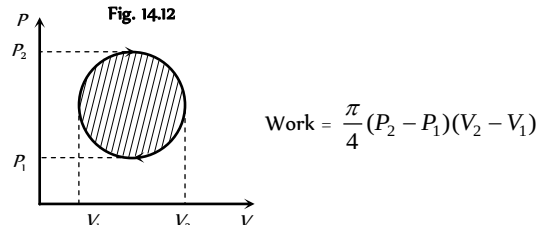


Fig. 14.13

First Law of Thermodynamics (FLOT)

(1) It is a statement of conservation of energy in thermodynamical process.

(2) According to it heat given to a system (ΔQ) is equal to the sum of increase in its internal energy (ΔU) and the work done (ΔW) by the system against the surroundings.

$$\Delta Q = \Delta U + \Delta W$$

(3) It makes no distinction between work and heat as according to it the internal energy (and hence temperature) of a system may be increased either by adding heat to it or doing work on it or both.

(4) ΔQ and ΔW are the path functions but ΔU is the point function.

(5) In the above equation all three quantities ΔQ , ΔU and ΔW must be expressed either in *Joule* or in *calorie*.

(6) The first law introduces the concept of internal energy.

(7) Limitation : First law of thermodynamics does not indicate the direction of heat transfer. It does not tell anything about the conditions, under which heat can be transformed into work and also it does not indicate as to why the whole of heat energy cannot be converted into mechanical work continuously.

Table 14.1 : Useful sign convention in thermodynamics

Quantity	Sign	Condition
ΔQ	+	When heat is supplied to a system
	-	When heat is drawn from the system
ΔW	+	When work done by the gas (expansion)
	-	When work done on the gas (compression)
ΔU	+	With temperature rise, internal energy increases
	-	With temperature fall, internal energy decreases

Isobaric Process

When a thermodynamic system undergoes a physical change in such a way that its pressure remains constant, then the change is known as isobaric process.

(1) **Equation of state** : In this process V and T changes but P remains constant. Hence Charles's law is obeyed in this process.

$$\text{Hence if pressure remains constant } V \propto T \Rightarrow \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

(2) **Indicator diagram** : Graph 1 represent isobaric expansion, graph 2 represent isobaric compression.

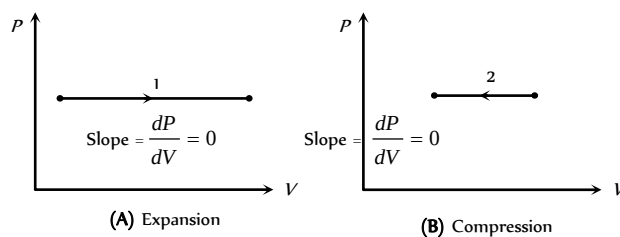


Fig. 14.14

(i) In isobaric expansion (Heating)

Temperature $\longrightarrow$ increases so ΔU is positive

Volume $\longrightarrow$ increases so ΔW is positive

Heat $\longrightarrow$ flows into the system so ΔQ is positive

(ii) In isobaric compression (Cooling)

Temperature $\longrightarrow$ decreases so ΔU is negative

Volume $\longrightarrow$ decreases so ΔW is negative

Heat $\longrightarrow$ flows out from the system so ΔQ is negative

(3) **Specific heat** : Specific heat of gas during isobaric process

$$C_p = \left(\frac{f}{2} + 1\right)R$$

(4) **Bulk modulus of elasticity** : $K = \frac{\Delta P}{-\frac{\Delta V}{V}} = 0$ [As $\Delta P = 0$]

(5) **Work done in isobaric process**

$$\Delta W = \int_{V_i}^{V_f} P dV = P \int_{V_i}^{V_f} dV = P[V_f - V_i] \quad [\text{As } P = \text{constant}]$$

$$\Rightarrow \Delta W = P(V_f - V_i) = \mu R[T_f - T_i] = \mu R \Delta T$$

(6) **FLOT in isobaric process** : From $\Delta Q = \Delta U + \Delta W$

$$\therefore \Delta U = \mu C_v \Delta T = \mu \frac{R}{(\gamma - 1)} \Delta T \quad \text{and} \quad \Delta W = \mu R \Delta T$$

$$\Rightarrow (\Delta Q)_p = \mu \frac{R}{(\gamma - 1)} \Delta T + \mu R \Delta T = \mu \left(\frac{\gamma}{\gamma - 1}\right) R \Delta T = \mu C_p \Delta T$$

(7) **Examples of isobaric process** : All state changes occurs at constant temperature and pressure.

Boiling of water

(i) Water $\longrightarrow$ vapours

(ii) Temperature $\longrightarrow$ constant

(iii) Volume $\longrightarrow$ increases

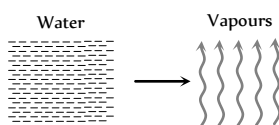


Fig. 14.15

(iv) A part of heat supplied is used to change volume (expansion) against external pressure and remaining part is used to increase its potential energy (kinetic energy remains constant)

$$(v) \text{ From FLOT } \Delta Q = \Delta U + \Delta W \Rightarrow mL = \Delta U + P(V_f - V_i)$$

Freezing of water

(i) Water $\longrightarrow$ ice

(ii) Temperature $\longrightarrow$ constant

(iii) Volume $\longrightarrow$ increases

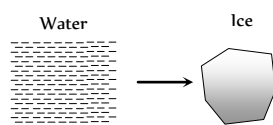


Fig. 14.16

(iv) Heat is given by water itself. It is used to do work against external atmospheric pressure and to decrease the internal potential energy.

$$(v) \text{ From FLOT } \Delta Q = \Delta U + \Delta W \Rightarrow -mL = \Delta U + P(V_f - V_i)$$

Isochoric or Isometric Process

When a thermodynamic process undergoes a physical change in such a way that its volume remains constant, then the change is known as isochoric process.

(i) **Equation of state** : In this process P and T changes but $V =$ constant. Hence Gay-Lussac's law is obeyed in this process i.e. $P \propto T \Rightarrow$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \text{constant}$$

(2) **Indicator diagram** : Graph 1 and 2 represent isometric increase in pressure at volume V_1 and isometric decrease in pressure at volume V_2

respectively and slope of indicator diagram $\frac{dP}{dV} = \infty$

(i) Isometric heating

(a) Pressure $\longrightarrow$ increases

(b) Temperature $\longrightarrow$ increases

(c) $\Delta Q \longrightarrow$ positive

(d) $\Delta U \longrightarrow$ positive

(ii) Isometric cooling

(a) Pressure $\longrightarrow$ decreases

(b) Temperature $\longrightarrow$ decreases

(c) $\Delta Q \longrightarrow$ negative

(d) $\Delta U \longrightarrow$ negative

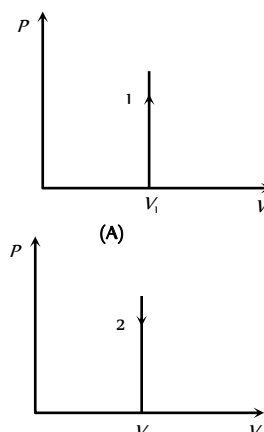


Fig. 14.17

(3) **Specific heat** : Specific heat of gas during isochoric process

$$C_v = \frac{f}{2}R$$

(4) **Bulk modulus of elasticity** : $K = \frac{\Delta P}{-\frac{\Delta V}{V}} = \frac{\Delta P}{0} = \infty$

(5) **Work done in isochoric process**

$$\Delta W = P\Delta V = P[V_f - V_i] = 0 \quad [\text{As } V = \text{constant}]$$

(6) **FLOT in isochoric process** : From $\Delta Q = \Delta U + \Delta W$

$$\therefore \Delta W = 0 \Rightarrow (\Delta Q)_v = \Delta U = \mu C_v \Delta T = \mu \frac{R}{\gamma - 1} \Delta T = \frac{P_f V_f - P_i V_i}{\gamma - 1}$$

Isothermal Process

When a thermodynamic system undergoes a physical change in such a way that its temperature remains constant, then the change is known as isothermal changes.

(i) **Essential condition for isothermal process**

(i) The walls of the container must be perfectly conducting to allow free exchange of heat between the gas and its surrounding.

(ii) The process of compression or expansion should be so slow so as to provide time for the exchange of heat.

Since these two conditions are not fully realised in practice, therefore, no process is perfectly isothermal.

(2) **Equation of state** : In this process, P and V change but $T =$ constant i.e. change in temperature $\Delta T = 0$.

Boyle's law is obeyed i.e. $PV = \text{constant} \Rightarrow P_f V_f = P_i V_i$

(3) **Example of isothermal process** : Melting of ice (at 0°C) and boiling of water (at 100°C) are common example of this process.

(4) **Indicator diagram** : According to $PV = \text{constant}$, graph between P and V is a part of rectangular hyperbola. The graphs at different temperature are parallel to each other are called isotherms.

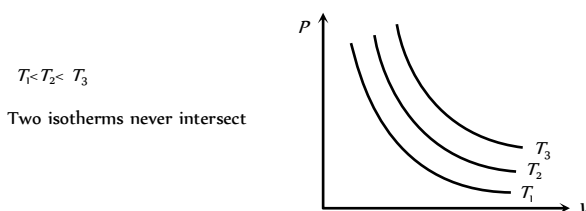


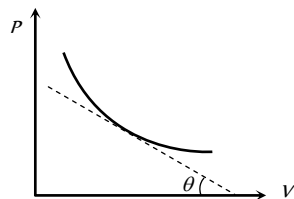
Fig. 14.18

(i) Slope of isothermal curve : By differentiating $PV = \text{constant}$. We get

$$P dV + V dP = 0$$

$$\Rightarrow P dV = -V dP$$

$$\Rightarrow \text{Slope} = \tan \theta = \frac{dP}{dV} = -\frac{P}{V}$$



(ii) Area between the isotherm and volume axis represents the work done in isothermal process.

If volume increases $\Delta W = +$ (Area under curve) and if volume decreases $\Delta W = -$ (Area under curve)

(5) **Specific heat** : Specific heat of gas during isothermal change is infinite. As $C = \frac{Q}{m \Delta T} = \frac{Q}{m \times 0} = \infty$ [As $\Delta T = 0$]

(6) **Isothermal elasticity (E_θ)** : For this process $PV = \text{constant}$.

$$\Rightarrow P dV = -V dP \Rightarrow P = \frac{dP}{-dV/V} = \frac{\text{Stress}}{\text{Strain}} = E_\theta$$

$$\Rightarrow E_\theta = P \text{ i.e. isothermal elasticity is equal to pressure}$$

At N.T.P., $E_\theta = \text{Atmospheric pressure} = 1.01 \times 10^5 \text{ N/m}^2$

(7) **Work done in isothermal process**

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{\mu RT}{V} dV \quad [\text{As } PV = \mu RT]$$

$$W = \mu RT \log_e \left(\frac{V_f}{V_i} \right) = 2.303 \mu RT \log_{10} \left(\frac{V_f}{V_i} \right)$$

$$\text{or } W = \mu RT \log_e \left(\frac{P_i}{P_f} \right) = 2.303 \mu RT \log_{10} \left(\frac{P_i}{P_f} \right)$$

(8) **FLOT in isothermal process** : From $\Delta Q = \Delta U + \Delta W$

$$\therefore \Delta U = 0 \quad [\text{As } \Delta T = 0] \Rightarrow \Delta Q = \Delta W$$

i.e. heat supplied in an isothermal change is used to do work against external surrounding.

or if the work is done on the system than equal amount of heat energy will be liberated by the system.

Adiabatic Process

When a thermodynamic system undergoes a change in such a way that no exchange of heat takes place between System and surroundings, the process is known as adiabatic process.

In this process P , V and T changes but $\Delta Q = 0$.

(1) **Essential conditions for adiabatic process**

(i) There should not be any exchange of heat between the system and its surroundings. All walls of the container and the piston must be perfectly insulating.

(ii) The system should be compressed or allowed to expand suddenly so that there is no time for the exchange of heat between the system and its surroundings.

Since, these two conditions are not fully realised in practice, so no process is perfectly adiabatic.

(2) **Some examples of adiabatic process**

(i) Sudden compression or expansion of a gas in a container with perfectly non-conducting walls.

(ii) Sudden bursting of the tube of bicycle tyre.

(iii) Propagation of sound waves in air and other gases.

(iv) Expansion of steam in the cylinder of steam engine.

(3) **FLOT in adiabatic process** : From $\Delta Q = \Delta U + \Delta W$

$$\text{For adiabatic process } \Delta Q = 0 \Rightarrow \Delta U = -\Delta W$$

If $\Delta W = \text{positive}$ then $\Delta U = \text{negative}$ so temperature decreases i.e. adiabatic expansion produce cooling.

If $\Delta W = \text{negative}$ then $\Delta U = \text{positive}$ so temperature increases i.e. adiabatic compression produce heating.

(4) **Equation of state** : In adiabatic change ideal gases do not obeys Boyle's law but obeys Poisson's law. According to it

$$PV^\gamma = \text{constant; where } \gamma = \frac{C_p}{C_v}$$

(i) For temperature and volume

$$TV^\gamma = \text{constant} \Rightarrow T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \text{ or } T \propto V^{1-\gamma}$$

(ii) For temperature and pressure

$$\frac{T^\gamma}{P^{\gamma-1}} = \text{const.} \Rightarrow T_1^\gamma P_1^{1-\gamma} = T_2^\gamma P_2^{1-\gamma} \text{ or } T \propto P^{\frac{\gamma-1}{\gamma}} \text{ or } P \propto T^{\frac{\gamma}{\gamma-1}}$$

Table 14.2 : Special cases of adiabatic process

Type of gas	$P \propto \frac{1}{V^\gamma}$	$P \propto T^{\frac{\gamma}{\gamma-1}}$	$T \propto \frac{1}{V^{\gamma-1}}$
Monoatomic $\gamma = 5/3$	$P \propto \frac{1}{V^{5/3}}$	$P \propto T^{5/2}$	$T \propto \frac{1}{V^{2/3}}$
Diatomic $\gamma = 7/5$	$P \propto \frac{1}{V^{7/5}}$	$P \propto T^{7/2}$	$T \propto \frac{1}{V^{2/5}}$
Polyatomic $\gamma = 4/3$	$P \propto \frac{1}{V^{4/3}}$	$P \propto T^4$	$T \propto \frac{1}{V^{1/3}}$

(5) **Indicator diagram**

(i) Curve obtained on PV graph are called adiabatic curve.

(ii) Slope of adiabatic curve : From $PV^\gamma = \text{constant}$

By differentiating, we get

$$dP V^\gamma + P \gamma V^{\gamma-1} dV = 0$$

$$\frac{dP}{dV} = -\gamma \frac{PV^{\gamma-1}}{V^\gamma} = -\gamma \left(\frac{P}{V} \right)$$

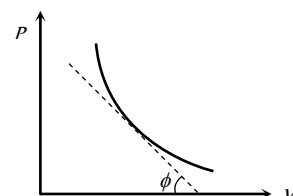


Fig. 14.20

$$\therefore \text{Slope of adiabatic curve } \tan \phi = -\gamma \left(\frac{P}{V} \right)$$

$$(iii) \text{ But we also know that slope of isothermal curve } \tan \theta = \frac{-P}{V}$$

$$\text{Hence } (\text{Slope})_{ad} = \gamma \times (\text{Slope})_{is} \text{ or } \frac{(\text{Slope})_{adi}}{(\text{Slope})_{iso}} > 1$$

(6) **Specific heat** : Specific heat of a gas during adiabatic change is

$$\text{zero As } C = \frac{Q}{m\Delta T} = \frac{0}{m\Delta T} = 0 \quad [\text{As } Q = 0]$$

(7) **Adiabatic elasticity (E_ϕ)** : $PV^\gamma = \text{constant}$

Differentiating both sides $d(PV^\gamma) + P\gamma V^{\gamma-1}dV = 0$

$$\gamma P = \frac{dP}{-dV/V} = \frac{\text{Stress}}{\text{Strain}} = E_\phi \Rightarrow E_\phi = \gamma P$$

i.e. adiabatic elasticity is γ times that of pressure

$$\text{Also isothermal elasticity } E_\theta = P \Rightarrow \frac{E_\phi}{E_\theta} = \gamma = \frac{C_p}{C_v}$$

i.e. the ratio of two elasticity of gases is equal to the ratio of two specific heats.

(8) **Work done in adiabatic process**

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{K}{V^\gamma} dV \Rightarrow W = \frac{[P_i V_i - P_f V_f]}{(\gamma - 1)} = \frac{\mu R(T_i - T_f)}{(\gamma - 1)}$$

(As $PV^\gamma = K$, $PV = \mu RT$, and $PV = \mu RT$)

(i) $W \propto$ quantity of gas (either M or μ)

(ii) $W \propto$ temperature difference ($T_i - T_f$)

$$(iii) W \propto \frac{1}{\gamma - 1} \quad \because \gamma_{mono} > \gamma_{di} > \gamma_{tri} \Rightarrow W_{-} < W_{-} < W_{-}$$

(9) **Comparison between isothermal and adiabatic indicator diagrams :**

Always remember that adiabatic curves are more steeper than isothermal curves

(i) Equal expansion : Graph 1 represent isothermal process and 2 represent adiabatic process

$$W_{isotherm} > W_{adiabatic}$$

$$P_{isotherm} > P_{adiabatic}$$

$$T_{isotherm} > T_{adiabatic}$$

$$(\text{Slope})_{isotherm} < (\text{Slope})_{adiabatic}$$

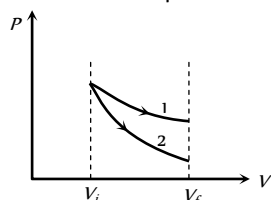


Fig. 14.21

(ii) Compression : Graph 1 represent adiabatic process and 2 represent isothermal process

$$W_{adiabatic} > W_{isotherm}$$

$$P_{adiabatic} > P_{isotherm}$$

$$T_{adiabatic} > T_{isotherm}$$

$$(\text{Slope})_{adiabatic} < (\text{Slope})_{isotherm}$$

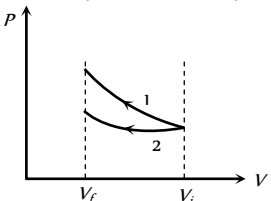


Fig. 14.22

(10) **Free expansion** : Free expansion is adiabatic process in which no work is performed on or by the system. Consider two vessels placed in a system which is enclosed with thermal insulation (asbestos-covered). One vessel contains a gas and the other is evacuated. When suddenly the stopcock is opened, the gas rushes into the evacuated vessel and expands freely.

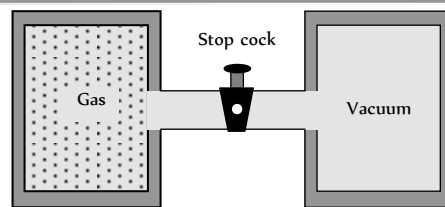


Fig. 14.23

$\Delta W = 0$ (Because walls are rigid)

$\Delta Q = 0$ (Because walls are insulated)

$\Delta U = U_f - U_i = 0$ (Because ΔQ and ΔW are zero. Thus the final and initial energies are equal in free expansion.)

Cyclic and Non-cyclic Process

A cyclic process consists of a series of changes which return the system back to its initial state.

In non-cyclic process the series of changes involved do not return the system back to its initial state.

$$(1) \text{ In case of cyclic process as } U_f = U_i \Rightarrow \Delta U = U_f - U_i = 0$$

i.e. change in internal energy for cyclic process is zero and also $\Delta U \propto \Delta T \Rightarrow \Delta T = 0$ i.e. temperature of system remains constant.

$$(2) \text{ From FLOT } \Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta W$$

i.e. heat supplied is equal to the work done by the system.

(3) For cyclic process P - V graph is a closed curve and area enclosed by the closed path represents the work done.

If the cycle is clockwise work done is positive and if the cycle is anticlockwise work done is negative.

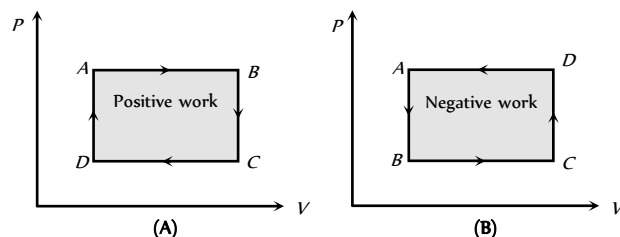


Fig. 14.24

(4) Work done in non cyclic process depends upon the path chosen or the series of changes involved and can be calculated by the area covered between the curve and volume axis on PV diagram.

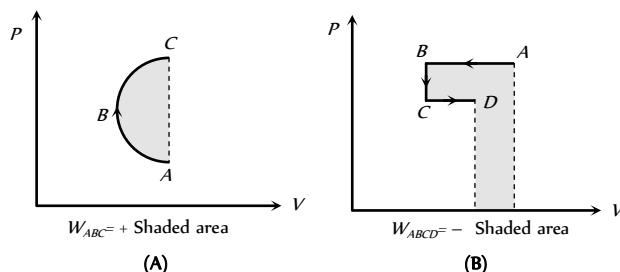


Fig. 14.25

Quasi Static Process

When we perform a process on a given system, its state is, in general, changed. Suppose the initial state of the system is described by the values P_1, V_1, T_1 and the final state by P_2, V_2, T_2 . If the process is performed in such a way that at any instant during the process, the system is very nearly in thermodynamic equilibrium, the process is called quasi-static. This means, we can specify the parameters P, V, T uniquely at any instant during such a process.

Actual processes are not quasi-static. To change the pressure of a gas, we can move a piston inside the enclosure. The gas near the piston is acted upon by piston. The pressure of the gas may not be uniform everywhere while the piston is moving. However, we can move the piston very slowly to make the process as close to quasi-static as we wish. Thus, a quasi-static process is an idealised process in which all changes take place infinitely slowly.

Reversible and Irreversible Process

(1) **Reversible process** : A reversible process is one which can be reversed in such a way that all changes occurring in the direct process are exactly repeated in the opposite order and inverse sense and no change is left in any of the bodies taking part in the process or in the surroundings. For example if heat is absorbed in the direct process, the same amount of heat should be given out in the reverse process, if work is done on the working substance in the direct process then the same amount of work should be done by the working substance in the reverse process. The conditions for reversibility are

(i) There must be complete absence of dissipative forces such as friction, viscosity, electric resistance etc.

(ii) The direct and reverse processes must take place infinitely slowly.

(iii) The temperature of the system must not differ appreciably from its surroundings.

Some examples of reversible process are

(a) All isothermal and adiabatic changes are reversible if they are performed very slowly.

(b) When a certain amount of heat is absorbed by ice, it melts. If the same amount of heat is removed from it, the water formed in the direct process will be converted into ice.

(c) An extremely slow extension or contraction of a spring without setting up oscillations.

(d) When a perfectly elastic ball falls from some height on a perfectly elastic horizontal plane, the ball rises to the initial height.

(e) If the resistance of a thermocouple is negligible there will be no heat produced due to Joule's heating effect. In such a case heating or cooling is reversible. At a junction where a cooling effect is produced due to Peltier effect when current flows in one direction and equal heating effect is produced when the current is reversed.

(f) Very slow evaporation or condensation.

It should be remembered that the conditions mentioned for a reversible process can never be realised in practice. Hence, a reversible process is only an ideal concept. In actual process, there is always loss of heat due to friction, conduction, radiation etc.

(2) **Irreversible process** : Any process which is not reversible exactly is an irreversible process. All natural processes such as conduction, radiation, radioactive decay etc. are irreversible. All practical processes such as free expansion, Joule-Thomson expansion, electrical heating of a wire are also irreversible. Some examples of irreversible processes are given below

(i) When a steel ball is allowed to fall on an inelastic lead sheet, its kinetic energy changes into heat energy by friction. The heat energy raises the temperature of lead sheet. No reverse transformation of heat energy occurs.

(ii) The sudden and fast stretching of a spring may produce vibrations in it. Now a part of the energy is dissipated. This is the case of irreversible process.

(iii) Sudden expansion or contraction and rapid evaporation or condensation are examples of irreversible processes.

(iv) Produced by the passage of an electric current through a resistance is irreversible.

(v) Heat transfer between bodies at different temperatures is also irreversible.

(vi) Joule-Thomson effect is irreversible because on reversing the flow of gas a similar cooling or heating effect is not observed.

Mixed Graphical Representation

(1) PV -graphs

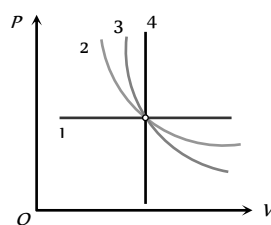


Fig. 14.26

1 $\longrightarrow$ Isobaric (P -constant)

2 $\longrightarrow$ Isothermal (Because $P \propto \frac{1}{V}$)

3 $\longrightarrow$ Adiabatic (Because $P \propto \frac{1}{V^\gamma}$)

4 $\longrightarrow$ Isochoric (V -constant)

(2) PT -graphs

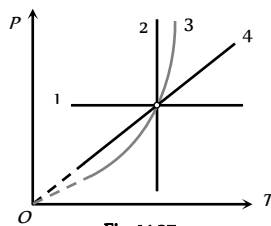


Fig. 14.27

1 $\longrightarrow$ Isobaric (P -constant)

2 $\longrightarrow$ Isothermal (T -constant)

3 $\longrightarrow$ Adiabatic (Because $P \propto T^{\frac{\gamma}{\gamma-1}}$)

4 $\longrightarrow$ Isochoric (In isochoric $P \propto T$)

(3) VT -graphs

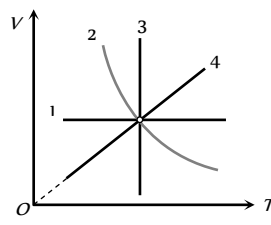


Fig. 14.28

1 $\longrightarrow$ Isochoric (V -constant)

2 $\longrightarrow$ Adiabatic (Because $V \propto T^{\frac{1}{1-\gamma}}$)

3 $\longrightarrow$ Isothermal (T -constant)

4 $\longrightarrow$ Isobaric (In isobaric $V \propto T$)

Heat Engine

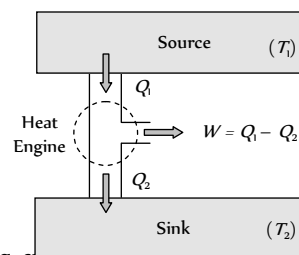
Heat engine is a device which converts heat into work continuously through a cyclic process.

The essential parts of a heat engine are

(1) **Source** : It is a reservoir of heat at high temperature and infinite thermal capacity. Any amount of heat can be extracted from it.

(2) **Working substance** : Steam, petrol etc.

(3) **Sink** : It is a reservoir of heat at low temperature and infinite thermal capacity. Any amount of heat can be given to the sink.



The working substance absorbs heat Q_1 from the source, does an amount of work W , returns the heat Q_2 to the sink and

Fig. 14.29

comes back to its original state and there occurs no change in its internal energy.

By repeating the same cycle over and over again, work is continuously obtained.

The performance of heat engine is expressed by means of "efficiency" η which is defined as the ratio of useful work obtained from the engine to the heat supplied to it.

$$\eta = \frac{\text{Work done}}{\text{Heat input}} = \frac{W}{Q_1}$$

For cyclic process $\Delta U = 0$ hence from FLOT $\Delta Q = \Delta W$

$$\text{So } W = Q_1 - Q_2 \Rightarrow \eta = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

A perfect heat engine is one which converts all heat into work *i.e.* $W = Q_1$ so that $Q_2 = 0$ and hence $\eta = 1$.

But practically efficiency of an engine is always less than 1.

Refrigerator or Heat Pump

A refrigerator or heat pump is basically a heat engine run in reverse direction.

It essentially consists of three parts

- (1) **Source** : At higher temperature T_1 .
- (2) **Working substance** : It is called refrigerant liquid ammonia and freon works as a working substance.
- (3) **Sink** : At lower temperature T_2 .

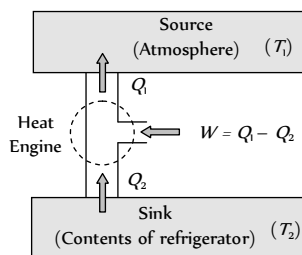


Fig. 14.30

The working substance takes heat Q_2 from a sink (contents of refrigerator) at lower temperature, has a net amount of work done W on it by an external agent (usually compressor of refrigerator) and gives out a larger amount of heat Q_1 to a hot body at temperature T_1 (usually atmosphere). Thus, it transfers heat from a cold to a hot body at the expense of mechanical energy supplied to it by an external agent. The cold body is thus cooled more and more.

The performance of a refrigerator is expressed by means of "coefficient of performance" β which is defined as the ratio of the heat extracted from the cold body to the work needed to transfer it to the hot body.

$$\text{i.e. } \beta = \frac{\text{Heat extracted}}{\text{Work done}} = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

A perfect refrigerator is one which transfers heat from cold to hot body without doing work

i.e. $W = 0$ so that $Q_1 = Q_2$ and hence $\beta = \infty$

- (1) **Carnot refrigerator** : For Carnot refrigerator $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

$$\Rightarrow \frac{Q_1 - Q_2}{Q_2} = \frac{T_1 - T_2}{T_2} \text{ or } \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

$$\text{So coefficient of performance } \beta = \frac{T_2}{T_1 - T_2}$$

where T_1 = temperature of surrounding, T_2 = temperature of cold body. It is clear that $\beta = 0$ when $T_2 = 0$

i.e. the coefficient of performance will be zero if the cold body is at the temperature equal to absolute zero.

(2) **Relation between coefficient of performance and efficiency of refrigerator**

$$\text{We know } \beta = \frac{Q_2}{Q_1 - Q_2} = \frac{Q_2 / Q_1}{1 - Q_2 / Q_1} \quad \dots (i)$$

$$\text{But the efficiency } \eta = 1 - \frac{Q_2}{Q_1} \text{ or } \frac{Q_2}{Q_1} = 1 - \eta \quad \dots (ii)$$

$$\text{From (i) and (ii) we get, } \beta = \frac{1 - \eta}{\eta}$$

Second Law of Thermodynamics

First law of thermodynamics merely explains the equivalence of work and heat. It does not explain why heat flows from bodies at higher temperatures to those at lower temperatures. It cannot tell us why the converse is possible. It cannot explain why the efficiency of a heat engine is always less than unity. It is also unable to explain why cool water on stirring gets hotter whereas there is no such effect on stirring warm water in a beaker. Second law of thermodynamics provides answers to these questions. Statement of this law is as follows

(1) **Clausius statement** : It is impossible for a self acting machine to transfer heat from a colder body to a hotter one without the aid of an external agency.

From Clausius statement it is clear that heat cannot flow from a body at low temperature to one at higher temperature unless work is done by an external agency. This statement is in fair agreement with our experiences in different branches of physics. For example, electrical current cannot flow from a conductor at lower electrostatic potential to that at higher potential unless an external work is done. Similarly, a body at a lower gravitational potential level cannot move up to higher level without work done by an external agency.

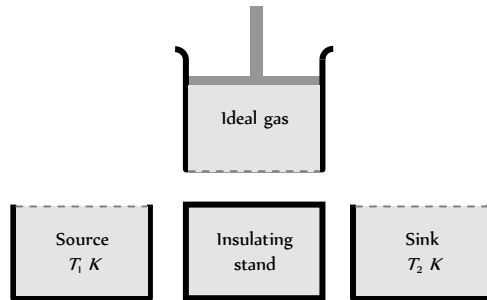
(2) **Kelvin's statement** : It is impossible for a body or system to perform continuous work by cooling it to a temperature lower than the temperature of the coldest one of its surroundings. A Carnot engine cannot work if the source and sink are at the same temperature because work done by the engine will result into cooling the source and heating the surroundings more and more.

(3) **Kelvin-Planck's statement** : It is impossible to design an engine that extracts heat and fully utilises into work without producing any other effect.

From this statement it is clear that any amount of heat can never be converted completely into work. It is essential for an engine to return some amount of heat to the sink. An engine essentially requires a source as well as sink. The efficiency of an engine is always less than unity because heat cannot be fully converted into work.

Carnot Engine

(i) Carnot designed a theoretical engine which is free from all the defects of a practical engine. This engine cannot be realised in actual practice, however, this can be taken as a standard against which the performance of an actual engine can be judged.



It consists of the following parts
Fig. 14.31

(i) A cylinder with perfectly non-conducting walls and a perfectly conducting base containing a perfect gas as working substance and fitted with a non-conducting frictionless piston

(ii) A source of infinite thermal capacity maintained at constant higher temperature T_1

(iii) A sink of infinite thermal capacity maintained at constant lower temperature T_2

(iv) A perfectly non-conducting stand for the cylinder.

(2) **Carnot cycle** : As the engine works, the working substance of the engine undergoes a cycle known as Carnot cycle. The Carnot cycle consists of the following four strokes

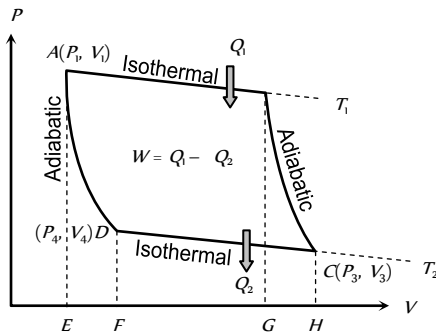


Fig. 14.32

(i) First stroke (Isothermal expansion) (curve AB) :

The cylinder containing ideal gas as working substance allowed to expand slowly at this constant temperature T_1 .

Work done = Heat absorbed by the system

$$W_1 = Q_1 = \int_{V_1}^{V_2} P dV = RT_1 \log_e \left(\frac{V_2}{V_1} \right) = \text{Area ABGE}$$

(ii) Second stroke (Adiabatic expansion) (curve BC) :

The cylinder is then placed on the non conducting stand and the gas is allowed to expand adiabatically till the temperature falls from T_1 to T_2 .

$$W_2 = \int_{V_2}^{V_3} P dV = \frac{R}{(\gamma - 1)} [T_1 - T_2] = \text{Area BCHG}$$

(iii) Third stroke (Isothermal compression) (curve CD) :

The cylinder is placed on the sink and the gas is compressed at constant temperature T_2 .

Work done = Heat released by the system

$$W_3 = Q_2 = - \int_{V_3}^{V_4} P dV = -RT_2 \log_e \frac{V_4}{V_3}$$

$$= RT_2 \log_e \frac{V_3}{V_4} = \text{Area CDFH}$$

(iv) Fourth stroke (adiabatic compression) (curve DA) : Finally the cylinder is again placed on non-conducting stand and the compression is continued so that gas returns to its initial stage.

$$W_4 = - \int_{V_4}^{V_1} P dV = - \frac{R}{\gamma - 1} (T_2 - T_1) = \frac{R}{\gamma - 1} (T_1 - T_2) = \text{Area ADFE}$$

(3) **Efficiency of Carnot cycle** : The efficiency of engine is defined as the ratio of work done to the heat supplied i.e.

$$\eta = \frac{\text{Work done}}{\text{Heat input}} = \frac{W}{Q_1}$$

Net work done during the complete cycle

$$W = W_1 + W_2 + (-W_3) + (-W_4) = W_1 - W_3 = \text{Area ABCD}$$

$$[\text{As } W_2 = W_4]$$

$$\therefore \eta = \frac{W}{Q_1} = \frac{W_1 - W_3}{W_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{W_3}{W_1} = 1 - \frac{Q_2}{Q_1}$$

$$\text{or } \eta = 1 - \frac{RT_2 \log_e (V_3 / V_4)}{RT_1 \log_e (V_2 / V_1)}$$

Since points B and C lie on same adiabatic curve

$$\therefore T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \text{ or } \frac{T_1}{T_2} = \left(\frac{V_3}{V_2} \right)^{\gamma-1} \quad \dots(i)$$

Also point D and A lie on the same adiabatic curve

$$\therefore T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1} \text{ or } \frac{T_1}{T_2} = \left(\frac{V_4}{V_1} \right)^{\gamma-1} \quad \dots(ii)$$

$$\text{From (i) and (ii), } \frac{V_3}{V_2} = \frac{V_4}{V_1} \text{ or } \frac{V_3}{V_4} = \frac{V_2}{V_1} \Rightarrow \log_e \left(\frac{V_3}{V_4} \right) = \log_e \left(\frac{V_2}{V_1} \right)$$

$$\text{So efficiency of Carnot engine } \eta = 1 - \frac{T_2}{T_1}$$

(i) Efficiency of a heat engine depends only on temperatures of source and sink and is independent of all other factors.

(ii) All reversible heat engines working between same temperatures are equally efficient and no heat engine can be more efficient than Carnot engine (as it is ideal).

(iii) As on Kelvin scale, temperature can never be negative (as 0 K is defined as the lowest possible temperature) and T_1 and T_2 are finite, efficiency of a heat engine is always lesser than unity, i.e., whole of heat can never be converted into work which is in accordance with second law.

(4) **Carnot theorem** : The efficiency of Carnot's heat engine depends only on the temperature of source (T) and temperature of sink (T), i.e.,

$$\eta = 1 - \frac{T_2}{T_1}$$

Carnot stated that no heat engine working between two given temperatures of source and sink can be more efficient than a perfectly reversible engine (Carnot engine) working between the same two

temperatures. Carnot's reversible engine working between two given temperatures is considered to be the most efficient engine.

Table 14.3 : Difference Between Petrol Engine and Diesel Engine

Petrol engine	Diesel engine
Working substance is a mixture of petrol vapour and air.	Working substance in this engine is a mixture of diesel vapour and air.
Efficiency is smaller (~47%).	Efficiency is larger (~55%).
It works with a spark plug.	It works with an oil plug.
It is associated with the risk of explosion, because petrol vapour and air is compressed. So, low compression ratio is kept.	No risk of explosion, because only air is compressed. Hence compression ratio is kept large.
Petrol vapour and air is created with spark plug.	Spray of diesel is obtained through the jet.

Entropy

Entropy is a measure of disorder of molecular motion of a system. Greater is the disorder, greater is the entropy.

The change in entropy *i.e.*

$$dS = \frac{\text{Heat absorbed by system}}{\text{Absolute temperature}} \text{ or } dS = \frac{dQ}{T}$$

The relation is called the mathematical form of Second Law of Thermodynamics.

(1) For solids and liquids

(i) When heat given to a substance changes its state at constant temperature, then change in entropy $dS = \frac{dQ}{T} = \pm \frac{mL}{T}$

where positive sign refers to heat absorption and negative sign to heat evolution.

(ii) When heat given to a substance raises its temperature from T_1 to T_2 , then change in entropy

$$dS = \int \frac{dQ}{T} = \int_{T_1}^{T_2} mc \frac{dT}{T} = mc \log_e \left(\frac{T_2}{T_1} \right)$$

$$\Rightarrow \Delta S = 2.303 mc \log_{10} \left(\frac{T_2}{T_1} \right)$$

(2) For a perfect gas : Perfect gas equation for n moles is $PV = nRT$

$$\Delta S = \int \frac{dQ}{T} = \int \frac{\mu C_V dT + P dV}{T} \quad [\text{As } dQ = dU + dW]$$

$$\Rightarrow \Delta S = \int \frac{\mu C_V dT + \frac{\mu RT}{V} dV}{T} \quad [\text{As } PV = \mu RT]$$

$$= \mu C_V \int_{T_1}^{T_2} \frac{dT}{T} + \mu R \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\therefore \Delta S = \mu C_V \log_e \left(\frac{T_2}{T_1} \right) + \mu R \log_e \left(\frac{V_2}{V_1} \right)$$

$$\text{In terms of } T \text{ and } P, \Delta S = \mu C_P \log_e \left(\frac{T_2}{T_1} \right) - \mu R \log_e \left(\frac{P_2}{P_1} \right)$$

$$\text{and in terms of } P \text{ and } V, \Delta S = \mu C_V \log_e \left(\frac{P_2}{P_1} \right) + \mu C_P \log_e \left(\frac{V_2}{V_1} \right)$$

✍ When a thermos bottle is vigorously shaken :

Heat transferred to the coffee $\Rightarrow \Delta Q = 0$

[As thermos flask is insulated from the surrounding]

Work is done on the coffee against viscous force

$\Delta W = (-)$

Internal energy of the coffee increases $\Delta U = (+)$

and temperature of the coffee also increases $\Delta T = (+)$



✍ Work done without the volume limits

$$W = \int_{V_i}^{V_f} P dV$$

From this equation it seems as if work done can be calculated only when P - V equation is known and limits V_i and V_f are known to us. But it is not so. We can calculate work done if we know the limits of temperature.

For μ moles of an ideal gas if $P = \frac{\alpha}{T}$ with temperature limits T_1 and $2T_1$,

$$\text{From } PV = \mu RT \Rightarrow V = \frac{\mu RT}{P} = \frac{\mu RT^2}{\alpha} \Rightarrow dV = \frac{2\mu RT}{\alpha} dT$$

$$\therefore W = \int_{V_i}^{V_f} P dV = \int_{T_0}^{2T_0} \left(\frac{\alpha}{T} \right) \left(\frac{2\mu RT}{\alpha} \right) dT = 2\mu RT_0$$

✍ Work done with spring : If mass less piston is attached to a spring of force constant K and a mass m is placed over the piston. If the external pressure is P_0 and due to expansion of gas the piston moves up through a distance x then

Total work done by the gas

$$W = W_1 + W_2 + W_3$$

where $W =$ Work done against

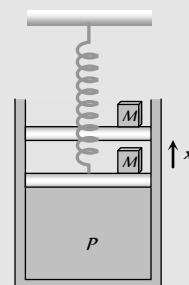
external pressure (P_0)

$W_1 =$ Work done against spring

force (Kx)

$W_2 =$ Work done against gravitational force (mg)

$$\therefore W = P_0 V + \frac{1}{2} Kx^2 + mgx$$



✍ The efficiency of an actual engine is much lesser than that of an ideal engine. Actually the practical efficiency of a steam engine is about (8-15)% while that of a petrol engine is 40%. The efficiency of a diesel engine is maximum and is about (50-55)%.

✍ When P and V bear the relation $PV^\gamma = \text{constant}$, where $\gamma \neq 1$ or γ the process is called a polytropic one. In this process the molar heat

capacity is, $C = C_V + \frac{R}{1-\gamma} = \frac{R}{\gamma-1} + \frac{R}{1-\gamma}$

✍ Enthalpy : Four quantities called "thermodynamic potentials" are useful in the chemical thermodynamics of reactions and non-cyclic processes. They are internal energy, the enthalpy, the Helmholtz free energy and the Gibbs free energy. Enthalpy is defined by

$$H = U + PV$$

where P and V are the pressure and volume, and U is internal energy, Enthalpy is somewhat parallel to the first law of thermodynamics for a constant pressure system $Q = \Delta U + P\Delta V$ since in this case $Q = \Delta H$.

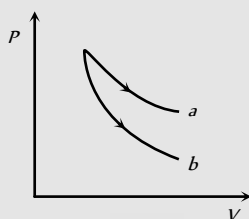
✍ Confusion about FLOT

It is typical for chemistry texts to write the first law as

$$\Delta U = \Delta Q + \Delta W$$

It is the same law, of course the thermodynamic expression of the conservation of energy principle. It is just that W is defined as the work done on the system instead of work done by the system. In the context of physics, the common scenario is one of adding heat to a volume of gas and using the expansion of that gas to do work, as in the pushing down of a piston in an internal combustion engine. In the context of chemical reactions and process, it may be more common to deal with situations where work is done on the system rather than by it.

✍ Possibilities



If $a \longrightarrow$ Isothermal then $b \longrightarrow$ Must be adiabatic

But If $b \longrightarrow$ adiabatic then it is not compulsory that a must be isothermal, it may be adiabatic also.

Ordinary Thinking

Objective Questions

First Law of Thermodynamics ($\Delta Q = \Delta U + \Delta W$)

- First law of thermodynamics is given by [CPMT 1977, 91]
 - $dQ = dU + PdV$
 - $dQ = dU \times PdV$
 - $dQ = (dU + dV)P$
 - $dQ = PdU + dV$
- The internal energy of an ideal gas depends upon [RPMT 1997; MP PMT 1999; CPMT 2003]
 - Specific volume
 - Pressure
 - Temperature
 - Density
- In changing the state of thermodynamics from A to B state, the heat required is Q and the work done by the system is W . The change in its internal energy is [MP PMT 1986; AMU (Med.) 2001]
 - $Q + W$
 - $Q - W$
 - Q
 - $\frac{Q - W}{2}$
- Heat given to a system is 35 joules and work done by the system is 15 joules. The change in the internal energy of the system will be
 - 50 J
 - 20 J
 - 30 J
 - 50 J
- The temperature of an ideal gas is kept constant as it expands. The gas does external work. During this process, the internal energy of the gas [MP PMT 1990]
 - Decreases
 - Increases
 - Remains constant
 - Depends on the molecular motion
- The first law of thermodynamics is concerned with the conservation of [MP PMT 1987; CBSE PMT 1990, 92; AFMC 1997; CPMT 1999; BHU 1999; DCE 2000; BCECE 2003]
 - Momentum
 - Energy
 - Mass
 - Temperature
- A thermodynamic system goes from states (i) P_1, V to $2P_1, V$ (ii) P, V to $P, 2V$. Then work done in the two cases is [MP PMT 1990]
 - Zero, Zero
 - Zero, PV_1
 - PV_1 , Zero
 - PV_1, P_1V_1
- If the amount of heat given to a system be 35 joules and the amount of work done by the system be -15 joules, then the change in the internal energy of the system is [MP PMT 1989]
 - 50 joules
 - 20 joules
 - 30 joules
 - 50 joules
- A system is given 300 calories of heat and it does 600 joules of work. How much does the internal energy of the system change in this process [MP PET 1991]
 $(J = 4.18 \text{ joules/cal})$
 - 654 Joule
 - 156.5 Joule
 - 300 Joule
 - 528.2 Joule
- Work done on or by a gas, in general depends upon the
 - Initial state only
 - Final state only
 - Both initial and final states only
 - Initial state, final state and the path
- If R = universal gas constant, the amount of heat needed to raise the temperature of 2 mole of an ideal monoatomic gas from 273K to 373K when no work is done [MP PET 1990]
 - 100 R
 - 150 R
 - 300 R
 - 500 R
- Find the change in internal energy of the system when a system absorbs 2 kilocalorie of heat and at the same time does 500 joule of work [EAMCET 1984]
 - 7900 J
 - 8200 J
 - 5600 J
 - 6400 J

[MP PET/PMT 1988]

13. A system performs work ΔW when an amount of heat is ΔQ added to the system, the corresponding change in the internal energy is ΔU . A unique function of the initial and final states (irrespective of the mode of change) is [CPMT 1981; J & KCET 2004]
- (a) ΔQ (b) ΔW
(c) ΔU and ΔQ (d) ΔU
14. A container of volume 1 m^3 is divided into two equal compartments by a partition. One of these compartments contains an ideal gas at 300 K . The other compartment is vacuum. The whole system is thermally isolated from its surroundings. The partition is removed and the gas expands to occupy the whole volume of the container. Its temperature now would be [Manipal MEE 1995]
- (a) 300 K (b) 239 K
(c) 200 K (d) 100 K
15. 110 J of heat is added to a gaseous system, whose internal energy change is 40 J , then the amount of external work done is [CBSE PMT 1993; DPMT 1996, 1997; AFMC 1999; JIPMER 2000; MH CET 2000; Pb. PMT 2003]
- (a) 150 J (b) 70 J
(c) 110 J (d) 40 J
16. Which of the following is not thermodynamical function [CBSE PMT 1993; CPMT 2001; DCE 1996; 2001]
- (a) Enthalpy (b) Work done
(c) Gibb's energy (d) Internal energy
17. When the amount of work done is 333 cal and change in internal energy is 167 cal , then the heat supplied is [AFMC 1998]
- (a) 166 cal (b) 333 cal
(c) 500 cal (d) 400 cal
18. First law thermodynamics states that [KCET 1999]
- (a) System can do work
(b) System has temperature
(c) System has pressure
(d) Heat is a form of energy
19. A thermo-dynamical system is changed from state (P_1, V_1) to (P_2, V_2) by two different process. The quantity which will remain same will be [RPET 1999]
- (a) ΔQ (b) ΔW
(c) $\Delta Q + \Delta W$ (d) $\Delta Q - \Delta W$
20. In thermodynamic process, 200 Joules of heat is given to a gas and 100 Joules of work is also done on it. The change in internal energy of the gas is [AMU (Engg.) 1999]
- (a) 100 J (b) 300 J
(c) 419 J (d) 24 J
21. A perfect gas contained in a cylinder is kept in vacuum. If the cylinder suddenly bursts, then the temperature of the gas
- (a) Remains constant (b) Becomes zero
(c) Increases (d) Decreases
22. If 150 J of heat is added to a system and the work done by the system is 110 J , then change in internal energy will be [AMU (Engg.) 1999; BHU 2000]
- (a) 260 J (b) 150 J
(c) 110 J (d) 40 J
23. If ΔQ and ΔW represent the heat supplied to the system and the work done on the system respectively, then the first law of thermodynamics can be written as [Roorkee 2000]
- (a) $\Delta Q = \Delta U + \Delta W$ (b) $\Delta Q = \Delta U - \Delta W$
(c) $\Delta Q = \Delta W - \Delta U$ (d) $\Delta Q = -\Delta W - \Delta U$
- where ΔU is the internal energy
24. For free expansion of the gas which of the following is true [AMU (Med.) 2000]
- (a) $Q = 0, W > 0$ and $\Delta E_{\text{int}} = -W$
(b) $Q = 0, W > 0$ and $\Delta E_{\text{int}} = -W$
(c) $W = 0, Q > 0$, and $\Delta E_{\text{int}} = Q$
(d) $W > 0, Q < 0$ and $\Delta E_{\text{int}} = 0$
25. Which of the following can not determine the state of a thermodynamic system [AFMC 2001]
- (a) Pressure and volume
(b) Volume and temperature
(c) Temperature and pressure
(d) Any one of pressure, volume or temperature
26. Which of the following is not a thermodynamics co-ordinate [AIIMS 2001]
- (a) P (b) T
(c) V (d) R
27. In a given process for an ideal gas, $dW = 0$ and $dQ < 0$. Then for the gas [IIT-JEE (Screening) 2001]
- (a) The temperature will decrease
(b) The volume will increase
(c) The pressure will remain constant
(d) The temperature will increase
28. The specific heat of hydrogen gas at constant pressure is $C_p = 3.4 \times 10^3\text{ cal/kg } ^\circ\text{C}$ and at constant volume is $C_v = 2.4 \times 10^3\text{ cal/kg } ^\circ\text{C}$. If one kilogram hydrogen gas is heated from 10°C to 20°C at constant pressure, the external work done on the gas to maintain it at constant pressure is
- (a) 10^5 cal (b) 10^4 cal
(c) 10^3 cal (d) $5 \times 10^3\text{ cal}$
29. Which of the following parameters does not characterize the thermodynamic state of matter [CPMT 2001; AIEEE 2003]
- (a) Volume (b) Temperature

- (c) Pressure (d) Work
30. In a thermodynamic system working substance is ideal gas, its internal energy is in the form of [MP PMT 2003]
 (a) Kinetic energy only
 (b) Kinetic and potential energy
 (c) Potential energy
 (d) None of these
31. Which of the following statements is correct for any thermodynamic system [AIEEE 2004]
 (a) The internal energy changes in all processes
 (b) Internal energy and entropy are state functions
 (c) The change in entropy can never be zero
 (d) The work done in an adiabatic process is always zero
32. A system is provided with 200 cal of heat and the work done by the system on the surrounding is 40 J. Then its internal energy
 (a) Increases by 600 J (b) Decreases by 800 J
 (c) Increases by 800 J (d) Decreases by 50 J
33. In a thermodynamic process, pressure of a fixed mass of a gas is changed in such a manner that the gas molecules gives out 20 J of heat and 10 J of work is done on the gas. If the initial internal energy of the gas was 40 J, then the final internal energy will be
 (a) 30 J (b) 20 J
 (c) 60 J (d) 40 J
34. Heat is not being exchanged in a body. If its internal energy is increased, then [RPMT 2002]
 (a) Its temperature will increase
 (b) Its temperature will decrease
 (c) Its temperature will remain constant
 (d) None of these
35. Out of the following which quantity does not depend on path
 (a) Temperature (b) Energy
 (c) Work (d) None of these
36. First law of thermodynamics is a special case of [CPMT 1985; RPET 2000; DCE 2000; CBSE PMT 2000; AIEEE 2002; AFMC 2002]
 (a) Newton's law
 (b) Law of conservation of energy
 (c) Charle's law
 (d) Law of heat exchange
37. One mole of an ideal monoatomic gas is heated at a constant pressure of one atmosphere from 0°C to 100°C . Then the change in the internal energy is [Pb. PMT 2001]
 (a) 6.56 joules (b) 8.32×10^2 joules
 (c) 12.48×10^2 joules (d) 20.80 joules
38. If the ratio of specific heat of a gas at constant pressure to that at constant volume is γ , the change in internal energy of a mass of gas, when the volume changes from V to $2V$ constant pressure p , is
 (a) $R/(\gamma-1)$ (b) pV
 (c) $pV/(\gamma-1)$ (d) $\gamma pV/(\gamma-1)$
39. If $C_V = 4.96 \text{ cal/mole } K$, then increase in internal energy when temperature of 2 moles of this gas is increased from 340 K to 342 K [RPET 1997]
 (a) 27.80 cal (b) 19.84 cal
 (c) 13.90 cal (d) 9.92 cal
40. Temperature is a measurement of coldness or hotness of an object. This definition is based on [RPET 2003]
 (a) Zeroth law of thermodynamics
 (b) First law of thermodynamics
 (c) Second law of thermodynamics
 (d) Newton's law of cooling
41. When heat energy of 1500 Joules, is supplied to a gas at constant pressure $2.1 \times 10^5 \text{ N/m}^2$, there was an increase in its volume equal to $2.5 \times 10^{-3} \text{ m}^3$. The increase in internal energy of the gas in Joules is [EAMCET (Engg.) 1999]
 (a) 450 [Orissa PMT 2004] (b) 525
 (c) 975 (d) 2025
42. If heat given to a system is 6 kcal and work done is 6 kJ. Then change in internal energy is [BHU Med. 2000]
 (a) 19.1 kJ (b) 12.5 kJ
 (c) 25 kJ [DPMT 2004] (d) Zero
43. In a thermodynamics process, pressure of a fixed mass of a gas is changed in such a manner that the gas releases 20 J of heat and 8 J of work is done on the gas. If the initial internal energy of the gas was 30 J. The final internal energy will be
 (a) 18 J (b) 9 J
 (c) 4.5 J (d) 36 J
44. A monoatomic gas of n -moles is heated from temperature T to T' under two different conditions (i) at constant volume and (ii) at constant pressure. The change in internal energy of the gas is [RPET 2002]
 (a) More for (i)
 (b) More for (ii)
 (c) Same in both cases
 (d) Independent of number of moles
45. The state of a thermodynamic system is represented by [MH CET 2004]
 (a) Pressure only
 (b) Volume only
 (c) Pressure, volume and temperature
 (d) Number of moles
46. A perfect gas goes from state A to another state B by absorbing $8 \times 10^5 \text{ J}$ of heat and doing $6.5 \times 10^5 \text{ J}$ of external work. It is now transferred between the same two states in another process in which it absorbs 10^5 J of heat. Then in the second process
 (a) Work done on the gas is $0.5 \times 10^5 \text{ J}$
 (b) Work done by gas is $0.5 \times 10^5 \text{ J}$
 (c) Work done on gas is 10^5 J [CBSE PMT 1998]
 (d) Work done by gas is 10^5 J
47. If a system undergoes contraction of volume then the work done by the system will be [BHU 1999]
 (a) Zero (b) Negligible

- (c) Negative (d) Positive
48. Which of the following is *incorrect* regarding the first law of thermodynamics [AIEEE 2005]
- (a) It introduces the concept of the internal energy
(b) It introduces the concept of the entropy
(c) It is not applicable to any cyclic process
(d) None of the above

Isothermal Process

1. For an ideal gas, in an isothermal process [BHU 1998]
- (a) Heat content remains constant
(b) Heat content and temperature remain constant
(c) Temperature remains constant
(d) None of the above
2. Can two isothermal curves cut each other
- (a) Never
(b) Yes
(c) They will cut when temperature is 0°C
(d) Yes, when the pressure is critical pressure
3. In an isothermal expansion [KCET 2000; AFMC 2001]
- (a) Internal energy of the gas increases
(b) Internal energy of the gas decreases
(c) Internal energy remains unchanged
(d) Average kinetic energy of gas molecule decreases
4. In an isothermal reversible expansion, if the volume of 96 gm of oxygen at 27°C is increased from 70 litres to 140 litres, then the work done by the gas will be
- (a) $300 R \log_{10} 2$ (b) $81 R \log_e 2$
(c) $900 R \log_{10} 2$ (d) $2.3 \times 900 R \log_{10} 2$
5. A vessel containing 5 litres of a gas at 0.8 m pressure is connected to an evacuated vessel of volume 3 litres. The resultant pressure inside will be (assuming whole system to be isolated)
- (a) $4/3$ m (b) 0.5 m
(c) 2.0 m (d) $3/4$ m
6. For an isothermal expansion of a perfect gas, the value of $\frac{\Delta P}{P}$ is equal [CPMT 1980]
- (a) $-\gamma^{1/2} \frac{\Delta V}{V}$ (b) $-\frac{\Delta V}{V}$
(c) $-\gamma \frac{\Delta V}{V}$ (d) $-\gamma^2 \frac{\Delta V}{V}$
7. The gas law $\frac{PV}{T} = \text{constant}$ is true for [MNR 1974; MP PMT 1984; BHU 1995, 98, 2000]
- (a) Isothermal changes only
(b) Adiabatic changes only
(c) Both isothermal and adiabatic changes
(d) Neither isothermal nor adiabatic changes
8. One mole of O_2 gas having a volume equal to 22.4 litres at 0°C and 1 atmospheric pressure is compressed isothermally so that its volume reduces to 11.2 litres. The work done in this process is
- (a) 1672.5 J (b) 1728 J
(c) -1728 J (d) -1572.5 J
9. If a gas is heated at constant pressure, its isothermal compressibility
- (a) Remains constant
(b) Increases linearly with temperature
(c) Decreases linearly with temperature
(d) Decreases inversely with temperature
10. Work done per mol in an isothermal change is [RPMT 2004; BCECE 2005]
- (a) $RT \log_{10} \frac{V_2}{V_1}$ (b) $RT \log_{10} \frac{V_1}{V_2}$
(c) $RT \log_e \frac{V_2}{V_1}$ (d) $RT \log_e \frac{V_1}{V_2}$
11. The isothermal Bulk modulus of an ideal gas at pressure P is [CPMT 1974, 81; UPSEAT 1998; IIT 1998]
- (a) P (b) γP
(c) P/γ (d) P/γ
12. In isothermal expansion, the pressure is determined by [AFMC 1995]
- (a) Temperature only
(b) Compressibility only
(c) Both temperature and compressibility
(d) None of these
13. The isothermal bulk modulus of a perfect gas at normal pressure is
- (a) $1.013 \times 10^5 \text{ N/m}^2$ (b) $1.013 \times 10^6 \text{ N/m}^2$
(c) $1.013 \times 10^{11} \text{ N/m}^2$ (d) $1.013 \times 10^{11} \text{ N/m}^2$
14. In an isothermal change, an ideal gas obeys [MP PMT 1993] [EAMCET 1994; CPMT 1999]
- (a) Boyle's law (b) Charles's law
(c) Gaylussac law (d) None of the above
15. In isothermic process, which statement is wrong [RPMT 1997]
- (a) Temperature is constant
(b) Internal energy is constant
(c) No exchange of energy
(d) (a) and (b) are correct
16. An ideal gas A and a real gas B have their volumes increased from V to $2V$ under isothermal conditions. The increase in internal energy [CBSE PMT 1993; JIPMER 2001, 02]
- (a) Will be same in both A and B
(b) Will be zero in both the gases
(c) Of B will be more than that of A
(d) Of A will be more than that of B
17. The specific heat of a gas in an isothermal process is [AFMC 1998]
- (a) Infinite (b) Zero
(c) Negative (d) Remains constant

18. A thermally insulated container is divided into two parts by a screen. In one part the pressure and temperature are P and T for an ideal gas filled. In the second part it is vacuum. If now a small hole is created in the screen, then the temperature of the gas will
- (a) Decrease (b) Increase
(c) Remain same (d) None of the above
19. A container that suits the occurrence of an isothermal process should be made of [Pb. PMT 2000]
- (a) Copper (b) Glass
(c) Wood (d) Cloth
20. In an isothermal process the volume of an ideal gas is halved. One can say that [MP PMT 2004]
- (a) Internal energy of the system decreases
(b) Work done by the gas is positive
(c) Work done by the gas is negative
(d) Internal energy of the system increases
21. A thermodynamic process in which temperature T of the system remains constant though other variable P and V may change, is called [Pb. PMT 2004]
- (a) Isochoric process (b) Isothermal process
(c) Isobaric process (d) None of these
22. If an ideal gas is compressed isothermally then [RPMT 2003]
- (a) No work is done against gas
(b) Heat is released by the gas
(c) The internal energy of gas will increase
(d) Pressure does not change
23. When an ideal gas in a cylinder was compressed isothermally by a piston, the work done on the gas was found to be 1.5×10^4 joules. During this process about [MP PMT 1987]
- (a) 3.6×10^3 cal of heat flowed out from the gas
(b) 3.6×10^3 cal of heat flowed into the gas
(c) 1.5×10^4 cal of heat flowed into the gas
(d) 1.5×10^4 cal of heat flowed out from the gas
24. When heat is given to a gas in an isothermal change, the result will be [MP PET 1995; RPMT 1997]
- (a) External work done
(b) Rise in temperature
(c) Increase in internal energy
(d) External work done and also rise in temp.
25. When 1 gm of water at 0°C and $1 \times 10^5 \text{ N/m}^2$ pressure is converted into ice of volume 1.091 cm^3 , the external work done will be
- (a) 0.0091 joule (b) 0.0182 joule
(c) - 0.0091 joule (d) - 0.0182 joule
26. The latent heat of vaporisation of water is 2240 J/gm . If the work done in the process of expansion of 1 g is 168 J, then increase in internal energy is [Pb. PET 1998; CPMT 2000]
- (a) 2408 J (b) 2240 J
(c) 2072 J (d) 1904 J
27. 540 calories of heat convert 1 cubic centimeter of water at 100°C into 1671 cubic centimeter of steam at 100°C at a pressure of one atmosphere. Then the work done against the atmospheric pressure is nearly [MP PET 1999]
- (a) 540 cal (b) 40 cal
(c) Zero cal (d) 500 cal
28. One mole of an ideal gas expands at a constant temperature of 300 K from an initial volume of 10 litres to a final volume of 20 litres. The work done in expanding the gas is
($R = 8.31 \text{ J/mole-K}$) [MP PMT 1995; UPSEAT 2000]
- (a) 750 joules (b) 1728 joules
(c) 1500 joules (d) 3456 joules
29. A cylinder fitted with a piston contains 0.2 moles of air at temperature 27°C . The piston is pushed so slowly that the air within the cylinder remains in thermal equilibrium with the surroundings. Find the approximate work done by the system if the final volume is twice the initial volume [BHU (Med.) 2000]
- (a) 543 J (b) 345 J
(c) 453 J (d) 600 J
30. The volume of an ideal gas is 1 litre and its pressure is equal to 72 cm of mercury column. The volume of gas is made 900 cm by compressing it isothermally. The stress of the gas will be
- (a) 8 cm (mercury) (b) 7 cm (mercury)
(c) 6 cm (mercury) (d) 4 cm (mercury)
31. During an isothermal expansion of an ideal gas [UPSEAT 2005]
- (a) Its internal energy decreases
(b) Its internal energy does not change
(c) The work done by the gas is equal to the quantity of heat absorbed by it
(d) Both (b) and (c) are correct

Adiabatic Process

1. If a cylinder containing a gas at high pressure explodes, the gas undergoes [MP PET/PMT 1988]
- (a) Reversible adiabatic change and fall of temperature
(b) Reversible adiabatic change and rise of temperature
(c) Irreversible adiabatic change and fall of temperature
(d) Irreversible adiabatic change and rise of temperature
2. The work done in an adiabatic change in a gas depends only on [CPMT 1971; MP PET 1995]
- (a) Change in pressure (b) Change in volume
(c) Change in temperature (d) None of the above
3. In adiabatic expansion [DPMT 1999]
- (a) $\Delta U = 0$ (b) $\Delta U = \text{negative}$
(c) $\Delta U = \text{positive}$ (d) $\Delta W = \text{zero}$
4. The pressure in the tyre of a car is four times the atmospheric pressure at 300 K. If this tyre suddenly bursts, its new temperature will be ($\gamma = 1.4$) [RPMT 1996; MP PMT 1990]
- (a) $300(4)^{1.4/0.4}$ (b) $300\left(\frac{1}{4}\right)^{-0.4/1.4}$
(c) $300(2)^{-0.4/1.4}$ (d) $300(4)^{-0.4/1.4}$

5. A gas at NTP is suddenly compressed to one-fourth of its original volume. If γ is supposed to be $\frac{3}{2}$, then the final pressure is
- (a) 4 atmosphere (b) $\frac{3}{2}$ atmosphere
(c) 8 atmosphere (d) $\frac{1}{4}$ atmosphere
6. A monoatomic gas ($\gamma = 5/3$) is suddenly compressed to $\frac{1}{8}$ of its original volume adiabatically, then the pressure of the gas will change to
[CPMT 1976, 83; MP PMT 1994; DPMT 1996; Roorkee 2000; KCET 2000; Pb. PMT 1999, 2001]
- (a) $\frac{24}{5}$
(b) 8
(c) $\frac{40}{3}$
(d) 32 times its initial pressure
7. The pressure and density of a diatomic gas ($\gamma = 7/5$) change adiabatically from (P, d) to (P', d') . If $\frac{d'}{d} = 32$, then $\frac{P'}{P}$ should be [CPMT 1982; EAMCET 2001]
- (a) $1/128$ (b) 32
(c) 128 (d) None of the above
8. An ideal gas at 27°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. If $\gamma = \frac{5}{3}$, then the rise in temperature is [CPMT 1984; CBSE PMT 1999; DPMT 2000; BHU 2001; Pb. PET 2001; UPSEAT 2002, 03; KCET 2003;]
- (a) 450 K (b) 375 K
(c) 225 K (d) 405 K
9. Two identical samples of a gas are allowed to expand (i) isothermally (ii) adiabatically. Work done is [MNR 1998]
- (a) More in the isothermal process
(b) More in the adiabatic process
(c) Neither of them
(d) Equal in both processes
10. Which is the correct statement [MP PMT 1993]
- (a) For an isothermal change $PV = \text{constant}$
(b) In an isothermal process the change in internal energy must be equal to the work done
(c) For an adiabatic change $\frac{P_2}{P_1} = \left(\frac{V_2}{V_1}\right)^\gamma$, where γ is the ratio of specific heats
(d) In an adiabatic process work done must be equal to the heat entering the system
11. The slopes of isothermal and adiabatic curves are related as
[CPMT 1971; BHU 1996; MH CET 1999; UPSEAT 2000; RPET 2003]
- (a) Isothermal curve slope = adiabatic curve slope
(b) Isothermal curve slope = $\gamma \times$ adiabatic curve slope
(c) Adiabatic curve slope = $\gamma \times$ isothermal curve slope
(d) Adiabatic curve slope = $\frac{1}{\gamma} \times$ isothermal curve slope [BHU 1995]
12. Pressure-temperature relationship for an ideal gas undergoing adiabatic change is ($\gamma = C_p / C_v$)
[CPMT 1992; MP PMT 1986, 87, 94, 97; Pb. PET 1998; DCE 2001; MP PET 2001; UPSEAT 1999, 2001; AFMC 2002]
- (a) $PT^\gamma = \text{constant}$ (b) $PT^{-1+\gamma} = \text{constant}$
(c) $P^{\gamma-1}T^\gamma = \text{constant}$ (d) $P^{1-\gamma}T^\gamma = \text{constant}$
13. The amount of work done in an adiabatic expansion from temperature T to T_1 is [MP PMT 1989]
- (a) $R(T - T_1)$ (b) $\frac{R}{\gamma - 1}(T - T_1)$
(c) RT (d) $R(T - T_1)(\gamma - 1)$
14. During the adiabatic expansion of 2 moles of a gas, the internal energy of the gas is found to decrease by 2 joules, the work done during the process on the gas will be equal to [CPMT 1988]
- (a) 2 J (b) -1 J
(c) 2 J (d) -2 J
15. The adiabatic elasticity of hydrogen gas ($\gamma = 1.4$) at NTP is [MP PMT 1990]
- (a) $1 \times 10^5 \text{ N/m}^2$ (b) $1 \times 10^{-8} \text{ N/m}^2$
(c) 1.4 N/m^2 (d) $1.4 \times 10^5 \text{ N/m}^2$
16. If γ denotes the ratio of two specific heats of a gas, the ratio of slopes of adiabatic and isothermal PV curves at their point of intersection is [NCERT 1990; MH CET 1999; MP PMT 2000]
- (a) $1/\gamma$ (b) γ
(c) $\gamma - 1$ (d) $\gamma + 1$
17. Air in a cylinder is suddenly compressed by a piston, which is then maintained at the same position. With the passage of time [NCERT 1971; DPMT 1990; KCET 2000; AIIMS 2000; MH CET 2001]
- (a) The pressure decreases
(b) The pressure increases
(c) The pressure remains the same
(d) The pressure may increase or decrease depending upon the nature of the gas
18. When a gas expands adiabatically [CPMT 1990]
- (a) No energy is required for expansion
(b) Energy is required and it comes from the wall of the container of the gas
(c) Internal energy of the gas is used in doing work
(d) Law of conservation of energy does not hold
19. One gm mol of a diatomic gas ($\gamma = 1.4$) is compressed adiabatically so that its temperature rises from 27°C to 127°C . The work done will be
- (a) 2077.5 joules (b) 207.5 joules
(c) 207.5 ergs (d) None of the above

20. Compressed air in the tube of a wheel of a cycle at normal temperature suddenly starts coming out from a puncture. The air inside [NCERT 1970]
- Starts becoming hotter
 - Remains at the same temperature
 - Starts becoming cooler
 - May become hotter or cooler depending upon the amount of water vapour present
21. The adiabatic Bulk modulus of a perfect gas at pressure is given by [CPMT 1982; MH CET 2001]
- P
 - $2P$
 - $P/2$
 - γP
22. An adiabatic process occurs at constant [MNR 1985; AFMC 1996; AIIMS 1999; UPSEAT 1999, 2000; Pb. PET 2004]
- Temperature
 - Pressure
 - Heat
 - Temperature and pressure
23. A polyatomic gas $\left(\gamma = \frac{4}{3}\right)$ is compressed to $\frac{1}{8}$ of its volume adiabatically. If its initial pressure is P_0 , its new pressure will be [MP PET 1994; BHU 1995]
- $8P_0$
 - $16P_0$
 - $6P_0$
 - $2P_0$
24. For adiabatic processes $\left(\gamma = \frac{C_p}{C_v}\right)$ [KCET 1999; MP PET 1995; CPMT 2003]
- $P^\gamma V = \text{constant}$
 - $T^\gamma V = \text{constant}$
 - $TV^{\gamma-1} = \text{constant}$
 - $TV^\gamma = \text{constant}$
25. An ideal gas is expanded adiabatically at an initial temperature of 300 K so that its volume is doubled. The final temperature of the hydrogen gas is ($\gamma = 1.40$) [MP PMT 1995; DPMT 1999]
- 227.36 K
 - 500.30 K
 - 454.76 K
 - -47°C
26. A given system undergoes a change in which the work done by the system equals the decrease in its internal energy. The system must have undergone an [Haryana CEE 1996; UPSEAT 2003]
- Isothermal change
 - Adiabatic change
 - Isobaric change
 - Isochoric change
27. During the adiabatic expansion of 2 moles of a gas, the internal energy was found to have decreased by 100 J . The work done by the gas in this process is [MP PET 1996, 97]
- Zero
 - -100 J
 - 200 J
 - 100 J
28. In an adiabatic expansion of a gas initial and final temperatures are T_1 and T_2 respectively, then the change in internal energy of the gas is [MP PET 1997]
- $\frac{R}{\gamma-1}(T_2 - T_1)$
 - $\frac{R}{\gamma-1}(T_1 - T_2)$
 - $R(T_1 - T_2)$
 - Zero
29. Helium at 27°C has a volume of 8 litres . It is suddenly compressed to a volume of 1 litre . The temperature of the gas will be [$\gamma = 5/3$] [CBSE PMT 1993; MP PMT 1999; Pb. PMT 2002]
- 108°C
 - 9327°C
 - 1200°C
 - 927°C
30. A cycle tyre bursts suddenly. This represents an [SCRA 1994]
- Isothermal process
 - Isobaric process
 - Isochoric process
 - Adiabatic process
31. One mole of helium is adiabatically expanded from its initial state (P_i, V_i, T_i) to its final state (P_f, V_f, T_f) . The decrease in the internal energy associated with this expansion is equal to [SCRA 1994; BHU 2002]
- $C_V(T_i - T_f)$
 - $C_P(T_i - T_f)$
 - $\frac{1}{2}(C_P + C_V)(T_i - T_f)$
 - $(C_P - C_V)(T_i - T_f)$
32. At N.T.P. one mole of diatomic gas is compressed adiabatically to half of its volume $\gamma = 1.41$. The work done on gas will be
- 1280 J
 - 1610 J
 - 1815 J
 - 2025 J
33. For adiabatic process, wrong statement is [RPMT 1997]
- $dQ = 0$
 - $dU = -dW$
 - $Q = \text{constant}$
 - Entropy is not constant
34. A diatomic gas initially at 18°C is compressed adiabatically to one-eighth of its original volume. The temperature after compression will be [Pb. PET 1995; CBSE PMT 1996; CPMT 1999]
- 10°C
 - 887°C
 - 668K
 - 144°C
35. A gas is being compressed adiabatically. The specific heat of the gas during compression is [SCRA 1996]
- Zero
 - Infinite
 - Finite but non-zero
 - Undefined
36. The process in which no heat enters or leaves the system is termed as [Pb. PET 1996; BHU 1998; BCECE 2003]
- Isochoric
 - Isobaric
 - Isothermal
 - Adiabatic
37. Two moles of an ideal monoatomic gas at 27°C occupies a volume of V . If the gas is expanded adiabatically to the volume $2V$, then the work done by the gas will be [$\gamma = 5/3, R = 8.31\text{ J/mol K}$] [RPET 1999]
- -2767.23 J
 - 2767.23 J
 - 2500 J
 - -2500 J

38. At 27°C a gas is suddenly compressed such that its pressure becomes $\frac{1}{8}$ th of original pressure. Temperature of the gas will be ($\gamma = 5/3$) [BHU 2000]
- (a) 420K (b) 327°C
(c) 300K (d) -142°C
39. $\Delta U + \Delta W = 0$ is valid for [RPMT 2000]
- (a) Adiabatic process (b) Isothermal process
(c) Isobaric process (d) Isochoric process
40. An ideal gas at a pressures of 1 atmosphere and temperature of 27°C is compressed adiabatically until its pressure becomes 8 times the initial pressure, then the final temperature is ($\gamma = 3/2$) [EAMCET (Engg.) 2000]
- (a) 627°C (b) 527°C
(c) 427°C (d) 327°C
41. Air is filled in a motor tube at 27°C and at a pressure of 8 atmospheres. The tube suddenly bursts, then temperature of air is [Given γ of air = 1.5] [MP PMT 2002]
- (a) 27.5°C (b) 75°C
(c) 150K (d) 150°C
42. If $\gamma = 2.5$ and volume is equal to $\frac{1}{8}$ times to the initial volume then pressure P' is equal to (Initial pressure = P) [RPET 2003]
- (a) $P' = P$ (b) $P' = 2P$
(c) $P' = P \times (2)^{15/2}$ (d) $P' = 7P$
43. In an adiabatic process, the state of a gas is changed from P_1, V_1, T_1 , to P_2, V_2, T_2 . Which of the following relation is correct [Orissa JEE 2003]
- (a) $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$ (b) $P_1 V_1^{\gamma-1} = P_2 V_2^{\gamma-1}$
(c) $T_1 P_1^\gamma = T_2 P_2^\gamma$ (d) $T_1 V_1^\gamma = T_2 V_2^\gamma$
44. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_p / C_v for the gas is [AIIEE 2003]
- (a) $\frac{3}{2}$ (b) $\frac{4}{3}$
(c) 2 (d) $\frac{5}{3}$
45. In adiabatic expansion of a gas [BCECE 2001; MP PET 2003]
- (a) Its pressure increases
(b) Its temperature falls
(c) Its density increases
(d) Its thermal energy increases
46. One mole of an ideal gas at an initial temperature of $T\text{K}$ does 6 J joules of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is 5/3, the final temperature of gas will be [CBSE PMT 2004]
- (a) $(T + 2.4)K$ (b) $(T - 2.4)K$
(c) $(T + 4)K$ (d) $(T - 4)K$
47. A gas is suddenly compressed to $1/4$ th of its original volume at normal temperature. The increase in its temperature is ($\gamma = 1.5$)
- (a) 273K (b) 573K
(c) 373K (d) 473K
48. A gas ($\gamma = 1.3$) is enclosed in an insulated vessel fitted with insulating piston at a pressure of 10^5N/m^2 . On suddenly pressing the piston the volume is reduced to half the initial volume. The final pressure of the gas is [RPET 2002]
- (a) $2^{0.7} \times 10^5$ (b) $2^{1.3} \times 10^5$
(c) $2^{1.4} \times 10^5$ (d) None of these
49. The internal energy of the gas increases in [MP PMT 1989; RPMT 2001]
- (a) Adiabatic expansion (b) Adiabatic compression
(c) Isothermal expansion (d) Isothermal compression
50. We consider a thermodynamic system. If ΔU represents the increase in its internal energy and W the work done by the system, which of the following statements is true [CBSE PMT 1998]
- (a) $\Delta U = -W$ in an adiabatic process
(b) $\Delta U = W$ in an isothermal process
(c) $\Delta U = -W$ in an isothermal process
(d) $\Delta U = W$ in an adiabatic process
51. A gas is suddenly compressed to one fourth of its original volume. What will be its final pressure, if its initial pressure is P [Pb. PET 2002]
- (a) Less than P (b) More than P
(c) P (d) Either (a) or (c)
52. A gas for which $\gamma = 1.5$ is suddenly compressed to $\frac{1}{4}$ th of the initial volume. Then the ratio of the final to the initial pressure is [EAMCET 2000]
- (a) 1 : 16 (b) 1 : 8
(c) 1 : 4 (d) 8 : 1
53. One mole of an ideal gas with $\gamma = 1.4$, is adiabatically compressed so that its temperature rises from 27°C to 35°C . The change in the internal energy of the gas is ($R = 8.3\text{J/mol.K}$) [EAMCET 2001]
- (a) -166J (b) 166J
(c) -168J (d) 168J
54. The volume of a gas is reduced adiabatically to $\frac{1}{4}$ of its volume at 27°C , if the value of $\gamma = 1.4$, then the new temperature will be
- (a) $350 \times 4^{0.4}\text{K}$ (b) $300 \times 4^{0.4}\text{K}$
(c) $150 \times 4^{0.4}\text{K}$ (d) None of these
55. During an adiabatic expansion of 2 moles of a gas, the change in internal energy was found -50J . The work done during the process is [Pb. PET 1996]
- (a) Zero (b) 100J
(c) -50J (d) 50J
56. Adiabatic modulus of elasticity of a gas is $2.1 \times 10^5\text{N/m}^2$. What will be its isothermal modulus of elasticity ($\frac{C_p}{C_v} = 1.4$)

[UPSEAT 1999]

- (a) $1.8 \times 10^5 \text{ N/m}^2$ (b) $1.5 \times 10^5 \text{ N/m}^2$
 (c) $1.4 \times 10^5 \text{ N/m}^2$ (d) $1.2 \times 10^5 \text{ N/m}^2$

57. For an adiabatic expansion of a perfect gas, the value of $\frac{\Delta P}{P}$ is equal to [CPMT 1983; MP PMT 1990]

- (a) $-\sqrt{\gamma} \frac{\Delta V}{V}$ (b) $-\frac{\Delta V}{V}$
 (c) $-\gamma \frac{\Delta V}{V}$ (d) $-\gamma^2 \frac{\Delta V}{V}$

Isobaric and Isochoric Processes

1. A gas expands under constant pressure P from volume V_1 to V_2 . The work done by the gas is

[CBSE PMT 1990; RPMT 2003]

- (a) $P(V_2 - V_1)$ (b) $P(V_1 - V_2)$
 (c) $P(V_1^\gamma - V_2^\gamma)$ (d) $P \frac{V_1 V_2}{V_2 - V_1}$

2. When heat is given to a gas in an isobaric process, then

[DPMT 2001]

- (a) The work is done by the gas
 (b) Internal energy of the gas increases
 (c) Both (a) and (b)
 (d) None from (a) and (b)

3. One mole of a perfect gas in a cylinder fitted with a piston has a pressure P , volume V and temperature T . If the temperature is increased by 1 K keeping pressure constant, the increase in volume is

- (a) $\frac{2V}{273}$ (b) $\frac{V}{91}$
 (c) $\frac{V}{273}$ (d) V

4. A gas is compressed at a constant pressure of 50 N/m^2 from a volume of 10 m^3 to a volume of 4 m^3 . Energy of 100 J then added to the gas by heating. Its internal energy is

[MNR 1994]

- (a) Increased by 400 J (b) Increased by 200 J
 (c) Increased by 100 J (d) Decreased by 200 J

5. Work done by air when it expands from 50 litres to 150 litres at a constant pressure of 2 atmospheres is

- (a) $2 \times 10^4 \text{ joules}$ (b) $2 \times 100 \text{ joules}$
 (c) $2 \times 10^5 \times 100 \text{ joules}$ (d) $2 \times 10^{-5} \times 100 \text{ joules}$

6. Work done by 0.1 mole of a gas at 27°C to double its volume at constant pressure is ($R = 2 \text{ cal/mol}^\circ \text{C}$)

[EAMCET 1994]

- (a) 54 cal (b) 600 cal
 (c) 60 cal (d) 546 cal

7. Unit mass of a liquid with volume V_1 is completely changed into a gas of volume V_2 at a constant external pressure P and temperature T . If the latent heat of evaporation for the given mass is L , then the increase in the internal energy of the system is

- (a) Zero (b) $P(V_2 - V_1)$
 (c) $L - P(V_2 - V_1)$ (d) L

8. A gas expands 0.25 m^3 at constant pressure 10^3 N/m^2 , the work done is

[CPMT 1997; UPSEAT 1999; JIPMER 2001, 02]

- (a) 2.5 ergs (b) 250 J
 (c) 250 W (d) 250 N

9. Two kg of water is converted into steam by boiling at atmospheric pressure. The volume changes from $2 \times 10^{-3} \text{ m}^3$ to 3.34 m^3 . The work done by the system is about

- (a) -340 kJ (b) -170 kJ
 (c) 170 kJ (d) 340 kJ

10. An ideal gas has volume V_0 at 27°C . It is heated at constant pressure so that its volume becomes $2V_0$. The final temperature is

[BCECE 2003]

- (a) 54°C (b) 32.6°C
 (c) 327°C (d) 150 K

11. If 300 ml of a gas at 27°C is cooled to 7°C at constant pressure, then its final volume will be

[Pb. PET 1999; BHU 2003; CPMT 2004]

- (a) 540 ml (b) 350 ml
 (c) 280 ml (d) 135 ml

12. Which of the following is correct in terms of increasing work done for the same initial and final state [RPMT 1996]

- (a) Adiabatic < Isothermal < Isobaric
 (b) Isobaric < Adiabatic < Isothermal
 (c) Adiabatic < Isobaric < Isothermal
 (d) None of these

13. A sample of gas expands from volume V_1 to V_2 . The amount of work done by the gas is greatest when the expansion is

[CBSE PMT 1997; AIIMS 1998; JIPMER 2000]

- (a) Isothermal (b) Isobaric
 (c) Adiabatic (d) Equal in all cases

14. Which of the following is a slow process [J & K CET 2000]

- (a) Isothermal (b) Adiabatic
 (c) Isobaric (d) None of these

15. How much work to be done in decreasing the volume of an ideal gas by an amount of $2.4 \times 10^{-4} \text{ m}^3$ at normal temperature and constant normal pressure of $1 \times 10^5 \text{ N/m}^2$

[UPSEAT 1999]

- (a) 28 joule (b) 27 joule
 (c) 25 joule (d) 24 joule

16. A Container having 1 mole of a gas at a temperature 27°C has a movable piston which maintains at constant pressure in container of 1 atm . The gas is compressed until temperature becomes 127°C . The work done is (C_p for gas is $7.03 \text{ cal/mol}^\circ \text{C}$)

- (a) 703 J (b) 814 J
 (c) 121 J (d) 2035 J

17. In a reversible isochoric change [NCERT 1990]

[Roorkee 1999]

- (a) $\Delta W = 0$ (b) $\Delta Q = 0$
(c) $\Delta T = 0$ (d) $\Delta U = 0$
18. Entropy of a thermodynamic system does not change when this system is used for [AIIMS 1995]
(a) Conduction of heat from a hot reservoir to a cold reservoir
(b) Conversion of heat into work isobarically
(c) Conversion of heat into internal energy isochorically
(d) Conversion of work into heat isochorically
19. The work done in which of the following processes is zero [UPSEAT 2003]
(a) Isothermal process (b) Adiabatic process
(c) Isochoric process (d) None of these
20. In which thermodynamic process, volume remains same [Orissa PMT 2004]
(a) Isobaric (b) Isothermal
(c) Adiabatic (d) Isochoric
21. In an isochoric process if $T_1 = 27^\circ C$ and $T_2 = 127^\circ C$, then P_1 / P_2 will be equal to [RPMT 2003]
(a) $9 / 59$ (b) $2 / 3$
(c) $3 / 4$ (d) None of these
22. Which is incorrect [DCE 2001]
(a) In an isobaric process, $\Delta p = 0$
(b) In an isochoric process, $\Delta W = 0$
(c) In an isothermal process, $\Delta T = 0$
(d) In an isothermal process, $\Delta Q = 0$
23. Which relation is correct for isometric process [RPMT 2001; BCECE 2003]
(a) $\Delta Q = \Delta U$ (b) $\Delta W = \Delta U$
(c) $\Delta Q = \Delta W$ (d) None of these
4. In a cyclic process, the internal energy of the gas
(a) Increases (b) Decreases
(c) Remains constant (d) Becomes zero
5. Irreversible process is
(a) Adiabatic process
(b) Joule-Thomson expansion
(c) Ideal isothermal process
(d) None of the above
6. For a reversible process, necessary condition is
(a) In the whole cycle of the system, the loss of any type of heat energy should be zero
(b) That the process should be too fast
(c) That the process should be slow so that the working substance should remain in thermal and mechanical equilibrium with the surroundings
(d) The loss of energy should be zero and it should be *quasistatic*
7. In a cyclic process, work done by the system is [BHU 2002]
(a) Zero
(b) Equal to heat given to the system
(c) More than the heat given to system
(d) Independent of heat given to the system
8. An ideal gas heat engine operates in a Carnot's cycle between $227^\circ C$ and $127^\circ C$. It absorbs $6 \times 10^4 J$ at high temperature. The amount of heat converted into work is [KCET 2004]
(a) $4.8 \times 10^4 J$ (b) $3.5 \times 10^4 J$
(c) $1.6 \times 10^4 J$ (d) $1.2 \times 10^4 J$
9. An ideal heat engine exhausting heat at $77^\circ C$ is to have a 30% efficiency. It must take heat at [BCECE 2004]
(a) $127^\circ C$ (b) $227^\circ C$
(c) $327^\circ C$ (d) $673^\circ C$
10. Efficiency of Carnot engine is 100% if [Pb. PET 2000]
(a) $T_2 = 273 K$ (b) $T_2 = 0 K$
(c) $T_1 = 273 K$ (d) $T_1 = 0 K$
11. A Carnot's engine used first an ideal monoatomic gas then an ideal diatomic gas. If the source and sink temperature are $411^\circ C$ and $69^\circ C$ respectively and the engine extracts 1000 J of heat in each cycle, then area enclosed by the PV diagram is
(a) 100 J (b) 300 J
(c) 500 J (d) 700 J
12. A Carnot engine absorbs an amount Q of heat from a reservoir at an absolute temperature T and rejects heat to a sink at a temperature of $T/3$. The amount of heat rejected is [UPSEAT 2004]
(a) $Q / 4$ (b) $Q / 3$
(c) $Q / 2$ (d) $2Q / 3$
13. The temperature of sink of Carnot engine is $27^\circ C$. Efficiency of engine is 25%. Then temperature of source is

Heat Engine, Refrigerator and Second Law of Thermodynamics

1. A Carnot engine working between $300 K$ and $600 K$ has work output of $800 J$ per cycle. What is amount of heat energy supplied to the engine from source per cycle [DPMT 1999; Pb. PMT 2002, 05; Kerala PMT 2004]
(a) $1800 J/cycle$ (b) $1000 J/cycle$
(c) $2000 J/cycle$ (d) $1600 J/cycle$
2. The coefficient of performance of a Carnot refrigerator working between $30^\circ C$ and $0^\circ C$ is [UPSEAT 2002]
(a) 10 (b) 1
(c) 9 (d) 0
3. If the door of a refrigerator is kept open, then which of the following is true [DPMT 2001; BHU 2001; JIPMER 2002; AIEEE 2002; CPMT 2003]
(a) Room is cooled
(b) Room is heated
(c) Room is either cooled or heated
(d) Room is neither cooled nor heated

[DCE 2002; CPMT 2002]

- (a) $227^{\circ}C$ (b) $327^{\circ}C$
(c) $127^{\circ}C$ (d) $27^{\circ}C$
14. The temperature of reservoir of Carnot's engine operating with an efficiency of 70% is 1000 K. The temperature of its sink is
(a) 300 K (b) 400 K
(c) 500 K (d) 700 K
15. In a Carnot engine, when $T_2 = 0^{\circ}C$ and $T_1 = 200^{\circ}C$, its efficiency is η_1 and when $T_1 = 0^{\circ}C$ and $T_2 = -200^{\circ}C$, its efficiency is η_2 , then what is η_1 / η_2 [DCE 2004]
(a) 0.577 (b) 0.733
(c) 0.638 (d) Can not be calculated
16. The efficiency of Carnot's engine operating between reservoirs, maintained at temperatures $27^{\circ}C$ and $-123^{\circ}C$, is
(a) 50% (b) 24%
(c) 0.75% (d) 0.4%
17. A Carnot engine operates between $227^{\circ}C$ and $27^{\circ}C$. Efficiency of the engine will be [DCE 1999; BHU 2004]
(a) $\frac{1}{3}$ (b) $\frac{2}{5}$
(c) $\frac{3}{4}$ (d) $\frac{3}{5}$
18. A measure of the degree of disorder of a system is known as [Pb. PET 1997; MH CET 1999]
(a) Isobaric (b) Isotropy
(c) Enthalpy (d) Entropy
19. A Carnot engine has the same efficiency between 800 K to 500 K and x K to 600 K. The value of x is [Pb. PMT 1996; CPMT 1996]
(a) 1000 K (b) 960 K
(c) 846 K (d) 754 K
20. A scientist says that the efficiency of his heat engine which operates at source temperature $127^{\circ}C$ and sink temperature $27^{\circ}C$ is 26%, then [CBSE PMT 2001]
(a) It is impossible
(b) It is possible but less probable
(c) It is quite probable
(d) Data are incomplete
21. A Carnot's engine is made to work between $200^{\circ}C$ and $0^{\circ}C$ first and then between $0^{\circ}C$ and $-200^{\circ}C$. The ratio of efficiencies of the engine in the two cases is [KCET 2002]
(a) 1.73 : 1 (b) 1 : 1.73
(c) 1 : 1 (d) 1 : 2
22. Efficiency of a Carnot engine is 50% when temperature of outlet is 500 K. In order to increase efficiency up to 60% keeping temperature of intake the same what is temperature of outlet [CBSE PMT 2002]
(a) 200 K (b) 400 K
(c) 600 K (d) 800 K
23. Even Carnot engine cannot give 100% efficiency because we cannot [AIIEE 2002]
(a) Prevent radiation
(b) Find ideal sources

(c) Reach absolute zero temperature

(d) Eliminate friction

24. "Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of [AIIEE 2003, EAM]
(a) Second law of thermodynamics
(b) Conservation of momentum [DCE 2003]
(c) Conservation of mass
(d) First law of thermodynamics
25. A Carnot engine takes 3×10^6 cal. of heat from a reservoir at $627^{\circ}C$ and gives it to a sink at $27^{\circ}C$. The work done by the engine is [AIIEE 2003]
(a) 4.2×10^6 J (b) 8.4×10^6 J
(c) 16.8×10^6 J (d) Zero
26. The first operation involved in a Carnot cycle is [AFMC 1998]
(a) Isothermal expansion [DPMT 2002, 03; BVP 2004]
(b) Adiabatic expansion
(c) Isothermal compression
(d) Adiabatic compression
27. For which combination of working temperatures the efficiency of Carnot's engine is highest [KCET 2000]
(a) 80 K, 60 K (b) 100 K, 80 K
(c) 60 K, 40 K (d) 40 K, 20 K
28. The efficiency of Carnot engine when source temperature is T_1 and sink temperature is T_2 will be [DCE 2000]
(a) $\frac{T_1 - T_2}{T_1}$ (b) $\frac{T_2 - T_1}{T_2}$
(c) $\frac{T_1 - T_2}{T_2}$ (d) $\frac{T_1}{T_2}$
29. An ideal heat engine working between temperature T_1 and T_2 has an efficiency η , the new efficiency if both the source and sink temperature are doubled, will be [DPMT 2000]
(a) $\frac{\eta}{2}$ (b) η
(c) 2η (d) 3η
30. An ideal refrigerator has a freezer at a temperature of $-13^{\circ}C$. The coefficient of performance of the engine is 5. The temperature of the air (to which heat is rejected) will be [BHU 2000; CPMT 2002]
(a) $325^{\circ}C$ (b) 325 K
(c) $39^{\circ}C$ (d) $320^{\circ}C$
31. In a mechanical refrigerator, the low temperature coils are at a temperature of $-23^{\circ}C$ and the compressed gas in the condenser has a temperature of $27^{\circ}C$. The theoretical coefficient of performance is [UPSEAT 2001]
(a) 5 (b) 8
(c) 6 (d) 6.5
32. An engine is supposed to operate between two reservoirs at temperature $727^{\circ}C$ and $227^{\circ}C$. The maximum possible efficiency of such an engine is [UPSEAT 2005]
(a) $1/2$ (b) $1/4$
(c) $3/4$ (d) 1
33. An ideal gas heat engine operates in Carnot cycle between $227^{\circ}C$ and $127^{\circ}C$. It absorbs 6×10^4 cal of heat at higher temperature. Amount of heat converted to work is [CBSE PMT 2005]
(a) 2.4×10^4 cal (b) 6×10^4 cal

- (c) $1.2 \times 10^4 \text{ cal}$ (d) $4.8 \times 10^4 \text{ cal}$

34. Which of the following processes is reversible

[CBSE PMT 2005]

- (a) Transfer of heat by radiation
(b) Electrical heating of a nichrome wire
(c) Transfer of heat by conduction
(d) Isothermal compression

Critical Thinking

Objective Questions

1. When an ideal diatomic gas is heated at constant pressure, the fraction of the heat energy supplied which increases the internal energy of the gas, is

[IIT 1990; UPSEAT 1998; RPET 2000]

- (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{3}{7}$ (d) $\frac{5}{7}$

2. 1 cm^3 of water at its boiling point absorbs 540 calories of heat to become steam with a volume of 1671 cm^3 . If the atmospheric pressure = $1.013 \times 10^5 \text{ N/m}^2$ and the mechanical equivalent of heat = 4.19 J/calorie , the energy spent in this process in overcoming intermolecular forces is

[MP PET 1999, 2001; Orissa JEE 2002]

- (a) 540 cal (b) 40 cal
(c) 500 cal (d) Zero

3. During the melting of a slab of ice at 273 K at atmospheric pressure

- (a) Positive work is done by ice-water system on the atmosphere
(b) Positive work is done on the ice-water system by the atmosphere
(c) The internal energy of the ice-water system increases
(d) The internal energy of the ice-water system decreases

4. Two identical containers A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume V. The mass of the gas in A is m_A and that in B is m_B . The gas in each cylinder is now allowed to expand isothermally to the same final volume $2V$. The changes in the pressure in A and B are found to be ΔP and $1.5 \Delta P$ respectively. Then

- (a) $4m_A = 9m_B$ (b) $2m_A = 3m_B$
(c) $3m_A = 2m_B$ (d) $9m_A = 3m_B$

5. A monoatomic ideal gas, initially at temperature T_1 , is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If L_1 and L_2 are the lengths of the gas column before and after expansion respectively, then T_1/T_2 is given by

[IIT-JEE (Screening) 2000]

- (a) $\left(\frac{L_1}{L_2}\right)^{2/3}$ (b) $\frac{L_1}{L_2}$

- (c) $\frac{L_2}{L_1}$ (d) $\left(\frac{L_2}{L_1}\right)^{2/3}$

6. A closed hollow insulated cylinder is filled with gas at 0°C and also contains an insulated piston of negligible weight and negligible thickness at the middle point. The gas on one side of the piston is heated to 100°C . If the piston moves 5 cm, the length of the hollow cylinder is

[EAMCET 2001]

- (a) 13.65 cm (b) 27.3 cm
(c) 38.6 cm (d) 64.6 cm

7. A mono atomic gas is supplied the heat Q very slowly keeping the pressure constant. The work done by the gas will be

- (a) $\frac{2}{3}Q$ (b) $\frac{3}{5}Q$
(c) $\frac{2}{5}Q$ (d) $\frac{1}{5}Q$

8. A gas mixture consists of 2 moles of oxygen and 4 moles argon at temperature T . Neglecting all vibrational modes, the total internal energy of the system is

[IIT 1999; UPSEAT 2003]

- (a) 4 RT (b) 15 RT
(c) 9 RT (d) 11 RT

9. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3 . The total work done is W . Then

- (a) $P_3 > P_1, W > 0$ (b) $P_3 < P_1, W < 0$
(c) $P_3 = P_1, W < 0$ (d) $P_3 = P_1, W = 0$

10. Work done by a system under isothermal change from a volume V_1 to V_2 for a gas which obeys Vander Waal's equation

$$(V - \beta n) \left(P + \frac{\alpha n^2}{V} \right) = nRT$$

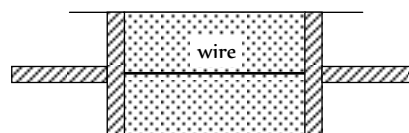
(a) $nRT \log_e \left(\frac{V_2 - n\beta}{V_1 - n\beta} \right) + \alpha n^2 \left(\frac{V_1 - V_2}{V_1 V_2} \right)$

(b) $nRT \log_{10} \left(\frac{V_2 - \alpha\beta}{V_1 - \alpha\beta} \right) + \alpha n^2 \left(\frac{V_1 - V_2}{V_1 V_2} \right)$

(c) $nRT \log_e \left(\frac{V_2 - n\alpha}{V_1 - n\alpha} \right) + \beta n^2 \left(\frac{V_1 - V_2}{V_1 V_2} \right)$

(d) $nRT \log_e \left(\frac{V_1 - n\beta}{V_2 - n\beta} \right) + \alpha n^2 \left(\frac{V_1 V_2}{V_1 - V_2} \right)$

11. A cylindrical tube of uniform cross-sectional area A is fitted with two air tight frictionless pistons. The pistons are connected to each other by a metallic wire. Initially the pressure of the gas is P and temperature is T , atmospheric pressure is also P . Now the temperature of the gas is increased to $2T$, the tension in the wire will be

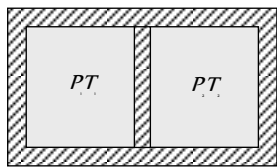


- (a) $2P_0A$ (b) P_0A
(c) $\frac{P_0A}{2}$ (d) $4P_0A$
12. The molar heat capacity in a process of a diatomic gas if it does a work of $\frac{Q}{4}$ when a heat of Q is supplied to it is
(a) $\frac{2}{5}R$ (b) $\frac{5}{2}R$
(c) $\frac{10}{3}R$ (d) $\frac{6}{7}R$
13. An insulator container contains 4 moles of an ideal diatomic gas at temperature T . Heat Q is supplied to this gas, due to which 2 moles of the gas are dissociated into atoms but temperature of the gas remains constant. Then
(a) $Q = 2RT$ (b) $Q = RT$
(c) $Q = 3RT$ (d) $Q = 4RT$
14. The volume of air increases by 5% in its adiabatic expansion. The percentage decrease in its pressure will be
(a) 5% (b) 6%
(c) 7% (d) 8%
15. The temperature of a hypothetical gas increases to $\sqrt{2}$ times when compressed adiabatically to half the volume. Its equation can be written as
(a) $PV^{3/2} = \text{constant}$ (b) $PV^{5/2} = \text{constant}$
(c) $PV^{7/3} = \text{constant}$ (d) $PV^{4/3} = \text{constant}$
16. Two Carnot engines A and B are operated in succession. The first one, A receives heat from a source at $T_1 = 800K$ and rejects to sink at T_2K . The second engine B receives heat rejected by the first engine and rejects to another sink at $T_3 = 300K$. If the work outputs of two engines are equal, then the value of T_2 is
(a) $100K$ (b) $300K$
(c) $550K$ (d) $700K$
17. When an ideal monoatomic gas is heated at constant pressure, fraction of heat energy supplied which increases the internal energy of gas, is [AIIMS 1995]
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{3}{7}$ (d) $\frac{3}{4}$
18. When an ideal gas ($\gamma = 5/3$) is heated under constant pressure, then what percentage of given heat energy will be utilised in doing external work [RPET 1999]
(a) 40% (b) 30%
(c) 60% (d) 20%
19. Which one of the following gases possesses the largest internal energy [SCRA 1998]
(a) 2 moles of helium occupying $1m^3$ at $300K$
(b) 56 kg of nitrogen at $107Nm^{-2}$ and $300K$
(c) 8 grams of oxygen at 8 atm and $300K$
(d) 6×10^{26} molecules of argon occupying $40m^3$ at $900K$
20. Two samples A and B of a gas initially at the same pressure and temperature are compressed from volume V to $V/2$ (A isothermally and adiabatically). The final pressure of A is [MP PET 1996, 99; MP PMT 1997, 99]
(a) Greater than the final pressure of B
(b) Equal to the final pressure of B
(c) Less than the final pressure of B
(d) Twice the final pressure of B
21. Initial pressure and volume of a gas are P and V respectively. First it is expanded isothermally to volume $4V$ and then compressed adiabatically to volume V . The final pressure of gas will be [CBSE PMT 1999]
(a) $1P$ (b) $2P$
(c) $4P$ (d) $8P$
22. A thermally insulated rigid container contains an ideal gas heated by a filament of resistance 100Ω through a current of $1A$ for 5 min then change in internal energy is [IIT-JEE (Screening) 2005]
(a) 0 kJ (b) 10 kJ
(c) 20 kJ (d) 30 kJ
23. A reversible engine converts one-sixth of the heat input into work. When the temperature of the sink is reduced by $62^\circ C$, the efficiency of the engine is doubled. The temperatures of the source and sink are [CBSE PMT 2000]
(a) $80^\circ C, 37^\circ C$ (b) $95^\circ C, 28^\circ C$
(c) $90^\circ C, 37^\circ C$ (d) $99^\circ C, 37^\circ C$
24. An engineer claims to have made an engine delivering 10 kW power with fuel consumption of 1 g/sec. The calorific value of the fuel is 2 kcal/g. Is the claim of the engineer [J & K CET 2000]
(a) Valid
(b) Invalid
(c) Depends on engine design
(d) Depends of the load
25. Find the change in the entropy in the following process 100 gm of ice at $0^\circ C$ melts when dropped in a bucket of water at $50^\circ C$ (Assume temperature of water does not change) [BHU (Med.) 2000]
(a) -4.5 cal/K (b) $+4.5 \text{ cal/K}$
(c) $+5.4 \text{ cal/K}$ (d) -5.4 cal/K
26. An ideal gas expands in such a manner that its pressure and volume can be related by equation $PV^2 = \text{constant}$. During this process, the gas is [UPSEAT 2002]
(a) Heated
(b) Cooled

- (c) Neither heated nor cooled
(d) First heated and then cooled
27. A Carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased

[UPSEAT 2005]

- (a) 840 K (b) 280 K
(c) 560 K (d) 380 K
28. P - V diagram of a diatomic gas is a straight line passing through origin. The molar heat capacity of the gas in the process will be
- (a) $4R$ (b) $2.5R$
(c) $3R$ (d) $\frac{4R}{3}$
29. Following figure shows an adiabatic cylindrical container of volume V_0 divided by an adiabatic smooth piston (area of cross-section = A) in two equal parts. An ideal gas ($C_p / C_v = \gamma$) is at pressure P_1 and temperature T_1 in left part and gas at pressure P_2 and temperature T_2 in right part. The piston is slowly displaced and released at a position where it can stay in equilibrium. The final pressure of the two parts will be (Suppose x = displacement of the piston)



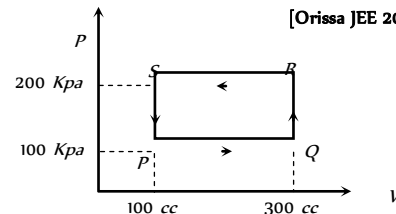
- (a) P_2 (b) P_1
(c) $\frac{P_1 \left(\frac{V_0}{2}\right)^\gamma}{\left(\frac{V_0}{2} + Ax\right)^\gamma}$ (d) $\frac{P_2 \left(\frac{V_0}{2}\right)^\gamma}{\left(\frac{V_0}{2} + Ax\right)^\gamma}$
30. Two cylinders A and B fitted with pistons contain equal amounts of an ideal diatomic gas at 300 K . The piston of A is free to move while that of B is held fixed. The same amount of heat is given to the gas in each cylinder. If the rise in temperature of the gas in A is 30 K , then the rise in temperature of the gas in B is
- (a) 30 K (b) 18 K
(c) 50 K (d) 42 K

- (d) Relation between ΔU_I and ΔU_{II} can not be determined

2. A thermodynamic system is taken through the cycle $PQRSP$ process. The net work done by the system is

[Orissa JEE 2002]

- (a) 20 J
(b) -20 J
(c) 400 J
(d) -374 J



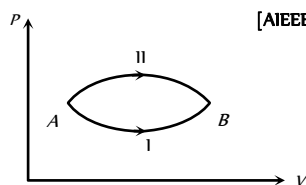
[IIT 1998]

Graphical Questions

1. A system goes from A to B via two processes I and II as shown in figure. If ΔU_I and ΔU_{II} are the changes in internal energies in the processes I and II respectively, then

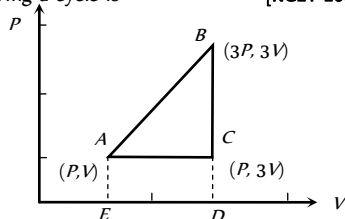
[AIEEE 2005]

- (a) $\Delta U_{II} > \Delta U_I$
(b) $\Delta U_{II} < \Delta U_I$
(c) $\Delta U_I = \Delta U_{II}$



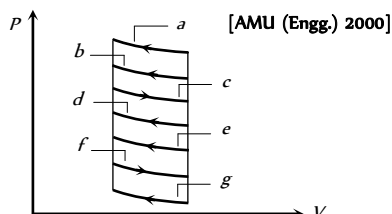
3. An ideal gas is taken around $ABCA$ as shown in the above P - V diagram. The work done during a cycle is [KCET 2001]

- (a) $2PV$
(b) PV
(c) $1/2 PV$
(d) Zero



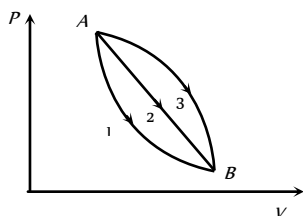
4. The P - V diagram shows seven curved paths (connected by vertical paths) that can be followed by a gas. Which two of them should be parts of a closed cycle if the net work done by the gas is to be at its maximum value

- (a) ac
(b) cg
(c) af
(d) cd

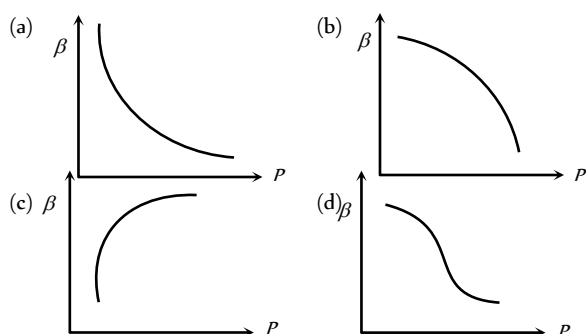


5. An ideal gas of mass m in a state A goes to another state B via three different processes as shown in figure. If Q_1 , Q_2 and Q_3 denote the heat absorbed by the gas along the three paths, then

- (a) $Q_1 < Q_2 < Q_3$
(b) $Q_1 < Q_2 = Q_3$
(c) $Q_1 = Q_2 > Q_3$
(d) $Q_1 > Q_2 > Q_3$



6. Which of the following graphs correctly represents the variation of $\beta = -(dV/dP)/V$ with P for an ideal gas at constant temperature [IIT-JEE (Screening) 2002]

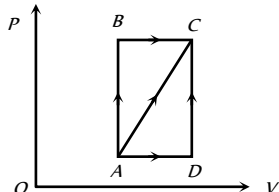


7. A thermodynamic process is shown in the figure. The pressures and volumes corresponding to some points in the figure are :

$$P_A = 3 \times 10^4 \text{ Pa}, P_B = 8 \times 10^4 \text{ Pa} \quad \text{and} \\ V_A = 2 \times 10^{-3} \text{ m}^3, V_D = 5 \times 10^{-3} \text{ m}^3$$

In process AB , 600 J of heat is added to the system and in process BC , 200 J of heat is added to the system. The change in internal energy of the system in process AC would be

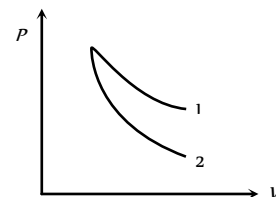
- (a) 560 J
(b) 800 J
(c) 600 J
(d) 640 J



8. P - V plots for two gases during adiabatic process are shown in the figure. Plots 1 and 2 should correspond respectively to

[IIT-JEE (Screening) 2001]

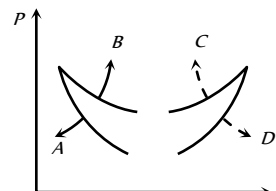
- (a) He and O_2
(b) O_2 and He
(c) He and Ar
(d) O_2 and N_2



9. Four curves A , B , C and D are drawn in the adjoining figure for a given amount of gas. The curves which represent adiabatic and isothermal changes are

[CPMT 1986; UPSEAT 1999]

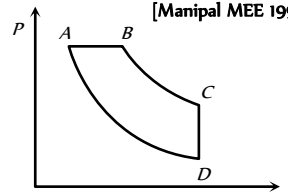
- (a) C and D respectively
(b) D and C respectively
(c) A and B respectively
(d) B and A respectively



10. In pressure-volume diagram given below, the isochoric, isothermal, and isobaric parts respectively, are

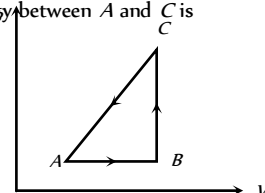
[Manipal MEE 1995]

- (a) BA, AB, DC [MP PET 1992]
(b) DC, CB, BA
(c) AB, BC, CD
(d) CD, DA, AB



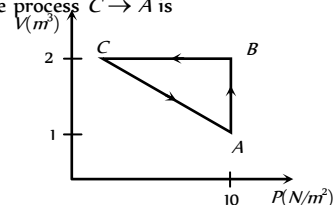
11. The P - V diagram of a system undergoing thermodynamic transformation is shown in figure. The work done on the system in going from $A \rightarrow B \rightarrow C$ is 50 J and 20 cal heat is given to the system. The change in internal energy between A and C is

- (a) 34 J
(b) 70 J
(c) 84 J
(d) 134 J



12. An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the work done by the gas in the process $C \rightarrow A$ is

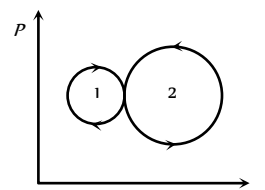
- (a) -5 J
(b) -10 J
(c) -15 J
(d) -20 J



13. In the following indicator diagram, the net amount of work done will be

- (a) Positive
(b) Negative
(c) Zero
(d) Infinity

[CBSE PMT 1992]



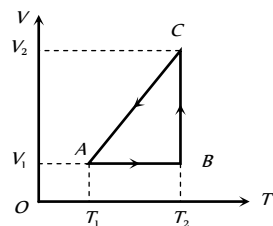
14. A cyclic process for 1 mole of an ideal gas is shown in figure in the V - T diagram. The work done in AB , BC and CA respectively

(a) $0, RT_2 \ln \left(\frac{V_1}{V_2} \right), R(T_1 - T_2)$

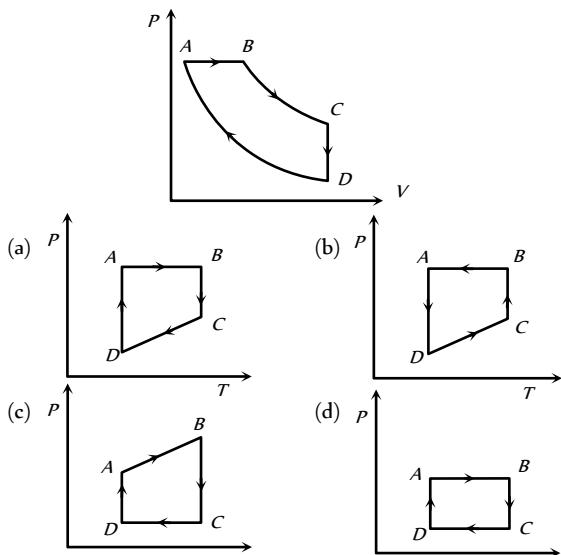
(b) $R(T_1 - T_2), 0, RT_1 \ln \frac{V_1}{V_2}$

(c) $0, RT_2 \ln \left(\frac{V_2}{V_1} \right), R(T_1 - T_2)$

(d) $0, RT_2 \ln \left(\frac{V_2}{V_1} \right), R(T_2 - T_1)$



15. A cyclic process $ABCD$ is shown in the figure P - V diagram. Which of the following curves represent the same process



16. Carnot cycle (reversible) of a gas represented by a Pressure-Volume curve is shown in the diagram

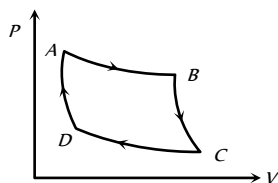
Consider the following statements

- I. Area $ABCD$ = Work done on the gas
- II. Area $ABCD$ = Net heat absorbed
- III. Change in the internal energy in cycle = 0

Which of these are correct

[AMU (Med.) 2001]

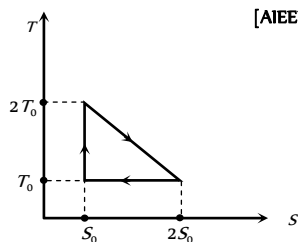
- (a) I only
- (b) II only
- (c) II and III
- (d) I, II and III



17. The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is

[AIEEE 2005]

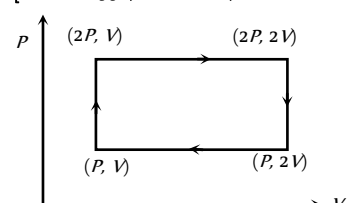
- (a) $1/3$
- (b) $2/3$
- (c) $1/2$
- (d) $1/4$



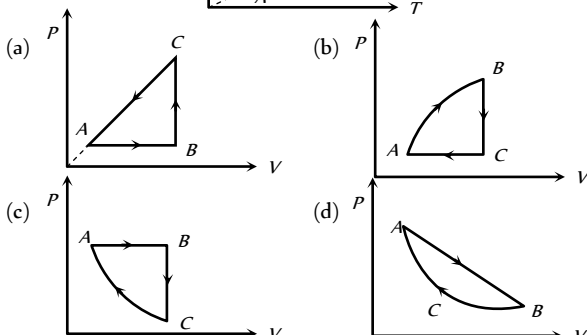
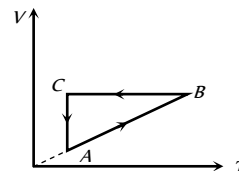
18. Work done in the given P - V diagram in the cyclic process is

[UPSEAT 1998; RPET 2000; Kerala PMT 2002]

- (a) PV
- (b) $2PV$
- (c) $PV/2$
- (d) $3PV$

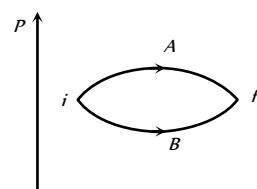


19. A cyclic process $ABCA$ is shown in the V - T diagram. Process on the P - V diagram is



20. In the figure given two processes A and B are shown by which a thermo-dynamical system goes from initial to final state F . If ΔQ_A and ΔQ_B are respectively the heats supplied to the systems then

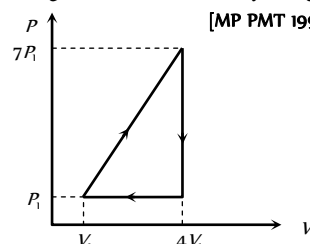
- (a) $\Delta Q_A = \Delta Q_B$
- (b) $\Delta Q_A \geq \Delta Q_B$
- (c) $\Delta Q_A < \Delta Q_B$
- (d) $\Delta Q_A > \Delta Q_B$



21. In the cyclic process shown in the figure, the work done by the gas in one cycle is

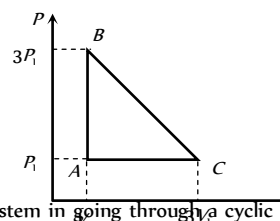
[MP PMT 1999]

- (a) $28 P_1 V_1$
- (b) $14 P_1 V_1$
- (c) $18 P_1 V_1$
- (d) $9 P_1 V_1$



22. An ideal gas is taken around the cycle $ABCA$ as shown in the P - V diagram. The net work done by the gas during the cycle is equal to

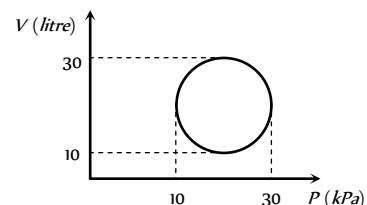
- (a) $12 P_1 V_1$
- (b) $6 P_1 V_1$
- (c) $3 P_1 V_1$
- (d) $2 P_1 V_1$



23. Heat energy absorbed by a system in going through a cyclic process shown in figure is

[AIIMS 1995; BHU 2002]

- (a) $10 \pi J$
- (b) $10 \pi J$



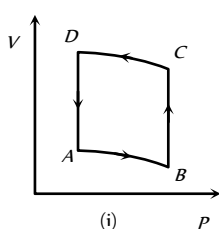
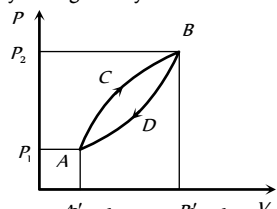
(c) $10\pi J$

 (d) $10^{-3}\pi J$

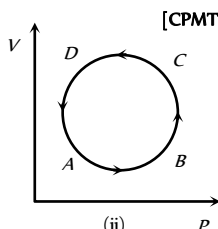
24. A thermodynamic system is taken from state A to B along ACB and is brought back to A along BDA as shown in the PV diagram. The net work done during the complete cycle is given by the area

- (a) $PACBPP$
 (b) $ACBB'A'A$
 (c) $ACBDA$
 (d) $ADBB'A'A$

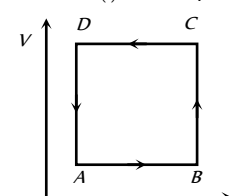
25. In the diagrams (i) to (iv) of variation of volume with changing pressure is shown. A gas is taken along the path $ABCD$. The change in internal energy of the gas will be



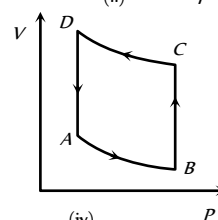
(i)



(ii)



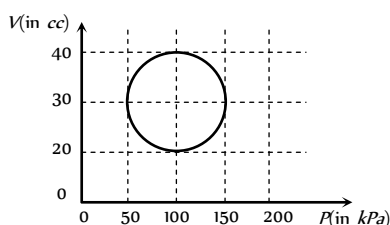
(iii)



(iv)

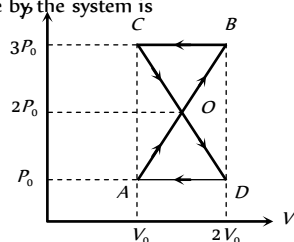
- (a) Positive in all cases (i) to (iv)
 (b) Positive in cases (i), (ii) and (iii) but zero in (iv) case
 (c) Negative in cases (i), (ii) and (iii) but zero in (iv) case
 (d) Zero in all four cases
26. A system is taken through a cyclic process represented by a circle as shown. The heat absorbed by the system is

- (a) $\pi \times 10^3 J$
 (b) $\frac{\pi}{2} J$
 (c) $4\pi \times 10^2 J$
 (d) πJ

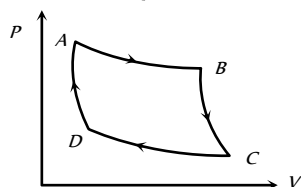


27. A thermodynamic system undergoes cyclic process $ABCD$ as shown in figure. The work done by the system is

- (a) $P_0 V_0$
 (b) $2P_0 V_0$
 (c) $\frac{P_0 V_0}{2}$
 (d) Zero



28. The P - V graph of an ideal gas cycle is shown here as below. The adiabatic process is described by

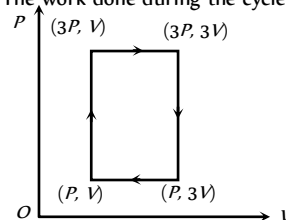


[CPMT 1985; UPSEAT 2003]

- (a) AB and BC
 (b) AB and CD
 (c) BC and DA
 (d) BC and CD

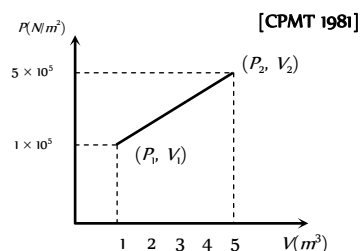
29. An ideal gas is taken round the cycle $ABCD$ as shown in following P - V diagram. The work done during the cycle is [IIT 1983; CPMT 1992]

- (a) PV
 (b) $2PV$
 (c) $4PV$
 (d) Zero



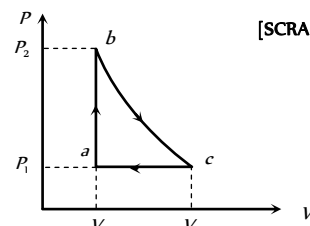
30. A system changes from the state (P_1, V_1) to (P_2, V_2) as shown in the figure. What is the work done by the system [CPMT 1981]

- (a) $7.5 \times 10^5 \text{ joule}$
 (b) $7.5 \times 10^5 \text{ erg}$
 (c) $12 \times 10^5 \text{ joule}$
 (d) $6 \times 10^5 \text{ joule}$



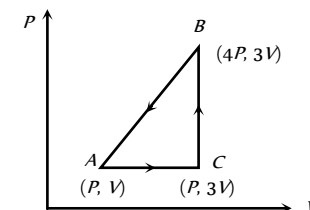
31. Carbon monoxide is carried around a closed cycle abc in which bc is an isothermal process as shown in the figure. The gas absorbs 7000 J of heat as its temperature increases from 300 K to 1000 K in going from a to b . The quantity of heat rejected by the gas during the process ca is [SCRA 1994]

- (a) 4200 J
 (b) 5000 J
 (c) 9000 J
 (d) 9800 J



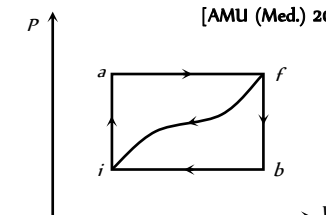
32. A sample of ideal monoatomic gas is taken round the cycle $ABCA$ as shown in the figure. The work done during the cycle is

- (a) Zero
 (b) $3PV$
 (c) $6PV$
 (d) $9PV$

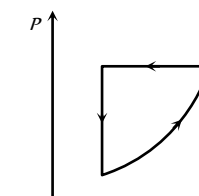


33. When a system is taken from state i to a state f along path iaf , $Q = 50 J$ and $W = 20 J$. Along path ibf , $Q = 35 J$. If $W = -13 J$ for the curved return path fi , Q for this path is [AMU (Med.) 2000]

- (a) 33 J
 (b) 23 J
 (c) -7 J
 (d) -43 J



34. For one complete cycle of a thermodynamic process on a gas as shown in the P - V diagram, Which of following is correct

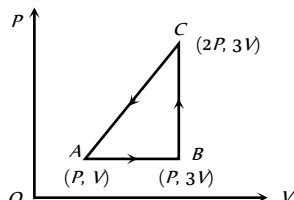


- (a) $\Delta E_{\text{int}} = 0, Q < 0$
 (b) $\Delta E_{\text{int}} = 0, Q > 0$
 (c) $\Delta E_{\text{int}} > 0, Q < 0$
 (d) $\Delta E_{\text{int}} < 0, Q > 0$

35. An ideal gas is taken around $ABCA$ as shown in the above P - V diagram. The work done during a cycle is

[UPSEAT 2001]

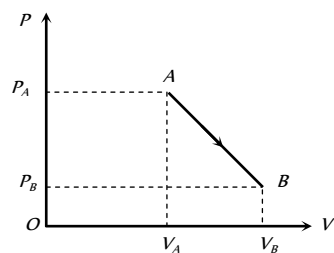
- (a) Zero
 (b) $\frac{1}{2} PV$
 (c) $2 PV$
 (d) PV



36. An ideal gas is taken from point A to the point B , as shown in the P - V diagram, keeping the temperature constant. The work done in the process is

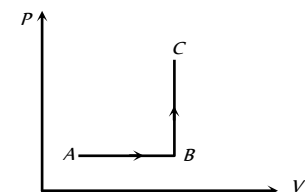
[UPSEAT 2005]

- (a) $(P_A - P_B)(V_B - V_A)$
 (b) $\frac{1}{2} (P_B - P_A)(V_B + V_A)$
 (c) $\frac{1}{2} (P_B - P_A)(V_B - V_A)$
 (d) $\frac{1}{2} (P_B + P_A)(V_B - V_A)$



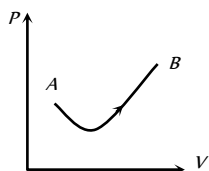
37. The P - V diagram of a system undergoing thermodynamic transformation is shown in figure. The work done by the system in going from $A \rightarrow B \rightarrow C$ is 30 J and 40 J heat is given to the system. The change in internal energy between A and C is

- (a) 10 J
 (b) 70 J
 (c) 84 J
 (d) 134 J



38. Consider a process shown in the figure. During this process the work done by the system

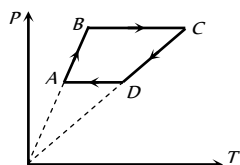
- (a) Continuously increases
 (b) Continuously decreases
 (c) First increases, then decreases
 (d) First decreases, then increases



39. Six moles of an ideal gas performs a cycle shown in figure. If the temperature are $T_i = 600 \text{ K}$, $T_c = 800 \text{ K}$, $T_d = 2200 \text{ K}$ and $T_f = 1200 \text{ K}$, the work done per cycle is

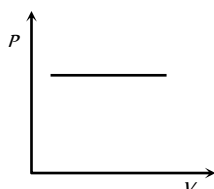
[BCECE 2005]

- (a) 20 kJ
 (b) 30 kJ
 (c) 40 kJ
 (d) 60 kJ



40. Which of the accompanying PV diagrams best represents an isothermal process

[MP PET 2005]

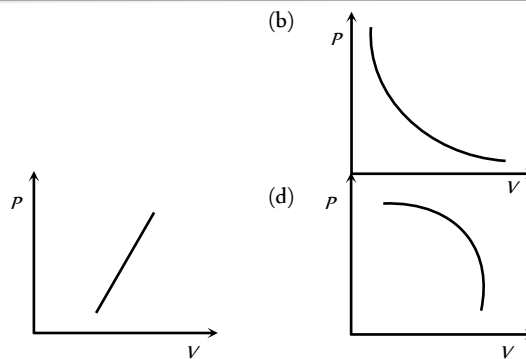


(a)

(b)

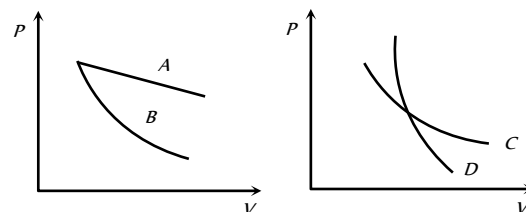
(c)

(d)



41. In the following figure, four curves A , B , C and D are shown. The curves are

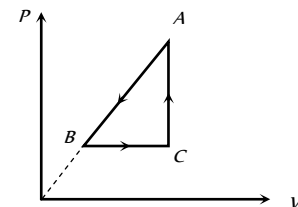
[DCE 2003]



- (a) Isothermal for A and D while adiabatic for B and C
 (b) Adiabatic for A and C while isothermal for B and D
 (c) Isothermal for A and B while adiabatic for C and D
 (d) Isothermal for A and C while adiabatic for B and D

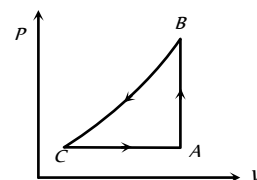
42. P - V diagram of a cyclic process $ABCA$ is as shown in figure. Choose the correct statement

- (a) $\Delta Q_{A \rightarrow B}$ = negative
 (b) $\Delta U_{B \rightarrow C}$ = positive
 (c) ΔW_{CAB} = negative
 (d) All of these

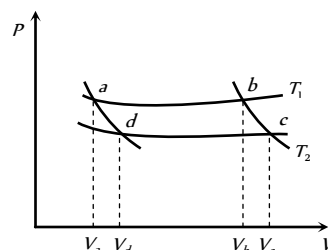


43. A sample of an ideal gas is taken through a cycle as shown in figure. It absorbs 50 J of energy during the process AB , no heat during BC , rejects 70 J during CA . 40 J of work is done on the gas during BC . Internal energy of gas at A is 1500 J, the internal energy at C would be

- (a) 1590 J
 (b) 1620 J
 (c) 1540 J
 (d) 1570 J



44. In the following P - V diagram two adiabats cut two isotherms at temperatures T_i and T_f (fig.). The value of $\frac{V_a}{V_d}$ will be



(a) $\frac{V_b}{V_c}$

(b) $\frac{V_c}{V_b}$

(c) $\frac{V_d}{V_a}$

(d) $V_b V_c$

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : Reversible systems are difficult to find in real world.
Reason : Most processes are dissipative in nature
[AIIMS 2005]
2. Assertion : Air quickly leaking out of a balloon becomes coolers
Reason : The leaking air undergoes adiabatic expansion.[AIIMS 2005]
3. Assertion : Thermodynamic process in nature are irreversible.
Reason : Dissipative effects can not be eliminated.
[AIIMS 2004]
4. Assertion : When a bottle of cold carbonated drink is opened, a slight fog forms around the opening.
Reason : Adiabatic expansion of the gas causes lowering of temperature and condensation of water vapours.[AIIMS 2003]
5. Assertion : The isothermal curves intersect each other at a certain point.
Reason : The isothermal change takes place slowly, so the isothermal curves have very little slope.
[AIIMS 2001]
6. Assertion : In adiabatic compression, the internal energy and temperature of the system get decreased.
Reason : The adiabatic compression is a slow process
[AIIMS 2001]
7. Assertion : In isothermal process whole of the heat energy supplied to the body is converted into internal energy.
Reason : According to the first law of thermodynamics
 $\Delta Q = \Delta U + p\Delta V$. [AIIMS 1997]
8. Assertion : We can not change the temperature of a body without giving (or taking) heat to (or from) it.
Reason : According to principle of conservation of energy, total energy of a system should remains conserved.
9. Assertion : The specific heat of a gas in an adiabatic process is zero and in an isothermal process is infinite.
Reason : Specific heat of a gas is directly proportional to change of heat in system and inversely proportional to change in temperature.
10. Assertion : Work and heat are two equivalent form of energy.
Reason : Work is the transfer of mechanical energy irrespective of temperature difference, whereas heat is the transfer of thermal energy because of temperature difference only.
11. Assertion : The heat supplied to a system is always equal to the increase in its internal energy.
- Reason : When a system changes from one thermal equilibrium to another, some heat is absorbed by it.
12. Assertion : A room can be cooled by opening the door of a refrigerator in a closed room.
Reason : Heat flows from lower temperature (refrigerator) to higher temperature (room).
13. Assertion : It is not possible for a system, unaided by an external agency to transfer heat from a body at lower temperature to another body at higher temperature.
Reason : According to Clausius statement, " No process is possible whose sole result is the transfer of heat from a cooled object to a hotter object.
14. Assertion : If an electric fan be switched on in a closed room, the air of the room will be cooled.
Reason : Fan air decrease the temperature of the room.
15. Assertion : The internal energy of an isothermal process does not change.
Reason : The internal energy of a system depends only on pressure of the system.
16. Assertion : In an adiabatic process, change in internal energy of a gas is equal to work done on or by the gas in the process.
Reason : Temperature of gas remains constant in a adiabatic process.
17. Assertion : An adiabatic process is an isentropic process.
Reason : Change in entropy is zero in case of adiabatic process.
18. Assertion : Work done by a gas in isothermal expansion is more than the work done by the gas in the same expansion, adiabatically.
Reason : Temperature remains constant in isothermal expansion and not in adiabatic expansion.
19. Assertion : First law of thermodynamics is a restatement of the principle of conservation
Reason : Energy is fundamental quantity.
20. Assertion : Zeroth law of thermodynamic explain the concept of energy.
Reason : Energy is dependent on temperature.
21. Assertion : Efficiency of a Carnot engine increase on reducing the temperature of sink.
Reason : The efficiency of a Carnot engine is defined as ratio of net mechanical work done per cycle by the gas to the amount of heat energy absorbed per cycle from the source.
22. Assertion : The entropy of the solids is the highest
Reason : Atoms of the solids are arranged in orderly manner.

Answers

First Law of Thermodynamics ($\Delta Q = \Delta U + \Delta W$)

1	a	2	c	3	b	4	b	5	c
---	---	---	---	---	---	---	---	---	---

6	b	7	b	8	d	9	a	10	d
11	c	12	a	13	d	14	a	15	b
16	b	17	c	18	d	19	d	20	b
21	a	22	d	23	b	24	a	25	d
26	d	27	a	28	b	29	d	30	a
31	b	32	c	33	c	34	a	35	a
36	b	37	c	38	c	39	b	40	a
41	c	42	a	43	a	44	c	45	c
46	a	47	c	48	b				

Isothermal Process

1	c	2	a	3	c	4	d	5	b
6	b	7	c	8	d	9	a	10	c
11	a	12	b	13	a	14	a	15	c
16	c	17	a	18	c	19	a	20	c
21	b	22	b	23	a	24	a	25	a
26	c	27	b	28	b	29	b	30	a
31	d								

Adiabatic Process

1	c	2	c	3	b	4	d	5	c
6	d	7	c	8	b	9	a	10	a
11	c	12	d	13	b	14	d	15	d
16	b	17	a	18	c	19	a	20	c
21	d	22	c	23	b	24	c	25	a
26	b	27	d	28	a	29	d	30	d
31	a	32	c	33	d	34	c	35	a
36	d	37	b	38	d	39	a	40	d
41	c	42	c	43	a	44	a	45	b
46	d	47	a	48	b	49	b	50	a
51	b	52	d	53	b	54	b	55	d
56	b	57	c						

Isoobaric and Isochoric Processes

1	a	2	c	3	c	4	a	5	a
6	c	7	c	8	b	9	d	10	c
11	c	12	a	13	b	14	a	15	d
16	b	17	a	18	d	19	c	20	d
21	d	22	d	23	a				

Heat Engine, Refrigerator and Second Law of Thermodynamics

1	d	2	c	3	b	4	c	5	b
6	d	7	b	8	d	9	b	10	b
11	c	12	b	13	c	14	a	15	a
16	a	17	b	18	d	19	b	20	a

21	b	22	b	23	c	24	a	25	b
26	a	27	d	28	a	29	b	30	c
31	a	32	a	33	c	34	d		

Critical Thinking Questions

1	d	2	c	3	bc	4	c	5	d
6	d	7	c	8	d	9	c	10	a
11	b	12	c	13	b	14	c	15	a
16	c	17	b	18	a	19	b	20	c
21	b	22	d	23	d	24	b	25	b
26	b	27	d	28	c	29	c	30	d

Graphical Questions

1	c	2	b	3	a	4	c	5	a
6	a	7	a	8	b	9	c	10	d
11	d	12	a	13	b	14	c	15	a
16	c	17	a	18	a	19	c	20	d
21	d	22	d	23	c	24	c	25	d
26	b	27	d	28	c	29	c	30	c
31	d	32	b	33	d	34	a	35	d
36	d	37	a	38	a	39	c	40	b
41	d	42	d	43	a	44	a		

Assertion & Reason

1	a	2	a	3	a	4	a	5	e
6	d	7	e	8	d	9	a	10	a
11	d	12	d	13	a	14	d	15	c
16	c	17	a	18	b	19	c	20	e
21	b	22	a						

AS Answers and Solutions

First Law of Thermodynamics ($\Delta Q = \Delta U + \Delta W$)

- (a) $\Delta Q = \Delta U + \Delta W$ and $\Delta W = P\Delta V$
- (c)
- (b) $\Delta Q = \Delta U + \Delta W$
 $\Rightarrow \Delta U = \Delta Q - \Delta W = Q - W$ (using proper sign)
- (b) $\Delta U = \Delta Q - W = 35 - 15 = 20 \text{ J}$
- (c) Internal energy depends only on the temperature of the gas.
- (b)

7. (b) (i) Case $\rightarrow$ Volume = constant $\Rightarrow \int P dV = 0$
- (ii) Case $\rightarrow P = \text{constant} \Rightarrow \int_{V_1}^{2V_1} P dV = P \int_{V_1}^{2V_1} dV = PV_1$
8. (d) $\Delta Q = \Delta W + \Delta U \Rightarrow 35 = -15 + \Delta U \Rightarrow \Delta U = 50 J$
9. (a) $J\Delta Q = \Delta U + \Delta W, \Delta U = J\Delta Q - \Delta W$
 $\Delta U = 4.18 \times 300 - 600 = 654 \text{ Joule}$
10. (d) Work done $= \int_1^2 P dV$, which is state dependent as well as path dependent.
11. (c) $\Delta Q = \Delta U + \Delta W \because \Delta W = 0 \Rightarrow \Delta Q = \Delta U = \frac{f}{2} \mu R \Delta T$
 $= \frac{3}{2} \times 2R(373 - 273) = 300R$
12. (a) $\Delta Q = 2k \text{ cal} = 2 \times 10^3 \times 4.2 J = 8400 J$ and $\Delta W = 500 J$.
Hence from $\Delta Q = \Delta U + \Delta W, \Delta W = \Delta Q - \Delta U = 8400 - 500 = 7900 J$
13. (d) Change in internal energy (ΔU) depends upon initial and final state of the function while ΔQ and ΔW are path dependent also.
14. (a) This is the case of free expansion and in this case $\Delta W = 0$, $\Delta U = 0$ so temperature remains same i.e. $300 K$.
15. (b) $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta W = \Delta Q - \Delta U = 100 - 40 = 70 J$
16. (b) Work done is not a thermodynamical function.
17. (c) $\Delta Q = \Delta U + \Delta W = 167 + 333 = 500 \text{ cal}$
18. (d) Heat always refers to energy in transit from one body to another because of temperature difference.
19. (d) Change in internal energy does not depend upon path so $\Delta U = \Delta Q - \Delta W$ remain constant.

20. (b) $\Delta Q = \Delta U + \Delta W$; $\Delta Q = 200J$ and $\Delta W = -100J$
 $\Rightarrow \Delta U = \Delta Q - \Delta W = 200 - (-100) = 300J$
21. (a) During free expansion of a perfect gas no, work is done and also no heat is supplied from outside. Therefore, no change in internal energy. Hence, temperature remain constant.
22. (d) $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = \Delta Q - \Delta W = 150 - 110 = 40J$
23. (b) From FLOT $\Delta Q = \Delta U + \Delta W$
 $\therefore$ Heat supplied to the system so $\Delta Q \rightarrow$ Positive
 and work is done on the system so $\Delta W \rightarrow$ Negative
 Hence $+\Delta Q = \Delta U - \Delta W$
24. (a)
25. (d) State of a thermodynamic state cannot determine by a single variable (P or V or T)
26. (d) R is the universal gas constant.
27. (a) From FLOT
 $\Rightarrow dU = dQ - dW \Rightarrow dU = dQ (< 0)$ ($\because dW = 0$)
 $\Rightarrow dU < 0$ So temperature will decrease.
28. (b) From FLOT $\Delta Q = \Delta U + \Delta W$
 Work done at constant pressure $(\Delta W)_P = (\Delta Q)_P - \Delta U$
 $(\Delta Q)_P - (\Delta Q)_V$ (As we know $(\Delta Q)_V = \Delta U$)
 Also $(\Delta Q)_P = mc_P \Delta T$ and $(\Delta Q)_V = mc_V \Delta T$
 $\Rightarrow (\Delta W)_P = m(c_P - c_V) \Delta T$
 $\Rightarrow (\Delta W)_P = 1 \times (3.4 \times 10^3 - 2.4 \times 10^3) \times 10 = 10^4 cal$
29. (d)
30. (a) Ideal gas possess only kinetic energy.
31. (b) The internal energy and entropy depend only on the initial and final states of the system and not on the path followed to attain that state.
32. (c) $\Delta Q = \Delta U + \Delta W$
 $\therefore \Delta Q = 200 cal = 200 \times 4.2 = 840J$ and $\Delta W = 40J$
 $\Rightarrow \Delta U = \Delta Q - \Delta W = 840 - 40 = 800J$
33. (c) $\Delta Q = \Delta U + \Delta W = (U_f - U_i) + \Delta W$
 $\Rightarrow 30 = (U_f - 40) + 10 \Rightarrow U_f = 60J$
34. (a) With rise in temperature, internal energy also increases.
35. (a)
36. (b) Heat supplied to a gas raise its internal energy and does some work against expansion, so it is a special case of law of conservation of energy.
37. (c) Change in internal energy is always equal to the heat supplied at constant volume.
 i.e. $\Delta U = (\Delta Q)_V = \mu C_V \Delta T$.
 For monoatomic gas $C_V = \frac{3}{2} R$
 $\Rightarrow \Delta U = \mu \left(\frac{3}{2} R \right) \Delta T = 1 \times \frac{3}{2} \times 8.31 \times (100 - 0)$
 $= 12.48 \times 10^2 J$
38. (c) $\Delta U = \mu C_V \Delta T = n \left(\frac{R}{\gamma - 1} \right) \Delta T$
 $\Rightarrow \Delta U = \frac{P \Delta V}{(\gamma - 1)} = \frac{P(2V - V)}{\gamma - 1} = \frac{PV}{(\gamma - 1)}$
39. (b) $\Delta U = \mu C_V \Delta T = 2 \times 4.96 \times (342 - 340) = 19.84 cal$
40. (a)
41. (c) According to FLOT
 $\Delta Q = \Delta U + P(\Delta V) \Rightarrow \Delta U = \Delta Q - P(\Delta V)$
 $= 1500 - (2.1 \times 10^5)(2.5 \times 10^{-3}) = 975 Joule$
42. (a) $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = \Delta Q - \Delta W$
 $= 6 \times 4.18 - 6 = 19.08 kJ \approx 19.1 kJ$
43. (a) Given $\Delta Q = -20J$, $\Delta W = -8J$ and $U_i = 30J$
 $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = (\Delta Q - \Delta W)$
 $\Rightarrow (U_f - U_i) = (U_f - 30) = -20 - (-8) \Rightarrow U_f = 18J$
44. (c) Change in internal energy $\Delta U = \mu C_V \Delta T$
 it doesn't depend upon type of process. Actually it is a state function
45. (c)
46. (a) In first process using $\Delta Q = \Delta U + \Delta W$
 $\Rightarrow 8 \times 10^5 = \Delta U + 6.5 \times 10^5 \Rightarrow \Delta U = 1.5 \times 10^5 J$
 Since final and initial states are same in both process
 So ΔU will be same in both process
 For second process using $\Delta Q = \Delta U + \Delta W$
 $\Rightarrow 10^5 = 1.5 \times 10^5 + \Delta W \Rightarrow \Delta W = -0.5 \times 10^5 J$
47. (c) $\Delta W = P \Delta V$; here ΔV is negative so ΔW will be negative
48. (b) Entropy is related to second law of thermodynamics.

Isothermal Process

1. (c) In isothermal process temperature remains constant.
2. (a) If isothermal curves cut each other then at equilibrium two temperature will be there which is impossible.
3. (c) In isothermal expansion temperature remains constant, hence no change in internal energy.
4. (d) $W = \mu RT \log_e \frac{V_2}{V_1}$
 $= \left(\frac{m}{M} \right) RT \log_e \frac{V_2}{V_1} = 2.3 \times \frac{m}{M} RT \log_{10} \frac{V_2}{V_1}$
 $= 2.3 \times \frac{96}{32} R (273 + 27) \log_{10} \frac{140}{70} = 2.3 \times 900 R \log_{10} 2$
5. (b) $0.8 \times 5 = P \times (3 + 5) \Rightarrow P = 0.5 m$
6. (b) Differentiate $PV = \text{constant}$ w.r.t V
 $\Rightarrow P \Delta V + V \Delta P = 0 \Rightarrow \frac{\Delta P}{P} = - \frac{\Delta V}{V}$

7. (c)
8. (d) $W = -\mu RT \log_e \frac{V_2}{V_1} = -1 \times 8.31 \times (273 + 0) \log_e \left(\frac{22.4}{11.2} \right)$
 $= -8.31 \times 273 \times \log_e 2 = -1572.5 J$ [$\because \log_e 2 = 0.693$]
9. (a) $E_\theta = P$, if $P = \text{constant}$, $E_\theta = \text{constant}$
10. (c) For isothermal process $PV = RT \Rightarrow P = \frac{RT}{V}$
 $\therefore W = PdV = \int_{V_1}^{V_2} \frac{RT}{V} dV = RT \log_e \frac{V_2}{V_1}$
11. (a) $E_\theta = P$
12. (b) For such a case, pressure = $\frac{1}{\text{Compressibility}}$
13. (a) $E_\theta = P = 1.013 \times 10^5 N/m^2$
14. (a) In isothermal process, compressibility $E_\theta = P$.
15. (c) In isothermal process, exchange of energy takes place between system and surrounding to maintain the system temperature constant.
16. (c) No change in the internal energy of ideal gas but for real gas internal energy increases because work is done against intermolecular forces.
17. (a) In isothermal process temperature remains constant. i.e., $\Delta T = 0$. Hence according to $C = \frac{Q}{m\Delta T} \Rightarrow C_{iso} = \infty$
18. (c) This is the case of free expansion of gas. In free expansion $\Delta U = 0 \Rightarrow \text{Temp. remains same.}$
19. (a) An isothermal process takes place at constant temperature, must be carried out in a vessel with conducting wall so that heat generated should go out at once.
20. (c) For isothermal process
 $dU = 0$ and work done $= dW = P(V_2 - V_1)$
 $\therefore V_2 = \frac{V_1}{2} = \frac{V}{2} \therefore dW = -\frac{PV}{2}$
21. (b) In isothermal process, temperature remains constant.
22. (b) In isothermal process, heat is released by the gas to maintain the constant temperature.
23. (a) In isothermal compression, there is always an increase of heat, which must flow out the gas.
 $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta W$ ($\because \Delta U = 0$)
 $\Rightarrow \Delta Q = -1.5 \times 10^4 J = \frac{1.5 \times 10^4}{4.18} cal = -3.6 \times 10^3 cal$
24. (a) In isothermal change, temperature remains constant, Hence $\Delta U = 0$.
 Also from $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta W$
25. (a) It is an isothermal process. Hence work done $= P(V_2 - V_1)$
 $= 1 \times 10^5 \times (1.091 - 1) \times 10^{-6} = 0.0091 J$
26. (c) $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = \Delta Q - \Delta W = 2240 - 168 = 2072 J$.
27. (b) Amount of heat given $= 540 \text{ calories}$
 Change in volume $\Delta V = 1670 \text{ c.c}$

Atmospheric pressure $P = 1.01 \times 10^6 \text{ dyne/cm}^2$

Work done against atmospheric pressure

$$W = P\Delta V = \frac{1.01 \times 10^6 \times 1670}{4.2 \times 10^7} \approx 40 \text{ cal}$$

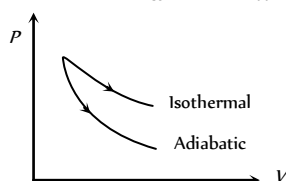
28. (b) $W_{iso} = \mu RT \log_e \frac{V_2}{V_1} = 1 \times 8.31 \times 300 \log_e \frac{20}{10} = 1728 J$
29. (b) $W = \mu RT \log_e \left(\frac{V_2}{V_1} \right) = 0.2 \times 8.3 \times \log_e 2 \times (27 + 273)$
 $= 0.2 \times 8.3 \times 300 \times 0.693 = 345 J$
30. (a) For isothermal process $P_1 V_1 = P_2 V_2$
 $\Rightarrow P_2 = \frac{P_1 V_1}{V_2} = \frac{72 \times 1000}{900} = 80 \text{ cm}$
 Stress $\Delta P = P_2 - P_1 = 80 - 72 = 8 \text{ cm}$
31. (d) During isothermal change $T = \text{constant} \Rightarrow \Delta U = 0$
 also from FLOT, $\Delta Q = \Delta W$.

Adiabatic Process

1. (c) Gas cylinder suddenly explodes is an irreversible adiabatic change and work done against expansion reduces the temperature.
2. (c) Work done in adiabatic change $= \frac{\mu R(T_1 - T_2)}{\gamma - 1}$
3. (b) In case of adiabatic expansion $\Delta W = \text{positive}$ and $\Delta Q = 0$
 from FLOT $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta U = -\Delta W$ i.e., ΔU will be negative.
4. (d) For adiabatic process $\frac{T^\gamma}{P^{\gamma-1}} = \text{constant}$
 $\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_1}{P_2} \right)^{\frac{1-\gamma}{\gamma}} \Rightarrow \frac{T_2}{300} = \left(\frac{4}{1} \right)^{\frac{(1-1.4)}{1.4}} \Rightarrow T_2 = 300(4)^{-\frac{0.4}{1.4}}$
5. (c) $PV^\gamma = \text{constant} \Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma \Rightarrow \frac{P_2}{1} = \left(\frac{V_1}{V_1/4} \right)^{3/2} = 8$
 $\Rightarrow P_2 = 8 \text{ atm.}$
6. (d) $PV^\gamma = \text{constant} \Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma \Rightarrow P_2 = (8)^{5/3} P_1 = 32 P_1$
7. (c) Volume of the gas $V = \frac{m}{d}$ and using $PV^\gamma = \text{constant}$
 We get $\frac{P'}{P} = \left(\frac{V}{V'} \right)^\gamma = \left(\frac{d'}{d} \right)^\gamma = (32)^{7/5} = 128$
8. (b) $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1} \Rightarrow T_2 = 300 \left(\frac{27}{8} \right)^{\frac{5}{3}-1} = 300 \left(\frac{27}{8} \right)^{\frac{2}{3}}$
 $= 300 \left\{ \left(\frac{27}{8} \right)^{1/3} \right\}^2 = 800 \left(\frac{3}{2} \right)^2 = 675 K$
 $\Rightarrow \Delta T = 675 - 300 = 375 K$
9. (a) In thermodynamic processes.
 Work done = Area covered by PV diagram with V -axis

From graph it is clear that $(\text{Area})_{iso} > (\text{Area})_{adi}$

$$\Rightarrow W_{iso} > W_{adi}$$



10. (a) Since $PV = RT$ and $T = \text{constant}$; $\therefore PV = \text{constant}$.

11. (c) For Isothermal process $PV = \text{constant}$

$$\Rightarrow \left(\frac{dP}{dV} \right) = \frac{-P}{V} = \text{Slope of Isothermal curve}$$

For adiabatic $PV^\gamma = \text{constant}$

$$\Rightarrow \frac{dP}{dV} = \frac{-\gamma P}{V} = \text{Slop of adiabatic curve slope}$$

$$\text{Clearly, } \left(\frac{dP}{dV} \right)_{\text{adiabatic}} = \gamma \left(\frac{dP}{dV} \right)_{\text{Isothermal}}$$

12. (d) $PV^\gamma = \text{constant} \Rightarrow P \left(\frac{RT}{P} \right)^\gamma = \text{constant}$

$$\Rightarrow P^{1-\gamma} T^\gamma = \text{constant.}$$

13. (b) $W_{adi} = \frac{R}{\gamma-1} (T_i - T_f) = \frac{R}{\gamma-1} (T - T_i)$

14. (d) $dQ = 0 = -2 + dW \Rightarrow dW = 2 \text{ J}$

$$\Rightarrow \text{Work done by the gas} = 2 \text{ J}$$

$$\Rightarrow \text{Work done on the gas} = -2 \text{ J}$$

15. (d) $E_\phi = \gamma P = 1.4 \times (1 \times 10^5) = 1.4 \times 10^5 \text{ N/m}^2$

16. (b) Slope of adiabatic curve = $\gamma \times (\text{Slope of isothermal curve})$

17. (a) Due to compression the temperature of the system increases to a very high value. This causes the flow of heat from system to the surroundings, thus decreasing the temperature. This decrease in temperature results in decrease in pressure.

18. (c) $\Delta Q = \Delta U + \Delta W = 0 \Rightarrow \Delta W = -\Delta U$

if ΔW is positive i.e., gas does work then ΔU should be negative meaning internal energy is used in doing work.

19. (a) $W = \frac{R}{\gamma-1} (T_1 - T_2)$

$$= \frac{8.31 \times \{(273+27) - (273+127)\}}{1.4-1} = -2077.5 \text{ joules}$$

20. (c) Pressure is reduced, so the temperature falls.

21. (d) Adiabatic Bulk modulus $E_\phi = \gamma P$

22. (c) In adiabatic process, no heat transfers between system and surrounding.

23. (b) $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma \Rightarrow P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = P_0 (8)^{4/3} = 16P_0.$

24. (c) In adiabatic process $PV^\gamma = \text{constant}$

$$\Rightarrow \left(\frac{RT}{V} \right)^\gamma \cdot V^\gamma = \text{constant} \Rightarrow TV^{\gamma-1} = \text{constant}$$

25. (a) $TV^{\gamma-1} = \text{constant} \Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^\gamma \Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^\gamma$

$$\Rightarrow T_2 = 300 \left(\frac{1}{2} \right)^{0.4} = 227.36 \text{ K}$$

26. (b) In adiabatic change $Q = \text{constant} \Rightarrow \Delta Q = 0$

$$\text{So } \Delta W = -\Delta U (\because \Delta Q = \Delta U + \Delta W)$$

27. (d) For adiabatic process from FLOT

$$\Delta W = -\Delta U \quad (\because \Delta Q = 0)$$

$$\Rightarrow \Delta W = -(-100) = +100 \text{ J}$$

28. (a) $\Delta U = -\Delta W = -\frac{R(T_1 - T_2)}{(\gamma-1)} = \frac{R(T_2 - T_1)}{\gamma-1}$

29. (d) $TV^{\gamma-1} = \text{constant} \Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 927^\circ \text{C}$

30. (d) The process is very fast, so the gas fails to gain or lose heat. Hence this process is adiabatic

31. (a) $\Delta U = \mu C_V \Delta T = 1 \times C_V (T_f - T_i) = -C_V (T_i - T_f)$

$$\Rightarrow |\Delta U| = C_V (T_i - T_f)$$

32. (c) $T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 273(2)^{0.41} = 273 \times 1.328 = 363 \text{ K}$

$$W = \frac{R(T_1 - T_2)}{\gamma-1} = \frac{8.31(273-363)}{1.41-1} = -1824$$

$$\Rightarrow |W| \approx 1815 \text{ J}$$

33. (d)

34. (c) $TV^{\gamma-1} = \text{constant}$

$$\Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = (273+18) \left(\frac{V}{V/8} \right)^{0.4} = 668 \text{ K}$$

35. (a) $\Delta Q = mc\Delta\theta$. Here $\Delta Q = 0$, hence $c = 0$

36. (d) In adiabatic process, no transfer of heat takes place between system and surrounding.

37. (b) $W = \frac{\mu R(T_1 - T_2)}{(\gamma-1)} = \frac{\mu R T_1}{(\gamma-1)} \left[1 - \frac{T_2}{T_1} \right]$

$$= \frac{\mu R T_1}{(\gamma-1)} \left[1 - \left(\frac{V_1}{V_2} \right)^{\gamma-1} \right]$$

$$= \frac{2 \times 8.31 \times 300}{\left(\frac{5}{3} - 1 \right)} \left[1 - \left(\frac{1}{2} \right)^{\frac{5}{3}-1} \right] = +2767.23 \text{ J}$$

38. (d) $T^\gamma P^{1-\gamma} = \text{constant} \Rightarrow T \propto P^{\frac{\gamma-1}{\gamma}}$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{1}{8} \right)^{\frac{5/3-1}{5/3}}$$

$$T_2 = 300 \times \left(\frac{1}{8}\right)^{0.4} = 131 K = -142^\circ C$$

39. (a) In adiabatic process $\Delta Q = 0 \Rightarrow \Delta U + \Delta W = 0$
 $(\because \Delta Q = \Delta U + \Delta W)$

40. (d) Using relation $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = (8)^{\frac{3/2-1}{3/2}} = 2$.
 $\Rightarrow T_2 = 2T_1 \Rightarrow T_2 = 2(273 + 27) = 600 K = 327^\circ C$

41. (c) $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{T_2}{T_1} = \left(\frac{1}{8}\right)^{\frac{1.5-1}{1.5}} = \left(\frac{1}{8}\right)^{\frac{1}{3}} = \frac{1}{2}$
 $\Rightarrow T_2 = \frac{T_1}{2} = \frac{300}{2} = 150 K$.

42. (c) $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} \Rightarrow \frac{P'}{P} = (8)^{5/2} \Rightarrow P' = P \times (2)^{15/2}$

43. (a)

44. (a) Given $P \propto T^3$, but we know for an adiabatic process, the pressure $P \propto T^{\gamma/\gamma-1}$

$$\text{So } \frac{\gamma}{\gamma-1} = 3 \Rightarrow \gamma = \frac{3}{2} \Rightarrow \frac{C_p}{C_v} = \frac{3}{2}$$

45. (b)

46. (d) $W = \frac{R(T_i - T_f)}{\gamma-1} \Rightarrow 6R = \frac{R(T - T_f)}{\left(\frac{5}{3} - 1\right)} \Rightarrow T_f = (T - 4)K$.

47. (a) $\because TV^{\gamma-1} = \text{constant} \Rightarrow T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$$\Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1} = T_1 (4)^{1.5-1} = 2T_1$$

$\therefore$ change in temperature

$$= T_2 - T_1 = 2T_1 - T_1 = T_1 = 273 K$$

48. (b) $\because PV^{\gamma} = k$ (constant) $\Rightarrow P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$

$$\Rightarrow P_2 = P_1 \left(\frac{V_1}{V_2}\right)^{\gamma} = 10^5 \times (2)^{1.3} \quad (\because V_2 = \frac{V_1}{2})$$

49. (b) In adiabatic process $\Delta U = -\Delta W$. In compression ΔW is negative, so ΔU is positive i.e. internal energy increases.

50. (a) According to the first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

In adiabatic process $\Delta Q = 0$, hence $\Delta U = -\Delta W$

51. (b) $PV^{\gamma} = \text{constant} \Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = \left(\frac{V_1}{V_1/4}\right)^{\gamma} = 4^{\gamma}$

$$\Rightarrow P_2 = 4^{\gamma} P$$

As γ is always greater than one so $4^{\gamma} > 4 \Rightarrow P_2 > 4P$

52. (d) $P_1 V_1^{\gamma} = P_2 V_2^{\gamma} \Rightarrow \frac{P_2}{P_1} = \left[\frac{V_1}{V_2}\right]^{\gamma} = \left[\frac{4}{1}\right]^{3/2} = \frac{8}{1}$

53. (b) Change in internal energy of the gas

$$\Delta U = -\Delta W \frac{R}{\gamma-1} [T_2 - T_1] = \frac{8.3}{(1.4-1)} [308 - 300] = 166 J$$

54. (b) For adiabatic change $TV^{\gamma-1} = \text{constant}$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \Rightarrow T_2 = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \times T_1$$

$$\Rightarrow T_2 = \left(\frac{V}{V/4}\right)^{1.4-1} \times 300 = 300 \times (4)^{0.4} K$$

55. (d) For adiabatic forces $\Delta W = -\Delta U$ ($\because \Delta Q = 0$)
 $\Rightarrow \Delta W = -(-50) = +50 J$

56. (b) $\frac{\text{Adiabatic elasticity } (E_{\phi})}{\text{Isothermal elasticity } (E_{\theta})} = \gamma \Rightarrow E_{\theta} = \frac{E_{\phi}}{\gamma}$

$$\Rightarrow E_{\theta} = \frac{2.1 \times 10^5}{1.4} = 1.5 \times 10^5 N/m^2$$

57. (c) $PV^{\gamma} = \text{constant}$: Differentiating both sides

$$P\gamma V^{\gamma-1} dV + V^{\gamma} dP = 0 \Rightarrow \frac{dP}{P} = -\gamma \frac{dV}{V}$$

Isobaric and Isochoric Processes

1. (a) Work done $= P\Delta V = P(V_2 - V_1)$

2. (c) When heat is supplied at constant pressure, a part of it goes in the expansion of gas and remaining part is used to increase the temperature of the gas which in turn increases the internal energy.

3. (c) For isobaric process $\frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow V_2 = V \times \frac{274}{273}$

$$\text{Increase} = \frac{274 V}{273} - V = \frac{V}{273}$$

4. (a) From FLOT $\Delta Q = \Delta U + \Delta W = \Delta U + P\Delta V$
 $\Rightarrow 100 = \Delta U + 50 \times (4 - 10) \Rightarrow \Delta U = 400 J$

5. (a) $W = P \times \Delta V = 2 \times 10^5 (150 - 50) \times 10^{-3} = 2 \times 10^4 J$

6. (c) $W = P\Delta V = nR\Delta T = 0.1 \times 2 \times 300 = 60 \text{ cal}$

7. (c) $\Delta Q = \Delta U + P\Delta V \Rightarrow mL = \Delta U + P(V_f - V_i)$
 $\Rightarrow \Delta U = L - P(V_f - V_i)$ ($\because m = 1$)

8. (b) $\Delta W = P\Delta V = 10^3 \times 0.25 = 250 J$

9. (d) $W = P\Delta V = 1.01 \times 10^5 (3.34 - 2 \times 10^{-3})$
 $= 337 \times 10^3 J \approx 340 KJ$

10. (c) $\frac{T_2}{T_1} = \frac{V_2}{V_1} = 2 \Rightarrow T_2 = 2 \times T_1 = 2 \times 300 = 600 K = 327^\circ C$

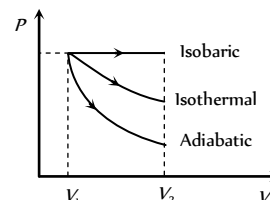
11. (c) $V \propto T$ at constant pressure

$$\Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow V_2 = \frac{V_1 T_2}{T_1} = \frac{300 \times 280}{300} = 280 \text{ ml.}$$

12. (a) In thermodynamic process, work done is equal to the area covered by the PV curve with volume axis.

Hence, according to graph shown

$$W_{\text{adiabatic}} < W_{\text{isothermal}} < W_{\text{isobaric}}$$



13. (b) (Similar to previous question)
14. (a)
15. (d) $W = P\Delta V = 2.4 \times 10^{-4} \times 1 \times 10^5 = 24 J$
16. (b) At constant pressure
 $W = P\Delta V = \mu R\Delta T = 1 \times 8.31 \times 100 = 831 \approx 814 J$
17. (a) $\Delta V = 0 \Rightarrow P\Delta V = 0 \Rightarrow \Delta W = 0$
18. (d) Entropy of a reversible process does not change.
19. (c) $W = P\Delta V = 0$ (As $\Delta V = 0$)
20. (d)
21. (d) At constant volume $P \propto T \Rightarrow \frac{P_1}{P_2} = \frac{T_1}{T_2} \Rightarrow \frac{P_1}{P_2} = \frac{300}{400} = \frac{3}{4}$
22. (d) In isothermal process $\Delta Q \neq 0$.
23. (a) For isochoric process $\Delta V = 0 \Rightarrow \Delta W = 0$
 From FLOT $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta U$

Heat Engine, Refrigerator and Second Law of Thermodynamics

1. (d) $\eta = \frac{T_1 - T_2}{T_1} - \frac{W}{Q} \Rightarrow Q = \left(\frac{T_1}{T_1 - T_2} \right) W$
 $= \frac{600}{(600 - 300)} \times 800 = 1600 J$
2. (c) Coefficient of performance
 $K = \frac{T_2}{T_1 - T_2} = \frac{273}{303 - 273} = \frac{273}{30} = 9$
3. (b) In a refrigerator, the heat dissipated in the atmosphere is more than that taken from the cooling chamber, therefore the room is heated if the door of a refrigerator is kept open.
4. (c) Internal energy is a state function.
5. (b)
6. (d) For a reversible process $\int \frac{dQ}{T} = 0$
7. (b) For cyclic forces $\Delta U = 0$ So, $\Delta Q = \Delta W$
8. (d) $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{400}{500} = \frac{1}{5} \therefore \eta = \frac{W}{Q} \Rightarrow \frac{1}{5} = \frac{W}{Q}$
 $\Rightarrow W = \frac{Q}{5} = \frac{6}{5} \times 10^4 = 1.2 \times 10^4 J$
9. (b) $\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{30}{100} = 1 - \frac{350}{T_1}$
 $\Rightarrow \frac{350}{T_1} = 1 - \frac{50}{100} = \frac{70}{100} = \frac{7}{10} \Rightarrow T_1 = 500 K = 227^\circ C$
10. (b) $\eta = 1 - \frac{T_2}{T_1}$ for 100% efficiency $\eta = 1$ which gives $T_2 = 0 K$
11. (c) $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{(273 + 69)}{(273 + 411)} = 0.5$
 $\Rightarrow \text{Work done} = \eta \times Q = 0.5 \times 1000 = 500 J$
12. (b) $\therefore \eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$
 where $Q_1 = \text{heat absorbed}$, $Q_2 = \text{heat rejected}$
 $\Rightarrow 1 - \frac{T/3}{T} = \frac{W}{Q_1} \Rightarrow \frac{2}{3} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$
 $\Rightarrow \frac{2}{3} = 1 - \frac{Q_2}{Q_1} \Rightarrow \frac{Q_2}{Q_1} = \frac{1}{3} \Rightarrow Q_2 = \frac{Q_1}{3} = \frac{Q}{3}$
13. (c) $\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{25}{100} = 1 - \frac{300}{T_1} \Rightarrow \frac{1}{4} = 1 - \frac{300}{T_1}$
 $T_1 = 400 K = 127^\circ C$
14. (a) $\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{70}{100} = 1 - \frac{T_2}{1000} \Rightarrow T_2 = 300 K$
15. (a) $\eta = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1} \Rightarrow \eta_1 = \frac{(473 - 273)}{473} = \frac{200}{473}$
 and $\eta_2 = \frac{273 - 73}{273} = \frac{200}{273}$
 So required ratio $\frac{\eta_1}{\eta_2} = \frac{273}{473} = 0.577$
16. (a) $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{(273 + 123)}{(273 + 27)} = 1 - \frac{150}{300} = \frac{1}{2} = 50\%$
17. (b) $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = \frac{2}{5}$
18. (d)
19. (b) In first case, $(\eta_1) = 1 - \frac{500}{800} = \frac{3}{8}$
 and in second case, $(\eta_2) = 1 - \frac{600}{x}$
 Since $\eta_1 = \eta_2$, therefore $\frac{3}{8} = 1 - \frac{600}{x}$
 or $\frac{600}{x} = 1 - \frac{3}{8} = \frac{5}{8}$ or $x = \frac{600 \times 8}{5} = 960 K$
20. (a) $\eta_{\max} = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{400} = \frac{1}{4} = 25\%$
 So 26% efficiency is impossible
21. (b) In first case $\eta_1 = 1 - \frac{T_2}{T_1} = 1 - \frac{(273 + 0)}{(273 + 200)} = \frac{200}{473}$
 In second case $\eta_2 = 1 - \frac{(273 - 200)}{(273 + 0)} = \frac{200}{273}$
 $\Rightarrow \frac{\eta_1}{\eta_2} = \frac{1}{\left(\frac{473}{273} \right)} = 1 : 1.73$
22. (b) $\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{1}{2} = 1 - \frac{500}{T_1} \Rightarrow \frac{500}{T_1} = \frac{1}{2}$ (i)
 $\frac{60}{100} = 1 - \frac{T_2'}{T_1} \Rightarrow \frac{T_2'}{T_1} = \frac{2}{5}$ (ii)

Dividing equation (i) by (ii), $\frac{500}{T_2'} = \frac{5}{4} \Rightarrow T_2' = 400K$

23. (c)
24. (a)

25. (b) $\eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q} \Rightarrow W = \left(1 - \frac{T_1}{T_2}\right) Q = \left\{1 - \frac{(273 + 27)}{(273 + 627)}\right\}$
 $\Rightarrow W = \left(1 - \frac{300}{900}\right) \times 3 \times 10^6 = 2 \times 10^6 \times 4.2 J = 8.4 \times 10^6 J$

26. (a)

27. (d) $\eta = 1 - \frac{T_2}{T_1}$; for η to be max. ratio $\frac{T_2}{T_1}$ should be min.

28. (a)

29. (b) In first case $\eta_1 = \frac{T_1 - T_2}{T_1}$
 In second case $\eta_2 = \frac{2T_1 - 2T_2}{2T_1} = \frac{T_1 - T_2}{T_1} = \eta$

30. (c) Coefficient of performance

$$K = \frac{T_2}{T_1 - T_2} \Rightarrow 5 = \frac{(273 - 13)}{T_1 - (273 - 13)} = \frac{260}{T_1 - 260}$$

$$\Rightarrow 5T_1 - 1300 = 260 \Rightarrow 5T_1 = 1560$$

$$\Rightarrow T_1 = 312K \rightarrow 39^\circ C$$

31. (a) Coefficient of performance $K = \frac{T_2}{T_1 - T_2}$
 $= \frac{(273 - 23)}{(273 + 27) - (273 - 23)} = \frac{250}{300 - 250} = \frac{250}{50} = 5$

32. (a) $\eta = \frac{T_1 - T_2}{T_1} = \frac{(273 + 727) - (273 + 227)}{273 + 727} = \frac{1000 - 500}{1000} = \frac{1}{2}$

33. (c) $\eta = \frac{T_1 - T_2}{T_1} = \frac{W}{Q} \Rightarrow W = \frac{Q(T_1 - T_2)}{T_1}$
 $= \frac{6 \times 10^4 [(227 + 273) - (273 + 127)]}{(227 + 273)}$
 $= \frac{6 \times 10^4 \times 100}{500} = 1.2 \times 10^4 \text{ cal}$

34. (d) Slow isothermal expansion or compression of an ideal gas is reversible process, while the other given process are irreversible in nature.

3. (b,c) There is a decrease in volume during melting on an ice slab at 273K. Therefore, negative work is done by ice-water system on the atmosphere or positive work is done on the ice-water system by the atmosphere. Hence option (b) is correct. Secondly heat is absorbed during melting (*i.e.* ΔQ is positive) and as we have seen, work done by ice-water system is negative (ΔW is negative). Therefore, from first law of thermodynamics $\Delta U = \Delta Q - \Delta W$.

Change in internal energy of ice-water system, ΔU will be positive or internal energy will increase.

4. (c) Process is isothermal. Therefore, $T = \text{constant}$,

$\left(P \propto \frac{1}{V}\right)$ volume is increasing, therefore pressure will decrease.

In chamber A :

$$\Delta P = P_i - P_f = \frac{\mu_A RT}{V} - \frac{\mu_A RT}{2V} = \frac{\mu_A RT}{2V} \quad \dots(i)$$

In chamber B :

$$1.5\Delta P = P_i - P_f = \frac{\mu_B RT}{V} - \frac{\mu_B RT}{2V} = \frac{\mu_B RT}{2V} \quad \dots(ii)$$

from equations (i) and (ii) $\frac{\mu_A}{\mu_B} = \frac{1}{1.5} = \frac{2}{3}$

$$\Rightarrow \frac{m_A / M}{m_B / M} = \frac{2}{3} \Rightarrow 3m_A = 2m_B.$$

5. (d) $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \Rightarrow \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{L_2 A}{L_1 A}\right)^{\frac{5}{3}-1} = \left(\frac{L_2}{L_1}\right)^{\frac{2}{3}}$

6. (d) Using Boyle's law, we have $\frac{V}{T} = \text{constant}$

$$\Rightarrow \frac{\frac{l}{2} + 5}{373} = \frac{\frac{l}{2} - 5}{273}$$

As the piston moves 5 cm, the length of one side will be

$\left(\frac{l}{2} + 5\right)$ and other side $\left(\frac{l}{2} - 5\right)$. On solving this equation,

we get $l = 64.6 \text{ cm}$.

Critical Thinking Questions

1. (d) Fraction of supplied energy which increases the internal energy is given by

$$f = \frac{\Delta U}{(\Delta Q)_p} = \frac{(\Delta Q)_v}{(\Delta Q)_p} = \frac{\mu C_v \Delta T}{\mu C_p \Delta T} = \frac{1}{\gamma}$$

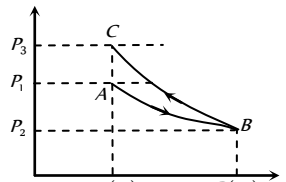
For diatomic gas $\gamma = \frac{7}{5} \Rightarrow f = \frac{5}{7}$

2. (c) $\Delta Q = \Delta U + \Delta W$

$$\therefore \Delta U = \Delta Q - \Delta W = 540 - \frac{P(V_2 - V_1)}{J}$$

$$= 540 - \frac{1.013 \times 10^5 \times [(1671 - 1) \times 10^{-6}]}{4.2}$$

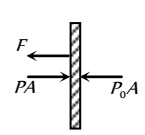
$$= 540 - 39.7 = 500 \text{ calories}$$

7. (c) $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta W = (\Delta Q)_p - \Delta U = (\Delta Q)_p \left[1 - \frac{(\Delta Q)_v}{(\Delta Q)_p} \right]$
 $= (\Delta Q)_p \left[1 - \frac{C_v}{C_p} \right] = Q = \left[1 - \frac{3}{5} \right] = \frac{2}{5} Q$
 $\therefore (\Delta Q)_p = Q$ and $\gamma = \frac{5}{3}$ for monatomic gas
8. (d) Oxygen is diatomic gas, hence its energy of two moles
 $= 2 \times \frac{5}{2} RT = 5RT$
 Argon is a monoatomic gas, hence its internal energy of 4 moles
 $= 4 \times \frac{3}{2} RT = 6RT$
 Total Internal energy $= (6 + 5)RT = 11RT$
9. (c) From graph it is clear that $P_3 > P_1$.
 Since area under adiabatic process (BCED) is greater than that of isothermal process (ABDE). Therefore net work done
 $W = W_i + (-W_A) \therefore W_A > W_i \Rightarrow W < 0$
- 
10. (a) According to given Vander Waal's equation

$$P = \frac{nRT}{V - n\beta} - \frac{an^2}{V^2}$$

 Work done, $W = \int_{V_1}^{V_2} P dV = nRT \int_{V_1}^{V_2} \frac{dV}{V - n\beta} - an^2 \int_{V_1}^{V_2} \frac{dV}{V^2}$

$$= nRT \left[\log_e (V - n\beta) \right]_{V_1}^{V_2} + an^2 \left[\frac{1}{V} \right]_{V_1}^{V_2}$$

$$= nRT \log_e \frac{V_2 - n\beta}{V_1 - n\beta} + an^2 \left(\frac{1}{V_1} - \frac{1}{V_2} \right)$$
11. (b) Volume of the gas is constant $V = \text{constant} \therefore P \propto T$
 i.e., pressure will be doubled if temperature is doubled
 $\therefore P = 2P_0$
 Now let F be the tension in the wire. Then equilibrium of any one piston gives
 $F = (P - P_0)A = (2P_0 - P_0)A = P_0 A$
- 
12. (c) $dU = C_v dT = \left(\frac{5}{2} R \right) dT$ or $dT = \frac{2(dU)}{5R}$
 From first law of thermodynamics
 $dU = dQ - dW = Q - \frac{Q}{4} = \frac{3Q}{4}$. Now molar heat capacity
 $C = \frac{dQ}{dT} = \frac{Q}{\frac{2(dU)}{5R}} = \frac{5RQ}{2 \left(\frac{3Q}{4} \right)} = \frac{10}{3} R$
13. (b) $Q = \Delta U = U_f - U_i = [\text{internal energy of 4 moles of a monoatomic gas} + \text{internal energy of 2 moles of a diatomic gas}] - [\text{internal energy of 4 moles of a diatomic gas}]$

$$= \left(4 \times \frac{3}{2} RT + 2 \times \frac{5}{2} RT \right) - \left(4 \times \frac{5}{2} RT \right) = RT$$

Note : (a) 2 moles of diatomic gas becomes 4 moles of a monoatomic gas when gas dissociated into atoms.

(b) Internal energy of μ moles of an ideal gas of degrees of freedom F is given by $U = \frac{f}{2} \mu RT$

$F = 3$ for a monoatomic gas and 5 for diatomic gas.

14. (c) $PV^\gamma = K$ or $P \gamma V^{\gamma-1} dV + dP \cdot V^\gamma = 0$
 or $\frac{dP}{P} = -\gamma \frac{dV}{V}$ or $\frac{dP}{P} \times 100 = -\gamma \left(\frac{dV}{V} \times 100 \right)$
 $= -1.4 \times 5 = -7\%$

15. (a) $TV^{\gamma-1} = \text{constant}$
 $\therefore \frac{T_1}{T_2} = \left(\frac{V_2}{V_1} \right)^{\gamma-1}$ or $\left(\frac{1}{2} \right)^{\gamma-1} = \sqrt{\frac{1}{2}}$
 $\therefore \gamma - 1 = \frac{1}{2}$ or $\gamma = \frac{3}{2}$ $\therefore PV^{3/2} = \text{constant}$

16. (c) $\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1} \Rightarrow \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$
 $\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2} \therefore W_A = W_B$
 $\therefore T_2 = \frac{T_1 + T_3}{2} = \frac{800 + 300}{2} = 550K$

17. (b) For monoatomic gas
 $\gamma = \frac{C_p}{C_v} = \frac{5}{3}$ we know $\Delta Q = nC_p \Delta T$
 and $\Delta U = nC_v \Delta T \Rightarrow \frac{\Delta U}{\Delta Q} = \frac{C_v}{C_p} = \frac{3}{5}$
 i.e. fraction of heat energy to increase the internal energy be 3/5.

18. (a) $\Delta Q = \Delta U + \Delta W \Rightarrow \frac{\Delta W}{\Delta Q} = 1 - \frac{\Delta U}{\Delta Q} = 1 - \frac{nC_v dT}{nC_p dT}$
 $\Rightarrow \frac{\Delta W}{\Delta Q} = 1 - \frac{C_v}{C_p} = 1 - \frac{3}{5} = \frac{2}{5} = 0.4$

19. (b) $\Delta U = \mu C_v \Delta T = \frac{m}{M} C_v \Delta T = \frac{N}{N_A} C_v \Delta T$
 $\Rightarrow (\Delta U)_N = \frac{56 \times 10^3}{14} \times \frac{5}{2} R \times 300$
 and $(\Delta U)_A = \frac{6 \times 10^{26}}{6 \times 10^{23}} \times \frac{3}{2} R \times 900 \Rightarrow (\Delta U)_N > (\Delta U)_A$

20. (c) A is compressed isothermally, hence
 $P_1 V_1 = P_2 V_2 \Rightarrow P_2 = 2P_1$
 and B is compressed adiabatically, hence
 $P_1 V_1^\gamma = P_2 V_2^\gamma \Rightarrow P_2 = (2)^\gamma P_1$
 Since $\gamma > 1$, hence $P_2' > P_2$ or $P_2 < P_2'$

21. (b) In isothermal process $P_1 V_1 = P_2 V_2$
 or $PV = P_2 \times 4V \therefore P_2 = \frac{P}{4}$
 In adiabatic process

$$P_2 V_2^\gamma = P_3 V_3^\gamma \Rightarrow \frac{P}{4} \times (4V)^{1.5} = P_2 V^{1.5} \Rightarrow P_2 = 2P$$

22. (d) Volume of the ideal gas is constant so $W = P\Delta V = 0$ using FLOT $\Delta Q = \Delta U \Rightarrow \Delta U = i^2 R t = 1^2 \times 100 \times 5 \times 60 = 30 \times 10^3 = 30 \text{ KJ}$

23. (d) Initially $\eta = \left(1 - \frac{T_2}{T_1}\right) = \frac{W}{Q} = \frac{1}{6}$ (i)

Finally $\eta' = \left(1 - \frac{T_2'}{T_1}\right) = \left(1 - \frac{(T_2 - 62)}{T_1}\right) = 1 - \frac{T_2}{T_1} + \frac{62}{T_1}$
 $= \eta + \frac{62}{T_1}$ (ii)

It is given that $\eta' = 2\eta$. Hence solving equation (i) and (ii)

$$\Rightarrow T_1 = 372 \text{ K} = 99^\circ\text{C} \text{ and } T_2 = 310 \text{ K} = 37^\circ\text{C}$$

24. (b) Input energy $= \frac{1g}{\text{sec}} \times \frac{2kcal}{g} = 2kcal/\text{sec}.$

Output energy $= 10 \text{ KW} = 10 \text{ KJ/S} = \frac{10}{4.2} kcal/\text{sec}.$

$$\Rightarrow \eta = \frac{\text{output energy}}{\text{input energy}} = \frac{10}{4.2 \times 2} > 1, \text{ it is impossible.}$$

25. (b) Gain of entropy of ice

$$S_1 = \frac{\Delta Q}{T} = \frac{mL}{T} = \frac{80 \times 100}{(0 + 273)} = \frac{8 \times 10^3}{273} cal/K$$

Loss of entropy of water $= S_2 = -\frac{\Delta Q}{T} = -\frac{mL}{T}$

$$= -\frac{80 \times 100}{(273 + 50)} = -\frac{8 \times 10^3}{323} cal/K$$

Total change of entropy

$$S_1 + S_2 = \frac{8 \times 10^3}{273} - \frac{8 \times 10^3}{323} = +4.5 cal/K$$

26. (b) $PV^2 = \text{constant}$ represents adiabatic equation. So during the expansion of ideal gas internal energy of gas decreases and temperature falls.

27. (d) Initially $\eta = \frac{T_1 - T_2}{T_1} \Rightarrow 0.5 = \frac{T_1 - (273 + 7)}{T_1}$

$$\Rightarrow \frac{1}{2} = \frac{T_1 - 280}{T_1} \Rightarrow T_1 = 560 \text{ K}$$

Finally $\eta_1' = \frac{T_1' - T_2}{T_1'} \Rightarrow 0.7 = \frac{T_1' - (273 + 7)}{T_1'} \Rightarrow T_1' = 933 \text{ K}$

$$\therefore \text{increase in temperature} = 933 - 560 = 373 \text{ K} \approx 380 \text{ K}$$

28. (c) P - V diagram of the gas is a straight line passing through origin.

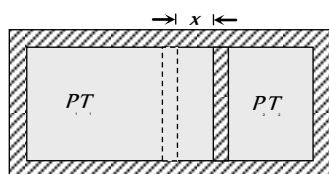
Hence $P \propto V$ or $PV^{-1} = \text{constant}$

Molar heat capacity in the process $PV^x = \text{constant}$ is

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - x}; \text{ Here } \gamma = 1.4 \text{ (For diatomic gas)}$$

$$\Rightarrow C = \frac{R}{1.4 - 1} + \frac{R}{1 - 1} \Rightarrow C = 3R$$

29. (c) As finally the piston is in equilibrium, both the gases must be at same pressure P_f . It is given that displacement of piston be in final state x and if A is the area of cross-section of the piston. Hence the final volumes of the left and right part finally can be given by figure as



$$V_L = \frac{V_0}{2} + Ax \text{ and } V_R = \frac{V_0}{2} - Ax$$

As it is given that the container walls and the piston are adiabatic in left side and the gas undergoes adiabatic expansion and on the right side the gas undergoes adiabatic compressive. Thus we have for initial and final state of gas on left side

$$P_1 \left(\frac{V_0}{2}\right)^\gamma = P_f \left(\frac{V_0}{2} + Ax\right)^\gamma \text{(i)}$$

Similarly for gas in right side, we have

$$P_2 \left(\frac{V_0}{2}\right)^\gamma = P_f \left(\frac{V_0}{2} - Ax\right)^\gamma \text{(ii)}$$

From eq. (i) and (ii)

$$\frac{P_1}{P_2} = \frac{\left(\frac{V_0}{2} + Ax\right)^\gamma}{\left(\frac{V_0}{2} - Ax\right)^\gamma} \Rightarrow Ax = \frac{V_0}{2} \left[\frac{P_1^{1/\gamma} - P_2^{1/\gamma}}{P_1^{1/\gamma} + P_2^{1/\gamma}} \right]$$

Now from equation (i) $P_f = \frac{P_1 \left(\frac{V_0}{2}\right)^\gamma}{\left[\frac{V_0}{2} + Ax\right]^\gamma}$

30. (d) In both cylinders A and B the gases are diatomic ($\gamma = 1.4$). Piston A is free to move i.e. it is isobaric process. Piston B is fixed i.e. it is isochoric process. If same amount of heat ΔQ is given to both then

$$(\Delta Q)_{\text{isobaric}} = (\Delta Q)_{\text{isochoric}} \Rightarrow \mu C_p (\Delta T)_A = \mu C_v (\Delta T)_B$$

$$\Rightarrow (\Delta T)_B = \frac{C_p}{C_v} (\Delta T)_A = \gamma (\Delta T)_A = 1.4 \times 30 = 42 \text{ K.}$$

Graphical Questions

- (c) As internal energy is a point function therefore change in internal energy does not depend upon the path followed i.e. $\Delta U_I = \Delta U_{II}$
- (b) Work done by the system = Area of shaded portion on P - V diagram
 $= (300 - 100)10^{-6} \times (200 - 10) \times 10^3 = 20 \text{ J}$
- (a) Work done = Area enclosed by triangle
 $ABC = \frac{1}{2} AC \times BC = \frac{1}{2} \times (3V - V) \times (3P - P) = 2 PV$
- (c) Area enclosed between a and f is maximum. So work done in closed cycles follows a and f is maximum.
- (a) Initial and final states are same in all the process.
Hence $\Delta U = 0$; in each case.
By FLOT; $\Delta Q = \Delta W = \text{Area enclosed by curve with volume axis.}$
 $\therefore (\text{Area})_c < (\text{Area})_b < (\text{Area})_a \Rightarrow Q_c < Q_b < Q_a.$
- (a) For an isothermal process $PV = \text{constant}$
 $\Rightarrow PdV + VdP = 0 \Rightarrow -\frac{1}{V} \left(\frac{dV}{dP}\right) = \frac{1}{P}$
So, $\beta = \frac{1}{P}$ $\therefore$ graph will be rectangular hyperbola.
- (a) By adjoining graph $W_{AB} = 0$ and

$$W_{BC} = 8 \times 10^4 [5 - 2] \times 10^{-3} = 240 \text{ J}$$

$$\therefore W_{AC} = W_{AB} + W_{BC} = 0 + 240 = 240 \text{ J}$$

$$\text{Now, } \Delta Q_{AC} = \Delta Q_{AB} + \Delta Q_{BC} = 600 + 200 = 800 \text{ J}$$

$$\text{From FLOT } \Delta Q_{AC} = \Delta U_{AC} + \Delta W_{AC}$$

$$\Rightarrow 800 = \Delta U_{AC} + 240 \Rightarrow \Delta U_{AC} = 560 \text{ J.}$$

8. (b) In adiabatic process, slope of PV -graph.

$$\frac{dP}{dV} = -\gamma \frac{P}{V} \Rightarrow |\text{Slope}| \propto \gamma$$

From the given graph (Slope)₁ > (Slope)₂ $\Rightarrow \gamma_2 > \gamma_1$

therefore 1 should correspond to O ($\gamma = 1.4$) and 2 should correspond to He ($\gamma = 1.66$)

9. (c) As we know that slope of isothermal and adiabatic curves are always negative and slope of adiabatic curve is always greater than that of isothermal curve
Hence in the given graph curve A and B represents adiabatic and isothermal changes respectively.

10. (d) Process CD is isochoric as volume is constant, Process DA is isothermal as temperature constant and Process AB is isobaric as pressure is constant.

11. (d) Heat given $\Delta Q = 20 \text{ cal} = 20 \times 4.2 = 84 \text{ J}$.

Work done $\Delta W = -50 \text{ J}$ [As process is anticlockwise]

By first law of thermodynamics

$$\Rightarrow \Delta U = \Delta Q - \Delta W = 84 - (-50) = 134 \text{ J}$$

12. (a) For cyclic process. Total work done $= W_{AB} + W_{BC} + W_{CA}$

$$\Delta W_x = P\Delta V = 10(2 - 1) = 10 \text{ J and } \Delta W_y = 0$$

(as $V = \text{constant}$)

$$\text{From FLOT, } \Delta Q = \Delta U + \Delta W$$

$$\Delta U = 0 \text{ (Process } ABCA \text{ is cyclic)}$$

$$\Rightarrow \Delta Q = \Delta W_x + \Delta W_y + \Delta W_z$$

$$\Rightarrow 5 = 10 + 0 + \Delta W_z \Rightarrow \Delta W_z = -5 \text{ J}$$

13. (b) The cyclic process 1 is clockwise where as process 2 is anticlockwise. Clockwise area represents positive work and anticlockwise area represents negative work. Since negative area (2) > positive area (1), hence net work done is negative.

14. (c) Process AB is isochoric, $\therefore W_{AB} = P\Delta V = 0$

$$\text{Process } BC \text{ is isothermal } \therefore W_{BC} = RT_2 \ln \left(\frac{V_2}{V_1} \right)$$

Process CA is isobaric

$$\therefore W_{CA} = -P\Delta V = -R\Delta T = -R(T_1 - T_2) = R(T_2 - T_1)$$

(Negative sign is taken because of compression)

15. (a) AB is isobaric process, BC is isothermal process, CD is isometric process and DA is isothermal process

These process are correctly represented by graph (a).

16. (c) Work done by the gas (as cyclic process is clockwise) $\therefore \Delta W = \text{Area } ABCD$

So from the first law of thermodynamics ΔQ (net heat absorbed) $= \Delta W = \text{Area } ABCD$

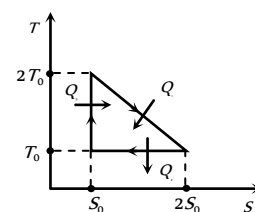
As change in internal energy in cycle $\Delta U = 0$.

17. (a) $Q_1 = T_0 S_0 + \frac{1}{2} T_0 S_0 = \frac{3}{2} T_0 S_0$

$$Q_2 = T_0 S_0 \text{ and } Q_3 = 0$$

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1} = 1 - \frac{2}{3} = \frac{1}{3}$$



18. (a) Work done = Area of closed PV diagram

$$= (2V - V) \times (2P - P) = PV$$

19. (c) From the given VT diagram,

In process AB , $V \propto T \Rightarrow$ Pressure is constant (As quantity of the gas remains same)

In process BC , $V = \text{Constant}$ and in process CA , $T = \text{constant}$

$\therefore$ These processes are correctly represented on PV diagram by graph (c).

20. (d) $\Delta Q = \Delta U + \Delta W$; ΔU does not depend upon path.

$$\therefore \Delta W_A > \Delta W_B \Rightarrow \Delta Q_A > \Delta Q_B$$

21. (d) Work done = Area under curve $= \frac{6P_1 \times 3V_1}{2} = 9 PV_1$

22. (d) Work done $= \frac{1}{2} \times 2P_1 \times 2V_1 = 2P_1 V_1$

23. (c) In a cyclic, $\Delta U = 0$

From FLOT, $\Delta Q = \Delta U + \Delta W = 0 + \Delta W = \text{Area of closed curve}$

$$\Rightarrow \Delta Q = \pi r \pi \left(\frac{20}{2} \right)^2 kP_a \times \text{litre}$$

$$= 100 \pi \times 10^3 \times 10^{-3} \text{ J} = 100 \pi \text{ J}$$

24. (c) The work done in cyclic process is equal to the area enclosed by the PV diagram

25. (d) In all given cases, process is cyclic and in cyclic process $\Delta U = 0$.

26. (b) In cyclic process $\Delta Q = \text{Work done} = \text{Area inside the closed curve.}$

$$\text{Treat the circle as an ellipse of area} = \frac{\pi}{4} (P_2 - P_1)(V_2 - V_1)$$

$$\Rightarrow \Delta Q = \frac{\pi}{4} \{(150 - 50) \times 10^3\} = \frac{\pi}{2} \text{ J}$$

27. (d) $W_{\text{net}} = - \text{Area of triangle } BCO = - \frac{P_0 V_0}{2}$

$$W_{\text{net}} = + \text{Area of triangle } AOD = + \frac{P_0 V_0}{2}$$

28. (c) AD and BC represent adiabatic process (more slope)
 AB and DC represent isothermal process (less slope)

29. (c) Work done = Area of curve enclosed

$$= 2V \times 2P = 4PV$$

30. (c) Work done = Area of PV graph (here trapezium)

$$= \frac{1}{2} (1 \times 10^5 + 5 \times 10^5) \times (5 - 1) = 12 \times 10^5 \text{ J}$$

31. (d) For path ab : $(\Delta U)_{ab} = 7000 \text{ J}$

By using $\Delta U = \mu C_V \Delta T$

$$7000 = \mu \times \frac{5}{2} R \times 700 \Rightarrow \mu = 0.48$$

For path ca :

$$(\Delta Q)_{ca} = (\Delta U)_{ca} + (\Delta W)_{ca} \quad \dots(i)$$

$$\therefore (\Delta U)_{ab} + (\Delta U)_{bc} + (\Delta U)_{ca} = 0$$

$$\therefore 7000 + 0 + (\Delta U)_{ca} = 0 \Rightarrow (\Delta U)_{ca} = -7000 \text{ J} \quad \dots(ii)$$

$$\text{Also } (\Delta W)_{ca} = P_1(V_1 - V_2) = \mu R(T_1 - T_2)$$

$$= 0.48 \times 8.31 \times (300 - 1000) = -2792.16 \text{ J} \quad \dots(iii)$$

on solving equations (i), (ii) and (iii)

$$(\Delta Q)_{ca} = -7000 - 2792.16 = -9792.16 \text{ J} = -9800 \text{ J}$$

32. (b) Work done = Area enclosed by indicator diagram

$$= \frac{1}{2} \times (3V - V)(4P - P) = 3PV$$

33. (d) ΔU , remains same for both path

$$\text{For path } iaf: \Delta U = \Delta Q - \Delta W = 50 - 20 = 30 \text{ J}.$$

$$\text{For path } fi: \Delta U = -30 \text{ J and } \Delta W = -13 \text{ J}$$

$$\Rightarrow \Delta Q = -30 - 13 = -43 \text{ J}.$$

34. (a) $\Delta E_{\text{int}} = 0$, for a complete cycle and for given cycle work done is negative, so from first law of thermodynamics Q will be negative i.e. $Q < 0$.

35. (d) Work done = Area enclosed by the curve

$$= \frac{1}{2} (3V - V)(2P - P) = PV$$

36. (d) W = Area bonded by the indicator diagram with V -axis

$$= \frac{1}{2} (P_A + P_B)(V_B - V_A)$$

37. (a) Heat given $\Delta Q = 40 \text{ J}$ and Work done $\Delta W = 30 \text{ J}$

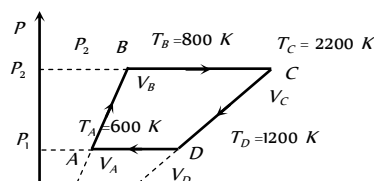
$$\Rightarrow \Delta U = \Delta Q - \Delta W = 40 - 30 = 10 \text{ J}.$$

38. (a) As the volume is continuously increasing and the work of expansion is always positive, so the work done by the system continuously increases.

39. (c) Processes A to B and C to D are parts of straight line graphs of the form $y = mx$

$$\text{Also } P = \frac{\mu R}{V} T \quad (\mu = 6)$$

$\Rightarrow P \propto T$. So volume remains constant for the graphs AB and CD



So no work is done during processes for A to B and C to D i.e. $W_a = 0$ and $W_c = P(V_c - V) = \mu R(T_c - T)$

$$= 6R(2200 - 800) = 6R \times 1400 \text{ J}$$

$$\text{Also } W_b = P(V_c - V) = \mu R(T_c - T)$$

$$= 6R(600 - 1200) = -6R \times 600 \text{ J}$$

Hence work done in complete cycle

$$W = W_a + W_b + W_c + W_d$$

$$= 0 + 6R \times 1400 + 0 - 6R \times 600$$

$$= 6R \times 900 = 6 \times 8.3 \times 800 \approx 40 \text{ kJ}$$

40. (b) In isothermal process $P \propto \frac{1}{V}$.

Hence graph between P and V is a hyperbola.

41. (d) Adiabatic curves are more steeper than isothermal curves.

42. (d) During process A to B , pressure and volume both are decreasing. Therefore, temperature and hence, internal energy of the gas will decrease ($T \propto PV$) or $\Delta U_{A \rightarrow B}$ = negative.

Further $\Delta W_{A \rightarrow B}$ is also negative as the volume of the gas is decreasing. Thus $\Delta Q_{A \rightarrow B}$ is negative.

In process B to C pressure of the gas is constant while volume is increasing. Hence temperature should increase or $\Delta U_{B \rightarrow C}$ = positive. During C to A volume is constant while pressure is increasing. Therefore, temperature and hence, internal energy of the gas should increase or $\Delta U_{C \rightarrow A}$ = positive. During process CAB volume of the gas is decreasing. Hence, work done by the gas is negative.

43. (a) $\Delta W_{AB} = 0$ as V = constant

$$\therefore \Delta Q_{AB} = \Delta U_{AB} = 50 \text{ J} \quad (\text{Given})$$

$$U_A = 1500 \text{ J} \therefore U_B = (1500 + 50) \text{ J} = 1550 \text{ J}$$

$$\Delta W_{BC} = -\Delta U_{BC} = -40 \text{ J} \quad (\text{Given})$$

$$\therefore \Delta U_{BC} = 40 \text{ J} \therefore U_C = (1550 + 40) \text{ J} = 1590 \text{ J}$$

44. (a) For adiabatic process $T_1 V_b^{\gamma-1} = \text{Constant}$

$$\text{For } bc \text{ curve } T_1 V_b^{\gamma-1} = T_2 V_c^{\gamma-1} \text{ or } \frac{T_2}{T_1} = \left(\frac{V_b}{V_c} \right)^{\gamma-1} \quad \dots(i)$$

$$\text{For } ad \text{ curve } T_1 V_d^{\gamma-1} = T_2 V_a^{\gamma-1} \text{ or } \frac{T_2}{T_1} = \left(\frac{V_a}{V_d} \right)^{\gamma-1} \quad \dots(ii)$$

$$\text{From equation (i) and (ii) } \frac{V_b}{V_c} = \frac{V_a}{V_d}$$

Assertion and Reason

- (a) In a perfectly reversible system, there is no loss of energy. Losses can be minimised, friction can be reduced, the resistance in L-C oscillating system can also be negligible. But one cannot completely eliminate energy losses. This makes a perfectly reversible system, an ideal.
- (a) Adiabatic expansion produces cooling.
- (a) In reversible process, there always occurs some loss of energy. This is because energy spent in working against the dissipative force is not recovered back. Some irreversible process occur in nature such as friction where extra work to cancel the effect of friction. Salt dissolves in water but a salt does not separate by itself into pure salt and pure water.
- (a) When a bottle of cold carbonated drink is opened. A slight fog forms around the opening. This is because of adiabatic expansion of gas causes lowering of temperature and condensation of water vapours.
- (e) As isothermal processes are very slow and so the different isothermal curves have different slopes so they cannot intersect each other.
- (d) Adiabatic compression is a rapid action and both the internal energy and the temperature increases.
- (e) As there is no change in internal energy of the system during an isothermal change. Hence, the energy taken by the gas is

utilised by doing work against external pressure. According to FLOT $\Delta Q = \Delta U + P\Delta V$

Hence $\Delta Q = \Delta U = P\Delta V$

Therefore, reason is true and assertion is false.

8. (d) We can change the temperature of a body without giving (or taking) heat to (or from) it. For example in an adiabatic compression temperature rises and in an adiabatic expansion temperature falls, although no heat is given or taken from the system in the respective changes.

9. (a) $c = \frac{Q}{m\Delta\theta}$; a gas may be heated by putting pressure, so it can have values for 0 to ∞ .

C_p and C_v are its two principle specific heats, out of infinite possible values.

In adiabatic process $C = 0$ and in isothermal process $C = \infty$.

10. (a) Heat is similar to work in that both represent ways of transferring energy. Neither heat nor work is an intrinsic property of a system, that is, we cannot say that a system contains a certain amount of heat or work.

11. (d) According to first law of thermodynamics, $\Delta Q = \Delta U + \Delta W = \Delta U + P\Delta V$. If heat is supplied in such a manner that volume does not change $\Delta V = 0$ i.e., isochoric process, then whole of the heat energy supplied to the system will increase internal energy only. But, in any other process it is not possible.

Also heat may be absorbed or evolved when state of thermal equilibrium changes.

12. (d) When the door of refrigerator is kept open, heat rejected by the refrigerator to the room will be more than the heat taken by the refrigerator from the room (by an amount equal to work done by the compressor). Therefore, temperature of room will increase and so it will be warmed gradually. As according to 2nd law of thermodynamics, heat cannot be transferred on its own, from a body at lower temperature to another at higher temperature.

13. (a) Second law of thermodynamics can be explained with the help of example of refrigerator, as we know that refrigerator, the working substance extracts heat from colder body and rejects a large amount of heat to a hotter body with the help of an external agency i.e., the electric supply of the refrigerator. No refrigerator can ever work without external supply of electric energy to it.

14. (d) If an electric fan is switched on in a closed room, the air will be heated because due to motion of the fan, the speed of air molecules will increase. In fact, we feel cold due to evaporation of our sweat.

15. (c) The internal energy of system depends only on its temperature. In isothermal process temperature does not change, therefore, internal energy of the system remains the same.

16. (c) In an adiabatic process, no exchange of heat is permissible i.e., $\Delta Q = 0$.

As, $\Delta Q = \Delta U + \Delta W = 0 \Rightarrow \Delta U = -\Delta W$.

Also in adiabatic process, temperature of gas changes.

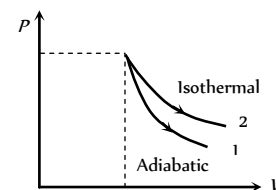
17. (a) Change in entropy, $\Delta S = \frac{\Delta Q}{T}$. In an adiabatic change, heat transfer $\Delta Q = 0$. $\therefore \Delta S = 0$, or $S = \text{constant}$ i.e., entropy remains constant in an adiabatic process, or an adiabatic process is an isentropic process.

18. (b) As we know, in thermodynamic processes work done = Area covered by P - V diagram with volume axis.

Hence, according to following graph.

$$(\text{Area})_1 < (\text{Area})_2 \Rightarrow W_1 < W_2$$

Also in isothermal changes temperature remains same but in adiabatic changes temperature also changes.



19. (c) First law of thermodynamics is restatement of the principle of conservation of energy as applied to heat energy.

20. (e) Zeroth law of thermodynamics explain the concept of temperature. According to which there exist a scalar quantity called temperature which is property of all thermodynamic system.

21. (b) Efficiency of Carnot cycle $\eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$, for Carnot

engine when T_2 decrease η increases.

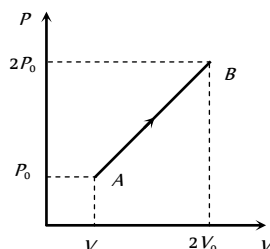
22. (a) Entropy is a measure of the disorder or randomness of the system. Greater the randomness, greater the entropy.

Thermodynamics

Self Evaluation Test -14

1. The P - V diagram of 2 gm of helium gas for a certain process $A \rightarrow B$ is shown in the figure. what is the heat given to the gas during the process $A \rightarrow B$

- (a) $4P_0V_0$
(b) $6P_0V_0$
(c) $4.5P_0V_0$
(d) $2P_0V_0$



2. A certain mass of gas at 273 K is expanded to 81 times its volume under adiabatic condition. If $\gamma = 1.25$ for the gas, then its final temperature is [Pb. PET 1997]

- (a) -235°C (b) -182°C
(c) -91°C (d) 0°C

3. In an adiabatic process 90 J of work is done on the gas. The change in internal energy of the gas is [CPMT 1996]

- (a) -90 J
(b) $+90\text{ J}$
(c) 0 J
(d) Depends on initial temperature

4. If a Carnot's engine functions at source temperature 127°C and at sink temperature 87°C , what is its efficiency [DCE 1997]

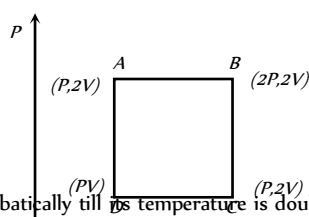
- (a) 10% (b) 25%
(c) 40% (d) 50%

5. In the case of diatomic gas, the heat given at constant pressure is that part of energy which is used for the expansion of gas, is

- (a) $\frac{2}{5}$ (b) $\frac{3}{7}$
(c) $\frac{2}{7}$ (d) $\frac{5}{7}$

6. An ideal monoatomic gas is taken round the cycle $ABCD$ shown in the PV diagram in the given fig. The work done during the cycle is

- (a) $\frac{1}{2}PV$
(b) $2PV$
(c) PV
(d) Zero



7. A gas is compressed adiabatically till its temperature is doubled. The ratio of its final volume to initial volume will be

- (a) $1/2$ (b) More than $1/2$
(c) Less than $1/2$ (d) Between 1 and 2

8. A tyre filled with air (27°C , and 2 atm) bursts, then what is temperature of air ($\gamma = 1.5$) [RPMT 2002]

- (a) -33°C (b) 0°C
(c) 27°C (d) 240°C

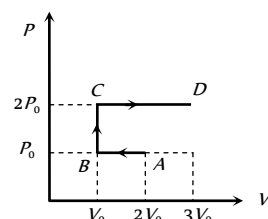
9. A gas expands adiabatically at constant pressure such that its temperature $T \propto \frac{1}{\sqrt{V}}$, the value of C_p / C_v of gas is

[RPMT 2002; MHCET 2004]

- (a) 1.30 (b) 1.50
(c) 1.67 (d) 2.00

10. P - V diagram of an ideal gas is as shown in figure. Work done by the gas in process $ABCD$ is

- (a) $4P_0V_0$
(b) $2P_0V_0$
(c) $3P_0V_0$
(d) P_0V_0



11. An engineer claims to have made an engine delivering 10 kW power with fuel consumption of 1 g s^{-1} . The calorific value of fuel is 2 kcal/g . His claim [J & K CET 2000]

- (a) Is non-valid (b) Is valid
(c) Depends on engine (d) Depends on load

12. An ideal gas heat engine operates in a Carnot cycle between 27°C and 127°C . It absorbs 6 kcal at the higher temperature. The amount of heat (in kcal) converted into work is equal to

- (a) 3.5 (b) 1.6
(c) 1.2 (d) 4.8

13. A gas expands with temperature according to the relation $V = kT^{2/3}$. What is the work done when the temperature changes by 30°C [UPSEAT 1998]

- (a) $10R$ (b) $20R$
(c) $30R$ (d) $40R$

14. An ideal gas ($\gamma = 1.5$) is expanded adiabatically. How many times has the gas to be expanded to reduce the root mean square velocity of molecules 2.0 times

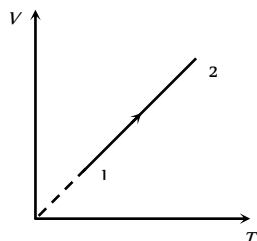
- (a) 4 times (b) 16 times
(c) 8 times (d) 2 times

15. Three samples of the same gas A, B and C ($\gamma = 3/2$) have initially equal volume. Now the volume of each sample is doubled. The process is adiabatic for A isobaric for B and isothermal for C. If the final pressures are equal for all three samples, the ratio of their initial pressures are [BHU 1997]

- (a) $2\sqrt{2} : 2 : 1$ (b) $2\sqrt{2} : 1 : 2$
(c) $\sqrt{2} : 1 : 2$ (d) $2 : 1 : \sqrt{2}$

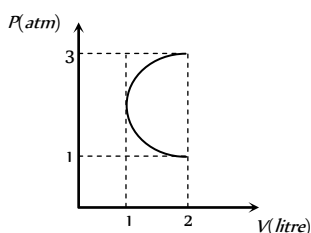
16. Volume versus temperature graph of two moles of helium gas is as shown in figure. The ratio of heat absorbed and the work done by the gas in process 1-2 is

- (a) 3
(b) $\frac{5}{2}$
(c) $\frac{5}{3}$
(d) $\frac{7}{2}$



17. In the P - V diagram shown in figure ABC is a semicircle. The work done in the process ABC is

- (a) Zero
(b) $\frac{\pi}{2} atm - lt$
(c) $-\frac{\pi}{2} atm - lt$
(d) $4 atm - lt$



18. Heat is supplied to a diatomic gas at constant pressure. The ratio of $\Delta Q : \Delta U : \Delta W$ is

- (a) 5 : 3 : 2 (b) 5 : 2 : 3
(c) 7 : 5 : 2 (d) 7 : 2 : 5

19. A gas undergoes a change of state during which 100 J of heat is supplied to it and it does 20 J of work. The system is brought back to its original state through a process during which 20 J of heat is released by the gas. The work done by the gas in the second process is

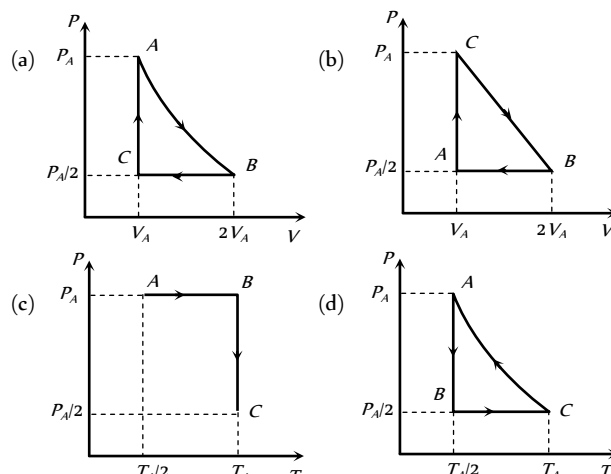
- (a) 60 J (b) 40 J
(c) 80 J (d) 20 J

20. N moles of an ideal diatomic gas are in a cylinder at temperature T . Suppose on supplying heat to the gas, its temperature remain constant but n moles get dissociated into atoms. Heat supplied to the gas is

- (a) Zero (b) $\frac{1}{2} nRT$
(c) $\frac{3}{2} nRT$ (d) $\frac{3}{2} (N - n)RT$

21. Three moles of an ideal gas $\left(C_p = \frac{7}{2} R\right)$ at pressure P_A and temperature T_A is isothermally expanded to twice its initial volume. It is then compressed at constant pressure to its original volume.

Finally the gas is compressed at constant volume to its original pressure P_A . The correct P - V and P - T diagrams indicating the process are



22. A cylinder of mass 1 kg is given heat of 20000 J at atmospheric pressure. If initially temperature of cylinder is $20^\circ C$, then work done by the cylinder will be (Given that Specific heat of cylinder = 400 J/kg, Coefficient of volume expansion = $9 \times 10^{-5} / ^\circ C$, Atmospheric pressure = 10^5 N/m² and density of cylinder 9000 kg/m³)

- (a) 0.02 J (b) 0.05 J
(c) 0.08 J (d) 0.1 J

23. In a thermodynamic process pressure of a fixed mass of a gas is changed in such a manner that the gas releases 30 joules of heat and 10 joules of work was done on the gas. If the initial internal energy of the gas was 30 joules, then the final internal energy will be [CPMT 1986]

- (a) 2 J (b) -18 J
(c) 10 J (d) 58 J

24. In an adiabatic change, the pressure P and temperature T of a monoatomic gas are related by the relation $P \propto T^c$, where c equals [CBSE PMT 1994; BHU 1997; AIIMS 2001; MH CET 2000]

- (a) 5/3 (b) 2/5
(c) 3/5 (d) 5/2

25. The internal energy of an ideal gas increases during an isothermal process when the gas is [SCRA 1998]

- (a) Expanded by adding more molecules to it
(b) Expanded by adding more heat to it
(c) Expanded against zero pressure
(d) Compressed by doing work on it

AS Answers and Solutions

(SET -14)

1. (b) Change in internal energy from $A \rightarrow B$ is

$$\Delta U = \frac{f}{2} \mu R \Delta T = \frac{f}{2} (P_f V_f - P_i V_i)$$

$$= \frac{3}{2}(2P_0 \times 2V_0 - P_0 \times V_0) = \frac{9}{2}P_0 V_0$$

Work done in process $A \rightarrow B$ is equal to the Area covered by the graph with volume axis i.e.,

$$W_{A \rightarrow B} = \frac{1}{2}(P_0 + 2P_0) \times (2V_0 - V_0) = \frac{3}{2}P_0 V_0$$

$$\text{Hence, } \Delta Q = \Delta U + \Delta W = \frac{9}{2}P_0 V_0 + \frac{3}{2}P_0 V_0 = 6P_0 V_0$$

2. (b) For adiabatic process $TV^{\gamma-1} = \text{constant}$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \Rightarrow T_2 = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \times T_1$$

$$\Rightarrow T_2 = \left(\frac{1}{81}\right)^{1.25-1} \times 273 = \left(\frac{1}{81}\right)^{0.25} \times 273$$

$$= \frac{273}{3} = 91K \rightarrow -182^\circ C$$

3. (b) For adiabatic process $\Delta Q = 0$

$$\text{From } \Delta Q = \Delta U + \Delta W \Rightarrow 0 = \Delta U - 90 \Rightarrow \Delta U = +90J$$

4. (d) $\eta = \frac{T_1 - T_2}{T_1} = \frac{(127 + 273) - (87 + 273)}{(127 + 273)}$

$$= \frac{400 - 360}{400} = 0.1 \rightarrow 10\%$$

5. (c) $\Delta W = \text{energy used for expansion} = PdV = RdT$

$$\Delta Q = \text{heat supplied to diatomic gas at constant } P$$

$$= C_p dT = \frac{7}{2}RdT \quad (\because C_p = \frac{7}{2}R) \quad \therefore \frac{\Delta W}{\Delta Q} = \frac{RdT}{\frac{7}{2}RdT} = \frac{2}{7}$$

6. (c)

7. (c) $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 2 \Rightarrow \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{1}{2} \Rightarrow \frac{V_2}{V_1} = \left(\frac{1}{2}\right)^{\frac{1}{\gamma-1}} < \frac{1}{2}$

$$\Rightarrow V_2 < \frac{V_1}{2}$$

8. (a) $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{T_2}{(273+27)} = \left(\frac{1}{2}\right)^{\frac{1.5-1}{1.5}} = \left(\frac{1}{2}\right)^{\frac{1}{3}} = \frac{1}{2.5}$

$$\Rightarrow T_2 = \frac{T_1}{1.25} = \frac{(273+27)}{1.25} = 238K = -34.8^\circ C$$

9. (b) $TV^{\gamma-1} = \text{constant} \Rightarrow T \propto V^{1-\gamma}$

$$\text{According to question } T \propto V^{-\frac{1}{2}}$$

$$\text{Hence } 1 - \gamma = -\frac{1}{2} \Rightarrow \gamma = \frac{3}{2} = 1.5$$

10. (c) $W_{AB} = -P_0 V_0$, $W_{BC} = 0$ and $W_{CD} = 4P_0 V_0$

$$\Rightarrow W_{ABCD} = -P_0 V_0 + 0 + 4P_0 V_0 = 3P_0 V_0$$

11. (a) Power = 10 KW = 10000 J/s = $\frac{10000}{4.2} = 2.38 \text{ k cal / gm}$

But the calorific value of fuel is only 2 k cal/gm. Hence claim is invalid.

12. (c) Efficiency of a carnot engine is given by $\eta = 1 - \frac{T_2}{T_1}$

$$\text{or } \frac{W}{Q} = 1 - \frac{T_2}{T_1} \Rightarrow \frac{W}{6} = 1 - \frac{(273+127)}{(273+227)} \Rightarrow W = 1.2 \text{ k cal}$$

13. (b) $W = \int PdV = \int \frac{RT}{V} dV$

$$\text{Since } V = kT^{2/3} \Rightarrow dV = \frac{2}{3}kT^{-1/3}dT$$

$$\text{Eliminating } K, \text{ we find } \frac{dV}{V} = \frac{2}{3} \frac{dT}{T}$$

$$\text{Hence } W = \int_{T_1}^{T_2} \frac{2}{3} \frac{RT}{T} dT = \frac{2}{3}R(T_2 - T_1) = \frac{2}{3}R(30) = 20R$$

14. (b) $v_{rms} = \sqrt{\frac{3RT}{M}} \Rightarrow v_{rms} \propto \sqrt{T}$

v_{rms} is to reduce two times i.e. temperature of the gas will

$$\text{have to reduce four times or } \frac{T'}{T} = \frac{1}{4}$$

During adiabatic process $TV^{\gamma-1} = T'V'^{\gamma-1}$

$$\Rightarrow \frac{V'}{V} = \left(\frac{T}{T'}\right)^{\frac{1}{\gamma-1}} = (4)^{\frac{1}{1.5-1}} = (4)^2 = 16 \Rightarrow V' = 16V$$

15. (b) Let the initial pressure of the three samples be P_A, P_B and

$$P_C, \text{ then } P_A(V)^{3/2} = (2V)^{3/2}P, P_B = P \text{ and}$$

$$P_C(V) = P(2V)$$

$$\Rightarrow P_A : P_B : P_C = (2)^{3/2} : 1 : 2 = 2\sqrt{2} : 1 : 2$$

16. (b) $V-T$ graph is a straight line passing through origin. Hence, $V \propto T$ or $P = \text{constant}$

$$\therefore \Delta Q = nC_p \Delta T \text{ and } \Delta U = nC_v \Delta T$$

$$\text{Also } \Delta W = \Delta Q - \Delta U = \mu(C_p - C_v)\Delta T$$

$$\therefore \frac{\Delta Q}{\Delta W} = \frac{nC_p \Delta T}{n(C_p - C_v)\Delta T} = \frac{C_p}{C_p - C_v} = \frac{1}{1 - \frac{C_v}{C_p}}$$

$$\frac{C_v}{C_p} = \frac{3}{5} \text{ for helium gas. Hence } \frac{\Delta Q}{\Delta W} = \frac{1}{1 - 3/5} = \frac{5}{2}$$

17. (b) W_{AB} is negative (volume is decreasing) and

$$W_{BC} \text{ is positive (volume is increasing) and}$$

$$\text{since, } |W_{BC}| > |W_{AB}|$$

$\therefore$ net work done is positive and area between semicircle

$$\text{which is equal to } \frac{\pi}{2} \text{ atm} \cdot \text{lt.}$$

18. (c) $\Delta Q = \mu C_p \Delta T = \frac{7}{2} \mu R \Delta T \quad \left(C_p = \frac{7}{2}R\right)$

$$\Delta U = \mu C_v \Delta T = \frac{5}{2} \mu R \Delta T \quad \left(C_v = \frac{5}{2}R\right)$$

$$\text{and } \Delta W = \Delta Q - \Delta U = \mu R \Delta T$$

$$\Rightarrow \Delta Q : \Delta U : \Delta W = 7 : 5 : 2$$

19. (a) In a cyclic process $\Delta U = 0 \Rightarrow \Delta Q = \Delta W$

$$\Rightarrow (100 - 20) = 20 + W_2 \Rightarrow W_2 = 60 \text{ J}$$

20. (b) Since the gas is enclosed in a vessel, therefore, during heating process, volume of the gas remains constant. Hence, no work is done by the gas. It means heat supplied to the gas is used to increase its internal energy only.

$$\text{Initial internal energy of the gas is } U_1 = N \left(\frac{5}{2} R \right) T$$

Since n moles get dissociated into atoms, therefore, after heating, vessel contains $(N - n)$ moles of diatomic gas and $2n$ moles of a mono-atomic gas. Hence the internal energy for the gas, after heating, will be equal to

$$U_2 = (N - n) \left(\frac{5}{2} R \right) T + 2n \left(\frac{3}{2} R \right) T = \frac{5}{2} NRT + \frac{1}{2} nRT$$

Hence, the heat supplied = increase in internal energy

$$= (U_2 - U_1) = \frac{1}{2} nRT$$

$$= (10^5 \text{ N/m}^2) \left(\frac{1}{9 \times 10^3} \text{ m}^3 \right) (9 \times 10^{-5} / ^\circ\text{C}) (50^\circ\text{C}) = 0.05 \text{ J}$$

23. (c) $\Delta Q = \Delta U + \Delta W = (U_f - U_i) + \Delta W$

$$\Rightarrow -30 = (U_f - 30) - 10 \Rightarrow U_f = 10 \text{ J}$$

24. (d) $T^\gamma P^{1-\gamma} = \text{constant} \Rightarrow P \propto T^{\frac{\gamma}{\gamma-1}}$

Comparing above equation with given equation

$$P \propto T^C \Rightarrow C = \frac{\gamma}{\gamma-1} = \frac{5/3}{5/3-1} = \frac{5}{2}$$

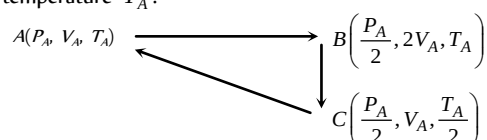
25. (a) Internal energy of an ideal gas is given by

$$U = \frac{f}{2} \mu RT = \frac{f}{2} \left(\frac{N}{N_A} \right) RT \Rightarrow U \propto NT$$

In isothermal process $T = \text{constant} \Rightarrow U \propto N$.

i.e. internal energy increases by increasing number of molecules (N).

21. (a) Let the process start from initial pressure P_A , volume V_A and temperature T_A .



- (i) Isothermal expansion ($PV = \text{constant}$) at temperature T_A to twice the initial volume V_A
- (ii) Compression at constant pressure $\frac{P_A}{2}$ to original volume V_A (i.e. $V \propto T$)
- (iii) Isochoric process (at volume V_A) to initial condition (i.e. $P \propto T$)

22. (b) $\Delta Q = mc\Delta T \Rightarrow \Delta T = \frac{20000 \text{ J}}{1 \text{ kg} \times (400 \text{ J/kg}^\circ\text{C})} = 50^\circ\text{C}$

$$\Rightarrow T_{\text{new}} = 70^\circ\text{C}$$

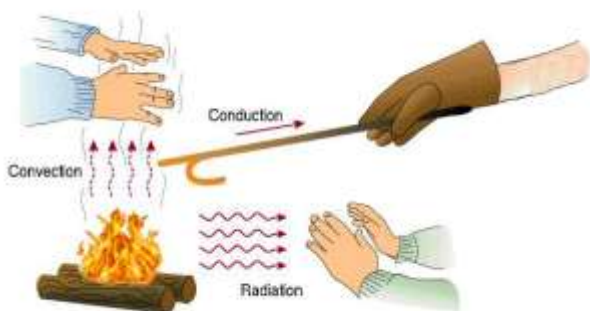
$$\text{Hence } W = P_{\text{atm}} \Delta V = P_{\text{atm}} V_0 \gamma \Delta T$$



Chapter 15 Transmission of Heat

Heat energy transfers from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to another may take place by one of the following modes.

Conduction, Convection and Radiation



Conduction

The process of transmission of heat energy in which the heat is transferred from one particle to other particle without dislocation of the particle from their equilibrium position is called conduction.

(1) Heat flows from hot end to cold end. Particles of the medium simply oscillate but do not leave their place.

(2) Medium is necessary for conduction

(3) It is a slow process

(4) The temperature of the medium increases through which heat flows

(5) Conduction is a process which is possible in all states of matter.

(6) When liquid and gases are heated from the top, they conduct heat from top to bottom.

(7) In solids only conduction takes place



Fig. 15.1

(8) In non-metallic solids and fluids the conduction takes place only due to vibrations of molecules, therefore they are poor conductors.

(9) In metallic solids free electrons carry the heat energy, therefore they are good conductor of heat.

Conduction in Metallic Rod

When one end of a metallic rod is heated, heat flows by conduction from the hot end to the cold end.

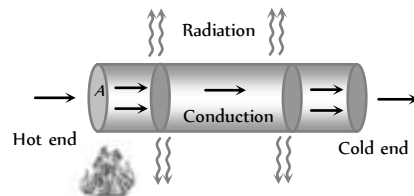


Fig. 15.2

(1) **Variable state** : In this state temperature of every part of the rod increases

Heat received by each cross-section of the rod from hotter end used in three ways.

- A part increases temperature of itself.
- Another part transferred to neighbouring cross-section.
- Remaining part radiates.

- $\theta_1 > \theta_2 > \theta_3 > \theta_4 > \theta_5$
- $\theta \rightarrow$ Changing

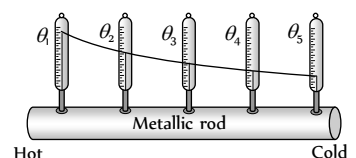


Fig. 15.3

(2) **Steady state** : After sometime, a state is reached when the temperature of every cross-section of the rod becomes constant. In this state, no heat is absorbed by the rod. The heat that reaches any cross-section is transmitted to the next except that a small part of heat is lost to surrounding from the sides by convection & radiation. This state of the rod is called steady state.

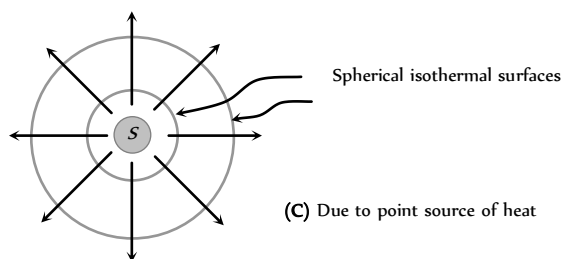
(3) **Isothermal surface** : Any surface (within a conductor) having its all points at the same temperature, is called isothermal surface. The direction

Plane isothermal surfaces

Cylindrical isothermal surfaces



of flow of heat through a conductor at any point is perpendicular to the isothermal surface passing through that point.



(4) **Temperature gradient (T.G.)**: The rate of change of temperature with distance between two isothermal surfaces is called temperature gradient. Hence

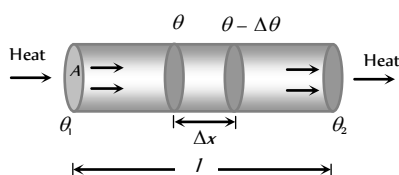


Fig. 15.5

(i) Temperature gradient = $\frac{-\Delta\theta}{\Delta x}$

(ii) The negative sign shows that temperature θ decreases as the distance x increases in the direction of heat flow.

(iii) For uniform temperature fall $\frac{\theta_1 - \theta_2}{l} = \frac{\Delta\theta}{\Delta x}$

(iv) Unit : K/m or $^{\circ}C/m$ (S.I.) and Dimensions $[L^{-1}\theta]$

(5) **Law of thermal conductivity** : Consider a rod of length l and area of cross-section A whose faces are maintained at temperature θ_1 and θ_2 respectively. The curved surface of rod is kept insulated from surrounding to avoid leakage of heat

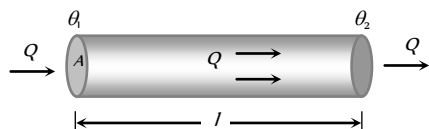


Fig. 15.6

(i) In steady state the amount of heat flowing from one face to the other face in time t is given by $Q = \frac{KA(\theta_1 - \theta_2)t}{l}$

where K is coefficient of thermal conductivity of material of rod.

(ii) Rate of flow of heat i.e. heat current $\frac{Q}{t} = H = \frac{KA(\theta_1 - \theta_2)}{l}$

(iii) In case of non-steady state or variable cross-section, a more general equation can be used to solve problems.

$$\frac{dQ}{dt} = -KA \frac{d\theta}{dx}$$

(6) **More about K** : It is the measure of the ability of a substance to conduct heat through it.

(i) Units : $Cal/cm\text{-}sec \cdot ^{\circ}C$ (in C.G.S.), $kcal/m\text{-}sec\text{-}K$ (in M.K.S.) and $W/m\text{-}K$ (in S.I.). Dimension : $[MLT^{-3}\theta^{-1}]$

(ii) The magnitude of K depends only on nature of the material.

(iii) Substances in which heat flows quickly and easily are known as good conductor of heat. They possess large thermal conductivity due to large number of free electrons e.g. Silver, brass etc. For perfect conductors, $K = \infty$.

(iv) Substances which do not permit easy flow of heat are called bad conductors. They possess low thermal conductivity due to very few free electrons e.g. Glass, wood etc. and for perfect insulators, $K = 0$.

(v) The thermal conductivity of pure metals decreases with rise in temperature but for alloys thermal conductivity increases with increase of temperature.

(vi) Human body is a bad conductor of heat (but it is a good conductor of electricity).

(vii) **Decreasing order of conductivity** : For some special cases it is as follows

(a) $K_{Ag} > K_{Cu} > K_{Al}$

(b) $K_{Solid} > K_{Liquid} > K_{Gas}$

(c) $K_{Metals} > K_{Non\text{-}metals}$

Table 15.1 : Thermal conductivity of some material

Substance	Thermal conductivity ($W/m\text{-}K$)	Substance	Thermal conductivity ($W/m\text{-}K$)
Aluminium	240	Concrete	0.9
Copper	400	Water	0.6
Gold	300	Glass wool	0.04
Iron	80	Air	0.024
Lead	35	Helium	0.14
Glass	0.9	Hydrogen	0.17
Wood	0.1-0.2	Oxygen	0.024

(7) **Relation between temperature gradient and thermal conductivity** :

In steady state, rate of flow of heat $\frac{dQ}{dt} = -KA \frac{d\theta}{dx} = -KA \times (\text{T.G.}) \Rightarrow$

(T.G.) $\propto \frac{1}{K}$ ($\frac{dQ}{dt} = \text{constant}$)

Temperature difference between the hot end and the cold end in steady state is inversely proportional to K , i.e. in case of good conductors temperature of the cold end will be very near to hot end.

In ideal conductor where $K = \infty$, temperature difference in steady state will be zero.

(8) **Thermal resistance (R)** : The thermal resistance of a body is a measure of its opposition to the flow of heat through it.

It is defined as the ratio of temperature difference to the heat current (= Rate of flow of heat)

(i) Hence $R = \frac{\theta_1 - \theta_2}{H} = \frac{\theta_1 - \theta_2}{KA(\theta_1 - \theta_2)/l} = \frac{l}{KA}$

(ii) **Unit** : $^{\circ}\text{C}\times\text{sec}/\text{cal}$ or $\text{K}\times\text{sec}/\text{kcal}$ and **Dimension** : $[M^{-1}L^{-2}T^3\theta]$

(9) **Wiedmann-Franz law** : At a given temperature T , the ratio of thermal conductivity to electrical conductivity is constant *i.e.*, $(K/\sigma T) = \text{constant}$, *i.e.*, a substance which is a good conductor of heat (*e.g.*, silver) is also a good conductor of electricity. Mica is an exception to above law.

(10) **Thermometric conductivity or diffusivity** : It is a measure of rate of change of temperature (with time) when the body is not in steady state (*i.e.*, in variable state)

It is defined as the ratio of the coefficient of thermal conductivity to the thermal capacity per unit volume of the material. Thermal capacity per

$$\text{unit volume} = \frac{mc}{V} = \rho c$$

$$(\rho = \text{density of substance}) \Rightarrow \text{Diffusivity } (D) = \frac{K}{\rho c}$$

Unit : m/sec and **Dimension** : $[L^2T^{-1}]$

Table 15.2 : Electrical Analogy for Thermal Conduction

Electrical conduction	Thermal conduction
Electric charge flows from higher potential to lower potential	Heat flows from higher temperature to lower temperature
The rate of flow of charge is called the electric current, <i>i.e.</i> $I = \frac{dq}{dt}$	The rate of flow of heat may be called as heat current <i>i.e.</i> $H = \frac{dQ}{dt}$
The relation between the electric current and the potential difference is given by Ohm's law, that is $I = \frac{V_1 - V_2}{R}$ where R is the electrical resistance of the conductor	Similarly, the heat current may be related with the temperature difference as $H = \frac{\theta_1 - \theta_2}{R}$ where R is the thermal resistance of the conductor
The electrical resistance is defined as $R = \frac{\rho l}{A} = \frac{l}{\sigma A}$ where ρ = Resistivity and σ = Electrical conductivity	The thermal resistance may be defined as $R = \frac{l}{KA}$ where K = Thermal conductivity of conductor
$\frac{dq}{dt} = I = \frac{V_1 - V_2}{R} = \frac{\sigma A}{l}(V_1 - V_2)$	$\frac{dQ}{dt} = H = \frac{\theta_1 - \theta_2}{R} = \frac{KA}{l}(\theta_1 - \theta_2)$

Applications of Conductivity in Daily Life

(1) Cooking utensils are provided with wooden handles, because wood is a poor conductor of heat. The hot utensils can be easily handled from the wooden handles and our hands are saved from burning.



Fig. 15.7

(2) We feel warmer in a fur coat. The air enclosed in the fur coat being bad conductor heat does not allow the body heat to flow outside. Hence we feel warmer in a fur coat.

(3) Eskimos make double walled houses of the blocks of ice. Air enclosed in between the double



Fig. 15.8

walls prevents transmission of heat from the house to the cold surroundings.

For exactly the same reason, two thin blankets are warmer than one blanket of their combined thickness. The layer of air enclosed in between the two blankets makes the difference.

(4) Wire gauze is placed over the flame of Bunsen burner while heating the flask or a beaker so that the flame does not go beyond the gauze and hence there is no direct contact between the flame and the flask. The wire gauze being a good conductor of heat, absorb the heat of the flame and transmit it to the flask.



Fig. 15.9

Davy's safety lamp has been designed on this principle. The gases in the mines burn inside the gauze placed around the flame of the lamp. The temperature outside the gauze is not high, so the gases outside the gauze do not catch fire.

(5) Birds often swell their feathers in winter. By doing so, they enclose more air between their bodies and the feathers. The air, being bad conductor of heat prevents the out flow of their body heat. Thus, birds feel warmer in winter by swelling their feathers.

Combination of Metallic Rods

(i) **Series combination** : Let n slabs each of cross-sectional area A , lengths $l_1, l_2, l_3, \dots, l_n$ and conductivities $K_1, K_2, K_3, \dots, K_n$ respectively be connected in the series

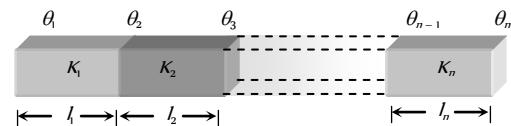


Fig. 15.10

(i) **Heat current** : Heat current is the same in all the conductors. *i.e.*, $\frac{Q}{t} = H_1 = H_2 = H_3 \dots \dots \dots = H_n$

$$\frac{K_1 A (\theta_1 - \theta_2)}{l_1} = \frac{K_2 A (\theta_2 - \theta_3)}{l_2} = \frac{K_n A (\theta_{n-1} - \theta_n)}{l_n}$$

(ii) **Equivalent thermal resistance** : $R = R_1 + R_2 + \dots R_n$

(iii) **Equivalent thermal conductivity** : It can be calculated as follows

From $R_S = R_1 + R_2 + R_3 + \dots$

$$\frac{l_1 + l_2 + \dots l_n}{K_s} = \frac{l_1}{K_1 A} + \frac{l_2}{K_2 A} + \dots + \frac{l_n}{K_n A}$$

$$\Rightarrow K_s = \frac{l_1 + l_2 + \dots l_n}{\frac{l_1}{K_1} + \frac{l_2}{K_2} + \dots \frac{l_n}{K_n}}$$

(a) For n slabs of equal length $K_s = \frac{n}{\frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} + \dots \frac{1}{K_n}}$

(b) For two slabs of equal length, $K_s = \frac{2K_1 K_2}{K_1 + K_2}$

(iv) **Temperature of interface of composite bar** : Let the two bars are arranged in series as shown in the figure.

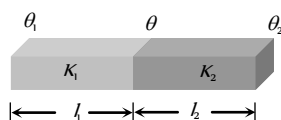


Fig. 15.11

Then heat current is same in the two conductors.

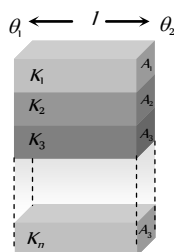
$$\text{i.e., } \frac{Q}{t} = \frac{K_1 A (\theta_1 - \theta)}{l_1} = \frac{K_2 A (\theta - \theta_2)}{l_2}$$

$$\text{By solving we get } \theta = \frac{\frac{K_1}{l_1} \theta_1 + \frac{K_2}{l_2} \theta_2}{\frac{K_1}{l_1} + \frac{K_2}{l_2}}$$

$$(a) \text{ If } l = l \text{ then } \theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$$

$$(b) \text{ If } K = K \text{ and } l = l \text{ then } \theta = \frac{\theta_1 + \theta_2}{2}$$

(2) **Parallel Combination** : Let n slabs each of length l , areas $A_1, A_2, A_3, \dots, A_n$ and thermal conductivities $K_1, K_2, K_3, \dots, K_n$ are connected in parallel then



$$(i) \text{ Equivalent resistance : } \frac{1}{R_s} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$\text{For two slabs } R_s = \frac{R_1 R_2}{R_1 + R_2}$$

(ii) **Temperature gradient** : Same across each slab.

(iii) **Heat current** : in each slab will be different. Net heat current will be the sum of heat currents through individual slabs. i.e., $H = H_1 + H_2 + H_3 + \dots + H_n$

$$\begin{aligned} & \frac{K(A_1 + A_2 + \dots + A_n)(\theta_1 - \theta_2)}{l} \\ &= \frac{K_1 A_1 (\theta_1 - \theta_2)}{l} + \frac{K_2 A_2 (\theta_1 - \theta_2)}{l} + \dots + \frac{K_n A_n (\theta_1 - \theta_2)}{l} \\ \Rightarrow K &= \frac{K_1 A_1 + K_2 A_2 + K_3 A_3 + \dots + K_n A_n}{A_1 + A_2 + A_3 + \dots + A_n} \end{aligned}$$

$$(a) \text{ For } n \text{ slabs of equal area } K = \frac{K_1 + K_2 + K_3 + \dots + K_n}{n}$$

$$(b) \text{ For two slabs of equal area } K = \frac{K_1 + K_2}{2}$$

Ingen-Hauz Experiment

It is used to compare thermal conductivities of different materials. If l_1, l_2 and l_3 are the lengths of wax melted on rods as

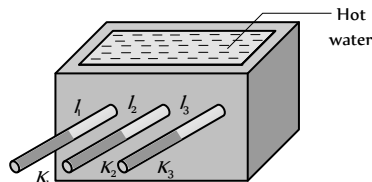


Fig. 15.13

shown in the figure, then the ratio of thermal conductivities is

$$K_1 : K_2 : K_3 = l_1^2 : l_2^2 : l_3^2$$

$$\Rightarrow \text{Thermal conductivity } (K) \propto (\text{Melted length } l)$$

Searle's Experiment

It is a method of determination of K of a metallic rod.

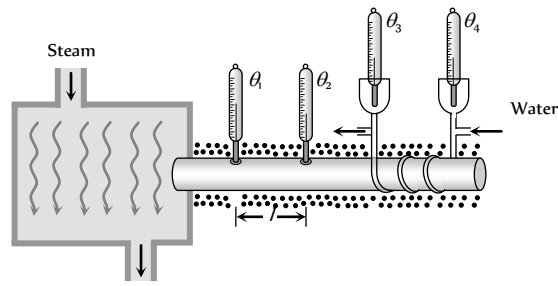


Fig. 15.14

(i) In this experiment a temperature difference $(\theta_1 - \theta_2)$ is maintained across a rod of length l and area of cross section A . If the thermal conductivity of the material of the rod is K , then the amount of heat transmitted by the rod from the hot end to the cold end in time t is

$$\text{given by, } Q = \frac{KA(\theta_1 - \theta_2)t}{l} \dots\dots(i)$$

(2) In Searle's experiment, this heat reaching the other end is utilized to raise the temperature of certain amount of water flowing through pipes circulating around the other end of the rod. If temperature of the water at the inlet is θ_3 and at the outlet is θ_4 , then the amount of heat absorbed by water is given by, $Q = mc(\theta_4 - \theta_3)$

$\dots\dots(ii)$

(3) Where, m is the mass of the water which has absorbed this heat and temperature is raised and c is the specific heat of the water

$$\text{Equating (i) and (ii), } K \text{ can be determined i.e., } K = \frac{mc(\theta_4 - \theta_3)l}{A(\theta_1 - \theta_2)t}$$

(4) In numericals we may have the situation where the amount of heat travelling to the other end may be required to do some other work *e.g.*, it may be required to melt the given amount of ice. In that case equation (i) will have to be equated to mL .

$$\text{i.e. } mL = \frac{KA(\theta_1 - \theta_2)t}{l}$$

Growth of Ice on Lake

(1) Water in a lake starts freezing if the atmospheric temperature drops below 0°C . Let y be the thickness of ice layer in the lake at any instant t and atmospheric temperature is $-\theta^\circ\text{C}$.

(2) The temperature of water in contact with lower surface of ice will be zero.

(3) If A is the area of lake, heat escaping through ice in time dt is $dQ_1 = \frac{KA[0 - (-\theta)]dt}{y}$

(4) Suppose the thickness of ice layer increases by dy in time dt , due to escaping of above heat. Then $dQ_2 = mL = \rho(dy)AL$

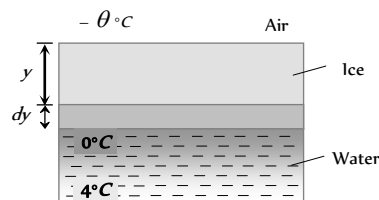


Fig. 15.15

(5) As $dQ_1 = dQ_2$, hence, rate of growth of ice will be $(dy/dt) = (K\theta / \rho Ly)$

So, the time taken by ice to grow to a thickness y is

$$t = \frac{\rho L}{K\theta} \int_0^y y dy = \frac{\rho L}{2K\theta} y^2$$

(6) If the thickness is increased from y_1 to y_2 then time taken

$$t = \frac{\rho L}{K\theta} \int_{y_1}^{y_2} y dy = \frac{\rho L}{2K\theta} (y_2^2 - y_1^2)$$

(7) Take care and do not apply a negative sign for putting values of temperature in formula and also do not convert it to absolute scale.

(8) Ice is a poor conductor of heat, therefore the rate of increase of thickness of ice on ponds decreases with time.

(9) It follows from the above equation that time taken to double and triple the thickness, will be in the ratio of

$$t_1 : t_2 : t_3 :: 1^2 : 2^2 : 3^2, \text{ i.e., } t_1 : t_2 : t_3 :: 1 : 4 : 9$$

(10) The time intervals to change the thickness from 0 to y , from y to $2y$ and so on will be in the ratio

$$\Delta t_1 : \Delta t_2 : \Delta t_3 :: (1^2 - 0^2) : (2^2 - 1^2) : (3^2 - 2^2)$$

$$\Rightarrow \Delta t_1 : \Delta t_2 : \Delta t_3 :: 1 : 3 : 5$$

Convection



Mode of transfer of heat by means of migration of material particles of medium is called convection. It is of two types.

(1) **Natural convection** : This arises due to difference of densities at two places and is a consequence of gravity because on account of gravity the hot light particles rise up and cold heavy particles try settling down. It mostly occurs on heating a liquid/fluid.

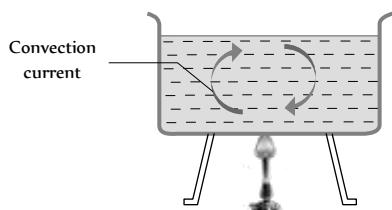


Fig. 15.16

(2) **Forced convection** : If a fluid is forced to move to take up heat from a hot body then the convection process is called forced convection. In this case Newton's law of cooling holds good. According to which rate of loss of heat from a hot body due to moving fluid is directly

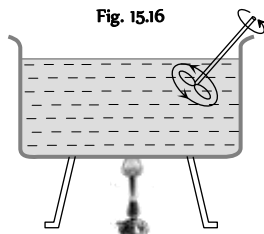


Fig. 15.17

proportional to the surface area of body and excess temperature of body over its surroundings i.e.

$$\frac{Q}{t} \propto A(T - T_0) \Rightarrow \frac{Q}{t} = hA(T - T_0)$$

where h = Constant of proportionality called convection coefficient, T = Temperature of body and T_0 = Temperature of surrounding

Convection coefficient (h) depends on properties of fluid such as density, viscosity, specific heat and thermal conductivity.

(3) Natural convection takes place from bottom to top while forced convection in any direction.

(4) In case of natural convection, convection currents move warm air upwards and cool air downwards. That is why heating is done from base, while cooling from the top.

(5) Natural convection plays an important role in ventilation, in changing climate and weather and in forming land and sea breezes and trade winds.

(6) Natural convection is not possible in a gravity free region such as a free falling lift or an orbiting satellite.

(7) The force of blood in our body by heart helps in keeping the temperature of body constant.

(8) If liquids and gases are heated from the top (so that convection is not possible) they transfer heat (from top to bottom) by conduction.

(9) Mercury though a liquid is heated by conduction and not by convection.

Radiation

(1) The process of the transfer of heat from one place to another place without heating the intervening medium is called radiation.



Fig. 15.18

(2) Precisely it is electromagnetic energy transfer in the form of electromagnetic wave through any medium. It is possible even in vacuum e.g. the heat from the sun reaches the earth through radiation.

(3) The wavelength of thermal radiations ranges from $7.8 \times 10^{-7} \text{ m}$ to $4 \times 10^{-4} \text{ m}$. They belong to *infra-red* region of the electromagnetic spectrum. That is why thermal radiations are also called *infra-red* radiations.

(4) Medium is not required for the propagation of these radiations.

(5) They produce sensation of warmth in us but we can't see them.

(6) Every body whose temperature is above zero Kelvin emits thermal radiation.

(7) Their speed is equal to that of light i.e. $(= 3 \times 10^8 \text{ m/s})$.

(8) Their intensity is inversely proportional to the square of distance of point of observation from the source (i.e. $I \propto 1/d^2$).

(9) Just as light waves, they follow laws of reflection, refraction, interference, diffraction and polarisation.

(10) When these radiations fall on a surface then exert pressure on that surface which is known as radiation pressure.

(11) While travelling these radiations travel just like photons of other electromagnetic waves. They manifest themselves as heat only when they are absorbed by a substance.

(12) Spectrum of these radiations can not be obtained with the help of glass prism because it absorbs heat radiations. It is obtained by quartz or rock salt prism because these materials do not have free electrons and interatomic vibrational frequency is greater than the radiation frequency, hence they do not absorb heat radiations.

(13) **Diathermanous Medium** : A medium which allows heat radiations to pass through it without absorbing them is called diathermanous medium. Thus the temperature of a diathermanous medium does not increase irrespective of the amount of the thermal radiations passing through it *e.g.*, dry air, SO_2 , rock salt ($NaCl$).

(i) Dry air does not get heated in summers by absorbing heat radiations from sun. It gets heated through convection by receiving heat from the surface of earth.

(ii) In winters heat from sun is directly absorbed by human flesh while the surrounding air being diathermanous is still cool. This is the reason that sun's warmth in winter season appears very satisfying to us.

(14) **Athermanous medium** : A medium which partly absorbs heat rays is called a thermous medium. As a result temperature of an athermanous medium increases when heat radiations pass through it *e.g.*, wood, metal, moist air, simple glass, human flesh *etc.*

Colour of Heated Object

When a body is heated, all radiations having wavelengths from zero to infinity are emitted.

(1) Radiations of longer wavelengths are predominant at lower temperature.

(2) The wavelength corresponding to maximum emission of radiations shifts from longer wavelength to shorter wavelength as the temperature increases. Due to this the colour of a body appears to be changing.

(3) A blue flame is at a higher temperature than a yellow flame

Table 15.3 : Variation of colour of a body on heating

Temperature	Colour
$525^\circ C$	Dull red
$900^\circ C$	Cherry red
$1100^\circ C$	Orange red
$1200^\circ C$	Yellow
$1600^\circ C$	White

Interaction of Radiation with Matter

When thermal radiations (Q) fall on a body, they are partly reflected, partly absorbed and partly transmitted.

$$(1) Q = Q_a + Q_r + Q_t$$

$$(2) \frac{Q_a}{Q} + \frac{Q_r}{Q} + \frac{Q_t}{Q} = a + r + t = 1$$

$$(3) a = \frac{Q_a}{Q} = \text{Absorptance or absorbing power}$$

$$r = \frac{Q_r}{Q} = \text{Reflectance or reflecting power}$$

$$t = \frac{Q_t}{Q} = \text{Transmittance or transmitting power}$$

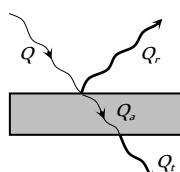


Fig. 15.19

(4) r , a and t all are the pure ratios so they have no unit and dimension.

(5) **Different bodies**

(i) If $a = t = 0$ and $r = 1 \rightarrow$ body is perfect reflector

(ii) If $r = t = 0$ and $a = 1 \rightarrow$ body is perfectly black body

(iii) If, $a = r = 0$ and $t = 1 \rightarrow$ body is perfect transmitter

(iv) If $t = 0 \Rightarrow r + a = 1$ or $a = 1 - r$ *i.e.* good reflectors are bad absorbers.

Emissive Power, Absorptive Power and Emissivity

If temperature of a body is more than it's surrounding then body emits thermal radiation

(1) **Monochromatic Emittance or Spectral emissive power (e_λ)** : For a given surface it is defined as the radiant energy emitted per sec per unit area of the surface with in a unit wavelength around λ *i.e.* lying between $\left(\lambda - \frac{1}{2}\right)$ to $\left(\lambda + \frac{1}{2}\right)$.

$$\text{Spectral emissive power } (e_\lambda) = \frac{\text{Energy}}{\text{Area} \times \text{time} \times \text{wavelength}}$$

$$\text{Unit : } \frac{\text{Joule}}{m^2 \times \text{sec} \times \text{\AA}} \quad \text{and} \quad \text{Dimension : } [ML^{-1}T^{-3}]$$

(2) **Total emittance or total emissive power (e)** : It is defined as the total amount of thermal energy emitted per unit time, per unit area of the body for all possible wavelengths.

$$e = \int_0^\infty e_\lambda d\lambda$$

$$\text{Unit : } \frac{\text{Joule}}{m^2 \times \text{sec}} \quad \text{or} \quad \frac{\text{Watt}}{m^2} \quad \text{and} \quad \text{Dimension : } [MT^{-3}]$$

(3) **Monochromatic absorptance or spectral absorptive power (a_λ)** : It is defined as the ratio of the amount of the energy absorbed in a certain time to the total heat energy incident upon it in the same time, both in the unit wavelength interval. It is dimensionless and unit less quantity. It is represented by a_λ .

(4) **Total absorptance or total absorbing power (a)** : It is defined as the total amount of thermal energy absorbed per unit time, per unit area of the body for all possible wavelengths.

$$a = \int_0^\infty a_\lambda d\lambda$$

(5) **Emissivity (ε)** : Emissivity of a body at a given temperature is defined as the ratio of the total emissive power of the body (e) to the total emissive power of a perfect black body (E) at that temperature *i.e.* $\varepsilon = \frac{e}{E}$

($\varepsilon \rightarrow$ read as epsilon)

(i) For perfectly black body $\varepsilon = 1$

(ii) For highly polished body $\varepsilon = 0$

(iii) But for practical bodies emissivity (ε) lies between zero and one ($0 < \varepsilon < 1$).

Perfectly Black Body

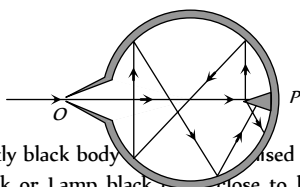
(1) A perfectly black body is that which absorbs completely the radiations of all wavelengths incident on it.

(2) As a perfectly black body neither reflects nor transmits any radiation, therefore the absorptance of a perfectly black body is unity i.e. $t = 0$ and $r = 0 \Rightarrow a = 1$.

(3) We know that the colour of an opaque body is the colour (wavelength) of radiation reflected by it. As a black body reflects no wavelength so, it appears black, whatever be the colour of radiations incident on it.

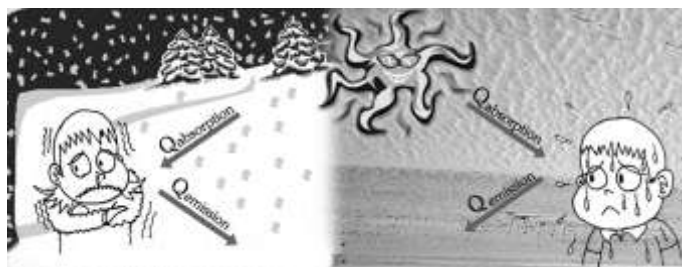
(4) When perfectly black body is heated to a suitable high temperature, it emits radiation of all possible wavelengths. For example, temperature of the sun is very high (6000 K approx.) it emits all possible radiation so it is an example of black body.

(5) **Ferry's black body** : A perfectly black body can't be realised in practice. The nearest example of an ideal black body is the Ferry's black body. It is a doubled walled evacuated spherical cavity whose inner wall is blackened. The space between the wall is evacuated to prevent the loss of heat by conduction and radiation. There is a fine hole in it. All the radiations incident upon this hole are absorbed by this black body. If this black body is heated to high temperature then it emits radiations of all wavelengths. It is the hole which is to be regarded as a black body and not the total enclosure



(6) A perfectly black body is not realised in practice but materials like Platinum black or Lamp black come close to being ideal black bodies. Such materials absorb 96% to 85% of incident radiations.

Prevost Theory of Heat Exchange



(1) Every body emits heat radiations at all finite temperature (Except 0 K) as well as it absorbs radiations from the surroundings.

(2) Exchange of energy among various bodies takes place via radiation.

(3) The process of heat exchange among various bodies is a continuous phenomenon.

(4) At absolute zero temperature (0 K or -273°C) this law is not applicable because at this temperature the heat exchange among various bodies ceases.

(5) If $Q_{\text{em}} > Q_{\text{absorbed}} \rightarrow$ temperature of body decreases and consequently the body appears colder.

If $Q_{\text{em}} < Q_{\text{absorbed}} \rightarrow$ temperature of body increases and it appears hotter.

If $Q_{\text{em}} = Q_{\text{absorbed}} \rightarrow$ temperature of body remains constant (thermal equilibrium)

Kirchoff's Law

According to this law the ratio of emissive power to absorptive power is same for all surfaces at the same temperature and is equal to the emissive power of a perfectly black body at that temperature. Hence

$$\frac{e_1}{a_1} = \frac{e_2}{a_2} = \dots \left(\frac{E}{A} \right)_{\text{Perfectly black body}}$$

But for perfectly black body $A = 1$ i.e. $\frac{e}{a} = E$

If emissive and absorptive powers are considered for a particular wavelength λ , $\left(\frac{e_\lambda}{a_\lambda} \right) = (E_\lambda)_{\text{black}}$

Now since $(E_\lambda)_{\text{black}}$ is constant at a given temperature, according to this law if a surface is a good absorber of a particular wavelength it is also a good emitter of that wavelength.

This in turn implies that a good absorber is a good emitter (or radiator)

Applications of Kirchoff's Law

(1) Sand is rough black, so it is a good absorber and hence in deserts, days (when radiation from the sun is incident on sand) will be very hot. Now in accordance with Kirchoff's law, good absorber is a good emitter so nights (when sand emits radiation) will be cold. This is why days are hot and nights are cold in desert.

(2) Sodium vapours, on heating, emit two bright yellow lines. These are called D_1 , D_2 lines of sodium. When continuous white light from an arc lamp is made to pass through sodium vapours at low temperature, the continuous spectrum is intercepted by two dark lines exactly in the same places as D_1 and D_2 lines. Hence sodium vapours when cold, absorb the same wavelength, as they emit while hot. This is in accordance with Kirchoff's law.

(3) When a shining metal ball having some black spots on its surface is heated to a high temperature and is seen in dark, the black spots shine brightly and the shining ball becomes dull or invisible. The reason is that the black spots on heating absorb radiation and so emit these in dark while the polished shining part reflects radiations and absorb nothing and so does not emit radiations and becomes invisible in the dark.

(4) When a green glass is heated in furnace and taken out, it is found to glow with red light. This is because red and green are complementary colours. At ordinary temperatures, a green glass appears green, because it transmits green colour and absorb red colour strongly. According to Kirchoff's law, this green glass, on heating must emit the red colour, which is absorbed strongly. Similarly when a red glass is heated to a high temperature it will glow with green light.

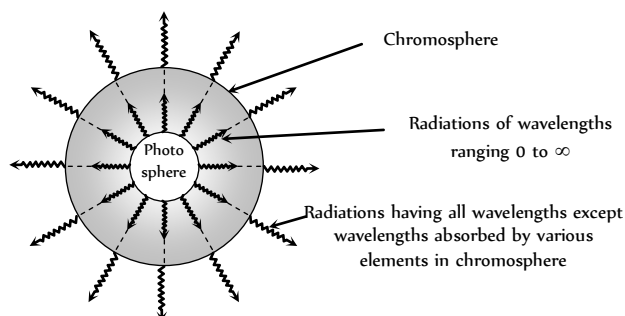
(5) A person with black skin experiences more heat and more cold as compared to a person of white skin because when the outside temperature is greater, the person with black skin absorbs more heat and when the outside temperature is less the person with black skin radiates more energy.

(6) **Kirchoff law also explains the Fraunhofer lines :**

(i) Sun's inner most part (photosphere) emits radiation of all wavelength at high temperature.

(ii) When these radiation enters in outer part (chromosphere) of sun, few wavelength are absorbed by some terrestrial elements (present in vapour form at lower temperature)

(iii) These absorbed wavelengths, which are missing appear as dark lines in the spectrum of the sun called **Fraunhofer lines**.



(iv) During total solar eclipse **Fraunhofer lines** appear bright because the gases and vapour present in the chromosphere start emitting those radiation which they had absorbed.

Stefan's Law

According to it the radiant energy emitted by a perfectly black body per unit area per sec (*i.e.* emissive power of black body) is directly proportional to the fourth power of its absolute temperature, *i.e.* $E \propto T^4$
 $\Rightarrow E = \sigma T^4$

where σ is a constant called Stefan's constant having dimension $[MT^{-3}\theta^{-4}]$ and value $5.67 \times 10^{-8} W/m^2 K^4$.

(i) For ordinary body : $e = \epsilon E = \epsilon \sigma T^4$
 (ii) Radiant energy : If Q is the total energy radiated by the ordinary body then $e = \frac{Q}{A \times t} = \epsilon \sigma T^4 \Rightarrow Q = A \epsilon \sigma T^4 t$

(iii) Radiant power (P) : It is defined as energy radiated per unit area *i.e.* $P = \frac{Q}{t} = A \epsilon \sigma T^4$.

(iv) If an ordinary body at temperature T is surrounded by a body at temperature T_0 , then Stefan's law may be put as

$$e = \epsilon \sigma (T^4 - T_0^4)$$

Rate of Loss of Heat (R_H) and Rate of Cooling (R_C)

(i) **Rate of loss of heat (or initial rate of loss of heat)** : If an ordinary body at temperature T is placed in an environment of temperature T_0 ($T > T_0$) then heat loss by radiation is given by

$$\Delta Q = Q_{\text{emission}} - Q_{\text{absorption}} = A \epsilon \sigma (T^4 - T_0^4)$$

(2) Rate of loss of heat (R_H) = $\frac{dQ}{dt} = A \epsilon \sigma (T^4 - T_0^4)$

(i) If two bodies are made of same material, have same surface finish

$$\text{and are at the same initial temperature then } \frac{dQ}{dt} \propto A \Rightarrow \left(\frac{dQ}{dt} \right)_1 = \frac{A_1}{A_2} \left(\frac{dQ}{dt} \right)_2$$

(3) **Initial rate of fall in temperature (Rate of cooling)**: If m is the body and c is the specific heat then

$$\frac{dQ}{dt} = mc \cdot \frac{dT}{dt} = mc \frac{d\theta}{dt} \quad (\because Q = mc \Delta T \text{ and } dT = d\theta)$$

$$(i) \text{ Rate of cooling } (R_C) = \frac{d\theta}{dt} = \frac{(dQ/dt)}{mc} = \frac{A \epsilon \sigma}{mc} (T^4 - T_0^4)$$

$$= \frac{A \epsilon \sigma}{V \rho c} (T^4 - T_0^4); \text{ where } m = \text{density } (\rho) \times \text{volume } (V)$$

(ii) for two bodies of the same material under identical environments, the ratio of their rate of cooling is $\frac{(R_C)_1}{(R_C)_2} = \frac{A_1}{A_2} \cdot \frac{V_2}{V_1}$

(4) **Dependence of rate of cooling** : When a body cools by radiation the rate of cooling depends on

(i) Nature of radiating surface *i.e.* greater the emissivity, faster will be the cooling.

(ii) Area of radiating surface, *i.e.* greater the area of radiating surface, faster will be the cooling.

(iii) Mass of radiating body *i.e.* greater the mass of radiating body slower will be the cooling.

(iv) Specific heat of radiating body *i.e.* greater the specific heat of radiating body slower will be cooling.

(v) Temperature of radiating body *i.e.* greater the temperature of body faster will be cooling.

(vi) Temperature of surrounding *i.e.* greater the temperature of surrounding slower will be cooling.

Table 15.4 : Comparison of rate of heat loss (R) and rate of cooling (R) for different bodies

Body	Condition	Rate of heat loss $R_H = \frac{dQ}{dt}$	Rate of cooling $R_C = \frac{dT}{dt}$ or $\frac{d\theta}{dt}$
Two solid sphere	T, T_0, c, ρ are same	$R_H \propto A \propto r^2$ $\Rightarrow \frac{(R_H)_1}{(R_H)_2} = \frac{r_1^2}{r_2^2}$	$R_C \propto \frac{A}{V} \propto \frac{r^2}{r^3} \propto \frac{1}{r}$
Two solid sphere of diff. material	T, T_0 - same	$R_H \propto A \propto r^2$	$R_C \propto \frac{A}{V \rho c} \propto \frac{1}{r \rho c}$
Different shape bodies like cube, sphere plate	T, T_0, c, ρ - same	$R_H \propto A$ $A_{\text{max}} \rightarrow \text{Plate}$ $A_{\text{min}} \rightarrow \text{sphere}$	$R_C \propto \frac{A}{V}$
Bodies of different materials	T, T_0, m, A are same but c diff.	$R_H \rightarrow$ same for all bodies	$R_C \propto \frac{1}{c}$

Newton's Law of Cooling

When the temperature difference between the body and its surrounding is not very large i.e. $T - T_0 = \Delta T$ then $T^4 - T_0^4$ may be approximated as $4T_0^3 \Delta T$

By Stefan's law, $\frac{dT}{dt} = \frac{A\sigma}{mc} [T^4 - T_0^4]$

Hence $\frac{dT}{dt} = \frac{A\sigma}{mc} 4T_0^3 \Delta T \Rightarrow \frac{dT}{dt} \propto \Delta T$ or $\frac{d\theta}{dt} \propto \theta - \theta_0$

i.e., if the temperature of body is not very different from surrounding, **rate of cooling is proportional to temperature difference** between the body and its surrounding. This law is called Newton's law of cooling.

(1) Greater the temperature difference between body and its surrounding greater will be the rate of cooling.

(2) If $\theta = \theta_0$, $\frac{d\theta}{dt} = 0$ i.e. a body can never be cooled to a temperature lesser than its surrounding by radiation.

(3) If a body cools by radiation from $\theta_1^\circ\text{C}$ to $\theta_2^\circ\text{C}$ in time t , then $\frac{d\theta}{dt} = \frac{\theta_1 - \theta_2}{t}$ and $\theta = \theta_{av} = \frac{\theta_1 + \theta_2}{2}$. The Newton's law of cooling becomes $\left[\frac{\theta_1 - \theta_2}{t} \right] = K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$.

This form of law helps in solving numericals.

(4) Practical examples

(i) Hot water loses heat in smaller duration as compared to moderate warm water.

(ii) Adding milk in hot tea reduces the rate of cooling.

Cooling Curves

(1) **Curve between $\log(\theta - \theta_0)$ and time**

As $\frac{d\theta}{dt} \propto -(\theta - \theta_0) \Rightarrow \frac{d\theta}{(\theta - \theta_0)} = -K dt$

Integrating $\log_e(\theta - \theta_0) = -Kt + C$

$\log_e(\theta - \theta_0) = -Kt + \log_e A$

This is a straight line with negative slope

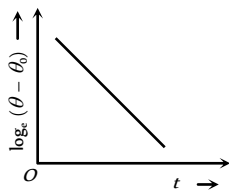


Fig. 15.22

(2) **Curve between temperature of body and time**

As $\log_e(\theta - \theta_0) = -Kt + \log_e A \Rightarrow \log_e \frac{\theta - \theta_0}{A} = -Kt$

$\Rightarrow \theta - \theta_0 = Ae^{-Kt}$

which indicates temperature decreases exponentially with increasing time.

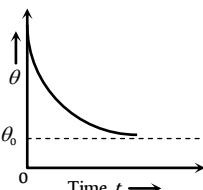


Fig. 15.23

(3) **Curve between the rate of cooling**

(R) and body temperature (θ).

$R = K(\theta - \theta_0) = K\theta - K\theta_0$

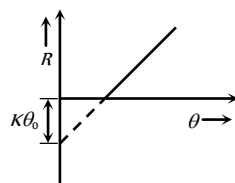


Fig. 15.24

This is a straight line intercept

R -axis at $-K\theta_0$

(4) **Curve between rate of cooling (R)**

and temperature difference between

body (θ) and surrounding (θ_0)

$R \propto (\theta - \theta_0)$. This is a straight line

passing through origin.

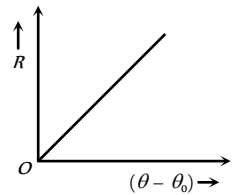
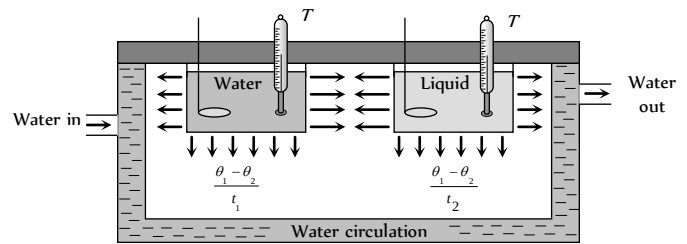


Fig. 15.25

Determination of Specific Heat of Liquid

If volume, radiating surface area, nature of surface, initial temperature and surrounding of water and given liquid are equal and they are allowed to cool down (by radiation) then rate of loss of heat and fall in temperature of both will be same.



i.e. $\left(\frac{dQ}{dt} \right)_{\text{water}} = \left(\frac{dQ}{dt} \right)_{\text{liquid}}$ Fig. 15.26

$(m_W c_W + W) \frac{(\theta_1 - \theta_2)}{t_1} = (m_l c_l + W) \frac{(\theta_1 - \theta_2)}{t_2}$

or $\left[\frac{m_W c_W + W}{t_1} \right] = \left[\frac{m_l c_l + W}{t_2} \right]$

$W = mc$ = Water equivalent of calorimeter, where m and c are mass and specific heat of calorimeter.

If density of water and liquid is ρ and ρ' respectively then $m_W = V\rho_W$ and $m_l = V\rho_l$

Specific heat of liquid $c_l = \frac{1}{m_l} \left[\frac{t_l}{t_W} (m_W c_W + W) - W \right]$

Distribution of Energy in the Spectrum of Black Body

A perfectly black body emits radiation of all possible wavelength.

Langley and later on Lummer and Pringsheim investigated the distribution of energy amongst the different wavelengths in the thermal spectrum of a black body radiation. The results obtained are shown in figure. From these curves it is clear that

(1) At a given temperature energy is not uniformly distributed among different wavelengths.

(2) At a given temperature intensity of heat radiation increases with wavelength, reaches a maximum at a particular wavelength and with further increase in wavelength it decreases.

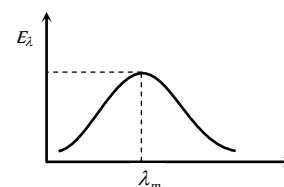


Fig. 15.27

(3) For all wavelengths an increase in temperature causes an increase in intensity.

(4) The area under the curve will represent the total intensity of radiation at a particular temperature i.e. $\text{Area} = E = \int E_{\lambda} d\lambda$

From Stefan's law $E = \sigma T \Rightarrow \text{Area under } E_{\lambda} - \lambda \text{ curve } (A) \propto T$

(5) The energy (E) emitted corresponding to the wavelength of maximum emission (λ_m) increases with fifth power of the absolute temperature of the black body i.e., $E_{\max} \propto T^5$

Wien's Displacement Law

According to Wien's law the product of wavelength corresponding to maximum intensity of radiation and temperature of body (in Kelvin) is constant, i.e. $\lambda_m T = b = \text{constant}$

where b is Wien's constant and has value $2.89 \times 10^{-3} \text{ m} \cdot \text{K}$.

As the temperature of the body increases, the wavelength at which the spectral intensity (E_{λ}) is maximum shifts towards left. Therefore it is also called Wien's displacement law.

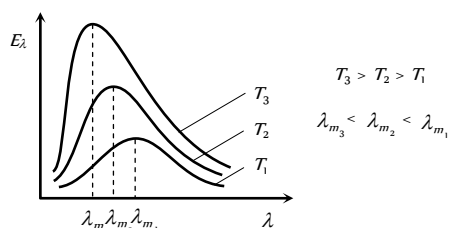


Fig. 15.28

This law is of great importance in 'Astrophysics' as through the analysis of radiations coming from a distant star, by finding λ_m the temperature of the star $T (= b / \lambda_m)$ is determined.

Law of Distribution of Energy (Planck's Hypothesis)

(1) The theoretical explanation of black body radiation was done by Planck.

(2) According to Planck's atoms of the walls of a uniform temperature enclosure behave as oscillators, each with a characteristic frequency of oscillation.

(3) These oscillations emit electromagnetic radiations in the form of photons (The radiation coming out from a small hole in the enclosure are called black body radiation). The energy of each photon is $h\nu$. Where ν is the frequency of oscillator and h is the Planck's constant. Thus emitted energies may be $h\nu, 2h\nu, 3h\nu, \dots, nh\nu$ but not in between.

According to Planck's law $E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{[e^{hc/\lambda kT} - 1]} d\lambda$

where c = speed of light and k = Boltzmann's constant. This equation is known as Planck's radiation law. It is correct and complete law of radiation

(4) This law is valid for radiations of all wavelengths ranging from zero to infinity.

(5) For radiations of short wavelength $\left(\lambda \ll \frac{hc}{KT} \right)$ Planck's law

reduces to Wien's energy distribution law $E_{\lambda} d\lambda = \frac{A}{\lambda^5} e^{-B/\lambda T} d\lambda$

(6) For radiations of long wavelength $\left(\lambda \gg \frac{hc}{KT} \right)$ Planck's law

reduces to Rayleigh-Jeans energy distribution law $E_{\lambda} d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$

Temperature of the Sun and Solar Constant

If R is the radius of the sun and T its temperature, then the energy emitted by the sun per sec through radiation in accordance with Stefan's law will be given by

$$P = A\sigma T^4 = 4\pi R^2 \sigma T^4$$

In reaching earth this energy will spread over a sphere of radius r (= average distance between sun and earth); so the intensity of solar radiation at the surface of earth (called solar constant S) will be given by

$$S = \frac{P}{4\pi r^2} = \frac{4\pi R^2 \sigma T^4}{4\pi r^2}$$

$$\text{i.e. } T = \left[\left(\frac{r}{R} \right)^2 \frac{S}{\sigma} \right]^{1/4}$$

$$= \left[\left(\frac{1.5 \times 10^8}{7 \times 10^5} \right)^2 \times \frac{1.4 \times 10^3}{5.67 \times 10^{-8}} \right]^{1/4} \approx 5800 \text{ K}$$

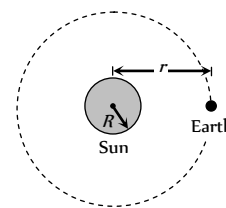


Fig. 15.29

As $r = 1.5 \times 10^8 \text{ km}$, $R = 7 \times 10^5 \text{ km}$,

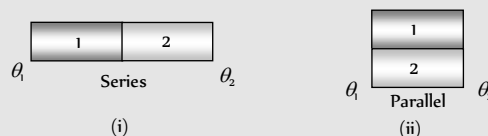
$$S = 2 \frac{\text{cal}}{\text{cm}^2 \text{min}} = 1.4 \frac{\text{kW}}{\text{m}^2} \text{ and } \sigma = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$$

This result is in good agreement with the experimental value of temperature of sun, i.e., 6000 K

Tips & Tricks

✍ Glass and water vapours transmit shorter wavelengths through them but reflects longer wavelengths. This concept is utilised in Green house effect. Glass transmits those waves which are emitted by a source at a temperature greater than 100°C . So, heat rays emitted from sun are able to enter through glass enclosure but heat emitted by small plants growing in the nursery gets trapped inside the enclosure.

✍ Suppose two metallic rods are first connected in series then in parallel.



If Q_s heat flows in time t_s in series combination and Q_p heat flows

in time t_p in parallel combine, then $\frac{Q_p}{Q_s} = \frac{t_p}{t_s} \times \frac{R_s}{R_p}$

If Rods are identical then $R_s = \frac{R}{2}$ and $R_p = 2R \Rightarrow \frac{Q_p}{Q_s} = 4 \left(\frac{t_p}{t_s} \right)$

✍ If temperature of a body becomes θ_1 to θ_2 in t time and it becomes θ_2 to θ_3 in next time then use

$$\frac{\theta_2 - \theta_0}{\theta_1 - \theta_0} = \frac{\theta_3 - \theta_0}{\theta_2 - \theta_0} \quad (\theta_0 = \text{temperature of environment})$$

✍ Newton's law of cooling can be used to compare the specific heat of the two liquids.

If equal masses of two liquids having same surface area and finish cool from same initial temperature to same final temperature with same

$$\text{surrounding then } \frac{t_1}{t_2} = \frac{K_2}{K_1} = \frac{C_1}{C_2}$$

✍ Radiations from sun take 8 min and 20 sec to reach earth.

✍ Suppose temperature of a body decreases $\theta_1^\circ\text{C}$ to $\theta_2^\circ\text{C}$ in time t_1 and $\theta_2^\circ\text{C}$ to $\theta_3^\circ\text{C}$ in time t_2 in the same environment

If $(\theta_1 - \theta_0) \geq (\theta_2 - \theta_0)$ then $t_1 \geq t_2$

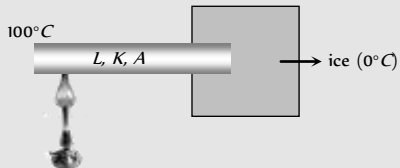
✍ Green glass is a good absorber of red light and a good reflector of green light. Consequently at lower temperature it is a good emitter of red light.

Hence Green $\rightleftharpoons$ Red
Also Yellow $\rightleftharpoons$ Blue

✍ While solving the problems of heat flow, remember the following equation

e.g. If we are interested in finding the mass of ice which transforms into water in unit time. For this we will take

$$\frac{T.D.}{R} = L_f \cdot \frac{dm}{dt}$$

$$\Rightarrow \frac{dm}{dt} = \frac{T.D.}{(L_f)(R)}$$


✍ Confusion

The rate of cooling has been used in many books, with double meanings. At some places, Rate of cooling $= \frac{dQ}{dt}$ and at other places,

rate of cooling $= \frac{d\theta}{dt}$. Our suggestion is that look for the units, if the

rate of cooling is in cal/m in or J/sec etc., then it is $\frac{dQ}{dt}$. But if rate of

cooling is in $^\circ\text{C}/\text{min}$ it means $\frac{d\theta}{dt}$.

Conduction

- In which case the thermal conductivity increases from left to right [NCERT 1974]
 - Al, Cu, Ag
 - Ag, Cu, Al
 - Cu, Ag, Al
 - Al, Ag, Cu

- Which of the following cylindrical rods will conduct most heat, when their ends are maintained at the same steady temperature [CPMT 1981; NCERT 1982]

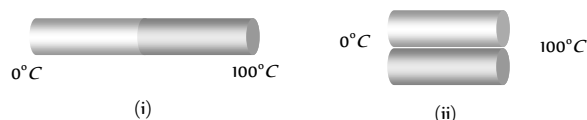
MP PMT 1987; CBSE PMT 1995]

- Length 1 m; radius 1 cm
 - Length 2 m; radius 1 cm
 - Length 2 m; radius 2 cm
 - Length 1 m; radius 2 cm
- The heat is flowing through two cylindrical rods of same material. The diameters of the rods are in the ratio 1 : 2 and their lengths are in the ratio 2 : 1. If the temperature difference between their ends is the same, the ratio of rate of flow of heat through them will be

[NCERT 1982; CBSE PMT 1995; EAMCET 1997]

- 1 : 1
 - 2 : 1
 - 1 : 4
 - 1 : 8
- Two identical square rods of metal are welded end to end as shown in figure (i), 20 calories of heat flows through it in 4 minutes. If the rods are welded as shown in figure (ii), the same amount of heat will flow through the rods in

[NCERT 1982]



- 1 minute
 - 2 minutes
 - 4 minutes
 - 16 minutes
- For cooking the food, which of the following type of utensil is most suitable

[MNR 1986; MP PET 1990; CPMT 1991;

SCRA 1998; MP PMT/PET 1998, 2000; RPET 2001]

- High specific heat and low conductivity
- High specific heat and high conductivity
- Low specific heat and low conductivity
- Low specific heat and high conductivity



Ordinary Thinking

Objective Questions

6. Under steady state, the temperature of a body [CPMT 1978]
 (a) Increases with time
 (b) Decreases with time
 (c) Does not change with time and is same at all the points of the body
 (d) Does not change with time but is different at different points of the body
7. The coefficient of thermal conductivity depends upon [MP PET/PMT 1984; AFMC 1996; Orissa JEE 2005]
 (a) Temperature difference of two surfaces
 (b) Area of the plate
 (c) Thickness of the plate
 (d) Material of the plate
8. When two ends of a rod wrapped with cotton are maintained at different temperatures and after some time every point of the rod attains a constant temperature, then [MP PET/PMT 1988]
 (a) Conduction of heat at different points of the rod stops because the temperature is not increasing
 (b) Rod is bad conductor of heat
 (c) Heat is being radiated from each point of the rod
 (d) Each point of the rod is giving heat to its neighbour at the same rate at which it is receiving heat
9. The length of the two rods made up of the same metal and having the same area of cross-section are 0.6 m and 0.8 m respectively. The temperature between the ends of first rod is 90°C and 60°C and that for the other rod is 150 and 110°C . For which rod the rate of conduction will be greater
 (a) First (b) Second
 (c) Same for both (d) None of the above
10. The ratio of thermal conductivity of two rods of different material is 5 : 4. The two rods of same area of cross-section and same thermal resistance will have the lengths in the ratio
 (a) 4 : 5 (b) 9 : 1
 (c) 1 : 9 (d) 5 : 4
11. The thermal conductivity of a material in CGS system is 0.4. In steady state, the rate of flow of heat 10 cal/sec-cm, then the thermal gradient will be [MP PMT 1989]
 (a) $10^\circ\text{C}/\text{cm}$ (b) $12^\circ\text{C}/\text{cm}$
 (c) $25^\circ\text{C}/\text{cm}$ (d) $20^\circ\text{C}/\text{cm}$
12. Two rectangular blocks A and B of different metals have same length and same area of cross-section. They are kept in such a way that their cross-sectional area touch each other. The temperature at one end of A is 100°C and that of B at the other end is 0°C . If the ratio of their thermal conductivity is 1 : 3, then under steady state, the temperature of the junction in contact will be
 (a) 25°C (b) 50°C
 (c) 75°C (d) 100°C
13. Two vessels of different materials are similar in size in every respect. The same quantity of ice filled in them gets melted in 20 minutes and 30 minutes. The ratio of their thermal conductivities will be [MP PMT 1989; CMEET Bihar 1995]
 (a) 1.5 (b) 1
 (c) 2/3 (d) 4
14. Two rods A and B are of equal lengths. Their ends are kept between the same temperature and their area of cross-sections are A_1 and A_2 and thermal conductivities K_1 and K_2 . The rate of heat transmission in the two rods will be equal, if [MP PMT 1991; CBSE PMT 2002]
 (a) $K_1 A_2 = K_2 A_1$ (b) $K_1 A_1 = K_2 A_2$
 (c) $K_1 = K_2$ (d) $K_1 A_1^2 = K_2 A_2^2$
15. In variable state, the rate of flow of heat is controlled by
 (a) Density of material (b) Specific heat
 (c) Thermal conductivity (d) All the above factors
16. If the ratio of coefficient of thermal conductivity of silver and copper is 10 : 9, then the ratio of the lengths upto which wax will melt in Ingen Hausz experiment will be [DPMT 2001]
 (a) 6 : 10 (b) $\sqrt{10} : 3$
 (c) 100 : 81 (d) 81 : 100
17. The thickness of a metallic plate is 0.4 cm. The temperature between its two surfaces is 20°C . The quantity of heat flowing per second is 50 calories from 5cm^2 area. In CGS system, the coefficient of thermal conductivity will be
 (a) 0.4 (b) 0.6
 (c) 0.2 (d) 0.5
18. In Searle's method for finding conductivity of metals, the temperature gradient along the bar [MP PMT 1984]
 (a) Is greater nearer the hot end
 (b) Is greater nearer to the cold end
 (c) Is the same at all points along the bar
 (d) Increases as we go from hot end to cold end
19. The dimensions of thermal resistance are [MP PET 1984; BVP 2003]
 (a) $M^{-1}L^{-2}T^3K$ (b) $ML^2T^{-2}K^{-1}$
 (c) $ML^2T^{-3}K$ (d) $ML^2T^{-2}K^{-2}$
20. A piece of glass is heated to a high temperature and then allowed to cool. If it cracks, a probable reason for this is the following property of glass [CPMT 1985]
 (a) Low thermal conductivity
 (b) High thermal conductivity
 (c) High specific heat
 (d) High melting point
21. Two walls of thicknesses d and d' and thermal conductivities k and k' are in contact. In the steady state, if the temperatures at the outer [MP PMT 1985] T_1 and T_2 , the temperature at the common wall is [MP PMT 1990; CBSE PMT 1999]
 (a) $\frac{k_1 T_1 d_2 + k_2 T_2 d_1}{k_1 d_2 + k_2 d_1}$ (b) $\frac{k_1 T_1 + k_2 T_2}{d_1 + d_2}$
 (c) $\left(\frac{k_1 d_1 + k_2 d_2}{T_1 + T_2} \right) T_1 T_2$ (d) $\frac{k_1 d_1 T_1 + k_2 d_2 T_2}{k_1 d_1 + k_2 d_2}$
22. A slab consists of two parallel layers of copper and brass of the same thickness and having thermal conductivities in the ratio 1 : 4. If

the free face of brass is at 100°C and that of copper at 0°C , the temperature of interface is

[IIT 1981; MP PMT 1987, 2001]

- (a) 80°C (b) 20°C
(c) 60°C (d) 40°C

23. The temperature gradient in a rod of 0.5 m long is $80^{\circ}\text{C}/\text{m}$. If the temperature of hotter end of the rod is 30°C , then the temperature of the cooler end is

- (a) 40°C (b) -10°C
(c) 10°C (d) 0°C

24. On heating one end of a rod, the temperature of whole rod will be uniform when

- (a) $K = 1$ (b) $K = 0$
(c) $K = 100$ (d) $K = \infty$

25. Snow is more heat insulating than ice, because

- (a) Air is filled in porous of snow
(b) Ice is more bad conductor than snow
(c) Air is filled in porous of ice
(d) Density of ice is more

26. Two thin blankets keep more hotness than one blanket of thickness equal to these two. The reason is

- (a) Their surface area increases
(b) A layer of air is formed between these two blankets, which is bad conductor
(c) These have more wool
(d) They absorb more heat from outside

27. Ice formed over lakes has

- (a) Very high thermal conductivity and helps in further ice formation
(b) Very low conductivity and retards further formation of ice
(c) It permits quick convection and retards further formation of ice
(d) It is very good radiator

28. Two rods of same length and material transfer a given amount of heat in 12 seconds, when they are joined end to end. But when they are joined lengthwise, then they will transfer same heat in same conditions in

[BHU 1998; UPSEAT 2002]

- (a) 24 s (b) 3 s
(c) 1.5 s (d) 48 s

29. Wires A and B have identical lengths and have circular cross-sections. The radius of A is twice the radius of B i.e. $r_A = 2r_B$. For a given temperature difference between the two ends, both wires conduct heat at the same rate. The relation between the thermal conductivities is given by

- (a) $K_A = 4K_B$ (b) $K_A = 2K_B$
(c) $K_A = K_B/2$ (d) $K_A = K_B/4$

30. Two identical plates of different metals are joined to form a single plate whose thickness is double the thickness of each plate. If the coefficients of conductivity of each plate are 2 and 3 respectively, then the conductivity of composite plate will be

- (a) 5 (b) 2.4
(c) 1.5 (d) 1.2

31. If the radius and length of a copper rod are both doubled, the rate of flow of heat along the rod increases

- (a) 4 times (b) 2 times
(c) 8 times (d) 16 times

32. The coefficients of thermal conductivity of copper, mercury and glass are respectively K_c , K_m and K_g such that $K_c > K_m > K_g$. If the same quantity of heat is to flow per second per unit area of each and corresponding temperature gradients are X_c , X_m and X_g , then

- (a) $X_c = X_m = X_g$ (b) $X_c > X_m > X_g$
(c) $X_c < X_m < X_g$ (d) $X_m < X_c < X_g$

33. If two metallic plates of equal thicknesses and thermal conductivities K_1 and K_2 are put together face to face and a common plate is constructed, then the equivalent thermal conductivity of this plate will be [MP PMT 1991]



- (a) $\frac{K_1 K_2}{K_1 + K_2}$ (b) $\frac{2K_1 K_2}{K_1 + K_2}$
(c) $\frac{(K_1^2 + K_2^2)^{3/2}}{K_1 K_2}$ (d) $\frac{(K_1^2 + K_2^2)^{3/2}}{2K_1 K_2}$

34. The quantity of heat which crosses unit area of a metal plate during conduction depends upon

[MP PMT 1992; JIPMER 1997]

- (a) The density of the metal
(b) The temperature gradient perpendicular to the area
(c) The temperature to which the metal is heated
(d) The area of the metal plate

35. The ends of two rods of different materials with their thermal conductivities, radii of cross-sections and lengths all are in the ratio 1 : 2 are maintained at the same temperature difference. If the rate of flow of heat in the larger rod is 4 cal/sec , that in the shorter rod in cal/sec will be

[EAMCET 1986]

- (a) 1 (b) 2
(c) 8 (d) 16

36. Two spheres of different materials one with double the radius and one-fourth wall thickness of the other, are filled with ice. If the time taken for complete melting ice in the large radius one is 25 minutes and that for smaller one is 16 minutes, the ratio of thermal conductivities of the materials of larger sphere to the smaller sphere is [EAMCET 1991]

- (a) 4 : 5 (b) 5 : 4
(c) 25 : 1 (d) 1 : 25

37. The ratio of the diameters of two metallic rods of the same material is 2 : 1 and their lengths are in the ratio 1 : 4. If the temperature difference between their ends are equal, the rate of flow of heat in them will be in the ratio [MP PET 1994]

- (a) 2 : 1 (b) 4 : 1
(c) 8 : 1 (d) 16 : 1

38. Two cylinders P and Q have the same length and diameter and are made of different materials having thermal conductivities in the ratio $2 : 3$. These two cylinders are combined to make a cylinder. One end of P is kept at 100°C and another end of Q at 0°C . The temperature at the interface of P and Q is [MP PMT 1994; EAMCET 2000]

(a) 30°C (b) 40°C
(c) 50°C (d) 60°C

39. Two identical rods of copper and iron are coated with wax uniformly. When one end of each is kept at temperature of boiling water, the length upto which wax melts are 8.4cm and 4.2cm respectively. If thermal conductivity of copper is 0.92 , then thermal conductivity of iron is [MP PET 1995]

(a) 0.23 (b) 0.46
(c) 0.115 (d) 0.69

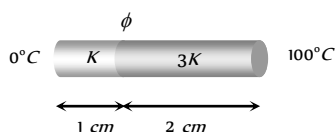
40. Mud houses are cooler in summer and warmer in winter because [BVP 2003]

(a) Mud is superconductor of heat
(b) Mud is good conductor of heat
(c) Mud is bad conductor of heat
(d) None of these

41. The temperature of hot and cold end of a 20cm long rod in thermal steady state are at 100°C and 20°C respectively. Temperature at the centre of the rod is [MP PMT 1996]

(a) 50°C (b) 60°C
(c) 40°C (d) 30°C

42. Two bars of thermal conductivities K and $3K$ and lengths 1cm and 2cm respectively have equal cross-sectional area, they are joined lengths wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature ϕ of the interface is



(a) 50°C (b) $\frac{100}{3}^\circ\text{C}$
(c) 60°C (d) $\frac{200}{3}^\circ\text{C}$

43. A heat flux of 4000 J/s is to be passed through a copper rod of length 10cm and area of cross-section 100cm^2 . The thermal conductivity of copper is $400\text{ W/m}^\circ\text{C}$. The two ends of this rod must be kept at a temperature difference of

(a) 1°C (b) 10°C
(c) 100°C (d) 1000°C

44. On a cold morning, a metal surface will feel colder to touch than a wooden surface because [AIIMS 1998]

(a) Metal has high specific heat
(b) Metal has high thermal conductivity
(c) Metal has low specific heat

(d) Metal has low thermal conductivity

45. In order that the heat flows from one part of a solid to another part, what is required [Pb. PMT 1999; EAMCET 1998]

(a) Uniform density (b) Density gradient
(c) Temperature gradient (d) Uniform temperature

46. At a common temperature, a block of wood and a block of metal feel equally cold or hot. The temperatures of block of wood and block of metal are [AIIMS 1999]

(a) Equal to temperature of the body
(b) Less than the temperature of the body
(c) Greater than temperature of the body
(d) Either (b) or (c)

47. According to the experiment of Ingen Hausz the relation between the thermal conductivity of a metal rod is K and the length of the rod whenever the wax melts is [UPSEAT 1999]

(a) $K/l = \text{constant}$ (b) $K^2/l = \text{constant}$
(c) $K/l^2 = \text{constant}$ (d) $Kl = \text{constant}$

48. Temperature of water at the surface of lake is -20°C . Then temperature of water just below the lower surface of ice layer is

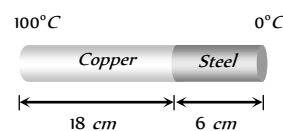
(a) -4°C (b) 0°C
(c) 4°C (d) -20°C

49. One end of a metal rod of length 1.0 m and area of cross section 100cm^2 is maintained at 100°C . If the other end of the rod is maintained at 0°C , the quantity of heat transmitted through the rod per minute is (Coefficient of thermal conductivity of material of rod $= 100\text{ W/m-K}$) [EAMCET (Engg.) 2000]

(a) $3 \times 10^3\text{ J}$ (b) $6 \times 10^3\text{ J}$
(c) $9 \times 10^3\text{ J}$ (d) $12 \times 10^3\text{ J}$

50. The coefficient of thermal conductivity of copper is nine times that of steel. In the composite cylindrical bar shown in the figure. What will be the temperature at the junction of copper and steel

(a) 75°C
(b) 67°C
(c) 33°C
(d) 25°C



51. The lengths and radii of two rods made of same material are in the ratios $3 : 4$ and $4 : 3$ respectively. If the temperature difference between the ends for the two rods be the same, then in the steady state, the amount of heat flowing per second through them will be in the ratio [MP PET 2000]

(a) $1 : 3$ (b) $4 : 3$
(c) $8 : 9$ (d) $3 : 2$

52. A slab consists of two parallel layers of two different materials of same thickness having thermal conductivities K_1 and K_2 . The equivalent conductivity of the combination is [BHU 2001]

- (a) $K_1 + K_2$ (b) $\frac{K_1 + K_2}{2}$
(c) $\frac{2K_1K_2}{K_1 + K_2}$ (d) $\frac{K_1 + K_2}{2K_1K_2}$

53. There are two identical vessels filled with equal amounts of ice. The vessels are of different metals. If the ice melts in the two vessels in 20 and 35 minutes respectively, the ratio of the coefficients of thermal conductivity of the two metals is

[AFMC 1998; MP PET 2001]

- (a) 4 : 7 (b) 7 : 4
(c) 16 : 49 (d) 49 : 16

54. Surface of the lake is at 2°C . Find the temperature of the bottom of the lake

[Orissa JEE 2002]

- (a) 2°C (b) 3°C
(c) 4°C (d) 1°C

55. The heat is flowing through a rod of length 50 cm and area of cross-section 5cm^2 . Its ends are respectively at 25°C and 125°C . The coefficient of thermal conductivity of the material of the rod is $0.092\text{ kcal/m}\cdot\text{s}\cdot^\circ\text{C}$. The temperature gradient in the rod is

[MP PET 2002]

- (a) 2°C/cm (b) 2°C/m
(c) 20°C/cm (d) 20°C/m

56. In the Ingen Hauz's experiment the wax melts up to lengths 10 and 25 cm on two identical rods of different materials. The ratio of thermal conductivities of the two materials is

[MP PET 2002]

- (a) 1 : 6.25 (b) 6.25 : 1
(c) $1 : \sqrt{2.5}$ (d) 1 : 2.5

57. Heat current is maximum in which of the following (rods are of identical dimension)

[Orissa JEE 2003]

- (a)  (b) 
(c)  (d) 

58. Two rods of same length and cross section are joined along the length. Thermal conductivities of first and second rod are K_1 and K_2 . The temperature of the free ends of the first and second rods are maintained at θ_1 and θ_2 respectively. The temperature of the common junction is

[MP PET 2003]

- (a) $\frac{\theta_1 + \theta_2}{2}$ (b) $\frac{K_2K_2}{K_1 + K_2}(\theta_1 + \theta_2)$
(c) $\frac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$ (d) $\frac{K_2\theta_1 + K_1\theta_2}{K_1 + K_2}$

59. Consider a compound slab consisting of two different materials having equal thickness and thermal conductivities K and $2K$ respectively. The equivalent thermal conductivity of the slab is

- (a) $\sqrt{2K}$ (b) $3K$
(c) $\frac{4}{3}K$ (d) $\frac{2}{3}K$

60. Two rods having thermal conductivity in the ratio of 5 : 3 having equal lengths and equal cross-sectional area are joined by face to face. If the temperature of the free end of the first rod is 100°C and

free end of the second rod is 20°C . Then temperature of the junction is

[CPMT 1996; DPMT 1997, 03; BVP 2004]

- (a) 70°C (b) 50°C
(c) 50°C (d) 90°C

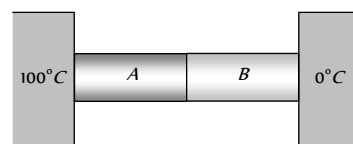
61. Woollen clothes are used in winter season because woollen clothes

- (a) Are good sources for producing heat
(b) Absorb heat from surroundings
(c) Are bad conductors of heat
(d) Provide heat to body continuously

62. Two metal cubes A and B of same size are arranged as shown in the figure. The extreme ends of the combination are maintained at the indicated temperatures. The arrangement is thermally insulated. The coefficients of thermal conductivity of A and B are $300\text{ W/m}\cdot^\circ\text{C}$ and $200\text{ W/m}\cdot^\circ\text{C}$, respectively. After steady state is reached, the temperature of the interface will be

[IIT 1996]

- (a) 45°C
(b) 90°C
(c) 30°C
(d) 60°C



63. A cylindrical rod having temperature T_1 and T_2 at its ends. The rate of flow of heat is $Q_1\text{ cal/sec}$. If all the linear dimensions are doubled keeping temperature constant then rate of flow of heat Q_2 will be

[CBSE PMT 2001]

- (a) $4Q_1$ (b) $2Q_1$
(c) $\frac{Q_1}{4}$ (d) $\frac{Q_1}{2}$

64. A body of length 1m having cross sectional area 0.75m^2 has heat flow through it at the rate of 6000 Joule/sec . Then find the temperature difference if $K = 200\text{ Jm}^{-1}\text{K}^{-1}$

[CPMT 2001]

- (a) 20°C (b) 40°C
(c) 80°C (d) 100°C

65. A wall has two layers A and B made of different materials. The thickness of both the layers is the same. The thermal conductivity of A and B are K and K' such that $K = 3K'$. The temperature across the wall is 20°C . In thermal equilibrium

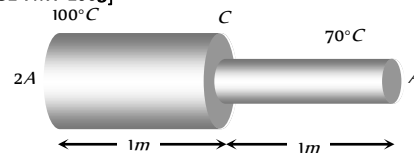
- (a) The temperature difference across $A = 15^\circ\text{C}$
(b) The temperature difference across $A = 5^\circ\text{C}$
(c) The temperature difference across $A = 10^\circ\text{C}$
(d) The rate of transfer of heat through A is more than that through B .

66. A metal rod of length 2m has cross sectional areas $2A$ and A as shown in figure. The ends are maintained at temperatures 100°C and 70°C . The temperature at middle point C is

[CPMT 2000]

- (a) 80°C
(b) 85°C
(c) 90°C
(d) 95°C

[CBSE PMT 2003]



67. The ratio of the coefficient of thermal conductivity of two different materials is 5 : 3. If the thermal resistance of the rod of same

thickness resistance of the rods of same thickness of these materials is same, then the ratio of the length of these rods will be

- (a) 3 : 5 (b) 5 : 3
(c) 3 : 4 (d) 3 : 2

68. Which of the following circular rods. (given radius r and length l) each made of the same material as whose ends are maintained at the same temperature will conduct most heat

[CBSE PMT 2005]

- (a) $r = 2r_0; l = 2l_0$ (b) $r = 2r_0; l = l_0$
(c) $r = r_0; l = l_0$ (d) $r = r_0; l = 2l_0$

Convection

1. It is hotter for the same distance over the top of a fire than it is in the side of it, mainly because

[NCERT 1976, 79, 80; AIIMS 2000]

- (a) Air conducts heat upwards
(b) Heat is radiated upwards
(c) Convection takes more heat upwards
(d) Convection, conduction and radiation all contribute significantly transferring heat upwards

2. One likes to sit under sunshine in winter season, because

- (a) The air surrounding the body is hot by which body gets heat
(b) We get energy by sun
(c) We get heat by conduction by sun
(d) None of the above

3. Air is bad conductor of heat or partly conducts heat, still vacuum is to be placed between the walls of the thermos flask because

- (a) It is difficult to fill the air between the walls of thermos flask
(b) Due to more pressure of air, the thermos can get crack
(c) By convection, heat can flow through air
(d) On filling the air, there is no advantage

4. While measuring the thermal conductivity of a liquid, we keep the upper part hot and lower part cool, so that

[CPMT 1985; MP PMT/PET 1988]

- (a) Convection may be stopped
(b) Radiation may be stopped
(c) Heat conduction is easier downwards
(d) It is easier and more convenient to do so

5. For proper ventilation of building, windows must be open near the bottom and top of the walls so as to let pass

- (a) In more air
(b) In cool air near the bottom and hot air out near the roof
(c) In hot air near the roof and cool air out near the bottom
(d) Out hot air near the roof

6. The layers of atmosphere are heated through

[MP PET 1986]

- (a) Convection (b) Conduction
(c) Radiation (d) (b) and (c) both

7. Mode of transmission of heat, in which heat is carried by the moving particles, is

[KCET 1999]

- (a) Radiation (b) Conduction

- (c) Convection (d) Wave motion

8. In a closed room, heat transfer takes place by [BHU 2001]

- (a) Conduction (b) Convection
(c) Radiation (d) All of these

9. In heat transfer, which method is based on gravitation

[CBSE PMT 2000]

- (a) Natural convection (b) Conduction
(c) Radiation (d) Stirring of liquids

10. When fluids are heated from the bottom, convection currents are produced because [UPSEAT 2000]

- (a) Molecular motion of fluid becomes aligned
(b) Molecular collisions take place within the fluid
(c) Heated fluid becomes more dense than the cold fluid above it
(d) Heated fluid becomes less dense than the cold fluid above it

11. If a liquid is heated in weightlessness, the heat is transmitted through [RPMT 1996]

- (a) Conduction
(b) Convection
(c) Radiation
(d) Neither, because the liquid cannot be heated in weightlessness

12. The rate of loss of heat from a body cooling under conditions of forced convection is proportional to its (A) heat capacity (B) surface area (C) absolute temperature (D) excess of temperature over that of surrounding : state if

[NCERT 1982]

- (a) A, B, C are correct (b) Only A and C are correct
(c) Only B and D are correct (d) Only D is correct

13. In which of the following process, convection does not take place primarily [IIT-JEE (Screening) 2005]

- (a) Sea and land breeze
(b) Boiling of water
(c) Warming of glass of bulb due to filament
(d) Heating air around a furnace

Radiation (General, Kirchoff's law, Black body)

1. On a clear sunny day, an object at temperature T is placed on the top of a high mountain. An identical object at the same temperature is placed at the foot of mountain. If both the objects are exposed to sun-rays for two hours in an identical manner, the object at the top of the mountain will register a temperature

- (a) Higher than the object at the foot
(b) Lower than the object at the foot
(c) Equal to the object at the foot
(d) None of the above

2. The velocity of heat radiation in vacuum is

[EAMCET 1982; KCET 1998]

- (a) Equal to that of light (b) Less than that of light
(c) Greater than that of light (d) Equal to that of sound

3. In which process, the rate of transfer of heat is maximum

[EAMCET 1977; MP PMT 1994; MH CET 2001]

- (a) Conduction
(b) Convection
(c) Radiation
(d) In all these, heat is transferred with the same velocity

4. Which of the following is the correct device for the detection of thermal radiation [Manipal MEE 1995, UPSEAT 2000]
 (a) Constant volume thermometer
 (b) Liquid-in-glass thermometer
 (c) Six's maximum and minimum thermometer
 (d) Thermopile
5. A thermos flask is polished well [AFMC 1996]
 (a) To make attractive
 (b) For shining
 (c) To absorb all radiations from outside
 (d) To reflect all radiations from outside
6. Heat travels through vacuum by [AIIMS 1998; CPMT 2003]
 (a) Conduction (b) Convection
 (c) Radiation (d) Both (a) and (b)
7. The energy supply being cut-off, an electric heater element cools down to the temperature of its surroundings, but it will not cool further because [CPMT 2001]
 (a) Supply is cut off
 (b) It is made of metal
 (c) Surroundings are radiating
 (d) Element & surroundings have same temp.
8. We consider the radiation emitted by the human body. Which of the following statements is true [CBSE PMT 2003]
 (a) The radiation is emitted only during the day
 (b) The radiation is emitted during the summers and absorbed during the winters
 (c) The radiation emitted lies in the ultraviolet region and hence is not visible
 (d) The radiation emitted is in the infra-red region
9. The earth radiates in the infra-red region of the spectrum. The spectrum is correctly given by [RPET 2002; AIEEE 2003]
 (a) Wien's law (b) Rayleigh jeans law
 (c) Planck's law of radiation (d) Stefan's law of radiation
10. Infrared radiation is detected by [AIEEE 2002]
 (a) Spectrometer (b) Pyrometer
 (c) Nanometer (d) Photometer
11. Pick out the statement which is not true [KCET 2002]
 (a) IR radiations are used for long distance photography
 (b) IR radiations arise due to inner electron transitions in atoms
 (c) IR radiations are detected by using a bolometer
 (d) Sun is the natural source of IR radiation
12. A hot and a cold body are kept in vacuum separated from each other. Which of the following cause decrease in temperature of the hot body [AFMC 2005]
 (a) Radiation
 (b) Convection
 (c) Conduction
 (d) Temperature remains unchanged
13. Good absorbers of heat are [J & K CET 2002]
 (a) Poor emitters (b) Non-emitters
 (c) Good emitters (d) Highly polished
14. For a perfectly black body, its absorptive power is [MP PMT 1989, 92; RPMT 2001; RPET 2001, 03; AFMC 2003]
 (a) 1 (b) 0.5
 (c) 0 (d) Infinity
15. Certain substance emits only the wavelengths $\lambda_1, \lambda_2, \lambda_3$ and λ_4 when it is at a high temperature. When this substance is at a colder temperature, it will absorb only the following wavelengths
 (a) λ_1 (b) λ_2
 (c) λ_1 and λ_2 (d) $\lambda_1, \lambda_2, \lambda_3$ and λ_4
16. As compared to the person with white skin, the person with black skin will experience [CPMT 1988]
 (a) Less heat and more cold (b) More heat and more cold
 (c) More heat and less cold (d) Less heat and less cold
17. Relation between emissivity e and absorptive power a is (for black body)
 (a) $e = a$ (b) $e = \frac{1}{a}$
 (c) $e = a^2$ (d) $a = e^2$
18. Which of the following statements is wrong [BCECE 2001]
 (a) Rough surfaces are better radiators than smooth surface
 (b) Highly polished mirror like surfaces are very good radiators
 (c) Black surfaces are better absorbers than white ones
 (d) Black surfaces are better radiators than white
19. Half part of ice block is covered with black cloth and rest half is covered with white cloth and then it is kept in sunlight. After some time clothes are removed to see the melted ice. Which of the following statements is correct
 (a) Ice covered with white cloth will melt more
 (b) Ice covered with black cloth will melt more
 (c) Equal ice will melt under both clothes
 (d) It will depend on the temperature of surroundings of ice
20. If between wavelength λ and $\lambda + d\lambda$, e_λ and a_λ be the emissive and absorptive powers of a body and E_λ be the emissive power of a perfectly black body, then according to Kirchhoff's law, which is true [RPMT 1998; MP PET 1991]
 (a) $e_\lambda = a_\lambda = E_\lambda$ (b) $e_\lambda E_\lambda = a_\lambda$
 (c) $e_\lambda = a_\lambda E_\lambda$ (d) $e_\lambda a_\lambda E_\lambda = \text{constant}$
21. When p calories of heat is given to a body, it absorbs q calories; then the absorbtion power of body will be
 (a) p/q (b) q/p
 (c) p^2/q^2 (d) q^2/p^2
22. Distribution of energy in the spectrum of a black body can be correctly represented by [MP PMT 1989]
 (a) Wien's law (b) Stefan's law
 (c) Planck's law (d) Kirchhoff's law
23. In rainy season, on a clear night the black seat of a bicycle becomes wet because

- (a) It absorbs water vapour
(b) Black seat is good absorber of heat
(c) Black seat is good radiator of heat energy
(d) None of the above
24. There is a rough black spot on a polished metallic plate. It is heated upto 1400 K approximately and then at once taken in a dark room. Which of the following statements is true
[NCERT 1984; CPMT 1998]
(a) In comparison with the plate, the spot will shine more
(b) In comparison with the plate, the spot will appear more black
(c) The spot and the plate will be equally bright
(d) The plate and the black spot can not be seen in the dark room
25. At a certain temperature for given wave length, the ratio of emissive power of a body to emissive power of black body in same circumstances is known as [RPMT 1997]
(a) Relative emissivity (b) Emissivity
(c) Absorption coefficient (d) Coefficient of reflection
26. The cause of Fraunhofer lines is [RPMT 1996; EAMCET 2001]
(a) Reflection of radiations by chromosphere
(b) Absorption of radiations by chromosphere
(c) Emission of radiations by chromosphere
(d) Transmission of radiations by chromosphere
27. Two thermometers A and B are exposed in sun light. The valve of A is painted black, But that of B is not painted. The correct statement regarding this case is [BHU (Med.) 1999; MH CET 1999]
(a) Temperature of A will rise faster than B but the final temperature will be the same in both
(b) Both A and B show equal rise in beginning
(c) Temperature of A will remain more than B
(d) Temperature of B will rise faster
28. There is a black spot on a body. If the body is heated and carried in dark room then it glows more. This can be explained on the basis of [RPET 2000]
(a) Newton's law of cooling (b) Wien's law
(c) Kirchoff's law (d) Stefan's
29. When red glass is heated in dark room it will seem [RPET 2000]
(a) Green (b) Purple
(c) Black (d) Yellow
30. A hot body will radiate heat most rapidly if its surface is [UPSEAT 1999, 2000]
(a) White & polished (b) White & rough
(c) Black & polished (d) Black & rough
31. A body, which emits radiations of all possible wavelengths, is known as [CPMT 2001; Pb. PET 2002]
(a) Good conductor (b) Partial radiator
(c) Absorber of photons (d) Perfectly black-body
32. Which of the following is the example of ideal black body [AIEEE 2002; CBSE PMT 2002]
(a) Kajal (b) Black board
(c) A pin hole in a box (d) None of these
33. An ideal black body at room temperature is thrown into a furnace. It is observed that [IIT-JEE (Screening) 2002]
(a) Initially it is the darkest body and at later times the brightest
(b) It is the darkest body at all times
(c) It cannot be distinguished at all times
(d) Initially it is the darkest body and at later times it cannot be distinguished
34. Absorption co-efficient of an open window is... [KCET 2004]
(a) Zero (b) 0.5
(c) 1 (d) 0.25
35. Which of the prism is used to see infra-red spectrum of light [RPMT 2000]
(a) Rock-salt (b) Nicol
(c) Flint (d) Crown
36. Which of the following statement is correct [RPMT 2001]
(a) A good absorber is a bad emitter
(b) Every body absorbs and emits radiations at every temperature
(c) The energy of radiations emitted from a black body is same for all wavelengths
(d) The law showing the relation of temperatures with the wavelength of maximum emission from an ideal black body is Plank's law
37. A piece of blue glass heated to a high temperature and a piece of red glass at room temperature, are taken inside a dimly lit room then [KCET 2005]
(a) The blue piece will look blue and red will look as usual
(b) Red look brighter red and blue look ordinary blue
(c) Blue shines like brighter red compared to the red piece
(d) Both the pieces will look equally red.
38. Which of the following law states that "good absorbers of heat are good emitters" [Orissa JEE 2005]
(a) Stefan's law (b) Kirchoff's law
(c) Planck's law (d) Wein's law

Radiation (Wein's law)

1. According to Wein's law [DCE 1995, 96; MP PET/PMT 1988
DPMT 1999; AIIMS 2002; CBSE PMT 2004]
(a) $\lambda_m T = \text{constant}$ (b) $\frac{\lambda_m}{T} = \text{constant}$
(c) $\frac{T}{\lambda_m} = \text{constant}$ (d) $T + \lambda_m = \text{constant}$
2. On investigation of light from three different stars A , B and C , it was found that in the spectrum of A the intensity of red colour is maximum, in B the intensity of blue colour is maximum and in C the intensity of yellow colour is maximum. From these observations it can be concluded that [CPMT 1989]
(a) The temperature of A is maximum, B is minimum and C is intermediate

- (b) The temperature of A is maximum, C is minimum and B is intermediate
(c) The temperature of B is maximum, A is minimum and C is intermediate
(d) The temperature of C is maximum, B is minimum and A is intermediate
3. If wavelengths of maximum intensity of radiations emitted by the sun and the moon are $0.5 \times 10^{-6} m$ and $10^{-4} m$ respectively, the ratio of their temperatures is [MP PMT 1990]
(a) $1/100$ (b) $1/200$
(c) 100 (d) 200
4. The wavelength of radiation emitted by a body depends upon
(a) The nature of its surface
(b) The area of its surface
(c) The temperature of its surface
(d) All the above factors
5. If black wire of platinum is heated, then its colour first appear red, then yellow and finally white. It can be understood on the basis of
(a) Wien's displacement law
(b) Prevost theory of heat exchange
(c) Newton's law of cooling
(d) None of the above
6. Colour of shining bright star is an indication of its [AIIMS 2001; RPMT 1999; BCECE 2005]
(a) Distance from the earth (b) Size
(c) Temperature (d) Mass
7. The wavelength of maximum emitted energy of a body at $700 K$ is $4.08 \mu m$. If the temperature of the body is raised to $1400 K$, the wavelength of maximum emitted energy will be
(a) $1.02 \mu m$ (b) $16.32 \mu m$
(c) $8.16 \mu m$ (d) $2.04 \mu m$
8. A black body at $200 K$ is found to emit maximum energy at a wavelength of $14 \mu m$. When its temperature is raised to $1000 K$, the wavelength at which maximum energy is emitted is [RPMT 1998; MP PET 1991; BVP 2003]
(a) $14 \mu m$ (b) $70 \mu m$
(c) $2.8 \mu m$ (d) $2.8 mm$
9. Two stars emit maximum radiation at wavelength $3600 \text{ \AA}$ and $4800 \text{ \AA}$ respectively. The ratio of their temperatures is [MP PMT 1991]
(a) $1 : 2$ (b) $3 : 4$
(c) $4 : 3$ (d) $2 : 1$
10. A black body emits radiations of maximum intensity at a wavelength of $5000 \text{ \AA}$, when the temperature of the body is $1227^\circ C$. If the temperature of the body is increased by $1000^\circ C$, the maximum intensity of emitted radiation would be observed at
(a) $2754.8 \text{ \AA}$ (b) $3000 \text{ \AA}$
(c) $3500 \text{ \AA}$ (d) $4000 \text{ \AA}$
11. Four pieces of iron heated in a furnace to different temperatures show different colours listed below. Which one has the highest temperature [MP PET 1992]
(a) White (b) Yellow
(c) Orange (d) Red
12. If a black body is heated at a high temperature, it seems to be
(a) Blue (b) White
(c) Red (d) Black
13. If the temperature of the sun becomes twice its present temperature, then [MP PET 1989; RPMT 1996]
(a) Radiated energy would be predominantly in infrared
(b) Radiated energy would be predominantly in ultraviolet [MP PMT 1992]
(c) Radiated energy would be predominantly in X-ray region
(d) Radiated energy would become twice the present radiated energy
14. The maximum energy in the thermal radiation from a hot source occurs at a wavelength of $11 \times 10^{-5} cm$. According to Wein's law, the temperature of the source (on Kelvin scale) will be n times the temperature of another source (on Kelvin scale) for which the wavelength at maximum energy is $5.5 \times 10^{-5} cm$. The value n is [CPMT 1991]
(a) 2 (b) 4
(c) $\frac{1}{2}$ (d) 1
15. The wavelength of maximum energy released during an atomic explosion was $2.93 \times 10^{-10} m$. Given that Wein's constant is $2.93 \times 10^{-3} m - K$, the maximum temperature attained must be of the order of [Haryana CEE 1996; MH CET 2002; Pb. PET 2000]
(a) $10^{-7} K$ (b) $10^7 K$
(c) $10^{-13} K$ (d) $5.86 \times 10^7 K$ [MP PET 1990]
16. The maximum wavelength of radiation emitted at $2000 K$ is $4 \mu m$. What will be the maximum wavelength of radiation emitted at [MP PMT/PET 1998; DPMT 2000]
(a) $3.33 \mu m$ (b) $0.66 \mu m$
(c) $1 \mu m$ (d) $1 m$
17. How is the temperature of stars determined by [BHU 1999, 02; DCE 2000, 03]
(a) Stefan's law (b) Wein's displacement law
(c) Kirchhoff's law (d) Ohm's law
18. On increasing the temperature of a substance gradually, which of the following colours will be noticed by you [Pb. PMT 1995; Pb. PET 1996; CPMT 1995, 98; KCET 2000]
(a) White (b) Yellow
(c) Green (d) Red
19. A black body has maximum wavelength λ_m at temperature $2000 K$. Its corresponding wavelength at temperature $3000 K$ will be [CBSE PMT 2001; MP PET 1992]
(a) $\frac{3}{2} \lambda_m$ (b) $\frac{2}{3} \lambda_m$
(c) $\frac{4}{9} \lambda_m$ (d) $\frac{9}{4} \lambda_m$

20. Relation between the colour and the temperature of a star is given by [Kerala PET 2001]
 (a) Wein's displacement law
 (b) Planck's law
 (c) Hubble's law
 (d) Fraunhofer diffraction law
21. A black body at a temperature of 1640 K has the wavelength corresponding to maximum emission equal to $1.75\text{ }\mu$. Assuming the moon to be a perfectly black body, the temperature of the moon, if the wavelength corresponding to maximum emission is $14.35\text{ }\mu$ is [Kerala (Med.) 2002]
 (a) 100 K (b) 150 K
 (c) 200 K (d) 250 K
22. The maximum wavelength of radiations emitted at 900 K is $4\text{ }\mu\text{m}$. What will be the maximum wavelength of radiations emitted at 1200 K [BHU 2002]
 (a) $3\text{ }\mu\text{m}$ (b) $0.3\text{ }\mu\text{m}$
 (c) $1\text{ }\mu\text{m}$ (d) 1 m
23. Solar radiation emitted by sun resembles that emitted by a black body at a temperature of 6000 K . Maximum intensity is emitted at a wavelength of about $4800\text{ \AA}$. If the sun were to cool down from 6000 K to 3000 K then the peak intensity would occur at a wavelength [UPSEAT 2002]
 (a) $4800\text{ \AA}$ (b) $9600\text{ \AA}$
 (c) $7200\text{ \AA}$ (d) $6400\text{ \AA}$
24. What will be the ratio of temperatures of sun and moon if the wavelengths of their maximum emission radiations rates are $140\text{ \AA}$ and $4200\text{ \AA}$ respectively [J & K CET 2004]
 (a) $1 : 30$ (b) $30 : 1$
 (c) $42 : 14$ (d) $14 : 42$
25. The radiation energy density per unit wavelength at a temperature T has a maximum at a wavelength λ . At temperature $2T$, it will have a maximum at a wavelength [UPSEAT 2004]
 (a) 4λ (b) 2λ
 (c) $\lambda/2$ (d) $\lambda/4$
26. The absolute temperatures of two black bodies are 2000 K and 3000 K respectively. The ratio of wavelengths corresponding to maximum emission of radiation by them will be
 (a) $2 : 3$ (b) $3 : 2$
 (c) $9 : 4$ (d) $4 : 9$
27. The temperature of sun is 5500 K and it emits maximum intensity radiation in the yellow region ($5.5 \times 10^{-7}\text{ m}$). The maximum radiation from a furnace occurs at wavelength $11 \times 10^{-7}\text{ m}$. The temperature of furnace is [J & K CET 2000]
 (a) 1125 K (b) 2750 K
 (c) 5500 K (d) 11000 K
28. A particular star (assuming it as a black body) has a surface temperature of about $5 \times 10^4\text{ K}$. The wavelength in nanometers at which its radiation becomes maximum is [EAMCET (Med.) 2003]
 ($b = 0.0029\text{ mK}$)
 (a) 48 (b) 58
 (c) 60 (d) 70
29. The maximum energy in thermal radiation from a source occurs at the wavelength $4000\text{ \AA}$. The effective temperature of the source is
 (a) 7000 K (b) 80000 K
 (c) 10^4 K (d) 10^6 K
30. The intensity of radiation emitted by the sun has its maximum value at a wavelength of 510 nm and that emitted by the north star has the maximum value at 350 nm . If these stars behave like black bodies, then the ratio of the surface temperature of the sun and north star is [IIT 1997 Cancelled; JIPMER 2000; AIIMS 2000]
 (a) 1.46 (b) 0.69
 (c) 1.21 (d) 0.83

Radiation (Stefan's law)

1. The amount of radiation emitted by a perfectly black body is proportional to [AFMC 1995; Pb. PMT 1997; CPMT 1974, 98, 02; AIIMS 2000; DPMT 1995, 98, 02]
 (a) Temperature on ideal gas scale
 (b) Fourth root of temperature on ideal gas scale
 (c) Fourth power of temperature on ideal gas scale
 (d) Source of temperature on ideal gas scale
2. A metal ball of surface area 200 cm^2 and temperature 527°C is surrounded by a vessel at 27°C . If the emissivity of the metal is 0.4, then the rate of loss of heat from the ball is ($\sigma = 5.67 \times 10^{-8}\text{ J/m}^2\text{ s}^{-1}\text{K}^{-4}$) [MP PMT/PET 1988]
 (a) 108 joules approx. (b) 168 joules approx.
 (c) 182 joules approx. (d) 192 joules approx.
3. The rate of radiation of a black body at 0°C is E/sec . The rate of radiation of this black body at 273°C will be [MP PMT 1989; Kerala PET 2002; UPSEAT 2001]
 (a) $16 E$ (b) $8 E$
 (c) $4 E$ (d) E
4. A black body radiates energy at the rate of $E\text{ W/m}^2$ at a high temperature TK . When the temperature is reduced to $\frac{T}{2}\text{ K}$, the [RPM 2003] energy will be [CPMT 1988; UPSEAT 1998; MNR 1993; SCRA 1996; MP PMT 1992; DPMT 2001; MH CET 2001]
 (a) $\frac{E}{16}$ (b) $\frac{E}{4}$
 (c) $4E$ (d) $16E$
5. An object is at a temperature of 400°C . At what temperature would it radiate energy twice as fast? The temperature of the surroundings may be assumed to be negligible [MP PMT 1990; DPMT 2002]
 (a) 200°C (b) 200 K
 (c) 800°C (d) 800 K
6. A black body at a temperature of 227°C radiates heat energy at the rate of $5\text{ cal/cm}^2\text{-sec}$. At a temperature of 727°C , the rate of heat radiated per unit area in cal/cm^2 will be
 (a) 80 (b) 160

- (c) 250 (d) 500
7. Energy is being emitted from the surface of a black body at 127°C temperature at the rate of $1.0 \times 10^6 \text{ J/sec-m}^2$. Temperature of the black body at which the rate of energy emission is $16.0 \times 10^6 \text{ J/sec-m}^2$ will be
[MP PMT 1991; AFMC 1998]
- (a) 254°C (b) 508°C
(c) 527°C (d) 727°C
8. In MKS system, Stefan's constant is denoted by σ . In CGS system multiplying factor of σ will be
(a) 1 (b) 10^3
(c) 10^5 (d) 10^2
9. If temperature of a black body increases from 7°C to 287°C , then the rate of energy radiation increases by
[AIIMS 1997; Haryana PMT 2000; RPMT 2003]
- (a) $\left(\frac{287}{7}\right)^4$ (b) 16
(c) 4 (d) 2
10. The temperature of a piece of iron is 27°C and it is radiating energy at the rate of $Q \text{ kWm}^{-2}$. If its temperature is raised to 151°C , the rate of radiation of energy will become approximately
(a) $2Q \text{ kWm}^{-2}$ (b) $4Q \text{ kWm}^{-2}$
(c) $6Q \text{ kWm}^{-2}$ (d) $8Q \text{ kWm}^{-2}$
11. The temperatures of two bodies A and B are 727°C and 127°C . The ratio of rate of emission of radiations will be
[MP PET 1986]
- (a) $727/127$ (b) $625/16$
(c) $1000/400$ (d) $100/16$
12. The temperature at which a black body of unit area loses its energy at the rate of 1 joule/second is
(a) -65°C (b) 65°C
(c) 65 K (d) None of these
13. The area of a hole of heat furnace is 10^{-4} m^2 . It radiates 1.58×10^5 calories of heat per hour. If the emissivity of the furnace is 0.80, then its temperature is
(a) 1500 K (b) 2000 K
(c) 2500 K (d) 3000 K
14. Two spheres P and Q, of same colour having radii 8 cm and 2 cm are maintained at temperatures 127°C and 527°C respectively. The ratio of energy radiated by P and Q is
(a) 0.054 (b) 0.0034
(c) 1 (d) 2
15. A body radiates energy 5W at a temperature of 127°C . If the temperature is increased to 927°C , then it radiates energy at the rate of
[MP PET 1994; BHU 1995; CPMT 1998; AFMC 2000]
- (a) 410W (b) 81W
(c) 405W (d) 200W
16. A thin square steel plate with each side equal to 10 cm is heated by a blacksmith. The rate of radiated energy by the heated plate is 1134 W. The temperature of the hot steel plate is (Stefan's constant $\sigma = 5.67 \times 10^{-8} \text{ watt m}^{-2} \text{ K}^{-4}$, emissivity of the plate = 1)
(a) 1000 K (b) 1189 K
(c) 2000 K (d) 2378 K
17. The temperatures of two bodies A and B are respectively 727°C and 327°C . The ratio $H_A : H_B$ of the rates of heat radiated by them is
[UPSEAT 1999; MP PET 1999; MH CET 2000; AIIMS 2000]
- (a) 727 : 327 (b) 5 : 3
(c) 25 : 9 (d) 625 : 81
18. The energy emitted per second by a black body at 27°C is 10 J. If the temperature of the black body is increased to 327°C , the energy emitted per second will be
[CPMT 1999; DCE 1999]
- (a) 20 J (b) 40 J
(c) 80 J (d) 160 J
[MP PET 1992]
19. The radiant energy from the sun incident normally at the surface of earth is $20 \text{ kcal/m}^2 \text{ min}$. What would have been the radiant energy incident normally on the earth, if the sun had a temperature twice of the present one
[CBSE PMT 1998; Pb. PET 2001]
- (a) $160 \text{ kcal/m}^2 \text{ min}$ (b) $40 \text{ kcal/m}^2 \text{ min}$
(c) $320 \text{ kcal/m}^2 \text{ min}$ (d) $80 \text{ kcal/m}^2 \text{ min}$
20. A spherical black body with a radius of 12 cm radiates 440 W power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be
[IIT 1997 Re-Exam]
- (a) 225 (b) 450
(c) 900 (d) 1800
21. If the temperature of the sun (black body) is doubled, the rate of energy received on earth will be increased by a factor of [CBSE PMT 1993; BHU 2000]
- (a) 2 (b) 4
(c) 8 (d) 16
22. The ratio of energy of emitted radiation of a black body at 27°C and 927°C is
[MP PMT 1994] [Pb. PMT 1995; CPMT 1997, 2000; CBSE PMT 2000; DPMT 1998, 02, 03]
- (a) 1 : 4 (b) 1 : 16
(c) 1 : 64 (d) 1 : 256
23. If the temperature of a black body be increased from 27°C to 327°C the radiation emitted increases by a fraction of
[Pb. PET 1997; JIPMER 1999]

- (a) 16 (b) 8
(c) 4 (d) 2
24. The rectangular surface of area $8\text{ cm} \times 4\text{ cm}$ of a black body at a temperature of 127°C emits energy at the rate of E per second. If the length and breadth of the surface are each reduced to half of the initial value and the temperature is raised to 327°C , the rate of emission of energy will become [MP PET 2000]
- (a) $\frac{3}{8}E$ (b) $\frac{81}{16}E$
(c) $\frac{9}{16}E$ (d) $\frac{81}{64}E$
25. At temperature T , the power radiated by a body is Q watts. At the temperature $3T$ the power radiated by it will be [MP PET 2000]
- (a) $3Q$ (b) $9Q$
(c) $27Q$ (d) $81Q$
26. Two spherical black bodies of radii r_1 and r_2 and with surface temperature T_1 and T_2 respectively radiate the same power. Then the ratio of r_1 and r_2 will be [KCET 2001; UPSEAT 2001]
- (a) $\left(\frac{T_2}{T_1}\right)^2$ (b) $\left(\frac{T_2}{T_1}\right)^4$
(c) $\left(\frac{T_1}{T_2}\right)^2$ (d) $\left(\frac{T_1}{T_2}\right)^4$
27. Temperature of a black body increases from 327°C to 927°C , the initial energy possessed is 2 KJ , what is its final energy
- (a) 32 KJ (b) 320 KJ
(c) 1200 KJ (d) None of these
28. The original temperature of a black body is 727°C . The temperature at which this black body must be raised so as to double the total radiant energy, is [Pb. PMT 2001]
- (a) 971 K (b) 1190 K
(c) 2001 K (d) 1458 K
29. Two black metallic spheres of radius 4 m , at 2000 K and 1 m at 4000 K will have ratio of energy radiation as [RPET 2000; AIEEE 2002]
- (a) $1:1$ (b) $4:1$
(c) $1:4$ (d) $2:1$
30. The energy spectrum of a black body exhibits a maximum around a wavelength λ_o . The temperature of the black body is now changed such that the energy is maximum around a wavelength $\frac{3\lambda_o}{4}$. The power radiated by the black body will now increase by a factor of [KCET 2002]
- (a) $256/81$ (b) $64/27$
(c) $16/9$ (d) $4/3$
31. A black body is at a temperature 300 K . It emits energy at a rate, which is proportional to [Pb. PMT 1998; AIIMS 2002; MH CET 2003]
- (a) 300 (b) $(300)^2$
(c) $(300)^3$ (d) $(300)^4$
32. If the temperature of a hot body is increased by 50% then the increase in the quantity of emitted heat radiation will be [RPET 1998; EAMCET 2001; MP PMT 2003]
- (a) 125% (b) 200%
(c) 300% (d) 400%
33. Two identical metal balls at temperature 200°C and 400°C kept in air at 27°C . The ratio of net heat loss by these bodies is
- (a) $1/4$ (b) $1/2$
(c) $1/16$ (d) $\frac{473^4 - 300^4}{673^4 - 300^4}$
34. Two spheres made of same material have radii in the ratio $1:2$. Both are at same temperature. Ratio of heat radiation energy emitted per second by them is [MP PMT 2002; MH CET 2004]
- (a) $1:2$ (b) $1:8$
(c) $1:4$ (d) $1:16$
35. A black body at a temperature of 127°C radiates heat at the rate of $1\text{ cal/cm} \times \text{sec}$. At a temperature of 527°C the rate of heat radiation from the body in $(\text{cal/cm} \times \text{sec})$ will be [MP PET 2002]
- (a) 16.0 (b) 10.45
(c) 4.0 (d) 2.0
36. A black body radiates 20 W at temperature 227°C . If temperature of the black body is changed to 727°C then its radiating power will be [DCE 2001]
- [CBSE PMT 2002; DCE 1999, 03; AIIMS 2003]
- (a) 120 W (b) 240 W
(c) 320 W (d) 360 W
37. Two spheres of same material have radius 1 m and 4 m and temperature 4000 K and 2000 K respectively. The energy radiated per second by the first sphere is [Pb. PMT 2002]
- (a) Greater than that by the second
(b) Less than that by the second
(c) Equal in both cases
(d) The information is incomplete
38. The radiation emitted by a star A is $10,000$ times that of the sun. If the surface temperatures of the sun and the star A are 6000 K and 2000 K respectively, the ratio of the radii of the star A and the sun is
- (a) $300:1$ (b) $600:1$
(c) $900:1$ (d) $1200:1$
39. A black body radiates at the rate of W watts at a temperature T . If the temperature of the body is reduced to $T/3$, it will radiate at the rate of (in Watts) [BHU 1998; MP PET 2003]
- (a) $\frac{W}{81}$ (b) $\frac{W}{27}$
(c) $\frac{W}{9}$ (d) $\frac{W}{3}$

40. Star A has radius r surface temperature T while star B has radius $4r$ and surface temperature $T/2$. The ratio of the power of two stars, $P_1 : P_2$ is [MP PMT 2004]
 (a) 16 : 1 (b) 1 : 16
 (c) 1 : 1 (d) 1 : 4
41. Suppose the sun expands so that its radius becomes 100 times its present radius and its surface temperature becomes half of its present value. The total energy emitted by it then will increase by a factor of [AIIMS 2004]
 (a) 10 (b) 625
 (c) 256 (d) 16
42. If the temperature of the sun were to be increased from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on the earth to what it was previously will be
 (a) 4 (b) 16
 (c) 32 (d) 64
43. At 127°C radiates energy is $2.7 \times 10^7 \text{ J/s}$. At what temperature radiated energy is $4.32 \times 10^7 \text{ J/s}$ [BCECE 2004]
 (a) 400 K (b) 4000 K
 (c) 80000 K (d) 40000 K
44. If the initial temperatures of metallic sphere and disc, of the same mass, radius and nature are equal, then the ratio of their rate of cooling in same environment will be [J & K CET 2004]
 (a) 1 : 4 (b) 4 : 1
 (c) 1 : 2 (d) 2 : 1
45. A black body radiates energy at the rate of $1 \times 10^7 \text{ J/s} \times m$ at temperature of 227°C . The temperature to which it must be heated so that it radiates energy at rate of $1 \times 10^8 \text{ J/sm}$, is
 (a) 5000 K (b) 5000°C
 (c) 500 K (d) 500°C
46. The temperature of the body is increased from -73°C to 327°C , the ratio of energy emitted per second is : [CPMT 2001; Pb. PET 2001]
 (a) 1 : 3 (b) 1 : 81
 (c) 1 : 27 (d) 1 : 9
47. If the temperature of the body is increased by 10%, the percentage increase in the emitted radiation will be [RPMT 2001, 02]
 (a) 46% (b) 40%
 (c) 30% (d) 80%
48. If the sun's surface radiates heat at $6.3 \times 10^7 \text{ Wm}^{-2}$. Calculate the temperature of the sun assuming it to be a black body ($\sigma = 5.7 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$) [BHU (Med.) 2000]
 (a) $5.8 \times 10^3 \text{ K}$ (b) $8.5 \times 10^3 \text{ K}$
 (c) $3.5 \times 10^8 \text{ K}$ (d) $5.3 \times 10^8 \text{ K}$
49. A sphere at temperature 600K is placed in an environment of temperature is 200K . Its cooling rate is H . If its temperature reduced to 400K then cooling rate in same environment will become [CBSE PMT 1999; BHU 2001]
 (a) $(3/16)H$ (b) $(16/3)H$
 (c) $(9/27)H$ (d) $(1/16)H$
50. The value of Stefan's constant is [RPMT 2002]
 (a) $5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ (b) $5.67 \times 10^{-5} \text{ W/m}^2 \cdot \text{K}^4$
 (c) $5.67 \times 10^{-11} \text{ W/m}^2 \cdot \text{K}^4$ (d) None of these
51. Rate of cooling at 600K , if surrounding temperature is 300K is R . The rate of cooling at 900K is [DPMT 2002]
 (a) $\frac{16}{3}R$ (b) $2R$
 (c) $3R$ (d) $\frac{2}{3}R$
52. A black body of surface area 10cm^2 is heated to 127°C and is suspended in a room at temperature 27°C . The initial rate of loss of heat from the body at 127°C temperature will be [BCECE 2004]
 (a) 2.99 W (b) 1.89 W
 (c) 1.18 W (d) 0.99 W
53. Two identical objects A and B are at temperatures T_1 and T_2 respectively. Both objects are placed in a room with perfectly absorbing walls maintained at temperatures T ($T_A > T > T_B$). The objects A and B attain temperature T eventually which one of the following is correct statement [CPMT 1997]
 (a) ' A ' only emits radiations while B only absorbs them until both attain temperature
 (b) A loses more radiations than it absorbs while B absorbs more radiations than it emits until temperature T is attained
 (c) Both A and B only absorb radiations until they attain temperature T
 (d) Both A and B only emit radiations until they attain temperature T
54. When the body has the same temperature as that of surroundings [UPSEAT 1999]
 (a) It does not radiate heat
 (b) It radiates the same quantity of heat as it absorbs
 (c) It radiates less quantity of heat as it receives from surroundings
 (d) It radiates more quantity of heat as it receives heat from surroundings
55. The ratio of radiant energies radiated per unit surface area by two bodies is 16 : 1, the temperature of hotter body is 1000K , then the temperature of colder body will be [UPSEAT 2001]
 (a) 250 K (b) 500 K
 (c) 1000 K (d) 62.5 K
56. The spectral energy distribution of star is maximum at twice temperature as that of sun. The total energy radiated by star is
 (a) Twice as that of the sun
 (b) Same as that of the sun
 (c) Sixteen times as that of the sun
 (d) One sixteenth of sun

Radiation (Newton's Law of Cooling)

- hot water cools from 60°C to 50°C in the first 10 minutes and to 42°C in the next 10 minutes. The temperature of the surrounding is [MP PET 1993]
 (a) 5°C (b) 10°C

- (c) 15°C (d) 20°C
2. A bucket full of hot water cools from 75°C to 70°C in time T_1 , from 70°C to 65°C in time T_2 and from 65°C to 60°C in time T_3 , then [NCERT 1980; MP PET 1989; CBSE PMT 1995; KCET 2003; MH CET 1999]
- (a) $T_1 = T_2 = T_3$ (b) $T_1 > T_2 > T_3$
(c) $T_1 < T_2 < T_3$ (d) $T_1 > T_2 < T_3$
3. Consider two hot bodies B_1 and B_2 which have temperatures 100°C and 80°C respectively at $t = 0$. The temperature of the surroundings is 40°C . The ratio of the respective rates of cooling R_1 and R_2 of these two bodies at $t = 0$ will be
- (a) $R_1 : R_2 = 3 : 2$ (b) $R_1 : R_2 = 5 : 4$
(c) $R_1 : R_2 = 2 : 3$ (d) $R_1 : R_2 = 4 : 5$
4. Newton's law of cooling is a special case of
- (a) Stefan's law (b) Kirchhoff's law
(c) Wien's law (d) Planck's law
5. Equal masses of two liquids are filled in two similar calorimeters. The rate of cooling will [MP PMT 1987]
- (a) Depend on the nature of the liquids
(b) Depend on the specific heats of liquids
(c) Be same for both the liquids
(d) Depend on the mass of the liquids
6. In Newton's experiment of cooling, the water equivalent of two similar calorimeters is 10 gm each. They are filled with 350 gm of water and 300 gm of a liquid (equal volumes) separately. The time taken by water and liquid to cool from 70°C to 60°C is 3 min and 95 sec respectively. The specific heat of the liquid will be
- (a) $0.3\text{ Cal/gm}^{\circ}\text{C}$ (b) $0.5\text{ Cal/gm}^{\circ}\text{C}$
(c) $0.6\text{ Cal/gm}^{\circ}\text{C}$ (d) $0.8\text{ Cal/gm}^{\circ}\text{C}$
7. Newton's law of cooling is used in laboratory for the determination of the [CPMT 1973; CPMT 2002]
- (a) Specific heat of the gases (b) The latent heat of gases
(c) Specific heat of liquids (d) Latent heat of liquids
8. A body cools from 60°C to 50°C in 10 minutes when kept in air at 30°C . In the next 10 minutes its temperature will be
- (a) Below 40°C (b) 40°C
(c) Above 40°C (d) Cannot be predicted
9. Liquid is filled in a vessel which is kept in a room with temperature 20°C . When the temperature of the liquid is 80°C , then it loses heat at the rate of 60 cal/sec . What will be the rate of loss of heat when the temperature of the liquid is 40°C
- (a) 180 cal/sec (b) 40 cal/sec
(c) 30 cal/sec (d) 20 cal/sec
10. Which of the following statements is true/correct [Manipal MEE 1995]
- (a) During clear nights, the temperature rises steadily upward near the ground level
- (b) Newton's law of cooling, an approximate form of Stefan's law, is valid only for natural convection
- (c) The total energy emitted by a black body per unit time per unit area is proportional to the square of its temperature in the Kelvin scale
- (d) Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The energy radiated per second by the first sphere is greater than that radiated per second by the second sphere
11. A body takes 4 minutes to cool from 100°C to 70°C . To cool from 70°C to 40°C it will take (room temperature is 15°C)
- (a) 7 minutes (b) 6 minutes
(c) 5 minutes (d) 4 minutes
12. A cup of tea cools from 80°C to 60°C in one minute. The ambient temperature is 30°C . In cooling from 60°C to 50°C it will take [MP PET 1990; MP PMT 1995; UPSEAT 2000; MH CET 2002]
- (a) 30 seconds (b) 60 seconds
(c) 90 seconds (d) 50 seconds
13. A liquid cools down from 70°C to 60°C in 5 minutes . The time taken to cool it from 60°C to 50°C will be [MP PET 1992, 2000; MP PMT 1996]
- (a) 5 minutes
(b) Lesser than 5 minutes
(c) Greater than 5 minutes
(d) Lesser or greater than 5 minutes depending upon the density of the liquid
14. If a metallic sphere gets cooled from 62°C to 50°C in 10 minutes and in the next 10 minutes gets cooled to 42°C , then the temperature of the surroundings is [MP PET 1997]
- (a) 30°C (b) 36°C
(c) 26°C (d) 20°C
15. The rates of cooling of two different liquids put in exactly similar calorimeters and kept in identical surroundings are the same if
- (a) The masses of the liquids are equal
(b) Equal masses of the liquids at the same temperature are taken
(c) Different volumes of the liquids at the same temperature are taken
(d) Equal volumes of the liquids at the same temperature are taken
16. A body cools from 60°C to 50°C in 10 minutes . If the room temperature is 25°C and assuming Newton's law of cooling to hold good, the temperature of the body at the end of the next 10 minutes will be [MP PMT 1994] [MP PMT/PET 1998; BHU 2000; Pb. PMT 2001]
- (a) 38.5°C (b) 40°C
(c) 42.85°C (d) 45°C
17. The temperature of a liquid drops from 365 K to 361 K in 2 minutes . Find the time during which temperature of the liquid drops from 344 K to 342 K . Temperature of room is 293 K
- (a) 84 sec (b) 72 sec

- (c) 66 sec (d) 60 sec
18. A body cools from 50.0°C to 49.9°C in 5 s. How long will it take to cool from 40.0°C to 39.9°C ? Assume the temperature of surroundings to be 30.0°C and Newton's law of cooling to be valid [CBSE PMT 1994]
- (a) 2.5 s (b) 10 s
(c) 20 s (d) 5 s
19. A container contains hot water at 100°C . If in time T_1 temperature falls to 80°C and in time T_2 temperature falls to 60°C from 80°C , then [CPMT 1997]
- (a) $T_1 = T_2$ (b) $T_1 > T_2$
(c) $T_1 < T_2$ (d) None
20. Hot water kept in a beaker placed in a room cools from 70°C to 60°C in 4 minutes. The time taken by it to cool from 69°C to 59°C will be [JIPMER 1999]
- (a) The same 4 minutes (b) More than 4 minutes
(c) Less than 4 minutes (d) We cannot say definitely
21. Newton's law of cooling, holds good only if the temperature difference between the body and the surroundings is [BHU 2000]
- (a) Less than 10°C (b) More than 10°C
(c) Less than 100°C (d) More than 100°C
22. In a room where the temperature is 30°C , a body cools from 61°C to 59°C in 4 minutes. The time (in min.) taken by the body to cool from 51°C to 49°C will be [UPSEAT 2000]
- (a) 4 min (b) 6 min
(c) 5 min (d) 8 min
23. According to 'Newton's Law of cooling', the rate of cooling of a body is proportional to the [MP PET 2001]
- (a) Temperature of the body
(b) Temperature of the surrounding
(c) Fourth power of the temperature of the body
(d) Difference of the temperature of the body and the surroundings
24. A body cools in 7 minutes from 60°C to 40°C . What time (in minutes) does it take to cool from 40°C to 28°C if the surrounding temperature is 10°C ? Assume Newton's Law of cooling holds [Kerala (Engg.) 2001]
- (a) 3.5 (b) 11
(c) 7 (d) 10
25. A body takes 5 minutes for cooling from 50°C to 40°C . Its temperature comes down to 33.33°C in next 5 minutes. Temperature of surroundings is [MP PMT 2002]
- (a) 15°C (b) 20°C
(c) 25°C (d) 10°C
26. The temperature of a body falls from 50°C to 40°C in 10 minutes. If the temperature of the surroundings is 20°C . Then temperature of the body after another 10 minutes will be
- (a) 36.6°C (b) 33.3°C
(c) 35°C (d) 30°C
27. It takes 10 minutes to cool a liquid from 61°C to 59°C . If room temperature is 30°C then time taken in cooling from 51°C to 49°C is
- (a) 10 min (b) 11 min
(c) 13 min (d) 15 min
28. A calorimeter of mass 0.2 kg and specific heat 900 J/kg-K . Containing 0.5 kg of a liquid of specific heat 2400 J/kg-K . Its temperature falls from 60°C to 55°C in one minute. The rate of cooling is [MP PET 2003]
- (a) 5 J/s (b) 15 J/s
(c) 100 J/s (d) 115 J/s
29. According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surroundings, and n is equal to
- (a) One (b) Two
(c) Three (d) Four
30. The initial temperature of a body is 80°C . If its temperature falls to 64°C in 5 minutes and in 10 minutes to 52°C then the temperature of surrounding will be [MP PMT 2003]
- (a) 26°C (b) 49°C
(c) 35°C (d) 42°C
31. A liquid cools from 50°C to 45°C in 5 minutes and from 45°C to 41.5°C in the next 5 minutes. The temperature of the surrounding is
- (a) 27°C (b) 40.3°C
(c) 23.3°C (d) 33.3°C
32. A cup of tea cools from 65.5°C to 62.5°C in one minute in a room of 22.5°C . How long will the same cup of tea take, in..... minutes, to cool from 46.50°C to 40.5°C in the same room? (choose nearest value) [Kerala PMT 2004]
- (a) 1 (b) 2
(c) 3 (d) 4
33. The temperature of a body falls from 62°C to 50°C in 10 minutes. If the temperature of the surroundings is 26°C , the temperature in next 10 minutes will become [RPMT 2002]
- (a) 42°C (b) 40°C
(c) 56°C (d) 55°C
34. A body takes 5 minutes to cool from 90°C to 60°C . If the temperature of the surroundings is 20°C , the time taken by it to cool from 60°C to 30°C will be. [RPMT 2003]
- (a) 5 min (b) 8 min
(c) 11 min (d) 12 min
35. An object is cooled from 75°C to 65°C in 2 minutes in a room at 30°C . The time taken to cool another object from 55°C to 45°C in the same room in minutes is [EAMCET (Med.) 1996]
- (a) 4 (b) 5

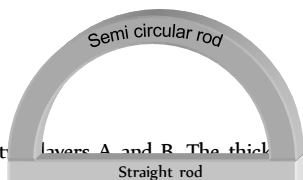
- (c) 6 (d) 7
36. A body takes 5 minute to cool from 80°C to 50°C . How much time it will take to cool from 60°C to 30°C , if room temperature is 20°C .
 (a) 40 minute (b) 9 minute
 (c) 30 minute (d) 20 minute
37. A cane is taken out from a refrigerator at 0°C . The atmospheric temperature is 25°C . If t_1 is the time taken to heat from 0°C to 5°C and t_2 is the time taken from 10°C to 15°C , then
 (a) $t_1 > t_2$ (b) $t_1 < t_2$
 (c) $t_1 = t_2$ (d) There is no relation

Critical Thinking

Objective Questions

1. Two rods (one semi-circular and other straight) of same material and of same cross-sectional area are joined as shown in the figure. The points A and B are maintained at different temperature. The ratio of the heat transferred through a cross-section of a semi-circular rod to the heat transferred through a cross section of the straight rod in a given time is [UPSEAT 2002]

- (a) $2 : \pi$
 (b) $1 : 2$
 (c) $\pi : 2$
 (d) $3 : 2$



2. A wall is made up of two layers A and B . The thickness of the two layers is the same. The thermal conductivity of A is double than that of B . In thermal equilibrium the temperature difference between the two ends is 36°C . Then the difference of temperature at the two surfaces of A will be [BHU 1997; MP PET 1996, 99; DPMT 2000]

- (a) 6°C (b) 12°C
 (c) 18°C (d) 24°C

3. Ice starts forming in lake with water at 0°C and when the atmospheric temperature is -10°C . If the time taken for 1 cm of ice be 7 hours , then the time taken for the thickness of ice to change from 1 cm to 2 cm is [NCERT 1971; MP PMT/PET 1988; UPSEAT 1996]

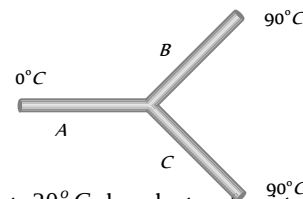
- (a) 7 hours (b) 14 hours
 (c) Less than 7 hours (d) More than 7 hours

4. A cylinder of radius R made of a material of thermal conductivity K_1 is surrounded by a cylindrical shell of inner radius R and outer radius $2R$ made of material of thermal conductivity K_2 . The two ends of the combined system are maintained at two different temperatures. There is no loss of heat across the cylindrical surface and the system is in steady state. The effective thermal conductivity of the system is

- (a) $K_1 + K_2$ (b) $\frac{K_1 K_2}{K_1 + K_2}$
 (c) $\frac{K_1 + 3K_2}{4}$ (d) $\frac{3K_1 + K_2}{4}$

5. Three rods made of the same material and having the same cross section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of the junction of the three rods will be [IIT-JEE (Screening) 2001]

- (a) 45°C
 (b) 60°C [Orissa JEE 2005]
 (c) 30°C
 (d) 20°C



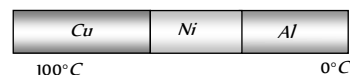
6. A room is maintained at 20°C by a heater of resistance 20 ohm connected to 200 volt mains. The temperature is uniform through out the room and heat is transmitted through a glass window of area 1 m^2 and thickness 0.2 cm . What will be the temperature outside? Given that thermal conductivity K for glass is $0.2\text{ cal/m}^{\circ}\text{C/sec}$ and $J = 4.2\text{ J/cal}$ [IIT 1978]

- (a) 15.24°C (b) 15.00°C
 (c) 24.15°C (d) None of the above

7. There is formation of layer of snow $x\text{ cm}$ thick on water, when the temperature of air is $-\theta^{\circ}\text{C}$ (less than freezing point). The thickness of layer increases from x to y in the time t , then the value of t is given by

- (a) $\frac{(x+y)(x-y)\rho L}{2k\theta}$ (b) $\frac{(x-y)\rho L}{2k\theta}$
 (c) $\frac{(x+y)(x-y)\rho L}{k\theta}$ (d) $\frac{(x-y)\rho L k}{2\theta}$ [IIT 1980; CPMT 1991]

8. A composite metal bar of uniform section is made up of length 25 cm of copper, 10 cm of nickel and 15 cm of aluminium. Each part being in perfect thermal contact with the adjoining part. The copper end of the composite rod is maintained at 100°C and the aluminium end at 0°C . The whole rod is covered with belt so that there is no heat loss occurs at the sides. If $K_{\text{Cu}} = 2K_{\text{Al}}$ and $K_{\text{Al}} = 3K_{\text{Ni}}$, then what will be the temperatures of $\text{Cu} - \text{Ni}$ and $\text{Ni} - \text{Al}$ junctions respectively



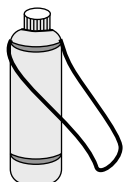
- (a) 23.33°C and 78.8°C (b) 83.33°C and 20°C
 (c) 50°C and 30°C (d) 30°C and 50°C

9. Three rods of identical area of cross-section and made from the same metal form the sides of an isosceles triangle ABC , right angled at B . The points A and C are maintained at temperatures T and $\sqrt{2}T$ respectively. In the steady state the temperature of the point C is T_C . Assuming that only heat conduction takes place, $\frac{T_C}{T}$ is equal to [IIT 1995]

- (a) $\frac{1}{(\sqrt{2}+1)}$ (b) $\frac{3}{(\sqrt{2}+1)}$
(c) $\frac{1}{2(\sqrt{2}-1)}$ (d) $\frac{1}{\sqrt{3}(\sqrt{2}-1)}$

10. The only possibility of heat flow in a thermos flask is through its cork which is 75 cm in area and 5 cm thick. Its thermal conductivity is 0.0075 cal/cmsec°C. The outside temperature is 40°C and latent heat of ice is 80 cal/g. Time taken by 500 g of ice at 0°C in the flask to melt into water at 0°C is [CPMT 1974, 78; MNR 1983]

- (a) 2.47 hr
(b) 4.27 hr
(c) 7.42 hr
(d) 4.72 hr



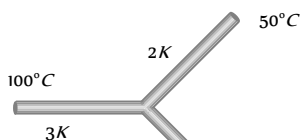
11. A sphere, a cube and a thin circular plate, all made of the same material and having the same mass are initially heated to a temperature of 1000°C. Which one of these will cool first

[J & K CET 2000; MH CET 2000; UPSEAT 2001]

- (a) Plate (b) Sphere
(c) Cube (d) None of these

12. Three rods of the same dimension have thermal conductivities 3K, 2K and K. They are arranged as shown in fig. Given below, with their ends at 100°C, 50°C and 20°C. The temperature of their junction is

- (a) 60°C
(b) 70°C
(c) 50°C
(d) 35°C



13. Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0°C. In the second case, the rods are joined end to end and connected to the same vessels. Let q_1 and q_2 g/s be the rate of melting of ice in two cases respectively. The ratio of q_1/q_2 is

[IIT-JEE (Screening) 2004]

- (a) $\frac{1}{2}$ (b) $\frac{2}{1}$
(c) $\frac{4}{1}$ (d) $\frac{1}{4}$

14. A solid cube and a solid sphere of the same material have equal surface area. Both are at the same temperature 120°C, then [MP PET 1992, 96; MP PMT 2000]

- (a) Both the cube and the sphere cool down at the same rate
(b) The cube cools down faster than the sphere
(c) The sphere cools down faster than the cube
(d) Whichever is having more mass will cool down faster

15. Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies emit total radiant power at the same rate. The wavelength λ_B corresponding to maximum spectral radiance in the radiation from B is shifted from the wavelength corresponding to

maximum spectral radiance in the radiation from A, by $1.00\mu\text{m}$. If the temperature of A is 5802 K

- (a) The temperature of B is 1934 K
(b) $\lambda_B = 1.5\mu\text{m}$
(c) The temperature of B is 11604 K
(d) The temperature of B is 2901 K

16. A black body is at a temperature of 2880 K. The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1 , between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3 . The Wein's constant $b = 2.88 \times 10^6 \text{ nm K}$. Then

[IIT 1998]

- (a) $U_1 = 0$ (b) $U_3 = 0$
(c) $U_1 > U_2$ (d) $U_2 > U_1$

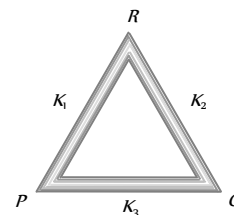
17. A black metal foil is warmed by radiation from a small sphere at temperature T and at a distance d. It is found that the power received by the foil is P. If both the temperature and the distance are doubled, the power received by the foil will be

- (a) 16P (b) 4P
[UPSEAT 2002] (c) 2P (d) P

18. Three rods of same dimensions are arranged as shown in figure they have thermal conductivities K_1, K_2 and K_3 . The points P and Q are maintained at different temperatures for the heat to flow at the same rate along PRQ and PQ then which of the following option is correct

[KCET 2001]

- (a) $K_3 = \frac{1}{2}(K_1 + K_2)$
(b) $K_3 = K_1 + K_2$
(c) $K_3 = \frac{K_1 K_2}{K_1 + K_2}$
(d) $K_3 = 2(K_1 + K_2)$



19. Two metallic spheres S_1 and S_2 are made of the same material and have identical surface finish. The mass of S_1 is three times that of S_2 . Both the spheres are heated to the same high temperature and placed in the same room having lower temperature but are thermally insulated from each other. The ratio of the initial rate of cooling of S_1 to that of S_2 is [IIT 1995]

- (a) 1/3 (b) $(1/3)^{1/3}$
(c) $1/\sqrt{3}$ (d) $\sqrt{3}/1$

20. Three discs A, B and C having radii 2m, 4m, and 6m respectively are coated with carbon black on their other surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm, respectively. The power radiated by them are Q , Q , and Q respectively

[IIT-JEE (Screening) 2004]

- (a) Q is maximum (b) Q is maximum
(c) Q is maximum (d) $Q = Q_1 = Q_2$

21. The total energy radiated from a black body source is collected for one minute and is used to heat a quantity of water. The temperature of water is found to increase from 20°C to 20.5°C . If the absolute temperature of the black body is doubled and the experiment is repeated with the same quantity of water at 20°C , the temperature of water will be [UPSEAT 2004]

- (a) 21°C (b) 22°C
(c) 24°C (d) 28°C

22. A solid sphere and a hollow sphere of the same material and size are heated to the same temperature and allowed to cool in the same surroundings. If the temperature difference between each sphere and its surroundings is T , then [Manipal MEE 1995]

- (a) The hollow sphere will cool at a faster rate for all values of T
(b) The solid sphere will cool at a faster rate for all values of T
(c) Both spheres will cool at the same rate for all values of T
(d) Both spheres will cool at the same rate only for small values of T

23. A solid copper cube of edges 1 cm is suspended in an evacuated enclosure. Its temperature is found to fall from 100°C to 99°C in 100 s . Another solid copper cube of edges 2 cm , with similar surface nature, is suspended in a similar manner. The time required for this cube to cool from 100°C to 99°C will be approximately [MP PMT 1997]

- (a) 25 s (b) 50 s
(c) 200 s (d) 400 s

24. A body initially at 80°C cools to 64°C in 5 minutes and to 52°C in 10 minutes . The temperature of the body after 15 minutes will be [UPSEAT 2000; Pb. PET 2004]

- (a) 42.7°C (b) 35°C
(c) 47°C (d) 40°C

25. A 5 cm thick ice block is there on the surface of water in a lake. The temperature of air is -10°C ; how much time it will take to double the thickness of the block

$$(L = 80\text{ cal/g}, K_{\text{ice}} = 0.004\text{ Erg/s-k}, d_{\text{ice}} = 0.92\text{ g cm}^{-3})$$

[RPET 1998]

- (a) 1 hour (b) 191 hours
(c) 19.1 hours (d) 1.91 hours

26. Four identical rods of same material are joined end to end to form a square. If the temperature difference between the ends of a diagonal is 100°C , then the temperature difference between the ends of other diagonal will be [MP PET 1989; RPMT 2002]

- (a) 0°C
(b) $\frac{100}{l}^\circ\text{C}$; where l is the length of each rod
(c) $\frac{100}{2l}^\circ\text{C}$
(d) 100°C

27. A cylindrical rod with one end in a steam chamber and the other end in ice results in melting of 0.1 gm of ice per second. If the rod is replaced by another with half the length and double the radius of

the first and if the thermal conductivity of material of second rod is

 $\frac{1}{4}$ that of first, the rate at which ice melts in gm/sec will be [EAMCET 1987]

- (a) 3.2 (b) 1.6
(c) 0.2 (d) 0.1

28. One end of a copper rod of length 1.0 m and area of cross-section 10^{-3} is immersed in boiling water and the other end in ice. If the coefficient of thermal conductivity of copper is $92\text{ cal/m-s-}^\circ\text{C}$ and the latent heat of ice is $8 \times 10^4\text{ cal/kg}$, then the amount of ice which will melt in one minute is [MNR 1994]

- (a) $9.2 \times 10^{-3}\text{ kg}$ (b) $8 \times 10^{-3}\text{ kg}$
(c) $6.9 \times 10^{-3}\text{ kg}$ (d) $5.4 \times 10^{-3}\text{ kg}$

29. An ice box used for keeping eatable cold has a total wall area of 1 metre^2 and a wall thickness of 5.0 cm . The thermal conductivity of the ice box is $K = 0.01\text{ joule/metre-}^\circ\text{C}$. It is filled with ice at 0°C along with eatables on a day when the temperature is 30°C . The latent heat of fusion of ice is $334 \times 10^3\text{ joules/kg}$. The amount of ice melted in one day is

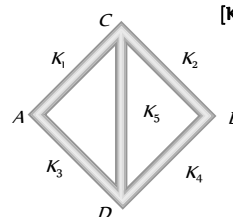
$$(1\text{ day} = 86,400\text{ seconds})$$

[MP PMT 1995]

- (a) 776 gms (b) 7760 gms
(c) 11520 gms (d) 1552 gms

30. Five rods of same dimensions are arranged as shown in the figure. They have thermal conductivities K_1, K_2, K_3, K_4 and K_5 . When points A and B are maintained at different temperatures, no heat flows through the central rod if

- (a) $K_1 = K_4$ and $K_2 = K_3$
(b) $K_1 K_4 = K_2 K_3$
(c) $K_1 K_2 = K_3 K_4$
(d) $\frac{K_1}{K_4} = \frac{K_2}{K_3}$



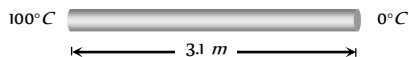
[KCET 2002]

31. A hot metallic sphere of radius r radiates heat. Its rate of cooling is
(a) Independent of r (b) Proportional to r
(c) Proportional to r^2 (d) Proportional to $1/r$

32. A solid copper sphere (density ρ and specific heat capacity c) of radius r at an initial temperature 200 K is suspended inside a chamber whose walls are at almost 0 K . The time required (in $\mu\text{ s}$) for the temperature of the sphere to drop to 100 K is

- (a) $\frac{72}{7} \frac{r\rho c}{\sigma}$ (b) $\frac{7}{72} \frac{r\rho c}{\sigma}$
(c) $\frac{27}{7} \frac{r\rho c}{\sigma}$ (d) $\frac{7}{27} \frac{r\rho c}{\sigma}$

33. One end of a copper rod of uniform cross-section and of length 3.1 m is kept in contact with ice and the other end with water at 100°C . At what point along its length should a temperature of 200°C be maintained so that in steady state, the mass of ice melting be equal to that of the steam produced in the same interval of time. Assume that the whole system is insulated from the surroundings. Latent heat of fusion of ice and vaporisation of water are 80 cal/gm and 540 cal/gm respectively



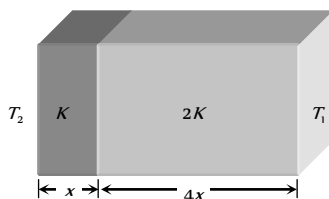
- (a) 40 cm from 100°C end (b) 40 cm from 0°C end
(c) 125 cm from 100°C end (d) 125 cm from 0°C end

34. A sphere and a cube of same material and same volume are heated upto same temperature and allowed to cool in the same surroundings. The ratio of the amounts of radiations emitted will be

- (a) 1 : 1 (b) $\frac{4\pi}{3} : 1$
(c) $\left(\frac{\pi}{6}\right)^{1/3} : 1$ (d) $\frac{1}{2}\left(\frac{4\pi}{3}\right)^{2/3} : 1$

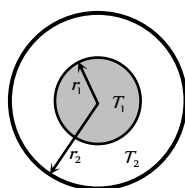
35. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)f$, with f which equal to [AIEEE 2004]

- (a) 1
(b) $\frac{1}{2}$
(c) $\frac{2}{3}$
(d) $\frac{1}{3}$



36. The figure shows a system of two concentric spheres of radii r_1 and r_2 and kept at temperatures T_1 and T_2 respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to [AIEEE 2005]

- (a) $\frac{r_1 r_2}{(r_1 - r_2)}$
(b) $(r_2 - r_1)$
(c) $(r_2 - r_1)(r_1 r_2)$
(d) $\ln\left(\frac{r_2}{r_1}\right)$



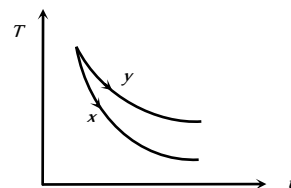
37. Four rods of identical cross-sectional area and made from the same metal form the sides of square. The temperature of two diagonally opposite points are T and $\sqrt{2}T$ respectively in the steady state. Assuming that only heat conduction takes place, what will be the temperature difference between other two points

- (a) $\frac{\sqrt{2}+1}{2}T$ (b) $\frac{2}{\sqrt{2}+1}T$
(c) 0 (d) None of these

1. The graph, shown in the adjacent diagram, represents the variation of temperature (T) of two bodies, x and y having same surface area, with time (t) due to the emission of radiation. Find the correct relation between the emissivity (e) and absorptivity (a) of the two bodies

[IIT-JEE (Screening) 2003]

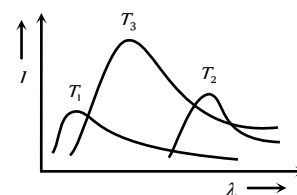
- (a) $e_x > e_y$ & $a_x < a_y$
(b) $e_x < e_y$ & $a_x > a_y$
(c) $e_x > e_y$ & $a_x > a_y$
(d) $e_x < e_y$ & $a_x < a_y$



2. The plots of intensity versus wavelength for three black bodies at temperatures T_1 , T_2 and T_3 respectively are as shown. Their temperature are such that

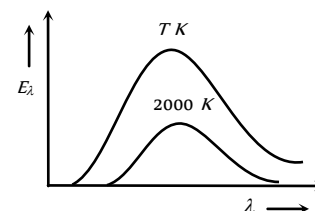
[IIT-JEE (Screening) 2000]

- (a) $T_1 > T_2 > T_3$
(b) $T_1 > T_3 > T_2$
(c) $T_2 > T_1 > T_3$
(d) $T_2 > T_3 > T_1$



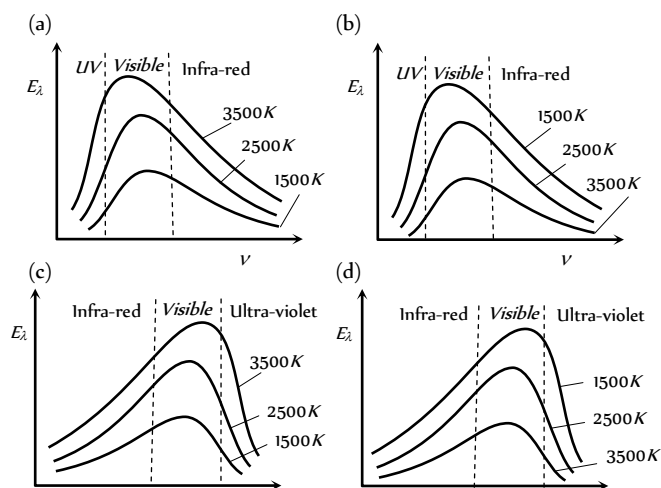
3. The adjoining diagram shows the spectral energy density distribution E_λ of a black body at two different temperatures. If the areas under the curves are in the ratio 16 : 1, the value of temperature T is [DCE 1999]

- (a) 32,000 K
(b) 16,000 K
(c) 8,000 K
(d) 4,000 K

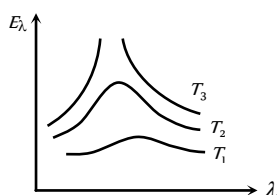


[BCECE 2005]

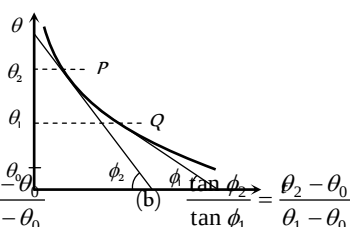
4. Following graph shows the correct variation in intensity of heat radiations by black body and frequency at a fixed temperature



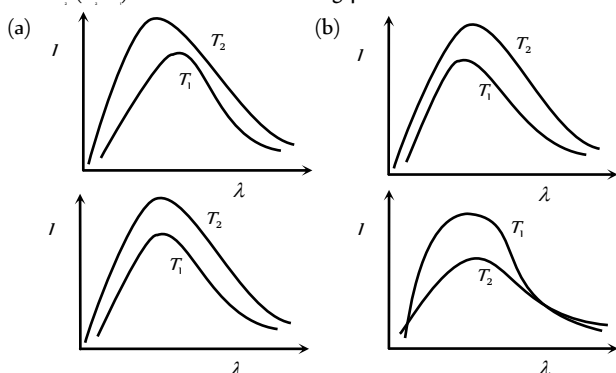
5. Variation of radiant energy emitted by sun, filament of tungsten lamp and welding arc as a function of its wavelength is shown in figure. Which of the following option is the correct match



- (a) Sun- T_1 , tungsten filament - T_2 , welding arc - T_3
 (b) Sun - T_2 , tungsten filament - T_1 , welding arc - T_3
 (c) Sun - T_3 , tungsten filament - T_2 , welding arc - T_1
 (d) Sun - T_1 , tungsten filament - T_3 , welding arc - T_2
6. A body cools in a surrounding which is at a constant temperature of θ_0 . Assume that it obeys Newton's law of cooling. Its temperature θ is plotted against time t . Tangents are drawn to the curve at the points $P(\theta = \theta_1)$ and $Q(\theta = \theta_2)$. These tangents meet the time axis at angles of ϕ_2 and ϕ_1 , as shown

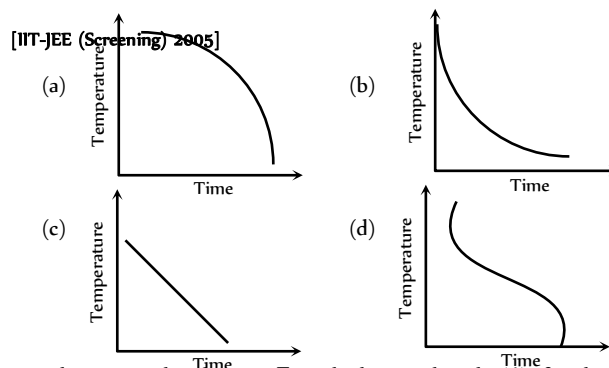


- (a) $\frac{\tan \phi_2}{\tan \phi_1} = \frac{\theta_1 - \theta_0}{\theta_2 - \theta_0}$ (b) $\frac{\tan \phi_1}{\tan \phi_2} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$
 (c) $\frac{\tan \phi_1}{\tan \phi_2} = \frac{\theta_1}{\theta_2}$ (d) $\frac{\tan \phi_1}{\tan \phi_2} = \frac{\theta_2}{\theta_1}$
7. Shown below are the black body radiation curves at temperatures T_1 and T_2 ($T_2 > T_1$). Which of the following plots is correct

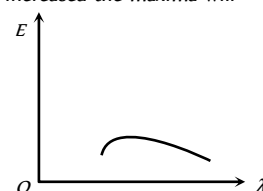


- (c) (d)

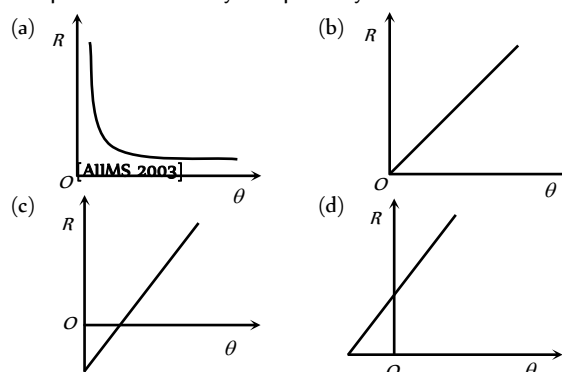
8. The spectrum of a black body at two temperatures 27°C and 327°C is shown in the figure. Let A_1 and A_2 be the areas under the two curves respectively. The value of $\frac{A_2}{A_1}$ is
- (a) 1 : 16
 (b) 4 : 1
 (c) 2 : 1
 (d) 16 : 1
9. A block of metal is heated to a temperature much higher than the room temperature and allowed to cool in a room free from air currents. Which of the following curves correctly represents the rate of cooling [Manipal MEE 1995]



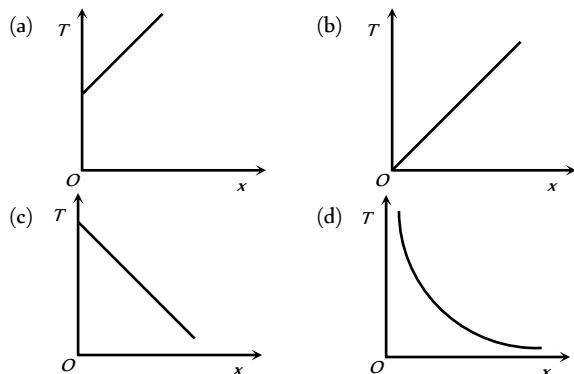
10. The energy distribution E with the wavelength (λ) for the black body radiation at temperature T Kelvin is shown in the figure. As the temperature is increased the maxima will



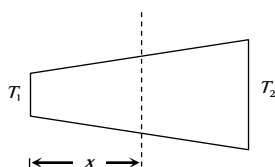
- (a) Shift towards left and become higher
 (b) Rise high but will not shift
 (c) Shift towards right and become higher
 (d) Shift towards left and the curve will become broader
11. For a small temperature difference between the body and the surroundings the relation between the rate of loss heat R and the temperature of the body is depicted by



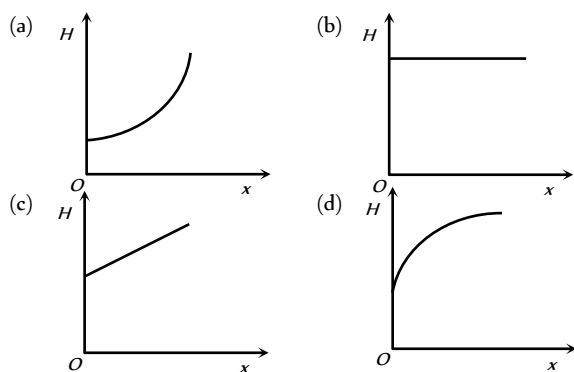
12. Heat is flowing through a conductor of length l from $x = 0$ to $x = l$. If its thermal resistance per unit length is uniform, which of the following graphs is correct



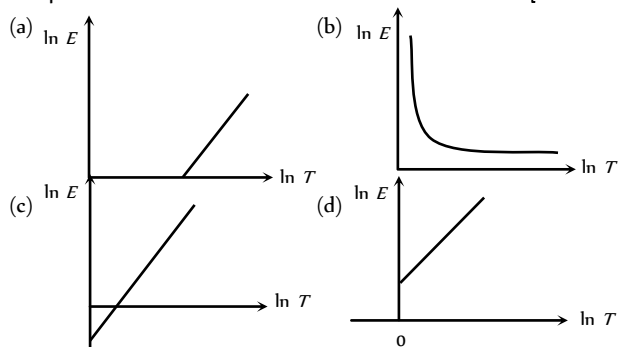
13. Radius of a conductor increases uniformly from left end to right end as shown in fig.



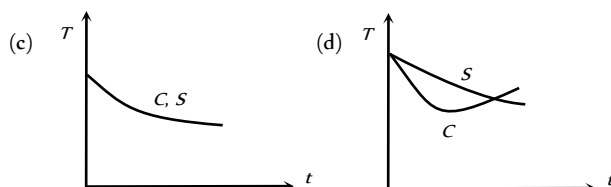
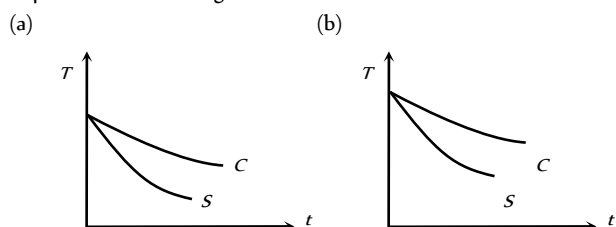
Material of the conductor is isotropic and its curved surface is thermally isolated from surrounding. Its ends are maintained at temperatures T_1 and T_2 ($T_1 > T_2$). If, in steady state, heat flow rate is equal to H , then which of the following graphs is correct



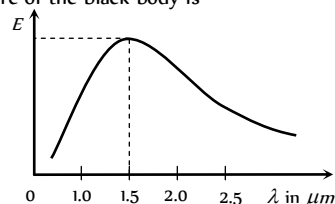
14. Which of the following graphs correctly represents the relation between $\ln E$ and $\ln T$ where E is the amount of radiation emitted per unit time from unit area of a body and T is the absolute temperature [DCE 2002]



15. A hollow copper sphere S and a hollow copper cube C , both of negligible thin walls of same area, are filled with water at 90°C and allowed to cool in the same environment. The graph that correctly represents their cooling is



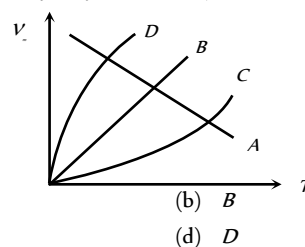
16. In the figure, the distribution of energy density of the radiation emitted by a black body at a given temperature is shown. The possible temperature of the black body is



- (a) 1500 K (b) 2000 K
(c) 2500 K (d) 3000 K

17. Which of the following is the $\nu_m = T$ graph for a perfectly black body (ν_m = maximum frequency of radiation)

[RPMT 1996]



- (a) A
(c) C

- (b) B
(d) D

A Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : A body that is a good radiator is also a good absorber of radiation at a given wavelength.
 Reason : According to Kirchoff's law the absorptivity of a body is equal to its emissivity at a given wavelength. [AIIMS 2005]
2. Assertion : For higher temperature, the peak emission wavelength of a black body shifts to lower wavelengths.
 Reason : Peak emission wavelength of a blackbody is proportional to the fourth power of temperature. [AIIMS 2005]
3. Assertion : Temperatures near the sea coast are moderate.
 Reason : Water has a high thermal conductivity. [AIIMS 2003]
4. Assertion : It is hotter over the top of a fire than at the same distance on the sides.
 Reason : Air surrounding the fire conducts more heat upwards. [AIIMS 2003]
5. Assertion : Bodies radiate heat at all temperatures.
 Reason : Rate of radiation of heat is proportional to the fourth power of absolute temperature. [AIIMS 1999, 2002]
6. Assertion : Woolen clothes keep the body warm in winter.
 Reason : Air is a bad conductor of heat. [AIIMS 2002]
7. Assertion : The equivalent thermal conductivity of two plates of same thickness in contact (series) is less than the smaller value of thermal conductivity.
 Reason : For two plates of equal thickness in contact (series) the equivalent thermal conductivity is given by

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$
 [AIIMS 1997]
8. Assertion : A hollow metallic closed container maintained at a uniform temperature can act as a source of black body radiation.
 Reason : All metals acts as a black body [AIIMS 1996]
9. Assertion : If the temperature of a star is doubled then the rate of loss of heat from it becomes 16 times.
 Reason : Specific heat varies with temperature. [AIIMS 1996]
10. Assertion : The radiation from the sun's surface varies as the fourth power of its absolute temperature.
 Reason : The sun is not a black body. [AIIMS 1999]
11. Assertion : Blue star is at high temperature than red star.
 Reason : Wein's displacement law states that $T \propto (1/\lambda_m)$. [AIIMS 2002]
12. Assertion : The S.I. unit of thermal conductivity is watt $m K$.
 Reason : Thermal conductivity is a measure of ability of the material to allow the passage of heat through it.
13. Assertion : A brass tumbler feels much colder than a wooden tray on a chilly day.
 Reason : The thermal conductivity of brass is less than that of wood.
14. Assertion : Like light radiations, thermal radiations are also electromagnetic radiation.
 Reason : The thermal radiations require no medium for propagation.
15. Assertion : Snow is better insulator than ice.
 Reason : Snow contain air packet and air is good insulator of heat.
16. Assertion : Water can be boiled inside satellite by convection.
 Reason : Convection is the process in which heat is transmitted from a place of higher temperature to a place of lower temperature by means of particles with their migrations from one place to another.
17. Assertion : The absorbance of a perfect black body is unity.
 Reason : A perfect black body when heated emits radiations of all possible wavelengths at that temperature.
18. Assertion : A man would feel iron or wooden balls equally hot at $98.4^\circ F$.
 Reason : At $98.4^\circ F$ both iron and wood have same thermal conductivity.
19. Assertion : As temperature of a black body is raised, wavelength corresponding to maximum energy reduces.
 Reason : Higher temperature would mean higher energy and hence higher wavelength.
20. Assertion : All black coloured objects are considered black bodies.
 Reason : Black colour is a good absorber of heat.
21. Assertion : Greater is the coefficient of thermal conductivity of a material, smaller is the thermal resistance of a rod of that material.
 Reason : Thermal resistance is the ratio of temperature difference between the ends of the conductor and rate of flow of heat.
22. Assertion : Radiation is the speediest mode of heat transfer.
 Reason : Radiation can be transmitted in zig-zag motion.
23. Assertion : Two thin blankets put together are warmer than a single blanket of double the thickness.

Reason : Thickness increases because of air layer enclosed between the two blankets.

24. Assertion : Animals curl into a ball, when they feel very cold.

Reason : Animals by curling their body reduces the surface area.

Answers

Conduction

1	a	2	d	3	d	4	a	5	d
6	d	7	d	8	d	9	c	10	d
11	c	12	a	13	a	14	b	15	d
16	b	17	c	18	c	19	a	20	a
21	a	22	a	23	b	24	d	25	a
26	b	27	b	28	d	29	d	30	b
31	b	32	c	33	b	34	b	35	a
36	d	37	d	38	b	39	a	40	c
41	b	42	c	43	c	44	b	45	c
46	a	47	c	48	b	49	b	50	a
51	c	52	b	53	b	54	c	55	a
56	a	57	a	58	c	59	c	60	a
61	c	62	d	63	b	64	b	65	b
66	c	67	b	68	b				

Convection

1	c	2	a	3	c	4	a	5	b
6	a	7	c	8	b	9	a	10	d
11	a	12	c	13	c				

Radiation (General, Kirchoff's law, Black body)

1	b	2	a	3	c	4	d	5	d
6	c	7	d	8	d	9	c	10	b
11	b	12	a	13	c	14	a	15	d
16	b	17	a	18	b	19	b	20	c
21	b	22	c	23	c	24	a	25	b
26	b	27	a	28	c	29	a	30	d
31	d	32	c	33	a	34	c	35	a
36	d	37	c	38	b				

Radiation (Wein's law)

1	a	2	c	3	d	4	c	5	a
6	c	7	d	8	c	9	c	10	b
11	a	12	b	13	b	14	c	15	b
16	a	17	b	18	a	19	b	20	a

21	c	22	a	23	b	24	b	25	c
26	b	27	b	28	b	29	a	30	b

Radiation (Stefan's law)

1	c	2	c	3	a	4	a	5	d
6	a	7	c	8	b	9	b	10	b
11	b	12	c	13	c	14	c	15	c
16	b	17	d	18	d	19	c	20	d
21	d	22	d	23	a	24	d	25	d
26	a	27	a	28	b	29	a	30	a
31	d	32	d	33	d	34	c	35	a
36	c	37	c	38	c	39	a	40	c
41	b	42	d	43	c	44	d	45	a
46	b	47	a	48	a	49	a	50	a
51	a	52	d	53	b	54	b	55	b
56	c								

Radiation (Newton's Law of Cooling)

1	b	2	c	3	a	4	a	5	b
6	c	7	c	8	c	9	d	10	b
11	b	12	d	13	c	14	c	15	d
16	c	17	a	18	b	19	c	20	b
21	a	22	b	23	d	24	c	25	b
26	b	27	d	28	d	29	a	30	b
31	d	32	d	33	a	34	c	35	a
36	b	37	b						

Critical Thinking Questions

1	a	2	b	3	d	4	c	5	b
6	a	7	a	8	b	9	b	10	a
11	a	12	b	13	c	14	b	15	ab
16	d	17	b	18	c	19	b	20	b
21	d	22	a	23	c	24	a	25	c
26	a	27	c	28	c	29	d	30	b
31	d	32	b	33	a	34	c	35	d
36	a	37	c						

Graphical Questions

1	c	2	b	3	d	4	c	5	c
6	b	7	a	8	d	9	b	10	a
11	c	12	c	13	b	14	d	15	c
16	b	17	b						

Assertion & Reason

1	a	2	c	3	b	4	c	5	e
6	a	7	d	8	c	9	b	10	c

11	a	12	b	13	c	14	b	15	a
16	e	17	b	18	c	19	c	20	e
21	b	22	c	23	c	24	a		

AS Answers and Solutions

Conduction

1. (a) *Cu* is better conductor than *Al* and *Ag* is better conductor than *Cu*. Hence conductivity in increasing order is $Al < Cu < Ag$.

2. (d) $\frac{Q}{t} = \frac{KA \Delta \theta}{l} \Rightarrow \frac{Q}{t} \propto \frac{A}{l} \propto \frac{r^2}{l}$
 $\therefore \frac{r^2}{l}$ is maximum in option (d), hence it will conduct more heat.

3. (d) $\frac{Q}{t} = \frac{KA \Delta \theta}{l} \Rightarrow \frac{Q}{t} \propto \frac{A}{l} \propto \frac{d^2}{l}$ (d = Diameter of rod)
 $\Rightarrow \frac{(Q/t)_1}{(Q/t)_2} = \left(\frac{d_1}{d_2}\right)^2 \times \frac{l_2}{l_1} = \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right) = \frac{1}{8}$

4. (a) $\frac{Q}{t} = \frac{KA \Delta \theta}{l} = \frac{\Delta \theta}{(l / KA)} = \frac{\Delta \theta}{R}$ (R = Thermal resistance)
 $\Rightarrow t \propto R$ ($\because Q$ and $\Delta \theta$ are same)
 $\Rightarrow \frac{t_p}{t_s} = \frac{R_p}{R_s} = \frac{R/2}{2R} = \frac{1}{4} \Rightarrow t_p = \frac{t_s}{4} = \frac{4}{4} = 1 \text{ min.}$

(Series resistance $R_s = R_1 + R_2$ and parallel resistance

$$R_p = \frac{R_1 R_2}{R_1 + R_2})$$

5. (d) For cooking utensils, low specific heat is preferred for it's material as it should need less heat to raise it's temperature and it should have high conductivity, because, it should transfer heat quickly.
6. (d) In steady state there is no absorption of heat in any position. Heat passes on or is radiated from it's surface. Therefore, in steady state the temperature of the body does not change with time but can be different at different points of the body.
7. (d) It is the property of material.
8. (d) Because steady state has been reached.
9. (c) $\frac{Q_1}{t} = \frac{KA(90-60)}{0.6} = 50 \text{ KA}$
and $\frac{Q_2}{t} = \frac{KA(150-110)}{0.8} = 50 \text{ KA}$

10. (d) Given $A_1 = A_2$ and $\frac{K_1}{K_2} = \frac{5}{4}$

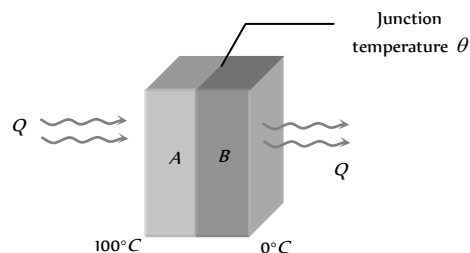
$$\therefore R_1 = R_2 \Rightarrow \frac{l_1}{K_1 A} = \frac{l_2}{K_2 A} \Rightarrow \frac{l_1}{l_2} = \frac{K_1}{K_2} = \frac{5}{4}.$$

11. (c) $\frac{\Delta Q}{\Delta t} = \frac{KA \Delta \theta}{\Delta x} \Rightarrow \text{Thermal gradient } \frac{\Delta \theta}{\Delta x}$
 $= \frac{(\Delta Q / \Delta t)}{KA} = \frac{10}{0.4} = 25^\circ \text{C} / \text{cm}$

12. (a) It is given that $\frac{K_1}{K_2} = \frac{1}{3} \Rightarrow K_1 = K$ then $K_2 = 3K$

the temperature of the junction in contact

$$\theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2} = \frac{1 \times 100 + 3 \times 0}{1 + 3} = \frac{100}{4} = 25^\circ \text{C}$$



13. (a) $Q = \frac{KA(\theta_1 - \theta_2)t}{l}$; in both the cases, A , l and $(\theta_1 - \theta_2)$ are same so $Kt = \text{constant} \Rightarrow \frac{K_1}{K_2} = \frac{t_1}{t_2} = \frac{30}{20} = \frac{3}{2} = 1.5.$

14. (b) $\left(\frac{Q}{t}\right)_1 = \frac{K_1 A_1 (\theta_1 - \theta_2)}{l}$ and $\left(\frac{Q}{t}\right)_2 = \frac{K_2 A_2 (\theta_1 - \theta_2)}{l}$

given $\left(\frac{Q}{t}\right)_1 = \left(\frac{Q}{t}\right)_2 \Rightarrow K_1 A_1 = K_2 A_2$

15. (d) In variable state $\frac{Q}{t} \propto K$ and $\frac{Q}{t} \propto \frac{1}{\rho c} \Rightarrow \frac{Q}{t} \propto \frac{K}{\rho c}$

(K = thermal conductivity, ρ = density, c = specific heat)

16. (b) $K_1 : K_2 = l_1^2 : l_2^2 \Rightarrow \frac{l_1}{l_2} = \sqrt{\frac{K_1}{K_2}} = \sqrt{\frac{10}{9}} = \frac{\sqrt{10}}{3}$

17. (c) $\frac{Q}{t} = \frac{KA(\Delta\theta)}{l} \Rightarrow 50 = \frac{5 \times 20 K}{0.4} \Rightarrow K = \frac{1}{5} = 0.2$

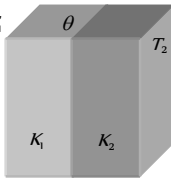
18. (c)

19. (a) Thermal resistance

$$= \frac{l}{KA} = \left[\frac{L}{MLT^{-3}K^{-1} \times L^2} \right] = [M^{-1}L^{-2}T^3K]$$

20. (a) When a piece of glass is heated, due to low thermal conductivity it does not conduct heat fast. Hence unequal expansion of its layers crack the glass.

21. (a) In series both walls have same rate of heat flow. Therefore

$$\frac{dQ}{dt} = \frac{K_1 A (T_1 - \theta)}{d_1} = \frac{K_2 A (\theta - T_2)}{d_2}$$


$$\Rightarrow K_1 d_2 (T_1 - \theta) = K_2 d_1 (\theta - T_2)$$

$$\Rightarrow \theta = \frac{K_1 d_2 T_1 + K_2 d_1 T_2}{K_1 d_2 + K_2 d_1}$$

22. (a) Temperature of interface $\theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$

($\because \frac{K_1}{K_2} = \frac{1}{4} \Rightarrow$ If $K = K$ then $K_1 = 4K$)

$$\Rightarrow \theta = \frac{K \times 0 + 4K \times 100}{5K} = 80^\circ C$$

23. (b) $\frac{\theta_1 - \theta_2}{l} = 80 \Rightarrow \frac{30 - \theta_2}{0.5} = 80 \Rightarrow \theta_2 = -10^\circ C$

24. (d) $\frac{dQ}{dt} = -KA \frac{d\theta}{dx}$; when $K = \infty$, $\frac{d\theta}{dx} = 0$

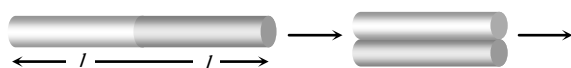
i.e. θ is independent of x i.e. constant or uniform.

25. (a) Air is poor conductor of heat.

26. (b)

27. (b)

28. (d) Let the heat transferred be Q .



When rods are joined end to end. Heat transferred by each

rod $= Q = \frac{KA\Delta\theta}{l} \times 12$ (i)

When rods are joined lengthwise, $Q = \frac{KA\Delta\theta}{2l} t$ (ii)

From equation (i) and (ii) we get $t = 48 s$

29. (d) $\frac{Q}{t} = \frac{KA\Delta\theta}{l} \Rightarrow \frac{K_A}{K_B} = \frac{A_B}{A_A} = \left(\frac{r_B}{r_A}\right)^2 = \frac{1}{4} \Rightarrow K_A = \frac{K_B}{4}$

30. (b) Thermal conductivity of composite plate

$$K_{eq} = \frac{2K_1 K_2}{K_1 + K_2} = \frac{2 \times 2 \times 3}{2 + 3} = \frac{12}{5} = 2.4$$

31. (b) $Q \propto \frac{A}{l} \propto \frac{r^2}{l} \Rightarrow \frac{Q_2}{Q_1} = \frac{r_2^2}{r_1^2} \times \frac{l_1}{l_2}$

$$\Rightarrow \frac{Q_2}{Q_1} = \frac{4}{1} \times \frac{1}{2} \Rightarrow Q_2 = 2Q_1$$

32. (c) $\frac{Q}{At} = K \frac{\Delta\theta}{l} \Rightarrow K \frac{\Delta\theta}{l} = \text{constant} \Rightarrow \frac{\Delta\theta}{l} \propto \frac{1}{K}$

Hence If $K_c > K_m > K_g$, then

$$\left(\frac{\Delta\theta}{l}\right)_c < \left(\frac{\Delta\theta}{l}\right)_m < \left(\frac{\Delta\theta}{l}\right)_g \Rightarrow X_c < X_m < X_g$$

because higher K implies lower value of the temperature gradient.

33. (b) In series $R_{eq} = R_1 + R_2 \Rightarrow \frac{2l}{K_{eq}A} = \frac{l}{K_1A} + \frac{l}{K_2A}$

$$\Rightarrow \frac{2}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2} \Rightarrow K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

34. (b) $\frac{dQ}{dt} = KA \frac{d\theta}{dl} \Rightarrow \frac{dQ}{dt} \propto \frac{d\theta}{dl}$ (Temperature gradient)

35. (a) $\frac{dQ}{dt} = \frac{K(\pi r^2)d\theta}{dl} \Rightarrow \left(\frac{dQ}{dt}\right)_s = \frac{K_s \times r_s^2 \times l_l}{K_l \times r_l^2 \times l_s} = \frac{1}{2} \times \frac{1}{4} \times \frac{2}{1}$

$$\Rightarrow \left(\frac{dQ}{dt}\right)_s = \left(\frac{dQ}{dt}\right)_l = \frac{4}{4} = 1$$

36. (d) $Q = \frac{KA(\Delta\theta)t}{l}$

$\because Q$ and $\Delta\theta$ are same for both spheres hence

$$K \propto \frac{l}{At} \propto \frac{l}{r^2 t} \Rightarrow \frac{K_{\text{larger}}}{K_{\text{smaller}}} = \frac{l_l}{l_s} \times \left(\frac{r_s}{r_l}\right)^2 \times \frac{t_s}{t_l}. \text{ It is given}$$

that $r_l = 2r_s$, $l_l = \frac{1}{4}l_s$ and $t_l = 25 \text{ min}$, $t_s = 16 \text{ min}$.

$$\Rightarrow \frac{K_{\text{larger}}}{K_{\text{smaller}}} = \left(\frac{1}{4}\right) \left(\frac{1}{2}\right)^2 \times \frac{16}{25} = \frac{1}{25}$$

37. (d) $\frac{Q}{t} = \frac{KA(\Delta\theta)}{l} \Rightarrow \frac{Q}{t} \propto \frac{A}{l} \propto \frac{r^2}{l}$

$$\Rightarrow \frac{(Q/t)_1}{(Q/t)_2} = \left(\frac{r_1}{r_2}\right)^2 \times \frac{l_2}{l_1} = \left(\frac{2}{1}\right)^2 \times \left(\frac{4}{1}\right) = \frac{16}{1}$$

38. (b) Temperature of interface $\theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$

where $K = 2K$ and $K = 3K$ ($\because \frac{K_1}{K_2} = \frac{2}{3}$)

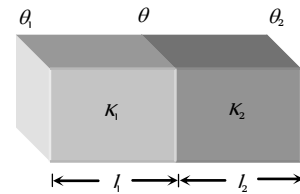
$$\Rightarrow \theta = \frac{2K \times 100 + 3K \times 0}{2K + 3K} = \frac{200K}{5K} = 40^\circ C$$

39. (a) $\frac{K_1}{K_2} = \frac{l_1^2}{l_2^2} \therefore K_2 = \frac{K_1 l_1^2}{l_2^2} = \frac{0.92 \times (4.2)^2}{(8.4)^2} = 0.23$
40. (c) Mud is bad conductor of heat. So it prevents the flow of heat between surroundings and inside.
41. (b) Temperature gradient $= \frac{100 - 20}{20} = 4^\circ\text{C}/\text{cm}$
temperature at centre $= 100 - 4 \times 10 = 60^\circ\text{C}$
42. (c) Temperature of interface
$$\theta = \frac{K_1 \theta_1 l_2 + K_2 \theta_2 l_1}{K_1 l_2 + K_2 l_1} = \frac{K \times 0 \times 2 + 3K \times 100 \times 1}{K \times 2 + 3K \times 1}$$
$$= \frac{300K}{5K} = 60^\circ\text{C}$$
43. (c) $\Delta\theta = \frac{Q \times l}{KA t} = \frac{4000 \times 0.1}{400 \times 10^{-2}} = 100^\circ\text{C}$
44. (b) Heat passes quickly from the body into the metal which leads to a cold feeling.
45. (c) Heat energy always flow from higher temperature to lower temperature. Hence, temperature difference *w.r.t.* length (temperature gradient) is required to flow heat from one part of a solid to other part.
46. (a) When the temperature of an object is equal to that of human body, no heat is transferred from the object to body and vice versa, Therefore block of wood and block of metal feel equally cold and hot if they have same temperature as human body.
47. (c)
48. (b) Temperature of water just below the lower surface of ice layer is 0°C .
49. (b) $\frac{Q}{t} = \frac{KA(\theta_1 - \theta_2)}{l} = \frac{100 \times 100 \times 10^{-4}(100 - 0)}{1}$
 $\Rightarrow \frac{Q}{t} = 100 \text{ Joule/sec} = 6 \times 10^3 \text{ Joule/min}$
50. (a) Temperature of interface $\theta = \frac{K_1 \theta_1 l_2 + K_2 \theta_2 l_1}{K_1 l_2 + K_2 l_1}$
It is given that $K_{Cu} = 9K_S$. So if $K_S = K_1 = K$ then $K_{Cu} = K_2 = 9K$
 $\Rightarrow \theta = \frac{9K \times 100 \times 6 + K \times 0 \times 18}{9K \times 6 + K \times 18} = \frac{5400K}{72K} = 75^\circ\text{C}$
51. (c) $\frac{Q}{t} = \frac{KA(\theta_1 - \theta_2)}{l} \Rightarrow \frac{Q}{t} \propto \frac{A}{l} \propto \frac{r^2}{l}$
[As $(\theta_1 - \theta_2)$ and K are constants]
$$\Rightarrow \left(\frac{Q}{t}\right)_1 = \frac{r_1^2}{r_2^2} \times \frac{l_2}{l_1} = \frac{4}{9} \times \frac{2}{1} = \frac{8}{9}$$
52. (b) In parallel combination equivalent conductivity
$$K = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2} = \frac{K_1 + K_2}{2} \quad (\text{As } A_1 = A_2)$$
53. (b) $Q = \frac{KA(\theta_1 - \theta_2)}{l} t \Rightarrow K_1 t_1 = K_2 t_2 \Rightarrow \frac{K_1}{K_2} = \frac{t_2}{t_1} = \frac{35}{20} = \frac{7}{4}$

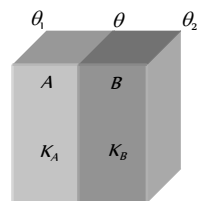
(As Q, l, A and $(\theta_1 - \theta_2)$ are same)

54. (c) A lake cools from the surface down. Above 4°C , the cooled water at the surface flows to the bottom because of its greater density. But when the surface temperature drops below 4°C (here it is 2°C), the water near the surface is less dense than the warmer water below. Hence the downward flow ceases, the water at the bottom remains at 4°C until nearly the entire lake, is frozen.
55. (a) Temperature gradient $\frac{d\theta}{dx} = \frac{(125 - 25)^\circ\text{C}}{50 \text{ cm}} = 2^\circ\text{C}/\text{cm}$
56. (a) $K \propto l^2 \Rightarrow \frac{K_1}{K_2} = \frac{l_1^2}{l_2^2} = \left(\frac{10}{25}\right)^2 = \frac{1}{6.25}$
57. (a) Thermal resistance of Cu is lesser than the thermal resistance of steel. Hence only in option (a) thermal resistance is minimum so heat current is maximum.
58. (c) At steady state, rate of heat flow for both blocks will be same
i.e., $\frac{K_1 A(\theta_1 - \theta)}{l_1} = \frac{K_2 A(\theta - \theta_2)}{l_2}$ (given $l_1 = l_2$)

$$\Rightarrow K_1 A(\theta_1 - \theta) = K_2 A(\theta - \theta_2) \Rightarrow \theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$$



59. (c) $K = \frac{2K_1 K_2}{K_1 + K_2} = \frac{2 \cdot K \cdot 2K}{K + 2K} = \frac{4}{3} K$
60. (a) Temperature of interface $\theta = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$
It is given that $\frac{K_1}{K_2} = \frac{5}{3} \Rightarrow K_1 = 5K$ and $K_2 = 3K$
$$\theta = \frac{5K \times 100 + 3K \times 20}{5K + 3K} = \frac{560K}{8K} = 70^\circ\text{C}$$
61. (c) In winter, the temperature of surrounding is low compared to the body temperature (37.4°C). Since woolen clothes are bad conductors of heat, so they keep the body warm.
62. (d) Temperature of interface $T = \frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$
$$= \frac{300 \times 100 + 200 \times 0}{300 + 200} = 60^\circ\text{C}$$
63. (b) Rate of heat flow $\left(\frac{Q}{t}\right) = \frac{k\pi r^2(\theta_1 - \theta_2)}{L} \propto \frac{r^2}{L}$
$$\therefore \frac{Q_1}{Q_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{l_2}{l_1}\right) = \left(\frac{1}{2}\right)^2 \times \left(\frac{2}{1}\right) = \frac{1}{2} \Rightarrow Q_2 = 2Q_1$$
64. (b) $\frac{Q}{t} = \frac{KA\Delta\theta}{l} \Rightarrow 6000 = \frac{200 \times 0.75 \times \Delta\theta}{1}$
$$\therefore \Delta\theta = \frac{6000 \times 1}{200 \times 0.75} = 40^\circ\text{C}$$
65. (b) In series rate of flow of heat is same



$$\Rightarrow \frac{K_A A(\theta_1 - \theta)}{l} = \frac{K_B A(\theta - \theta_2)}{l}$$

$$\Rightarrow 3K_B(\theta_1 - \theta) = K_B(\theta - \theta_2)$$

$$\Rightarrow 3(\theta_1 - \theta) = (\theta - \theta_2)$$

$$\Rightarrow 3\theta_1 - 3\theta = \theta - \theta_2 \Rightarrow 4\theta_1 - 4\theta = \theta_1 - \theta_2$$

$$\Rightarrow 4(\theta_1 - \theta) = (\theta_1 - \theta_2)$$

$$\Rightarrow 4(\theta_1 - \theta) = 20 \Rightarrow (\theta_1 - \theta) = 5^\circ\text{C}$$

66. (c) Let θ be temperature middle point C and in series rate of heat flow is same $\Rightarrow K(2A)(100 - \theta) = KA(\theta - 70)$

$$\Rightarrow 200 - 2\theta = \theta - 70 \Rightarrow 3\theta = 270 \Rightarrow \theta = 90^\circ\text{C}$$

67. (b) Thermal resistances are same

$$\Rightarrow \frac{l_1}{K_1 A_1} = \frac{l_2}{K_2 A_2} \Rightarrow \frac{l_1}{K_1} = \frac{l_2}{K_2} (\because A_1 = A_2)$$

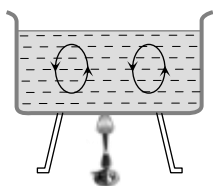
$$\Rightarrow \frac{l_1}{l_2} = \frac{K_1}{K_2} = \frac{5}{3}$$

68. (b) $\frac{Q}{t} \propto \frac{r^2}{l}$; from the given options, option (b) has higher value of $\frac{r^2}{l}$.

Convection

1. (c) Convection significantly transferring heat upwards (Gravity effect).
2. (a) Heat flows from hot air to cold body so person feels comfort.
3. (c) No flow of heat by convection in vacuum.
4. (a)
5. (b) Density of hot air is lesser than the density of cold air so hot air rises up.

6. (a)
7. (c) In convection hot particles moves up ward (due to low density) and light particle moves downward (due to high density).



8. (b)
9. (a) Natural convection arises due to difference of density at two places and is a consequence of gravity.
10. (d)
11. (a) Convection is not possible in weightlessness. So the liquid will be heated through conduction.
12. (c) In forced convection rate of loss of heat $\frac{Q}{t} \propto A(T - T_0)$
13. (c)

Radiation (General, Kirchoff's law, Black body)

1. (b) Because of uneven surfaces of mountains, most of it's parts remain under shadow. So, most of the mountains. Land is not heated up by sun rays. Besides this, sun rays fall slanting on the mountains and are spread over a larger area. So, the heat received by the mountains top per unit area is less and they are less heated compared to planes (Foot).
2. (a) The velocity of heat radiation in vacuum is equal to that of light.

3. (c) Radiation is the fastest mode of heat transfer.
4. (d) A thermopile is a sensitive instrument, used for detection of heat radiation and measurement of their intensity.
5. (d) The polished surface reflects all the radiation.
6. (c) Heat radiations are electromagnetic waves of high wavelength.
7. (d) When element and surrounding have same temperature. There will be no temperature difference, hence heat will not flow from the filament and it's temperature remains constant.
8. (d) Every body at all time, at all temperatures emits radiation (except at $T = 0$). The radiation emitted by the human body is in the infra-red region.
9. (c)
10. (b) Infrared radiations are detected by pyrometer.
11. (b)
12. (a) In vacuum heat flows by the radiation mode only.
13. (c) Good absorbers are always good emitters of heat.
14. (a) A perfectly black body is a good absorber of radiations falls on it. So it's absorptive power is 1.
15. (d) According to Kirchoff's law in spectroscopy. If a substance emit certain wavelengths at high temperature, it absorbs the same wavelength at comparatively lower temperature.
16. (b) A person with dark skin absorbs more heat radiation and feels more heat. It also radiates more heat and feels more cold.
17. (a) For a black body emissivity = absorptive power.
18. (b) Highly polished mirror like surfaces are good reflectors, but not good radiators.
19. (b) Black cloth is a good absorber of heat, therefore ice covered by black cloth melts more as compared to that covered by white cloth.
20. (c) According to Kirchoff's law, the ratio of emissive power to absorptive power is same for all bodies is equal to the emissive power of a perfectly black body i.e.,

$$\left(\frac{e}{a}\right)_{\text{body}} = E_{\text{Blackbody}} \text{ for a particular wave length}$$

$$\left(\frac{e_\lambda}{a_\lambda}\right)_{\text{body}} = (E_\lambda)_{\text{Blackbody}} \Rightarrow e_\lambda = a_\lambda E_\lambda$$

21. (b) Absorption power = $\frac{\text{Heat absorbed}}{\text{Total heat given}}$
22. (c) Because Planck's law explains the distribution of energy correctly at low temperature as well as at high temperature.
23. (c)
24. (a) The black spot on heating absorbs radiations and so emits them in the dark room while the polished shining part reflects radiation and absorbs nothing and so does not emit radiations and becomes invisible in the dark.
25. (b)
26. (b) When the light emitted from the sun's photosphere passes through it's outer part Chromosphere, certain wave lengths are absorbed. In the spectrum of sunlight, a large number of dark lines are seen called Fraunhofer lines.
27. (a) As for a black body rate of absorption of heat is more. Hence thermometer A shows faster rise in temperature but finally both will acquire the atmospheric temperature.
28. (c) According to Kirchoff's law, a good emitter is also a good absorber.
29. (a) Red and green colours are complementary to each other. When red glass is heated it absorbs green light strongly, hence

according to Kirchhoff's law, the emissive power of red glass should be maximum for green light. That's why when this heated red glass is taken in dark room it strongly emits green light and looks greenish.

30. (d) Black and rough surfaces are good absorber that's why they emit well. (Kirchhoff's law).
31. (d)
32. (c) When light incident on pin hole, enters into the box and suffers successive reflection at the inner wall. At each reflection some energy is absorbed. Hence the ray once it enters the box can never come out and pin hole acts like a perfect black body.
33. (a) Initially black body absorbs all the radiant energy incident on it. So it is the darkest one. Black body radiates maximum energy if all other condition are same. So when the temperature of the black body becomes equal to the temperature of furnace it will be brightest of all.
34. (c) Open window behaves like a perfectly black body.
35. (a) Ordinary glass prism (crown, flint) absorbs the infrared radiation but rock salt prism transmit them. Hence it is used to obtain the spectrum of infrared radiation.
36. (d) A good absorber is a good emitter hence option (a) is wrong. Every body stops absorbing and emitting radiation at 0 K hence option (b) is wrong.
The energy of radiation emitted from a black body is not same for all wavelength hence option (c) is wrong.
Planck's law relates the wavelength (λ) and temperature (T) according to the relation $E_{\lambda} d_{\lambda} = \frac{8\pi hc}{\lambda^5} \frac{1}{[e^{hc/kT} - 1]} d_{\lambda}$. Hence option (d) is correct.
37. (c) When blue glass is heated at high temperature, it absorbs all the radiation of, higher wavelength except blue. If it is taken inside a dark room, it emits all the radiation of higher wavelength, hence it looks brighter red as compared to the red piece.
38. (b)

Radiation (Wein's law)

1. (a)
2. (c) According to Wein's law, $\lambda_m T = \text{constant}$
 $\lambda_r > \lambda_y > \lambda_b \Rightarrow T_r < T_y < T_b$ or $T_A < T_C < T_B$
3. (d) $\lambda_m T = \text{constant} \Rightarrow \frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{10^{-4}}{0.5 \times 10^{-5}} = 200$.
4. (c) $\lambda_m T = \text{constant}$
5. (a) According to Wein's law $\lambda_m T = \text{constant}$, on heating up to ordinary temperatures, only long wavelength (red) radiation is emitted. As the temperature rises, shorter wavelengths are also emitted in more and more quantity. Hence the colour of radiation emitted by the hot wire shifts from red to yellow, then to blue and finally to white.
6. (c) According to Wein's displacement law.
7. (d) $\lambda_{m_1} T_1 = \lambda_{m_2} T_2 \Rightarrow \lambda_{m_2} = \frac{\lambda_{m_1} T_1}{T_2} = 4.08 \times \frac{700}{1400} = 2.04 \text{ m}$
8. (c) $\lambda_{m_1} T_1 = \lambda_{m_2} T_2 \Rightarrow \lambda_{m_2} = \frac{\lambda_{m_1} T_1}{T_2} = \frac{14 \times 200}{1000} = 2.8 \text{ } \mu\text{m}$

9. (c) $\frac{T_1}{T_2} = \frac{\lambda_{m_2}}{\lambda_{m_1}} = \frac{4800}{3600} \Rightarrow \frac{48}{36} = \frac{4}{3}$
10. (b) $\lambda_{m_2} = \frac{T_1}{T_2} \times \lambda_{m_1} = \frac{1500}{2500} \times 5000 = 3000 \text{ } \text{\AA}$
11. (a) At low temperature short wavelength radiation is emitted. As the temperature rise colour of emitted radiation are in the following order
Red $\rightarrow$ Yellow $\rightarrow$ Blue $\rightarrow$ White (at highest temperature)
12. (b) Similar to Q. 11
13. (b) The wavelength corresponding to maximum emission of radiation from the sun is $\lambda_{\text{max}} = 4753 \text{ } \text{\AA}$ (close to the wavelength of violet colour of visible region). Hence if temperature is doubled λ_c is decreased $\left(\lambda_m \propto \frac{1}{T} \right)$ i.e. mostly ultraviolet radiations emits.
14. (c) $\frac{T_1}{T_2} = \frac{\lambda_{m_2}}{\lambda_{m_1}} = \frac{5.5 \times 10^5}{11 \times 10^5} = \frac{1}{2} \Rightarrow n = \frac{1}{2}$.
15. (b) $\therefore T = \frac{b}{\lambda_m} = \frac{2.93 \times 10^{-3}}{2.93 \times 10^{-10}} = 10^7 \text{ K}$
16. (a) $\therefore \frac{\lambda_{m_2}}{\lambda_{m_1}} = \frac{T_1}{T_2} \Rightarrow \lambda_{m_2} = \frac{2000}{2400} \times 4 = 3.33 \text{ } \mu\text{m}$
17. (b)
18. (a)
19. (b) $\lambda_{m_2} = \frac{T_1}{T_2} \times \lambda_{m_1} = \frac{2000}{3000} \times \lambda_{m_1} = \frac{2}{3} \lambda_{m_1} = \frac{2}{3} \lambda_m$
20. (a)
21. (c) $\frac{T_2}{T_1} = \frac{\lambda_{m_1}}{\lambda_{m_2}} = \frac{1.75}{14.35} \Rightarrow T_2 = \frac{1.75}{14.35} \times 1640 = 200 \text{ K}$
22. (a) $\frac{\lambda_2}{\lambda_1} = \frac{T_1}{T_2} \Rightarrow \lambda_2 = \frac{T_1}{T_2} \times \lambda_1 = \frac{900}{1200} \times 4 = 3 \text{ } \mu\text{m}$
23. (b) $\lambda_{m_2} = \frac{\lambda_{m_1} T_1}{T_2} = \frac{4800 \times 6000}{3000} = 9600 \text{ } \text{\AA}$
24. (b) $\frac{T_1}{T_2} = \frac{\lambda_{m_2}}{\lambda_{m_1}} = \frac{4200}{140} = \frac{30}{1}$
25. (c) $\therefore \lambda_m T = \lambda'_m T' \Rightarrow \lambda_0 T = \lambda' \times 2T \Rightarrow \lambda' = \frac{\lambda_0}{2}$
26. (b) $\lambda_m T = \lambda'_m T' \Rightarrow \frac{\lambda_m}{\lambda'_m} = \frac{T'}{T} = \frac{3000}{2000} = \frac{3}{2}$
27. (b) $\lambda_{m_1} T = \lambda_{m_2} T_2 \Rightarrow 5.5 \times 10^{-7} \times 5500 = 11 \times 10^{-7} T$
 $T = 550 \times 5 \text{ K} = 2750 \text{ K}$
28. (b) According to Wein's displacement law
 $\lambda_m T = b$ or $\lambda_m = \frac{b}{T} = \frac{0.0029}{5 \times 10^4} = 58 \times 10^{-9} \text{ m} = 58 \text{ nm}$
29. (a) $\lambda_m = \frac{b}{T} \Rightarrow T = \frac{b}{\lambda_m} = \frac{2.93 \times 10^{-3}}{4000 \times 10^{-10}} = 7325 \text{ K}$
30. (b) $\frac{T_S}{T_N} = \frac{(\lambda_N)_{\text{max}}}{(\lambda_S)_{\text{max}}} = \frac{350}{510} = 0.69$

Radiation (Stefan's law)

1. (c) $E \propto T^4$ (Stefan's law)
2. (c) Rate of heat loss $E = \sigma eA(T^4 - T_0^4)$
 $= 5.67 \times 10^{-8} \times 0.4 \times 200 \times 10^{-4} \times [(273 + 527)^4 - (273 + 27)^4]$
 $= 5.67 \times 10^{-8} \times 0.4 \times 200 \times 10^{-4} \times (800)^4 - (300)^4 = 182 \text{ J/sec}$
3. (a) $\frac{E_1}{E_2} = \left(\frac{T_1}{T_2}\right)^4 \Rightarrow \frac{E}{E_2} = \left(\frac{273 + 0}{273 + 273}\right)^4 \Rightarrow E_2 = 16 E$
4. (a) $E \propto T^4 \Rightarrow \frac{E_1}{E_2} = \frac{T^4}{T^4} \times 2^4 \Rightarrow E_2 = \frac{E}{16}$
5. (d) $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{2}{1} = \left(\frac{420 + 273}{T}\right)^4 = \left(\frac{673}{T}\right)^4$
 $\Rightarrow T = 2^{1/4} \times 673 = 800 \text{ K}$
6. (a) $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{273 + 727}{237 + 227}\right)^4 = \left(\frac{1000}{500}\right)^4 = 16 \Rightarrow E_2 = 80$
7. (c) $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow T_2 = \left(\frac{E_2}{E_1}\right)^{1/4} \times T_1 = (16)^{1/4} \times (273 + 127)$
 $\Rightarrow T_2 = 800 \text{ K} = 527^\circ \text{C}$
8. (b) In M.K.S. system unit of σ is $\frac{J}{m^2 \times \text{sec} \times K^4}$
 $\Rightarrow 1 \frac{J}{m^2 \times \text{sec} \times K^4} = \frac{10^7 \text{ erg}}{10^4 \text{ cm}^2 \times \text{sec} \times K^4}$
 $= 10^3 \frac{\text{erg}}{\text{cm}^2 \times \text{sec} \times K^4}$
9. (b) For a black body rate of energy $\frac{Q}{t} = P = A\sigma T^4$
 $\Rightarrow P \propto T^4 \Rightarrow \frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4 = \left\{\frac{(273 + 7)}{(273 + 287)}\right\}^4 = \frac{1}{16}$
10. (b) $E_2 = E_1 \frac{T_2^4}{T_1^4} = Q \times \frac{(273 + 151)^4}{(273 + 27)^4} = \left(\frac{424}{300}\right)^4 = 3.99Q \approx 4Q$
11. (b) $\frac{E_1}{E_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{727 + 273}{127 + 273}\right)^4 = \left(\frac{1000}{400}\right)^4 = \frac{10^4}{4^4} = \frac{625}{16}$
12. (c) $E = \sigma T^4 \Rightarrow 5.6 \times 10^{-8} \times T^4 = 1$
 $\Rightarrow T = \left[\frac{1}{5.6 \times 10^{-8}}\right]^{1/4} = 65 \text{ K}$
13. (c) According to Stefan's law $E = \sigma \varepsilon AT^4$
 $\Rightarrow \frac{1.58 \times 10^5 \times 4.2}{60 \times 60} = 5.6 \times 10^{-8} \times 10^{-4} \times 0.8 \times T^4$
 $T \approx 2500 \text{ K}$
14. (c) Total energy radiated from a body $Q = A\sigma T^4 t$
 $\Rightarrow Q \propto AT^4 \propto r^2 T^4 \quad (\because A = 4\pi r^2)$
 $\Rightarrow \frac{Q_P}{Q_Q} = \left(\frac{r_P}{r_Q}\right)^2 \left(\frac{T_P}{T_Q}\right)^4 = \left(\frac{8}{2}\right)^2 \left\{\frac{(273 + 127)}{(273 + 527)}\right\}^4 = 1$
15. (c) Rate of energy $\frac{Q}{t} = P = A\sigma T^4 \Rightarrow P \propto T^4$
 $\Rightarrow \frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{927 + 273}{127 + 273}\right)^4 \Rightarrow P_1 = 405 \text{ W}$
16. (b) The rate of radiated energy $\frac{Q}{t} = P = A\sigma T^4$
 $\Rightarrow 1134 = 5.67 \times 10^{-8} \times (0.1)^2 T^4 \Rightarrow T = 1189 \text{ K}$
17. (d) $Q \propto T^4 \Rightarrow \frac{H_A}{H_B} = \left(\frac{273 + 727}{273 + 327}\right)^4 = \left(\frac{10}{6}\right)^4 = \left(\frac{5}{3}\right)^4 = \frac{625}{81}$
18. (d) $(Q)_{\text{Blackbody}} = A\sigma T^4 t \Rightarrow Q \propto T^4$
 $\Rightarrow Q_2 = Q_1 \left(\frac{T_2}{T_1}\right)^4 = 10 \left(\frac{273 + 327}{273 + 27}\right)^4 = 10 \left(\frac{600}{300}\right)^4 = 160 \text{ J}$
19. (c) $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{E_2}{20} = \left(\frac{2T}{T}\right)^4 = 16 \Rightarrow E_2 = 320 \text{ kcal/m}^2 \text{ min.}$
20. (d) Radiated power by blackbody $P = \frac{Q}{t} = A\sigma T^4$
 $\Rightarrow P \propto AT^4 \propto r^2 T^4 \Rightarrow \frac{P_1}{P_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{T_1}{T_2}\right)^4$
 $\Rightarrow \frac{440}{P_2} = \left(\frac{12}{6}\right)^2 \left(\frac{500}{1000}\right)^4 \Rightarrow P_2 = 1760 \text{ W} \approx 1800 \text{ W}$
21. (d) Amount of energy radiated $\propto$ (Temperature).
22. (d) $\frac{Q_1}{Q_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{273 + 27}{273 + 927}\right)^4 = \left(\frac{1}{4}\right)^4 = \frac{1}{256}$
23. (a) $\frac{E_2}{E_1} = \frac{T_2^4}{T_1^4} = \left(\frac{237 + 227}{273 + 27}\right)^4 = \left(\frac{600}{300}\right)^4 = 16$
24. (d) $(Q)_{\text{Blackbody}} = A\sigma T^4 t \Rightarrow \frac{Q}{t} \propto P = A\sigma T^4$
 Breadth are halved so area becomes one fourth.
 $\Rightarrow \frac{P_1}{P_2} = \frac{A_1}{A_2} \times \left(\frac{T_1}{T_2}\right)^4 \Rightarrow \frac{A_1}{(A_1/4)} \times \left(\frac{273 + 327}{273 + 127}\right)$
 $\Rightarrow P_2 = \frac{81}{64} E$
25. (d) Power radiated $P \propto T^4 \Rightarrow \frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4$
 $\Rightarrow \frac{Q}{P_2} = \left(\frac{T}{3T}\right)^4 \Rightarrow P_2 = 81Q$
26. (a) For black body, $P = A\sigma T^4$. For same power $A \propto \frac{1}{T^4}$
 $\Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{r_1}{r_2} = \left(\frac{T_2}{T_1}\right)^2$
27. (a) $\frac{Q_2}{Q_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{273 + 927}{273 + 327}\right)^4 = \left(\frac{1200}{600}\right)^4 = 16$

$$\Rightarrow Q = 32 \text{ KJ}$$

28. (b) $\frac{Q_2}{Q_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{2}{1} = \left(\frac{T_2}{T_1}\right)^4$
 $\Rightarrow T_2^4 = 2 \times T_1^4 = 2 \times (273 + 727)^4 \Rightarrow T_2 = 1190 \text{ K}$

29. (a) $\frac{Q_1}{Q_2} = \frac{r_1^2 T_1^4}{r_2^2 T_2^4} = \frac{4^2}{1^2} \times \left(\frac{2000}{4000}\right)^4 = 1$

30. (a) According to Wein's law $\lambda T = \text{constant}$

$$\Rightarrow \lambda_{m_1} T_1 = \lambda_{m_2} T_2 \Rightarrow T_2 = \frac{\lambda_{m_1}}{\lambda_{m_2}} T_1 = \frac{\lambda_0}{3\lambda_0/4} \times T_1 = \frac{4}{3} T_1$$

$$\text{Now } P \propto T^4 \Rightarrow \frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{P_2}{P_1} = \left(\frac{4/3 T_1}{T_1}\right)^4 = \frac{256}{81}$$

31. (d) $E \propto T^4$

32. (d) $Q \propto T^4 \Rightarrow \frac{Q_1}{Q_2} = \left(\frac{T_1}{T_2}\right)^4$
 $\Rightarrow \frac{Q_1}{Q_2} = \left(\frac{T}{T + T/2}\right)^4 = \frac{16}{81} \Rightarrow Q_2 = \frac{81}{16} Q_1$
 $\% \text{ increase in energy} = \frac{Q_2 - Q_1}{Q_1} \times 100 = 400\%$

33. (d) If temperature of surrounding is considered then
 net loss of energy of a body by radiation
 $Q = A\sigma(T^4 - T_0^4)t \Rightarrow Q \propto (T^4 - T_0^4) \Rightarrow \frac{Q_1}{Q_2} = \frac{T_1^4 - T_0^4}{T_2^4 - T_0^4}$
 $= \frac{(273 + 200)^4 - (273 + 27)^4}{(273 + 400)^4 - (273 + 27)^4} = \frac{(473)^4 - (300)^4}{(673)^4 - (300)^4}$

34. (c) $Q = A\sigma T^4 \Rightarrow Q \propto A \propto r^2$ ($\because T = \text{constant}$)
 $\Rightarrow \frac{Q_1}{Q_2} = \frac{r_1^2}{r_2^2} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

35. (a) $\frac{Q_2}{Q_1} = \frac{T_2^4}{T_1^4} = \left(\frac{273 + 527}{273 + 127}\right)^4 = \left(\frac{800}{400}\right)^4 \Rightarrow Q_2 = 16 \frac{\text{cal}}{\text{cm}^2 \times \text{s}}$

36. (c) For a black body $\frac{Q}{t} = P = A\sigma T^4$
 $\Rightarrow \frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{P_2}{20} = \left(\frac{273 + 727}{273 + 227}\right)^4$
 $\Rightarrow \frac{P_2}{20} = (2)^4 \Rightarrow P_2 = 320 \text{ W}$

37. (c) Energy radiated per sec $\frac{Q}{t} = P = A\sigma T^4$
 $P \propto r^2 T^4 \Rightarrow \frac{P_2}{P_1} = \frac{r_2^2}{r_1^2} \cdot \frac{T_2^4}{T_1^4} = \frac{4^2}{1^2} \times \left(\frac{2000}{4000}\right)^4 = 1$

38. (c) $Q \propto AT^4 \propto r^2 T^4 \Rightarrow \frac{Q_{\text{star}}}{Q_{\text{sun}}} = \frac{r_{\text{star}}^2 T_{\text{star}}^4}{r_{\text{sun}}^2 T_{\text{sun}}^4}$

$$\Rightarrow \frac{10000}{1} = \frac{r_{\text{star}}^2}{r_{\text{sun}}^2} \times \left(\frac{6000}{2000}\right)^4 \Rightarrow \frac{r_{\text{star}}}{r_{\text{sun}}} = \frac{100 \times 9}{1} = \frac{900}{1}$$

39. (a) $P = \left(\frac{Q}{t}\right) \propto T^4 \Rightarrow \frac{W}{P_2} = \left(\frac{T}{T/3}\right)^4 \Rightarrow P_2 = \frac{W}{81}$

40. (c) Power $P \propto AT^4 \propto r^2 T^4$
 $\Rightarrow \frac{P_2}{P_1} = \left(\frac{r_2}{r_1}\right)^2 \times \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{4r}{r}\right)^2 \times \left(\frac{T/2}{T}\right)^4 = 1$

41. (b) $\frac{Q_2}{Q_1} = \left(\frac{r_2^2}{r_1^2}\right) \times \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{100}{1}\right)^2 \times \left(\frac{1}{2}\right)^4 = 625$

42. (d) $Q \propto r^2 T^4 \Rightarrow \frac{Q_2}{Q_1} = \left(\frac{r_2}{r_1}\right)^2 \times \left(\frac{T_2}{T_1}\right)^4 = (2)^2 \times (2)^4 = 64$

43. (c) Energy radiated from a body $Q = A\sigma T^4 t$
 $\Rightarrow \frac{Q_2}{Q_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{T_2}{T_1} = \left(\frac{Q_2}{Q_1}\right)^{1/4} = \left(\frac{4.32 \times 10^6}{2.7 \times 10^{-3}}\right)^{1/4}$
 $= \left(\frac{16 \times 27}{27} \times 10^8\right)^{1/4} = 2 \times 10^2$
 $\Rightarrow T_2 = 200 \times T_1 = 80000 \text{ K}$

44. (d) $E \propto AT^4 \Rightarrow \frac{E_{\text{sphere}}}{E_{\text{disc}}} = \frac{4\pi r^2}{2\pi r^2} \times \left(\frac{T}{T}\right)^4 = \frac{2}{1}$

45. (a) $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow \frac{T_2}{T_1} = \left(\frac{E_2}{E_1}\right)^{1/4} = \left(\frac{10^9}{10^5}\right)^{1/4} = 10$
 $\Rightarrow T_2 = 10T_1 = 10 \times (273 + 227) = 5000 \text{ K}$

46. (b) Energy per second $P = \left(\frac{Q}{t}\right) \propto T^4$
 $\frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{273 - 73}{273 + 327}\right)^4 = \left(\frac{200}{600}\right)^4 = \frac{1}{81}$

47. (a) $Q \propto T^4 \Rightarrow \frac{Q_1}{Q_2} = \left(\frac{T_1}{T_2}\right)^4$
 If $T_1 = T$ then $T_2 = T + \frac{10}{100} T = 1.1T$
 $\Rightarrow \frac{Q_1}{Q_2} = \left(\frac{T}{1.1T}\right)^4 \Rightarrow Q_2 = 1.46 Q_1$
 $\Rightarrow \% \text{ increase in energy} = \frac{Q_2 - Q_1}{Q_1} \times 100 = 46\%$

48. (a) From Stefan's law $E = \sigma T^4$
 $T^4 = \frac{E}{\sigma} = \frac{6.3 \times 10^7}{5.7 \times 10^8} = 1.105 \times 10^{15} = 0.1105 \times 10^{16}$
 $T = 0.58 \times 10^4 \text{ K} = 5.8 \times 10^3 \text{ K}$

49. (a) Rate of cooling $\propto (T^4 - T_0^4)$
 $\Rightarrow \frac{H}{H'} = \frac{(T_1^4 - T_0^4)}{(T_2^4 - T_0^4)} = \frac{400^4 - 200^4}{600^4 - 200^4}$

$$\text{or } H' = \frac{(16+4)(16-4)H}{(36+4)(36-4)} = \frac{3}{16}H$$

50. (a)

51. (a) Rate of cooling $\propto (T^4 - T_0^4) \Rightarrow \frac{R_1}{R_2} = \frac{(T_1^4 - T_0^4)}{(T_2^4 - T_0^4)}$

$$\Rightarrow \frac{R}{R_2} = \frac{(600)^4 - (300)^4}{(900)^4 - (300)^4} \text{ or } R_2 = \frac{16}{3}R$$

52. (d) Loss of heat $\Delta Q = A\sigma(T^4 - T_0^4)t$

$$\Rightarrow \text{Rate of loss of heat } \frac{\Delta Q}{t} = A\sigma(T^4 - T_0^4)$$

$$= 10 \times 10^{-4} \times 1 \times 5.67 \times 10^{-8} \{273 + 127\}^4 - (273 + 27)^4\}$$

$$= 0.99 \text{ W}$$

53. (b) According to Prevost theory every body radiate heat at all temperature (except 0K) and also absorbs heat from surroundings.

$\therefore T_A > T \Rightarrow$ Object A emits radiations more than the radiations it absorbs.

and $T_B < T \Rightarrow$ Object B absorbs more radiations than it emits.

After a certain time all bodies attains a common temperature.

54. (b) According to Prevost theory

55. (b) $Q \propto T^4 \Rightarrow \frac{Q_1}{Q_2} = \frac{T_1^4}{T_2^4} \Rightarrow T_2^4 = \left(\frac{E_2}{E_1}\right)T_1^4$

$$\Rightarrow T_2^4 = \frac{1}{16} \times (1000)^4 = \left(\frac{1000}{2}\right)^4 \Rightarrow T_2 = 500K$$

56. (c) $Q \propto T^4$

Radiation (Newton's Law of Cooling)

1. (b) According to Newton's law of cooling

$$\frac{\theta_1 - \theta_2}{t} = K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$$

$$\text{In the first case, } \frac{(60-50)}{10} = K \left[\frac{60+50}{2} - \theta_0 \right]$$

$$1 = K(55 - \theta) \quad \dots(i)$$

$$\text{In the second case, } \frac{(50-42)}{10} = K \left[\frac{50+42}{2} - \theta_0 \right]$$

$$0.8 = K(46 - \theta_0) \quad \dots(ii)$$

$$\text{Dividing (i) by (ii), we get } \frac{1}{0.8} = \frac{55 - \theta_0}{46 - \theta_0}$$

$$\text{or } 46 - \theta_0 = 44 - 0.8\theta_0 \Rightarrow \theta_0 = 10^\circ C$$

2. (c) According to Newton's law of cooling

Rate of cooling $\propto$ Mean temperature difference

$$\Rightarrow \frac{\text{Fall in temperature}}{\text{Time}} \propto \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

$$\therefore \left(\frac{\theta_1 + \theta_2}{2} \right)_1 > \left(\frac{\theta_1 + \theta_2}{2} \right)_2 > \left(\frac{\theta_1 + \theta_2}{2} \right)_3$$

$$\Rightarrow T_1 < T_2 < T_3$$

3. (a) Initially at $t = 0$

Rate of cooling (R) $\propto$ Fall in temperature of body ($\theta - \theta_0$)

$$\Rightarrow \frac{R_1}{R_2} = \frac{\theta_1 - \theta_0}{\theta_2 - \theta_0} = \frac{100 - 40}{80 - 40} = \frac{3}{2}$$

4. (a) For small difference of temperature, it is the special case of Stefan's law.

5. (b) Liquid having more specific heat has slow rate of cooling because for equal masses rate of cooling $\frac{d\theta}{dt} \propto \frac{1}{c}$.

6. (c) $S_l = \frac{1}{m_l} \left[\frac{t_l}{t_w} (m_w C_w + W) - W \right]$

$$= \frac{1}{300} \left[\frac{95}{3 \times 60} (350 \times 1 + 10) - 10 \right] = 0.6 \text{ Cal/gm}^\circ C$$

7. (c) Newton's law of cooling is used for the determination of specific heat of liquids.

8. (c) By Newton's law of cooling.

9. (d) Rate of loss of heat $\left(\frac{\Delta Q}{t} \right) \propto$ temperature difference $\Delta \theta$

$$\left(\frac{\Delta Q}{t} \right)_1 = \frac{\Delta \theta_2}{\Delta \theta_1} \Rightarrow \frac{60}{\left(\frac{\Delta Q}{t} \right)_2} = \frac{80 - 60}{40 - 20} \Rightarrow \left(\frac{\Delta Q}{t} \right)_2 = \frac{20 \text{ cal}}{\text{sec}}$$

10. (b) During clear nights object on surface of earth radiate out heat and temperature falls. Hence option (a) is wrong.

The total energy radiated by a body per unit time per unit area $E \propto T$. Hence option (c) is wrong.

Energy radiated per second is given by $\frac{Q}{t} = PA\sigma T^4$

$$\Rightarrow \frac{P_1}{P_2} = \frac{A_1}{A_2} \cdot \left(\frac{T_1}{T_2} \right)^4 = \left(\frac{r_1}{r_2} \right)^2 \cdot \left(\frac{T_1}{T_2} \right)^4 = \left(\frac{1}{4} \right)^2 \left(\frac{4000}{200} \right) = \frac{1}{1}$$

$\therefore P = P$, hence option (d) is wrong.

Newton's law is an approximate form of Stefan's law of radiation and works well for natural convection. Hence option (b) is correct.

11. (b) $\frac{\theta_1 - \theta_2}{t} = K \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$

$$\therefore \frac{100 - 70}{4} = K \left(\frac{100 + 70}{2} - 15 \right) = 60K \Rightarrow K = \frac{1}{8}$$

$$\text{Again } \frac{70 - 40}{t} = \frac{1}{8} \left(\frac{70 + 40}{2} - 15 \right) = 5 \Rightarrow t = 6 \text{ min.}$$

12. (d) $\frac{80 - 60}{1} = K \left(\frac{80 + 60}{2} - 30 \right) \Rightarrow K = \frac{1}{2}$

$$\text{Again } \frac{60 - 50}{t} = \frac{1}{2} \left(\frac{60 + 50}{2} - 30 \right) \Rightarrow t = 0.8 \times 60 = 48 \text{ sec.}$$

13. (c) According to Newton's law of cooling

Rate of cooling $\propto$ mean temperature difference.

Initially, mean temperature difference

$$= \left(\frac{70 + 60}{2} - \theta_0 \right) = (65 - \theta_0)$$

Finally, mean temperature difference

$$= \left(\frac{60 + 50}{2} - \theta_0 \right) = (55 - \theta_0)$$

In second case mean temperature difference decreases, so rate of fall of temperature decreases, so it takes more time to cool through the same range.

$$14. \quad (c) \quad \frac{\theta_1 - \theta_2}{t} = K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$$

In the first 10 minute

$$\frac{62 - 50}{10} = K \left[\frac{62 + 50}{2} - \theta_0 \right] \Rightarrow 1.2 = K[56 - \theta_0] \quad \dots (i)$$

$$\frac{50 - 42}{10} = K \left[\frac{50 + 42}{2} - \theta_0 \right] \Rightarrow 0.8 = K[46 - \theta_0] \quad \dots (ii)$$

$$\text{from equations (i) and (ii)} \quad \frac{1.2}{0.8} = \frac{(56 - \theta_0)}{(46 - \theta_0)} \Rightarrow \theta_0 = 26^\circ\text{C}$$

$$15. \quad (d) \quad \frac{d\theta}{dt} = \frac{\sigma A}{mc} (T^4 - T_0^4). \text{ If the liquids put in exactly similar calorimeters and identical surrounding then we can consider } T, \text{ and } A \text{ constant then } \frac{d\theta}{dt} \propto \frac{(T^4 - T_0^4)}{mc}$$

If we consider that equal masses of liquid (m) are taken at the same temperature then $\frac{d\theta}{dt} \propto \frac{1}{c}$

So for same rate of cooling c should be equal which is not possible because liquids are of different nature. Again from equation (i)

$$\frac{d\theta}{dt} \propto \frac{(T^4 - T_0^4)}{mc} \Rightarrow \frac{d\theta}{dt} \propto \frac{(T^4 - T_0^4)}{V\rho c}$$

Now if we consider that equal volume of liquid (V) are taken at the same temperature then $\frac{d\theta}{dt} \propto \frac{1}{\rho c}$.

So for same rate of cooling multiplication of $\rho \times c$ for two liquid of different nature can be possible. So option (d) may be correct.

$$16. \quad (c) \quad \frac{60 - 50}{10} = K \left[\frac{60 + 50}{2} - 25 \right] \quad \dots (i)$$

$$\frac{50 - \theta}{10} = K \left[\frac{50 + \theta}{2} - 25 \right] \quad \dots (ii)$$

$$\text{On dividing, we get } \frac{10}{50 - \theta} = \frac{60}{\theta} \Rightarrow \theta = 42.85^\circ\text{C}$$

$$17. \quad (a) \quad \frac{365 - 361}{2} = K \left[\frac{365 + 361}{2} - 293 \right] = 70 \text{ K} \Rightarrow K = \frac{1}{35}$$

$$\text{Again } \frac{344 - 342}{t} = \frac{1}{35} \left[\frac{344 + 342}{2} - 293 \right] = \frac{10}{7}$$

$$\Rightarrow t = \frac{14}{10} \text{ min} = \frac{14}{10} \times 60 = 84 \text{ sec.}$$

$$18. \quad (b) \quad \frac{50 - 49.9}{5} = K \left[\frac{50 + 49.9}{2} - 30 \right] \quad \dots (i)$$

$$\frac{40 - 39.9}{t} = K \left[\frac{40 + 39.9}{2} - 30 \right] \quad \dots (ii)$$

from equations (i) and (ii) we get $t \approx 10 \text{ sec.}$

$$19. \quad (c) \quad \text{Rate of loss of heat is directly proportional to the temperature difference between water and the surroundings.}$$

$$20. \quad (b) \quad \text{Rate of cooling} = \frac{-d\theta}{dt} \propto \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

In second case average temperature will be less hence rate of cooling will be less. Therefore time taken will be more than 4 minutes.

$$21. \quad (a)$$

$$22. \quad (b) \quad \text{First case, } \frac{61 - 59}{4} = K \left[\frac{61 + 59}{2} - 30 \right] \quad \dots (i)$$

$$\text{Second case, } \frac{51 - 49}{t} = K \left[\frac{51 + 49}{2} - 30 \right] \quad \dots (ii)$$

By solving equation (i) and (ii) we get $t = 6 \text{ min.}$

$$23. \quad (d)$$

$$24. \quad (c) \quad \text{In first case } \frac{60 - 40}{7} = K \left[\frac{60 + 40}{2} - 10 \right] \quad \dots (i)$$

$$\text{In second case } \frac{40 - 28}{t} = K \left[\frac{40 + 28}{2} - 10 \right] \quad \dots (ii)$$

.....(i)

By solving $t = 7 \text{ minutes}$

$$25. \quad (b) \quad \text{In first case } \frac{50 - 40}{5} = K \left[\frac{50 + 40}{2} - \theta_0 \right] \quad \dots (i)$$

$$\text{In second case } \frac{40 - 33.33}{5} = K \left[\frac{40 + 33.33}{2} - \theta_0 \right] \quad \dots (ii)$$

By solving $\theta_0 = 20^\circ\text{C}$.

$$26. \quad (b) \quad \text{In first case } \frac{50 - 40}{10} = K \left[\frac{50 + 40}{2} - 20 \right] \quad \dots (i)$$

$$\text{In second case } \frac{40 - \theta_2}{10} = K \left[\frac{40 + \theta_2}{2} - 20 \right] \quad \dots (ii)$$

By solving $\theta_2 = 33.3^\circ\text{C}$.

$$27. \quad (d) \quad \text{In first case } \frac{61 - 59}{10} = K \left[\frac{61 + 59}{2} - 30 \right] \quad \dots (i)$$

$$\text{In second case } \frac{51 - 49}{10} = K \left[\frac{51 + 49}{2} - 30 \right] \quad \dots (ii)$$

By solving $t = 15 \text{ min.}$

$$28. \quad (d) \quad \text{Rate of cooling (here it is rate of loss of heat)}$$

$$\frac{dQ}{dt} = (mc + W) \frac{d\theta}{dt} = (m_l c_l + m_c c_c) \frac{d\theta}{dt}$$

$$\Rightarrow \frac{dQ}{dt} = (0.5 \times 2400 + 0.2 \times 900) \left(\frac{60 - 55}{60} \right) = 115 \frac{\text{J}}{\text{sec}}$$

$$29. \quad (a) \quad \text{According to Newton's law}$$

Rate of cooling $\propto$ temperature difference $\Delta\theta$

$$30. \quad (b) \quad \text{According to Newton's law } \frac{\theta_1 - \theta_2}{t} = k \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$$

Initially,

$$\frac{(80-64)}{5} = K \left(\frac{80+64}{2} - \theta_0 \right) \Rightarrow 3.2 = K[72 - \theta_0] \dots (i)$$

Finally

$$\frac{(64-52)}{10} = K \left[\frac{64+52}{2} - \theta_0 \right] \Rightarrow 1.2 = K[58 - \theta_0] \dots (ii)$$

On solving equation (i) and (ii) $\theta_0 = 49^\circ\text{C}$.

$$31. (d) \frac{50-45}{5} = K \left(\frac{50+45}{2} - \theta_0 \right) \dots (i)$$

$$\frac{45-41.5}{5} = K \left(\frac{45+41.5}{2} - \theta_0 \right) \dots (ii)$$

Solving equation (i) and (ii) we set $\theta_0 = 33.3^\circ\text{C}$.

$$32. (d) \frac{65.5-62.5}{1} = K \left(\frac{65.5+62.5}{2} - 22.5 \right) \Rightarrow K = \frac{3}{41.5}$$

$$\text{And again } \frac{46.5-40.5}{t} = \frac{3}{41.5} \left(\frac{46.5+40.5}{2} - 22.5 \right)$$

$$\Rightarrow \frac{6}{t} = \frac{3}{41.5} \times 21 \Rightarrow t = \frac{82}{21} \approx 4 \text{ minute.}$$

$$33. (a) \frac{62-50}{10} = K \left(\frac{62+50}{2} - 26 \right) \Rightarrow \frac{6}{5} = K \times 30 \Rightarrow K = \frac{1}{25}$$

$$\text{And, } \frac{50-\theta}{10} = \frac{1}{25} \left(\frac{50+\theta}{2} - 26 \right) \Rightarrow \theta = 42^\circ\text{C}$$

$$34. (c) \frac{90-60}{5} = K \left(\frac{90+60}{2} - 20 \right) \Rightarrow 6 = K \times 55 \Rightarrow K = \frac{6}{55}$$

$$\text{And, } \frac{60-30}{t} = \frac{6}{55} \left(\frac{60+30}{2} - 20 \right) \Rightarrow t = 11 \text{ minute.}$$

35. (a) According to Newton's law of cooling

$$\text{in first case, } \frac{75-65}{t} = K \left[\frac{75+65}{2} - 30 \right] \dots (i)$$

$$\text{in second case, } \frac{55-45}{t} = K \left[\frac{55+45}{2} - 30 \right] \dots (ii)$$

$$\text{Dividing eq. (i) by (ii) we get } \frac{5t}{10} = \frac{40}{20} \Rightarrow t = 4 \text{ minutes}$$

36. (b) According to Newton's law of cooling

$$\text{in first case, } \frac{80-50}{5} = K \left[\frac{80+50}{2} - 20 \right] \dots (i)$$

$$\text{in second case, } \frac{60-30}{t} = K \left[\frac{60+30}{2} - 20 \right] \dots (ii)$$

$$\text{Dividing equation (i) by (ii) we get, } \frac{t}{2} = \frac{45}{25} \Rightarrow t = 9 \text{ min.}$$

37. (b) According to Newton's law of cooling.

Critical Thinking Questions

$$1. (a) \frac{dQ}{dt} = \frac{KA\Delta\theta}{l}, \text{ For both rods } K, A \text{ and } \Delta\theta \text{ are same } \Rightarrow$$

$$\frac{dQ}{dt} \propto \frac{1}{l} \text{ So } \frac{(dQ/dt)_{\text{semi circular}}}{(dQ/dt)_{\text{straight}}} = \frac{l_{\text{straight}}}{l_{\text{semicircular}}} = \frac{2r}{\pi r} = \frac{2}{\pi}$$

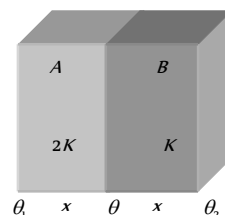
$$2. (b) \text{ Suppose thickness of each wall is } x \text{ then } \left(\frac{Q}{t} \right)_{\text{combination}} = \left(\frac{Q}{t} \right)_A \Rightarrow \frac{K_S A (\theta_1 - \theta_2)}{2x} = \frac{2KA(\theta_1 - \theta_2)}{x}$$

$$\therefore K_S = \frac{2 \times 2K \times K}{(2K + K)} = \frac{4}{3}K \text{ and } (\theta_1 - \theta_2) = 36^\circ$$

$$\Rightarrow \frac{\frac{4}{3}KA \times 36}{2x} = \frac{2KA(\theta_1 - \theta_2)}{x}$$

Hence temperature difference across wall A is

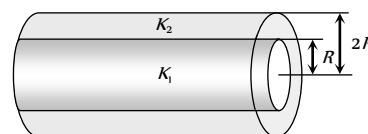
$$(\theta_1 - \theta) = 12^\circ\text{C}$$



$$3. (d) t = \frac{\rho L}{2K\theta} (x_2^2 - x_1^2) \Rightarrow t \propto (x_2^2 - x_1^2)$$

$$\Rightarrow \frac{t}{t'} = \frac{(x_2^2 - x_1^2)}{(x_2'^2 - x_1'^2)} \Rightarrow \frac{9}{t'} = \frac{(1^2 - 0^2)}{(2^2 - 1^2)} \Rightarrow t' = 21 \text{ hours}$$

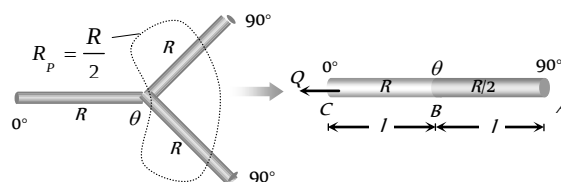
4. (c) Both the cylinders are in parallel, for the heat flow from one end as shown.



Hence $K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$; where A_1 = Area of cross-section of inner cylinder = πR^2 and A_2 = Area of cross-section of cylindrical shell = $\pi \{(2R)^2 - (R)^2\} = 3\pi R^2$

$$\Rightarrow K_{eq} = \frac{K_1(\pi R^2) + K_2(3\pi R^2)}{\pi R^2 + 3\pi R^2} = \frac{K_1 + 3K_2}{4}$$

5. (b) Let the temperature of junction be θ . Since rods B and C are parallel to each other (because both having the same temperature difference). Hence given figure can be redrawn as follows



$$\therefore \frac{Q}{t} = \frac{(\theta_1 - \theta_2)}{R} \text{ and } \left(\frac{Q}{t} \right)_{AB} = \left(\frac{Q}{t} \right)_{BC}$$

$$\Rightarrow \frac{(90 - \theta)}{R/2} = \frac{(\theta - 0)}{R} \Rightarrow 180 - 2\theta = \theta \Rightarrow \theta = 60^\circ\text{C}$$

$$6. (a) \text{ Heat developed by the heater } H = \frac{V^2}{R} \cdot \frac{t}{J} = \frac{(200)^2 \times t}{20 \times 4.2}$$

$$\text{Heat conducted by the glass } H = \frac{0.2 \times 1 \times (20 - \theta)t}{0.002}$$

$$\text{Hence } \frac{(200)^2 \times t}{20 \times 4.2} = \frac{0.2 \times (20 - \theta)t}{0.002} \Rightarrow \theta = 15.24^\circ\text{C}$$

7. (a) Since $t = \frac{\rho L}{2k\theta}(x_2^2 - x_1^2)$

$$\therefore t = \frac{\rho L}{2k\theta}(x^2 - y^2) = \frac{\rho L(x+y)(x-y)}{2K\theta}$$

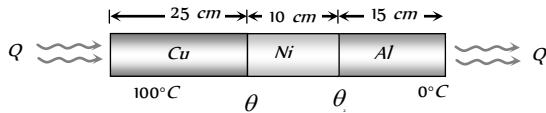
8. (b) If suppose $K_{Ni} = K \Rightarrow K_{Al} = 3K$ and $K_{Cu} = 6K$.

Since all metal bars are connected in series

$$\text{So } \left(\frac{Q}{t}\right)_{\text{Combination}} = \left(\frac{Q}{t}\right)_{Cu} = \left(\frac{Q}{t}\right)_{Al} = \left(\frac{Q}{t}\right)_{Ni}$$

$$\text{and } \frac{3}{K_{eq}} = \frac{1}{K_{Cu}} + \frac{1}{K_{Al}} + \frac{1}{K_{Ni}} = \frac{1}{6K} + \frac{1}{3K} + \frac{1}{K} = \frac{9}{6K}$$

$$\Rightarrow K_{eq} = 2K$$



$$\text{Hence, if } \left(\frac{Q}{t}\right)_{\text{Combination}} = \left(\frac{Q}{t}\right)_{Cu}$$

$$\Rightarrow \frac{K_{eq} A(100 - 0)}{l_{\text{Combination}}} = \frac{K_{Cu} A(100 - \theta_1)}{l_{Cu}}$$

$$\Rightarrow \frac{2K A(100 - 0)}{(25 + 10 + 15)} = \frac{6K A(100 - \theta_1)}{25} \Rightarrow \theta_1 = 83.33^\circ\text{C}$$

$$\text{Similar if } \left(\frac{Q}{t}\right)_{\text{Combination}} = \left(\frac{Q}{t}\right)_{Al}$$

$$\Rightarrow \frac{2K A(100 - 0)}{50} = \frac{3K A(\theta_2 - 0)}{15} \Rightarrow \theta_2 = 20^\circ\text{C}$$

9. (b) $\because T_B > T_A \Rightarrow$ Heat will flow B to A via two paths (i) B to A (ii) and along BCA as shown.

Rate of flow of heat in path BCA will be same

$$\text{i.e. } \left(\frac{Q}{t}\right)_{BC} = \left(\frac{Q}{t}\right)_{CA}$$

$$\Rightarrow \frac{k(\sqrt{2}T - T_C)A}{a} = \frac{k(T_C - T)A}{\sqrt{2}a}$$

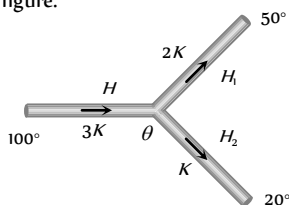
$$\Rightarrow \frac{T_C}{T} = \frac{3}{1 + \sqrt{2}}$$

10. (a) $mL = \frac{KA\Delta\theta}{\Delta x} t \Rightarrow 500 \times 80 = \frac{0.0075 \times 75 \times (40 - 0)t}{5}$

$$\Rightarrow t = 8.9 \times 10^3 \text{ sec} = 2.47 \text{ hr.}$$

11. (a) Rate of cooling $\frac{\Delta\theta}{t} = \frac{A\sigma(T^4 - T_0^4)}{mc} \Rightarrow \frac{\Delta\theta}{t} \propto A$. Since area of plate is largest so it will cool fastest.

12. (b) Let the temperature of junction be θ then according to following figure.

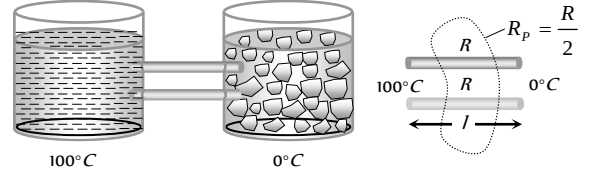


$$H = H_1 + H_2$$

$$\Rightarrow \frac{3K \times A \times (100 - \theta)}{l} = \frac{2KA(\theta - 50)}{l} + \frac{KA(\theta - 20)}{l}$$

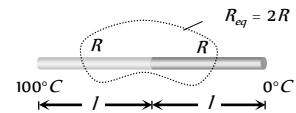
$$\Rightarrow 300 - 3\theta = 3\theta - 120 \Rightarrow \theta = 70^\circ\text{C}$$

13. (c) Initially the rods are placed in vessels as shown below



$$\frac{Q}{t} = \frac{(\theta_1 - \theta_2)}{R} \Rightarrow \left(\frac{Q}{t}\right)_1 = \frac{mL}{t} = q_1 L = \frac{(100 - 0)}{\frac{R}{2}} \quad \dots (i)$$

Finally when rods are joined end to end as shown



$$\Rightarrow \left(\frac{Q}{t}\right)_2 = \frac{mL}{t} = q_2 L = \frac{(100 - 0)}{2R} \quad \dots (ii)$$

From equation (i) and (ii), $\frac{q_1}{q_2} = \frac{4}{1}$

14. (b) Rate of cooling of a body $R = \frac{\Delta\theta}{t} = \frac{A\sigma(T^4 - T_0^4)}{mc}$

$$\Rightarrow R \propto \frac{A}{m} \propto \frac{\text{Area}}{\text{Volume}}$$

$$\Rightarrow \text{For the same surface area, } R \propto \frac{1}{\text{Volume}}$$

$\therefore$ Volume of cube < Volume of sphere

$\Rightarrow R_{\text{Cube}} > R_{\text{Sphere}}$ i.e. cube, cools down with faster rate.

15. (a,b) According to Stefan's law

$$E = eA\sigma T^4 \Rightarrow E_1 = e_1 A \sigma T_1^4 \text{ and } E_2 = e_2 A \sigma T_2^4$$

$$\therefore E_1 = E_2 \therefore e_1 T_1^4 = e_2 T_2^4$$

$$\Rightarrow T_2 = \left(\frac{e_1}{e_2} T_1^4\right)^{\frac{1}{4}} = \left(\frac{1}{81} \times (5802)^4\right)^{\frac{1}{4}} \Rightarrow T_2 = 1934 \text{ K}$$

And, from Wein's law $\lambda_A \times T_A = \lambda_B \times T_B$

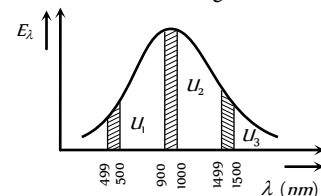
$$\Rightarrow \frac{\lambda_A}{\lambda_B} = \frac{T_B}{T_A} \Rightarrow \frac{\lambda_B - \lambda_A}{\lambda_B} = \frac{T_A - T_B}{T_A}$$

$$\Rightarrow \frac{1}{\lambda_B} = \frac{5802 - 1934}{5802} = \frac{3968}{5802} \Rightarrow \lambda_B = 1.5 \mu\text{m}$$

16. (d) Wein's displacement law is $\lambda_m T = b$

$$\Rightarrow \lambda_m = \frac{b}{T} = \frac{2.88 \times 10^6}{2880} = 1000 \text{ nm.}$$

Energy distribution with wavelength will be as follows



From the graph it is clear that $U > U$.

17. (b) Energy received per second i.e., power $P \propto (T^4 - T_0^4)$

$$\Rightarrow P \propto T^4 \quad (\because T_0 \ll T)$$

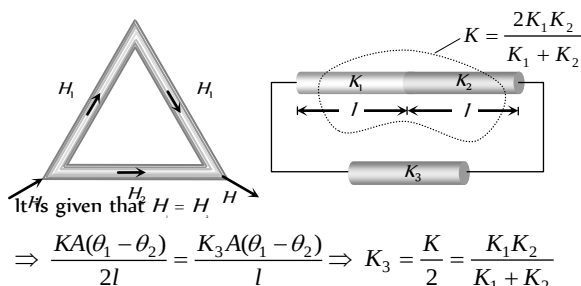
Also energy received per sec $(p) \propto \frac{1}{d^2}$

(inverse square law)

$$\Rightarrow P \propto \frac{T^4}{d^2} \Rightarrow \frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4 \times \left(\frac{d_2}{d_1}\right)^2$$

$$\Rightarrow \frac{P}{P_2} = \left(\frac{T}{2T}\right)^2 \times \left(\frac{2d}{d}\right)^2 = \frac{1}{4} \Rightarrow P_2 = 4P.$$

18. (c) The given arrangement of rods can be redrawn as follows



19. (b) Rate of cooling $(R) = \frac{\Delta\theta}{t} = \frac{A\sigma(T^4 - T_0^4)}{mc}$

$$\Rightarrow R \propto \frac{A}{m} \propto \frac{\text{Area}}{\text{volume}} \propto \frac{r^2}{r^3} \propto \frac{1}{r}$$

$$\Rightarrow \text{Rate } (R) \propto \frac{1}{r} \propto \frac{1}{m^{1/3}} \left[\because m = \rho \times \frac{4}{3}\pi r^3 \Rightarrow r \propto m^{1/3} \right]$$

$$\Rightarrow \frac{R_1}{R_2} = \left(\frac{m_2}{m_1}\right)^{1/3} = \left(\frac{1}{3}\right)^{1/3}$$

20. (b) Radiated power $P = A\sigma T^4 \Rightarrow P \propto AT^4$

From Wein's law, $\lambda_m T = \text{constant} \Rightarrow T \propto \frac{1}{\lambda_m}$

$$\therefore P \propto \frac{A}{(\lambda_m)^4} \propto \frac{r^2}{(\lambda_m)^4}$$

$$\Rightarrow Q_A : Q_B : Q_C = \frac{2^2}{(300)^4} : \frac{4^2}{(400)^4} : \frac{6^2}{(500)^4}$$

$\therefore Q_B$ will be maximum.

21. (d) The total energy radiated from a black body per minute.

$$Q \propto T^4 \Rightarrow \frac{Q_2}{Q_1} = \left(\frac{2T}{T}\right)^4 = 16 \Rightarrow Q_2 = 16Q_1$$

If m be mass of water taken and S be its specific heat capacity, then $Q_1 = ms(20.5 - 20)$ and $Q_2 = ms(\theta - 20)$

$\theta^\circ\text{C}$ = Final temperature of water

$$\Rightarrow \frac{Q_2}{Q_1} = \frac{\theta - 20}{0.5} \Rightarrow \frac{16}{1} = \frac{\theta - 20}{0.5} \Rightarrow \theta = 28^\circ\text{C}$$

22. (a) Rate of cooling $\frac{\Delta\theta}{t} = \frac{A\sigma(T^4 - T_0^4)}{mc}$

As surface area, material and temperature difference are same, so rate of loss of heat is same in both the spheres. Now in this case rate of cooling depends on mass.

$$\Rightarrow \text{Rate of cooling } \frac{\Delta\theta}{t} \propto \frac{1}{m}$$

$\therefore m_{\text{solid}} > m_{\text{hollow}}$. Hence hollow sphere will cool fast.

23. (c) Rate of cooling $\frac{\Delta\theta}{t} = \frac{A\sigma(T^4 - T_0^4)}{mc}$

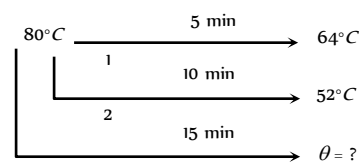
$$\Rightarrow t \propto \frac{m}{A} \quad [\because \Delta\theta, t, \sigma, (T^4 - T_0^4) \text{ are constant}]$$

$$\Rightarrow t \propto \frac{m}{A} \propto \frac{\text{Volume}}{\text{Area}} \propto \frac{a^3}{a^2} \Rightarrow t \propto a \Rightarrow \frac{t_1}{t_2} = \frac{a_1}{a_2}$$

$$\Rightarrow \frac{100}{t_2} = \frac{1}{2} \Rightarrow t_2 = 200\text{sec}.$$

24. (a) According to Newton law of cooling

$$\frac{\theta_1 - \theta_2}{t} = K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$$



$$\text{For first process : } \frac{(80 - 64)}{5} = K \left[\frac{80 + 64}{2} - \theta_0 \right] \quad \dots(i)$$

$$\text{For second process : } \frac{(80 - 52)}{10} = K \left[\frac{80 + 52}{2} - \theta_0 \right] \quad \dots(ii)$$

$$\text{For third process : } \frac{(80 - \theta)}{15} = K \left[\frac{80 + \theta}{2} - \theta_0 \right] \quad \dots(iii)$$

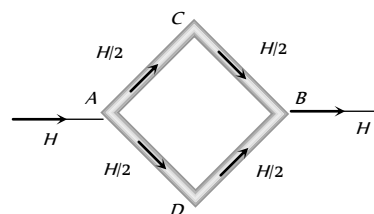
On solving equation (i) and (ii) we get $K = \frac{1}{15}$ and

$\theta_0 = 24^\circ\text{C}$. Putting these values in equation (iii) we get $\theta = 42.7^\circ\text{C}$

25. (c) $t = \frac{Ql}{KA(\theta_1 - \theta_2)} = \frac{mLl}{KA(\theta_1 - \theta_2)} = \frac{V\rho Ll}{KA(\theta_1 - \theta_2)}$

$$= \frac{5 \times A \times 0.92 \times 80 \times \frac{5+10}{2}}{0.004 \times A \times 10 \times 3600} = 19.1 \text{ hours}.$$

26. (a) Suppose temperature difference between A and B is 100°C and $\theta_A > \theta_B$.



Heat current will flow from A to B via path ACB and ADB. Since all the rod are identical so $(\Delta\theta)_+ = (\Delta\theta)_-$

(Because heat current $H = \frac{\Delta\theta}{R}$; here R = same for all.)

$$\Rightarrow \theta_A - \theta_C = \theta_A - \theta_D \Rightarrow \theta_C = \theta_D$$

i.e. temperature difference between C and D will be zero.

$$27. \quad (c) \quad \frac{Q}{t} = \frac{KA\Delta\theta}{l} \Rightarrow \frac{mL}{t} = \frac{K(\pi r^2)\Delta\theta}{l}$$

$$\Rightarrow \text{Rate of melting of ice} \left(\frac{m}{t}\right) \propto \frac{Kr^2}{l}$$

Since for second rod K becomes $\frac{1}{4}$ th r becomes double and length becomes half, so rate of melting will be twice i.e.

$$\left(\frac{m}{t}\right)_2 = 2\left(\frac{m}{t}\right)_1 = 2 \times 0.1 = 0.2 \text{ gm/sec.}$$

$$28. \quad (c) \quad \text{Heat transferred in one minute is utilised in melting the ice so,}$$

$$\frac{KA(\theta_1 - \theta_2)t}{l} = m \times L$$

$$\Rightarrow m = \frac{10^{-3} \times 92 \times (100 - 0) \times 60}{1 \times 8 \times 10^4} = 6.9 \times 10^{-3} \text{ kg}$$

$$29. \quad (d) \quad \frac{dQ}{dt} = \frac{KA}{l} d\theta = \frac{0.01 \times 1}{0.05} \times 30 = 6/\text{sec}$$

Heat transferred in one day (86400 sec)

$$\theta = 6 \times 86400 = 518400 \text{ J}$$

$$\text{Now } Q = mL \Rightarrow m = \frac{Q}{L} = \frac{518400}{334 \times 10^3}$$

$$= 1.552 \text{ kg} = 1552 \text{ g.}$$

$$30. \quad (b) \quad \text{For no current flow between } C \text{ and } D$$

$$\left(\frac{Q}{t}\right)_{AC} = \left(\frac{Q}{t}\right)_{CB} \Rightarrow \frac{K_1 A(\theta_A - \theta_C)}{l} = \frac{K_2 A(\theta_C - \theta_B)}{l}$$

$$\Rightarrow \frac{\theta_A - \theta_C}{\theta_C - \theta_B} = \frac{K_2}{K_1} \quad \dots(i)$$

$$\text{Also } \left(\frac{Q}{t}\right)_{AD} = \left(\frac{Q}{t}\right)_{DB} \Rightarrow \frac{K_3 A(\theta_A - \theta_D)}{l} = \frac{K_4 A(\theta_D - \theta_B)}{l}$$

$$\Rightarrow \frac{\theta_A - \theta_D}{\theta_D - \theta_B} = \frac{K_4}{K_3} \quad \dots(ii)$$

It is given that $\theta_C = \theta_D$, hence from equation (i) and (ii) we

$$\text{get } \frac{K_2}{K_1} = \frac{K_4}{K_3} \Rightarrow K_1 K_4 = K_2 K_3$$

$$31. \quad (d) \quad \text{Rate of cooling } R_C = \frac{d\theta}{dt} = \frac{A\epsilon\sigma(T^4 - T_0^4)}{mc}$$

$$\Rightarrow \frac{d\theta}{dt} \propto \frac{A}{V} \propto \frac{r^2}{r^3} \Rightarrow \frac{d\theta}{dt} \propto \frac{1}{r}$$

$$32. \quad (b) \quad \frac{dT}{dt} = \frac{\sigma A}{mcJ} (T^4 - T_0^4) \quad [\text{In the given problem fall in temperature of body } dT = (200 - 100) = 100\text{K, temp. of surrounding } T_0 = 0\text{K, Initial temperature of body } T = 200\text{K].}$$

$$\frac{100}{dt} = \frac{\sigma 4\pi r^2}{\frac{4}{3}\pi r^3 \rho c J} (200^4 - 0^4)$$

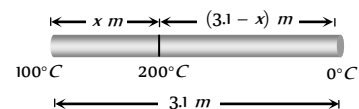
$$\Rightarrow dt = \frac{r\rho c J}{48\sigma} \times 10^{-6} \text{ s} = \frac{r\rho c}{\sigma} \cdot \frac{4.2}{48} \times 10^{-6}$$

$$= \frac{7}{80} \frac{r\rho c}{\sigma} \mu\text{s} \approx \frac{7}{72} \frac{r\rho c}{\sigma} \mu\text{s} \quad [\text{As } J = 4.2]$$

$$33. \quad (a) \quad \text{Rate of flow of heat is given by } \frac{dQ}{dt} = \frac{\Delta\theta}{l/K A} \text{ also}$$

$$\frac{dQ}{dt} = L \frac{dm}{dt} \quad (\text{where } L = \text{Latent heat})$$

$$\Rightarrow \frac{dm}{dt} = \frac{KA}{l} \left(\frac{\Delta\theta}{L}\right). \text{ Let the desired point is at a distance } x \text{ from water at } 100^\circ\text{C.}$$



$\therefore$ Rate of ice melting = Rate at which steam is being produced

$$\Rightarrow \left(\frac{dm}{dt}\right)_{\text{Steam}} = \left(\frac{dm}{dt}\right)_{\text{Ice}} \Rightarrow \left(\frac{\Delta\theta}{Ll}\right)_{\text{Steam}} = \left(\frac{\Delta\theta}{Ll}\right)_{\text{Ice}}$$

$$\Rightarrow \frac{(200 - 100)}{540 \times x} = \frac{(200 - 0)}{80(3.1 - x)} \Rightarrow x = 0.4 \text{ m} = 40 \text{ cm}$$

$$34. \quad (c) \quad Q = \sigma A t (T - T_0)$$

If T , T_0 , σ and t are same for both bodies then

$$\frac{Q_{\text{sphere}}}{Q_{\text{cube}}} = \frac{A_{\text{sphere}}}{A_{\text{cube}}} = \frac{4\pi r^2}{6a^2} \quad \dots(i)$$

But according to problem, volume of sphere = Volume of cube

$$\Rightarrow \frac{4}{3}\pi r^3 = a^3 \Rightarrow a = \left(\frac{4}{3}\pi\right)^{1/3} r$$

Substituting the value of a in equation (i) we get

$$\frac{Q_{\text{sphere}}}{Q_{\text{cube}}} = \frac{4\pi r^2}{6a^2} = \frac{4\pi r^2}{6\left\{\left(\frac{4}{3}\pi\right)^{1/3} r\right\}^2}$$

$$= \frac{4\pi r^2}{6\left(\frac{4}{3}\pi\right)^{2/3} r^2} = \left(\frac{\pi}{6}\right)^{1/3} : 1$$

$$35. \quad (d) \quad \text{Equation of thermal conductivity of the given combination}$$

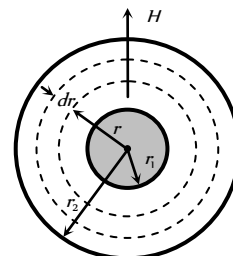
$$K_{eq} = \frac{l_1 + l_2}{\frac{l_1}{K_1} + \frac{l_2}{K_2}} = \frac{x + 4x}{\frac{x}{K} + \frac{4x}{2K}} = \frac{5}{3} K. \text{ Hence rate of flow of}$$

heat through the given combination is

$$\frac{Q}{t} = \frac{K_{eq} A (T_2 - T_1)}{(x + 4x)} = \frac{\frac{5}{3} K A (T_2 - T_1)}{5x} = \frac{1}{3} \frac{K A (T_2 - T_1)}{x}$$

On comparing it with given equation we get $f = \frac{1}{3}$

$$36. \quad (a) \quad \text{Consider a concentric spherical shell of radius } r \text{ and thickness } dr \text{ as shown in fig.}$$



The radial rate of flow of heat through this shell in steady state

$$\text{will be } H = \frac{dQ}{dt} = -KA \frac{dT}{dr} = -K(4\pi r^2) \frac{dT}{dr}$$

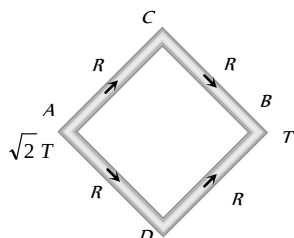
$$\Rightarrow \int_{r_1}^{r_2} \frac{dr}{r^2} = -\frac{4\pi K}{H} \int_{T_1}^{T_2} dT$$

Which on integration and simplification gives

$$H = \frac{dQ}{dt} = \frac{4\pi K r_1 r_2 (T_1 - T_2)}{r_2 - r_1} \Rightarrow \frac{dQ}{dt} \propto \frac{r_1 r_2}{(r_2 - r_1)}$$

37. (c) Similar to Q.No.26

Temperature difference between C and D is zero.



Graphical Questions

1. (c) Rate of cooling $\left(-\frac{dT}{dt}\right) \propto \text{emissivity } (e)$

$$\text{From graph, } \left(-\frac{dT}{dt}\right)_x > \left(-\frac{dT}{dt}\right)_y \Rightarrow e_x > e_y$$

Further emissivity $(e) \propto \text{Absorptive power } (a) \Rightarrow a_x > a_y$
($\because$ good absorbers are good emitters).

2. (b) According to Wien's law $\lambda_m \propto \frac{1}{T}$ and from the figure

$$(\lambda_m)_1 < (\lambda_m)_3 < (\lambda_m)_2 \text{ therefore } T_1 > T_3 > T_2$$

3. (d) $\frac{A_T}{A_{2000}} = \frac{16}{1}$ (given)

Area under $e_\lambda - \lambda$ curve represents the emissive power of body and emissive power $\propto T^4$

(Hence area under $e_\lambda - \lambda$ curve) $\propto T^4$

$$\Rightarrow \frac{AT}{A_{2000}} = \left(\frac{T}{2000}\right)^4 \Rightarrow \frac{16}{1} = \left(\frac{T}{2000}\right)^4 \Rightarrow T = 4000K.$$

4. (c) According to Wein's law $\lambda_m \propto \frac{1}{T} \Rightarrow \nu_m \propto T$. As the temperature of body increases, frequency corresponding to maximum energy in radiation (ν) increases this is shown in graph (c).

5. (c) According to Wein's displacement law.

6. (b) For $\theta-t$ plot, rate of cooling $= \frac{d\theta}{dt}$ = slope of the curve.

$$\text{At } P, \frac{d\theta}{dt} = \tan \phi_2 = k(\theta_2 - \theta_0), \text{ where } k = \text{constant.}$$

$$\text{At } Q, \frac{d\theta}{dt} = \tan \phi_1 = k(\theta_1 - \theta_0) \Rightarrow \frac{\tan \phi_2}{\tan \phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$$

7. (a) According to Wein's displacement law

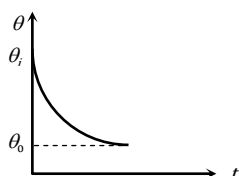
$$\lambda_m \propto \frac{1}{T} \Rightarrow \lambda_{m_2} < \lambda_{m_1} \quad (\because T_1 < T_2)$$

There fore $\lambda - \lambda$ graph for T have lesser wavelength (λ) and so curve for T_1 will shift towards left side.

8. (d) Area under given curve represents emissive power and emissive power $\propto T \Rightarrow A \propto T^4$

$$\Rightarrow \frac{A_2}{A_1} = \frac{T_2^4}{T_1^4} = \frac{(273 + 327)^4}{(273 + 27)^4} = \left(\frac{600}{300}\right)^4 = \frac{16}{1}$$

9. (b) According to Newton's law of cooling



Rate of cooling $\propto$ Temperature difference

$$\Rightarrow -\frac{d\theta}{dt} \propto (\theta - \theta_0) \Rightarrow -\frac{d\theta}{dt} = \alpha(\theta - \theta_0) \quad (\alpha = \text{constant})$$

$$\Rightarrow \int_{\theta_i}^{\theta} \frac{d\theta}{(\theta - \theta_0)} = -\alpha \int_0^t dt \Rightarrow \theta = \theta_0 + (\theta_i - \theta_0)e^{-\alpha t}$$

This relation tells us that, temperature of the body varies exponentially with time from θ_i to θ_0

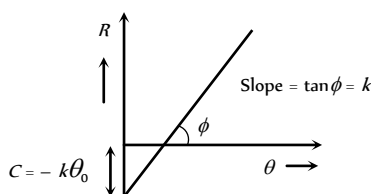
Hence graph (b) is correct.

10. (a) According to Wein's displacement law $\lambda_m \propto \frac{1}{T}$. Hence, if temperature increases λ_m decreases i.e., peak of the $E - \lambda$ curve shift towards left.

11. (c) Rate of loss of heat (R) $\propto$ temperature difference

$$\Rightarrow R \propto (\theta - \theta_0) \Rightarrow R = k(\theta - \theta_0) = k\theta - k\theta_0 \quad (k = \text{constant})$$

on comparing it with $y = mx + c$ it is observed that, the graph between R and θ will be straight line with slope $= k$ and intercept $= -k\theta_0$



12. (c) $\frac{dQ}{dt} = -KA \frac{d\theta}{dx}$

$$\therefore \frac{dQ}{dt}, K \text{ and } A \text{ are constants for all points}$$

$$\Rightarrow d\theta \propto -dx; \text{ i.e. temperature will decrease linearly with } x$$

13. (b) Since the curved surface of the conductor is thermally insulated, therefore, in steady state, the rate of flow of heat at every section will be the same. Hence the curve between H and x will be straight line parallel to x -axis.

14. (d) According to Stefan's law $E = \sigma T^4$

$$\Rightarrow \log E = \log \sigma + 4 \log T \Rightarrow \log E = 4 \log T + \log \sigma$$

on comparing this equations with $y = mx + C$

we find that graph between $\log E$ and $\log T$ will be a straight line, having positive slope ($m = 4$) and intercept on $\log E$ axis equal to $\log \sigma$

15. (c) $\frac{d\theta}{dt} = \frac{\varepsilon A \sigma}{mc} 4\theta_0^3 \Delta\theta$

For given sphere and cube $\frac{\varepsilon A \sigma}{mc} 4\theta_0^3 \Delta\theta$ is constant so for

both rate of fall of temperature $\frac{d\theta}{dt} = \text{constant}$

16. (b) $\lambda_m T = b$ where $b = 2.89 \times 10^{-3} \text{ mK}$

$$\Rightarrow T = \frac{b}{\lambda_m} = \frac{2.89 \times 10^{-3}}{1.5 \times 10^{-6}} \approx 2000 \text{ K}$$

17. (b) Wein's law $\lambda_m \propto \frac{1}{T}$ or $\nu_m \propto T$

ν_m increases with temperature. So the graph will be straight line.

Assertion and Reason

1. (a) According to Kirchhoff's law $\frac{e_\lambda}{a_\lambda} = E_\lambda$

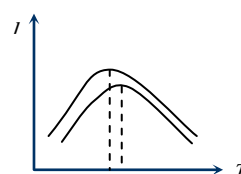
If for a particular wave length $E_\lambda = 1 \Rightarrow e_\lambda = a_\lambda$ i.e., absorptivity of a body is equal to its emissivity. This statement also reveals that a good radiator is also a good absorber and vice versa.

2. (c) According to Weins law $\lambda_m T = \text{constant}$ i.e., peak emission

wavelength $\lambda_m \propto \frac{1}{T}$. Also

as T increases λ_m decreases.

Hence assertion is true but reason is false.



3. (b) During the day when water is cooler than the land, the wind blows off the water onto the land (as warm air rises and cooler air fills the place). Also at night, the effect is reversed (since the water is usually warmer than the surrounding air on land). Due to this wind flow the temperature near the sea coast remains moderate.

4. (c) Heat is carried away from a fire sideways mainly by radiations. Above the fire, heat is carried by both radiation and by convection of air. The latter process carries much more heat.

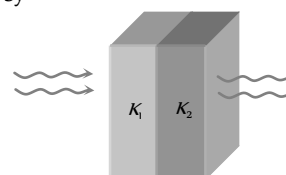


5. (e) Assertion is false because at absolute zero (0 K), heat is neither radiates nor absorbed. Reason is the statement of Stefan's law, as $E \propto T^4$.

6. (a) Woolen fibres encloses a large amount of air in them. Both wool and air are the bad conductors of heat and the coefficient of thermal conductivity is small. So, they prevent any loss of heat from our body.

7. (d) Equivalent thermal conductivity of two equally thick plates in series combination is given by

$$\frac{2}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$



If $K_1 < K_2$

then $K_1 < K < K_2$

Hence assertion and reason both are false.

8. (c) Hollow metallic closed container maintained at a uniform temperature can act as source of black body. It is also well-known that all metals cannot act as black body because if we take a highly metallic polished surface. It will not behave as a perfect black body.

9. (b) This is in accordance with the Stefan's law $E \propto T^4$.

10. (c) At a high temperature (6000 K), the sun acts like a perfect blackbody emitting complete radiation. That's why the radiation coming from the sun's surface follows Stefan's law $E = \sigma T^4$.

11. (a) From Wein's displacement law, temperature $(T) \propto 1/\lambda_m$ (where λ_m is the maximum wavelength). Thus temperature of a body is inversely proportional to the wavelength. Since blue star has smaller wavelength and red star has maximum wavelength, therefore blue star is at higher temperature than red star.

12. (b) From the definition heat flow, $Q = \frac{KA \Delta \theta t}{l}$

$$\text{Thermal conductivity } K = \frac{\theta \times l}{A \times \Delta \theta \times t}$$

$$\Rightarrow K = \frac{J \times m}{m^2 \times K \times \text{sec}} = \frac{\text{watt}}{m \times K}$$

If thermal conductivity of a substance is high, it will pass more heat.

13. (c) The thermal conductivity of brass is high *i.e.*, brass is a good conductor of heat. So, when a brass tumbler is touched, heat quickly flows from human body to tumbler. Consequently, the tumbler appears colder, on the other hand wood is a bad conductor. So, heat does not flow from the human body to the wooden tray in this case. Thus it appears comparatively hotter.
14. (b) Light radiations and thermal radiations both belong to electromagnetic spectrum. Light radiations belong to visible region while thermal radiation belongs to infrared region of EM spectrum. Also EM radiations require no medium for propagation.
15. (a) When the temperature of the atmosphere reaches below 0°C , then the water vapours present in air, instead of condensing, freeze directly in the form of minute particles of ice. Many particles coalesce and take cotton-like shape which is called snow. Thus snow contains air packets in which convection currents cannot be formed. Hence snow is a good heat insulator. In ice there is no air, so it is a bad insulator.
16. (e) In the process of convection, the liquid at the bottom, becoming lighter, rises up. Thus the basis of convection is the difference in weight and upthrust. In weightlessness, this difference does not exist. So convection is not possible.
17. (b) Both assertion and reason are true but reason is not correctly explaining the assertion.

18. (c) The 98.4°F is the standard body temperature of a man. If a man touch a iron or wooden ball at 98.4°F , no heat transfer takes place between ball and man, so both the balls would feel equally hot for the man.

19. (c) According to Wien's displacement law the $\lambda_m \propto \frac{1}{T}$.

Hence assertion is true but reason is false.

20. (e) It is not necessary that all black coloured objects are black bodies. For example, if we take a black surface which is highly polished, it will not behave as a perfect black body.

A perfectly black body absorbs all the radiations incident on it.

21. (b) By definition, $R = \frac{(\theta_1 - \theta_2)}{Q/t} = \frac{l}{KA} \Rightarrow R \propto \frac{1}{K}$.

22. (c) Actually, the process of radiation does not require any material for transmission of heat.

Thermal radiation travels with the velocity of light and hence the fastest mode of the transfer. Thermal radiation is always transmitted in a straight line.

23. (c) Two thin blankets put together are more warm because an insulating layer of air (as air is good insulator of heat) is enclosed between two blankets due to which it gives more warmth.

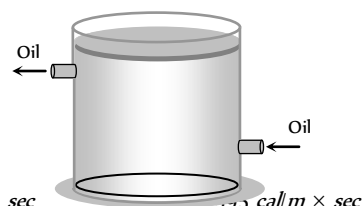
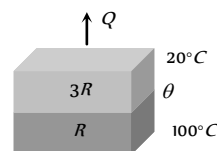
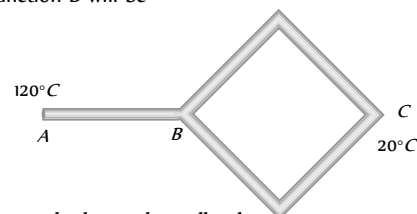
24. (a) When the animals feel cold, they curl their bodies into a ball so as to decrease the surface area of their bodies. As total energy radiated by body varies directly as the surface area of the body, the loss of heat due to radiation would be reduced.



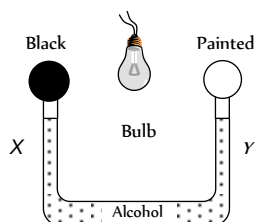
Transmission of Heat

Self Evaluation Test -15

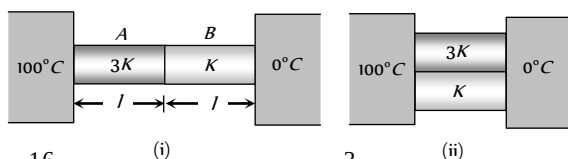
- A rod of 40 cm in length and temperature difference of 80°C at its two ends. Another rod B of length 60 cm and of temperature difference 90°C , having the same area of cross-section. If the rate of flow of heat is the same, then the ratio of their thermal conductivities will be
 (a) 3 : 4 (b) 4 : 3
 (c) 1 : 2 (d) 2 : 1
- Two vessels of different materials are similar in size in every respect. The same quantity of ice filled in them gets melted in 20 minutes and 40 minutes respectively. The ratio of thermal conductivities of the materials is [AFMC 1998]
 (a) 5 : 6 (b) 6 : 5
 (c) 3 : 1 (d) 2 : 1
- In a steady state of thermal conduction, temperature of the ends A and B of a 20 cm long rod are 100°C and 0°C respectively. What will be the temperature of the rod at a point at a distance of 6 cm from the end A of the rod
 (a) -30°C (b) 70°C
 (c) 5°C (d) None of the above
- Four rods of silver, copper, brass and wood are of same shape. They are heated together after wrapping a paper on it, the paper will burn first on
 (a) Silver (b) Copper
 (c) Brass (d) Wood
- The two opposite faces of a cubical piece of iron (thermal conductivity = 0.2 CGS units) are at 100°C and 0°C in ice. If the area of a surface is 4cm^2 , then the mass of ice melted in 10 minutes will be
 (a) 30 gm (b) 300 gm
 (c) 5 gm (d) 50 gm
- Wein's constant is 2892×10^{-6} MKS unit and the value of λ_m from moon is 14.46 microns. What is the surface temperature of moon
 (a) 100 K (b) 300 K
 (c) 400 K (d) 200 K
- If at temperature $T_1 = 1000\text{K}$, the wavelength is $1.4 \times 10^{-6}\text{m}$, then at what temperature the wavelength will be $2.8 \times 10^{-6}\text{m}$
 (a) 2000 K (b) 500 K
 (c) 250 K (d) None of these
- The wavelength of maximum intensity of radiation emitted by a star is 289.8 nm. The radiation intensity for the star is : (Stefan's constant $5.67 \times 10^{-8}\text{Wm}^{-2}\text{K}^{-4}$, constant $b = 2898\mu\text{mK}$) -
 (a) $5.67 \times 10^8\text{W/m}^2$ (b) $5.67 \times 10^{12}\text{W/m}^2$
 (c) $10.67 \times 10^7\text{W/m}^2$ (d) $10.67 \times 10^{14}\text{W/m}^2$
- Two friends A and B are waiting for another friend for tea. A took the tea in a cup and mixed the cold milk and then waits. B took the tea in the cup and then mixed the cold milk when the friend comes. Then the tea will be hotter in the cup of
 (a) A
 (b) B
 (c) Tea will be equally hot in both cups
 (d) Friend's cup
- There are two spherical balls A and B of the same material with same surface, but the diameter of A is half that of B. If A and B are heated to the same temperature and then allowed to cool, then
 (a) Rate of cooling is same in both
 (b) Rate of cooling of A is four times that of B
 (c) Rate of cooling of A is twice that of B
 (d) Rate of cooling of A is $\frac{1}{4}$ times that of B
- Five identical rods are joined as shown in figure. Point A and C are maintained at temperature 120°C and 20°C respectively. The temperature of junction B will be
 (a) 100°C
 (b) 80°C
 (c) 70°C
 (d) 0°C
- Can we boil water inside the earth satellite by convection
 (a) Yes
 (b) No
 (c) Nothing can be said
 (d) In complete information is given
- In the following figure, two insulating sheets with thermal resistances R and 3R as shown in figure. The temperature θ is
 (a) 20°C
 (b) 60°C
 (c) 75°C
 (d) 80°C [RPMT 2004]
- The top of insulated cylindrical container is covered by a disc having emissivity 0.6 and thickness 1 cm. The temperature is maintained by circulating oil as shown in figure. If temperature of upper surface of disc is 127°C and temperature of surrounding is 27°C , then the radiation loss to the surroundings will be (Take $\sigma = \frac{17}{3} \times 10^{-8}\text{W/m}^2\text{K}^4$) [IIT-JEE 2001]
 (a) $595\text{J/m} \times \text{sec}$
 (c) $991.0\text{J/m} \times \text{sec}$



15. The following figure shows two air-filled bulbs connected by a U-tube partly filled with alcohol. What happens to the levels of alcohol in the limbs X and Y when an electric bulb placed midway between the bulbs is lighted



- (a) The level of alcohol in limb X falls while that in limb Y rises
(b) The level of alcohol in limb X rises while that in limb Y falls
(c) The level of alcohol falls in both limbs
(d) There is no change in the levels of alcohol in the two limbs
16. Two conducting rods A and B of same length and cross-sectional area are connected (i) In series (ii) In parallel as shown. In both combination a temperature difference of 100°C is maintained. If thermal conductivity of A is $3K$ and that of B is K then the ratio of heat current flowing in parallel combination to that flowing in series combination is



- (a) $\frac{16}{3}$ (b) $\frac{3}{16}$
(c) $\frac{1}{1}$ (d) $\frac{1}{3}$
17. The area of the glass of a window of a room is 10 m^2 and thickness 2 mm . The outer and inner temperature are 40°C and 20°C respectively. Thermal conductivity of glass in MKS system is 0.2 . The heat flowing in the room per second will be

- (a) $3 \times 10^4\text{ joules}$ (b) $2 \times 10^4\text{ joules}$
(c) 30 joules (d) 45 joules
18. The spectrum from a black body radiation is a

[MP PMT 1989; RPET 2000]

- (a) Line spectrum
(b) Band spectrum
(c) Continuous spectrum
(d) Line and band spectrum both
19. The Wien's displacement law express relation between

[CBSE PMT 2002]

- (a) Frequency and temperature
(b) Temperature and amplitude
(c) Wavelength and radiating power of black body
(d) Wavelength corresponding to maximum energy and temperature

20. A black body is heated from 27°C to 127°C . The ratio of their energies of radiations emitted will be

[AIIMS 2001]

- (a) 3 : 4 (b) 9 : 16
(c) 27 : 64 (d) 81 : 256

21. A body takes T minutes to cool from 62°C to 61°C when the surrounding temperature is 30°C . The time taken by the body to cool from 46°C to 45.5°C is

[MP PET 1999]

- (a) Greater than T minutes
(b) Equal to T minutes
(c) Less than T minutes
(d) Equal to $T/2$ minutes

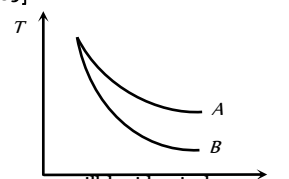
22. A partition wall has two layers A and B in contact, each made of a different material. They have the same thickness but the thermal conductivity of layer A is twice that of layer B . If the steady state temperature difference across the wall is 60 K , then the corresponding difference across the layer A is

[SCRA 1994; JIPMER 2001]

- (a) 10 K (b) 20 K
(c) 30 K (d) 40 K

23. Water and turpentine oil (specific heat less than that of water) are both heated to same temperature. Equal amounts of these placed in identical calorimeters are then left in air

[MP PMT 1989]



- (a) Their cooling curves will be identical
(b) A and B will represent cooling curves of water and oil respectively
(c) B and A will represent cooling curves of water and oil respectively
(d) None of the above

AS Answers and Solutions

(SET -15)

1. (a) $\frac{dQ}{dt} = \frac{KA(\theta_1 - \theta_2)}{d}$
 $\Rightarrow \frac{K_1 \Delta \theta_1}{l_1} = \frac{K_2 \Delta \theta_2}{l_2}$ ($\because \frac{dQ}{dt}$ and A are same)
 $\Rightarrow \frac{K_1 \times 80}{40} = \frac{K_2 \times 90}{60} \Rightarrow \frac{K_1}{K_2} = \frac{3}{4}$

2. (d) $\frac{Q}{t} = \frac{KA(\theta_1 - \theta_2)}{l} \Rightarrow \frac{mL}{t} = \frac{KA(\theta_1 - \theta_2)}{l}$
 $\Rightarrow K \propto \frac{1}{t}$ ($\because$ remaining quantities are same)
 $\Rightarrow \frac{K_1}{K_2} = \frac{t_2}{t_1} = \frac{40}{20} = \frac{2}{1}$

3. (b) In steady state, temperature gradient = constant